GOVERNMENT OF MADHYA PRADESH

PUBLIC WORKS DEPARTMENT







SCHEDULE OF RATES (SOR)
For BUILDING WORK
(VOLUME-II)

In Force From January, 1st 2024

Issued by :-

Engineer-in-Chief (Building)

MP Public Works Department

BHOPAL



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FOREWORD

Schedule of Rates for Building is issued by the Engineer-in-Chief, (Building) Public Works Department, Building from time to time. Since its inception the Public Work Department has issued Schedule of Rates for Building works at several occasion. Last such exercise was done by the Engineer-in-Chief, (Building) PWD, on 01-01-2024. Since this date, various amendments were required to be issued over the time. A part from this introduction of new item using new technologies and newer materials have necessitated the issuance of this new revised SOR. Due to increased size SOR is being released in two volumes **(Volume I & II).** Volume -I containing chapters 1 to 13 and Volume-II containing chapters 14 to 31.

Due care has been taken to frame this Schedule of Rates as correctly as possible. It is however, possible that some errors might have crept in. In case any error or omission is noticed, the same may be brought to the noticed of this office.

In order to simplify understanding of SOR, item name to each Item has been assigned. It is advised that cost abstract of the estimate may include only item number, item name, unit, rate, quantity and amount. This will enable easy at a glance reading of the estimate. However it always required to include full description of items in detailed estimate & BOQ.

Schedule of Testing Charges for Building Material (Civil) has also been added, which includes rates of testing of material from Departmental/Government Laboratory/National Accreditation Board for Testing and Calibration Laboratories (NAB).

The effort of **Mr. S. L. Suryavanshi, Chief Engineer (Building), Bhopal, Mr. Anand Prakash Rane, Chief Engineer Planning, Mr. Praveen Nigam Executive Engineer (Design), Mr. Ajay Bain Executive Engineer, Mrs. Shireen Khan Executive Engineer, Ms. Shivani Budhauliya (A.E.), Mr. Rajeev Shrivastav, Sub Engineer,** is highly appreciated who have taken this challenge of bringing out this Schedule of Rates (SOR) into effect in view of modern technologies and invention in the fields of Building & related fields as per latest National Building Code & other Standards.

I also extend my gratitude and thanks to **Mr. V. K. Aarakh Chief Engineer (Building) Gwaliar, Mr. R. K. Ahirwar Chief Engineer (Building) Jabalpur, Mr. J.S. Baghel Chief Engineer (Building) Rewa, Mr. G.P. Verma Chief Engineer Indore, Mr. Surendra Rao Gourkhede, Chief Engineer (Building), Mr. Mukesh Manhar Chief Architect, Mr. R. D. Choudhary, Superintendent Engineer (Building).**

Suggestion, if any, towards further improvement to this schedule of Rates for Building works shall be highly appreciated.

This Schedule of Rates (SOR) is available on the departmental website www.mppwd.gov.in and shall be enforced **w.e.f. 01-01-2024**

Place: Bhopal
Date: **01-01-2024**

1/1/24

(Er. G.P. Mehra)
Engineer-in-Chief, (Building)
MP Public Works Department Building
Bhopal

GENERAL NOTES

1. Reference mentioned herein shall be applicable to all sections to the extent the context permits and are intended to supplement the provisions in the particular section. In case of any discrepancy/ deviation, the provisions in the particular section shall take precedence.
2. In absence of any stipulation to the contrary, unit rates for various items of works are for completing works to the requirements of the specifications including full compensation for all operations and inclusive of all labour, materials, royalties, lease rent, wastages, temporary work, plant, equipment over head charges and profit, unless specified otherwise.

3. Interpretations

3.1 Engineer-In-Chief (Building) Public Works Department MP PWD Government of M.P. shall be the sole deciding authority as to the meaning, interpretation and implications for various provisions in this publication. His decision shall be final and binding.

3.2 Wherever any reference is made to any Indian Standard, it shall be taken as reference to the latest edition with all amendments issued thereto.

3.3 In the event of any variation between the CPWD specifications (adopted) and theIndian Standard, the former shall take precedence over the later.

3.4 Precedence of specification in SOR: - In case of any contradiction in the provisions of the CPWD specifications (adopted) and this document the provisions of this document would take precedence.

3.5 Precedence of Rate in SOR: - The rates as given in this schedule for all items are final, binding and conclusive. In case of doubt and printing mistakes if any, the decision of the Engineer-In-Chief (Building) Public Works Department MP PWD, Government of M.P. shall be final and binding.

4. Definitions

The following terms and expressions in the SOR shall have the meaning or implicationhereby assigned to them unless otherwise specified elsewhere.

4.1 Specifications would refer to the Central Public Works Department (CPWD) specification as applicable as on date along with up to date amendments, addendums and errata.

4.2 Engineer-in-Charge: The "Engineer-in-Charge" means the engineer officer of Executive Engineer rank in Building and Road (B&R) Division, who shall supervise and In-charge of the works and who shall sign the contract on behalf of the Governor of M.P.

4.3 IS: The standards, specification and code of practices issued by the Bureau of Indian Standards along with up to date amendments.

4.4 Superintendent Engineer means officer in-charge of the circle.

4.5 Chief Engineer means officer in-charge of Zone.

4.6 Engineer-in-chief (Building) means head of the department, who heads Building wing of MPPWD.

4.7 Engineer-in-Chief means head of department who heads Building and Road wing of M.P. PWD.

4.8 Competent Authority i.e as per PWD Manual/ NIT / Agreement/ time to time instruction issued by government or higher authority.

4.9 Site: The 'site' shall mean the land/or other places on, in, into or through which the work is to be executed under the contract or any adjacent land, path or street through which the work is to be executed under the contract, or any adjacent land, path or street which may be allotted or used for the purpose of carrying out the contract.

4.10 Best: The word 'best' when used shall mean that in the opinion of the Engineer- in-Charge, there is no superior material/ article and workmanship obtainable in the market and trade respectively. As far as possible the standard required shall be specified in preference to the word 'best'.

4.11 Department: "Department" mean Public Works Department of Govt. of M.P.

5. Floor and Levels of Building

5.1 **Floor 1:-** It is the lowest floor above the ground level in the building unless otherwise specified in a particular case. The floors above floor 1 shall be numbered in sequence as floor 2, floor 3 and so on the number shall increase upwards.

5.2 **Floor Level :** - For floor 1, top level of finished floor shall be the floor level and for all other floors above floor 1, top level of the structural slabs shall be the floor level.

5.3 **Plinth level: -**Floor 1 level or 1.2 m above the ground level whichever is lower shall be the plinth level.

6. Floor Levels of Special Structures

For structures like retaining walls, wing walls, chimneys, reservoirs/ tanksand other elevated structures, where elevations/ heights above a defined datum level have not been specified and identification of floors cannot be done as in case of building. Level, at 1.2 m above the ground level shall bethe floor 1 level as well as plinth level.Level at a height of 3.6 m above floor1 level will be reckoned as floor 2 level and level at a height of 3.6 m above the floor 2 level will be floor 3 level and so on.Where the total height above floor 1 level is not a whole number multiple of 3.6 metre, top most floor level shall be the next in sequence to the floor level below even if the difference in height between the two upper most floor levels is less than 3.6 metres.

7. Foundation and Plinth

The work in foundation and plinth shall include:

7.1 For buildings: All works up to 1.2 metre above ground level or up to floor 1(one) level whichever is lower.

7.2 For abutments, piers and well staining: All works upto 1.2 metre above the bed level.

7.3 For retaining wall, wing walls, compound walls, chimneys, over head reservoirs/ tanks and other elevated structures: All works upto 1.2 metre above the ground level:

7.4 For reservoirs/tanks (other than overhead reservoirs /tanks): All works upto 1.2 metre above the ground level.

7.5 For basements: All works upto 1.2 metre above ground level or upto floor 1 (one) level whichever is lower.

7.6 For boundary walls: All works upto plinth beam or upto 1.2 metre above ground level whichever is lower.

Note: Specific provision shall be made in the estimate for such situations where the foundation level is more than 3 (three) metre depth from the plinth for all types of structures mentioned above.

8. Measurements

Mode of measurement shall be as per the provisions of the contract and as specified in B.O.Q.

8.1 The order of dimensions shall be consistent and in the sequence of length,width and height or depth or thickness (LBD).

8.2 Rounding off: Rounding off where required shall be done in accordance with IS:2-1960.

9. **Rates: -**The rates include elements of hire and running charges of all types of plants, machinery and equipment required to complete the work, unless specified otherwise. Due diligence for safety and health of workers shall be observed by thecontractor by observing all necessary protocol as per law; cost of which shall be deemed to have been included in rates of SOR. The rates provided in thisschedule of rates include cost of all materials lead, lift, water and electricity charges performance of all workmanship; providing erecting and removing centering, forms, scaffolding, benching ladders, templates, safety equipment tools and plants and all other applications etc, required for the proper execution of workunless otherwise specified inclusive charges of duty, royalty and insurance etc.as may be applicable but excluding GST, since all taxes such as VAT, entry taxes etc. have been subsumed now in GST, rate shall not include GST which shall be payable to the contractor at prevailing rate on gross amount of contractor's bill. Provisions for covering etc. necessary to protect the work/ structure from inclement weather etc. and damage arising from falling materials or other cause such as rain, fire etc. shall be the contractor's responsibility and cost of all such arrangements shall be deemed included in SOR rates. SOR rates are also inclusiveof curing wherever required including arrangement of water and its lead or lift whatsoever. If there is any source of water like tube well, hand pump, well etc. under Govt. custody and if water from this source is provided free of cost for construction activity by the contractor then water charge will be deducted at the rate of 1% of the amount paid to the contractor from the items involving use of water. SOR rates are also inclusive of testing charges and provision for laboratory at site as well SOR rate are also inclusive of any certification to be obtained from statutory authority before putting the work machineries equipment fitment to use.

10. **Materials**

10.1 **Only Approved Material: -** The contract shall get approval from Engineer-in-Charge for Samples of all materials to be used on the work. The approved samples duly authenticated and sealed shall be kept in the custody of the Engineer-in-Charge till the completion of the work. All materials to be provided by the contractor shall be brand new andas per the samples approved by the Engineer-in-Charge. Contractor shall bring to site for use in construction only from list of approved make issued by the department. Engineer-in-Charge shall be final authority to make final selection from amongst branded makes of item. Items for which list of approved are not declared brand make are not by the department that shall be selected by the CE/Engineer- in-Charge.

10.2 **Temporary Accommodation at site: -**For work, if the amount of contract exceeds Rs. 100.00 lacs, the contractor shall provide a temporary accommodation of at least 20 sqm within boundaries of site where all the approved material shall be kept for display under lock and key of Engineer-in-Charge the after completion of work this accommodation could be further used as maintenance office. If the contract amount is more than Rs.500.00 lacs additional temporary site accommodation of 20 sqm shall also be provided by the contractor for SQC (Services for Quality Control) office use related to work.

10.3 **Vigilance by CCTV : -** For work, if the amount of contract exceeds Rs. 1000.00 lacs contractor shall install a CCTV setup (Adequate number to cover whole site) before start of work, the access to the vision should be given to Engineer-in-Charge along with its supervision team. In case of non compliance, penalty at the rate of 1000 rupees per days will be imposed to contractor.

10.4 **Materials from approved source only:** -Materials obtained by the contractor from the sources approved by the department shall be subjected to the mandatory tests. Where such

materials do not conform to the relevantspecifications, the material shall be rejected not withstanding the fact that the material / make is the approved make by the department.

10.5 **Testing:-** The contractor shall at his own expense, provide a material for tests (routine field test of materials/ cube test /core cutting /slab deflection/ nondestructive test/ etc. for any component of building), at his cost, make all arrangements and shall provide all facilities as the Engineer-in-Charge may require for collecting and preparing the required number of samples for such tests at such time and to such place or places as may be directed by the Engineer in-Charge and bear all charges including testing charges. The Contractor shall not be eligible for any claim or compensation either arising out of any delay in the work or due to any corrective measures required to be taken on account of and as a result of testing of materials The Engineer -in- Charge or his authorized representative shall always have access to the works and to all workshops and places where work component is being prepared or from where materials, manufactured articles or machinery are being obtained for the works and the contractor shall afford every facility and every assistance in obtaining the right to such access.

10.6 **Protection of Material at site: -** Materials stored at site, depending upon the individual characteristics, shall be protected from atmospheric effects due to rain, sun, wind and moisture to avoid deterioration.

10.7 **Assistance for lease/permits: -** It will not be obligatory on the part of Engineer-In-Charge to provide any assistance in obtaining lease/permits forexcavation of minor minerals. The contractor shall not be entitled to any excuse, whatsoever on account of any delay in obtaining permit etc. for minor minerals to be used on the work.

10.8 **Warranty:-** Warranty card shall be given of all the items which covered under warranty to Engineer-in-Charge at the time of execution, payment shall not be made for such items if warranty card/certificate is not given. The warranty card/certificate shall be given to the concern department at the time of handover of the building.

1. Testing

1.1 **SOR Rates include testing charges:-** The rates also include the element oftesting of samples of various materials brought by the contractor for use on the work, as well as other necessary tests for items of work as stipulated in the specifications.

1.2 **Testing Frequency: -**Frequency of such tests to be carried out must not be less than the prescribed frequencies.

1.3 **Field Laboratory: -**The contractor shall have to establish a field laboratoryat the site of work, if the amount of contract exceeds Rs. 100.00 lacs.

1.4 **Testing in Laboratory other than field laboratory: -**In other cases, testing of construction materials should be got done from any of the testing laboratories of the government works departments, and technical institutes, engineering college, etc. In case the contract provides for testing of material through third party like NABL accredited lab or any other lab directed by the Engineer-In-Charge, the cost of testing shall be borne by thecontractor.

1.5 **Payment only after testing: -**The work shall not be accepted in any case, if the contractor fails to observe the instructions of the department, regarding testing of materials. Before making any payment, it will be the responsibility of the officers making payment to ensure that all testing asper prescribed frequencies, have been carried out, and shall certify that the test results are in conformity with the requirement. If tests are not conducted to the prescribed frequency, the Engineer-In-Charge should reject that part of the work. Duly certified (by the representative of Engineer-in-Charge) copies of registers, containing records of tests shall have to be presented along with running account bills. Register (in original) shall have to be

submitted along with the final bill. Tests shall have to be conducted by the contractor's engineer under the supervision of the Engineer-In-Charge or his authorized representatives.

2. **Acceptance at Reduced Rate: -**If any item of work is found not upto the prescribed standard but the Engineer-In-Charge is of the opinion that the same is structurally adequate and can be accepted at a reduced rate, then in such cases, the Engineer-In-Charge shall have to submit proposals for appropriate reduction of rates supported by rate analysis, in justification thereof, through a D.O. letter to the territorial Superintending Engineer or Chief Engineer to obtain his approval expeditiously (ordinarily within 15 days).

3. **The Site Order Book: -** The "Site Order Book" shall be in custody of departmental supervisory staff. In case of a work for which a separate agency for supervision of work has been engaged by the department, the site order book shallbe in possession of the field staff of SQC agency. The Engineer-In-Charge or his authorized representative may record his instructions in this book, which shall be noted by the contractor or his authorized representative for compliance.

4. **Cement at Site: -** A duplicate cement register as per format here under shall be maintained at site of work. Extract certified copies of the entries for each month/each running bill shall be submitted to the Engineer-in-Charge by the Contractor. The original register shall also be submitted to the Engineer-in- Charge on completion of the work by the Contractor.

S. No.	Date of receipt of cement	No. of bags	Name and address of firm, from whom purchased	Signature of contractor or his authorised representative	Signature of authorised representative of Engineer-In- Charge	Result of test for initial and final setting time	Result of tests for compressive strength of cement	Remarks

15. **Only Tested Cement: -**Where contract provides for cement to be arranged by the contractor himself, only I.S. marked cement of relevant I.S. specifications as per IS: 4031 PART- 5, IS 269-2013, IS 1489, IS 8112 . IS 12269 etc. shall be allowed to be used in the work.

NOTE : The arrangement for necessary equipment and testing shall have to be made by the contractor himself at site, to be decided by the Engineer-in-Charge. All testing expenses shall be borne by the contractor. Any lot of cement brought to site by the contractor, would be permitted to be used in the work only after the satisfactory results of the tests, under the supervision of the Engineer-in-Charge or his authorized representative. The record of the tests result shall be maintained in the register referred in subsequent para.

16. **Volumetric use of cement:-**When cement is used for mortar/volumetric mix etc. batch of one bag mix cement or its multiples shall be mixed and the measurement of box for metal, sand to be of 35 liters or multiples of it (convenient box size 40x35x25cm.). For design mix concrete batching shall be done by weight.

17. **Aggregates: -**The aggregates to be used for all cement concrete items shall be crushed in the mechanical crushers. All concrete shall have to be mixed in power driven mixers having hoppers and capacity of mixing concrete mix of at least one bag of cement.

18. **Sand: -**Nothing extra shall be payable on account of use of Narmada sand or sand brought from any other quarry/river. Sand should conform to the requirements of IS: 1542 and IS: 2166 for using it in construction work.

19. **Rubbles from excavation: -**Rubble available from excavation of hard rock shall be the property of the contractor subject to recovery of Rs. 150/- per cum. of the quantity of the rock excavated. This excavated material should be removed from site by contractor at its own cost.

If after the instruction issued by engineer in charge for shifting of material from site contractor has bind to shift this excavated material from site within 15 days otherwise after this penalty will imposed on contractor at the rate of Rs 50 per cum per day.

20. **Ban on Teak wood:** -Teak wood shall not be used in any building work unless specific sanction of the State Government is obtained. Provision of teak wood shall not be made **in** any of the estimate without prior sanction of the state Government. Item such as teak wood beading, veneering, style and rail of flush door shall not be banned.

21. **Approval of Non SOR item** - If any Non S.O.R. Item is to be executed in any proposed project or in ongoing project, the rate analysis with full justification thereof must be approved by the Superintending Engineer of respective circle (B & R) or Chief Engineer, (Building) .

22. **In case** of any difference in item description and rate between BOQ and SOR, the rate and description of BOQ shall be prevail, on the other hand if there is difference in between rates only while description being the same in BOQ and SOR, the rate of SOR with up to date amendment shall be prevail.

23. **Safety In Construction** - The contractor shall employ only such methods of construction, tools and plant as are appropriate for the type of work or as approved by Engineer-in-Charge in writing. The contractor shall take all precautions and measures to ensure safety of works and workman and shall be fully responsible for the same. Safety pertaining to construction works such as excavation, centering and shuttering, trenching, blasting, demolition, electric connections, scaffolds, ladders, working platforms, gangway, mixing of bituminous materials, electric and gas welding, use of hoisting and construction machinery shall be governed by CPWD safety code, relevant safety codes and the direction of Engineer-in-Charge.

The Contractor must ensure safety of workmen as well as safety for the general public during construction in and around work-site. He must follow the laws, codes and standards laid down in this regard. The work-men must be trained and provided protective gear, life-saving equipment like safety shoes , safety net ,safety jacket , safety helmet and appropriate tools for their jobs and also make ensure that all safety equipment are being used properly without any excuse. Special precautions must be used if hazardous chemicals are used or stored at workplace (lead, silica, asbestos and wood/stone that will be cut and generate dust, construction materials containing zinc, cadmium, beryllium and mercury).Besides protection from noise and environmental pollution, public must also be safeguarded from falling through dug-up area, electrocution, flooding, falling objects, bridge-span dropping/ failures, crane falling/ overturning and damage to building from vibrations/ cave-ins from construction activities. Engineer-in- charge must ensure that contractor does not adopt any short-cut in this regard. Contractor is only responsible to provide sufficient required barricading material and properly barricade the project site to avoid any kind of external hindrance at project site. Type of barricading will choose by project type and location by Engineer-in-charge and contractor has to provide same up to the satisfaction level of engineer-in-charge.

24. **Black cotton (BC) soil**- If BC soil having swelling index more than 50 % , that should not be used for refilling under plinth, and under floor filling same shall be replaced by good earth.

25. **Mix design approval:** Proposal of mix design has to be submitted by contractor at its own cost before execution well with in time to start of work and the same shall be approved by competent authority.

26. **Samples approval-** Sample of all materials to be used on the work shall be got approved by the contractor from the Competent Authority well in time by producing at least four sample. The approved samples duly authenticated and sealed shall be kept in the custody of the Engineer-in-Charge till the completion of the work. All materials to be provided by the contractor shall be brand new and as per the samples approved by the Competent Authority.

27. The Competent Authority / Engineer-in-Charge shall have full powers to require the removal from the premises of all materials which in his opinion are not in accordance with the specifications and in case of default, the Engineer-in-Charge shall be at liberty to employ at the expense of the contractor, other persons to remove the same without being answerable or accountable for any loss or damage that may happen or arise to such materials.

11/1/24

(Er. G.P. Mehra)
Engineer-in-Chief (Building)
MP Public Works Department (Building)
Bhopal

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CHAPTER – 14.0

REPAIRS TO BUILDINGS AND STRUCTURAL REHABILITATION

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	**2**	**3**
1.	IS 419	Specifications for Putty for use in Window Frames
2.	IS 14900	Specifications for Transparent Float Glass

PREAMBLE
REPAIRS TO BUILDINGS AND STRUCTURAL REHABILITATION
(CHAPTER: 14.0)

14.1 Repairs to Plaster

The work includes cutting the patch and preparing the wall surface. Patches of 2.50 square metres and less in area shall be measured under item of 'Repairs to Plaster' under this sub-head. Plastering in patches over 2.5 square metres in area shall be paid for at the rate as applicable to new work under sub head Finishing.

14.2 Cutting

The mortar of the patch, where the existing plaster has cracked, crumbled or sounds hollow when gently tapped on the surface, shall be removed. The patch shall be cut out to a square or rectangular shape at position marked on the wall as directed by the Engineer-in-Chargeor his authorized representative. The edges shall be slightly under cut to provide a neat joint.

14.3 Preparation of Surface

The masonry joints which become exposed after removal of old plaster shall be raked out to a minimum depth of 10 mm in the case of brick work and 20 mm in the case of stone work. The raking shall be carried out uniformly with a raking tool and not with a basuli, and loose mortar dusted off. The surface shall then be thoroughly washed with water, and kept wet till plastering is commenced. In case of concrete surfaces, the same shall be thoroughly scrubbed with wire brushes after the plaster had been cut out. The surface shall be washed and cleaned and kept wet till plastering is commenced.

14.4 Fixing Door, Window or Clerestory Window Chowkhats in Existing Opening

For embedding hold fasts of doors, windows or clerestory windows, the requisite number of holes at the correct positions shall be cut out in the masonry. The size of the holes shall be such that the chowkhats with the hold-fasts can be conveniently erected in position. Where necessary, masonry shall be chipped uniformly to facilitate easy insertion of the frame in the opening.

Special care shall be taken when holes are made in load bearing pillars or wall portions separated by openings to ensure that beams etc. supported by them are properly propped up. In such portions cutting holes shall be done on one side at a time. The sides of the holes shall be truly parallel and perpendicular to the plane of the wall. Due care shall be taken, not to disturb the adjoining masonry and the masonry under the bearings of lintels and arches etc. spanning the opening. The holes shall then be cleaned of all dust, mortar and brick bats or stone pieces and thoroughly wetted.

14.5 Making Opening in the Masonry Construction and Fixing Chowkhats for Doors, Windows and Clerestory Windows

Before making opening it is necessary to examine that the wall exclusive of opening is adequate to take the load coming on the structure. All the structural members supported on the walls which have direct bearing over the area in which opening is to be made, shall be properly supported with props to relieve the load from masonry wall till the lintel over the opening is strong enough to take the load. Care should also be taken not to disturb the adjoining masonry. All precautions as explained in Chapter 15.0 (Demolition and Dismantling) should be followed in case of dismantling the external walls. The portion to be dismantled may be clearly marked on both sides of the wall. Dismantling shall be carried out from top to bottom within the marked area. The sides of the opening shall be as far as possible, parallel and perpendicular to the plane of wall.

RENEWING FLOATING GLASS PANES WITH PUTTY AND NAILS

14.6 Removing Broken Glass Panes

Old putty shall be raked out with hack knife. The brad (small nails without head) and pieces of broken glass shall be removed from the rebates of the sash bars. The pieces of glass panes as found useful shall be handed over to the Engineer-in-Charge of the work. No glass shall be inserted in frames until they have been primed and prepared for painting so that the wood may not draw oil out of the putty.

14.7 Removing Broken Glass Panes

The specifications shall be the same as described above except that the wooden fillets including nails shall be taken out carefully.

14.8 Glazing

The specifications for glass panes and their fixing shall be the same as per IS 14900. The fillet shall either be fixed flush or projected uniformly to match with the existing work by means of nails (brads). The new fillet provided shall be painted or finished otherwise to match with the existing finish of the joinery work. The glass panes shall be cleaned with methylated spirit of all sorts of splashing and droppings of wash and paints. All rubbish and unserviceable materials shall be disposed off in the dumping ground promptly as per the direction of Engineer-in-Charge.

FIXING FAN CLAMPS IN EXISTING R.C.C. SLABS

14.9 Fixing

A 15 x 7.5 cm size chase shall be cut from the ceiling to expose the reinforcement and upto 2.5 cm clear round the reinforcement bar as directed. This shall be done without any damage to adjoining portion of the ceiling. The two arms at the ends of the clamps shall be passed through the space over the reinforcement bar from the bottom of the slab. Then the two arms shall be bent down about 1.5 cm by means of a crow bar. The clamp shall be held in position and chase in the ceiling filled with cement concrete 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 20 mm nominal size). The ceiling shall then be finished to match the existing surface and properly cured. The exposed portion of the clamp shall be given two or more coats of paint including one priming coat of shade as directed by the Engineer-in- Charge.

WHITE WASH WITH LIME

14.10 Preparation of Surface

Before new work is white washed, the surface shall be thoroughly brushed free from mortar droppings and foreign matter. In case of old work, all loose particles and scales shall be scrapped off and holes in plaster as well as patches of less than 50 cm area shall be filled up with mortar of the same mix. Where so specifically ordered by the Engineer-in-Charge, the entire surface of old white wash shall be thoroughly removed by scrapping and this shall be paid for separately. Where efflorescence is observed the deposits may be brushed clean and washed. The surface shall then be allowed to dry for at least 48 hours before white washing is done.

14.11 Structural audit and diagnosis of existing building

Structural audit and diagnosis of existing building to find present serviceability and structural viability of structure, to suggest appropriate repairs and retrofitting measurerequired for the building to perform better in its service life.

Stages in Carrying Out Structural Audit

1. Study of architectural and structural drawings, design criteria, design calculations, structural stability certificate of existing structures
2. If architectural and structural drawings are not available, as built drawings can beprepared by an engineer
3. **Inspection:-** To recognize the types of structural defects, to identify any signs of material deterioration, to identify any signs of structural distress and deformation, to identify any alteration and addition in the structure, misuse which may result in overloading. The inspection report should reveal the following listings along with photographs and sketches.

a) General information of the building

- Name and address of the building
- Number of stories in each block of building
- Description of main usage of building viz. residential, commercial, institutional
- Maintenance history of the building

b) Structural System of the building

- Sub structure: Settlement of columns or foundations, Settlement of walls and floors, Deflection and cracks in Retaining wall, Soil bearing capacity through trial pits or From adjacent soil data
- Super structure: Materials used and framing system of structure, identification of the Critical structural members like floating columns, transfer beams, slender members, Rusting of exposed steel and its extent.
- Mention the status of all building elements like beams, slabs, columns, balconies, Canopy, false ceiling, chajja, parapet and railings with respect to parameters Deflection, cracks, leakages and spalling of concrete

- Likewise, verify the status of water tank, staircase, and lift and lift machine room.

c) Addition or Alterations in the building

- Identification of change of occupancy
- Alteration or addition of partition walls
- Alteration or addition in loadings- stacking
- Alteration or addition of toilets, water tank
- Alteration or addition of balcony

d) Dampness and leakages

- Detect the dampness in walls • Identify the leakages in Terrace, toilets, plumbing lines, drainage lines and overhead tanks.

4. **Testing** (nondestructive and destructive) the real strength and quality of a concrete structure to be checked with non-destructive tests. A number of non-destructive tests (NDT) for concrete members are available to determine present strength and quality of concrete. Some of these tests are very useful in assessment of damage to RCC structures subjected to corrosion, chemical attack, and fire and due to other reasons. These tests have been put under four categories depending on the purpose of test as under:

4.1 Concrete Strength

- Rebound Hammer Test: To measure surface hardness of concrete
- Ultrasonic Pulse Velocity Test: To assess homogeneity of concrete, to assess strength Of concrete qualitatively, to determine structural integrity
- Core Sampling and Testing: To measure strength, permeability, density of concrete.

4.2 Chemical Attack

- Carbonation Test: To assess depth of carbonation and pH of concrete
- Chloride Test: To assess total water/acid soluble chloride contents
- Sulphate Test: To assess total water/water soluble sulphate contents of concrete

4.3 Corrosion Potential Assessment

- Cover Meter: To measure cover of reinforcement, diameter of reinforcement and Spacing of reinforcement
- Half Cell Method: To assess probability of corrosion in the embedded steel
- Permeability Test: To assess permeability of concrete due to water and air

4.4 Homogeneity and integrity Assessment

- Ultrasonic pulse velocity for determination of cracks and discontinuities

4.5 Core Testing

This is direct method of assessing strength of concrete. In this method cylindrical core samples are taken from existing structures. The cores are visually inspected and tested in laboratory to check its compressive strength.

4.6 Pushover Analysis

Generally Push over analysis is used to understand the existing capacity of structure for seismic and gravity loading which will show different occupancy levels like immediate occupancy, life safety and collapse prevention. The seismic evaluation of existing buildings compares their capacity against earthquake demand at specific site and concerns the potential earthquake-caused risk to building systems and elements that are closely related tohuman life safety.

4.7 Report

Based on the above inspection, analysis and test results, the report should prepare as per annexure - that, conclude the critical areas that need immediate repairs and retrofitting. For example: number of columns requiring immediate repair and strengthening, repair of critical slab and beams, water proofing of terrace, toilet blocks, cracks in walls or structural elements etc.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.1		Repairs to plaster	Repairs to plaster of thickness 12mm to 20mm in patches of area 2.5 sq. metres and underincluding cutting the patch inproper shape, raking out joints and preparing and plastering the surface of the walls completeincluding disposal of rubbish tothe dumping ground within 50metres lead : With cement mortar 1:4 (1 cement : 4 sand).	sqm	180.00
14.2		Fixing chowkhats with holdfasts	Fixing chowkhats in existing opening including embedding chowkhats in floors or walls cutting masonry for holdfasts, embedding hold fasts in cement concrete blocks of size 15 x 10 x 10 cm with cement concrete 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 20 mm nominal size), painting two coats of approved wood preservative to sides of chowkhats and making good the damages to walls and floors as required complete including disposal of rubbish tothe dumping ground within 50 meters lead :		
	14.2.1	Door	Door chowkhats	each	544.00
	14.2.2	Window	Window chowkhats	each	325.00
	14.2.3	Clerestory	Clerestory window chowkhats	each	238.00
14.3		Fixing chowkhat with dash fasteners/ chemical fastener	Fixing chowkhat in existing opening in brick / RCC wall with dash fasteners/ chemical fastener of appropriate size (3nos on each vertical member of door chowkhat and 2 nos. on each vertical member of window chowkhats) including cost of dash fasteners/ chemical fastener.	each	146.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.4		Making opening in brick masonry	Making the opening in brick masonry including dismantling in floor or walls by cutting masonry and making good the damages to walls, flooring and jambs complete to match existing surface i/c disposal of mulba/ rubbish as directed by Engineer-in-charge. For door/ window/ clerestory window.	sqm	323.00
14.5		Renewing Float glass panes with putty	Renewing glass panes, with putty and nails wherever necessary including racking out the oldputty:		
	14.5.1	4.0mm	Float glass panes of thickness 4 mm	sqm	679.00
	14.5.2	5.5mm	Float glass panes of thickness 5.5 mm	sqm	759.00
14.6		Renewing float glass panes, with wooden fillets	Renewing glass panes with now wooden fillets wherevernecessary:		
	14.6.1	4 mm	Float glass panes of thickness 4 mm	sqm	801.00
	14.6.2	5.5mm	Float glass panes of thickness 5.5 mm	sqm	882.00
14.7		Renewing glass panes and refixing existing wooden fillet	Renewing glass panes and refixing existing wooden fillets:		
	14.7.1	4 mm thick glass	Float glass panes of thickness 4 mm	sqm	697.00
	14.7.2	5.5 mm thick glass	Float glass panes of thickness 5.5 mm	sqm	778.00
14.8		New wooden fillet	Supplying and fixing new wooden fillets wherevernecessary:		
	14.8.1	T.W.	2nd class teak wood fillets	meter	30.00
	14.8.2	H.W.	Haldu/Bija/Kail wood fillets.	meter	26.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.9		Renewal of old putty	Renewal of old putty of glass panes (length).	meter	15.00
14.10		Refixing old glass panes with putty	Refixing old glass panes with putty and nails.	sqm	199.00
14.11		Fixing old glass panes with wooden fillet	Fixing old glass panes with wooden fillets (excluding cost of fillets).	sqm	162.00
14.12		M.S. Fan clamps	Providing and fixing 16 mm M.S. Fan clamps of standard shape and size in existing R.C.C. slab including cutting chase, anchoring clamp to reinforcement bar, including cleaning, refilling, making good the chase with matching concrete, plastering and painting the exposed portion of the clamps complete.	each	165.00
14.13		Replacing sand stone slabs	Replacing sand stone slabs in roofing laid in cement mortar 1:4 (1 cement : 4 sand) including necessary repairs and cement pointing with same mortar complete including disposal of rubbish to dumping ground within 50 metres of lead :		
	14.13.1	Red/ white sand stone	Red/ white sand stone slabs 30 to 50 mm thick.	sqm	631.00
14.14		Renewing wooden battens in roofs	Renewing wooden battens in roofs, including making good the holes in wall and painting with oil type wood preservative of approved brand and manufacture complete including removal of rubbish to the dumping ground within 50 metres lead :		
	14.14.1	Sal wood	Sal wood battens.	cum	74943.00
14.15		Renewing wooden beams in roofs	Renewing wooden beams in roofs including making good the holes in walls and painting with oil type wood preservative of approved brand and manufacture complete including removal of rubbish to the dumping ground within 50 metres lead :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	14.15.1	Upto 4.0m	Not exceeding 4.00 metres in length.		
	14.15.1.1	Sal	Sal wood beams	cum	76349.00
	14.15.1.2	H.W.	Haldu/Bija/Kail wood fillets.	cum	76349.00
	14.15.2	4.0m upto 5m	Above 4.00 metres and upto 5.00 metres length.		
	14.15.2.1	Sal	Sal wood beams	cum	78106.00
	14.15.2.2	H.W.	Haldu/Bija/Kail wood fillets.	cum	78101.00
14.16		Repointing or Replastering	Raking out joints in lime orcement mortar and preparing the surface for re-pointing or replastering including disposal of rubbish to the dumping ground within 50 metres lead.	sqm	17.00
14.17		Taking out wind ties	Taking out wind ties from roof including cutting out rusted bolts, nuts etc. and removing materials to any distance within compound and stacking.	kg	1.2
14.18		Fixing of old wind tie	Fixing of old wind tie with new fittings including painting two or more coats with anticorrosive bitumastic paint of approvedbrand and manufacturer over and including priming coat of ready mixed zinc chromate yellow primer of approved brand.	meter	61.00
14.19		Renewing bottom rail and/or top runner of collapsible gate	Renewing bottom rail and/or top runner of collapsible gate including making good all damages and applying priming coat of zinc chromate yellow primer of approved brand and manufacturer.	kg	119.00
14.20		Renewing wheel or roller of steel door or gate	Renewing Wrought iron or M.S. Wheel or roller of steel door or gate and fitting and fixing the same with necessary clamps, nuts and bolts/welding and erection		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			etc. complete.		
	14.20.1	Upto 50mm dia	Wheel 50 mm dia. and below.	per wheel	136.00
	14.20.2	>50mm	Wheel above 50 mm dia.	per wheel	211.00
14.21		Cupboard shutters	Providing and fixing 25 mm thick shutters for cupboard etc. :		
	14.21.1	Panelled shutter	Panelled or panelled and glazed shutters :		
	14.21.1.1	Superior T.W.	Superior class teak wood including nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	3794.00
	14.21.1.2	1st Class T.W.	Ist class teak wood including nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	3413.00
	14.21.2	Glazed shutters	Glazed shutters :		
	14.21.2.1	Superior T.W.	Superior class teak wood including nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	4021.00
	14.21.2.2	1st Class T.W.	Ist class teak wood including nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	3611.00
14.22		Plain jaffri door and window shutters	Providing and fixing plain jaffri door and window shutters including bright or/and black enamelled M.S. butt hinges with necessary screws 35x10mm laths placed 35mm apart (frames to be paid separately) including fixing 50x12mm beading complete with:		
	14.22.1	T.W.	Second class teak wood.	sqm	3105.00
	14.22.2	H.W.	Hard wood, finger jointed	sqm	1340.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.23		Brass curtain rods	Providing and fixing curtain rods of 1.25mm thick brass plates with two brass brackets fixed withbrass screws and wooden plugs etc. wherever necessary complete.		
	14.23.1	20mm	20 mm diameter.	meter	265.00
	14.23.2	25mm	25 mm diameter.	meter	294.00
14.24		M.S. round or square bars in wooden frames	Providing and fixing M.S. round or square bars with M.S. flats at required spacing in wooden frames of windows and clerestory windows.	kg	67.00
14.25		Joists hoisting & fixing	Providing joists (karries) including hoisting fixing in position and applying wood preservative on unexposed surface etc. complete with:		
	14.25.1	Sal	Sal wood.	cum	73829.00
	14.25.2	H.W.	Haldu/Bija/Kail wood.	cum	73829.00
14.26		White washing old work	White washing with lime old work to give an even shade :		
	14.26.1	Two coats	two or more coats	sqm	7.00
	14.26.2	One coats	one or more coats	sqm	4.00
14.27		Scrapping	Removing white or colour wash by scrapping and sand papering and preparing the surface smooth including necessary repairs to scratches etc. complete.	sqm	5.00
14.28		Dry distemper	Distempering with dry distemper of approved brand and manufacture (one or more coats) and of required shade on old work to give an even shade.	sqm	19.00
14.29		Oil bound washable distemper	Distempering with oil bound washable distemper of approved brand and manufacture to give an even shade: Old work (one or more coats).	sqm	21.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.30		Removing dry or oil bound distemper	Removing dry or oil bound distemper, water proofing cement paint and the like by scrapping, sand papering and preparing the surface smooth including necessary repairs to scratches etc. complete.	sqm	6.00
14.31		Painting G.S. sheet with synthetic enamel paint	Painting on G.S. sheet with synthetic enamel paint of approved brand and manufacture of required colour to give an even shade: Old work (one or more coats).	sqm	28.00
14.32		Pipe Painting bitumastic paint (Old work)	Painting (one or more coats) on rain water, soil, waste and vent pipes and fittings with black anticorrosive bitumastic paint of approved brand and manufacture on old work :		
	14.32.1	75mm	75 mm diameter pipes	meter	9.00
	14.32.2	100mm	100 mm diameter pipes	meter	11.00
	14.32.3	150mm	150 mm diameter pipes	meter	17.00
14.33		Pipe Painting with aluminium paint (Old work)	Painting (one or more coats) on rain water, soil, waste and vent pipes and fittings with aluminium paint of approved brand and manufacture on old work :		
	14.33.1	75mm	75 mm diameter pipes	meter	21.00
	14.33.2	100mm	100 mm diameter pipes	meter	31.00
	14.33.3	150mm	150 mm diameter pipes	meter	46.00
14.34		Pipe Painting with synthetic enamel paint	Painting (one or more coats) on rain water, soil, waste and vent pipes and fittings with synthetic enamel paint of approved brand and manufacture and required colour on old work :		
	14.34.1	75mm	75 mm diameter pipes	meter	10.00
	14.34.2	100mm	100 mm diameter pipes	meter	13.00
	14.34.3	150mm	150 mm diameter pipes	meter	19.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.35		Painting with oil type wood preservative (Old work)	Painting with oil type wood preservative of approved brand and manufacture: Old work (one or more coats)	sqm	20.00
14.36		Painting wall with plastic emulsion paint (Old work)	Wall painting with plastic emulsion paint of approved brand and manufacture to give an even shade: One or more coats on old work.	sqm	41.00
14.37		Painting with synthetic enamel paint (Old work)	Painting with synthetic enamel paint of approved brand and manufacture of required colour to give an even shade: One or more coats on old work.	sqm	35.00
14.38		Painting with aluminium paint(Old work)	Painting with aluminium paint of approved brand and manufacture to give an even shade: One or more coats on old work.	sqm	32.00
14.39		Painting with acid proof paint (Old work)	Painting with acid proof paint of approved brand and manufacture of required colour to give an even shade: One or more coats on old work.	sqm	42.00
14.40		Painting with black anti-corrosive bitumastic paint (Old work)	Painting with black anti-corrosive bitumastic paint of approved brand and manufacture to give an even shade: One or more coats on old work.	sqm	28.00
14.41		French spirit polishing	French spirit polishing: One or more coats on old work.	sqm	59.00
14.42		Melamine polishing	Mealamine polishing on wood work by preparing the surface applying putty base course. Prepare the surface smooth by sand papering and then applying Mealamine polish with air compressor. One or more coats on old work.	sqm	807.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.43		Lacquer polishing on wood work	Lacquer polishing on wood work by preparing the surface applying putty base course. prepare the surface smooth by sand papering and then applying Lacquer polish with air compressor		
	14.43.1		One or more coats on old work.	sqm	674.00
14.44		Polishing on wood work with wax polish	Polishing on wood work with readymade wax polish of approved brand and manufacture :		
	14.44.1		Old work	sqm	28.00
14.45		Re-lettering with black Japan paint	Re-lettering with black Japan paint of approved brand and manufacture.	per letter per cm height	0.9
14.46		Painting with black Japan paint (Old work)	Painting (one or more coats) with black Japan paint of approved brand and manufacture to give an even shade.	sqm	30.00
14.47		C.P. brass chain and rubber plug	Providing and fixing C.P. brass chain and rubber plug completefor sink or wash basin :		
	14.47.1	32mm dia	32 mm dia	each	72.00
	14.47.2	40mm dia	40 mm dia	each	74.00
14.48		Distempering with acrylic washable distemper (Old work)	Distempering with 1st quality acrylic washable distemper (ready mixed) of approved manufacturer and of required shade and colour complete. As per manufacturer's specification. One or more coats on old work.	sqm	20.00
14.49		Water proofing cement paint on wall (Old work)	Finishing walls with water proofing cement paint of required shade :		
	14.49.1	With Primer	Old work (one or more coats applied @ 2.20 kg/10 sqm) over priming coat of primer applied @ 0.80 liters/10 sqm complete including cost of Priming coat.	sqm	42.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	14.49.2	Without primer	Old work (one or more coats @ 2.20 kg/10 sqm) complete.	sqm	30.00
14.50		Textured exterior paint (Old work)	Finishing walls with textured exterior paint of required shade :		
	14.50.1	@3.28 litre/10 sqm	Old work (Two or more coats on existing cement paint surface applied @ 3.28 ltr/10 sqm.	sqm	112.00
	14.50.2	@1.82litre/10 sqm	Old work (One or more coats) applied @ 1.82 ltr/10 sqm.	sqm	67.00
14.51		Acrylic Smooth exterior paint	Finishing walls with Acrylic Smooth exterior paint of required shade :		
	14.51.1	@ 1.67 ltr/ 10 sqm	Old work (Two or more coat applied @ 1.67 ltr/ 10 sqm) on existing cement paint surface).	sqm	87.00
	14.51.2	@ 0.90 ltr/10 sqm	Old work (One or more coat applied @ 0.90 ltr/10 sqm).	sqm	52.00
14.52		Premium Acrylic Smooth exterior paint with Silicone additives	Finishing walls with Premium Acrylic Smooth exterior paint with Silicone additives of required shade		
	14.52.1	@ 1.43 ltr/ 10 sqm	Old work (Two or more coats applied @ 1.43 ltr/ 10 sqm) over existing cement paint surface.	sqm	65.00
	14.52.2	@ 0.83 ltr/10 sqm	Old work (one or more coats applied @ 0.83 ltr/10 sqm).	sqm	41.00
14.53		Deluxe Multi surface paint interiors and exteriors (Old work)	Finishing walls with DeluxeMulti surface paint system for interiors and exteriors using Primer as per manufacturersspecifications :		
	14.53.1	On wall	One or more coats applied on walls @ 0.85 ltr/10 sqm.	sqm	58.00
	14.53.2	On Wood	Painting wood work with Deluxe Multi Surface Paint of required shade. One or more coat applied @0.70 ltr/10 sqm	sqm	52.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	14.53.3	On steel	Painting Steel work with Deluxe Multi Surface Paint to give an even shade. One or more coat applied @0.70 ltr/10 sqm.	sqm	52.00
14.54		Varnishing (Old work)	Varnishing with varnish of approved brand and manufacture:		
	14.54.1	Copal	One or more coats with copal varnish.	sqm	32.00
	14.54.2	Spar	One or more coats with spar varnish.	sqm	33.00
14.55		Melamine polishing (Old work)	Melamine polishing on wood work (one or more coat). Old work	sqm	90.00
14.56		Flatting varnish (old work)	Varnishing with flatting varnishof approved brand and manufacture one or more coats onold work.	sqm	30.00
14.57		Floor painting with floor enamel paint (old work)	Floor painting with floor enamel paint of approved brand and manufacture of required colour to give an even shade. One or more coat on old work.	sqm	50.00
14.58		Stone flooring stone to be supplied by the department	40 mm thick stone flooring over 20 mm (average)thick base of cement mortar 1:5 (1 cement : 5 sand) including pointing with cement mortar 1:2 (1 cement : 2 sand) with an admixture of pigment to match the shade of stone. Red sand stone / White sand stone) stone to be supplied by the department.		
	14.58.1	Rough dressed	Rough chiseled dressed stone	sqm	177.00
	14.58.2	Fine dressed	Fine dressed stone	sqm	207.00
14.59		C.R. masonry stone supplied by the department	Coursed rubble masonry (second sort) with hard stone in Cement mortar 1:6 (1 cement : 6 sand) (Stone shall be supplied by the department)	cum	2112.00
14.60		Laying paver blocks (CC paver blocks supplied by the	Laying old cement concrete interlocking paver blocks of any design/shape laid in required line,	sqm	142.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		department)	level, curvature, colour and pattern over and including 50mm thick compacted bed of sand, filling the joints with sand etc. all complete as per the direction of Engineer-in-charge. (Old CCpaver blocks shall be supplied by the department free of cost).		
14.61		Laying kerb stone (stones supplied by the department)	Laying at or near ground level old kerb stones of all types in position to the required line, level and curvature, jointed with cement mortar 1:3 (1 cement: 3 sand) including making joints with or without grooves (thickness of joints, except at sharp curve, shall not be more than 5mm) including making drainage openingwherever required etc. complete as per direction of Engineer-in- charge. (Length of finished kerb edging shall be measured for payment). (Old kerb stones shall be supplied by the department free of cost).	meter	27.00
14.62		Colour washing with lime (Old work)	Colour washing with lime to give an even shade:		
	14.62.1	One coat	(one or more coats)	sqm	9.00
	14.62.2	Two Coat	(two or more coats)	sqm	11.00
14.63		Scanning to detect the steel reinforcement in existing RCC elements	Conducting adequate no's. of scanning to detect the steel reinforcement in existing RCC elements by using ground/ Wall penetrating radar system to avoid the cutting of Main bars during drilling for inserting bolts etc. complete.	sqm	800.00
14.64		Removing loose plaster and Concrete work by low impact Hammer	Chipping, Removing, Carefully loose plaster and Concrete work by low impact Hammer, Breaker without damaging the structure as Instructed by Engineer-in-Charge.	sqm	120.00
14.65		Casting & Repairing of excessively damaged members	Providing & casting & Repairing of excessively damaged members with micro-concrete at all levels	kg	61.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		with micro-concrete	with all leads & lift or as directed including formwork, centering, scaffolding, and mixing, laying, levelling, and finishing including curing etc. complete (excluding steel reinforcement)		
14.66		Strengthening structural elements with non-metallic composite fiber wrapping system	Strengthening structural elements with non-metallic composite fiber wrapping system comprise of fiber sheet and compatiblesaturate, by dry/wet layup system including, Wrapping the fiber sheet to structural element at desired orientation using tamping roller to avoid any air voids etc. repeat the same procedure for multiple layers (if any) with the interval of 8 hrs. Sand pasting: Applying second coat of saturant after min. 12 hrs, rectify air voids if any paste the river sand on it to make surface rough to take any further finishes (Mode of measurement: surface area of Fiber Consumed).		
	14.66.1	300 GSM	300 GSM Carbon UD	sqm	3670.00
	14.66.2	400 GSM	400 GSM Carbon UD	sqm	4245.00
	14.66.3	600 GSM	600 GSM Carbon UD	sqm	5200.00
	14.66.4	900 GSM	900 GSM Carbon UD	sqm	7200.00
	14.66.5	1200 GSM	1200 GSM Carbon UD	sqm	8500.00
	14.66.6	400 GSM	400 GSM Carbon Angle layup ±45°	sqm	4480.00
14.67		Applying procured laminated with compatible structural adhesive	Providing Applying procured laminate with compatible structural adhesive including, Surface preparation: Grinding concrete substrate, cleaning it with wire brush removing oil, laitance if present, etc. complete. Profiling: Applying compatible		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			primer on prepared substrate, Filling the holes and uneven surface with thixotropic putty etc. complete Application of plate: Marking the application area on structural element, cutting the plate to require size, applying compatible structural adhesive on plate in parabolic manner, pastethe plate on desire area by using tamping roller to avoid any air voids etc. complete. Sand pasting: Applying second coat of saturant after min. 12 hrs, rectify air voids if any paste the river sand on it to make surface rough to take any further finishes (Mode of measurement: Per running Mt. of plate applied).		
	14.67.1	50mm wide(1.4mm thick)	50mm wide and 1.4 mm Thick Carbon Laminate NPS	meter	1900.00
	14.67.2	100mm wide (1.4mm thick)	100mm wide and 1.4 mm Thick Carbon Laminate NPS	meter	3400.00
	14.67.3	100mm wide (3mm thick)	100mm wide and 3 mm Thick Carbon Laminate NPS	meter	4900.00
	14.67.4	100mm wide (4.2mm thick)	100mm wide and 4.2 mm Thick Carbon Laminate NPS	meter	6300.00
	14.67.5	Steel Plate	Providing, Supplying Aluminum Steel Plate (High grade) Including Cutting, Fixing etc. complete. (100mm wide & 6 mm thick.)	meter	1800.00
	14.67.6	Epoxy putty	Providing and applying structural epoxy adhesive putty for levelling of surfaces and connecting aluminium plate which ever required.	kg	525.00
14.68		Mechanical prestress anchor	Providing and fixing mechanical prestress anchor Aluminum Plate (High grade) system for fixing of carbon laminates at specify location at the end of carbon laminate as per design. (Plate	each	4910.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			size 200x270x12 mm with 6 nos M12 and M16 anchors each).		
14.69		Fiber bars with adhesive	Providing and placing in position procured fiber bars with compatible structural adhesive including, groove making,Surface preparation, cleaning it with wire brush removing oil, laitance if present, etc. complete. Profiling: Applying compatible primer on prepared substrate, Filling the holes and uneven surface with thixotropic putty etc. complete. Application of plate: cutting the bars to require size, applying compatible structural adhesive on bars, place the bars in desired grooves etc. complete. rectify air voids if any paste the river sand on it to make surface rough to take any further finishes (Mode of measurement: Per running metre of bar applied) 12 mm dia Carbon rod	meter	1500.00
14.70		Rebaring of TMT bars	Rebaring of TMT bars by drilling hole & embedding bars of different diameter using epoxy grouting compound including all tools and tackles complete.		
	14.70.1	8 mm dia bar	8 mm dia bar 10 mm dia hole	Per 100 mm	73.00
	14.70.2	10 mm dia bar	10 mm dia bar 12 mm dia hole	Per 100 mm	88.00
	14.70.3	12 mm dia bar	12 mm dia bar 16 mm dia hole	Per 100 mm	102.00
	14.70.4	16 mm dia bar	16 mm dia bar 20 mm dia hole	Per 100 mm	131.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	14.70.5	20 mm dia bar	20 mm dia bar 25 mm dia hole	Per 100 mm	265.00
	14.70.6	25 mm dia bar	25 mm dia bar 32 mm dia hole	Per 100 mm	364.00
	14.70.7	32 mm dia bar	32 mm dia bar 40 mm dia hole	Per 100 mm	525.00
14.71		Chromium plated brass butt hinges	Providing and fixing chromium plated brass butt hinges with necessary chromium plated brass screws etc. complete.		
	14.71.1	125x70x4 mm	125x70x4 mm (ordinary type)	each	44.00
	14.71.2	100x70x4 mm	100x70x4 mm (ordinary type)	each	105.00
	14.71.3	75x65x4 mm	75x65x4 mm (heavy type)	each	122.00
	14.71.4	75x40x2.5 mm	75x40x2.5 mm (ordinary type)	each	74.00
	14.71.5	50x40x2.5 mm	50x40x2.5 mm (ordinary type)	each	30.00
14.72		Chromium plated brass pull bolt lock	Providing and fixing 85x42 mm chromium plated brass pull bolt lock with necessary chromium plated brass screws, nuts, bolts and washers etc. complete.	each	185.00
14.73		Repair to plaster	Repair to plaster of thickness 12mm to 20 mm in patches of area 2.5 sqm and under, including cutting the patch in proper shape, raking out joints and preparing plastering the wall surface with white cement based polymer modified self curing mortar, including disposal of rubbish, all complete as per the direction of Engineer-In-Charge.	sqm	355.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.74		Cleaning of terrace/loft storage tank	Cleaning of terrace/loft water storage tank (inside surface area) upto 2000 litre capacity at all heights with coconut brushes, duster etc., removal of silt,rubbish from the tank and cleaning the tank with fresh waterdisinfecting with bleaching powder @ 0.5gm per litre capacity of tank including marking the date of cleaning on the side of tank body with the help of stencil and paint and disposing of malba all complete as per direction of Engineer-in-Charge. (The old date already written on tank should be removed with paint remover or black paint and if date is not written with the stencil or old date is not removed deduction will be made @ Rs. 0.10 per litre) (if during cleaning any GI fittings or ball cock is damaged that is to be repaired by contractor at his own cost and nothing extra will be paid on this account)	litre	0.15
14.75		Cleaning of gully trap chamber	Cleaning and desilting of gully trap chamber, including removal of rubbish mixed with earth etc. and disposal of same, all as per the direction of Engineer- in-charge.	each	48.00
14.76		Chipping of concrete from slabs, beams, columns etc	Chipping of unsound/weak concrete material from slabs, beams, columns etc. with manual Chisel and/ or by standard power driven percussion type or of approved make including tapering of all edges, making square shoulders of cavities including cleaning the exposed concrete surface and reinforcement with wire brushes etc. and disposal of debris for all lead and lifts all complete as per direction of Engineer-In-Charge		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	14.76.1	75 mm	75 mm average thickness	sqm	190.00
	14.76.2	50 mm	50 mm average thickness	sqm	128.00
	14.76.3	25 mm	25 mm average thickness	sqm	64.00
14.77		Cleaning of reinforcement	Cleaning of reinforcement from rust from the reinforcing bars to give it a total rust free steel surface by using alkaline chemical rust remover of approved make with paint brush and removing loose particles after 24 hours of its application with wire brush and thoroughlywashing with water and allowing it to dry, all complete as per direction of Engineer-In-Charge.		
	14.77.1	Upto 12 mm diameter	Bars upto 12 mm diameter	metre	4.00
	14.77.2	Above 12 mm diameter	Bars above 12 mm diameter	metre	9.00
14.78		Drilling holes in concrete	Drilling suitable holes in reinforced or plain cement concrete with power driven drill machine to a minimum depth of 100mm upto 200mm in RCC beams, lintels, columns and slabs to introduce steel bars for sunshades/balconies including fixing the steel bars in position using epoxy resin anchor grout of approved make but excluding the cost of reinforcement, all complete as per direction of Engineer-In-Charge.		
	14.78.1	Upto and including 12mm dia	Upto and including 12mm dia	metre	78.00
14.79		Bonding coat on chipped portion of RCC	Providing, mixing and applying bonding coat of approved adhesive on chipped portion of RCC as per specifications and direction of Engineer-In-charge complete in all respect.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	14.79.1	SBR Polymer bond coat	SBR Polymer (@10% of cement weight) modified cementitious bond coat @ 2.2 kg cement per sqm of surface area mixed with specified proportion of approved polymer	sqm	81.00
	14.79.2	Epoxy bonding adhesive	Epoxy bonding adhesive having coverage 2.20 sqm/kg ofapproved make	sqm	318.00
14.80		SBR polymer modified cement mortar	Providing, mixing and applying SBR polymer (of approved make) modified Cement mortar in proportion of 1:4 (1 cement: 4 graded coarse sand with polymer minimum 2% by wt. of cement used) as per specifications and directions of Engineer-in-charge. **Note**: Measurement and payment: The pre-measurement of thickness shall be done just after the surface preparation is completed and Payment underthis item shall be made only after proper wet curing has been done and surface has been satisfactorily evaluated by sounding / tapping with a blunt metal instrument and/or the 75mm size cube crushing strength at the end of 28 days to be not less than 30 N/Sqmm2)		
	14.80.1	12 mm	12 mm average thickness	sqm	195.00
	14.80.2	25 mm	25 mm average thickness in 2 layers.	sqm	324.00
	14.80.3	50 mm	50 mm average thickness in 3 layers.	sqm	648.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.81		SBR polymer modified cement concrete	Providing, mixing and applying SBR polymer (of approved make @ minimum 2% by wt. of cement used) modified plain/reinforced cement concrete for structural members having minimum characteristic compressive strength [with ordinary portland cement, coarse sand and graded stone aggregate of 10mm maximum size in proportion as per design criteria] with specified average thickness. **Note**: Rates shall be for finished surface area of concrete and shall include the cost of labour, concrete and appropriate approved Super-Plasticiser for rendering concrete as flowable and SBR polymer but shall exclude cost of reinforcement, bond coat, Shear Keys, centering and shuttering, strutting, propping etc (Payment under this item shallbe made only after proper wet curing has been done and surface has been satisfactorily evaluated by sounding/tapping with a blunt metal instrument)		
	14.81.1	50 mm thick in M 25	50mm thick in Grade M 25 with cement content not less than 330 kg per cum	sqm	486.00
	14.81.2	75 mm thick in M 25	75mm thick in Grade M 25 with cement content not less than 330 kg per cum	sqm	729.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.82		SBR polymer modified concrete jacket for structural members	Providing and laying SBR Polymer modified (of approved make @ minimum 2% by wt. of cement used) plain/reinforced concrete jacket for the structural members e.g. columns, pillars, piers, beams etc with concrete having the specified minimum characteristic compressive strength [with ordinary portland cement, coarse sand and graded stone aggregate of 10mm maximum size in proportion as per design criteria] with specified average thickness all-round existing core of RCC member. **Note**: Rates shall be for finished surface area of concrete and shall include the cost of making holes in existing RCC slab, if required, for pouring concrete in shuttering mould of jacket and appropriate approved Super-Plasticiser for rendering concrete as flowable self compacting and SBR polymer but shall exclude cost of reinforcement, bond coat, Shear Keys, centering and shuttering, strutting, propping etc (Payment under this item shall be made only after proper wet curing has been done and surface has been satisfactorily evaluated by sounding/tapping with a blunt metal instrument)		
	14.82.1	50 mm thick in M 25	50mm thick in Grade M 25 with cement content not less than 330 kg per cum	sqm	486.00
	14.82.2	75 mm thick in M 25	75mm thick in Grade M 25 with cement content not less than 330 kg per cum	sqm	729.00
	14.82.3	100 mm thick in M 25	100mm thick in Grade M 25 with cement content not less than 330 kg per cum	sqm	972.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.83		Injecting grout in cracks/honeycomb of concrete/masonary	Providing and injecting approved grout in proportion recommended by the manufacturer into cracks/honey-comb area of concrete/masonry by suitable gun/ pump at required pressure including cutting of nipples after curing etc. complete as per directions of Engineer-in-Charge. (The payment shall be made on the basis of actual weight ofapproved grout injected.)		
	14.83.1	Acrylic Polymer	Stirrer mixed Acrylic Polymer of approved make @ 2% of weight of cement used) modified Cement slurry made with non shrink compound in concrete/RCC work	kg	65.00
	14.83.2	SBR Polymer	Stirrer mixed SBR Polymer (of approved make) modified Cement slurry made with Shrinkage Compensating Cement in concrete/RCC work.	kg	68.00
	14.83.3	Epoxy	Epoxy injection grout in concrete/RCC work of approved make	kg	658.00
14.84		Removing temporaty protective screeens	Providing, erecting, maintaining and removing temporary protective screens made out of specified fabric with all necessary fixing arrangement to ensure that it remains in position for the work duration as required by the Engineer-in-charge		
	14.84.1	Wooven PVC cloth	Wooven PVC cloth	kg	28.00
14.85		Cleaning exposed concrete surface	Cleaning of exposed concrete surface of sticking material including loose and foreign material by sand blasting with coarse sand followed by and including cleaning with oil freeair blast as per direction of Engineer in charge.	sqm	242.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
14.86		Shotcreting RCC members	Shotcreting R.C.C. columns, beams and slabs etc. in layers with approved design mix concrete having the specified minimum characteristic compressive strength [with ordinary portland cement, coarse sand and graded stone aggregate of 10 mm maximum size in proportion as per design criteria] including the cost of centering and shuttering at edges and corners etc. as directed by Engineer- inCharge **Note**: Rates shall include the providing necessary ground wires etc. The levelling gauges, if used, shall be paid for separately. Payment under this item shall be made only after proper wet curing has been done and surface has been satisfactorily evaluated by sounding/tapping with a blunt metal instrument.		
	14.86.1	25 mm thick	25mm thick in Grade M 25 with cement content not less than 330 kg per cum	Sqm	452.00
	14.86.2	50 mm thick	50mm thick in Grade M 25 with cement content not less than 330 kg per cum	sqm	742.00
	14.86.3	75 mm thick	75mm thick in Grade M 25 with cement content not less than 330 kg per cum	sqm	1079.00
14.87		Galvanised steel injection nipple	Providing and inserting 12mm dia galvanised steel injection nipplein honey comb area and along crack line including drilling of holes of required diametre (20mm to 30mm) up to depth from 30mmto 80mm at required spacing and making the hole & crack dust free by blowing compressed air, sealing the distance between injection nipple with adhesive chemical of approved make and allow it to cure complete as per	each	136.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			direction of Engineer-In-Charge.		
14.88		Steel wire febric jacketing	Providing and fixing hard drawn steel wire fabric of size 75 x25 mm mesh or other suitable size wire mesh to be fixed & firmly anchored to the concrete surface by means of "L" shaped mild steel shear key welded with existing reinforcement includingthe cost of materials, labour, tool & plants as approved by Engineer-incharge.	sqm	291.00

CHAPTER – 15.0

DISMANTLING AND DEMOLISHING

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	**2**	**3**
1.	IS 1200 (Part - XVIII)	Method of Measurements of Building and Civil Engineering Works (Part -XVIII) Demolition and Dismantling
2.	IS 4130	Demolition of Buildings-Code of Safety

PREAMBLE
DISMANTLING AND DEMOLISHING
(CHAPTER: 15.0)

15.1 Dismantling:

The term 'Dismantling' implies carefully separating the parts without damage and removing. This may consist of dismantling one or more parts of the building as specified or shown on the drawings.

15.2 Demolition:

The term 'Demolition' implies breaking up. This shall consist of demolishing whole or part of work including all relevant items as specified or shown on the drawings.

15.3 Precautions

All materials obtained from dismantling or demolition shall be the property of the Government unless otherwise specified and shall be kept in safe custody until they are handed over to the Engineer-in- Charge/authorized representative.

The demolition shall always be well planned before hand and shall generally be done in reverse order of the one in which the structure was constructed. The operations shall be got approved from the Engineer-in-Charge before starting the work. Due care shall be taken to maintain the safety measures prescribed in IS 4130.

Necessary propping, shoring and or under pinning shall be provided to ensure the safety of the adjoining work or property before dismantling and demolishing is taken up and the work shall be carried out in such a way that no damage is caused to the adjoining work or property. Wherever specified, temporary enclosures or partitions and necessary scaffolding with suitable double scaffolding and proper cloth covering shall also be provided, as directed by the Engineer-in-Charge.

Necessary precautions shall be taken to keep noise and dust nuisance to the minimum. All work needs to be done under the direction of Engineer-in-Charge. Helmets, goggle, safety belts etc. should be used whenever required and as directed by the Engineer-in-Charge. The demolition work shall be proceeded with in such a way that it causes the least damage and nuisance to the adjoining building and the public.

Dismantling shall be done in a systematic manner. All materials which are likely to be damaged by dropping from a height or by demolishing roofs, masonry etc. shall be carefully removed first. Chisels and cuters may be used carefully as directed. The dismantled articles shall be removed manually or otherwise, lowered to the ground (and not thrown) and then properly stacked as directed by the Engineer-in-Charge.

Where existing fixing is done by nails, screws, bolts, rivets, etc., dismantling shall be done by taking out the fixing with proper tools and not by tearing or ripping off.

Any serviceable material, obtained during dismantling or demolition, shall be separated out and stacked properly as directed by the Engineer-in-Charge within a lead of 50 metres. All unserviceable materials, rubbish etc. shall be disposed off as directed by the Engineer-in-Charge.

The contractor shall maintain/disconnect existing services, whether temporary or permanent, where required by the Engineer-in-Charge.

No demolition work should be carried out at night especially when the building or structure to be demolished is in an inhabited area.

Screens shall be placed where necessary to prevent injuries due to falling pieces.
Water may be used to reduce dust while tearing down plaster from brick work.
Safety belts shall be used by labourers while working at higher level to prevent falling from the structure. First-aid equipment shall be got available at all demolition works of any magnitude.

RECOMMENDATIONS FOR DEMOLITION OF CERTAIN SPECIAL TYPES AND ELEMENTS OF STRUCTURES

15.4 Roof Trusses

If a building has a pitched roof, the roof structure should be removed to wall plate level by hand method. Sufficient purlins and bracing should be retained to ensure stability of the remaining roof trusses while each individual truss is removed progressively.

Temporary bracing should be added, where necessary, to maintain stability. The end frame opposite to the end where dismantling is commenced, or a convenient intermediate frame should be independently and securely guyed in both directions before work starts.

On no account should the bottom tie of roof trusses be cut until the principal rafters are prevented from making outward movement.

15.5 Heavy Floor Beams

Heavy bulks of timber and steel beams should be supported before cutting at the extremities and should then be lowered to a safe working place. A single span arch. can be demolished by hand by cutting narrow segments progressively from each springing parallel to the span of the arch until the width of the arch has been reduced to a minimum which can then be collapsed.

15.6 In-situ Reinforced Concrete

Before commencing demolition, the nature and condition of the concrete, the condition and position of reinforcement, and the possibility of lack of continuity of reinforcement should be ascertained. Attention should be paid to the principles of the structural design to determine which parts of the structure depend on each other to maintain overall stability. Demolition should be commenced by removing partitions and external non-load bearing cladding. It should be noted that in some buildings the frame may rely on the panel wallsfor stability. Where hard demolition methods are to be used, the following procedures should be used.

(i) Reinforced Concrete Beams: For beams, a supporting rope should be attached to the beam. Then the concrete should be removed from both ends by pneumatic drill and the reinforcement exposed. The reinforcement should then be cut in such a way as to allow the beam to be lowered under control to the floor.

(ii) Reinforced Concrete Columns: For columns, the reinforcement should be exposed at the base after restraining wire guy ropes have been placed round the member at the top. The reinforcement should then be cut in such a way as to allow the column to be pulled down to the floor under control.

(iii) Reinforced Concrete Walls: Reinforced concrete walls should be cut into strips and demolished as for columns.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
15.1		Demolishing lime concrete	Demolishing lime concrete manually/by mechanical means and disposal of material within 50 metres lead as per direction of Engineer-in-Charge.	cum	202.00
15.2		Demolishing cement concrete	Demolishing cement concrete manually/by mechanical means including disposal of material within 50 metres lead as per direction of Engineer-in-Charge.		
	15.2.1	1:3:6 or richer	Nominal concrete 1:3:6 or richer mix (i/c equivalent design mix).	cum	578.00
	15.2.2	1:4:8 or leaner	Nominal concrete 1:4:8 or leaner mix (i/c equivalent design mix).	cum	356.00
15.3		Demolishing R.C.C.	Demolishing R.C.C. work manually/ by mechanical means including stacking of steel bars and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-Charge.	cum	844.00
15.4		Cutting reinforcement bar	Extra for cutting reinforcement bars manually/ by mechanical means in R.C.C. work (Payment shall be made on the cross sectional area of R.C.C. work) as per direction of Engineer-in-Charge.	sqm	276.00
15.5		Scrapping, cleaning straightening reinforcement	Extra for scrapping, cleaning and straightening reinforcement from R.C.C. work.	kg	2.25
15.6		Demolishing brick work	Demolishing brick work manually/ by mechanical means including stacking ofserviceable material and disposal of unserviceable material within 50 metres leadas per direction of Engineer-in- Charge.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	15.6.1	Mughal bricks	In lime mortar with old mughal bricks.	cum	421.00
	15.6.2	In lime mortar	In lime mortar.	cum	202.00
	15.6.3	In cement mortar	In cement mortar.	cum	487.00
15.7		Removing mortar from bricks	Removing mortar from bricks and cleaning bricks including stacking within a lead of 50 m (stacks of cleaned bricks shallbe measured) :		
	15.7.1	In lime mortar	From brick work in lime mortar.	1000 nos	1254.00
	15.7.2	In cement mortar	From brick work in cement mortar.	1000 nos	1560.00
15.8		Demolishing stone rubble masonry	Demolishing stone rubble masonry manually/ by mechanical means including stacking of serviceable material and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-Charge:		
	15.8.1	In lime mortar	In lime mortar.	cum	275.00
	15.8.2	In cement mortar	In cement mortar.	cum	582.00
15.9		Dismantling dressed stone work ashlar face stone work	Dismantling dressed stone work ashlar face stone work, marble work or precast concrete work manually/ by mechanical means including stacking ofserviceable and disposal of unserviceable material within50 metres lead as per direction of Engineer-in-Charge:		
	15.9.1	In lime mortar	In lime mortar.	cum	348.00
	15.9.2	In cement mortar	In cement mortar.	cum	680.00
15.10		Removing mortar from and cleaning stones	Removing mortar from and cleaning stones and concrete articles (net quantity of stacks of cleaned materials will be measured) :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	15.10.1	In lime mortar	In lime mortar.	cum	112.00
	15.10.2	In cement mortar	In cement mortar.	cum	162.00
15.11		Dismantling tile/ kota /marble/ Granite work	Dismantling tile/ Kota stone/ Marble / Granite work in floors/walls and roofs laid in cement mortar including stacking material within 50 metres lead.		
	15.11.1	Upto 25mm thick	For thickness of tiles 10 mm to 25 mm.	sqm	17.00
	15.11.2	Above 25mm thick	For thickness of tiles above 25 mm and upto 40 mm.	sqm	26.00
15.12		Dismantling stone slab flooring	Dismantling stone slab flooring laid in cement mortar including stacking of serviceable material and disposal of unserviceable material within 50 metres lead.	sqm	64.00
15.13		Taking out paver blocks	Taking out existing CC interlocking paver blocks from footpath/ central verge,including removal of rubbish etc., disposal of unserviceablematerial to the dumping ground for which payment shall be made separately and stacking of serviceable material within 50 metre lead as per direction of Engineer-in-Charge.	sqm	31.00
15.14		Dismantling doors, windows, shutter including chowkhats,	Dismantling doors, windows and clerestory windows (steel or wood) shutter including chowkhats, architrave, holdfasts etc. complete and stackingwithin 50 metres lead :		
	15.14.1	Area upto 3 sq.m.	Of area 3 sq. metres and below.	each	85.00
	15.14.2	Area>3.0 sq.m.	Of area beyond 3 sq. metres.	each	117.00
15.15		Taking out doors, windows shutters	Taking out doors, windows and clerestory window shutters (steel or wood) including stacking within 50 metres lead:		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	15.15.1	Area up to 3.0 sq.m.	Of area 3 sq. metres and below.	each	33.00
	15.15.2	Area>3.0 sq.m.	Of area beyond 3 sq. metres.	each	44.00
15.16		Dismantling wood work in frames, trusses, purlins and rafters upto 10 metres span	Dismantling wood work in frames, trusses, purlins and rafters upto 10 metres span and 5 metres height including stacking the material within 50 metres lead:		
	15.16.1	Section up to 40 sqm.cm	Of sectional area below 40 square centimetres.	metre	4.00
	15.16.2	Sectional area > 40 sqm	Of sectional area 40 square centimetres and above.	cum	1023.00
15.17		Dismantling trusses, rafters, purlins etc. of wood work beyond 10 span	Extra for dismantling trusses, rafters, purlins etc. of wood work for every additional span of one metre or part thereof beyond 10 metres :		
	15.17.1	Section up to 40 sqm.cm	Of sectional area below 40 square centimetres.	metre per metre span	0.45
	15.17.2	Sectional area > 40 sqm	Of sectional area 40 square centimetres and above.	cum per metre span	151.00
15.18		Dismantling trusses, rafters, purlins etc. of wood work for additional height	Extra for dismantling trusses, rafters, purlins etc. of wood work for every additional height of one metre or part thereof beyond 5 metres :		
	15.18.1	Section up to 40 sqm.cm	Of sectional area below 40 square centimetres.	metre per metre span	0.80
	15.18.2	Sectional area > 40 sqm	Of sectional area 40 square centimetres and above.	cum per metre span	213.00
15.19		Dismantling steel work in single sections	Dismantling steel work in single sections including dismembering and stacking within 50 metres lead in:		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	15.19.1	Joist	R.S. Joists	kg	1.33
	15.19.2	Angle,Tees etc	Channels, angles, tees and flats	kg	0.92
15.20		Dismantling steel work in built up sections with span upto 10 meter including dismembering	Dismantling steel work in built up sections with span upto 10 meter and hieght 5 meter in angles, tees, flats and channels including all gusset plates,bolts, nuts, cutting rivets, welding etc. including dismembering and stacking within 50metres lead.	kg	2.28
15.21		Dismantling steel work in built up sections with span upto 10 meter without dismembering	Dismantling steel work manually/ by mechanical means in built up sections with span upto 10 meter and hieght 5 meter without dismembering and stacking within 50 metres lead as per direction of Engineer-in-Charge.	kg	0.90
15.22		Extra for dismantling steel work of beyond 10 metres	Extra for dismantling trusses, rafters, purlins etc. of steel work for every additional span of one metre or part thereof beyond 10 metres	kg per metre span	0.20
15.23		Extra for dismantling steel work beyond 5 metres height	Extra for dismantling trusses, rafters, purlins etc. of steel work for every additional height of one metre or part thereofbeyond 5 metres.	kg per metre height	0.20
15.24		Extra for marking of structural steel	Extra for marking of structural steel work required to be re-erected.	kg	1.00
15.25		Removal of old Tarfelt	Removal of old Tarfelt i/c disposal as directed by Engineer in charge.	per sqm	8.70
15.26		Demolishing brick tile in terracing	Demolishing brick tile covering in terracing including stackingof serviceable material and disposal of unserviceable material within 50 metres lead.	sqm	24.00
15.27		Dismantling roofing sheet	Dismantling roofing including ridges, hips valleys and gutters etc., and stacking the material within 50 metres lead of:		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	15.27.1	G.S.	G.S. Sheet	sqm	39.00
	15.27.2	AC & Non A.C.	Asbestos/ Non asbestors sheet	sqm	18.00
15.28		Dismantling stone slab roofing	Dismantling stone slab roofing over wooden karries or R.C.C. battens (dismantling karries and battens to be paid for separately) including stacking of serviceable material and disposal of unserviceable material within 50 metres lead.	cum	636.00
15.29		Dismantling tiled roofing	Dismantling tiled roofing with battens boarding etc. complete including stacking ofserviceable material and disposal of unserviceable material within 50 metres lead.	sqm	82.00
15.30		Dismantling wooden ballies in posts and struts	Dismantling wooden ballies in posts and struts including stacking within 50 metres lead.	metre	5.00
15.31		Dismantling fencing posts or struts	Dismantling and stacking within 50 metres lead, fencing posts or struts including allearth work and dismantling ofconcrete etc. in base of :		
	15.31.1	Iron or pipe	T' or 'L' iron or pipe	each	73.00
	15.31.2	R.C.C.	R.C.C.	each	87.00
15.32		Cutting ballies or wooden posts of fencing	Cutting ballies or wooden posts of fencing at the point of projection above the concrete or ground and stacking the same within 50 metres lead.	each	8.00
15.33		Dismantling barbed wire or flexible wire rope in fencing	Dismantling barbed wire or flexible wire rope in fencing including making rolls and stacking within 50 metres lead.	kg	9.00
15.34		Dismantling ornamental design work	Dismantling wooden trellis work excluding frames but including stacking the serviceable material within 50 metres lead.	sqm	15.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
15.35		Dismantling expanded metal	Dismantling expanded metal or I.R.C. fabrics with necessary battens and beading including stacking the serviceable material within 50 metres lead.	sqm	18.00
15.36		Dismantling wooden board in wall lining	Dismantling wooden boardings in lining of walls and partitions, excluding supporting members but including stacking within 50 metres lead :		
	15.36.1	Thickness Upto 10 mm	Upto 10 mm thick	sqm	15.00
	15.36.2	Thickness 10-25mm	Thickness above 10 mm upto 25 mm	sqm	19.00
	15.36.3	Thickness 25-40 mm thick	Thickness above 25 mm upto 40 mm	sqm	22.00
15.37		Dismantling precast concrete or stone slabs	Dismantling precast concrete or stone slabs in walls, partition walls etc. including stacking within 50 metres lead:		
	15.37.1	Thickness Upto 40 mm	Thickness upto 40 mm	sqm	69.00
	15.37.2	Thickness 40-75mm	Thickness above 40 mm upto 75 mm	sqm	104.00
15.38		Dismantling cement asbestos or other hard board ceiling or partition	Dismantling cement asbestos or other hard board ceiling or partition walls including stacking of serviceable materials and disposal of unserviceable materials within 50 metres lead.	sqm	13.00
15.39		Dismantling C.I. or asbestos rain water pipe with fittings	Dismantling C.I. or asbestosrain water pipe with fittings and clamps including stacking the material within 50 metres lead :		
	15.39.1	75 to 80mm ϕ pipe	75 to 80 mm dia pipe.	metre	18.00
	15.39.2	100mm ϕ	100 mm dia pipe	metre	19.00
	15.39.3	150mm ϕ	150 mm dia pipe	metre	19.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
15.40		Dismantling G.I. pipes (external work)	Dismantling G.I. pipes (external work) including excavation and refilling trenches after taking out the pipes manually/ by mechanical means including stacking of pipes within 50 metres lead as per direction of Engineer-in-Charge :		
	15.40.1	15 -40mm ϕ pipe	15 mm to 40 mm nominal bore	metre	40.00
	15.40.2	Above 40mm ϕ pipe	Above 40 mm nominal bore	metre	47.00
15.41		Dismantling C.I. pipes	Dismantling C.I. pipesincluding excavation and refilling trenches after taking out the pipes manually/ by mechanical means, breaking lead caulked joints, melting of lead and making into blocks including stacking of pipes and lead at site within 50 metre lead as per direction of Engineer-in-Charge:		
	15.41.1	Upto 150mm ϕ	Upto 150 mm diameter	metre	114.00
	15.41.2	> 150mm upto 300mm ϕ	Above 150 mm dia upto 300 mm dia.	metre	153.00
	15.41.3	> 300mm ϕ	Above 300 mm diameter	metre	198.00
15.42		Taking out C.I. cover with frame of manholes	Taking out C.I. cover with frame from R.C.C. top slab of manholes of various sizes including demolishing of R.C.C. work manually/ by mechanical means and stacking of useful materials near the site and disposal of unserviceable materials within 50 metres lead as per direction of Engineer-in-Charge.	each	174.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
15.43		Taking out C.I. cover of inspection chambers	Taking out C.I. cover with frame from R.C.C. top slab of inspection chambers of various sizes including demolishing of R.C.C. work manually/ by mechanical means and stacking of useful materials near the site and disposal of unserviceable materials within 50 metres lead as per direction of Engineer-in-Charge.	each	103.00
15.44		Dismantling of road gully chamber	Dismantling of road gully chamber of various sizes including C.I. grating withframe including stacking of useful materials near the site and disposal of unserviceable materials within 50 metres lead including refilling the excavated gap.	each	236.00
15.45		Dismantling of flushing cistern of all types	Dismantling of flushing cistern of all types (C.I./PVC/Vitrious China) including stacking of useful materials near the site and disposal of unserviceable materials within 50 metres lead.	each	223.00
15.46		Dismantling of C.I. sluice valve	Dismantling of C.I. sluice valve including stacking of useful materials within a lead of 50 metres		
	15.46.1	Upto 150mm ϕ	Upto 150 mm diameter	each	83.00
	15.46.2	>150mm ϕ	Above 150 mm diameter	each	276.00
15.47		Dismantling of spindle fire hydrant	Dismantling of spindle fire hydrant including stacking of useful materials within 50 metres lead.	each	163.00
15.48		Dismantling old plaster or skirting	Dismantling old plaster or skirting raking out joints and cleaning the surface for plaster including disposal of rubbish to the dumping ground within 50 metres lead.	sqm	13.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
15.49		Dismantling aluminium/ Gypsum partitions, doors, windows, fixed glazing and false ceiling	Dismantling aluminium/ Gypsum partitions, doors, windows, fixed glazing and false ceiling including disposal of unserviceable surplus material and stacking of serviceable material with in 50 meters lead as directed by Engineer-in-Charge.	sqm	13.00
15.50		Dismantling of old S.W. pipes	Dismantling of old S.W. pipes including breaking of joints and bed concrete stacking of useful materials near the site within 50 m lead and disposal of unserviceable materials into municipal dumps:		
	15.50.1	100 mmϕ	100 mm diameter	metre	21.00
	15.50.2	150 mmϕ	150 mm diameter	metre	23.00
	15.50.3	200 mmϕ	200 mm diameter	metre	25.00
	15.50.4	250 mmϕ	250 mm diameter	metre	26.00
	15.50.5	300 mmϕ	300 mm diameter	metre	28.00
	15.50.6	350 mmϕ	350 mm diameter	metre	32.00
	15.50.7	400 mmϕ	400 mm diameter	metre	35.00
	15.50.8	440 mmϕ	440 mm diameter	metre	36.00
15.51		Dismantling of manhole including R.C.C. top slab, C.I. cover with frame	Dismantling of manhole including R.C.C. top slab, C.I. cover with frame including stacking of useful materials near the site and disposal of unserviceable materials into municipal dumps within 50 m lead :		
	15.51.1	90x80 cm	Rectangular manhole 90x80 cm and 45 cm deep	each	611.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	15.51.2	120x90 cm	Rectangular manhole 120x90 cm and 90 cm deep.	each	1069.00
	15.51.3	140x90cm	Rectangular arch type manhole 140x90cm and 2.45m deep.	each	2023.00
	15.51.4	122 cm diameter	Circular manhole 122 cm diameter and 1.68 m deep	each	1553.00
15.52		Extra for depth of manholes dismantled	Extra for depth of manholes dismantled:		
	15.52.1	90x80 cm	Rectangular manhole 90x80 cm and beyond 45 cm depth.	each	486.00
	15.52.2	120x90 cm	Rectangular manhole 120x90 cm and beyond 90 cm depth.	each	579.00
	15.52.3	140x90cm	Rectangular arch type manhole 140x90 cm and beyond 2.45m deep (upto 4.25 m depth).	each	468.00
	15.52.4	122cm diameter	Circular manhole 122 cm diameter and beyond 1.68 m deep (upto 2.29 m depth).	each	322.00
15.53		Taking out existing kerb stones	Taking out existing kerb stones of all types from footpath/ central verge, including removal of mortar etc., disposal of unserviceable material to the dumping ground and stacking of serviceable material within 50 metre lead as per direction of Engineer-in- Charge.	metre	9.00
15.54		Demolishing dry brick pitching	Demolishing dry brick pitching in floors, drains etc. including stacking of serviceable material and disposal of unserviceable material within 50 metres lead :	cum	513.00
15.55		Demolishing mud phaska	Demolishing mud phaska in terracing and disposal ofmaterial within 50 metres lead.	cum	356.00
15.56		Dismantling jack arch roofing and floors	Dismantling jack arch roofing and floors including stacking of serviceable material and disposal of unserviceable material within 50 metres lead.	Sqm	99.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
15.57		Demolishing R.B. work	Demolishing R.B. work manually/ by mechanical means including stacking of steel bars and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-charge.	cum	1236.00
15.58		Dismantling of roads	Dismantling manually/ by mechanical means including stacking of serviceable material and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-charge :		
	15.58.1	W.B.M. road	Water bound macadam road	sqm	86.00
	15.58.2	Bituminous road	Bituminous road	sqm	168.00
15.59		Dismantling steel cylinder R.C. pipes	Dismantling steel cylinder R.C. pipes including excavation and refilling trenches after taking out the pipes, manually/ by mechanical means breaking lead caulked joints, melting of lead and making into blocks including stacking of pipes & lead at site within 50 metres lead as per direction of Engineer-in-charge :		
	15.59.1	Up to 600 mm diameter	Up to 600 mm diameter	per metre	285.00
	15.59.2	Above 600 mm diameter	Above 600 mm diameter	per metre	720.00
15.60		Dismantling asbestos cement pressure pipes	Dismantling asbestos cement pressure pipes including excavation and refillingtrenches after taking out the pipes manually/ by mechanical means and stacking the pipes within 50 metres lead as per direction of Engineer-in-charge		
	15.60.1	Up to 150 mm diameter	Up to 150 mm diameter	per metre	129.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	15.60.2	Above 150 mm diameter	Above 150 mm diameter	per metre	156.00
15.61		Dismantling of R.C.C. spun vent shaft	Dismantling of R.C.C. spun vent shaft including excavating the cement concrete pit completely, taking out the shaft, refilling the excavated gap, stacking the useful materialsnear the site and disposal of unserviceable materials within 50 metres lead.	each	1776.00
15.62		Dismantling of cement concrete platform	Dismantling of cement concrete platform along with curtain walls and base concrete etc. including stacking of useful materials near the site and disposal of unserviceable materials within 50 metres lead:		
	15.62.1	120 x 120 cm	120 x 120 cm (outside to outside)	each	429.00
	15.62.2	210 x 120 cm	210 x 120 cm (outside to outside)	each	932.00
15.63		Dismantling of flexible pavement	Dismantling of flexible pavement (bituminous courses) by mechanical means and disposal of dismantled material up to a lead of 1 kilometre, as per direction of Engineer-in-charge.	cum	203.00
15.64		Disposal of building rubbish / malba / similar unserviceable, dismantled or waste materials	Disposal of building rubbish / malba / similar unserviceable, dismantled or waste materials by mechanical means, including loading, transporting, unloading to approved municipal dumping ground or as approved by Engineer-in-charge, beyond 50 m initial lead, for all leads including all lifts involved.	cum	121.00

CHAPTER – 16.0

TUBE WELL AND RAIN WATER HARVESTING

PREAMBLE

TUBE WELL AND RAIN WATER HARVESTING (CHAPTER: 16.0)

16.1 General Specifications

- Tube wells drilled shall be perfectly vertical and the rates of drilling are inclusive of vertically test required to be conducted.
- Total depth of bore shall generally be 200 ft. (60 meters) but the same may be increased or decreased depending upon the strata and the yield.
- The size of the bore shall be determined by the internal dia of casing pipe in all cases.
- Non-perforated black mild steel casing pipe of specified diameter and thickness shall be lowered for a depth of 9 meters from ground level in all cases but if the strata beyond 9 meters continuous to be soft casing pipe shall be lowered upto the rock level or the hard stratum level to be decided by the Engineer-in Charge. The black casing shall protrude 0.3 meters to 0.75 meters above ground level and a M.S. cap shall be screwed at top.
- The diameter of the bore shall be 6mm less than internal dia of the casing pipe beyond the end of casing pipe upto rock level and in rock the size of the bore shall be reduced by another 6mm.
- In water bearing strata other than rock perforated or slotted pipes of suitable diameter and strainer shall have to be fitted below the casing pipe properly coupled with each other in continuation to the casing pipe.
- For bores in sandy strata suitable size say 3mm to 10mm of washed gravel will have to be shrouded around the annular space of the strainer and perforated or slotted pipes. This should be properly filled in layers and rammed so that the gravel may occupy the cavities inthe strata around and there is no further sinking. The space around the casing shall be filled up with earth from spoils and properly rammed.
- The gravel packing shall be applied for 30mm to 150mm all round the slotted or perforated pipe, depending upon the nature of strata encountered and the require size of finished bore.
- Measurements for the gravel packed bores shall be allowed as per the internal dia of the casing pipe.The rates for drilling for the gravel packed bores shall be inclusive of lowering and removal of the housing pipe. Drilling has to be done upto rock level.
- The rates for drilling provided in the schedule are inclusive of depreciation charges of all T & P required for drilling operation transportation of drilling machine, erection at site, removal of machine from site after completion, cost of water, cost of drilling, fuel labour and all other unforeseen item for drilling work and clearance of site after completion of work.
- The rates for other items of work provided in the schedule of rates are inclusive of cost of all material including taxes tec. And labour including transportation, erection operation, fueland all other unforeseen item pertaining to the execution of the work, clearance of site etc. complete.
- No payment will be made to the contractor for drilling in case any collapse of bore occurs during boring or during lowering the pipes etc. Any precautions, steps required to control caving of the bore shall also be the contractor's responsibility for which no extra charges will be paid for.

- If during any of the operation carried out for completion of the tube well, any tools, pipes etc. fall down in the well, the contractor shall carry out the necessary finishing operation of his own cost. He shall use his own equipment for such operation. If the tube becomes useless due to any fall of an article, it shall be treated as abandoned tube well and no payment shall be made.

- Requirement of casing pipe: The casing pipe generally conforming to IS specifications IS: 4260-1992 shall be used.

- Requirement of GI Pipes Fully galvanized GI pipes conforming to IS Specification IS: 1239 (part-I) 1990 medium quality shall be used.

- Deep well guide hand pump suitable for fixing on a bore having a minimum diameter of 100mm complete with brass cylinders and double buckets of suitable size for pumping 900 litres of water per hour at 40 strokes per minutes from a maximum suction of 50 meters hand pump to be equipped with M.S. handle and bracket and additional M.S. support to prevent guide flange from the breakage and thus prevent breakage of handle and brackettoo. The pump to be in two halves provided with flanged joint for case of erection. The pump to be fitted with two vertical bright turned guide rods and provided with gun metal gland nuts and to have detachable compression cock spout and to be tapped for 32mm and 40mm pipes.

16.2 Yield Test:

Yield test means and include the observation of discharge from the well for the specified period varying from 4 to 20 hours, obtained by continuous pumping, for small and big bores, the rates include provision and transport to site of works all necessary equipment required for conducting the yield test and observation of yield in comparison to draw down and recuperation.

The rates include transportation for drilling machine to and from. No extra charges for installation and removal of the drilling machine from site of works are payable.

Well Development For wells in unconsolidated or in consolidated formations development work shall consist of de-sanding by pumping the wall with a gradually increasing capacity pumping is started with an amount of 20% of the permanent capacity and carried on till the water has become clear, free of sand and silt. The rate of flow is then augmented to 40%and the process repeated. Descending is continued to an amount of 15% of the permanent capacity if for this rate of flow the water is free from all materials surrounding the wall, the water will be perfectly clear for the normal capacity, when air is used submergence ratio, (ratio of length of drop pipe submerged to total length) 60% to 75% shall be adopted.

Disinfection After the well is complete constructed, before putting it to use for human consumption, the well should be cleaned thoroughly of all foreign substances, the casing pipe swabbed with alkalis to remove oil, grease etc. and the well disinfected with a chlorine solution. The chlorine solution must be of such volume and strength that a concentration of at least 50 mg/L of chlorine is obtained in all parts of the well.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
16.1		Bore well upto 90 meter depth	Boring/drilling bore well of required dia perfectly vertical to receive casing/ strainer pipe, by suitable method prescribed in IS: 2800 (part I), including collecting samples from different strata, preparing and submitting strata chart/bore log, including hire and running charges of all equipments, tools, plants and machineries required for the job, all complete as per direction of Engineer–in-Charge, upto 90 metre depth below ground level.		
	16.1.1	All types of soil upto 90 meter	All types of soil upto 90 metre depth.		
	16.1.1.1	100 mm dia	100 mm dia.	meter	378.00
	16.1.1.2	150 mm dia	150 mm dia.	meter	413.00
	16.1.1.3	200 mm dia	200 mm dia.	meter	451.00
	16.1.2	Rocky strata upto 90 meter	Rocky strata including Boulders.Upto 90 metre depth.		
	16.1.2.1	100 mm dia	100 mm dia.	meter	554.00
	16.1.2.2	150 mm dia	150 mm dia.	meter	625.00
	16.1.2.3	200 mm dia	200 mm dia.	meter	690.00
	16.1.3	All soil beyond 90 meter	All types of soil beyond 90 metre depth.	meter	
	16.1.3.1	100 mm dia	100 mm dia.	meter	385.00
	16.1.3.2	150 mm dia	150 mm dia.	meter	428.00
	16.1.3.3	200 mm dia	200 mm dia	meter	495.50
	16.1.4	Rocky strata beyond 90 meter	Rocky strata including Boulders. Beyond 90 metre depth.		
	16.1.4.1	100 mm dia	100 mm dia.	meter	572.00
	16.1.4.2	150 mm dia	150 mm dia.	meter	656.00
	16.1.4.3	200 mm dia	200 mm dia	meter	716.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
16.2		Unplasticized PVC bore well casing	Supplying, assembling, lowering and fixing in vertical position in bore well, unplasticized PVC medium well casing (CM) pipe of required dia, conforming to IS: 12818, including requiredhire and labour charges, fittings and accessories etc. all complete, for all depths, as per direction of Engineer-in-Charge.		
	16.2.1	100 mm dia	100 mm nominal dia. (minimum wall thickness 5mm).	meter	376.00
	16.2.2	150 mm dia	150 mm nominal dia (minimum wall thickness 7.5mm).	meter	423.00
	16.2.3	200 mm dia	200 mm nominal dia(minimum wall thickness 10mm).	meter	559.00
16.3		Unplasticized PVC bore well screen	Supplying, assembling, lowering and fixing in vertical position in bore well unplasticized PVC medium well screen (RMS)pipes with ribs, conforming to IS: 12818, including hire and labour charges, fittings and accessories etc. All complete, for all depths, as per direction of Engineer-in-Charge.		
	16.3.1	100 mm dia	100 mm nominal size dia.	meter	446.00
	16.3.2	150 mm dia	150 mm nominal size dia.	meter	551.00
	16.3.3	200 mm dia	200 mm nominal size dia.	meter	878.00
16.4		Mild steel bore casing pipe	Supplying, assembling, lowering and fixing in vertical position in bore well, ERW (Electric Resistance Welded) FE 410 mild steel screwed and socketed/plain ended casing pipes of required dia, conforming to IS: 4270, of reputed and approved make, including painted with outside surface with two coats of anticorrosive paint of approved brand and manufacture, including required		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			hire and labour charges, fittings and accessories, all complete, for all depths, as per direction of Engineer- in-Charge.		
	16.4.1	100 mm dia	100 mm nominal size dia having minimum wall thickness 4.50 mm.	meter	795.00
	16.4.2	150 mm dia	150 mm nominal size dia having minimum wall thickness 4.85 mm.	meter	1191.00
	16.4.3	200 mm dia	200 mm nominal size dia having minimum wall thickness 5.40 mm.	meter	1611.00
16.5		Plain slotted mild steel bore pipe (type A)	Supplying, assembling, lowering and fixing in vertical position in bore well, ERW (Electric Resistance Welded) FE 410 plain slotted (having slot of size 1.6/3.2 mm) mild steel threaded and socketed /plain bevel ended pipe (type A) of required dia, conforming to IS: 8110, of reputed and approved make, having wall thickness not less than 5.40 mm, including painted with outside surface with two coats of anticorrosive bitumestic paint of approved brand and manufacture, including hire and labour charges, fittings and accessories, all complete, for all depths, as per direction of Engineer–in-Charge.		
	16.5.1	100 mm dia	100 mm nominal size dia.	meter	855.00
	16.5.2	150 mm dia	150 mm nominal size dia.	meter	1271.00
	16.5.3	200 mm dia	200 mm nominal size dia.	meter	1691.00
16.6		Gravel packing in tubewell	Gravel packing in tubewell construction in accordance with IS: 4097, including providing gravel fine/ medium/ coarse, in required grading and sizes as per actual requirement, all complete as per direction of Engineer-in-charge.	cum	947.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
16.7		Development of tube well	Development of tube well in accordance with IS : 2800 (part I) and IS: 11189, to establish maximum rate of usable water yield without sand content (beyond permissible limit), with required capacity air compressor, running the compressor for required time till well is fully developed, measuring yield of well by "V" notch method or any other approved method, measuring static level and draw down etc. by step draw down method, collecting water samples and getting tested in approved laboratory, i/c disinfection of tubewell, all complete, including hire and labour charges of air compressor, tools and accessories etc., all as per requirement and direction of Engineer-in-Charge.	hour	478.00
16.8		Mild steel cap or welded plate to the top of bore well pipe	Providing and fixing suitable size threaded mild steel cap or spot welded plate to the top of bore well housing/ casing pipe, removable as per requirement, all complete for borewell of:		
	16.8.1	100 mm dia	100 mm dia	each	244.00
	16.8.2	150 mm dia	150 mm dia	each	366.00
	16.8.3	200 mm dia	200 mm dia	each	488.00
16.9		M.S. clamp	Providing and fixing M.S. clamp of required dia made of 100 x10 mm flat to the top of casing/ housing pipe of tubewell as per IS: 2800 (part I), including necessary bolts and nuts of required size complete.		
	16.9.1	100 mm dia	100 mm clamp.	each	1068.00
	16.9.2	150 mm dia	150 mm clamp.	each	1129.00
	16.9.3	200 mm dia	200 mm clamp.	each	1281.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
16.10		Bail plug /Bottom plug	Providing and fixing Bail plug/ Bottom plug of required dia to the bottom of pipe assembly of tubewell as per IS: 2800 (part I).		
	16.10.1	100 mm dia	100 mm dia	each	196.00
	16.10.2	150 mm dia	150 mm dia	each	242.00
	16.10.3	200 mm dia	200 mm dia	each	265.00
16.11		Boring for hand pump/ tube well	Boring with 100 mm diameter casing pipe for hand pump/ tube well in all soils except ordinary hard rocks requiring blasting including lowering the casing pipe.		
	16.11.1	Upto 6 metres	Upto 6 metres depth.	meter	236.00
	16.11.2	>6 m and upto 12 m	Beyond 6 m and upto 12 m depth.	meter	280.00
	16.11.3	> 12 m and upto 18 m	Beyond 12 m and upto 18 m depth.	meter	326.00
16.12		Filters with brass strainer	Providing and placing in position filters to 40 mm diameter G.I. pipe with brass strainer of approved quality. Filter of 32 mm dia	meter	545.00
16.13		40 mm G.I. pipe for tube well	Providing and fixing to filter and lowering to proper levels 40 mm G.I. pipe for tube well including cleaning and priming the tube well. Filter of 32 mm dia meter.	meter	386.00
16.14		Hand pump of India mark II	Providing and placing in position hand pump of Indiamark II of approved quality for 40 mm diameter G.I. pipe complete with all accessories.	each	841.00
16.15		Stone boulders in recharge pit	Supplying, filling, spreading and leveling stone boulders of size range 5 cm to 20 cm, in recharge pit, in the required thickness, for all leads and lifts, all complete as per direction of Engineer-in-charge.	cum	760.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
16.16		Gravels of in the recharge pit	Supplying, filling, spreading and leveling gravels of size range 5 mm to 10 mm, in the recharge pit, over the existing layer of boulders, in required thickness, for all leads and lifts, all complete as per direction of Engineer-in-Charge.	cum	877.00
16.17		Coarse sand in recharge pit	Supplying, filling, spreading and leveling coarse sand of size range 1.5 mm to 2 mm inrecharge pit, in required thickness over gravel layer, for all leads and lifts, all complete as per direction of Engineer-in- Charge.	cum	819.00
16.18		Precast R.C.C. perforated drain covers	Providing and fixing factory made precast RCC perforated drain covers, having concrete of strength not less than M-25, of size 1000 x 450x50 mm, reinforced with 8 mm dia four no longitudinal and 9 nos cross sectional T.M.T. Hoop bars, including providing50 mmdia perforations @ 100 to 125 mm c/c, including providing edge binding with M.S. flats of size 50 mmx1.6 mm complete, all as per direction of Engineer-in- Charge.	each	959.00
16.19		Rain water harvesting filter kit unplasticised PVC	Providing and fixing of unplasticised rigid PVC rain water harvesting filter kit UV protected with filter, PVC valve, conforming to IS: 1392, working pressure 4 kg\sqcmTypeB, minimum capacity 20 kilolitre/ hour, of approved makeincluding all accessories like unplasicied PVC pipe cliips of approved design with bombay nail of required length including making good the wall etc. complete for designed catchent		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			area.		
	16.19.1	75 mm dia for 100 square meter area	75mm diameter PVC pipe, having wall thickness 3.2 mm for 100 m^2 catchment area	each	5858.00
	16.19.2	90 mm diafor150 square meter area	90 mm diameter PVC pipe, having wall thickness 3.2 mm for 150 m^2 catchment area	each	7100.00
	16.19.3	110 mm diafor250 square meter area	110 mm diameter PVC pipe, having wall thickness 3.2 mm for 250 m^2 catchment area	each	8900.00
	16.19.4	160 mm diafor450 square meter area	160 mm diameter PVC pipe, having wall thickness 3.2 mm for 450 m^2 catchment area	each	16500.00
16.20		Rain water harvesting filter kit unplasticised (filter dia 160mm)	Providing and fixing of unplasticised rigid PVC rain water harvesting filter kit including filter of 160 mm diameter wall thickness 3.2mm with PVC valve, 110 mm diameter UV protected conforming to ID: 13592 for working pressure 6 kg\sqcmTypeB capacity 40 kilo liter\hour including all accessories like unplasicied PVC pipe clips of approved design with bombay nail of required length including making good the wall etc complete for catchement area upto 200 sqm.	each	8679.00
16.21		Rain water harvesting kit filter (filter dia 90mm)	Rain water filter made out of ABS plastics material, designed to operate on gravity of rainwater from roof through down pipe, having inlet (id) of 110 mm, outlet (id) of 110 mm; provided with SS 304 wire mesh filter element having diameter of not less than 90 mm and height not less than 500 mm with cap of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			not less than 70 mm height with two holes of not less than 10 mm on cap as a safety feature. Element must have back wash cap at bottom for cleaning dried particles. Filter must have flush outlet of not less than 60 mm moulded opposite to inlet connection. Flush valve must be supplied along with filter. Rainwater filter must be manufactured from injection moulding process, blended with uv stabilizers. Rainwater filters width to be not less than 400 mm, depth to be not less than140 mm and height must not be less than 600 mm overall. Moulded projection to be given at bottom to fix on wall and a polymer clamp to fix at top.		
16.21.1		75 mm diafor100 square meter area	System size 75mm diameter (filterain, tee, door bend, and valve 75mm) For the catchment up to 100 square meter.	each	5900.00
16.21.2		110 mm diafor200 square meter area	System size 110mm diameter (filterain, tee, door bend, and valve 110mm) For the catchment up to 200 square meter.	each	7750.00
16.21.3		160 mm diafor300 square meter area	System size 160mm diameter (filterain, tee, door bend, and valve 110mm) For the catchment up to 300 square meter.	each	11900.00
16.22		Borewell adaptor	Borewell adaptor for connecting Rain Water Harvesting pipe line to borewell.		
	16.22.1	75 mm dia	Borewell adaptor 75mm	each	250.00
	16.22.2	110 mm dia	Borewell adaptor 110mm	each	300.00
	16.22.3	160 mm dia	Borewell adaptor 160mm	each	400.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
16.23		Rain Water Harvesting Stainless Steel 304 Filter (250 mm Length & 200 mm Width + 90 mm Dia) for 150 Square mtr. Area.	Providing & Installation of Rainwater filter made out of ABS plastics material, **with two stages of filtration and live monitoring facility,** designed to harvest rooftop rainwater without wasting water, must be designed to operate on gravity of rainwater from roof through down pipe, having inlet (id) of 110 mm. outlet (id) of 110 mm; provided with SS 304 non clogging wedge wire filter screen in curved shape, of not less than 250 mm length and width not less than 200 mm, having horizontal 0.5 mm v shaped continuous slots for filtration, supported on vertical welded supports at interval of not less than 15 mm, provided with heavy duty rubber gasket on all sides, built with handle from SS rod having diameter of not less 3 mm and height not less than 25 mm and width not less than 65 mm. The unit must have front outlet to drain impurities and also to act as safety feature. The unit must have second filter element of HDPE material, not less than 90 mm diameter and not less than 455 mm length, with 200 micron filtration capacity, to be placed at bottom inside the clean water outlet of the filter unit. Filter must have transparent, openable lid made out of ABS material for live monitoring of rainwater harvesting process. The drain outlet or safety feature must not be less than 40 mm diameter (OD) moulded in front bottom side of unit. Readymade rotational moulded, heavy duty "built in flush unit" to drain out first rain shower or flush valve 110 mm must be supplied along with filter. Rainwater filter must be manufactured from injection moulding process, blended with	Each	7064.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			UV stabilizers. Rainwater filter's depth to be not less than 300 mm, length to be not less than 250 mm and height must not be less than 400 mm overall, minimum four moulded clamps must be moulded both at top and bottom to fix the unit firmly on the wall with Steel Screws.		
16.24		Rain Water Harvesting Polymer Filter (250 mm Length & 200 mm Width + 90 mm dia) For 140 Sq. Meter Area.	Providing & Installation of Rainwater filter made out of ABS plastics material, **with two stages of filtration and live monitoring facility,** designed to harvest rooftop rainwater without wasting water, must be designed to operate on gravity of rainwater from roof through down pipe, having inlet (id) of 110 mm. outlet (id) of 110 mm; provided with HDPE double layered filter cloth having 400 micron filtration capacity, provided with well craw, wrapped on PU coated curved frame of not less than 250 mm length and width not less than 200 mm, supported by vertically welded two supports and a handle of PU coated rod having diameter of not less 3 mm and height not less than 15 mm and width not less than 65 mm. The unit must have front outlet to drain impurities and also to act as safety feature. The unit must have second filter element of HDPE material, not less than 90 mm diameter and not less than 455 mm length, with 200 micron filtration capacity, to be placed at bottom inside the clean water outlet of the filter unit. Filter must have transparent, openable lid made out of ABS material for live monitoring of rainwater harvesting process. The drain outlet or safety feature must not be less than 40 mm diameter (od) moulded in front bottom side of unit. Readymade rotational		5241.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			molded, heavy duty "built in flush unit" to drain out first rain shower or a flush valve 110 mm must be supplied along with filter. Rainwater filter must be manufactured from injection moulding process, blended with UV stabilizers. Rainwater filter's depth to be not less than 300 mm, length to be not less than 250 mm and height must not be less than 400 mm overall, minimum four moulded clamps must be moulded both at top and bottom to fix the unit firmly on the wall with Required Steel Screws.		

CHAPTER – 17.0

SANITARY INSTALLATIONS

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1.	IS 771 (Part-1)	Specification for glazed fire clay sanitary appliances: Part 1: General requirements.
2.	IS 771 (Part-2)	Specification for glazed fire clay sanitary appliances: Part 2: Specific requirements of kitchen and laboratory sink.
3.	IS 772	Specific action for general requirements for enameled cast iron sanitary appliances.
4.	IS 774	Flushing cisterns for water closets and urinals (Other than plastic cistern)-Specifications.
5.	IS 1300	Phenolic moulding materials.- Specifications
6.	IS 1703	Water fittings- copper alloy float valves (horizontal plunger type) - Specification.
7.	IS 1729	Cast iron /Ductile Iron Drainage Pipes and pipe fittings for Over ground non-pressure pipe line Socket and Spigot Series
8.	IS 1795	Specification for pillar taps for water supply purposes.
9.	IS 2267	Polystyrene moulding and extrusion materials - Specifications
10.	IS 2326	Specification for Automatic Flushing Cisterns for Urinals (Other than plastic cisterns)
11.	IS 2548 (Part-1)	Plastic seats and covers for water closets Part 1: Thermo set seats and covers - Specifications
12.	IS 2548 (Part-2)	Plastic seats and covers for water closets Part 2: Thermoplastic seats and covers.- Specifications
13.	IS 2556	Vitreous sanitary appliances (vitreous china) –Specifications
14.	IS 2556 (Part-1)	Part-1: General requirements
15.	IS 2556 (Part-2)	Part-2: Specific requirements of wash-down water closets.
16.	IS 2556 (Part-3)	Part-3: Specific squatting pans
17.	IS 2556 (Part-4)	Part-4: Specific requirements of wash basins
18.	IS 2556 (Part-5)	Part-5: Specific requirements of laboratory sinks
19.	IS 2556 (Part-6)	Part-6: Specific requirements of Urinals and Partition plates
20.	IS 2556 (Part-7)	Part-7: Specific requirements of accessories for sanitary appliances
21.	IS 2556 (Part-14)	Part-14: Specific requirements of integrated squatting pans.
22.	IS 2556 (Part -15)	Part-15: Specific requirements of universal water closets.
23.	IS 2963	Specification for Copper alloy waste fittings for wash basins and sinks.
24.	IS 3076	Specification for low density polyethylene pipes for potable water supplies
25.	IS 3389	Urea formaldehyde moulding materials. – Specifications
26.	IS 3989	Specification for centrifugally cast (spun) iron spigot and socket soil, waste and ventilating pipes fittings and accessories.
27.	IS 4827	Specification for electroplated coating of nickel and chromium on copper and copper alloys.
28.	IS 4984	Specification for high density polyethylene pipes for potable water supplies.
29.	IS 4985	Unplasticised P.V.C. pipes for potable water supply - Specifications.
30	IS 7231	Plastic flushing cisterns for water closets and urinals
31.	IS 13983	Specifications. Stainless steel sinks for domestic purposes - Specifications.

PREAMBLE

SANITARY INSTALLATIONS
(CHAPTER: 17.0)

All vitreous sanitary appliances (Vitreous China) shall conform to IS 2556 (Part-I) general Requirements

17.1 General Requirements for Installation of Sanitary Fittings

The work shall be carried cut, complying in all respects with the requirements of relevant by laws of the local body in whose jurisdiction the work is situated.

Any damage caused to the building, or to electric, sanitary, water supply or other, installations etc. therein, either due to negligence on the part of the contractor, or due to actual requirements of the work, shall be made good and the building or the installation shall be restored to its original condition by the contractor. Nothing extra shall be paid for such restoration works except where otherwise specified.

For making good the damage to the under mentioned items of work, the specifications as given in the following paras shall apply, unless directed otherwise.

(i) **Masonry Work:** The masonry work shall be made good by using the same class of bricks, tiles or stones as was damaged during the execution of the work. The mortar used shall be cement mortar 1:5 (1 cement: 5 sand) or as directed by the Engineer-in- Charge.

(i) **Plain Concrete Work :** Concrete work for sub-grade of the flooring, foundations and other plain concrete works shall be cement concrete 1:5:10 (1 cement : 5 sand : 10 graded stone aggregate 40 mm nominal size). A coat of neat cement slurry shall be applied at the junction with old work, before laying fresh concrete.

(ii) **Cement Concrete Flooring and R.C.C. Work :** Cement concrete 1:2:4 (1 Cement : 2 sand : 4 graded stone aggregate 20 mm nominal size) shall be used after applying a coat of neat cement slurry at the junction with old work, and the surface finished to match with the surrounding surface.

(iv) Urinals shall be installed at a height of 65 cm from standing floor level to top of lip of the urinal until otherwise specified.

(v) The floor slab shall be suitably sunk to receive the stall urinal. Where the floor slab is not sunk, the stall urinal shall be provided over a platform. The lip of the stall urinal shall be flush with the finished floor level adjacent to it.

(vi) Wash basin shall be fixed at height of 750-800 mm from the floor level

17.2 Flushing Cisterns

The flushing cisterns shall be automatic or manually operated high level or low level as specified, for water closets and urinals. A high level cistern is intended to operate with minimum height of 125 cm and a low level cistern with a maximum height of 30 cm between the top of the pan and the under side of the cistern. Cisterns shall be of following type

(i) Vitreous China (IS 774) for Flushing type
(**i**) Automatic Flushing Cistern (IS 2326) and
(iii)Plastic cisterns (IS 7231).

The cistern shall be provided with a removable cover which shall fit closely and shall be secured against displacement. In designs, where the operating mechanism is attached to the cover, the cover may be made in two sections, the section supporting the mechanism being

securely fixed or booked to the body. The flush pipe (except plastic flush pipe) shall have an internal diameter of 32 + 1 mm for high level cistern and 38 + 1 mm for low level cistern. The steel flush pipe shall be not less than 1 mm thick whereas the lead flush pipe shall have a minimum thickness of 3.5 mm. For high density polyethylene pipes, the outside diameter of the pipes shall be 40 mm. For unplasticised PVC plumbing pipes the outside diameter of the pipe shall be 40 mm for high level cisterns, and 50 mm for low level cisterns. In the case of high level flushing cisterns, a pipe clip fitted with a rubber buffer shall be fixed to the flush pipe to prevent damage either to the pipe or to the seat when the seat is raised. No flush pipe is requiredfor coupled cisterns.

OPERATIONAL AND PERFORMANCE REQUIREMENTS

17.3 Flushing Arrangement

The cistern under working conditions and with the float valve in closed position shall operate on a single operation of the operating mechanism/lever without calling for a sudden jerk in pulling. If a valve is used instead of siphon for flushing purposes, the valve shall be completely leak proof.

17.4 Working Water Level

The working water-level shall be a minimum of 6.5 cm. below the effective top edge of the cistern and shall be legibly and permanently marked on the inside of the cistern. Effective top edge shall be taken on edge after top of the body without considering bead.

17.5 Freedom from Self Siphonage

The siphonic system shall be capable of being rapidly brought into action when the water is at the working water level, but shall not self siphon or leak into the flush pipe when the water is up to 1 cm above the invert of the overflow pipe.

17.6 Reduced Water Level

The discharge shall operate satisfactorily when the cistern is filled to a level up to 1 cm. below the working water level.

17.7 Discharge Capacity

When tested in accordance with IS 7231, cistern of 5 litres and 10 litres capacities, when required to give a full flush, shall respectively discharge 5 litres and 10 litres with variation of ± 0.5 litres. Dual-flush cistern of 10 litres capacity shall discharge alternatively a short flush of 5 ± 0.5 litres. Dual flush cistern of 6/3 litres capacity shall discharge 6 ± 0.5 litres and alternatively a half flush of 3 ± 0.5 litres.

17.8 Discharge Rate

When tested in accordance with IS 7231, the discharge rate shall be 10 ± 0.5 litres within 6 seconds and 5 ± 0.5 litres within 3 seconds for cistern of capacities 10 litres and 5 litres and 6 ± 0.5 litres within 6 second and 3 ± 0.5 litres within 3 second for cistern of 6/3 litres capacity respectively. The cistern shall be so designed that there is no appreciable variation in the force of the flush during the discharge of the required quantity of water. For coupled cisterns, this test shall not be applicable.

17.9 Special Requirements

Distortion Resistance Test

The cisterns, complete with its fittings, shall be installed and filled with water to the marked water line and observed for any distortion. The cistern shall not budge more than 6 mm and the cover shall not be dislodged.

17.10 Dead Load Test

When the flushing mechanism incorporates chain pull or hand operated lever, the cistern, complete with its fittings, when installed and filled with water to the marked water line and tested by the application of a dead load of 230 N applied 6 mm from the end of the operating lever arm for 30 seconds, shall not distort to such an extent that any part becomes detached. In the case of other operating mechanism, the dead load applied shall be a mass equivalent to the operating force required to overcome the normal hydrostatic head; Thirty seconds after the load is removed, the function and appearance of the cistern shall not be impaired.

17.11 Front Thrust Test

The front thrust test shall be applied only to cisterns intended for low level use. The cistern complete with its fittings, when installed and filled with water to the marked water line and tested by the method described in IS 7231, shall not distort to such an extent as to be inoperable or unsightly when the load is removed.

17.12 Impact Test

The cistern, complete with its fittings, when installed and filled as described in IS 7231 shall show no defect after one impact. Repeat the test but with the cistern empty. The cistern shall show no defect after the further impact.

17.13 Plastic Seat and Covers for Water Closet.

The seat and cover shall be of thermosetting or thermoplastic conforming to IS 2548 as specified. Unless otherwise specified these shall be of closed pattern.

17.14 Strength:

The seats shall withstand without permanent distortion of the seat or hinge fittings or damage to any finish, a load of 1150 N for 30 minutes applied in the manner prescribed inIS 2548.

17.15 Sand Cast Iron or Centrifugally Cast (Spun) Iron Pipes and Fittings

Sand cast iron spigot and socket soil, waste and ventilating pipes, fittings and accessories shall conform to IS 1729. Centrifugally cast (Spun) iron spigot and socket soil, waste and ventilating pipes, fittings and accessories shall conform to IS 3989. The fittings shallconform to the same I.S. specifications to which the pipe itself conforms in which they are connected. The pipes shall have spigot and socket ends, with head on spigot end in case of sand cast iron pipes and without head on spigot end in case of cast iron (Spun) pipes. The pipes and fittings shall be true to shape, smooth and cylindrical, their inner and outer surface being as nearly as practicable concentric. They shall be sound and shall be free fromcracks, taps, pinholes and other imperfections and shall be neatly dressed and carefully fettled. All pipes and fittings shall ring clearly when struck with a light hand hammer. The ends of pipes and fittings shall be reasonably square to their axis. The sand cast iron pipes shall be 1.5/1.8/2.0 metre in length including socket ends, cast iron (Spun) pipes shall be 1.5/1.75/2.0/2.5/3.0 metre in length excluding socket ends, unless shorter lengths are either specified or required at junctions etc. The pipe and fittings shall be supplied without ears, unless specified or directed otherwise. All pipes and fittings shall be coated internally and externally with the same material at the factory, the fitting being preheated prior to total immersion in a bath containing a uniformly heated composition having a tar or othersuitable base. The coating material shall have good adherence and shall not scale off. In all instances where the coating material has tar or similar base it shall be smooth and tenacious and hard enough not to flow when exposed to a temperature of 77 degree centigrade but not so brittle at a temperature of 0 degree centigrade as to chip off when scribed lightly with a pen knife. The standard weights and thicknesses of pipes and their tolerances shall be as prescribed in.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
17.1		W.C. Indian type with flushing cistern	Providing and fixing water closet squatting pan (Indian type W.C. pan) with 100mm sand cast Iron P or Strap, UPVC P or S trap, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever) conforming to IS : 7231, with all fittings and fixtures completeincluding cutting and making good thewalls and floors wherever required :		
	17.1.1	Size 580x440mm	White Vitreous china Orissa pattern W.C. pan of size 580x440mm with integral type foot rests.	each	2995.00
	17.1.2	Size 580x480mm	Stainless Steel AISI-304(18/8) Orissa pattern W.C. pan of size 585x480 mm with flush pipe and integrated type foot rests.	each	7031.00
17.2		W.C. European type PVC flushing cistern	Providing and fixing white vitreous china pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS : 7231, with all fittings and fixtures complete including cutting andmaking good the walls and floors wherever required :		
	17.2.1	White plastic seat cover and lid	W.C. pan with ISI marked white solid plastic seat and lid	each	3036.00
	17.2.2	Black plastic seat cover and lid	W.C. pan with ISI marked black solid plastic seat and lid	each	3013.00
	17.2.3	Extra for S.S.304 european seat	Providing and fixing AISI-304 (18/8) European type WC instead of white vitreous china pedestal type water closet.	each	4536.00
17.3		W.C. European type china flushing cistern C.P. flush bend	Providing and fixing white vitreous china pedestal type water closet (European type) with seat and lid, 10 litre low level white vitreous china flushing cistern and C.P. flush bend with fittings and C.I.brackets, 40mm flush bend, overflow arrangement with specials of standard make and mosquito proof coupling of approved		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs)
1	2	3	4	5	6
			municipal design complete including painting of fittings and brackets, cutting and making good the walls and floors wherever required :		
	17.3.1	White plastic seat cover and lid	W.C. pan with ISI marked white solid plastic seat and lid.	each	4613.00
	17.3.2	Black plastic seat cover and lid	W.C. pan with ISI marked black solid plastic seat and lid.	each	4590.00
17.4		W.C. Anglo Indian type	Providing and fixing vitreous china dual purpose closet suitable for use as squatting pan or European type water closet (Anglo Indian W.C pan) with seat and lid fixed with C.P. brass hinges and rubber buffers, 10 litre low level flushing cistern with fitting and brackets, 40mm flush bend, 20mm over flow pipe, with specials of standard make and mosquito proof coupling of approved municipal design complete, including painting of fittings and brackets, cutting and making good the walls and floors wherever required White vitreous china dual purpose WC pan with white solid plastic seat and lid with 10liter low level flushing cistern i/c C.P. flush bend.	each	5120.00
17.5		W.C. wall mounted	Providing and fixing white vitreous china extended wall mounting water closet of size 780x370x690 mm of approved shape including providing and fixing white vitreous china cistern with dual flush fitting, of flushing capacity 3 litre/6 litre (adjustable to 4 litre/8 litres), including seat cover, and cistern fittings, nuts, bolts and gasket etc complete.	each	10761.00
17.6		Urinal flat back or wall corner type with automatic flushing cistern	Providing and fixing white vitreous china flat back or wall corner type lipped front urinal basin of 430x260x350mm and 340x410x265 mm sizes respectively with automatic flushing cistern with standard flush pipe and C.P. brass spreaders with brass unions and G.I clamps complete, including painting of fittings and brackets, cutting and making good the walls and floors wherever required :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
	17.6.1	One urinal	One urinal basin with 5 litre white P.V.C. automatic flushing cistern.	each	2754.00
	17.6.2	Range of two urinal	Range of two urinal basins with 5 litre white P.V.C. automatic flushing cistern.	each	4458.00
	17.6.3	Range of three urinal	Range of three urinal basins with 10 litre white P.V.C. automatic flushing cistern.	each	6312.00
	17.6.4	Range of four urinal	Range of four urinal basins with 10 litre white P.V.C. automatic flushing cistern.	each	8384.00
17.7		Urinal flat back half stall urinal with automatic flushing cistern	Providing and fixing white vitreous china flat back half stall urinal of size 580x380x350mm with white PVC automatic flushing cistern, with fittings, standard size C.P. brass flush pipe, spreaders with unions and clamps (all in C.P. brass) with waste fitting as per IS : 2556, C.I. trap with outlet grating and other couplings in C.P. brass including painting of fittings and cutting and making good the walls and floors wherever required		
	17.7.1	Single half stall urinal	Single half stall urinal with 5 litre P.V.C. automatic flushing cistern.	each	5567.00
	17.7.2	Range of two half stall urinals	Range of two half stall urinals with 5 litre P.V.C. automatic flushing cistern.	each	8660.00
	17.7.3	Range of three half stall urinals	Range of three half stall urinals with 10 litre P.V.C. automatic flushing cistern.	each	12764.00
	17.7.4	Range of four half stall urinals	Range of four half stall urinals with 10 litre P.V.C. automatic flushing cistern.	each	15547.00
17.8		Squatting plate with flushing cistern	Providing and fixing one piece construction white vitreous china squatting plate with an integral longitudinal flushing pipe, white P.V.C. automatic flushing cistern, with fittings, standard size G.I. / PVC flush pipe for back and front flush with standard spreader pipes with fittings, G.I clamps and C.P. brass coupling complete including painting of fittings and cutting and making good the walls and floors etc.wherever required		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.8.1	Single squatting plate	Single squatting plate with 5 litre P.V.C. automatic flushing cistern.	each	3478.00
	17.8.2	Range of two squatting plates	Range of two squatting plates with 5 litre P.V.C. automatic flushing cistern.	each	5128.00
	17.8.3	Range of three squatting plates	Range of three squatting plates with 10 litre P.V.C. automatic flushing cistern.	each	7151.00
	17.8.4	Range of four squatting plates	Range of four squatting plates with 10 litre P.V.C. automatic flushing cistern.	each	8689.00
17.9		Urinal china water less	Providing and fixing white vitreous china water less urinal of size 600x330x315 mm having antibacterial /germs free ceramic surface, fixed with cartridge having debris catcher and hygiene seal.	each	12300.00
17.10		Urinal infrared sensor operated	Providing and fixing white vitreous china battery based infrared sensor operated urinal of approx. size 610 x 390 x 370 mm having pre and post flushing with water (250 ml and 500 ml consumption), having water inlet from back side, including fixing to wall with suitable brackets all as per manufacturers specification and direction of Engineer-in-Charge.	each	5032.50
17.11		Wash basin	Providing and fixing wash basin with C.I. brackets, 15 mm C.P. brass pillar taps,32 mm C.P. brass waste of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever require :		
	17.11.1	630x450 mm flat back pair of pillar cock (chinaclay)	White Vitreous China Wash basin size 630x450 mm with a pair of 15 mm C.P. brass pillar taps.	each	2460.00
	17.11.2	630x450 mm flat back single pillar cock (china clay)	White Vitreous China Wash basin size 630x450 mm with a single 15 mm C.P. brass pillar tap.	each	2187.00
	17.11.3	550x400 mm flat back pair of pillar cock (china clay)	White Vitreous China Wash basin size 550x400 mm with a pair of 15 mm C.P. brass pillar taps.	each	1938.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.11.4	550x400 mm flat back single pillar cock (chinaclay)	White Vitreous China Flat back wash basin size 550x400 mm with single 15 mm C.P. brass pillar tap.	each	1664.00
	17.11.5	600x480 mm Angle back single pillar cock (china clay)	White Vitreous China Angle back wash basin size 600x480 mm with single 15 mm C.P. brass pillar tap.	each	1775.00
	17.11.6	400x400 mm Angle back single pillar cock (china clay)	White Vitreous China Angle back wash basin size 400x400 mm with single 15 mm C.P. brass pillar tap.	each	1502.00
	17.11.7	450x300 mm flat back single pillar cock (chinaclay)	White Vitreous China Flat back wash basin size 450x300 mm with single 15 mm C.P. brass pillar tap.	each	1525.00
	17.11.8	660x460 mm Surgeon type pair of Elbow pillar cock (china clay)	White Vitreous China Surgeon type wash basin of size 660x460 mm with a pair of 15 mm C.P. brass pillar taps with elbow operated lever ISI marked.	each	3761.00
	17.11.9	660x460 mm Surgeon type single of E-bow pillar cock (chinaclay)	White Vitreous China Surgeon type wash basin of size 660x460 mm with single 15 mm C.P. brass pillar tapswith elbow operated levers ISI marked.	each	2907.00
	17.11.10	Oval shap single pillar cock (china clay)	Oval shape wash basin over counter /under counter with 15 mm C.P brass pillar tap (Counter to be paidsepratly).	each	7153.00
	17.11.11	405x355 mm round single pillar cock (S.S)	Stainless Steel AISI-304(18/8) Round basin 405x355 mm with single 15mm C.P. brass pillar tap	each	3308.00
	17.11.12	530x345 mm single pillar cock (S.S)	Stainless Steel AISI-304(18/8) Wash basin 530x345 mm with single 15 mm C.P. brass pillar tap.	each	2948.00
17.12		Pedestal vitreous china	Providing and fixing white vitreous china pedestal for wash basin completely recessed at the back for the reception of pipes and fittings.	each	1065.00
17.13		Kitchen sink (White glazed fire clay)	Providing and fixing kitchen sink with C.I. brackets, C.P. brass chain with rubber plug, 40 mm C.P. brass waste complete, including painting the fittings and brackets, cutting and making good the walls wherever required : White glazed fire clay kitchen sink of size 600x450x250m.	each	2492.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
17.14		Kitchen sink (stainless Steel)	Providing and fixing Stainless Steel A ISI 304 (18/8) kitchen sink as per IS 13983 with C.I. brackets and stainless steel plug 40 mm including painting of fittings and brackets, cutting and making good the walls wherever required :		
	17.14.1	With drain board	Kitchen sink with drain board		
	17.14.1.1	Depth 250mm	510x1040 mm bowl depth 250mm.	each	8074.00
	17.14.1.2	Depth 225mm	510x1040 mm bowl depth 225mm.	each	7609.00
	17.14.1.3	Depth 200mm	510x1040 mm bowl depth 200mm.	each	7493.00
	17.14.1.4	Depth 178mm	510x1040 mm bowl depth 178mm.	each	5286.00
	17.14.2	Without drain board	Kitchen sink without drain board		
	17.14.2.1	Depth 200mm	610x510 mm bowl depth 200 mm.	each	4583.00
	17.14.2.2	Depth 200mm	610x460 mm bowl depth 200 mm.	each	4351.00
	17.14.2.3	Depth 178mm	470x420 mm bowl depth 178 mm.	each	3422.00
17.15		Laboratory sink vitreous china	Providing and fixing white vitreous china laboratory sink with C.I. brackets, C.P. brass chain with rubber plug 40mm C.P brass waste and40mm C.P. brass trap with necessary C.P. brass unions complete including painting of fittings and brackets, cutting and making good the wall wherever required :		
	17.15.1	Depth 150mm	Size450x300x150mm	each	2341.00
	17.15.2	Depth 200mm	Size600x450x200mm	each	3328.00
17.16		Draining board (White glazed fire clay)	Providing and fixing White glazed fire clay draining board of size 600x450x25mm with C.I. brackets including painting of brackets, cutting and making good the walls wherever required.	each	1043.00
17.17		Water closet Indian type	Providing and fixing white vitreous china water closet squatting pan (Indian type)		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.17.1	Long pattern	Long pattern W.C. pan of size 580 mm	each	765.00
	17.17.2	Orissa pattern	Orissa pattern W.C. pan of size 580x440 mm	each	1322.00
17.18		Extra for coloured W.C. pan	Extra for using coloured W.C. pan instead of white W.C. pan Orissa pattern W.C. pan 580x440 mm	each	842.00
17.19		W.C. Pedestal type European type	Providing and fixing white vitreous china pedestal type (European type/ wash down type) water closet pan.	each	1235.00
17.20		Extra for coloured pedestal type W.C pan	Extra for using coloured pedestal type W.C pan (European type) with low level cistern of same colour instead of white vitreous china W.C pan and cistern.	each	1452.00
17.21		Foot rests	Providing and fixing a pair of white vitreous china foot rests of standard pattern for squatting pan water closet:		
	17.21.1	250x130x30 mm	250x130x30 mm	pair	200.00
	17.21.2	250x125x25 mm	250x125x25 mm	pair	142.00
17.22		Flushing cistern low Level (P.V.C)	Providing and fixing P.V.C. low level flushing cistern with manually controlled device (handle lever) conforming to IS: 7231, with allfittings and fixtures complete. 10 litre capacity		
	17.22.1	White	White	each	945.00
	17.22.2	Coloured	coloured	each	1037.00
17.23		Controlled flush, low level cistern made of vitreous china	Providing and fixing controlled flush, low level cistern made of vitreous china with all fittings complete.10 litre (full flush capacity)		
	17.23.1	White	White	each	1615.00
	17.23.2	Coloured	Coloured	each	2196.00
17.24		Plastic seat cover with lid for pedestal type W.C.	Providing and fixing solid plastic seat with lid for pedestal type W.C. pan complete :		
	17.24.1	White	White	each	456.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.24.2	Black	Black	each	433.00
	17.24.3	Coloured	Coloured (other than black and white)	each	688.00
17.25		Inlet connection for flush pipe	Providing and fixing G.I. inlet connection for flush pipe connecting with W.C. pan.	each	114.00
17.26		Urinal (flat back or wall corner type)	Providing and fixing white vitreous china flat back or wall corner type lipped front urinal basin of 430x260x350mm or 340x410x265mm sizes respectively.	each	733.00
17.27		Urinal flat back (half stall)	Providing and fixing white vitreous china flat back half stall urinal of size 580x380x350mm.	each	2243.00
17.28		Squatting plate urinal	Providing and fixing white vitreous china squatting plate urinal with integral rim longitudinal flush pipe.	each	1564.00
17.29		Wash basin white vitreous china excluding the cost of fittings	Providing and fixing white vitreous china wash basin including making all connections but excluding the cost of fittings :		
	17.29.1	630x450mm. Flat back	Flat back wash basin of size 630x450mm.	each	1379.00
	17.29.2	550x400mm Flat back	Flat back wash basin of size 550x400mm.	each	857.00
	17.29.3	600x480mm. Angle back	Angle back wash basin of size 600x480mm.	each	967.00
	17.29.4	400x400mm. Angle back	Angle back wash basin of size 400x400mm.	each	694.00
	17.29.5	450x300mm. Flatback	Flat back wash basin of size 450x300mm.	each	717.00
	17.29.6	660x460mm. Surgeon type	Surgeon type wash basin of size 660x460mm.	each	1519.00
17.30		Kitchen sink glazed fire clay	Providing and fixing kitchen sink including making all connections excluding cost of fittings. White glazed fire clay sink of size 600x450x250mm.	each	1843.00
17.31		Laboratory sinkvitreous china	Providing and fixing white vitreous china laboratory sink including making all connections excluding cost of fittings:		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
	17.31.1	depth 150 mm.	Size 450x300x150 mm.	each	1236.00
	17.31.2	depth 200 mm.	Size 600x450x200 mm.	each	2025.00
17.32		soil, waste and vent pipes	Providing and fixing soil, waste and vent pipes:		
	17.32.1	CI Pipe (sand cast)	Sand cast iron socket and spigot (S&S) pipe as per IS: 1729		
	17.32.1.1	75 mm dia	75 mm dia with wall thickness of 5 mm	metre	790.00
	17.32.1.2	100 mm dia	100 mm dia with wall thickness of 5 mm	metre	970.00
	17.32.2 17.32.2.1	CI (Spun cast) 75 mm dia	Spun cast iron socket and spigot (S&S) pipe as per IS: 3989 75 mm dia with wall thickness of 3.5 mm	metre	840.00
	17.32.2.2	100 mm dia	100 mm dia with wall thickness of 4 mm	metre	1020.00
	17.32.2.3	150 mm dia	150mm dia with wall thickness of 5mm	metre	2460.00
	17.32.3	CI (Hubless)	Hubless Centrifugally cast (spun) iron pipe epoxy coated in side & out side including complete coupling assembly (EPDM rubber gasket stainless steel claim and external metallic/ protective shield as per IS:15905)		
	17.32.3.1	75 mm dia	75 mm dia with wall thickness of 3.5 mm	metre	773.00
	17.32.3.2	100 mm dia	100 mm dia with wall thickness of 3.5mm	metre	910.00
	17.32.3.3	150 mm dia	150mm dia with wall thickness of 4 mm	metre	2430.00
17.33		pipe joints with cement mortar 1:2	Providing and filling the joints with spun yarn cement slurry and cement mortar 1:2 (1 cement : 2 sand) in S.C.I./ C.I. Pipes :		
	17.33.1	75 mm dia	75 mm dia pipe	each	39.00
	17.33.2	100 mm dia	100 mm dia pipe	each	46.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
17.34		M.S. holder	Providing and fixing M.S. holder-bat clamps of approved design to Sand Cast iron/cast iron (spun) pipe embedded in and including cement concrete blocks 10x10x10cm of Cement Concrete 1:2:4 mix (1 cement : 2 sand : 4 graded stone aggregate 20mm nominal size) including cost of cutting holes and making good the walls etc. :		
	17.34.1	75 mm dia	For 75 mm dia pipe	each	37.00
	17.34.2	100 mm dia	For 100 mm dia pipe	each	66.00
	17.34.3	150 mm dia	For 150 mm dia pipe	each	105.00
17.35		Bend (with lid)	Providing and fixing bend of required degree with access door, insertion rubber washer 3 mm thick, bolts and nuts complete.		
	17.35.1	150 mm dia	150 mm dia		
	17.35.1.1	Sand cast	Sand cast iron as per IS – 1729 minimum wall thickness 6.5mm	each	400.00
	17.35.1.2	Spun cast	Spun cast iron as per IS – 3989 minimum wall thickness 5 mm	each	430.00
	17.35.1.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905 minimum wall thickness 3.5 mm	each	680.00
	17.35.2	100 mm dia	100 mm dia		
	17.35.2.1	Sand cast	Sand cast iron as per IS – 1729 minimum wall thickness 6.5mm	each	364.00
	17.35.2.2	Spun cast	Spun cast iron as per IS – 3989 minimum wall thickness 4 mm	each	408.00
	17.35.2.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905 minimum wall thickness 3 mm	each	450.00
	17.35.3	75mm dia	75 mm dia		
	17.35.3.1	Sand cast	Sand cast iron as per IS – 1729 minimum wall thickness 6.5 mm	each	94.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
	17.35.3.2	Spun cast	Spun cast iron as per IS – 3989 minimum wall thickness 3.5 mm	each	341.00
	17.35.3.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905 minimum wall thickness 3 mm	each	370.00
17.36		end (Plain)	Providing and fixing plain bend of required degree.		
	17.36.1	150 mm dia	150 mm dia		
	17.36.1.1	Sand cast	Sand cast iron as per IS – 1729 minimum wall thickness 6.5mm	each	702.00
	17.36.1.2	Spun cast	Spun cast iron as per IS – 3989 minimum wall thickness 5 mm	each	825.00
	17.36.1.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905 minimum wall thickness 3.5 mm	each	600.00
	17.36.2	100 mm dia	100 mm dia		
	17.36.2.1	Sand cast	Sand cast iron as per IS – 1729 minimum wall thickness 6.5mm	each	381.00
	17.36.2.2	Spun cast	Spun cast iron as per IS – 3989 minimum wall thickness 4 mm	each	390.00
	17.36.2.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905 minimum wall thickness 3 mm	each	344.00
	17.36.3	75 mm dia	75 mm dia		
	17.36.3.1	Sand cast	Sand cast iron as per IS – 1729 minimum wall thickness 6.5mm	each	230.00
	17.36.3.2	Spun cast	Spun cast iron as per IS – 3989 minimum wall thickness 3.5 mm	each	259.00
	17.36.3.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905 minimum wall thickness 3 mm	each	242.00
17.37		Heel rest sanitary bend of cast iron	Providing and fixing heel rest sanitary bend		
	17.37.1	100 mm dia	100 mm dia		
	17.37.1.1	Sand cast	Sand cast iron as per IS - 1729	each	329.00
	17.37.1.2	Spun cast	Spun cast iron as per IS - 3989	each	387.00
	17.37.2	75 mm dia	75 mm dia		
	17.37.2.1	Sand cast	Sand cast iron as per IS - 1729	each	288.00
	17.37.2.2	Spun cast	Spun cast iron as per IS - 3989	each	329.00
17.38		Double equal junction with access door	Providing and fixing double equal junction of required degree with access door, insertion rubber washer 3 mm thick, bolts and nuts complete :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
	17.38.1.1	150 mm dia	150X150X150X150		
	17.38.1.2	Sand cast	Sand cast iron as per IS - 3989	each	1080.00
	17.38.2	100mm dia	100x100x100x100mm		
	17.38.2.1	Sand cast	Sand cast iron as per IS - 1729	each	744.00
	17.38.2.2	Spun cast	Spun cast iron as per IS - 3989	each	820.00
	17.38.3	75mm dia	75x75x75x75mm		
	17.38.3.1	Sand cast	Sand cast iron as per IS - 1729	each	590.00
	17.38.3.2	Spun cast	Spun cast iron as per IS - 3989	each	654.00
17.39		Double equal plain junction	Providing and fixing double equal plain junction of required degree.		
	17.39.1	150 mm dia	150X150X150X150 mm		
	17.39.1.1	Sand cast	Sand cast iron as per IS - 3989	each	1106.00
	17.39.2	100mm dia	100x100x100x100mm		
	17.39.2.1	Sand cast	Sand cast iron as per IS – 1729	each	666.00
	17.39.2.2	Spun cast	Spun cast iron as per IS – 3989	each	782.00
	17.39.2.3	Hubless C.I.	Hubless cast(spun) iron as per IS-5905	each	600.00
	17.39.3	75mm dia	75x75x75x75 mm		
	17.39.3.1	Sand cast	Sand cast iron as per IS – 1729	each	434.00
	17.39.3.2	Spun cast	Spun cast iron as per IS – 3989	each	573.00
	17.39.3.3	Hubless C.I.	Hubless cast(spun) iron as per IS-5905	each	362.00
17.40		Single equal junction with access door	Providing and fixing single equal junction of required degree with access door, insertion rubber washer 3 mm thick, bolts and nuts complete.		
	17.40.1	150 mm	150X150X150 mm		
	17.40.1.1	Spun cast	Spun cast iron as per IS - 3989	each	1499.00
	17.40.2	100mm	100x100x100 mm		
	17.40.2.1	Sand cast	Sand cast iron as per IS - 1729	each	495.00
	17.40.2.2	Spun cast	Spun cast iron as per IS - 3989	each	640.00
	17.40.3	75mm	75x75x75 mm		
	17.40.3.1	Sand cast	Sand cast iron as per IS - 1729	each	387.00
	17.40.3.2	Spun cast	Spun cast iron as per IS - 3989	each	533.00
17.41		Single equal plain junction	Providing and fixing single equalplain junction of required degree :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
	17.41.1	150 mm	150X150X150 mm		
	17.41.1.1	Spun cast	Spun cast iron as per IS - 3989	each	891.00
	17.41.2	100mm	100x100x100 mm		
	17.41.2.1	Sand cast	Sand cast iron as per IS – 1729	each	504.00
	17.41.2.2	Spun cast	Spun cast iron as per IS – 3989	each	607.00
	17.41.2.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	450.00
	17.41.3	75mm	75x75x75 mm		
	17.41.3.1	Sand cast	Sand cast iron as per IS – 1729	each	352.00
	17.41.3.2	Spun cast	Spun cast iron as per IS – 3989	each	428.00
	17.41.3.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	290.00
17.42		Double unequal junction with access door (100X75mm dia)	Providing and fixing double unequal junction of required degree with access door, insertion rubber washer 3 mm thick, bolts and nutscomplete :100x100x75x75 mm		
	17.42.1	Sand cast	Sand cast iron as per IS - 1729	each	745.00
	17.42.2	Spun cast	Spun cast iron as per IS - 3989	each	1093.00
17.43		Double unequal plain junction of cast iron (100X75mm dia)	Providing and fixing double unequal plain junction of required degree: 100x100x75x75 mm		
	17.43.1	Sand cast	Sand cast iron as per IS – 1729	each	724.00
	17.43.2	Spun cast	Spun cast iron as per IS – 3989	each	1014.00
	17.43.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	535.00
17.44		Single unequal junction with access door (100X75mm)	Providing and fixing single unequal junction of required degree with access door, insertion rubber washer 3 mm thick, bolts and nuts complete : 100x100x75 mm		
	17.44.1	Sand cast	Sand cast iron as per IS - 1729	each	541.00
	17.44.2	Spun cast	Spun cast iron as per IS - 3989	each	832.00
17.45		Single unequal plain junction (100X75mm)	Providing and fixing single unequal plain junction of required degree : 100x100x75 mm		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.45.1	Sand cast	Sand cast iron as per IS – 1729	each	491.00
	17.45.2	Spun cast	Spun cast iron as per IS – 3989	each	724.00
	17.45.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	490.00
17.46		Double equal plain invert branch	Providing and fixing double equal plain invert branch of required degree:		
	17.46.1	100mm	100x100x100x100 mm		
	17.46.1.1	Sand cast	Sand cast iron as per IS - 1729	each	724.00
	17.46.1.2	Spun cast	Sand cast iron as per IS - 3989	each	700.00
	17.46.1.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	803.00
	17.46.2	75mm	75x75x75x75 mm		
	17.46.2.1	Sand cast	Sand cast iron as per IS - 1729	each	544.00
	17.46.2.2	Spun cast	Spun cast iron as per IS - 3989	each	573.00
17.47		Single equal plain invert branch	Providing and fixing single equal plain invert branch of required degree:		
	17.47.1	100mm	100x100x100 mm		
	17.47.1.1	Sand cast	Sand cast iron as per IS - 1729	each	549.00
	17.47.1.2	Spun cast	Sand cast iron as per IS - 3989	each	570.00
	17.47.1.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	512.00
	17.47.2	75mm	75x75x75 mm		
	17.47.2.1	Sand cast	Sand cast iron as per IS - 1729	each	428.00
	17.47.2.2	Spun cast	Sand cast iron as per IS - 3989	each	434.00
	17.47.2.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	345.00
17.48		Double unequal invert branch (100X75mm)	Providing and fixing double unequal invert branch of required degree : 100x100x75x75 mm		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.48.1	Sand cast	Sand cast iron as per IS - 1729	each	724.00
	17.48.2	Spun cast	Spun cast iron as per IS - 3989	each	927.00
17.49		Unequal plain invert branch	Providing and fixing single unequal plain invert branch of required degree : 100x100x75 mm		
	17.49.1	Sand cast	Sand cast iron as per IS - 1729	each	636.00
	17.49.2	Spun cast	Spun cast iron as per IS - 3989	each	695.00
	17.49.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905 minimum wall thickness	each	575.00
17.50		Off sets (Sand cast)	Providing and fixing sand cast iron S&S off sets as per IS: 1729		
	17.50.1	76 mm off sets	76 mm off sets		
	17.50.1.1	75 mm dia	With 75 mm dia pipe	each	300.00
	17.50.1.2	100 mm dia	With 100 mm dia pipe	each	474.00
	17.50.2	114 mm off sets	114 mm off sets		
	17.50.2.1	75 mm dia	With 75 mm dia pipe	each	410.00
	17.50.2.2	100 mm dia	With 100 mm dia pipe	each	503.00
	17.50.3	152 mm off sets	152 mm off sets		
	17.50.3.1	75 mm dia	With 75 mm dia pipe	each	477.00
	17.50.3.2	100 mm dia	With 100 mm dia pipe	each	605.00
17.51		Off sets Sand cast iron	Providing and fixing sand cast iron S&S off sets as per IS: 3989.		
	17.51.1	75 mm off sets	75 mm off sets		
	17.51.1.1	75 mm dia	With 75 mm dia pipe	each	312.00
	17.51.1.2	100 mm dia	With 100 mm dia pipe	each	370.00
	17.51.1.3	150 mm dia	With 150 mm dia pipe	each	410.00
	17.51.2	150 mm off sets	150 mm off sets		
	17.51.2.1	75 mm dia	With 75 mm dia pipe	each	392.00
	17.51.2.2	100 mm dia	With 100 mm dia pipe	each	508.00
	17.51.2.3	150 mm dia	With 150 mm dia pipe	each	625.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
17.52		Fixing door piece, C.I. pipe fittings	Providing and fixing door piece, insertion rubber washer 3mm thick, bolts and nuts complete :		
	17.52.1	100mm dia	100 mm		
	17.52.1.1	Sand cast	Sand cast iron as per IS - 1729	each	541.00
	17.52.1.2	Spun cast	Spun cast iron as per IS - 3989	each	547.00
	17.52.2	75 mm dia	75 mm		
	17.52.2.1	Sand cast	Sand cast iron as per IS - 1729	each	387.00
	17.52.2.2	Spun cast	Spun cast iron as per IS - 3989	each	387.00
17.53		Terminal guard	Providing and fixing terminal guard:		
	17.53.1	150mm	150mm		
	17.53.1.1	Sand cast	Sand cast iron as per IS - 1729	each	322.00
	17.53.1.2	Spun cast	Spun cast iron as per IS - 3989	each	380.00
	17.53.1.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	410.00
	17.53.2	100 mm	100 mm		
	17.53.2.1	Sand cast	Sand cast iron as per IS - 1729	each	259.00
	17.53.2.2	Spun cast	Spun cast iron as per IS - 3989	each	300.00
	17.35.2.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	320.00
	17.53.3	75 mm	75 mm		
	17.53.3.1	Sand cast	Sand cast iron as per IS - 1729	each	196.00
	17.53.3.2	Spun cast	Spun cast iron as per IS - 3989	each	210.00
	17.53.3.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	250.00
17.54		Collar	Providing and fixing collar:		
	17.54.1	100 mm	100 mm		
	17.54.1.1	Sand cast	Sand cast iron S & S as per IS - 1729	each	201.00
	17.54.1.2	Spun cast	Sand cast iron S & S as per IS - 3989	each	317.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.54.2	75 mm	75 mm		
	17.54.2.1	Sand cast	Sand cast iron S & S as per IS - 1729	each	149.00
	17.54.2.2	Spun cast	Sand cast iron S & S as per IS- 3989	each	219.00
17.55		Lead caulked joints to C.I. pipes & fittings	Providing lead caulked joints to sand cast iron/ centrifugally cast (spun) iron pipes and fittings of diameter:		
	17.55.1	150mm	150mm	each	488.00
	17.55.2	100 mm	100 mm	each	234.00
	17.55.3	75 mm	75 mm	each	201.00
17.56		M.S. stays and clamps for C.I. pipes	Providing and fixing M.S. stays and clamps for sand cast iron/ centrifugally cast (spun) iron pipes of diameter :		
	17.56.1	100 mm dia	100 mm dia	each	62.00
	17.56.2	75 mm dia	75 mm dia	each	50.00
17.57		Trap of self cleansing design	Providing and fixing trap of self cleansing design with screwed down or hinged grating with or without vent arm complete, including cost of cutting and making good the walls and floors :		
	17.57.1	100X100mm	100 mm inlet and 100 mm outlet		
	17.57.1.1	Spun cast	Spun cast iron S & S as per IS: 3989.	each	825.00
	17.57.1.2	Sand cast	Sand Cast iron S & S as per IS: 1729.	each	610.00
	17.57.1.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	310.00
	17.57.2	100X75mm	100 mm inlet and 75 mm outlet		
	17.57.2.1	Sand cast	Sand cast iron S & S as per IS - 3989	each	854.00
	17.57.2.2	Spun cast	Spun Cast iron S & S as per IS- 1729.	each	540.00
	17.57.2.3	Hubless C.I.	Hubless cast(spun) iron as per IS-15905	each	260.00
17.58		Unplasticised rigid PVC soil and waste pipes	Providing and fixing unplasticised Rigid PVC soil and waste pipes conforming to IS: 13592 Type B including jointing with seal ring conforming to IS : 5382 leaving 10		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			mm gap for thermal expansion. Single socketed pipes for working pressure of 4 kg/sqcm. (minimum wall thickness 3.2mm)		
	17.58.1	75mm dia	75 mm diameter	meter	136.00
	17.58.2	110mm dia	110 mm diameter	meter	238.00
17.59		Unplasticised PVC pipe fittings	Providing and fixing unplasticised PVC moulded fittings/ accessories for unplasticised Rigid PVC soil and waste pipes conforming to IS: 13592 Type A including jointing with seal ring conforming to IS : 5382 leaving 10 mm gap for thermal expansion.		
	17.59.1	Coupler	Coupler		
	17.59.1.1	75 mm	75 mm	each	117.00
	17.59.1.2	110 mm	110 mm	each	154.00
	17.59.2	Pushfit Coupler	Single pushfit Coupler		
	17.59.2.1	75 mm	75 mm	each	140.00
	17.59.2.2	110 mm	110 mm	each	195.00
	17.59.3	T with door	Single tee with door		
	17.59.3.1	75 mm	75x75x75 mm	each	219.00
	17.59.3.2	110 mm	110x110x110 mm	each	346.00
	17.59.4	T without door	Single tee without door		
	17.59.4.1	75 mm	75x75x75 mm	each	201.00
	17.59.4.2	110 mm	110x110x110 mm	each	266.00
	17.59.5	Bend	Bend 87.50		
	17.59.5.1	75 mm	75 mm bend	each	117.00
	17.59.5.2	110 mm	110 mm bend	each	172.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
17.60		Unplasticised - PVC pipe clips	Providing and fixing Unplasticised - PVC pipe clips of approved design to Unplasticised - PVC soil and waste pipes by means of bombay nail of required length including and making good the wall etc. complete.		
	17.60.1	75 mm	75 mm	each	33.00
	17.60.2	110 mm	110 mm	each	50.00
17.61		UPVC trap of self cleaning design	Providing and fixing UPVC trap ofself cleaning design complete. Including cost of cutting and making good the wall and floors. 100 mm inlet and 75 mm outlet	each	517.00
17.62		M.S. holder bat clamp to C.I. Pipes	Providing and fixing M.S. holder bat clamp of approved design to sand cast iron/ cast iron (spun) pipes comprising of M.S. flat brackets made of 50x5mm flat of specified shape, projecting 75mm outside the wall surface and fixed on wall with 4nos, 6mm dia expansion hold fasteners including drilling necessary holes in brick wall/ CC/ RCC surface and the cost of bolts etc. The pipes shall be fixed to the already fixed brackets with the help of 30mm x1.6mm galvanised M.S. flats of specified shape and of total length 420mm and shall be fixed with M.S. nuts, bolts, and washers of size 25x6mm, one bolts on each side of the pipe.		
	17.62.1	580 mm	Total bracket length 580mm of approved shape and design (for single, 100mm dia pipe).	each	180.00
	17.62.2	810mm	Total bracket length 810mm of approved shape and design (for two, 100mm dia pipes).	each	224.00
	17.62.3	1040mm	Total bracket length 1040mm of approved shape and design (for three, 100mm dia pipes).	each	265.00
17.63		Chases in brick masonry walls	Cutting chases in brick masonry walls (clay brick/flyash brick/AAC blocks) for following diameter sand cast iron/		

Item	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			centrifugally cast (spun) iron pipes and making good the same with cement concrete 1:3:6 (1 cement : 3 sand :6 graded stone aggregate 12.5 mm nominal size) including necessary plaster and pointing in cement mortar 1:4 (1 cement : 4 sand) :		
	17.63.1	100 mm	100 mm dia	meter	222.00
	17.63.2	75 mm	75 mm dia	meter	159.00
	17.63.3	50 mm	50 mm dia	meter	101.00
17.64		P.V.C. waste pipe for sink or wash basin	Providing and fixing P.V.C. waste pipe for sink or wash basin including P.V.C. waste fittings complete.		
	17.64.1	Rigid	Semi rigid pipe		
	17.64.1.1	32 mm	32 mm dia	each	87.00
	17.64.1.2	40 mm	40 mm dia	each	98.00
	17.64.2	Flexible	Flexible pipe (only in residential & complaxes)		
	17.64.2.1	32 mm	32 mm dia	each	87.00
	17.64.2.2	40mm	40 mm dia	each	98.00
17.65		Sand cast Iron grating for gully trap.	Providing and fixing 100 mm sand cast Iron grating for gully trap.	each	43.00
17.66		Mosquito proof coupling	Providing and fixing in position 25mm diameter mosquito proof coupling of approved municipal design.	each	44.00
17.67		Mirror (open edges)	Providing and fixing bevelled edge mirror of superior glass (of approved quality) complete with 6 mm thick hard board ground fixed to wooden cleats with C.P. brass screws and washers complete.	Sqm	2440.00
17.68		Mirror (Plastic Frame)	Providing and fixing mirror of superior glass (of approved quality) and of required shape and size with plastic moulded frame of approved make and shape with 6 mm thick hard board backing.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.68.1	450mm dia	Circular shape 450mm dia.	each	806.00
	17.68.2	453x357mm	Rectangular shape 453x357mm	each	642.00
	17.68.3	Oval 450x350mm	Oval shape 450x350mm (outer dimensions)	each	697.00
	17.68.4	1500x450 mm	Rectangular shape 1500x450 mm	each	1311.00
17.69		Glass shelf	Providing and fixing glass shelf with edges round of supported on anodised aluminium angle frame with C.P.brass brackets and guard rail completefixed with 40 mm long screws, rawl plugs etc., complete.	sqm	474.00
17.70		Toilet paper holder	Providing and fixing toilet paper holder:		
	17.70.1	C.P. brass	C.P. brass	each	340.00
	17.70.2	Vitreous china	Vitreous china	each	324.00
17.71		Waste coupling CP brass	Providing and fixing CP brass waste coupling of approved quality and make for sink and wash basin.		
	17.71.1	31mm dia	Waste coupling 31mm of 79mm length and 62mm breadth	each	330.00
	17.71.2	38 mm dia	Waste coupling 38mm of 83mm length and 77mm breadth	each	348.00
17.72		Bottle trap CP brass	Providing and fixing CP brass bottle trap of approved quality and make for sink and wash basin.	each	853.00
17.73		Towel rail CP brass	Providing and fixing CP brass towel rail of approved quality and make		
	17.73.1	600mm	600x20mm	each	952.00
	17.73.2	450mm	450x20mm	each	836.00
17.74		Towel ring CP brass	Providing and fixing CP brass towel ring of approved quality and make 200mm dia ring	each	679.00
17.75		Soap dish CP brass	Providing and fixing CP brass soap dish of approved quality and make	each	422.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
17.76		Urinal spreader CP brass	Providing and fixing CP brass Urinal spreader of approved quality and make	each	304.00
17.77		Soap container C.P. brass	Providing and fixing C.P. brass soap container 109 mm wide, 125 high and 112 mm distance from wall of standard shape with bracket of the same material with all fittings etc. of approved quality and conforming toIS: specification.	each	1175.00
17.78		Grating C.P. brass	Providing and fixing C.P. brass grating of approved quality and make conforming to IS: specification.		
	17.78.1	100 mm dia	100 mm dia.	each	74.00
17.79		Waste Coupling PTMT	Providing and fixing PTMT Waste Coupling for wash basin and sink, of approved quality and colour.		
	17.79.1	31mm dia	Waste coupling 31mm dia of 79mm length and 62mm breadth weighing not less than 45gms.	each	113.00
	17.79.2	38mm dia	Waste coupling 38mm dia of 83mm length and 77mm breadth, weighing not less than 60gms.	each	131.00
17.80		Bottle Trap PTMT	Providing and fixing PTMT Bottle Trap for Wash basin and sink.		
	17.80.1	31mm dia	Bottle trap 31mm single piece moulded with height of 270mm, effective length of tail pipe 260mmfrom the centre of the waste coupling 77mm breadth with 25mm minimum water seal, weighing not less than 260gms.	each	499.00
	17.80.2	38mm dia	Bottle trap 38mm single piece moulded with height of 270mm, effective length of tail pipe 260mmfrom the centre of the waste coupling 77mm breadth with 25mm minimum water seal, weighting not less than 263gms.	each	522.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
17.81		Liquid soap container PTMT	Providing and fixing PTMT liquid soap container 109mm wide, 125mm high and 112mm distance from wall of standard shape with bracket of the same materials with snap fittings of approved quality and colour. Weighing not less than 105 gms.	each	217.00
17.82		Towel ring PTMT	Providing and fixing PTMT towel ring trapezoidal shape 215mm long,200mm wide with a minimum distances of 37mm from wall face with concealed fittings arrangement of approved quality and colour. Weighing not less than 88 gms.	each	203.00
17.83		Towel rail PTMT	Providing and fixing PTMT towel rail complete with brackets fixed to wooden cleats with CP brass screws with concealed fitting arrangement of approved quality and colour.		
	17.83.1	450mm	450mm long towel rail with total length of 495mm, 78mm wide and effective height of 88mm, weighingnot less than 170gms.	each	412.00
	17.83.2	600mm	600mm long towel rail with total length of 645mm, width 78mm and effective height of 88mm, weighingnot less than 190gms.	each	459.00
17.84		Shelf PTMT	Providing and fixing PTMT shelf 440 mm long, 124 mm width and 36 mm height of approved quality and colour. Weighing not less than 300 gms.	each	511.00
17.85		Urinal spreader PTMT	Providing and fixing PTMT 15 mm Urinal spreader size 95x69x100 mm with 1/2" BSP thread and shapes. Weighing not less than 60 gms.	each	176.00
17.86		Urinal cock PTMT	Providing and fixing PTMT urinal cock of approved quality and colour. 15 mm nominal bore, 80mm long. 42 mm high and 30mm wide with BSP female threads weighing not less than 48 gms.	each	144.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
17.87		Soap Dish Holder PTMT	Providing and fixing PTMT soap Dish Holder having length of 138mm, breadth 102mm, height of 75mm with concealed fitting arrangements. Weighing not less than 106 gms.	each	119.00
17.88		Painting C.I. cistern with bitumastic	Painting C.I. cistern with bitumasticor any other anti-corrosive paint inside and white paint over a coat of zinc chromate yellow primer (of approved quality) on the outside surface of the cistern, flush pipe, other fittings, etc. complete for new work.	each	327.00
17.89		Re-painting C.I. cistern with bitumastic	Re-painting C.I. cistern with bitumastic or any other anti-corrosive paint inside and white paint on the outside surface of the cistern, flush pipe, other fittings, etc. complete including polishing of wooden seat and lid and cleaning of W.C. pan with acid wherever necessary.	each	235.00
17.90		Repainting C.I. cistern with synthetic enamel paint	Repainting C.I. cistern with synthetic enamel paint of approved colour brand and manufacture on the outside surface of cistern flush pipe, other fittings etc. complete.	each	98.00
17.91		Painting C.I. waste vent pipes and fittings with synthetic enamel paint	Painting sand cast iron/ centrifugally cast (spun) iron soil, waste vent pipes and fittings with two coats ofsynthetic enamel paint of any colour such as chocolate grey, or buff etc.over a coat of primer (of approved quality) for new work :		
	17.91.1	75 mm dia	75 mm diameter pipe	meter	22.00
	17.91.2	100 mm dia	100 mm diameter pipe	meter	29.00
	17.91.3	150mm dia	150 mm diameter pipe	meter	36.00
17.92		Re-painting C.I. waste, vent pipes and fittings	Repainting sand cast iron/ centrifugally cast iron (spun) iron, soil, waste, vent pipes and fittings with one coat of synthetic enamel paint of any colour such as chocolate, grey or buff etc :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.92.1	75mm dia	75 mm diameter pipe	meter	11.00
	17.92.2	100mm dia	100 mm diameter pipe	meter	14.00
	17.92.3	150mm dia	150 mm diameter pipe	meter	302.00
17.93		Repainting bath tub	Repainting bath tub of size 1700x730x430mm with enamel paint.	each	302.00
17.94		Ceramic glazed half round pipe drain	Providing and laying Ceramic glazed half round pipe drain (having thickness 9 to 10mm) of approved make in colours such as White, Ivory, Grey, or any other colour, laid on 20 mm thick Cement Mortar 1:4 (1 Cement : 4 sand) including pointing the joints with white cement andmatching pigment etc., complete.		
	17.94.1	100mm dia	100 mm dia half round pipe	meter	195.00
	17.94.2	150mm dia	150 mm dia half round pipe	meter	260.00
	17.94.3	200mm dia	200mm dia half round pipe	meter	378.00
17.95		Coupler for PVC pipes & fittings (40mm diameter)	Providing and fixing coupler for PVC pipes & fittings waste/rain water/drainage disposal of superior Indian make (as approved by Engineer incharge) i/c jointing with rubber rings, adhesive, lubricant,solvent cement PVC clips or burried underground i/c making necessary chases and holes in walls, floors or trenching complete with making good or i/c excavation and refilling as per site requirement with testing etc.complete. FOR WASH BASIN CONNECTION (40mm diameter)	each	25.00
17.96		Wash basin pipe (40 mm diameter)	Providing and fixing PVC pipes & fittings waste/rain water/drainage disposal of superior Indian make (as approved by Engineer incharge) i/c jointing with rubber rings, adhesive, lubricant,solvent cement PVC clips or burried underground i/c making necessary chases and holes in walls, floors or trenching complete with making good or i/c excavation and refilling as per site requirement with testing etc. complete. FOR WASH BASIN CONNECTION (40mm diameter)	meter	62.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
17.97		90 degree bend (40mm diameter)	Providing and fixing 90 degree bend PVC pipes & fittings waste/rain water/drainage disposal of superior Indian make (as approved by Engineer incharge) i/c jointing with rubber rings, adhesive, lubricant,solvent cement PVC clips or burried underground i/c making necessary chases and holes in walls, floors or trenching complete with making good or i/c excavation and refilling as per site requirement with testing etc.complete. FOR WASH BASIN CONNECTION (40mm diameter)	each	22.00
17.98		45 degree bend (40mm diameter)	Providing and fixing 45 degree bend PVC pipes & fittings waste/rain water/drainage disposal of superior Indian make (as approved by Engineer incharge) i/c jointing with rubber rings, adhesive, lubricant,solvent cement PVC clips or burried underground i/c making necessary chases and holes in walls, floors or trenching complete with making good or i/c excavation and refilling as per site requirement with testing etc.complete. FOR WASH BASIN CONNECTION (40mm diameter)	each	30.00
17.99		Incinerator for girls-toilet	Construction of manually operated fire based incinerator for girls-toileti/c (i) Fire and ash collection chamber 0.60x0.60 m, made of calcium silicate bricks, having 12mm dia steel bars grating @10 cm c/c, i/c MS mesh weldedon top including 300 mm x 450 mm MS openable door of half grill and half MS plate (3mm thick). (ii) 0.80 m X 0.80 m incinerator chamber made of calcium silicate bricks with R.C.C.top having opening of size 0.40m X 0.40 m, for stacking of sanitary napkins. (ii) 0.40 m X 0.40 m duct made of 1.6 mm thick MS sheet for dropping napkins from the toilet opening. (iv) 150mm dia Class 'C' heavy duty MS pipe conforming to IS3589/1239 for gas emission with		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			cowel, up to 2.10 meter height above terrace level. v) Opening in wall having MS flap (1.6mm thick) of size 0.25X 0.25 m, finished with granite. etc complete as per standard drawing no. 5313-B-NB4-9919-R1 dated 21.02.2017		
	17.99.1	G.F only	For ground floor buildings.	nos	24946.00
	17.99.2	G.F. & Ist floor	For Ground floor & I st floor buildings.	nos	32115.00
	17.99.3	Additional floor	Add extra for every additional floorabove Ist floor.	nos	8552.00
17.100		High-performance sound-insulating waste-water piping system	Providing & Fixing Complete item of high-performance sound-insulating waste-water piping system (low Noise Drainage System) pipe having following properties 1. Density of pipe material is 1.9 g/cm^3 2. Tensile strength of 20n/mm^2 3. 3 layer of polypropylene with modulus of elasticity 1500 n/mm^2 4. Single (1) layer of polypropylene with modulus of elasticity 3800 n/mm^2, 5. Pipe handle the waste liquid with ph value 2 to 12. 6. Hot water resistant - short term 95°c / 203°f and long term 90°c / 194°f. Co-efficient of linear expansion .09 mm/mk.		
	17.100.1.	Pipe	Pipe		
	17.100.1.1	160mm	160mm	meter	1270.00
	17.100.1.2	110mm	110mm	meter	802.00
	17.100.1.3	75 mm	75 mm	meter	429.00
	17.100.1.4	40 mm	40 mm	meter	275.00
	17.100.2	Saddle With Rubber	Heavy Duty Saddle With Rubber		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.100.2.1	160mm	160mm	each	138.00
	17.100.2.2	110mm	110mm	each	87.00
	17.100.2.3	75mm	75mm	each	65.00
	17.100.2.4	40mm	40mm	each	42.00
	17.100.3	Bend Without door	Bend without door		
	17.100.3.1	160mm	160mm	each	730.00
	17.100.3.2	110mm	110mm	each	440.00
	17.100.3.3	75mm	75mm	each	320.00
	17.100.3.4	40mm	40mm	each	181.00
	17.100.4	Bend with door	Bend with door		
	17.100.4.1	160mm	160 mm	each	1369.00
	17.100.4.2	110mm	110mm	each	740.00
	17.100.4.3	75 mm	75 mm	each	480.00
	17.100.4.4	40 mm	40 mm	each	240.00
	17.100.5	Tee with door	Tee with door		
	17.100.5.1	160mm	160 mm	each	1380.00
	17.100.5.2	110mm	110mm	each	846.00
	17.100.5.3	75 mm	75 mm	each	510.00
	17.100.5.4	40 mm	40 mm	each	265.00
	17.100.6	Tee without door	Tee without door		
	17.100.6.1	160mm	160 mm	each	1220.00
	17.100.6.2	110mm	110 mm	each	630.00
	17.100.6.3	75 mm	75 mm	each	480.00
	17.100.6.4	40 mm	40 mm	each	215.00
	17.100.7	Y Tee without door	Y Tee without door		
	17.100.7.1	160mm	160 mm	each	1370.00
	17.100.7.2	110mm	110mm	each	672.00
	17.100.7.3	75 mm	75 mm	each	525.00
	17.100.7.4	40 mm	40 mm	each	275.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	17.100.8	Double Y Tee without door	Double Y Tee without door		
	17.100.8.1	160mm	160 mm	each	1385.00
	17.100.8.2	110mm	110 mm	each	637.00
.	17.100..3	75 mm	75 mm	each	545.00
	17.100.8.4	40 mm	40 mm	each	290.00
	17.100.9	Coupler			
	17.100.9.1	160mm	160 mm	each	570.00
	17.100.9.2	110mm	110 mm	each	380.00
	17.100.9.3	75 mm	75 mm	each	247.00
	17.100.9.4	40 mm	40 mm	each	190.00

CHAPTER – 18.0

WATER SUPPLY

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	2	3
1	IS 554	Pipe threads where pressure tight joints are required on the threads-Dimensions, tolerances and designation.
2	IS 778	Specification for copper alloy gate, and check valves for water works purposes
3	IS 779	Water meters (domestic type) -Specification
4	IS 780	Specification for sluice valves for water works purposes (50 to 300 mm)
5	IS 781	Specification for cast copper alloy screw down bib taps and stop valves for water services
6	IS 782	Specification for caulking lead
7	IS 909	Underground fire hydrant, sluice valve type-Specification
8	IS 1239 (Part-1)	Steel tubes tubular and other wrought steel fittings, Part 1- Steel tubes-Specification
9	IS 1239 (Part-2)	Specification for mild steel tubes tubular and other wrought steel fittings, Part 2-Mild street tubular and other wrought steel pipe fittings
10	IS 1536	Centrifugally cast (spun) iron pressure pipes for water gas and sewage- Specification
11	IS 1537	Specification for vertically cast iron pressure pipes for water, gas and sewage
12	IS 1538	Cast iron fittings for pressure pipes for water, gas and sewage - Specification
13	IS 1703	Water fittings - copper alloy float valves (horizontal plunger type) Specification
14	IS 2692	Ferrules for water services- Specification
15	IS 3950	Specification for surface boxes for sluice valves
16	IS 4736	Specification for Hot-dip Zinc Coatings on mild steel tubes
17	IS 5312 (Part-1)	Swing type reflex (non return) valves for water works purposes. Part 1-Single door pattern
18	IS 5312 (Part-2)	Swing type reflex (non return) valves for water works purposes. Part 2- Multi door pattern
19	IS 5382	Rubber sealing rings for gas mains, water mains and sewers
20	IS 9762	Specification for polyethylene floats (spherical) for float valves
21	IS 9763	Plastic Bib taps and stop valves (rising spindle) for cold water services-specifications
22	IS 15450	PE-AL-PE Pipes for hot and cold water supplies-Specifications
23	IS 15778	Chlorinated Polyvinyl Chloride (CPVC) pipes for potable hot and cold water distribution supplies-specifications.
24	IS 15801	Polypropylene- Random Copolymer Pipes for hot and cold water supplies-Specifications

PREAMBLE
WATER SUPPLY
(CHAPTER: 18.0)

18.1 GENERAL REQUIREMENTS

- Any damage caused to the building, or to electric, sanitary water supply or other installations etc. therein either due to negligence on the part of the contractor, or due to actual requirements of the work, shall be made good and the building or the installations shall be restored to its original condition by the contractor. Nothing extra shall be paid for it, except where otherwise specified.

- All water supply installation work shall be carried out through licensed plumbers.

- It is most important to ensure that wholesome water supply provided for drinking and culinary purposes, is in no way liable to contamination from any less satisfactory water. There shall, therefore, be no cross connection whatsoever between a pipe or fitting for conveying or containing wholesome water and a pipe or fitting for conveying or containing impure water or water liable to contamination or of uncertain quality of water which has been used for any purpose. The provision of reflux or non-return valves or closed and sealed valves shall not be construed a permissible substitute for complete absence of cross- connection.

- Where a supply of wholesome water is required as an alternative or standby to supply of less satisfactory water or is required to be mixed with the latter, it shall be delivered only into a cistern, and by a pipe or fitting discharging into the air gap at a height above the top edge of the cistern equal to twice its nominal bore, and in no case less than 15 cm.

- No piping shall be laid or fixed so as to pass into, through or adjoining any sewer, scour outlet or drain or any manhole connected therewith nor through any ash pit or manure-pit orany material of such nature that can cause undue deterioration of the pipe.
- Where the laying of any pipe through fouled soil or previous material is unavoidable, the piping shall be properly protected from contact with such soil or material by being carried through an exterior cast iron tube or by some other suitable means. Any piping or fitting laid or fixed which does not comply with the above requirements, shall be removed and re- laid in conformity with the above requirements.

- The design of the pipe work shall be such that there is no possibility of backflow towards the source of supply from any cistern or appliance whether by siphonage or otherwise, and reflux or nonreturn valves shall not be relied upon to prevent such back flow.
- All pipe work shall be so designed, laid or fixed, and maintained so that it remains completely watertight, thereby avoiding wastage of water, damage to property and the risk of contamination of the water conveyed.
- In designing and planning the layout of the pipe work, due attention shall be given to the maximum rate of discharge, required economy in labour and materials, protection against damage and corrosion, protection from frost, if required, and to avoidance of airlocks, noise transmission and unsightly arrangement.
- To reduce frictional losses, piping shall be as smooth as possible inside. Methods of jointing shall be such as to avoid internal roughness and projection at the joints, whether of the jointing materials or otherwise.

- Change in diameter and in direction shall preferably be gradual rather than abrupt to avoid undue loss of head. No bend or curve in piping shall be made so as to materially reduce or alter the cross-section.
- Underground piping shall be laid at such a depth that it is unlikely to be damaged by frost or traffic loads and vibrations. It shall not be laid in ground liable to subsidence, but where such ground cannot be avoided; special precautions shall be taken to avoid damage to the piping. Where piping has to be laid across recently disturbed ground, the ground shall be thoroughly consolidated so as to provide a continuous and even support.
- Where the service pipe is of diameter less than 50 mm the stop valves shall be of the screwdown type and shall have loose washer plates to act as non-return valves. Other stop valves in the service line may be of the gate type.
- In flats and tenements supplied by a common service pipe a stop valve shall be fixed to control the each branch separately. In large buildings a sufficient number of stop valves shall be fixed on branch pipes, and to control groups of ball valves and draw off taps, so as to minimize interruption of the supply during repairs, all such stop valves shall be fixed in accessible positions and properly protected from being tampered with, they may be of the gate type to minimize loss of head by friction.
- Water for drinking or for culinary purposes as far as possible shall be on branch pipes connected directly to the service pipe.
- Pumps shall not be allowed on the service pipe as they cause a drop of pressure on the suction side thereby affecting the supply to the adjoining properties. In cases where pumping is required, a properly protected storage tank of adequate capacity shall be provided to feed the pump.
- Service pipes shall be so designed and constructed as to avoid air-locks, so that all piping and fittings above ground can be completely emptied of water to facilitate repairs. There shall be draining taps or draw-off taps (not underground) at the lowest points, from which the piping shall rise continuously to draw-off taps, ball valves, cisterns, or vents (where provided at the high points).
- Service pipes shall be designed so as to reduce the production and transmission of noise as much as possible. Appliances which create noise shall be installed as far distant as possible from the living rooms of the house. High velocity of water in piping and fittings shall be avoided.Piping shall be confined, as far as possible, to rooms where appliances are fixed, it shall have easy bends, and where quietness is particularly desired, holder bats or clamps shall be insulated from the piping by suitable pads.
- The rising pipe to the storage cistern, if any, or to any feed cistern shall be taken as directly as possible to the cistern and shall be fixed away from windows or ventilators.
- All pipe work shall be planned so that the piping is accessible for inspection, replacement nd repair. To avoid its being unsightly, it is usually possible to arrange it in or adjacent to cupboards, recesses, etc. provided there is sufficient space to work on the piping with the usual tools. Piping shall not be buried in walls or solid floors. Where unavoidable, piping may be buried for short distances provided that adequate protection is given against damageand that no joints are buried. If piping is laid in ducts or chases, these shall be roomy enough to facilitate repairs and shall be so constructed as to prevent the entry of vermin. To facilitate removal of pipe casing, floor boards covering piping shall be fixed with screws or bolts.

- When it is necessary for a pipe to pass through a wall or floor, a sleeve shall be fixed therein for insertion of the pipe and to allow freedom for expansion, contraction and other movement. Piping laid in wood floors shall, where possible, be parallel with the joists.

- Where storage tanks are provided to meet overall requirements of water connection of service pipe with any distributing pipe shall not be permitted except one direct connection for culinary or drinking requirements.

- No service pipe shall be connected to any water closet or urinal. All such supplies shall be from flushing cisterns which shall have supply from storage tank.

- No service or supply pipe shall be connected directly to any hot-water system or to any apparatus used for heating other than through a feed cistern thereof.

18.2 Pipes and Specials

(i) Pipes and specials may be of any of the following types as specified:
(ii) Galvanised steel – IS 1239 and IS 4736
(iii) PE-AL-PE Pipes – IS 15450
(iv) PP-R Pipes – IS 15801
(v) CPVC pipes – IS 15778

In choosing the material for piping and fittings, account shall be taken of the character of the water to be conveyed through it, the nature of the ground in which the pipes are to belaid and the relative economics. The lengths shall be measured in running metre correct to a cm for the finished work, which shall include pipe and fittings such as bends, tees, elbows reducers, crosses, plugs, sockets, nipples and nuts, but exclude brass or gun metal taps (cocks), valves, unions, lead connection pipes and shower rose.Unions shall be paid for separately in external work as well as in internal work. For pipes 15 mm diameter, the holes in the walls and floors shall be made by drilling with chisel or jumper and not by dismantling the brick work or concrete. However, for bigger dimension pipes the holes shallbe carefully made of the smallest size as directed by the Engineer-in-Charge. After fixingthe pipes the holes shall be made good with cement mortar 1:3 (1 cement: 3 sand) and properly finished to match the adjacent surface.

18.3 Concealed Piping:

Pipes can be concealed in chases. The pipes and fitting are to be pressure tested prior to concealing the chases. To maintain alignment of CP fittings while joining, all alignment of fittings and pipe shall be done correctly.

18.4 Do Not Use Nails for Holding of Pipes in the Chases.

External Installations: For pipes fixed in the shafts, ducts etc. there should be sufficient space to work on the pipes. Pipes sleeves shall be fixed at a place the pipe is passing through a wall or floor so as to allow freedom for expansion and contraction. Clamping of the pipe is done to support it while allowing the freedom for movement.

18.5 Rate

External Work: The rate shall include the cost of labour and materials involved in all the operations described above. The rate shall not include excavation in trenches, painting of pipes and sand filling all around the pipe unless otherwise specified for G.I pipe line however for other categories of pipes rates are inclusive of excavation in trenches and backfilling of same.

18.6 Internal Work

The rate shall also include the cost of labour and material involved in all the operations described above. The rate shall include the cost of cutting holes in walls and floors and making good the same. This shall not however, include concealed pipe work in which case cutting of chase and making good shall be paid separately. It shall not include painting of pipes and providing sleeves, unless specified otherwise. It will also not include union which shall be paid for separately. However for other categories of pipes rates are inclusive of cost for cutting of chase and making good the same...

CUTTING CHASES IN MASONRY WALLS

18.7 Making Chases

Chases are made in the walls for housing G.I. Pipes etc.

18.8 Cuttingof Chases in One Brick Thick and Above Load Bearing Walls:

(i) As far as possible services should be planned with the help of vertical chases. Horizontal chases should be avoided.

(i) The depths of vertical chases and horizontal chases shall not exceed one third and one sixth of the thickness of the masonry respectively.

(ii) When narrow stretches of masonry (or short lengths of walls) such as between doors and windows, cannot be avoided, they should not be pierced with openings for soil pipes or waste pipes or timber joints, etc. Where there is a possibility of load concentration, such narrow lengths of walls shall be checked for stresses and high strength bricks mortar or concrete walls provided, if required.

(iii) Horizontal chases when unavoidable should be located in the upper or lower one third of height of storey and not more than three chases should be permitted in any stretch of a wall. No continuous horizontal chase shall exceed one metre in length. Where unavoidable, stresses in the affected area should be checked and kept within the permissible limits.

(iv) Vertical chases should not be closer than 2 m in any stretch of a wall. These shall be kept away from bearings of beams and lintels. If unavoidable, stresses in the affected area should be checked and kept within permissible limits.

(v) Masonry directly above a recess, if under than 30 cm (Horizontal dimension) should be supported on lintel. Holes in masonry may be provided up to 30 cm width x 30 cm height without any lintel. In the case of circular holes in masonry, no lintel should be provided up to 40 cm in diameter.

18.9 Cutting of chases in half brick load bearing walls

No chase shall be permitted in a half brick load bearing wall and as such no recessed conduits and concealed pipes shall be provided in half brick thick load bearing walls.

18.10 Cutting of chases in half brick non-loading bearing walls

In case of non load bearing half brick walls services should be planned with the help of vertical chases. Horizontal chases should be provided only when unavoidable.

18.11 Cutting of chases in stone masonry walls

The provision (i) to (v) described for chasing in brick work are equally applicable to stone masonry walls also.

Note:

(i) No inclined chase shall be permitted in brick masonry or stone masonry walls. In case inclined chases are unavoidable these shall be cut with written approval of the Engineer-in-Charge, and shall be repaired properly to his satisfaction. However, in half brick masonry wall, no inclined chase will be permitted.

(ii) Chases shall be made by chiseling out the masonry to proper line and depth. Any damage to the adjoining portion or to any other item shall be made good, as decided by the Engineer-in-Charge, for which no extra payment shall be made. All dismantled material shall be removed from site.

18.12 Filling Chases

After G.I. Pipes etc. are fixed in chases, the chases shall be filled with cement concrete 1:3:6 (1 cement: 3 sand: 6 graded stone aggregate 20 mm nominal size) or cement mortar 1:4 (1 cement: 4 sand) as may be specified or otherwise directed by the Engineer-in-Charge and made flush with the masonry surface. The concrete surface shall be roughened with wire brushes to provide a key for plastering.

18.13 Testing

All water supply systems shall be tested to hydrostatic pressure test. The pressure tests are similar to the test pressure used for other plastic/metal pipes. System may be tested in sections and such section shall be entirely checked on completion of connection to the overhead tank or pumping system or mains.

TABLE 18.1
For Water Storage Tank

Sl. No.	Capacity litres	Minimum Wall Thickness mm	Minimum Weight of Empty Tank kg
1	**2**	**3**	**4**
1.	200	4.4	7.8
2.	300	4.4	9.0
3.	400	5.5	15.0
4.	500	6.0	18.0
5.	700	6.6	23.5
6.	1000	7.0	33.0
7.	1250	7.0	40.0
8.	1500	7.0	47.0
9.	1700	7.0	54.0
10.	2000	8.2	64.0
11.	2500	8.2	81.0
12.	3000	8.8	96.0
13.	4000	10.4	138.0
14.	5000	10.7	191.0
15.	6000	10.7	209.0
16.	7500	10.7	250.0
17.	10000	11.5	363.0
18.	15000	11.5	550.0
19.	20000	13.2	814.0

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.1		Polyethelene Aluminium-Polyethelene Composite pressure pipes (PE-AL-PE) internal work	Providing and fixing Polyethelene-Aluminium-Polyethelene (PE-AL-PE) Composite Pressure Pipes conforming to IS - 15450 U.V. stabilised with carbon black having thermal stability for hot and cold water supply, capable to withstand temperature up to 80° C including all special fittings of composite material (engineering plastic blend and brass inserts wherever required) e.g. elbows, tees, reducers, couplers and connectors etc. with clamps at 1.00 metre spacing. This includes testing of joints complete as per direction of the Engineer-in-Charge.		
18.1.1		Exposed on wall	Internal work - Exposed on wall		
	18.1.1.1	16 mm	1216 (16 mm OD) pipe (minimum wall thickness 1.75mm)	metre	193.00
	18.1.1.2	20 mm	1620 (20 mm OD) pipe (minimum wall thickness 2mm)	metre	242.00
	18.1.1.3	25 mm	2025 (25 mm OD) pipe (minimum wall thickness 2.45mm)	metre	307.00
	18.1.1.4	32 mm	2532 (32 mm OD) pipe (minimum wall thickness 2.8mm)	metre	398.00
	18.1.1.5	40 mm	3240 (40 mm OD) pipe (minimum wall thickness 3.4mm)	metre	618.00
	18.1.1.6	50 mm	4050 (50 mm OD) pipe (minimum wall thickness 4mm)	metre	679.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.1.2		Concealed	Concealed work including cutting chases and making good the wall etc.		
	18.1.2.1	16 mm	1216 (16 mm OD) pipe (minimum wall thickness 1.75mm)	metre	299.00
	18.1.2.2	20 mm	1620 (20 mm OD) pipe (minimum wall thickness 2mm)	metre	360.00
	18.1.2.3	25 mm	2025 (25 mm OD) pipe (minimum wall thickness 2.45mm)	metre	441.00
	18.1.2.4	32 mm	2532 (32 mm OD) pipe (minimum wall thickness 2.8mm)	metre	553.00
18.2		PolyetheleneAluminium-PolyetheleneComposite Pressure Pipes External work	Providing and fixing Polyethelene-Aluminium-Polyethelene (PE-AL-PE) Composite Pressure Pipes conforming to IS - 15450 U.V. stabilized with carbon black having thermal stability for hot and cold water supply, capable to withstand temperature up to 80° C including all special fittings of composite material (engineering plastic blend and brass inserts wherever required) e.g. elbows ,tees ,reducers, couplers and connectors etc. with trenching, refilling and testing of joints complete as per direction of the engineer in charge. External work		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.2.1	16 mm	1216 (16 mm OD) pipe (minimum wall thickness 1.75mm)	metre	190.00
	18.2.2	20 mm	1620 (20 mm OD) pipe (minimum wall thickness 2mm)	metre	235.00
	18.2.3	25 mm	2025 (25 mm OD) pipe (minimum wall thickness 2.45mm)	metre	295.00
	18.2.4	32 mm	2532 (32 mm OD) pipe (minimum wall thickness 2.8mm)	metre	378.00
	18.2.5	40 mm	3240 (40 mm OD) pipe (minimum wall thickness 3.4mm)	metre	588.00
	18.2.6	50 mm	4050 (50 mm OD) pipe (minimum wall thickness 4mm)	metre	649.00
18.3		3 layer PP-R pipes internal work	Providing and fixing 3 layer PP-R (Poly propylene Random copolymer) pipes SDR 7.4 U V stabilized and anti - microbial fusion welded, having thermal stability for hot and cold water supply including allPP - R plain and brass threaded polypropylene random fittings i/c fixing the pipe with clamps at 1.00 m spacing. This includes testing of joints complete as per direction of Engineer in Charge. Internal work - Exposed on wall		
	18.3.1	16 mm	PN - 16 Pipe, 16 mm OD	metre	102.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.3.2	20 mm	PN - 16 Pipe, 20 mm OD	metre	137.00
	18.3.3	25 mm	PN - 16 Pipe, 25 mm OD	metre	186.00
	18.3.4	32 mm	PN - 16 Pipe, 32 mm OD	metre	285.00
	18.3.5	40 mm	PN - 16 Pipe, 40 mm OD	metre	437.00
	18.3.6	50 mm	PN - 16 Pipe, 50 mm OD	metre	596.00
18.4		3 layer PP-R pipes concealed work	Providing and fixing 3 layer PP-R (Poly propylene Random copolymer) pipes SDR 7.4 UV stabilized and anti - microbial fusion welded, having thermal stability for hot and cold water supply including allPP - R plain and brass threaded polypropylene random fittings i/c fixing the pipe with clamps at 1.00 m spacing. This includes the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer-in- Charge. Concealed work including cutting chases and making good the walls etc.		
	18.4.1	16 mm	PN - 16 Pipe, 16 mm OD	metre	177.00
	18.4.2	20 mm	PN - 16 Pipe, 20 mm OD	metre	217.00
	18.4.3	25 mm	PN - 16 Pipe, 25 mm OD	metre	278.00
	18.4.4	32 mm	PN - 16 Pipe, 32 mm OD	metre	400.00
18.5		3 layer PP-R pipes external work	Providing and fixing 3 layer PP-R (Poly propylene Random copolymer) pipes U V stabilized and anti - microbial fusion welded, having thermal stability for hot and cold water supply		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			including all PP - R plain and brass threaded polypropylene random fittings i/c trenching, refilling and testing of joints complete as per direction of Engineer-in-Charge. External work		
	18.5.1	16 mm	PN - 16 Pipe, 16 mm OD (SDR 7.4)	metre	99.00
	18.5.2	20 mm	PN - 16 Pipe, 20 mm OD (SDR 7.4)	metre	129.00
	18.5.3	25 mm	PN - 16 Pipe, 25 mm OD (SDR 7.4)	metre	178.00
	18.5.4	32 mm	PN - 16 Pipe, 32 mm OD (SDR 7.4)	metre	268.00
	18.5.5	40 mm	PN - 16 Pipe, 40 mm OD (SDR 7.4)	metre	407.00
	18.5.6	50 mm	PN - 16 Pipe, 50 mm OD (SDR 7.4)	metre	566.00
	18.5.7	63mm	PN - 16 Pipe, 63mm OD (SDR 7.4)	metre	863.00
	18.5.8	75 mm	PN - 16 Pipe, 75 mm OD (SDR 7.4)	metre	1225.00
	18.5.9	90 mm	PN - 16 Pipe, 90 mm OD (SDR 7.4)	metre	1960.00
	18.5.10	110 mm	PN - 10 Pipe, 110 mm OD (SDR - 11)	metre	2066.00
	18.5.11	160 mm	PN - 10 Pipe, 160 mm OD (SDR - 11)	metre	4264.00
18.6		CPVC pipes internal work	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot and cold water supply including all CPVC plain and brass threaded fittings		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes and fittings with one step CPVC solvent cement and testing of jointscomplete as per direction of Engineer-in-Charge.		
	18.6.1	Exposed on wall	Internal work - Exposed on wall		
	18.6.1.1	15 mm	15 mm nominal outer dia .Pipes.	metre	129.00
	18.6.1.2	20 mm	20 mm nominal outer dia .Pipes.	metre	153.00
	18.6.1.3	25 mm	25 mm nominal outer dia .Pipes.	metre	177.00
	18.6.1.4	32 mm	32 mm nominal outer dia .Pipes.	metre	254.00
	18.6.1.5	40 mm	40 mm nominal outer dia .Pipes.	metre	354.00
	18.6.1.6	50 mm	50 mm nominal outer dia .Pipes.	metre	528.00
	18.6.2	Concealed	Concealed work including cutting chases and making good the walls etc.		
	18.6.2.1	15 mm	15 mm nominal outer dia .Pipes.	metre	207.00
	18.6.2.2	20 mm	20 mm nominal outer dia .Pipes.	metre	234.00
	18.6.2.3	25 mm	25 mm nominal outer dia .Pipes.	metre	288.00
	18.6.2.4	32 mm	32 mm nominal outer dia .Pipes.	metre	360.00
18.7		CPVC pipes forexternal work	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes,		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			having thermal stability for hot and cold water supply including all CPVC plainand brass threaded fittings. This included jointing of pipes and fittings with onestep CPVC solvent cement, trenching, refilling andtesting of joints complete as per direction of Engineer-in- Charge. **External Work**		
	18.7.1	15 mm	15 mm nominal outer dia .Pipes.	metre	122.00
	18.7.2	20 mm	20 mm nominal outer dia .Pipes.	metre	141.00
	18.7.3	25 mm	25 mm nominal outer dia .Pipes.	metre	185.00
	18.7.4	32 mm	32 mm nominal outer dia .Pipes.	metre	238.00
	18.7.5	40 mm	40 mm nominal outer dia .Pipes.	metre	324.00
	18.7.6	50 mm	50 mm nominal outer dia .Pipes.	metre	498.00
	18.7.7	62.50 mm	62.50 mm nominal inner dia Pipes.	metre	1429.00
	18.7.8	75 mm	75 mm nominal inner dia .Pipes.	metre	2018.00
	18.7.9	100mm	100mm nominal inner dia .Pipes.	metre	2791.00
	18.7.10	150 mm	150 mm nominal inner dia .Pipes.	metre	3358.00
18.8		G.I. pipes	Providing and fixing G.I. pipes complete with G.I. fittings and clamps, including cutting and making good the walls etc.		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.8.1	Internal Exposed on wall	Internal work - Exposed on wall		
	18.8.1.1	15 mm	15 mm dia. nominal bore	metre	211.00
	18.8.1.2	20 mm	20 mm dia. nominal bore	metre	255.00
	18.8.1.3	25 mm	25 mm dia. nominal bore	metre	300.00
	18.8.1.4	32 mm	32 mm dia. nominal bore	metre	361.00
	18.8.1.5	40 mm	40 mm dia. nominal bore	metre	479.00
	18.8.1.6	50 mm	50 mm dia. nominal bore	metre	596.00
	18.8.2	Concealed	Concealed pipe including painting with anti corrosivebitumastic paint, cutting chases and making good the wall.		
	18.8.2.1	15 mm	15 mm dia nominal bore	metre	269.00
	18.8.2.2	20 mm	20 mm dia nominal bore	metre	309.00
	18.8.3	External	External work : including trenching and refilling etc.		
	18.8.3.1	15 mm	15 mm dia. nominal bore	metre	188.00
	18.8.3.2	20 mm	20 mm dia. nominal bore	metre	223.00
	18.8.3.3	25 mm	25 mm dia. nominal bore	metre	263.00
	18.8.3.4	32 mm	32 mm dia. nominal bore	metre	310.00
	18.8.3.5	40 mm	40 mm dia. nominal bore	metre	402.00
	18.8.3.6	50 mm	50 mm dia. nominal bore	metre	492.00
	18.8.3.7	65 mm	65 mm dia. nominal bore	metre	533.00
	18.8.3.8	80 mm	80 mm dia. nominal bore	metre	622.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.9		S.S. Pipe Internal	Providing and fixing Stainless Steel pipe and fitting of grade AISI 304 as per JIS standard 3448 complete with press type fitting (fitting shall be paid for separately) i/c fixing of the pipe with clamps at 1.00 m spacing including cutting and making good the walls including testing of joints complete as per direction of Engineer -in-charge. (The pipe length inserted in the fitting shall not be measured for payment)		
	18.9.1	Exposed	Internal work - Exposed on wall		
	18.9.1.1	15.88 mmdia	15.88 mm outer diapipe	metre	243.00
	18.9. 1.2	22.22 mm	22.22 mm outer diapipe	metre	375.00
	18.9.1.3	28.58 mm	28.58 mm outer diapipe	metre	467.00
	18.9.1.4	34.00 mm	34.00 mm outer dia Pipe	metre	636.00
	18.9.1.5	42.70 mm	42.70 mm outer dia Pipe	metre	774.00
	18.9.1.6	48.60 mm	48.60 mm outer dia Pipe	metre	896.00
	18.9.2	Concealed	Internal work - Concealed Pipe		
	18.9.2.1	15.88 mm	15.88 mm outer dia .Pipes.	metre	297.00
	18.9.2.2	22.22 mm	22.22 mm Outer diaPipes	metre	428.00
18.10		Stainless Steel Fitting of press fit type	Providing and fixing required Stainless Steel Fitting of press fit design of grade AISI 304 conforming to JWWA G116 standard with V-profile or M-profile and with O-ring sealing gasket of EPDM material of required diaas per dirction of Engineer-in-Charge.		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.1	Coupling/Socket	Coupling/Socket		
	18.10.1.1	15.88 mm	For 15.88 mm outer diaPipe	each	66.00
	18.10.1.2	22.22 mm	For 22.22 mm outer dia Pipe	each	102.00
	18.10.1.3	28.58 mm	For 28.58 mm outer dia Pipe	each	125.00
	18.10.1.4	34.00 mm	For 34.00 mm outer dia Pipe	each	268.00
	18.10.1.5	42.70 mm	For 42.70 mm outer dia Pipe	each	383.00
	18.10.1.6	48.60 mm	For 48.60 mm outer dia Pipe	each	518.00
	18.10.2	Reducer	Reducer		
	18.10.2.1	22.22 mm	For 22.22 mm x 15.88 mm outer dia pipe	each	120.00
	18.10.2.2	28.58 mm	For 28.58 mm x 15.88 mm outer dia pipe	each	164.00
	18.10.2.3	28.58 mm	For 28.58 mm x 22.22 mm outer dia pipe	each	168.00
	18.10.2.4	34.00 mm	For 34.00 mm x 15.88 mm outer dia pipe	each	272.00
	18.10.2.5	34.00 mm	For 34.00 mm x 22.22 mm outer dia pipe	each	275.00
	18.10.2.6	34.00 mm	For 34.00 mm x 28.58 mm outer dia pipe	each	275.00
	18.10.2.7	42.70 mm	For 42.70 mm x 15.88 mm outer dia pipe	each	417.00
	18.10.2.8	42.70 mm	For 42.70 mm x 22.22 mm outer dia pipe	each	420.00
	18.10.2.9	42.70 mm	For 42.70 mm x 28.58 mm outer dia pipe	each	420.00
	18.10.2.10	42.70 mm	For 42.70 mm x 34.00 mm outer dia pipe	each	450.00
	18.10.2.11	48.60 mm	For 48.60 mm x 15.88 mm outer dia pipe	each	544.00
	18.10.2.12	48.60 mm	For 48.60 mm x 22.22 mm outer dia pipe	each	546.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.2.13	48.60 mm	For 48.60 mm x 28.58 mm outer dia pipe	each	551.00
	18.10.2.14	48.60 mm	For 48.60 mm x 34.00 mm outer dia pipe	each	556.00
	18.10.2.15	48.60 mm	For 48.60 mm x 42.70 mm outer dia pipe	each	606.00
	18.10.3	Elbow 90°	Elbow 90°		
	18.10.3.1	15.88 mm	For 15.88 mm outer dia pipe	each	100.00
	18.10.3.2	22.22 mm	For 22.22 mm outer dia Pipe	each	178.00
	18.10.3.3	28.58 mm	For 28.58 mm outer dia Pipe	each	245.00
	18.10.3.4	34.00 mm	For 34.00 mm outer dia Pipe	each	498.00
	18.10.3.5	42.70 mm	For 42.70 mm outer dia Pipe	each	763.00
	18.10.3.6	48.60 mm	For 48.60 mm outer dia Pipe	each	938.00
	18.10.4	Reducing Elbow 90°	Reducing Elbow 90°		
	18.10.4.1	22.22mm	For22.22 mm x 15.88 mm outer dia pipe	each	172.00
	18.10.4.2	28.58 mm 15.88 mm	For 28.58 mm x 15.88 mm outer dia pipe	each	251.00
	18.10.4.3	28.58 mmx 22.22mm	For 28.58 mm x 22.22 mm outer dia pipe	each	257.00
	18.10.4.4	34.00 mm	For 34.00 mm x 22.22 mm outer dia pipe	each	374.00
	18.10.4.5	34.00 mmx 28.58mm	For 34.00 mm x 28.58 mm outer dia pipe	each	460.00
	18.10.4.6	42.70 mmx 34.00mm	For 42.70 mm x 34.00 mm outer dia pipe	each	811.00
	18.10.5	Equal Tee	Equal Tee		
	18.10.5.1	15.88 mm	For 15.88 mm outer dia pipe	each	185.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.5.2	22.22 mm	For 22.22 mm outer dia Pipe	each	268.00
	18.10.5.3	28.58 mm	For 28.58 mm outer dia Pipe	each	344.00
	18.10.5.4	34.00 mm	For 34.00 mm outer dia Pipe	each	717.00
	18.10.5.5	42.70 mm	For 42.70 mm outer dia Pipe	each	1004.00
	18.10.5.6	48.60 mm	For 48.60 mm outer dia Pipe	each	1194.00
	18.10.6	Reducing Tee	Reducing Tee		
	18.10.6.1	22.22 mm x 15.88 mm	For 22.22 mm x 15.88 mm outer dia pipe	each	261.00
	18.10.6.2	28.58 mm x 15.88 mm	For 28.58 mm x 15.88 mm outer dia pipe	each	325.00
	18.10.6.3	28.58 mm x 22.22mm	For 28.58 mm x 22.22 mm outer dia pipe	each	338.00
	18.10.6.4	34.00 mm x 15.88 mm	For 34.00 mm x 15.88 mm outer dia pipe	each	637.00
	18.10.6.5	34.00 mm x 22.22mm	For 34.00 mm x 22.22 mm outer dia pipe	each	648.00
	18.10.6.6	34.00 mm x 28.58mm	For 34.00 mm x 28.58 mm outer dia pipe	each	655.00
	18.10.6.7	42.70 mm x 15.88mm	For 42.70 mm x 15.88 mm outer dia pipe	each	951.00
	18.10.6.8	42.70 mm x 22.22mm	For 42.70 mm x 22.22 mm outer dia pipe	each	954.00
	18.10.6.9	42.70 mm x 28.58mm	For 42.70 mm x 28.58 mm outer dia pipe	each	955.00
	18.10.6.10	42.70 mm x 34.00mm	For 42.70 mm x 34.00 mm outer dia pipe	each	964.00
	18.10.6.11	48.60 mm x 15.88mm	For 48.60 mm x 15.88 mm outer dia pipe	each	1038.00
	18.10.6.12	48.60 mm x 22.22mm	For 48.60 mm x 22.22 mm outer dia pipe	each	1060.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.6.13	48.60 mm x 28.58mm	For 48.60 mm x 28.58 mm outer dia pipe	each	1069.00
	18.10.6.14	48.60 mm x 34.00mm	For 48.60 mm x 34.00 mm outer dia pipe	each	1113.00
	18.10.6.15	48.60 mm x 42.70mm	For 48.60 mm x 42.70 mm outer dia pipe	each	1153.00
	18.10.7	Male Thread Tee	Male Thread Tee		
	18.10.7.1	15.88 mm x 15 mm	For 15.88 mm outer dia x 15 mm nominal dia threaded	each	332.00
	18.10.7.2	22.22 mm x 15 mm	For 22.22 mm outer dia x 15 mm nominal dia threaded	each	400.00
	18.10.7.3	22.22 mm x 20 mm	For 22.22 mm outer dia x 20 mm nominal dia threaded	each	427.00
	18.10.7.4	28.58 mm x 15 mm	For 28.58 mm outer dia x 15 mm nominal dia threaded	each	481.00
	18.10.7.5	28.58 mm x 20 mm	For 28.58 mm outer dia x 20 mm nominal dia threaded	each	502.00
	18.10.7.6	28.58 mm x 25 mm	For 28.58 mm outer dia x 25 mm nominal dia threaded	each	561.00
	18.10.7.7	34.00 mm x 15 mm	For 34.00 mm outer dia x 15 mm nominal dia threaded	each	699.00
	18.10.7.8	34.00 mm x 20 mm	For 34.00 mm outer dia x 20 mm nominal dia threaded	each	756.00
	18.10.7.9	34.00 mm x 25 mm	For 34.00 mm outer dia x 25 mm nominal dia threaded	each	902.00
	18.10.7.10	34.00 mm x 32 mm	For 34.00 mm outer dia x 32 mm nominal dia threaded	each	1139.00
	18.10.7.11	42.70 mm x 15 mm	For 42.70 mm outer dia x 15 mm nominal dia threaded	each	1067.00
	18.10.7.12	42.70 mm x 20 mm	For 42.70 mm outer dia x 20 mm nominal dia threaded	each	1092.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.7.13	42.70 mm x 25 mm	For 42.70 mm outer dia x 25 mm nominal dia threaded	each	1189.00
	18.10.7.14	42.70 mm 32 mm	For 42.70 mm outer dia x 32 mm nominal dia threaded	each	1321.00
	18.10.7.15	42.70 mm 40 mm	For 42.70 mm outer dia x 40 mm nominal dia threaded	each	1604.00
	18.10.7.16	48.60 mm x 15 mm	For 48.60 mm outer dia x 15 mm nominal dia threaded	each	1162.00
	18.10.7.17	48.60 mm x 20 mm	For 48.60 mm outer dia x 20 mm nominal dia threaded	each	1196.00
	18.10.7.18	48.60 mm x 25 mm	For 48.60 mm outer dia x 25 mm nominal dia threaded	each	1247.00
	18.10.7.19	48.60 mm x 32 mm	For 48.60 mm outer dia x 32 mm nominal dia threaded	each	1498.00
	18.10.7.20	48.60 mm x 40 mm	For 48.60 mm outer dia x 40 mm nominal dia threaded	each	1662.00
	18.10.7.21	48.60 mm x 50 mm	For 48.60 mm outer dia x 50 mm nominal dia threaded	each	2031.00
	18.10.8	Female Thread Tee	Female Thread Tee		
	18.10.8.1	15.88 mm x 15 mm	For 15.88 mm outer dia x 15 mm nominal dia threaded	each	339.00
	18.10.8.2	22.22 mm x 15 mm	For 22.22 mm outer dia x 15 mm nominal dia threaded	each	400.00
	18.10.8.3	22.22 mm x 20 mm	For 22.22 mm outer dia x 20 mm nominal dia threaded	each	413.00
	18.10.8.4	28.58 mm x 15 mm	For 28.58 mm outer dia x 15 mm nominal dia threaded	each	517.00
	18.10.8.5	28.58 mm x 20 mm	For 28.58 mm outer dia x 20 mm nominal dia threaded	each	528.00
	18.10.8.6	28.58 mm x 25 mm	For 28.58 mm outer dia x 25 mm nominal dia threaded	each	578.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.8.7	34.00 mm x 15 mm	For 34.00 mm outer dia x 15 mm nominal dia threaded	each	776.00
	18.10.8.8	34.00 mm x 20 mm	For 34.00 mm outer dia x 20 mm nominal dia threaded	each	789.00
	18.10.8.9	34.00 mm x 25 mm	For 34.00 mm outer dia x 25 mm nominal dia threaded	each	890.00
	18.10.8.10	34.00 mm x 32 mm	For 34.00 mm outer dia x 32 mm nominal dia threaded	each	1057.00
	18.10.8.11	42.70 mm x 15 mm	For 42.70 mm outer dia x 15 mm nominal dia threaded	each	1038.00
	18.10.8.12	42.70 mm x 20 mm	For 42.70 mm outer dia x 20 mm nominal dia threaded	each	1057.00
	18.10.8.13	42.70 mm x 25 mm	For 42.70 mm outer dia x 25 mm nominal dia threaded	each	1187.00
	18.10.8.14	42.70 mm x 32 mm	For 42.70 mm outer dia x 32 mm nominal dia threaded	each	1239.00
	18.10.8.15	42.70 mm x 40 mm	For 42.70 mm outer dia x 40 mm nominal dia threaded	each	1350.00
	18.10.8.16	48.60 mm x 15 mm	For 48.60 mm outer dia x 15 mm nominal dia threaded	each	1237.00
	18.10.8.17	48.60 mm x 20 mm	For 48.60 mm outer dia x 20 mm nominal dia threaded	each	1250.00
	18.10.8.18	48.60 mm x 25 mm	For 48.60 mm outer dia x 25 mm nominal dia threaded	each	1295.00
	18.10.8.19	48.60 mm x 32 mm	For 48.60 mm outer dia x 32 mm nominal dia threaded	each	1375.00
	18.10.8.20	48.60 mm x 40 mm	For 48.60 mm outer dia x 40 mm nominal dia threaded	each	1432.00
	18.10.8.21	48.60 mm x 50 mm	For 48.60 mm outer dia x 50 mm nominal dia threaded	each	1432.00
	18.10.9	Female Connector/ Adapter	Female Thread Connector/ Adapter		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.9.1	15.88 mm x 15 mm	For 15.88 mm outer dia x 15 mm nominal dia threaded	each	221.00
	18.10.9.2	22.22 mm x 15 mm	For 22.22 mm outer dia x 15 mm nominal dia threaded	each	267.00
	18.10.9.3	22.22 mm x 20 mm	For 22.22 mm outer dia x 20 mm nominal dia threaded	each	275.00
	18.10.9.4	28.58 mm x 15 mm	For 28.58 mm outer dia x 15 mm nominal dia threaded	each	322.00
	18.10.9.5	28.58 mm x 20 mm	For 28.58 mm outer dia x 20 mm nominal dia threaded	each	332.00
	18.10.9.6	28.58 mm x 25 mm	For 28.58 mm outer dia x 25 mm nominal dia threaded	each	391.00
	18.10.9.7	34.00 mm x 25 mm	For 34.00 mm outer dia x 25 mm nominal dia threaded	each	475.00
	18.10.9.8	34.00 mm x 32 mm	For 34.00 mm outer dia x 32 mm nominal dia threaded	each	625.00
	18.10.9.9	42.70 mm x 32 mm	For 42.70 mm outer dia x 32 mm nominal dia threaded	each	669.00
	18.10.9.10	42.70 mm x 40 mm	For 42.70 mm outer dia x 40 mm nominal dia threaded	each	791.00
	18.10.9.11	48.60 mm x 40 mm	For 48.60 mm outer dia x 40 mm nominal dia threaded	each	973.00
	18.10.9.12	48.60 mm x 50 mm	For 48.60 mm outer dia x 50 mm nominal dia threaded	each	973.00
	18.10.10	Male Thread Connector/ Adapter	Male Thread Connector/ Adapter		
	18.10.10.1	15.88 mm x 15 mm	For 15.88 mm outer dia x 15 mm nominal dia threaded	each	224.00
	18.10.10.2	22.22 mm x 15 mm	For 22.22 mm outer dia x 15 mm nominal dia threaded	each	264.00
	18.10.10.3	22.22 mm x 20 mm	For 22.22 mm outer dia x 20 mm nominal dia threaded	each	288.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.10.4	28.58 mm x 20 mm	For 28.58 mm outer dia x 20 mm nominal dia threaded	each	364.00
	18.10.10.5	28.58 mm x 25 mm	For 28.58 mm outer dia x 25 mm nominal dia threaded	each	372.00
	18.10.10.6	34.00 mm x 25 mm	For 34.00 mm outer dia x 25 mm nominal dia threaded	each	535.00
	18.10.10.7	34.00 mm x 32 mm	For 34.00 mm outer dia x 32 mm nominal dia threaded	each	656.00
	18.10.10.8	42.70 mm x 32 mm	For 42.70 mm outer dia x 32 mm nominal dia threaded	each	754.00
	18.10.10.9	42.70 mm x 40 mm	For 42.70 mm outer dia x 40 mm nominal dia threaded	each	843.00
	18.10.10.10	48.60 mm x 40 mm	For 48.60 mm outer dia x 40 mm nominal dia threaded	each	976.00
	18.10.10.11	48.60 mm x 50 mm	For 48.60 mm outer dia x 50 mm nominal dia threaded	each	1322.00
	18.10.11	Valve Connector	Valve Connector		
	18.10.11.1	15.88 mm x 15 mm	For 15.88 mm outer dia x 15 mm nominal dia threaded	each	272.00
	18.10.11.2	22.22 mm x 15 mm	For 22.22 mm outer dia x 15 mm nominal dia threaded	each	321.00
	18.10.11.3	22.22 mm x 20 mm	For 22.22 mm outer dia x 20 mm nominal dia threaded	each	343.00
	18.10.11.4	28.58 mm x 25 mm	For 28.58 mm outer dia x 25 mm nominal dia threaded	each	492.00
	18.10.11.5	34.00 mm x 32 mm	For 34.00 mm outer dia x 32 mm nominal dia threaded	each	731.00
	18.10.11.6	42.70 mm x 40 mm	For 42.70 mm outer dia x 40 mm nominal dia threaded	each	1016.00
	18.10.11.7	48.60 mm x 50 mm	For 48.60 mm outer dia x 50 mm nominal dia threaded	each	1366.00
	18.10.12	Female Elbow 90°	Female Threaded Elbow 90°		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.12.1	15.88 mm x 15 mm	For 15.88 mm outer dia x 15 mm nominal dia threaded	each	238.00
	18.10.12.2	22.22 mm x 15 mm	For 22.22 mm outer dia x 15 mm nominal dia threaded	each	319.00
	18.10.12.3	22.22 mm x 20 mm	For 22.22 mm outer dia x 20 mm nominal dia threaded	each	330.00
	18.10.12.4	28.58 mm x 25 mm	For 25.58 mm outer dia x 25 mm nominal dia threaded	each	489.00
	18.10.12.5	34.00 mm x 32 mm	For 34.00 mm outer dia x 32 mm nominal dia threaded	each	851.00
	18.10.12.6	42.70 mm x 32 mm	For 42.70 mm outer dia x 32 mm nominal dia threaded	each	1069.00
	18.10.12.7	42.70 mm x 40 mm	For 42.70 mm outer dia x 40 mm nominal dia threaded	each	1156.00
	18.10.12.8	48.60 mm x 40 mm	For 48.60 mm outer dia x 40 mm nominal dia threaded	each	1281.00
	18.10.12.9	48.60 mm x 50 mm	For 48.60 mm outer dia x 50 mm nominal dia threaded	each	1494.00
	18.10.13	Male Threaded Elbow 90°	Male Threaded Elbow 90°		
	18.10.13.1	15.88 mm x 15 mm	For 15.88 mm outer dia x 15 mm nominal dia threaded	each	251.00
	18.10.13.2	22.22 mm x 15 mm	For 22.22 mm outer dia x 15 mm nominal dia threaded	each	317.00
	18.10.13.3	22.22 mm x 20 mm	For 22.22 mm outer dia x 20 mm nominal dia threaded	each	335.00
	18.10.13.4	28.58 mm x 25 mm	For 28.58 mm outer dia x 25 mm nominal dia threaded	each	438.00
	18.10.13.5	34.00 mm x 25 mm	For 34.00 mm outer dia x 25 mm nominal dia threaded	each	703.00
	18.10.13.6	34.00 mm x 32 mm	For 34.00 mm outer dia x 32 mm nominal dia threaded	each	855.00
	18.10.13.7	42.70 mm x 32 mm	For 42.70 mm outer dia x 32 mm nominal dia threaded	each	1085.00
	18.10.13.8	42.70 mm x 40 mm	For 42.70 mm outer dia x 40 mm nominal dia threaded	each	1231.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.10.13.9	48.60 mm x 40 mm	For 48.60 mm outer dia x 40 mm nominal dia threaded	each	1382.00
	18.10.13.10	48.60 mm x 50 mm	For 48.60 mm outer dia x 50 mm nominal dia threaded	each	1683.00
	18.10.14	Cap	Cap		
	18.10.14.1	15.88 mm	For 15.88 mm outer dia pipe	each	52.00
	18.10.14.2	22.22 mm	For 22.22 mm outer dia Pipe	each	74.00
	18.10.14.3	28.58 mm	For 28.58 mm outer dia Pipe	each	98.00
	18.10.14.4	34.00 mm	For 34.00 mm outer dia Pipe	each	201.00
	18.10.14.5	42.70 mm	For 42.70 mm outer dia Pipe	each	294.00
	18.10.14.6	48.60 mm	For 48.60 mm outer dia Pipe	each	383.00
	18.10.15	Pipe Bridge	Pipe Bridge		
	18.10.15.1	15.88 mm	For 15.88 mm outer dia pipe	each	249.00
	18.10.15.2	22.22 mm	For 22.22 mm outer dia Pipe	each	315.00
	18.10.15.3	28.58 mm	For 28.58 mm outer dia Pipe	each	473.00
	18.10.16	Slip Coupling	Slip Coupling/ Socket		
	18.10.16.1	15.88 mm	For 15.88 mm outer dia pipe	each	131.00
	18.10.16.2	22.22 mm	For 22.22 mm outer dia pipe	each	214.00
	18.10.16.3	28.58 mm	For 28.58 mm outer dia pipe	each	274.00
	18.10.16.4	34.00 mm	For 34.00 mm outer dia pipe	each	405.00
	18.10.16.5	42.70 mm	For 42.70 mm outer dia pipe	each	569.00
	18.10.16.6	48.60 mm	For 48.60 mm outer dia pipe	each	707.00
18.11		Fixing water meter &stop cock in GI pipe	Fixing water meter and stop cock in G.I. pipe line including cutting and threading the pipe and making long screws etc.	each	189.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			complete (cost of water meter and stop cock to be paid separately).		
18.12		Gun metal gate valve	Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) :		
	18.12.1	With CI wheel	With C.I. wheel of approved quality (screwed end)		
	18.12.1.1	25 mm dia	25 mm nominal bore	each	486.00
	18.12.1.2	32 mm dia	32 mm nominal bore.	each	558.00
	18.12.1.3	40 mm dia	40 mm nominal bore	each	677.00
	18.12.1.4	50 mm dia	50 mm nominal bore	each	976.00
	18.12.1.5	65 mm dia	65 mm nominal bore	each	1339.00
	18.12.1.6	80 mm dia	80 mm nominal bore	each	2302.00
	18.12.2	With brass lever handle	With brass lever handle of approved quality.		
	18.12.2.1	25 mm dia	25 mm nominal bore	each	341.00
	18.12.2.2	32 mm dia	32 mm nominal bore.	each	419.00
	18.12.2.3	40 mm dia	40 mm nominal bore	each	491.00
	18.12.2.4	50 mm dia	50 mm nominal bore	each	639.00
18.13		CPVC gate valve	Providing and fixing CPVC gate valve with knob of approved quality.		
	18.13.1	25 mm dia	25 mm nominal bore	each	428.00
	18.13.2	32 mm dia	32 mm nominal bore.	each	500.00
	18.13.3	40 mm dia	40 mm nominal bore	each	584.00
	18.13.4	50 mm dia	50 mm nominal bore	each	749.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.14		Ball float valve brass	Providing and fixing ball valve (brass) of approved quality, High or low pressure, with plastic floats complete :		
	18.14.1	15 mm dia	15 mm nominal bore	each	316.00
	18.14.2	20 mm dia	20 mm nominal bore	each	431.00
	18.14.3	25 mm dia	25 mm nominal bore	each	499.00
18.15		Gun metal non- return valve	Providing and fixing gun metal non- return valve of approved quality (screwed end) :		
	18.15.1	25 mmdia	25 mm nominal bore		
	18.15.1.1	Horizontal	Horizontal	each	462.00
	18.15.1.2	Vertical	Vertical	each	474.00
	18.15.2	32 mmdia	32 mm nominal bore		
	18.15.2.1	Horizontal	Horizontal	each	587.00
	18.15.2.2	Vertical	Vertical	each	715.00
	18.15.3	40 mmdia	40 mm nominal bore		
	18.15.3.1	Horizontal	Horizontal	each	746.00
	18.15.3.2	Vertical	Vertical	each	990.00
	18.15.4	50 mmdia	50 mm nominal bore		
	18.15.4.1	Horizontal	Horizontal	each	1138.00
	18.15.4.2	Vertical	Vertical	each	1353.00
	18.15.5	65 mmdia	65 mm nominal bore		
	18.15.5.1	Horizontal	Horizontal	each	2012.00
	18.15.5.2	Vertical	Vertical	each	2360.00
	18.15.6	80 mmdia	80 mm nominal bore		
	18.15.6.1	Horizontal	Horizontal	each	2886.00
	18.15.6.2	Vertical	Vertical	each	3699.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.16		Brass ferrule	Providing and fixing brass ferrule with C.I. mouthcover including boring andtapping the main :		
	18.16.1	15 mm dia	15 mm nominal bore	each	236.00
	18.16.2	20 mm dia	20 mm nominal bore	each	285.00
	18.16.3	25 mm dia	25 mm nominal bore	each	391.00
18.17		Uplasticised PVC connection	Providing and fixing uplasticised PVC connection pipe with brass unions :		
	18.17.1	30 cm length	30 cm length		
	18.17.1.1	15 mm dia	15 mm nominal bore	each	88.00
	18.17.1.2	20 mm dia	20 mm nominal bore	each	129.00
	18.17.2	45 cm length	45 cm length		
	18.17.2.1	15 mm dia	15 mm nominal bore	each	149.00
	18.17.2.2	20 mm dia	20 mm nominal bore	each	184.00
18.18		Masonry Chamber 30x30x50 for stop cock	Constructing masonry Chamber 30x30x50 cm inside, in brick work in cement mortar 1:4 (1 cement :4 sand) for stop cock, with C. I. surface box 100x100 x75 mm (inside) with hinged cover fixed in cement concrete slab 1:2:4 mix (1 cement :2 sand : 4 graded stone aggregate 20 mm nominal size), i/c necessary excavation, foundation concrete 1:5:10 (1 cement :5 sand:10 graded stone aggregate 40mm nominal size) and inside plastering with cement mortar 1:3 (1 cement :3 sand) 12mm thick finished with a floating coat of neat cement complete as per standard design :-		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.18.1	Clay bricks	With common burnt clay bricks of 25 kg /sqcm compressive strength.	each	916.00
	18.18.2	Flyash bricks	With well burnt flyash bricks of 50 kg/sqm compressive strength.	each	870.00
18.19		Masonry Chamber 60x60x75 for sluice valve	Constructing masonry Chamber 60x60x75 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 sand) for sluice valve, with C.I. surface box 100mm. top diameter, 160 mm bottom diameter and180 mm deep (inside) with chained lid and RCC top slab 1:2:4 mix (1 cement :2 sand : 4 graded stone aggregate 20mm nominal size) , i/c necessary excavation, foundation concrete 1:5:10 (1 cement : 5 sand : 10 graded stone aggregate 40 mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 sand) 12 mm thick finished with a floating coat of neat cement complete as per standard design		
	18.19.1	On clay bricks	With common burnt clay bricks of 25 kg /sqcm compressive strength.	each	5416.00
	18.19.2	On flyash bricks	With flyash bricks of 50 kg/cm^2 strength	each	4995.00
18.20		Masonry Chamber 60x60x75 cm for fire hydrants	Constructing masonry Chamber 60x60x75 cm, inside in brick work in cement mortar 1:4 (1 cement : 4 sand) for fire hydrants, with C.I. surface box 350x350 mm. top and 165 mm deep (inside) with		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			chained lid and RCC top slab 1:2:4 mix (1 cement : 2 sand : 4 graded stone aggregate 20 mm nominal size) , i/c necessary excavation, foundation concrete 1:5:10 (1 cement : 5 sand:10 graded stone aggregate 40 mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 sand) 12 mm thick finished with a floating coat of neat cementcomplete as per standard design :		
	18.20.1	Of clay bricks	With common burnt clay bricks of 25 kg/cm^2 compressive strength.	each	5168.00
	18.20.2	Of fly ash bricks	With flyash bricks of 50kg/cm^2 strength.	each	4851.00
18.21		Masonry Chamber 60x45x50 cm for water meter	Constructing masonry Chamber 60x45x50 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 sand) for water meter complete with C.I. double flap surface box 400x200x200 mm (inside) with locking arrangement and RCC top slab 1:2:4 mix (1 cement : 2 sand : 4 graded stone aggregate 20 mm nominal size) , i/c necessary excavation, foundation concrete 1:5:10 (1 cement : 5 sand:10 graded stone aggregate 40 mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 sand) 12 mm thick finished with a floating coat of neat cement complete as per standard design :		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.21.1	Of clay bricks	With common burnt clay bricks of 25 kg /sqcm compressive strength.	each	4870.00
	18.21.2	Of fly ash bricks	With flyash bricks of 50kg/sqm compressive strength.	each	4683.00
18.22		Painting G.I. pipes and fittings	Painting G.I. pipes and fittings with synthetic enamel white paint with two coats over a ready mixed priming coat, both of approved quality for new work :		
	18.22.1	15 mmdia	15 mm diameter pipe.	metre	8.00
	18.22.2	20 mm dia	20 mm diameter pipe.	metre	9.00
	18.22.3	25 mm dia	25 mm diameter pipe.	metre	12.00
	18.22.4	32 mm dia	32 mm diameter pipe.	metre	14.00
	18.22.5	40 mm dia	40 mm diameter pipe.	metre	16.00
	18.22.6	50 mm dia	50 mm diameter pipe.	metre	19.00
18.23		Repainting G.I. pipes and fittings	Repainting G.I. pipes and fittings with synthetic enamel white paint of approved quality :		
	18.23.1	15 mm dia	15 mm diameter pipe.	metre	4.00
	18.23.2	20 mm dia	20 mm diameter pipe.	metre	5.00
	18.23.3	25 mm dia	25 mm diameter pipe	metre	6.00
	18.23.4	32 mm dia	32 mm diameter pipe	metre	7.00
	18.23.5	40 mm dia	40 mm diameter pipe	metre	8.00
	18.23.6	50 mm dia	50 mm diameter pipe	metre	9.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.24		Painting two coats of anti-corrosive bitumastic paint	Painting G.I. pipes and fittings with two coats of anti-corrosive bitumastic paint of approved quality :		
	18.24.1	15 mm dia	15 mm diameter pipe	metre	4.00
	18.24.2	20 mm dia	20 mm diameter pipe	metre	5.00
	18.24.3	25 mm dia	25 mm diameter pipe	metre	7.00
	18.24.4	32 mm dia	32 mm diameter pipe	metre	8.00
	18.24.5	40 mm dia	40 mm diameter pipe	metre	9.00
	18.24.6	50 mm dia	50 mm diameter pipe	metre	10.00
	18.24.7	65 mmdia	65 mm diameter pipe	metre	13.00
	18.24.8	80 mmdia	80 mm diameter pipe	metre	15.00
18.25		Filling sand all-round G.I. pipes	Providing and filling sand all-round the G.I. pipes in external work.		
	18.25.1	15 mmdia	15 mm diameter pipe	metre	107.00
	18.25.2	20 mm dia	20 mm diameter pipe	metre	108.00
	18.25.3	25 mm dia	25 mm diameter pipe	metre	111.00
	18.25.4	32 mm dia	32 mm diameter pipe	metre	114.00
	18.25.5	40 mm dia	40 mm diameter pipe	metre	116.00
	18.25.6	50 mm dia	50 mm diameter pipe	metre	120.00
	18.25.7	65 mmdia	65 mm diameter pipe	metre	189.00
	18.25.8	80 mmdia	80 mm diameter pipe	metre	195.00
18.26		G.I. Union	Providing and fixing G.I. Union in G.I. pipe including cutting and threading the pipe and making long screws etc. complete (New work)		
	18.26.1	15 mm dia	15 mm nominal bore	each	117.00
	18.26.2	20 mm dia	20 mm nominal bore	each	148.00
	18.26.3	25 mm dia	25 mm nominal bore	each	158.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.26.4	32 mm dia	32 mm nominal bore	each	204.00
	18.26.5	40 mm dia	40 mm nominal bore	each	321.00
	18.26.6	50 mm dia	50mm nominal bore	each	428.00
	18.26.7	65 mmdia	65mm nominal bore	each	625.00
	18.26.8	80 mmdia	80 mm nominal bore	each	718.00
18.27		G.I. Union in existing G.I. pipe	Providing and fixing G.I. Union in existing G.I. pipe line, cutting and threading the pipe and making long screws including excavation, refilling the earth or cutting of wall and making good the same complete wherever required :		
	18.27.1	15 mm dia	15 mm nominal bore.	each	228.00
	18.27.2	20 mm dia	20 mm nominal bore.	each	252.00
	18.27.3	25 mm dia	25 mm nominal bore.	each	269.00
	18.27.4	32 mm dia	32 mm nominal bore.	each	316.00
	18.27.5	40 mm dia	40 mm nominal bore.	each	432.00
	18.27.6	50 mm dia	50 mm nominal bore.	each	579.00
	18.27.7	65 mmdia	65 mm nominal bore.	each	777.00
	18.27.8	80 mmdia	80 mm nominal bore.	each	870.00
18.28		Polyethylene water storage tank	Providing and placing on terrace (at all floor levels) polyethylene water storage tank ISI : 12701 marked with cover and suitable locking arrangement and making necessary holes for inlet, outlet and overflowpipes but without fittings and the base support for tank	per litre	7.00
18.29		Brass bib cock	Providing and fixing brass bib cock of approved quality		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.29.1	15 mm dia	15 mm nominal bore	each	306.00
	18.29.2	20 mm dia	20 mm nominal bore	each	338.00
18.30		Brass stop cock	Providing and fixing brass stop cock of approved quality :		
	18.30.1	15 mm dia	15 mm nominal bore	each	318.00
	18.30.2	20 mm dia	20 mm nominal bore	each	344.00
18.31		C.P. brass bib cock 15 mm dia	Providing and fixing C.P. brass bib cock of approved quality conforming to IS: 8931and weighing not less than 400gm, 15 mm nominal bore.	each	840.00
18.32		C.P. brass long nose bib cock	Providing and fixing C.P. brass long nose bib cock of approved quality conforming to IS standards and weighing not less than 810 gm, 15 mm nominal bore.	each	1258.00
18.33		C.P. brass long body bib cock	Providing and fixing C.P. brass long body bib cock of approved quality conforming to IS standards andweighing not less than 690 gm, 15 mm nominal bore	each	1272.00
18.34		C.P. brass stop cock concealed	Providing and fixing C.P. brass stop cock (concealed) of standard design and of approved make conforming to IS: 8931. 15 mm nominal bore.	each	1155.00
18.35		C.P. brass angle valve for basin	Providing and fixing C.P. brass angle valve for basin mixer and geyser points of approved quality conforming to IS:893115 mm nominal bore	each	714.00
18.36		Sensor operated pillar cock	Providing and fixing chrome plated brass battery based infrared sensor operated pillar cock, having foam flow technology. 15 mm nominal bore.	each	8216.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.37		C.P. brass pillar cock	Providing and fixing C.P. brass pillar cock approved quality and make conforming to IS: specification. and weighing not less than 650gm, 15 mm nominal bore 125 mm long foam flow.	each	1188.00
18.38		C.P. brass basin mixer	Providing and fixing C.P. brass basin mixer of approved quality and make conforming to IS: specification.15 mmnominal bore.	each	2951.00
18.39		C.P. brass wall mixer	Providing and fixing C.P. brass wall mixer of approved quality and make conforming to IS: specification. 15 mm nominal bore.	each	4344.00
18.40		C.P. brass sink mixer	Providing and fixing C.P. brass sink mixer of approved quality and make conforming to IS: specification. 15 mm nominal bore.	each	2951.00
18.41		C.P. brass elbow operated bibcock	Providing and fixing C.P. brass elbow operated bibcock of approved quality and make conforming to IS: specification. 15 mm nominal bore.	each	1118.00
18.42		C.P. brass elbow operated pillar cock	Providing and fixing C.P. brass elbow operated pillar cock of approved quality and make conforming to IS: specification. 15 mm nominal bore.	each	1397.00
18.43		C.P. brass shower rose	Providing and fixing C.P. brass shower rose with 15 or 20 mm inlet :		
	18.43.1	100 mm dia	100 mm diameter	each	806.00
	18.43.2	150 mm dia	150 mm diameter	each	922.00
	18.43.3	200 mm dia	200 mm diameter	each	1336.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.44		PTMT bib cock 15mmdia	Providing and fixing PTMT bib cock of approved quality and colour 15mm nominal bore.		
	18.44.1	86 mm long	86 mm long weighing not less than 88 gm	each	173.00
	18.44.2	122mm long	122 mm long weighing not less than 99 gm	each	208.00
	18.44.3	165mm long	165mm long weighing not less than 110 gm	each	242.00
	18.44.4	90mm long	90mm long weighing not less than 93 gm	each	237.00
18.45		PTMT stop cock	Providing and fixing PTMT stop cock of approved quality and colour.		
	18.45.1	15mmdia	15 mm nominal bore, 86mm long. Weighing not less than 88 gms.	each	173.00
	18.45.2	20mmdia	20mm nominal bore, 89mm long. Weighing not less than 88 gms.	each	213.00
	18.45.3	15mmdia	Concealed stop cock, 15mm nominal bore, 108mm long. Weighing not less than 108 gms.	each	242.00
18.46		PTMT pillar cock	Providing and fixing PTMT pillar cock of approved quality and colour.		
	18.46.1	15mm x 107mm	15mm nominal bore, 107mm long. Weighing not less than 110 gms.	each	251.00
	18.46.2	15mm x 125mm	15mm nominal bore, 125mm long foam flow weighing not less than 120 gms.	each	332.00
18.47		PTMT, push cock	Providing and fixing PTMT, push cock of approved quality and colour.		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.47.1	15mm x 98mm	15 mm nominal bore, 98mm long. Weighing not less than 75 gms.	each	144.00
	18.47.2	15mm x 80mm	15 mm nominal bore, 80mm long weighing not less than 46 gms.	each	126.00
18.48		PTMT grating	Providing and fixing PTMT grating of approved quality and colour.		
	18.48.1	Circular	Circular type.		
	18.48.1.1	100 mmdia	100 mm nominal dia.	each	55.00
	18.48.1.2	125 mmdia	125 mm nominal dia with 25 mm waste hole.	each	60.00
	18.48.2	Rectangular	Rectangular type with openable circular lid.		
	18.48.2.1	150 mmdia	150 mm nominal size square 100 mm diameter of the inner hinged round grating.	each	148.00
18.49		PTMT angle stop cock	Providing and fixing PTMT angle stop cock 15 mm nominal bore. Weighing not less than 85 gms.	each	208.00
18.50		PTMT swivelling shower	Providing and fixing PTMT swivelling shower, 15mm nominal bore. Weighing not less than 40gms.	each	164.00
18.51		PTMT Ball cock	Providing and fixing PTMT Ball cock of approved quality, colour and make complete with Epoxy coated aluminium rod with L.P./ H.P.H.D. plastic ball.		
	18.51.1	15mmdia	15 mm nominal bore, 105 mm long. Weighing not less than 138 gms.	each	234.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	18.51.2	20mmdia	20 mm nominal bore, 120 mm long. Weighing not less than 198 gms.	each	314.00
	18.51.3	25mmdia	25 mm nominal bore, 152mm long. Weighing not less than 440 gms.	each	615.00
	18.51.4	40mmdia	40mm nominal bore, 206mm long. Weighing not less than 690 gms.	each	1016.00
	18.51.5	50mmdia	50mm nominal bore, 242mm long. Weighing not less than 1240 gms.	each	1469.00
18.52		Unplasticised P.V.C. connection pipe	Providing and fixing Unplasticised P.V.C. connection pipe with PTMT Nuts collar and bush of approved quality and colour 15 mm nominal bore		
	18.52.1	30cm	30cm length.	each	61.00
	18.52.2	45cm	45 cm length.	each	79.00
18.53		PTMT extension nipple	Providing and fixing PTMT extension nipple for water tank pipe, fittings of approved quality and colour.		
	18.53.1	15mmdia	15 mm nominal bore. Weighing not less than 32 gms.	each	59.00
	18.53.2	20mmdia	20mm nominal bore. Weighing not less than 40gms.	each	65.00
	18.53.3	25mmdia	25mm nominal bore. Weighing not less than 62 gms.	each	82.00
18.54		Enclosed type water meter bulk type	Providing and fixing enclosed type water meter (bulk type) conforming to IS: 2373 and tested by Municipal Board complete		

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			with bolts, nuts, rubber insertions etc. (The tail pieces if required will be paid separately) :		
	18.54.1	80 mm dia	80 mm nominal bore	each	3548.00
	18.54.2	100 mm dia	100 mm nominal bore	each	5200.00
	18.54.3	150 mm dia	150 mm nominal bore	each	7497.00
	18.54.4	200 mmdia	200 mm nominal bore	each	8472.00
18.55		C.I. dirt box strainer	Providing and fixing C.I. dirt box strainer for bulk type water meter with nuts, bolts, rubber insertions etc. complete conforming to IS : 2373 :		
	18.55.1	80 mm dia	80 mm nominal bore	each	4186.00
	18.55.2	100 mm dia	100 mm nominal bore	each	6562.00
	18.55.3	150 mm dia	150 mm nominal bore	each	8526.00
	18.55.4	200 mmdia	200 mm nominal bore	each	11766.00
18.56		Cutting holes in walls	Cutting holes upto 30x30 cm in walls including making good the same :		
			With common burnt clay bricks of 25 kg /sqcmor flyash bricks of 50 kg/cm^{2}compressive strength.	each	154.00
18.57		Cutting holes R.C.C. floors and roofs	Cutting holes upto 15x15 cm in R.C.C. floors and roofs for passing drain pipe etc. and repairing the hole after insertion of drain pipe etc. with cement concrete 1:2:4(1 cement : 2 sand : 4 graded stone aggregate 20 mm nominal size) including finishing complete so as to make it leak proof.	each	129.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
18.58		Chases in walls	Making chases upto 7.5x7.5 cm in walls includingmaking good and finishing with matching surface after housing G.I. pipe etc.	metre	54.00
18.59		Making hole embedding pipes	Making hole upto 20x20 cm and embedding pipes upto 150 mm diameter in masonry and filling with cement concrete 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 20 mm nominal size) including disposal of malba.	metre	94.00
18.60		Connection GI distribution	Making connection of G.I. distribution branch with G.I. main of following sizes by providing and fixing tee, including cutting and threading the pipe etc. complete :		
	18.60.1	25 to 40 mmdia	25 to 40 mm nominal bore	each	247.00
	18.60.2	50 to 80 mmdia	50 to 80 mm nominal bore	each	644.00
18.61		CP jet assembly	Providing and fixing CP jet assembly to including connection etc complete.	each	375.00
18.62		Ferro-cement water storage tank	Providing and fixing ferro-cement water storage tank with cover, having minimum thickness of 10mm. The wall to be reinforced with two layers of double galvanized annealed square wooven wire mesh of 20 gauge wire spaced at 15mm for 300 litre capacity and 12mm for 450 to 2500 litres capacity finish with 3 layers of Cement mortar 1:2 duly modified with polymer based plastizers and hardening chemicals (Burogrout and Buroseal) The tank should conform to the specification prescribed by Structural Engineering Research	litre	7.00

Item No.	Sub Item No.	Sub Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Centre Roorkee/Gaziabad. The tanks to be painted with drinking water paint inside and aluminimum paint out side with provision of outlet, inlet, overflow and cleaning including hoisting and placing in position upto floor two level or 5 M. height.		
18.63		Extra for additional floor	Add extra for additional Floor liftbeyond5.0M.height.	litre	0.2
18.64		C.P. brass two way bib cock	Providing and fixing C.P. brass two way bib cock with wall flange of approved quality conforming to IS standards, 15 mm nominal bore	each	1304.00
18.65		C.P. brass two way angle cock	Providing and fixing C.P. brass two way angle cock with wall flange of approved quality conforming to IS standards, 15 mm nominal bore	each	1188.00
18.66		C.P. brass sink cock	Providing and fixing C.P. brass wall mounting/table mounted sink cock with wall flange of approved quality conforming to IS standards, 15 mm nominal bore	each	1591.00
18.67		Non-return valve	Providing and fixing in position best Indian Health Faucet with 8mm dia 1.2 meter long Flexible Tube and wall bracket with N. R. V. (non-return wall).	each	1193.00
18.68		C.P. brass Shower arm	Providing and fixing C.P. brass Shower arm with wall flange of approved quality conforming to IS standards, 15 mm nominal bore.	each	1188.00
	18.68.1		300 mm long	each	450.00
	18.68.2		600 mm long	each	650.00

CHAPTER – 19.0

DRAINAGE

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	**2**	**3**
1.	IS 458	Pre-cast Concrete Pipes (with and without reinforcement).
2.	IS 651	Specification for Salt Glazed Stoneware Pipes and Fittings.
3.	IS 783	Code of Practice for Laying Concrete Pipes
4.	IS 1726	Specification for Cast Iron Manhole Covers and Frames
5.	IS 1729	Cast Iron /Ductile Iron Drainage Pipes and Pipe Fittings Socket and Spigot Series for Over-ground Non-pressure Pipe Line.
6.	IS 4127	Code of Practice for Laying of Glazed Stone Ware Pipes
7.	IS 4885	Specifications for Sewer Bricks
8.	IS 12592	Pre-cast Concrete Manhole Covers and Frames - Specifications

PREAMBLE
DRAINAGE
(CHAPTER: 19.0)

19.1 GENERAL

The drainage system shall be designed considering that the discharge of water through a domestic drain is intermittent and limited in quantity andtherefore, small accumulations of solid matter are liable to form in the drains between the building and the public sewer.There is usually a gradual shifting of these deposits as discharges take place.Gradients shall be sufficient to prevent these temporary accumulations building up and blocking the drains.

Normally, the sewer shall be designed for discharging three times the dry weather flow flowing half-full with a minimum self cleansing velocity of 0.75 metre per second. The approximate gradients which give this velocity for the sizes of pipes likely to be used in building drainage and the corresponding discharges when flowing half-full are given in Table as under. The sizes and slopes shall conform to LocalMunicipal Bye-laws.

Table
Gradients for Sewers

	Minimum Gradient		Maximum Gradient	
Diameter mm	Gradients	Discharge cum/Min.	Gradients	Discharge cum/Min.
100	1 in 57	0.18	1 in 5.6	0.59
150	1 in 100	0.42	1 in 9.7	1.32
200	1 in 145	0.73	1 in 14	2.4
230	1 in 175	0.93	1 in 17	2.98
250	1 in 195	1.10	1 in 19	3.60
300	1 in 250	1.70	1 in 24.5	5.30

In cases, where it is practically not possible to conform to the minimum gradients, a flatter gradient may be used but the minimum velocity in such cases shall on no account be less than 0.61 metres per second.

On the other hand, it is undesirable to employ gradients giving velocity of flow greater than 2.4 metres per second.

PIPES AND SPECIALS

19.2 Glazed Stone Ware Pipes and Fittings

All pipes with spigot and socket ends and fittings shall conform to class SP1 of IS 651. These shall be sound, free from visible defects such as fire cracks or hair cracks. The glaze of the pipes shall be free from crazing. The pipes shall give a sharp clear tone when struck with a light hammer. There shall be no broken blisters.

The length of pipes shall be 60, 75, 90 cm exclusive of the internal depth of the socket. The pipes shall be handled with sufficient care to avoid damage to them.

The pipes between two manholes shall be laid truly in a straight line without vertical or horizontal undulation. The pipes shall be laid with socket ends facing upstream. The body of the pipe shall for its entire length rest on an even bed of concrete and places shall be

excavated in the concrete to receive the socket of the pipe. Where pipes are not bedded on concrete, the trench floor shall be left slightly high and carefully bottomed up as pipe laying proceeds, so that the pipe barrels rest on firm and undisturbed ground. If the excavation has been carried too low, the desired levels shall be made up with concrete 1:5:10 (1 cement: 5 sand: 10 graded stone aggregate 40 mm nominal size) for which no extra payment shall be made. If the floor of the trench consists of rock or very hard ground that cannot easily be excavated to smooth surface the pipe shall be laid on a levelling course of concrete as desired.

The rate shall include the cost of materials and labour involved in all the operationsdescribed above excluding the cost o, excavation, refilling, shoring and timbering in trenches, and cement concreting wherever required be paid for separately.

19.3 Manhole Covers

The covers and frames shall conform to IS 1726 for cast Iron and IS 12592 for pre-cast concrete covers and shall be of the following grades and types.

TABLE 19.1

Specifications of Manhole covers

Grades	Grade Designation	Type/shape of cover
1	**2**	**3**
Light Duty	LD - 2.5	Rectangular, Square, Circular
Medium Duty Square	MD - 10	Rectangular, Circular and (for pre-cast concrete manhole covers)
Heavy Duty	HD - 20	Circular-Square, Rectangular, (Scrapper Manhole)
Extra Heavy Duty	EHD - 35	Circular, Square, Rectangular, (Scrapper Manhole)

Manhole covers and frame shall be manufactured from appropriate grade of grey cast iron not inferior than FG150 grade of IS 210.

They shall be cleanly cast and shall be free from air and sand holes, cold shuts and warping.

Covers shall have on its operative top a raised chequered design to provide for an adequate no-slip grip. The rise of chequers shall be not less than 4mm.

Key holes, keys and lifting devices shall be provided in the manhole covered to facilitate their placement in the frames and their operative maintenance.

Manhole covers and frames shall be coated with materials having base with a black bituminous composition. The coating shall be smooth and tenacious. It shall not flow when exposed to temperature of 63°C and shall not be so brittle as to chip off at temperature of 0°C.

Size and shape and performance requirement of manhole covers and frames shall conform to IS 1726.

The cover shall be gas tight and water tight.

The sizes of covers specified shall be taken as the clear internal dimensions of the frame.

The approximate weight of the various type of manhole covers and frames shall be as per IS 1726.

The cover shall be capable of easy opening and closing and it shall be fitted in the frame in workmanship like manner.

19.4 Manholes

At every change of alignment, gradient or diameter of a drain, there shall be a manhole or inspection chamber. Bends and junctions in the drains shall be grouped together in manhole as far as possible. The maximum distance between manholes shall be 30 m.

Manholes of different types and sizes as specified shall be constructed in the sewer line at such places and to such levels and dimensions as shown in the drawings or as directed by the Engineer-in- Charge. The size specified shall indicate the inside dimensions between brick faces of the manholes.

Where the diameter of the drain is increased, the crown of the pipe shall be fixed at the same level and necessary slope given in the invert of the manhole chamber. In exceptional cases and where unavoidable, the crown of the branch sewer may be fixed at lower level butin such cases the peak flow level of the two sewers shall be kept the same.

Sewers of unequal sectional area shall not be jointed at the same invert in a manhole. The invert of the smaller sewer at its junction with main shall be at least 2/3 the diameter of the main above the invert of the main. The branch sewers shall deliver sewage in the manhole in the direction of main flow and the junction must be made with care so that flow in main is not impeded.

No drain from house fittings, e.g. gully trap or soil pipe, etc. to manhole shall normally exceed a length of 6 m unless it is unavoidable.

Manholes 90 × 80 cm are generally constructed within compound for house drainage only and near the buildings for house drainage. Manholes 1.2 m × 90 cm are generally constructed for main drainage work for depths less than 1.5 m.

Manhole 1.4 m × 90 cm is of the arched type and is generally constructed for main drainage works where depth is 1.50 m or more. The width of manholes shall be increased more than 90 cm on bends or junctions or pipes with diameter greater than 450 mm and that the benching width on either side of the channel is minimum 20 cm.

Manholes 1.4 m internal diameter are generally constructed for main drainage works where depth is 2.45 m or more as an alternative to manholes of arch type. The diameter shall be increased suitably, for pipes with diameter greater than 450 mm in the same manner as in the case of rectangular manholes.

Before deciding size of manholes, Local Municipal Bye Laws shall be consulted.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.1		Glazed stoneware pipes SP-1	Providing, laying and jointing glazed stoneware pipes class SP-1 with stiff mixture of cement mortar in the proportion of 1:1 (1 cement : 1 sand) including testing of joints etc. complete : Excavation to be paid seperataly		
	19.1.1	100 mm dia	100 mm diameter	metre	195.00
	19.1.2	150 mm dia	150 mm diameter	metre	260.00
	19.1.3	200 mm dia	200 mm diameter	metre	337.00
	19.1.4	230 mm dia	230 mm diameter	metre	461.00
	19.1.5	250 mm dia	250 mm diameter	metre	507.00
	19.1.6	300 mm dia	300 mm diameter	metre	680.00
19.2		Cement concrete 1:5:10 all-round S.W. pipes	Providing and laying cement concrete 1:5:10 (1 cement : 5 sand : 10 graded stone aggregate 40 mm nominal size) all-round S.W. pipes including bed concrete as per standard design:		
	19.2.1	100 mm dia	100 mm diameter S.W. pipe	metre	475.00
	19.2.2	150 mm dia	150 mm diameter S.W. pipe	metre	580.00
	19.2.3	200 mm dia	200 mm diameter S.W. pipe	metre	677.00
	19.2.4	230 mm dia	230 mm diameter S.W. pipe	metre	728.00
	19.2.5	250 mm dia	250 mm diameter S.W. pipe	metre	782.00
	19.2.6	300 mm dia	300 mm diameter S.W. pipe	metre	877.00
19.3		Cement concrete 1:5:10 upto haunches of S.W. pipes	Providing and laying cement concrete 1:5:10 (1 cement : 5 sand : 10 graded stone aggregate 40 mm nominal size) upto haunches of S.W. pipes including bed concrete as per standard design :		
	19.3.1	100 mm dia	100 mm diameter S.W. pipe	metre	226.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	19.3.2	150 mm dia	150 mm diameter S.W. pipe	metre	366.00
	19.3.3	200 mm dia	200 mm diameter S.W. pipe	metre	430.00
	19.3.4	230 mm dia	230 mm diameter S.W. pipe	metre	465.00
	19.3.5	250 mm dia	250 mm diameter S.W. pipe	metre	500.00
	19.3.6	300 mm dia	300 mm diameter S.W. pipe	metre	577.00
19.4		Unplasticised PVC pipe Below ground	Providing and laying below ground unplasticised PVC pipe to with stand working pressure of 4 kg/cm2 soild waste pipes confirming to IS:13592 and IS:4985 including jointing with seal ring confirming to IS :5282 leaving 10mm gap for thermal expansion all necessary fittings etc. complete. Excavation to be paid seperataly.		
	19.4.1	110 mm dia	110 mm diameter OD (min. wall thickness 3.2mm)	metre	234.00
	19.4.2	160 mm dia	160 mm diameter OD (min. wall thickness 4.0mm)	metre	477.00
	19.4.3	200 mm dia	200 mm diameter OD (min. wall thickness 4.9mm)	metre	751.00
	19.4.4	250 mm dia	250 mm diameter OD (min. wall thickness 6.2mm)	metre	972.00
	19.4.5	300 mm dia	300 mm diameter OD (min. wall thickness 7.7mm)	metre	1232.00
19.5		Cement concrete 1:3:6 all-round unplasticised PVC pipes	Providing and laying cement concrete 1:3:6 (1 cement: 3sand: 6 graded stone aggregate 20 mm nominal size) all-round PVC. pipes including bed concrete as per standard design:		
	19.5.1	110 mm dia	110 mm diameter PVC. pipe	metre	576.00
	19.5.2	160 mm dia	160 mm diameter PVC. pipe	metre	704.00
	19.5.3	200 mm dia	200 mm diameter PVC. pipe	metre	821.00
	19.5.4	250 mm dia	250 mm diameter PVC. pipe	metre	949.00
	19.5.5	300 mm dia	300 mm diameter PVC pipe	metre	1078.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.6		Cement concrete 1:3:6 up to haunches of unplasticised PVC pipes or HDPE DWC pipes	Providing and laying cement concrete 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 20 mm nominal size) up to haunches of PVC. pipes including bed concrete as per standard design :		
	19.6.1	110 mm dia	110 mm diameter PVC. pipe	metre	274.00
	19.6.2	160 mm dia	160 mm diameter PVC. pipe	metre	444.00
	19.6.3	200 mm dia	200 mm diameter PVC. pipe	metre	521.00
	19.6.4	250 mm dia	250 mm diameter PVC. Pipe	metre	607.00
	19.6.5	300 mm dia	300 mm diameter PVC pipe	metre	700.00
19.7		S.W. gully trap	Providing and fixing square-mouth S.W. gully trap class SP1 complete with C.I. grating brick masonry chamber with water tight C.I. cover with frame of 300 x300 mm size (inside) the weight of cover to be not less than 4.50 kg and frame to be not less than 2.70kg as per standard design :		
	19.7.1	100X100mm	100x100 mm size P type		
	19.7.1.1	Well burnt bricks	With well burnt bricks	each	1764.00
	19.7.1.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	each	1668.00
	19.7.2	150 x 100 mm	150 x 100 mm size P type.		
	19.7.2.1	Well burnt bricks	With well burnt bricks	each	1803.00
	19.7.2.2	Sewer bricks	With sewer bricks conforming to IS : 4885	each	1707.00
	19.7.3	180x150 mm	180x150 mm size P type		
	19.7.3.1	Well burnt bricks	With well burnt bricks	each	1913.00
	19.7.3.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	each	1818.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.8		R.C.C. pipes	Providing and laying non-pressure NP2 class (light duty) R.C.C. pipes with collars jointed with stiff mixture of cement mortar in the proportion of 1:2 (1 cement : 2 sand) including testing of joints etc. complete :		
	19.8.1	100 mm dia	100 mm dia. R.C.C. pipe	metre	243.00
	19.8.2	150 mm dia	150 mm dia. R.C.C. pipe	metre	285.00
	19.8.3	250 mm dia	250 mm dia. R.C.C. pipe	metre	385.00
	19.8.4	300 mm dia	300 mm dia. R.C.C. pipe	metre	480.00
	19.8.5	450 mm dia	450 mm dia. R.C.C. pipe	metre	699.00
	19.8.6	500 mm dia	500 mm dia. R.C.C. pipe	metre	826.00
	19.8.7	600 mm dia	600 mm dia. R.C.C. pipe	metre	1278.00
19.9		Rectangular manhole	Constructing brick masonry manhole in cement mortar 1:4 (1 cement : 4 sand) R.C.C. top slab with Cement Concrete 1:2:4 mix (1 cement : 2 sand : 4 graded stone aggregate 20 mm nominal size), foundation concrete 1:2:4 mix (1 cement : 2 sand : 4 graded stoneaggregate 20mm nominal size) inside plastering 12mm thick with cement mortar 1:3 (1 cement : 3 sand) finished with floating coat of neat cement and making channels in cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 20mm nominal size) finished with a floating coat of neat cement complete as per standard design		
	19.9.1	90X80 cm	Inside size 90x80 cm and 45 cm deep including C.I. cover with frame (light duty) 455x610 mm internal dimensions total weight of cover and frame to be notless than 38 kg (weight of cover 23 kg and weight of frame 15 kg) :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	19.9.1.1	Well burnt bricks	With well burnt bricks	each	7997.00
	19.9.1.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	each	7747.00
	19.9.2	120x90 cm	Inside size 120x90 cm and 90 cm deep including C.I. cover with frame (medium duty) 500 mm internal diameter, total weight of cover and frame to be not less than 116 kg (weight of cover 58 kg and weight offrame 58 kg) :		
	19.9.2.1	Well burnt bricks	With well burnt bricks	each	18086.00
	19.9.2.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	each	17402.00
	19.9.3	120x90 cm	Inside size 120x90 cm and 90 cm deep including C.I. cover with frame (heavy duty) 560 mm internal diameter, total weight of cover and frame to be not less than 208 kg (weight of cover 108 kg and weight of frame 100 kg) :		
	19.9.3.1	Well burnt bricks	With well burnt bricks	each	21816.00
	19.9.3.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	each	21228.00
19.10		Extra depth for manholes	Extra for depth for manholes		
	19.10.1	90x80 cm	Size 90x80 cm		
	19.10.1.1	Well burnt bricks	With well burnt bricks	metre	6176.00
	19.10.1.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	metre	5448.00
	19.10.2	120x90 cm	Size 120x90 cm		
	19.10.2.1	Well burnt bricks	With well burnt bricks	metre	7380.00
	19.10.2.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	metre	6512.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
19.11		Manhole circular 0.91m deep	Constructing brick masonry circular type manhole 0.91m internal dia at bottom and 0.56m dia at top in cement mortar 1:4 (1 cement : 4 sand), inside cement plaster 12 mm thick with cement mortar 1:3 (1 cement : 3 sand) finished with a floating coat of neat cement, foundation concrete 1:3:6 mix(1 cement : 3 sand : 6 graded stone aggregate 40mm nominal size), and making necessary channel in cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 20mm nominal size) finished with a floating coat of neat cement all complete as per standard design : 0.91 m deep with S.F.R.C. cover and frame (heavy duty, HD-20 grade designation) 560mm internal diameter conforming to I.S. 12592, total weight of cover and frame to be not less than 182kg., fixed in cement concrete 1:2:4 (1cement : 2 sand : 4 graded stone aggregate 20 mm nominal size) including centering shuttering all complete. (Foot rests and 12mm thick cement plaster atthe external surface shall be paid for separately) :		
	19.11.1	Well burnt bricks	With well burnt bricks	each	8418.00
	19.11.2	Sewer bricks	With Sewer bricks conforming to IS : 4885	each	7958.00
19.12		Extra depth beyond 0.9 depth for manhole circular type 0.91m dia	Extra depth for circular type manhole 0.91m internal dia (at bottom) beyond 0.91m to 1.67m		
	19.12.1	Well burnt bricks	With well burnt bricks	metre	5257.00
	19.12.2	Sewer bricks	With Sewer bricks conforming IS : 4885	metre	4648.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.13		Brick masonry manhole circular 1.22m dia 1.68 m deep	Constructing brick masonry circular manhole 1.22m internal dia at bottom and 0.56m dia at top in cement mortar 1:4 (1 cement :4 sand) inside cement plaster 12mm thick with cement mortar 1:3 (1 cement :3 sand) finished with a floating coat of neat cement foundationconcrete 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 40mm nominal size) and making necessary channel in cement concrete 1:2:4 (1cement : 2 sand : 4 graded stoneaggregate 20mm nominal size) finished with a floating coat of neat cement all complete as per standard design : 1.68 m deep with SFRC Cover and frame (heavy duty HD-20 grade designation) 560mm internal diameter conforming to I.S. 12592, total weight of cover and frame to be not less than 182kg. fixed in cement concrete1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 20mm nominal size) including centering shuttering all complete. (Foot rests and 12 mm thick cement plaster at the external surface shall be paid for separately)		
	19.13.1	Well burnt bricks	With well burnt bricks	each	15915.00
	19.13.2	Sewer bricks	With Sewer bricks conforming IS : 4885	each	15067.00
19.14		Extra depth beyond 1.68m for circular type manhole 1.22m dia	Extra depth for circular type manhole 1.22m internal dia (at bottom) beyond 1.68 m to 2.29 m :		
	19.14.1	Well burnt bricks	With well burnt bricks	metre	6809.00
	19.14.2	Sewer bricks	With Sewer bricks conforming IS : 4885	metre	6014.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.15		Manhole circular 1.52 m dia	Constructing brick masonry circular manhole 1.52 m internal dia at bottom and 0.56 m dia at top in cement mortar 1:4 (1 cement : 4 sand) inside cement plaster 12mm thick withcement mortar 1:3 (1 cement : 3sand) finished with a floating coat of neat cement, foundation concrete 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 40mm nominal size) and making necessary channel in cement concrete 1:2:4 (1cement : 2 sand : 4 graded stoneaggregate 20mm nominal size) finished with a floating coat of neat cement all complete as per standard design :		
	19.15.1	2.30m deep	2.30m deep with SFRC Cover and frame (heavy duty HD-20 grade designation) 560mm internal diameter conforming to I.S. 12592, total weight of cover and frame to be not less than 182 kg. fixed in cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 20mm nominal size) including centering shuttering all complete. (Foot rests and 12 mm thick cement plaster at the external surface shall be paid for separately) :		
	19.15.1.1	Well burnt bricks	With well burnt bricks	each	35165.00
	19.15.1.2	Sewer bricks	With Sewer bricks conforming IS : 4885	each	32659.00
19.16		Extra depth beyond 2.3m depth for manhole circular type 1.52 m dia	Extra depth for circular type manhole 1.52 m internal dia (at bottom) beyond 2.30 m :		
	19.16.1	Well burnt bricks	With well burnt bricks	metre	16863.00
	19.16.2	Sewer bricks	With Sewer bricks conforming IS : 4885	metre	14806.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.17		M.S. foot rests	Providing M.S. foot rests including fixing in manholes with 20x20x10 cm cement concrete blocks 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 20 mm nominal size) as per standard design :		
	19.17.1	20x20 mm MS bar	With 20x20 mm square bar	each	226.00
	19.17.2	20 mm dia MS bar	With 20 mm diameter round bar	each	187.00
19.18		Orange colour safety foot rest	Providing orange colour safety foot rest of minimum 6 mm thick plastic encapsulated as per IS : 10910 on 12mm dia steel bar conforming to IS : 1786 having minimum cross section as 23 mmx25mm and over all minimum length 263 mm and width as 165mm with minimum 112 mm space between protruded legs having 2 mm tread on top surface by ribbing or chequering besides necessary and adequate anchoring projections on tail length on 138 mm as per standard drawing and suitable to withstand the bend test andchemical resistance test as per specifications and having manufacture's permanent identification mark to be visible even after fixing, including fixing in manholes with 30x20x15 cm cement concrete block 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 20 mm nominal size) complete as per design.	each	266.00
19.19		Replacement of M.S. foot rests	Replacement of M.S. foot rests in manholes including dismantling concrete blocks and fixing with 20x20x10 cm cement concrete blocks 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 20 mm nominal size):		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	19.19.1	20x20 mm MS sq. bar	With 20x20 mm square bar	each	254.00
	19.19.2	20 mm dia MS round bar	With 20 mm diameter round bar	each	214.00
19.20		C.I. cover without frame	Supplying and fixing C.I. cover without frame for manholes :		
	19.20.1	455x610 mm light duty	455x610 mm rectangular C.I. cover (light duty) the weight of the cover to be not less than 23 kg.	each	1437.00
	19.20.2	500 mm dia medium duty	500 mm diameter C.I. cover (medium duty) the weight of the cover to be not less than 58 kg.	each	3134.00
	19.20.3	560 mm dia heavy duty	560 mm diameter C.I. cover (heavy duty) the weight of the cover to be not less than 108 kg.	each	6333.00
19.21		Precast R.C.C. manhole cover	Providing and fixing in position precast R.C.C. manhole cover and frame of required shape and approved quality		
	19.21.1	Light duty	Light duty D- 2.5		
	19.21.1.1	Rectangular 600x450mm	Rectangular shape 600x450mm internal dimensions	each	1229.00
	19.21.1.2	Square 450mm	Square shape 450mm internal dimensions	each	1045.00
	19.21.1.3	Circular 450mm	Circular shape 450mm internal diameter	each	987.00
	19.21.2	Medium duty	Medium duty - 10		
	19.21.2.1	Square 450mm	Square shape 450mm internal dimension	each	1183.00
	19.21.2.2	Circular 500mm	Circular shape 500mm internal diameter	each	1098.00
	19.21.3	Heavy duty	Heavy duty D - 20		
	19.21.3.1	Circular	Circular shape 560 mm internal diameter	each	1711.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	19.21.4	Extra Heavy Duty	Extra Heavy Duty - 35		
	19.21.4.1	Circutlar 560mm dia	Circular shape 560 mm internal dia.	each	1885.00
19.22		C.I. cover 300x300 mm	Supplying and fixing C.I. cover 300x300 mm without frame for gully trap (standard pattern) the weight of cover to be not less than 4.5kg.	each	274.00
19.23		Connection of drain with existing manhole	Making connection of drain or sewer line with existing manhole including breakinginto and making good the walls, floors with cement concrete 1:2:4 mix (1 cement : 2 sand : 4 graded stone aggregate 20 mm nominal size) cement plastered on both sides with cement mortar 1:3 (1 cement : 3 sand) finished with a floating coat of neat cement and making necessary channels for the drain etc. complete :		
	19.23.1	100 to 230 mm dia	For pipes 100 to 230 mm diameter	each	238.00
	19.23.2	250 to 300 mm dia	For pipes 250 to 300 mm diameter	each	292.00
	19.23.3	350 to 450 mm dia	For pipes 350 to 450 mm diameter	each	424.00
19.24		Sand cast iron drop connection	Providing sand cast iron drop connection externally for 60 cm drop from branch sewer line to main sewer manhole including inspection and cleaning eye with chain and lid, sand cast iron drop pipe and bend encased all-round with cement concrete 1:5:10 (1 cement : 5 sand : 10 graded stoneaggregate 40 mm nominal size) with all centering and shuttering required, cutting holes in walls and making good with brick work in cement		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			mortar 1:4 (1 cement : 4 sand) plastered with cement mortar 1:3 (1 cement : 3 sand) on inside of the manhole wall lead caulked joints between sand cast iron pipes and fittings, stiff cement mortar 1:1 (1 cement : 1 sand) joints between sand cast iron tee and S.W. pipe, making required channels complete as per standard design and specifications :		
	19.24.1	100 mm dia	100 mm dia. sand cast iron drop connection	each	4915.00
	19.24.2	150 mm dia	150 mm dia. sand cast iron drop connection	each	6856.00
19.25		Extra for depths of sand cast iron drop	Extra for depths beyond 60 cm of sand cast iron drop connection complete :		
	19.25.1	100 mm dia	For 100 mm dia. sand cast iron drop connection	metre	2092.00
	19.25.2	150 mm dia	For 150 mm dia. sand cast iron drop connection	metre	2399.00
19.26		Raising manhole cover and frame	Raising manhole cover and frame slab to required level including dismantling existing slab and making good the damage as required (Brick work for Raising depth of manhole to be paid separately) :		
	19.26.1	Rectangular 90x80 cm Light duty	Rectangular manhole 90x80 cm with rectangular cover 600x450 mm of grade LD - 2.5	each	1126.00
	19.26.2	Rectangular 120x90 cm Medium duty	Rectangular manhole 120x90cm with circular cover 500 mm dia of grade MD - 10	each	1758.00
	19.26.3	Rectangular 120x90 cm Heavy duty	Rectangular manhole 120x90cm with circular cover 560 mm dia of grade HD - 20	each	1638.00
	19.26.4	Circular 140 cm dia Extra heavy duty	Circular manhole 140 cm dia with circular cover 600 mm dia of grade Extra heavy duty HD - 35	each	167.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
19.27		Road gully chamber 50x45x60cm	Constructing brick masonry road gully chamber 50x45x60cm with bricks in cement mortar 1:4 (1 cement : 4 sand) including 500x450 mm precast R.C.C. horizontal grating with frame complete as per standard design :		
	19.27.1	Well burnt bricks	With well burnt bricks	each	3941.00
	19.27.2	Fly ash bricks	With fly ash bricks 100 key/cm^2 strangth	each	4900.00
19.28		Road gully chamber 45x45x77.5cm	Constructing brick masonry road gully chamber 45x45x77.5cm with bricks in cement mortar 1:4 (1 cement : 4 sand) with precast R.C.C. vertical grating complete as per standard design :		
	19.28.1	Well burnt bricks	With well burnt bricks	each	4910.00
	19.28.2	Fly ash bricks	With fly ash bricks 100 kg/cm^2 strength	each	6983.00
19.29		Road gully chamber 110x50x77.5cm	Constructing brick masonry road gully chamber 110x50x77.5cm with bricks in cement mortar 1:4 (1 cement : 4 sand) including 500x450 mm precast R.C.C. horizontal grating with frame and vertical grating complete as per standard design :		
	19.29.1	Well burnt bricks	With well burnt bricks	each	6285.00
	19.29.2	Fly ash bricks	With fly ash bricks 100 kg/cm^2 strength	each	6485.00
19.30		Soak pit circular	Making soak pit 2.5 m diameter 3.0 metre deep with 45 x 45 cm dry brick honey comb shaft with bricks of class designation 75 and S.W. drain pipe 100 mm diameter, 1.8 m long complete as per standard design.		
	19.30.1	Well burnt bricks	With well burnt bricks	each	20609.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.31		Soak pit square	Constructing soak pit 1.20x1.20x1.20m filled with brickbats including S.W. drain pipe 100 mm diameter and 1.20m long complete as per standard design.	each	3172.00
19.32		S.W. intercepting trap	Providing and fixing S.W. intercepting trap in manholes with stiff mixture of cement mortar 1:1 (1 cement : 1 sand) including testing of joints etc. complete :		
	19.32.1	100 mm dia	100 mm dia	each	298.00
	19.32.2	150 mm dia	150 mm dia	each	411.00
19.33		Half round glazed stoneware pipes	Providing, laying and jointing half round glazed stoneware pipes class SP-1 with stiff mixture of cement mortar in the proportion of 1:1 (1 cement : 1 sand) including testing of joints etc. complete : Excavation to be paid seperataly		
	19.33.1	100 mm dia	100 mm dia	metre	163.00
	19.33.2	150 mm dia	150 mm dia	metre	221.00
	19.33.3	200 mm dia	200 mm dia	metre	284.00
	19.33.4	230 mm dia	230 mm dia	metre	386.00
	19.33.5	250 mm dia	250 mm dia	metre	422.00
	19.33.6	300 mm dia	300 mm dia	metre	552.00
19.34		Half round RCC pipes	Providing and laying half round RCC pipes class (light duty) with collars jointed with stiff mixture of cement mortar in the proportion of 1:2 (1 cement : 2 sand) including testing of joints etc. complete : Excavation to be paid seperataly		
	19.34.1	100 mm dia	100 mm dia. Half round R.C.C. pipe	metre	196.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (inRs.)
1	2	3	4	5	6
	19.34.2	150 mm dia	150 mm dia. Half round R.C.C. pipe	metre	233.00
	19.34.3	250 mm dia	250 mm dia. Half round R.C.C. pipe	metre	315.00
	19.34.4	300 mm dia	300 mm dia. Half round R.C.C. pipe	metre	387.00
19.35		Deduct for not providing C.I. manhole	Deduct for not provide C.I. manhole cover with frame		
	19.35.1	for item No. 19.9.1.1 & 19.9.1.2	for item No. 19.9.1.1 & 19.9.1.2	each	1975.00
	19.35.2	for item No. 19.9.2.1 & 19.9.2.2	for item No. 19.9.2.1 & 19.9.2.2	each	6185.00
	19.35.3	for item No. 19.9.3.1 & 19.9.3.2	for item No. 19.9.3.1 & 19.9.3.2	each	10454.00
19.36		Precast RCC drain cover of size 600x450 x70mm	Providing & Fixing of Precast RCC drain cover of size 600x 450 x70mm (thickness) with 4 holes, manufactured by using M-30 grade of concrete reinforced with 8mm dia Tor Steel @100mm c/c both ways, manufactured by vibro compaction process using joint less FRP moulds, so at to achieve shuttering finish on five faces @top face will have broom finish etc complete.	each	350.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.37		Precast RCC drain cover of size 750x600 x70mm	Providing & Fixing of Precast RCC drain cover of size 750x600 x70mm (thickness) with 6 holes, manufactured by using M-30 grade of concrete reinforced with 8mm dia Tor Steel @100mm c/c both ways, manufactured by vibro compaction process using joint less FRP moulds, so at to achieve shuttering finish on five faces @top face will have broom finish etc complete.	each	550.00
19.38		Precast RCC drain cover of size 900x450 x90mm	Providing & Fixing of Precast RCC drain cover of size 900x450 x90mm (thickness) with 8 holes, manufactured by using M-30 grade of concrete reinforced with two layers of 8mm dia Tor Steel @100mm c/c both ways, manufactured by vibro compaction process using joint less FRP moulds, so at to achieve shuttering finish on five faces @ top face will have broom finish etc complete.	each	900.00
19.39		Precast RCC drain cover of size 1150x400 x100mm	Providing & Fixing of Precast RCC drain cover of size 1150x400 x100mm (thickness) with 10 holes, manufactured by using M-30 grade of concrete reinforced with two layers of 8mm dia Tor Steel @100mm c/c both ways, manufactured by vibro compaction process using joint less FRP moulds, so at to achieve shuttering finish on five faces @top face will have broom finish etc complete.	each	1100.00
19.40		FRP/GRP composite manhole cover size 900X450mm	Supplying and fixing of FRP/GRP composite manhole cover with frame of clear opening size 900X450mm rectangular having weight of (complete unit) (weight of frame 11.7 kg + weight of cover 43.76kg ± 10%) (HD/25MT)	each	10130.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			having load bearing capacity (Point load 25MT) as per testing code BSEN 124:1994. Having lifting arrangement should have an insertion of 5mm double sealed closed key hole and high strength plastic alloy/stainless steel socketed arrangement.		
19.41		FRP/GRP composite manhole cover size 450X600mm	Supplying and fixing of FRP/GRP composite manhole cover with frame of clear opening size 450X600mm rectangular having weight of (complete unit) (weight of frame 1.84kg + weight of cover 14.2kg ± 10%) (LD/2.5MT) having load bearing capacity (Point load 2.5MT) as per testing code BSEN 124:1994. having lifting arrangement should have an insertion of 5mm double sealed closed key hole and high strength plastic alloy/stainless steel socketed arrangement.	each	3951.00
19.42		FRP/GRP composite manhole cover size 300X300mm	Supplying and fixing of FRP/GRP composite manhole cover with frame of clear opening size 300X300mm (square) having weight of (complete unit) (weight of frame 1.12kg + weight of cover 4.1kg ± 10%) (LD/2.5MT) having load bearing capacity (Point load 2.5MT) as per testing code BSEN 124:1994. having lifting arrangement should have an insertion of 5mm double sealed closed key hole and high strength plastic alloy/stainless steel socketed arrangement.	each	2035.00
19.43		FRP/GRP composite manhole cover size 600mm round	Supplying and fixing of FRP/GRP composite manhole cover with frame of clear	each	9448.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			opening size 600mm (round) having weight of (complete unit) (weight of frame 13.6kg + weight of cover 43.1kg + - 10%) (HD/25MT) having load bearing capacity (Point load 25MT) as per testing code BSEN 124:1994. having lifting arrangement should have an insertion of 5 mm double sealed closed key hole and high strength plastic alloy/stainless steel socketed arrangement		
19.44		FRP/GRP composite manhole cover size 500mm round	Supplying and fixing of FRP/GRP composite manhole cover with frame of clear opening size 500mm (round) having weight of (complete unit) (weight of frame 11.2kg + weight of cover 41.78kg ± 10%) (HD/25MT) having load bearing capacity (Point load 25MT) as per testing code BSEN 124:1994. Having lifting arrangement should have an insertion of 5mm double sealed closed key hole and high strength plastic alloy/stainless steel socketed arrangement	each	7744.00
19.45		Deductions for FRP/GRP composit manhole cover	Deduct for not providing and installing FRP/GRP manhole cover size		
	19.45.1	900mmx450mm	900mmx450mm (weight of cover 43.76kg ± 10%)	each	6999.00
	19.45.2	450mmx600mm	450mmx600mm (weight of cover 14.2kg ± 10%)	each	2617.00
	19.45.3	300mmx300mm	300mmx300mm (weight of cover 4.1kg ± 10%)	each	1113.00
	19.45.4	Size 600mm	600mm (weight of cover 41.78kg ± 10%)	each	7533.00
	19.45.5	Size 500mm round	500mm (weight of cover 43.1kg ± 10%)	each	5626.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
19.46		HDPE DWC pipes below ground	Providind, laying and jointing HDPE Double Wall corrugated (DWC) pipes (excavation to be paid seperately)		
	19.46.1	100 mm dia to 110 mm dia	100 mm dia to 110 mm dia	metre	160.00
	19.46.2	150 mm dia to 160 mm dia	150 mm dia to 160 mm dia	metre	300.00
	19.46.3	200 mm dia	200 mm dia	metre	608.00
	19.46.4	250 mm dia	250 mm dia	metre	802.00
	19.46.5	300 mm dia	300 mm dia	metre	915.00

CHAPTER – 20.0

PILE WORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	2	3
1	IS 1200 (Part 23)	Method of measurement of building and Civil Engineering Works
2	IS 2911 (Part 1/Sec. 1)	Code of practice for Design and Construction of pile foundation. Driven cast
3	IS 2911 (Part 1/Sec. 2)	Code of practice of Design and Construction of pile foundation. Bored Cast
4	IS 2911 (Part 1/Sec. 3)	Code of practice for Design and Construction of pile foundation. Driven pre
5	IS 2911 (Part 1/Sec. 4)	Code of practice for Design and Construction of pile foundation. Bored pre-cast concrete piles.
6	IS 2911 (Part 3)	Code of practice for Design and Construction of pile foundation. Under reamed piles.
7	IS 2911 (Part 4)	Code of practice for design and Construction of pile foundation. Load test on piles
8	IS 5112	Safety Code for piling and other deep foundations.
9	IS 6426	Specification for pile driving hammer.
10	IS 6427	Glossary of terms relating to pile driving.
11	IS 6428	Specification for pile frame.
12	IS 9716	Guide for lateral dynamic load test on piles.
13	IS 14362	Pile boring equipments. General requirements

PREAMBLE
PILE WORK
(CHAPTER: 20.0)

20.1 Pile Driving

Installation of Piles: Installation of piles shall be as accurate as possible and as per design and drawings. The vertically or the required batter should be correctly maintained. Particular care shall be taken in respect of installing either single pile or piles in two pile groups.

20.2 Deviation /Tolerance

(i) The deviation/tolerance should be as per IS 2911 (Part 1/Sec.1). The piles should not deviate more than 75 mm or D/4 whichever is less (75 mm or D/10 whichever is more in case of piles having diameter more than 600 mm) from their designed position at the working level.

(ii) In case of a single pile under a column, the positional deviation should not be more than 50 mm or D/4 whichever is less (100 mm in case of piles having diameter more than 600 mm. Greater tolerance may be prescribed for piles driven over water and for raking piles.

Sequence of Installation: Normal sequence of installation of pile group is from the centre to the periphery of the group or from one side to the other. Particular care shall be taken to avoid damaging the already cast pile while driving a fresh tube nearby before the concrete has sufficiently set. The possibility of the pile getting damaged is more in compact soilsthan in loose soils.

20.3 Driving a Group of Friction Piles

(i) The skin friction increases considerably when the pile bore is driven in the loose sand as the pile tends to compact the sand. Therefore in such cases the order of installation shall be altered so that a compact block is not created where driving further pile bore will not be possible. Similar precaution will have to be taken where stiff clay or compact sand layers will have to be penetrated.

(ii) However driving the pile bore from centre outwards or commencing at a particular selected edge or even working across the group the problem pointed out in Para (I) above can be avoided.

(iii) In case of very soft soil it is advisable to start driving the bore hole from outside to inside so that the soil gets restrained from flowing out during operation.

20.4 Workability of Concrete for pile work:

The minimum slump shall be 100 mm when the concrete for the piles is being vibrated and when the concrete is not vibrated the maximum permitted slump is 150mm.The degree of workability in both the cases is considered as very high

20.5 Bored Cast in Situ Pile

In case permanent/temporary casing is not used then bored pile is stablised with drilling fluid. Bentonite supplied to site shall conform to IS 2720 (Part V). A certificate shall be obtained by the contractor from the manufacturer showing properties of each consignment and should be submitted to the Engineer-in-charge. Bentonite shall be mixed thoroughly with fresh clean water to make a suspension which will maintain the stability of the pile excavation for the period necessary to place concrete and complete construction. The temperature of the water used in mixing the bentonite suspension and when supplied to bore hole shall not be lower than 5ºC. Consistency of the drilling fluid suspension and when controlled throughout the boring as well as in concreting operations in order to keep the hole stabilized as well as to avoid concrete getting mixed up with thick suspension of mud.

Note: - For piles in high ground water table & unstable soil condition, boring & under Reaming shall be carried out with bentonite slurry.

20.6 Defective Pile

(i) In case defective piles are formed they shall be removed or left in place whichever is convenient without affecting performance of the adjacent piles or cap as a whole. Additional piles shall be provided to replace them as directed.

(ii) Any deviation from the designed location alignment or load capacity of any pile shall be noted and adequate measures taken well before concreting of the pile cap and plinth beam, if the deviations are beyond permissible limit.

(iii) During chipping of the pile, top manual chipping may be permitted after three days of pile casting pneumatic tools for chipping shall not be used before seven days after pile casting.

(iv) After concreting the actual quantity of concrete shall be compared with average obtained from observations actually made in the case of a few piles initially cast. If the actual quantity is found to be considerably less, special investigations shall be conducted and appropriate measures taken.

20.7 Pile load test

Test Pile: - Pile chosen for pile load test only. It may or may not be a working pile. It can work as a working pile if routine load test is carried out on it is up to half of the safe load.

Initial Test: - It is done on test piles to determine the safe load or allowable load or ultimate load bearing capacity.

Trial Pile: - Minimum two initial test should be conducted and half percentage of the total number of piles used in a project must be subjected to routine test. For largely varying strata and for important structure, it is up to 2%.

1- Initial Load Tests on Piles

This test is performed to confirm the design load calculations and to provide guidelines for setting up the limits of acceptance for routine tests. It also gives an idea of the suitability of the piling system. Initial Test on piles are to be carried out at one or more locations depending on the number of piles required. Load applied for the initial (cyclic) load test is
2.5 times the safe carrying capacity of the pile. Loading for Initial Tests is conducted as per Appendix 'A' Clause 6.3 of IS-2911 Part IV.

2- Routine Load Tests on Piles

Selection of piles for the Routine Test is done based on number of piles required subject to maximum of ½% of total number of piles required. The number of tests may be increased to 2% depending on the nature / type of structure. The test load applied is 1½ times the safe carrying capacity of the pile. The Maintained load method as described in Clause 6.2 of IS-2911 (Part IV) shall be followed for loading for the Routine Tests.

3- Different types of pile load test

A- Maintained Load Test:

Load is maintained in 10 or 8 steps and maintained for some time to exact settlement.(step load test).To test a single pile at least 3 dial gauges should be used and 4 dial gauges for group piles. Loading rate is 0.1mm/hr. For clay it is 0.02mm/hr. load is maintained for at least 2hrs. safe load is maintained for 24 hrs.

B- Constant Rate Of Penetration:

Test generally used for soft clays and penetration is done for constant rate of 0.75mm/min.The degree of expansiveness and consequent damage to the structures with light loading may be qualitatively judged as shown below:

TABLE 20.1

Degree Of Expansiveness

Degree of Expensiveness	Differential free swell, percent
Low	Less than 20
Moderate	20 to 35
High	35 to 50
Very high	Greater than 50

Note: In areas of soil showing high or very high differential free swell values, conventional shallow strips footings may not be adequate.

TABLE 20.2

Safe Load for Vertical Bored Cast in Situ Under-Reamed Piles in Sandy and Clayey Soils Including Black Cotton Soils

Size		Length		Mild Steel			Compression				SAFS Loads IN Uplift Resistance				Lateral Thrust	
Diametre of pile	under-reamed diameter	Single under-reamed	Double under-reamed	Longitudinal Reinforcement		Rings spacing of 6mm dia rings	Single under-reamed	Double under-reamed	Increase per 30 cm Length	Decrease per 30 cm Length	Single under-reamed	Double under-reamed	Increase per 30 cm length	Decrease per 3 cm length	Single under-reamed	Double under-reamed
cm	cm	m	m	No.	Dia mm	cm	t	t	t	t	t	t	t	t	t	t
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
20	50	3.5	3.5	3	10	18	8	12	0.9	0.7	4	6	0.65	0.55	1.0	1.2
25	62.5	3.5	3.5	4	10	22	12	18	1.15	0.9	6	9	0.85	0.70	1.5	1.8
30	75	3.5	3.5	4	12	25	16	24	1.4	1.1	8	12	1.05	0.85	2.0	2.4
37.5	94	3.5	3.75	5	12	30	24	36	1.8	1.4	12	18	1.35	1.10	3.0	3.6
40	100	3.5	4.0	6	12	30	23	42	1.9	1.5	14	21	1.45	1.15	3.4	4.0
45	112.5	3.5	4.5	7	12	30	35	52.5	2.15	1.7	17.5	25.75	1.60	1.30	4.0	4.8
50	125	3.5	5.0	9	12	30	42	63	2.4	1.9	21	31.5	1.80	1.45	4.5	5.4

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
20.1		R.C.C. piles Driven cast-in-situ plain	Providing, driving with hydraulic piling rigs with power units and installing driven cast-in-situ reinforced cement concrete piles of grade M-25 of specified diameter and length below the pile cap, to carry safe working load not less than specified, excluding the cost of steel reinforcement but including the cost of shoe and the length of pile to be embedded in the pile cap etc. all complete. (Length of pile for payment shall be measured from top of shoe to the bottom of pile cap) :		
	20.1.1	400 mm dia	400 mm dia piles	metre	1666.00
	20.1.2	450 mm dia	450 mm dia piles	metre	2061.00
	20.1.3	500 mm dia	500 mm dia piles	metre	2503.00
	20.1.4	550 mm dia	550 mm dia piles	metre	2737.00
	20.1.5	750 mm dia	750 mm dia piles.	metre	3519.00
	20.1.6	1000 mm dia	1000 mm dia piles.	metre	7787.00
	20.1.7	1200 mm dia	1200 mm dia piles.	metre	10154.00
	20.1.8	1500 mm dia	1500 mm dia piles.	metre	14472.00
20.2		RCC pile Bored cast-in-situ plain with temporary casing	Boring, providing and installation bored cast-in-situ reinforced cement concrete piles of grade M-25 of specified diameter and length below the pile cap, to carry a safe working load not less than specified, excluding the cost of steel reinforcement but including the cost of boring with bentonite solution and temporary casing of appropriate length for setting out and removal of same and the length of the pile to be embedded in the pile cap etc. by percussion drilling using Direct mud circulation (DMC) or Bailer and chisel technique by tripod and		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			mechanical Winch Machine all complete, including removal of excavated earth with all its lifts and leads (length of pile for payment shall be measured up to bottom of pile cap). **Note:** Truck Mounted rotary/ TMR/Tubewell boring machine shall not be used.		
	20.2.1	300 mm dia	300 mm dia piles	metre	901.00
	20.2.2	400 mm dia	400 mm dia piles	metre	1209.00
	20.2.3	450 mm dia	450 mm dia piles	metre	1375.00
	20.2.4	500 mm dia	500 mm dia. piles	metre	1617.00
	20.2.5	600 mm dia	600 mm dia piles	metre	2203.00
	20.2.6	750 mm dia	750 mm dia piles.	metre	3395.00
	20.2.7	1000 mm dia	1000 mm dia piles.	metre	6168.00
	20.2.8	1200 mm dia	1200 mm dia piles.	metre	8466.00
	20.2.9	1500 mm dia	1500 mm dia piles.	metre	12725.00
20.3		RCC piles cast in situ single under reamed boring with bentonite	Boring with hydraulic piling rigs with power units, providing and installing cast in situ single under reamed piles of specified diameter and length below pile cap in M-25 cement concrete, to carry a safe working load not less than specified, excluding the cost of steel reinforcement but including the cost of boring with bentonite solution and the length of the pile to be embedded in pile cap etc. all complete. (Length of pile for payment shall be measured upto to the bottom of pile cap) :		
	20.3.1	300 mm dia	300 mm dia piles.	metre	1893.00
	20.3.2	400 mm dia	400 mm dia piles	metre	2316.00
	20.3.3	450 mm dia	450 mm dia piles	metre	2568.00
	20.3.4	550 mm dia	550 mm dia piles	metre	2867.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
20.4		R.C.C. piles additional bulb in under reamed piles	Extra over item No. 20.3 for providing additional bulb inunder reamed piles, under specified diameter (Only the quantity of extra bulbs are to be paid).		
	20.4.1	300 mm dia	300mm dia piles.	each	1379.00
	20.4.2	400 mm dia	400mm dia piles.	each	1567.00
	20.4.3	450 mm dia	450 mm dia piles.	each	1679.00
	20.4.4	550 mm dia	550 mm dia piles.	each	1849.00
20.5		R.C.C. piles driven Pre-cast	Providing, driving (with vibrating pile driving hammer complete with power units & accessories) and installing driven Pre-cast reinforced cement concrete piles of specified diameter and length below the pile cap in M-25 cement concrete to carry safe working load not less than specified with a central through preformed hole with M.S. black pipe of dia, 40 mm for grouting with cement sand grouting of mix 1:2 (1cement : 2 coarse sand) under sufficient positive pressure to ensure complete filling including centring, shuttering,driving and removing the steel casing pipe and lifting casing etc. complete but excluding the cost of steel reinforcement. (Length of pile for payment shall be measured from top of the shoe to the bottom of pile cap).		
	20.5.1	400 mm dia	400 mm dia piles.	metre	2273.00
	20.5.2	450 mm dia	450 mm dia piles.	metre	2826.00
	20.5.3	500 mm dia	500 mm dia piles.	metre	3031.00
	20.5.4	550 mm dia	550 mm dia piles.	metre	3460.00
	20.5.5	750 mm dia	750 mm dia piles.	metre	6651.00
	20.5.6	1000 mm dia	1000 mm dia piles.	metre	9160.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
20.6		R.C.C. piles vertical load testing	Vertical load testing of piles in accordance with IS 2911 (Part IV) including installation of loading platform and preparation of pile head or construction of test cap and dismantling of test cap after test etc. complete as per specification and the direction of Engineer in-Charge. **Note**: 1. Initial and Routine Load Test shall not be carried out by Dynamic method of testing. **Note**: 2. Testing agency shall submit the design of loading platform for the approval of Engineer-in-charge.		
	20.6.1	Single pile (upto 50T)	Single pile upto 50 tonne capacity		
	20.6.1.1	Initial test.	Initial test.	per test	38562.00
	20.6.1.2	Routine test	Routine test	per test	17423.00
	20.6.2	Single pile (> 50T)	Single pile above 50 tonne and upto 100 tonne capacity		
	20.6.2.1	Initial test	Initial test	per test	46808.00
	20.6.2.2	Routine test.	Routine test.	per test	26715.00
	20.6.3	Group pile (upto 50T)	Group of two or more piles upto 50 tonne capacity		
	20.6.3.1	Initial test	Initial test	per test	56333.00
	20.6.3.2	Routine test	Routine test	per test	34264.00
20.7		R.C.C. piles cyclic vertical load testing	Cyclic vertical load testing of pile in accordance with IS Code of practice IS: 2911 (part IV) including preparation of pile head etc for:		
	20.7.1	Single pile.	Single pile.		
	20.7.1.1	Upto 50 T	Upto 50 tonne capacity pile.	per test	17423.00
	20.7.1.2	> 50T	Above 50 tonne and upto 100 tonne capacity pile.	per test	26715.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	20.7.2	Group pile	Group of two piles.		
	20.7.2.1	upto 50 T	Upto 50 tonne capacity each.	per test	34264.00
20.8		R.C.C.Pile lateral load testing of single	Lateral load testing of single pile in accordance with IS Code of practice IS : 2911 (Part IV) for determining safe allowablelateral load on pile :		
	20.8.1	Upto 50 T	Upto 50 tonne capacity pile.	per test	17423.00
	20.8.2	> 50T	Above 50 tonne and upto 100 tonne capacity pile.	per test	26715.00
20.9		R.C.C. pile Integrity testing	Integrity testing of Pile using Low Strain/ Sonic Integrity Test/ Sonic Echo Test method in accordance with IS 14893 including surface preparation of pile top by removing soil, mud, dust and chipping lean concrete lumps etc. and use of computerised equipment and high skill trained personal for conducting the test and submission of results, all complete as per direction of Engineer-in-Charge. **Note** :- The inclusion of the above item in the schedule of work shall be judiciously decided by the technical sanctioning authority, keeping in view the quality control, type of soil strata & importance of the project	per test	897.00
20.10		Single under reamed piles of specified diameter and length below pile cap	Boring, Providing and installing cast in situ single under reamed piles of specified diameter and length below pile cap in M 25 cement concrete, to carry a safe working load not less than specified, excluding the cost of steel reinforcement but including the cost of boring, cement concrete and the length of the pile to be embedded in pile cap etc. all complete. (Length of pile for payment shall be measured		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			upto to the bottom of pile cap) : **Note**: - The minimum grade of concrete to be used for piling shall be M-20 and the minimum cement content shall be 400 kg/m3 in all condition.		
	20.10.1	250 mm dia	250 mm diameter piles	metre	516.00
	20.10.2	300 mm dia	300 mm diameter piles	metre	644.00
	20.10.3	400 mm dia	400 mm diameter piles	metre	941.00
	20.10.4	450 mm dia	450 mm diameter piles	metre	1129.00
	20.10.5	600 mm dia	600 mm diameter piles	metre	1794.00
20.11		Extra for providing blub	Extra for providing bulb for item no.20.10		
	20.11.1	250 mm dia	250 mm diameter	each	419.00
	20.11.2	300 mm dia	300 mm diameter	each	657.00
	20.11.3	400 mm dia	400 mm diameter piles	each	1437.00
	20.11.4	450 mm dia	450 mm diameter piles	each	2076.00
	20.11.5	600 mm dia	600 mm diameter piles	each	4707.00

CHAPTER – 21.0

ALUMINIUM WORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	2	3
1.	IS 733	Wrought Aluminium and Aluminium Alloys, Bars, Rods and Sections (For General Engineering Purposes)
2.	IS 737	Wrought Aluminium and Aluminium alloy sheet and strip for general engineering purposes
3.	IS 1285	Wrought Aluminium and Aluminium Alloy, Extruded Round Tube and Hollow sections (For General Engineering Purposes)
4.	IS 1868	Anodic coating on Aluminium and its Alloys-Specification
5.	IS 1948	Specification for Aluminium Doors, Windows and Ventilators
6.	IS 3908	Specification for Aluminium equal leg angles
7.	IS 3909	Specification for Aluminium unequal leg angles
8.	IS 3965	Dimensions for wrought Aluminium and Aluminium Alloys bars, rods and sections
9.	IS 5523	Method of testing anodic coating on aluminium and its alloys.
10.	IS 6012	Measurement of coating thickness by Eddy Current Method
11.	IS 6315	Floor springs (Hydraulically regulated) for heavy doors-Specifications
12.	IS 6477	Dimensions of extruded hollow section and tolerances
13.	IS 12823	Wood products- Pre-laminated particle board -Specifications
14.	IS 14900	Transparent Float glass- Specifications

PREAMBLE

ALUMINIUM WORK
(CHAPTER: 21.0)

21.1 Aluminium Sections

Aluminium sections used for fixed/openable windows, ventilators, partitions, frame work and doors etc. shall be suitable for use to meet architectural designs to relevant works and shall be subject to approval of the Engineer-in-Charge for technical, structural, functional and visual considerations. The aluminium extruded sections shall conform to IS 733 and IS 1285 for chemical composition and mechanical properties. The stainless steel screws shall be of grade AISI 304. The permissible dimensional tolerances of the extruded sections shall be as per IS 6477 and shall be such as not to impair the proper and smooth functioning/operation and appearance of door and windows. Aluminium glazed doors, windows etc. shall be of sizes, sections and details as shown in the drawings. The details shown in the drawings may be varied slightly to suit the standards adopted by the manufacturers of the aluminium work, with the approval of Engineer-in-Charge. Before proceeding with any fabrication work, the contractor shall prepare and submit, complete fabrication and installation drawings for each type of glazing doors, windows, ventilators and partition etc. for the approval of the Engineer-in-Charge. If the sections are varied, the contractor shall obtain prior approval of Engineer-in- Charge and nothing extra shall be paidon this account.

21.2 Powder Coating

The powder used for powder coating shall be Epoxy/polyester powder of make approved by the Engineer-in-Charge. The contractor shall give detailed programme for powder coating in advance, to facilitate the inspection by Engineer-in-Charge or his authorized representative.

21.3 Performance Requirements for the Finish

(i) **Surface appearance**: The finish on significant surfaces shall show no scratches when illuminated and is examined at an oblique angle, no blisters, craters; pinholes orscratches shall be visible from a distance of about 1 m. There shall not be any visible variation in the colour of finished surfaces of different sections and between thecolours of different surfaces of same section.

(ii) **Adhesion**: When a coated test piece is tested using a spacing of 2 mm between each of the six parallel cuts (the cut is made through the full depth of powder coating so that metal surface is visible) and a piece of adhesive tape, approximately 25 mm x 150 mm approved by the Engineer-in-Charge is applied firmly to the cut area and then removed rapidly by pulling at right angles to the test area, no pieces of the finish other than debris from the cutting operation shall be removed from the surface of the finish.

21.4 PANELING MATERIAL

Physical and Mechanical Properties of paneling material are given in the as under:

TABLE 21.1

Physical and Mechanical Properties

Sl.No.	Properties	Flat Pressed Three Layer, Multilayer and Graded	
		Grade-I	Grade-II
(i)	Density variation (Max.) Percent	+ 10	+ 10
(ii)	Water absorption (Max)		
	(a) 2 hours	7.0	15.0
	(b) 24 hours	15.0	30.0
(iii)	Thickness swelling (Max.), percent, 2 hours	5.0	8.0
(iv)	Modulus of rupture (Min) N/mm^2		
	(a) Up to 20 mm thickness	15.0	11.0
	(b) Above 20 mm thickness	12.5	11.0
(v)	Tensile strength perpendicular to surface (Min.) N/m^2		
	(a) Up to 20 mm thickness	0.45	0.3
	(b) Above 20 mm thickness	0.4	0.3
(vi)	Tensile strength perpendicular to surface (Min.) N/mm^2		
	(a) After cyclic test*	0.2	-
	(b) After accelerated water resistance test**	0.15	-
(vii)	Screw withdrawal strength (Min.), N:		
	(a) Face	1250	1250
	(b) Edge	850	750
(viii)	Abrasion resistance (Min.) in number of revolutions		
	(a) Type I	450	450
	(b) Type II	250	250
	(c) Type III	80	80

Drying in air at 27 ± 2°C for 24 hours and then heating in dry air at 70°C for 72 hours. Three such cycles are to be followed and then specimens are tested for tensile strength perpendicularto the surface.

**** Accelerated Water Resistance Test:** Specimens are immersed in water at 27±2°C and water is brought to boiling and kept at boiling temperature for two hours. Specimens are then cooled in water to 27±2°C and tested for tensile strength perpendicular to the surface.

21.5 Float Glass

The glass shall be clear float glass and should be approved by the Engineer in Charge. It shall be clear, float transparent and free from cracks subject to allowable defects. The float glassshall conform to the IS 14900.

Allowable Defects: The allowable defects shall be as per Table below

TABLE 21.2
Specifications for Glass

S.No.	Defects	Central	Outer	Remarks
1.	Gaseous inclusion. Max size, mm	3.0	6.0	Separated by at least 30.0 cm
2.	Opaque gaseous inclusion. Max size. Mm	3.0	6.0	Separated by at least 60.0 cm
3.	Knots, dirt and stones, Max size. mm	1.0	1.0	Separated by at least 30.0 cm
4.	Scratches, Rubs and Crush	Faint	Light	Separated by at least 30.0 cm
5.	Bow, percent. Max	0.5	0.5	See 21.2.4.3
6.	Reams, Strings and lines	Light	Light	See 21.2.4.4
7.	Waviness	Nil	Nil	See 21.2.4.5
8.	Sulphur stains	Nil		
9.	Corner breakage and chip	Not more than nominal thickness of float glass		

21.6 Scratches, Rubs and Crush (Float Glass):

Place the sample of float glass in a vertical position approximately 50 cm from the viewer's position and look through it using either day light without direct sunlight or a background light suitable for observing each type of defect.

21.7 Bow

Depending on the side on which bow is present, stand the sample vertically on a wooden plank. Stretch a thread edge to edge. Measure the longest perpendicular. Distance from the thread to the surface of float glass facing the thread and express it as percentage of the length of float glass from edge along the thread.

21.8 EPDM- Gaskets

The EPDM gasket shall meet the requirements as given in Table below

Table - 21.3
Standards : EDPM Gaskets

S. No.	Description	Standard Follow	Specification
1	Tensile strength Kg.f/cm^2	ASTM-D 412	70 Min.
2	Elongation at break %	ASTM-D 412	250 Min.
3	Modulus 100% Kgf/cm^2	ASTM-D 412	22 Min.
4	Compression set % at 0° CC 22 Hrs.	ASTM-D 395	50 Max.
5	Ozone resistance	ASTM-D 1149	No visible cracks

21.9 Sealant

The sealants of approved grade and colour shall only be used. The silicone for perimeter joints (between Aluminium section and RCC/Stone masonry) shall be of make approved by the Engineer in Charge.

21.10 Method of Application

Surface Preparation: Clean all joints and glazing pockets by removing all foreign matter and contaminants such as grease, oil, dust, water, frost, surface dirt, old sealants or glazing compounds and protective coatings.

21.11 Masking

Areas adjacent to joints shall be masked to ensure neat sealant lines. Masking tape shall not be allowed to touch clean surfaces to which the silicone sealant is to adhere. Tooling shall be completed in one continuous stroke immediately after sealant application and before a skin forms and masking shall be removed immediately after tooling.

21.12 Application

Install backer rod of appropriate size and apply silicone sealant in a continuous operation using a positive pressure adequate to properly fill and seal the joint. The silicone sealant shall be tooled with light pressure to spread the sealant against backing material and thejoint surfaces before a skin forms. A tool with convex profile shall be used to keep the sealant within the joint. Soap or water shall not be used as a tooling aid. Remove masking tape as soon as silicone joint is tooled.

Tolerance: A tolerance of + 3 mm shall be allowed in the width of silicone joints. The depth of the joints at throat shall not be less than 6 mm.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.1		Aluminium work for doors, windows, ventilators & partitions	Providing and fixing aluminium work for doors, windows, ventilators and partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top,bottom and sides with required EPDM rubber/ neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass / stainless steel screws, all complete as per architectural drawings and the directions of Engineer-in-charge. (Glazing, paneling and dash fasteners to be paid for separately) :		
	21.1.1	Fixed	For fixed portion.		
	21.1.1.1	Anodised	Anodised aluminium (anodised transparent or dyed to required shade according to IS: 1868, Minimum anodic coating of grade AC15)	kg	378.00
	21.1.1.2	Powder coated	Powder coated aluminium (minimum thickness of powder coating 50 micron)	kg	416.00
	21.1.1.3	Polyester powder coated	Polyester powder coated aluminium (minimum thicknessof polyester powder coating 50 micron)	kg	426.00
	21.1.2	Openable	For shutters of doors, windows and ventilators including providing and fixing hinges/ pivots and making provision for fixing of fittings whereverrequired including the cost of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			EPDM rubber / neoprene gasket required (Fittings shall be paid for separately).		
	21.1.2.1	Anodised	Anodised aluminium (anodised transparent or dyed to required shade according to IS: 1868, Minimum anodic coating of grade AC15)	kg	446.00
	21.1.2.2	Powder coated	Powder coated aluminium (minimum thickness of powder coating 50 micron)	kg	484.00
	21.1.2.3	Polyester powder coated	Polyester powder coated aluminium (minimum thicknessof polyester powder coating 50 micron)	kg	494.00
	21.1.3	Deduct for not providing gasket	Deduct for not providing EDPM rubber/neoprene gasket all around.	kg	80.00
21.2		Partical Board panelling in aluminium doors, windows & shutters	Providing and fixing 12mm thick Pre-laminated particle board flat pressed three layer or graded wood particle board conformingto IS: 12823 Grade l Type ll, in panelling fixed in aluminium doors, windows shutters and partition frames with C.P. brass / stainless steel screws etc.complete as per architectural drawings and directions of engineer-in-charge.		
	21.2.1	One side decorative lamination	Pre-laminated particle board with decorative lamination on one side and balancing lamination on other side.	sqm	802.00
	21.2.2	Both side decorative lamination	Pre-laminated particle board with decorative lamination on both sides.	sqm	907.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.3		Glazing in aluminium door, window, ventilator shutters and partitions	Providing and fixing glazing in aluminium door, window, ventilator shutters and partitions etc. with EPDM rubber/ neoprene gasket etc. complete as per the architectural drawings and the directions of engineer-in-charge. (Cost of aluminium snap beading shall be paid in basic item) :		
	21.3.1	4.0 mm	With float glass panes of 4.0 mm thickness (weight not less than 10 kg/sqm)	sqm	774.00
	21.3.2	5.0 mm	With float glass panes of 5.0 mm thickness (weight not less than 12.50 kg/sqm)	sqm	948.00
	21.3.3	8.0 mm	With float glass panes of 8 mm thickness (weight not less than 20 kg/sqm).	sqm	1182.00
21.4		Extra for froasted glass	Add extra for providing and fixing froasted glass instead of float glass		
	21.4.1	4.0 mm	With glass panes of 4.0 mm thickness (weight not less than 10 kg/sqm)	sqm	259.00
	21.4.2	5.0 mm	With glass panes of 5.0 mm thickness (weight not less than 12.50 kg/sqm)	sqm	262.00
	21.4.3	8.0 mm	With glass panes of 8 mm thickness (weight not less than 20 kg/sqm).	sqm	262.00
21.5		Floor spring hydraulic	Providing and fixing double action hydraulic floor spring of approved brand and manufacture conforming to IS : 6315, having brand logo embossed on the body /plate with double spring mechanism and door weight upto 125 kg., for doors, including cost of cutting floors , embedding in		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			floors as required and making good the same matching to the existing floor finishing and cover plates with brass pivot and single piece M.S. sheet outer box with slide plate etc. complete as per the direction of Engineer-in-charge :		
	21.5.1	With SS cover plate	With stainless steel cover plate - minimum 1.25 mm thickness	each	1890.00
	21.5.2	With barss cover plate	With brass cover plate - minimum 1.25 mm thickness.	each	2072.00
21.6		Aluminium work powder coated in false ceiling frame	Providing and fixing powder coated aluminium work (minimum thickness of powder coating 50 micron) consisting of tee/ angle sections, of approved make conforming to IS : 733 in frames of false ceiling including aluminium angle cleats with necessary C.P. brass/ stainless steel sunk screws, aluminium perimeter angles fixed to wall with stainless steel rawl plugs @ 450 mm centre to centre and fixing the frame work to G.I. level adjusting hangers 6 mm dia. with necessary cadmium plated machine screws all complete as per approved architectural drawings and direction of the Engineer-in-charge (level adjusting hangers, ceiling cleats and expansion hold fasteners to be paid for separately).	kg	521.00
21.7		G.I. Ceiling Hangers in false ceiling	Providing and fixing 6 mm dia. G.I. level adjusting hangers (upto 1200mm length), fixed to roof slabs by means of ceiling cleats made out of G.I. flat 40x3mm size 60 mm long and stainless steel expandable dash fastener of 12.5 mm dia and 50 mm long, complete as per direction of Engineer -in-charge.	each	57.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.8		Aluminium cover sheet over expansion joints	Providing and fixing machine moulded aluminium covering of approved pattern and design,made out of machine cut aluminium sheet and machine holed for receiving dash fastener, over expansion joints on vertical surfaces/ceiling floors, the fixing on plate in one row on one side of joint only shall be done with stainless steel dash fasteners of 8 mm dia and 75 mm long bolt including providing aluminium washers 2 mm thick and 15 mm dia, at a staggered pitch of 200mm centre to centre including drilling holes in the receivingsurface and providing expandable plastic sleeves in holes etc. complete as per direction of Engineer-in-Charge :		
	21.8.1	Anodised	Anodised aluminium sheet 2.5mm thick (anodised transparent or dyed to required shade according to IS: 1868, Minimum anodic coating of grade AC 15)	kg	439.00
	21.8.2	Powder coated	Powder coated aluminium sheet 2.5mm thick (minimum thickness of powder coating 50 micron)	kg	461.00
21.9		Silicon sealant	Filling the gap in between aluminium frame and adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod of approved quality as per architectural drawings and direction of Engineer-in-charge complete.Upto 5 mm depth and 5 mm width	metre	53.00
21.10		Anodic coating	Extra for applying additional anodic coating AC 25 instead of AC 15 to aluminium extruded	kg	12.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			sections. For fixed portion & for shutters of doors, windows and ventilators.		
21.11		Double glazed hermetically sealed glazing	Providing and fixing double glazed hermetically sealed glazing in aluminium windows, ventilators and partition etc. with 6 mm thick clear float glass both side having 12 mm air gap including providing EPDM gasket, perforated aluminium spacers, desiccants, sealant (Both primary and secondary sealant) etc. as per specifications,drawings and direction of Engineer-in-Charge complete.	sqm	3103.00
21.12		Aluminium Tubular handle bar	Providing and fixing aluminium tubular handle bar 32 mm outer dia, 3.0 mm thick and length as required with SS screws etc .complete as per direction of Engineer-in-Charge.		
	21.12.1	Anodized	Anodized (AC 15) aluminium tubular handle bar	metre	423.00
	21.12.2	Powder coated	Powder coated minimum thickness 50 micron aluminium tubular handle bar.	metre	458.00
	21.12.3	Polyester powder coated	Polyester powder coated minimum thickness 50 micron aluminium tubular handle bar.	metre	474.00
21.13		Brass locks	Providing and fixing Brass 100 mm mortice latch and lock with 6 levers without pair of handles(best make of approved quality) for aluminium doors including necessary cutting and making good etc. complete.	each	400.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.14		Anodised aluminium sub frame work	Providing and fixing anodised aluminium (anodised transparentor dyed to required shade according to IS: 1868. Minimum anodic coating of grade AC15) sub frame work for windows and ventilators with extruded built up standard tubular sections of approved make conforming to IS: 733 and IS: 1285, fixed with dash fastener of required dia and size (Dash fastener to be paid forseparately).	kg	322.00
21.15		Aluminium casement windows fastener	Providing and fixing aluminium casement windows fastener of required length for aluminium windows with necessary screws etc. complete.		
	21.15.1	Anodized	Anodized (AC15) aluminium	each	54.00
	21.15.2	Powder coated	Powder coated minimum thickness 50 micron aluminium.	each	57.00
	21.15.3	Polyester powder coated	Polyester powder coated minimum thickness 50 micron aluminium.	each	56.00
21.16		Aluminium round shape handle	Providing and fixing aluminium round shape handle of outer dia 100mm with SS screws etc. Complete as per direction of Engineer-in-Charge		
	21.16.1	Anodized	Anodized (AC15) aluminium	each	62.00
	21.16.2	Powder coated	Powder coated minimum thickness 50 micron aluminium	each	65.00
	21.16.3	Polyester powder coated	Polyester powder coated minimum thickness 50 micron aluminium.	each	69.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.17		Anodised aluminium grill	Providing and fixing anodised aluminium grill (anodised transparent or dyed to required shade according to IS: 1868 with minimum anodic coating of grade AC15) of approved design/pattern, with approved standard section and fixed to the existing window frame with C.P. brass/ stainless steel screws @ 200mm centre to centre, including cutting the grill to proper opening size for fixing and operation of handles and fixing approved anodised aluminium standard section around the opening, all complete as per requirement and direction of Engineer-in-Charge. (Only weight of grill to be measured for payment).	kg	521.00
21.18		Frameless toughened glass door shutter	Providing and fixing 12 mm thick frameless toughened glass door shutter of approved brand and manufacture, including providing and fixing top & bottom pivot & double action hydraulic floor spring type fixing arrangementand making necessary holes etc. for fixing required door fittings, all complete as per direction of Engineer-in-Charge (Door handle, lock and stopper etc.to be paid separately).	sqm	4958.00
21.19		Heat /light control film reflective one way type to plain glass	Providing and fixing 50 micron thick reflective one way type heat /light control film to window/door glazing with complete accessories as per direction of Engineer-in-charge of approved brand and make.	sqm	448.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.20		Heat /light control film to frosted glass	Providing and fixing 50 micron thick Frosted heat /light control film to window/door glazing with complete accessories as per direction of Engineer-in-charge of approved brand and make.	sqm	518.00
21.21		Heat /light control film non reflective	Providing and fixing 25 micron thick Non reflective heat /light control film to window/door glazing with complete accessories as per direction of Engineer-in-charge of approved brand and make.	sqm	352.00
21.22		Toughened glass for partitions	Providing and fixing 12 mm thick toughened glass including all connectors and fixing accessories all complete for partitions, cubicals etc. (excluding frame work) As per direction of Engineer-in-Charge.	sqm	2554.00
21.23		Toughened glass railing	Providing and fixing 12mm thick toughened glass railing complete height of 1400mm with top 50m X 50mm teak wood hand railing finish with PU polish glass fitted with 45 mm dia SS glass holding studs.	metre	11765.00
21.24		Extra for eching film	Extra for froasted glass (with eching film) in item No. 21.22	sqm	837.00
21.25		Mil finish powder coated System aluminum partition/door/windows	Providing, section design pattern and fixing/installation of mil finish powder coated System aluminum partition/door/windows outer frame and shutter sections use(The profile should be of alloy 6063 T6 and dimensional tolerance should be within the limit specified in EN 12020) infill glass Dgu-5 mm ctg+ 1Omm Air gap +5 mm ctg-1,5 for shutter pleated mesh use for fittings Handle, (Casement for double side door handle)Locking system (MVD Locking), hinge type (Door hinges)All hardware and		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			accessories fitted must have corrosion protection to met requirement of minimum 480 hours of salt spray test as per ASTMB117. The drainage system should be built in to the 1.0 system with drainage cap equipped with non return valve. The wool pile should be enhanced with silent fin. Corner joints should mandatorily have spring loaded corner cleats to provide added strength to the corner. All accessories essential for smooth performance, durability should be used in line system . Product should have manufacturer and installation warranty of at least 10 years. All work complete as per directed by the Architect Engineer in-Charge Note:- After fixing the outer frame, the gap between frame and adjacent finish walls shall be filled with weather proof neutral cure silicon sealant on outside as well as inside with backer rod to provide the essential control of sealant depth and two sided adhesion of sealant.		
	21.25.1	3000X2600 mm	3000X2600 mm	sqm	16783.00
	21.25.2	3000X2400mm	3000X2400mm	sqm	17061.00
	21.25.3	1200X2600mm	1200X2600mm	sqm	27088.00
	21.25.4	1200X2700mm	1200X2700mm	sqm	26891.00
	21.25.5	4000X2600mm	4000X2600mm	sqm	18938.00
	21.25.6	4900X2600mm	4900X2600mm	sqm	17298.00
	21.25.7	1500X2600mm	1500X2600mm	sqm	23623.00
	21.25.8	2000X2600mm	2000X2600mm	sqm	20224.00
	21.25.9	2400X2700mm	2400X2700mm	sqm	18387.00
	21.25.10	3500X2600mm	3500X2600mm	sqm	15828.00
	21.25.11	4900X2600mm	4900X2600mm	sqm	17298.00
	21.25.12	4000X2600mm	4000X2600mm	sqm	18938.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.26		Electromechanical Swing double door sliding system, opening above 6.5 sqm	Supply, Installation, Testing & Commissioning of electromechanical Touch free operated with 12mm toughen glass swing door in Contour design , having double door m opening , sharply dropping opening torque during manual access. Reliable function to wind load control with motor-supported closing action. Integrated Energy Saving Mode for Infrared safety sensors. Modular operator system with upgrade interface also for retrofitting purposes. Radar motion detector inside & outside , Opening speed adjustable up to 250-600 mm /sec. including all cabling of the door system , 230VAC mains supply, complete with all accessories as required All work complete as per directed by the Architect/Engineer-in-Charge.	Sqm	61832.00
21.27		Electromechanical Swing double door sliding system, opening upto 6.5 sqm	Supply, Installation, Testing & Commissioning of electromechanical Touch free operated with 12mm toughen glass swing door in Contour design , having double door opening , sharply dropping opening torque during manual access. Reliable function to wind load control with motor-supported closing action. Integrated Energy Saving Mode for Infrared safety sensors. Modular operator system with upgrade interface also for retrofitting purposes. Radar motion detector inside & outside , Opening speed adjustable up to 250-600 mm /sec. including all cabling of the door system , 230VAC mains supply, complete with all accessories as required All work complete as per directed by the Architect/Engineer-in-Charge.	Sqm	69054.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
21.28		Mill finish powder coated System Profile glass partition and doors, system	Providing and fixing mill finish powder coated System Profile 10 mm thick toughened glass partition with 12 mm thick toughened glass doors, system profile size (25X45mm) and every glass joint use I-Profile frame for glass partition, with all necessary accessories like door handle, pivot, hydraulic floor spring, door lock etc. All work complete as per directed by the Architect/Engineer-in-Charge.	Sqm	11285.00
21.29		Mill finish powder coated System Profile glass partition without doors system	Providing and fixing mill finish powder coated System Profile 10 mm thick toughened glass partition having system profile size (25X45mm) with every glass joint use I-Profile frame for glass partition, All work complete as per directed by the Architect/Engineer-in-Charge.	Sqm	9305.00
21.30		10mm thick toughened glass railing with top 75mm X 50mm teak wood	Providing and fixing 10mm thick toughened glass railing of height up to 1200 mm with top 75mm X 50mm teak wood hand railing finish with PU polish glass fitted with45 mm dia SS glass holding studs all work complete as per directed by architect/engineer in charge.	meter	12115.00
21.31		10mm thick toughened glass railing with top 50mm X 50mm SS	Providing and fixing 10mm thick toughened glass railing of height up to 1200 mm with top 50mm X 50mm S.S hand railing glass fitted with45 mm dia SS glass holding studs all work complete as per directed by architect/engineer in charge.	meter	9430.00
21.32		12mm thick toughened glass Partition	Providing and fixing 12 mm thick toughened glass partition with patch fitting All work complete as per directed by the Architect/Engineer-in-Charge.	Sqm	3810.00

CHAPTER – 22.0

WATER PROOFING WORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	2	3
1.	IS 73	Paving Bitumen Specifications
2.	IS 702	Specifications for Industrial Bitumen
3.	IS 1322	Specifications for Bitumen felts for Water Proofing and Damp Proofing.
4.	IS 2645	Specifications for Integral Cement Water Proofing Compounds
5.	IS 3370 (Part -1)	Code of Practice for Concrete Structures for the Storage of Liquid: Part -1 General Requirements.
6.	IS 3384	Specifications for Bitumen Primer for Water Proofing and Damp Proofing
7.	IS 7193	Specification for Glass Fibre Bitumen Felts
8.	IS 12200	Provision of Water Stops at Transfers Construction Joints in Masonry and Concrete Dams - Code of Practice.
9.	IS 12432 (Part-3)	Application for Spray Applied Insulation - Code of Practice Part-3 Polyurethane/ Polyisocyarurate

PREAMBLE

WATER PROOFING WORK
(CHAPTER: 22.0)

22.1 Water Proofing Treatment

Blending Cement/Water with Water Proofing Compound shall be as described here under the required quantity of cement bags to be used for a particular portion of work should be emptied on a dry platform. Water proofing compound bearing ISI mark and conforming to IS 2645 should then be mixed properly with the cement. The quantity of water proofing compound to be mixed should be as prescribed by the manufacturer but not exceeding 3% by weight of cement. The quantity of cement and water proofing compound thus mixed should be thoroughly blended and the blended cement should again be packed in bags. For the water proofing compound in liquid form, the blending is to be done with water. This can be done by taking the just required quantity of water to be mixed in the particular batch of dry cement mortar. The required quantity of water thus collectedper batch of dry cement mortar to be prepared should be mixed with liquid water proofing compound from sealed tins with ISI mark. The water thus mixed with water proofing compound shall be thoroughly stirred so that the water is blended with water proofing compound properly.

Where ever cement slurry is to used it shall be prepared by using 2.2 kg of blended cement per sqm. area. Each time only that much quantity shall be prepared which can be covered on the surface and the surface in turn would be covered with 25 mm thick cement mortar base within half an hour. Slurry prepared and remained unused for more than half an hour shall be totally rejected. Where ever cement mortar 1:3 (1 blended cement: 3 coarse sand) shall be prepared with cement/ water duly blended as specified. Only that much quantity of cement mortar which can be consumed within half an hour, shall be prepared. Any cement mortar that is prepared and remains unused for more than half an hour shall not be used in the work and shall be rejected.

The surface to be treated with felt shall have a minimum slope of 1 to 120. This grading shall be carried out with cement concrete or cement plaster with sand, as desired, to the average thickness required and finished smooth. Such grading shall be paid for separately In existing roof where gola and drip course are provided at the junction of roof and vertical face of parapet wall, these shall be dressed suitably and finished smooth so as to ensure an easy and gradual turning of the flashing. Any dismantlement or forming and finishing smooth the junction for forming the base of the flashing shall not be measured or paid for separately and shall be deemed to form part of the preparation of the surface in the water proofing treatment While the grading of roof surface is beingdone, it shall be ensured that the outlet drain pipe have been fixed and mouth at the entrance have been eased and rounded off properly for easy flow of water.

When any pipe passes through the roof to be treated, angular fillet shall be built around it for the water proofing treatment to be taken over it. These fillets shall not be measured or paid for separately. Tucking in the water proofing felt will be required where the parapet wall exceeds 45cm in the height from the graded surface. Where the height is 45 cm or less, no groove will be required as the water proofing treatment will be carried over the top of the parapet wall to its full thickness. In the case of low dividing walls of height 30 cm or less, outlets therein shall be cut openfor full height and the bottom and sides shall be rendered smooth and corners rounded and such treatment shall not be measured and paid for separately.

For carrying over and tucking in the water proofing felts into the parapet walls, chimney stacks etc. a horizontal groove 6.5 cm deep, 7.5 cm wide section with its lower edge at not less than 15 cm above the graded roof surface shall be left on the inner face of the same during construction if possible.

When such groove has not been left, the same shall be cut out neatly and the base at rear of the groove shall be finished smooth with cement plaster 1:4 (1 cement: 4 sand). Such cutting of the groove and its finishing smooth shall be deemed to be part of the water proofing item and shall not be measured or paid for separately. No deduction shall be made either for not making the groove orwhen the later has already been left in the masonry by the construction agency. The graded surface of the roof and concrete fillets and the faces of walls shall be thoroughly cleaned with wire brushes and all loose scales etc. removed. The surface shall then be dusted off. Any crack in the roof shall be cut to 'V' section, cleaned and filled up flush with cement mortar slurry 1:4 (1 cement: 4 sand)or blown type petroleum bitumen of IS grade 85/25, or approved quality conforming to IS 702. Such cleaning of the surface or treating the cracks shall not be paid for separately.

When the first strip of felt has been bonded to the roof, Subsequent strips shall overlap the preceeding one by at least 7.5 cm at the longitudinal edges and 10 cm at the ends. All overlaps shallbe firmly bonded with hot bitumen. Streaks and trailings of bitumen near edges of laps shall be levelled by heating the overlap with a blow lamp and levelling down unevenness.

22.2 Water Proofing Treatment with Intergral Crystalline Admixture

22.2.1 General

One method that can simplify the protective process is to make concrete with Integral Crystalline Admixtures that reduce its permeability in effect to make the concrete itself waterproof. Apart from the regular workability admixtures, Integral Crystalline Durability admixtures shall be added to all concrete, structural and otherwise, to waterproof & enhance the Concrete Durability. The Crystalline Admixture shall be added either at the time of batching at the batching plant or in the drum of the transit mixer, when the concrete arrives the point of pouring.

The concrete water proofing industry redefined their terminology a short time ago. American Concrete Institute (ACI) 212-3R-10 "Report on Chemical Admixture for Concrete" documents devoted chapter 15 / page 46 to permeability reducing admixtures (PRA's) that outlines PRAH & PRAN classification or Differentiates between those suitable for concrete exposed to Non Hydrostatic Conditions (PRAN) and concrete exposed to Hydrostatic Conditions (PRAH). Besides reducing permeability some PRA's impart other beneficial characteristics such as reduced drying shrinkage, reduced chloride-ion penetration, improved freeze thaw resistance and enhanced autogenous sealing.

22.2.2 Material

Integral crystalline water-proofing admixture is one part cementitious powder added to the concrete mix at the time of batching. Integral crystalline water-proofing admixture consists of hydrophilic chemicals such as Portland cement, very fine treated silica sand and various active, proprietary chemicals. These active chemicals react with moisture in fresh concrete and with the by-products of cement hydration to cause a catalytic reaction, which generates a non-soluble crystalline formation throughout the pores and capillary tracts of the concrete. Thus, the concrete becomes permanently sealed against the penetration of water or liquids from any direction. The concrete is also protected from deterioration due to harsh environmental conditions. Integral crystalline water-proofing admixture is specially formulated to meet varying project and temperature conditions. This reaction continuous over the life of the concrete serving to seal not only initial shrinkage cracks, but also cracks that occur over time.

22.2.3 Technical Specifications /Parameters

The specifications of the materials should match or exceed the requirements mentioned in nomenclature of the item the product should be stored under dry condition. The shelf life of the integral crystalline admixture should be treated as one year when stored under normal conditions.
The water proofing compound used in integral crystalline water proofing treatment shall satisfy all the requirements indicated in relevant standards or as specified in concerned relevant codes etc. and the same shall be got tested and get approved from the Engineer-in-charge before its use.
The integral crystalline waterproofing admixtures of hydrophilic in nature shall confirm to the following requirements:

1. Dosage as specified in the item or higher as recommended by manufacturer's specification, material shall fulfill the requirements of American Concrete Institute Guidelines ACI-212- 3R-10 Chapter 15 and fall under PRAH (Permeability reducing Admixtures for HYDROSTATIC conditions) and must reduce Coefficient of Permeability of concrete by more than 90% (preferably ZERO Permeability), when compared to control concrete and tested as per DIN 1048 Part 5 by carrying out 4 cycles each of 5 bar hydrostatic pressure for 72 hours and drying for 48 hours between the cycles & co-efficient of Permeability calculated as per Darcy's Formula/ Valenta equation by incorporating penetration values obtained at the end of fourth cycle pressure.

2. The crystalline admixture shall be compatible with any other concrete admixture confirming to ASTM C494 and IS 9103.

3. The performance of the crystalline admixture must not be restricted by water-cement ratio of concrete mix. In other words, the crystalline admixture must perform at any water-cement ratio of the concrete mix.

4. Is shall possess CE mark approval as per EN934-2 and shall be procured from CE mark approved manufacturing unit. It shall also possess approval certificate from any national apex institution mandated to issue design codes. The product has no corrosion effect on reinforcement steel according to test norm DINV18998. The maximum chloride content lies less than 0.1% and maximum alkali content less than 9.3%.

5. The material must be capable of withstanding high hydrostatic pressure. It is important that the civil contractor chooses the best material available in the market and the product meeting all the required criteria and demonstrating highest resistance to the high hydrostatic pressure will be given preference.

In addition to recognizing the use of Integrate Crystalline Admixtures. ACI 212-3R-10 has also provided some guideline in para 15.3-"Selection and Evaluation" to select the best quality PRAH's materials as maximum permeability reducing admixtures at maximum Hydrostatic Pressure. Para 15.3 stated that the effect of the admixture can be evaluated by testing the permeability of concrete both directly and indirect methods. The US Army Corps of Engineers CRC C48-92 (1992) test method is a direct measurement of concrete permeability resistance during exposure to water under 200 psi or 13.8 bars or 1.38 MPa of hydrostatic pressure.

6. The material must reduce Chloride diffusion Co-efficient by minimum 45% when tested as per ASTM C 1556-4 and compared with the control concrete, thereby prolonging the durability & service life of the treated concrete structure.

7. The material must demonstrate minimum reduction of 20% in shrinkage cracks as compared to control concrete when tested as per standard BSISO1920-8:2009 & no internal expansion under Sulphate attack, when tested as per ASTM C-1012-12.

8. The crystalline admixture must be capable of self-healing of cracks up to a width of 0.5mm.

9. Product must possess third party assurance, confirming that the product, when used in the concrete, will have no detrimental side effects in terms of Alkali Silica Reaction (ASR), corrosion of steel re-enforcement etc.

10. The product performance shall not be affected by wear abrasion of the treated concrete surface and crystalline treated concrete shall not require protection layer.

11. The crystalline admixture shall be non-toxic and shall confirm to NSF-61 USA.

Note: - The manufacturer shall produce relevant test certificates of Integral Crystalline Admixture Material from reputed laboratories as per relevant codes stated above. To support the claim of crystal formation, national/ international test report of scanning electrons microscopic (SEM)Photographs dandified and mature crystal formation to plug all the capillary track and pores of the concrete shall be provided. Total quantity of Integral Crystalline Admixture Material required shall be arranged only after obtaining the prior approval of the Engineer-in-Charge in writing. Materials shall be kept under double lock and key and proper account of water proofing compound used inthe work shall be maintained. It shall be ensured that the consumption of the compound is as per specified requirements. Contractor shall associate himself with anyone of the specialist firmsmentioned in approved list of specialized agencies for the work relating to the Water Proofing Treatment.

22.2.4 Recommended uses

- Foundations/Rafts Reservoirs, Bridges and Dams
- Sewage and Water Treatment Plants Secondary Containment Structures
- Tunnels and Subway Systems Underground Vaults
- Parking Structures Swimming Pools and water tanks
- Pre-Cast, Cast-in-Place and Shotcrete applications Basement Retaining Walls

22.2.5 Dosage

Dosage 0.8% (minimum) to the weight of cement content per cubic meter of concrete or higher dosage as recommended by the manufacturer's specification.

22.2.6 Mixing

The integral crystalline admixture shall be used @ 0.8% (minimum) to the weight of cement content per cum of concrete or higher as recommended by the manufacturer's specification desired to meet water proofing and durability criteria. Integral crystalline Admixture must be added to the concrete at the time of batching at the batching plant or in the drum of the transit mixer, when the concrete arrives the point of pouring. The sequence of procedures for addition will vary according to the type of batch plant operation and equipment. The mixing shall be followed as below unless& until specified otherwise. For any detailing and mixing guidelines the manufacturer's specifications should be followed.

22.2.7 Ready mix plant- Dry batch operation

Add integral crystalline admixture in power form to the drum of the ready-mix truck under the batch plant and add 60%- 70% of the required water, along with required aggregate. Mix the materials for 2-3 minutes to ensure that the integral crystalline admixture is distributed evenly throughout the mix water. Add the balance of materials to the ready-mix truck in accordance with standard batching practices.

22.2.8 Ready mix plant- Central mix operation

Mix integral crystalline admixture with water to form a very thin slurry (e.g. 18kg of powder mixed with 22.7 litre of water). Pour the required amount of material into the drum of the ready-mix truck. The aggregate, cement, sand and water should be batched and mixed in the plant in accordance with standard practices (taking into account the quality of water that has already been placed in the ready-mix truck). Pour the concrete into the truck and mix for at least 5 minutes to ensure even distribution of integral crystalline admixture throughout the concrete.

22.2.9 Precast batch plant

Add integral crystalline admixture to the aggregate and sand, then mix thoroughly for 2-3 minutes before adding the cement and water. The total concrete mass should be blended using standard practices. It is important to obtain a homogeneous mixture of integral crystalline admixture with the concrete. Therefore dry integral crystalline admixture powder should never be mixed directly to wet concrete as this may cause clumping and hence thorough dispersion in the concrete mix may not occur.

22.2.10 Setting time and strength

The setting time of concrete mix is directly affected by the chemical and physical composition of ingredients, temperature of the concrete and prevailing climatic conditions. Retardation of set may occur when using integral crystalline admixture. The amount of retardation will depend upon the concrete mix design and the dosage rate of integral crystalline admixture. However, under normal conditions, integral crystalline admixture will provide a normal set concrete. Concrete containing integral crystalline admixture may develop higher ultimate strengths than plain concrete. Trial mixes of the concrete should be carried out under project conditions to determine setting time and strength of the concrete.

22.2.11 Application

Concrete treated with Integral Crystalline Admixture should be placed and finished in accordance with good concrete practices. ACI guidelines and recommendations should be observed.

22.2.12 Precaution / Special Consideration

It is important to obtain a homogeneous mixture of Crystalline Admixture with the concrete. Therefore, do not add dry Crystalline Admixture power directly to wet concrete as this may cause clumping and through dispersion will not occur. When incorporating Integral Crystalline Admixture, the temperature of the concrete mix should be above 4º C.

22.2.13 Storage / Self Life

Integral Crystalline Admixture must be stored dry at a minimum temperature of 7º C and its selflife is one year when stored under proper conditions.

22.2.14 Measurement

The quantity of Integral Crystalline Admixture will be measured in kg correct to two places of decimal as per the dosage as specified to the weight of cement content per cubic meter of reinforced cement concrete. The total weight of cement in kg is to be calculated as per the mix design of reinforced cement concrete.

22.2.15 Rate

The rate shall include the cost of all labour and materials involved in all the operations described above.

22.3 Water Proofing Treatment with Integral Crystalline Water Proofing Coating / Slurry

22.3.1 General

This Integral crystalline water proofing coating / slurry of hydrophilic in nature is applied to surface of the concrete to water proof and protect the concrete in-depth. It consists of Portland cement, specially treated quartz sand and a compound of active chemicals. Integral crystalline water proofing coating material needs only to be mixed with water prior to application. When integral crystalline water proofing material is applied to a concrete surface, the active chemicals react with moisture and the by-products of cement hydration to cause a catalytic reaction that generates an insoluble, crystalline structure. These crystals fill the pores and minor shrinkagecracks in the concrete to prevent any further water ingress (even under pressure). However, integralcrystalline water proofing material will still allow the passage of vapor through the structure (i.e. the concrete will be able to "breathe"). Even after the concrete has cured, integral crystalline water proofing material remains dormant in the concrete and will reactivate in the presence of moisture toseal capillary tracts and hairline cracks.In addition to water proofing the structure, integral crystalline water proofing Slurry protects concrete against seawater, wastewater, aggressive groundwater and many other aggressive chemical solutions. Integral crystalline water proofing material is approved for use in contact with potable water and is therefore suitable for use in water storage tanks, reservoirs, water treatment plants, etc. Integral Crystalline Slurry is not a decorative material.

22.3.2 Material

This Integral crystalline water proofing material consists of Portland cement, specially treated quartz sand and a compound of active chemicals. The water proofing compound used in integral crystalline water proofing treatment shall satisfy all the requirements indicated in relevant standards or as specified in concerned relevant codes etc. and the same shall be got tested and get approved from the Engineer-in-charge before its use. Technical Specification/ Parameters. The integral crystalline slurry / coating material of hydrophilic in natural shall confirm to the following requirements:

1. Dosage as specified or at the manufacturer's recommended reduction of co-efficient of Permeability of concrete more than 90% at 56 days curing, when tested as per DIN 1048 Part 5 (after applying 4 cycle each 5 bar hydrostatic pressure) and permeability co-efficient calculated as per Valenta / Darcy's formula.

2. It shall confirm to EN1504-3 (For structural repairs-R3, Compressive Strength > 25 Mpa) supplied from an approved manufacturing unit having CE approval confirming to EN1504-3R3.

3. The product has no corrosion effect on reinforcement steel according to test norm DINV18998. The maximum chloride content lies less than 0.1% and maximum alkali content less than 9.3%.

4. The material must be capable of withstanding high hydrostatic pressure. It is important that the civil contractor chooses the best material available in the market and the product meeting all the required criteria and demonstrating highest resistance to the high hydrostatic will be given preference.

5. The Integral Crystalline Slurry must be capable of self-healing of cracks up to a width of 0.5mm.

6. Product must possess third party assurance, confirming that the product, when used in the concrete, will have no detrimental side effects in terms of Alkali Silica Reaction (ASR), corrosion of Steel Reinforcement etc.

7. The product performance shall not be affected by wear abrasion of the treated concrete surface and crystalline treated concrete shall not require protection layer.

8. Potable Water Compatibility: Nontoxic & suitable for use in potable water facilities- NSF listed as per ANSI 61 listing.

Note: The manufacturer shall produce relevant test certificates of Integral Crystalline water proofing slurry material from reputed laboratories as per relevant codes as stated above. Total quantity of the Integral Crystalline water proofing slurry material required shall be arranged only after obtaining the prior approval of the Engineer-in-Charge in writing. Materials shall be kept under double lock and key and proper account of water proofing compound used in the work shall be maintained. It shall be ensured that the consumption of the compound is as per specified requirements. Contractor shall associate himself with anyone of the specialist firms mentioned in approved list of specialized agencies for the work relating to the Water Proofing Treatment.

22.3.3 Recommended uses

- Foundations/Rafts Reservoirs, Bridges and Dams
- Sewage and Water Treatment Plants Secondary Containment Structures
- Tunnels and Subway Systems Underground Vaults
- Parking Structures Swimming Pools and water tanks
- Pre-Cast, Cast-in-Place and Shotcrete applications Basement Retaining Walls

22.3.4 Preparation of surface

All concrete to be treated with integral crystalline water proofing slurry material must be clean and have an “open” capillary surface. Remove laitance, dirt, grease, etc. by means of high pressure water jetting, wet sandblasting or wire brushing. Faulty concrete in the form of cracks, honeycombing, etc. must be chased out, treated with the same material and filled flush with the mortar mixture as specified by the manufacturer. Surface must be carefully pre-watered prior to the application of integral crystalline water proofing material. The concrete surface mush be damp but with no wet sheen on the surface.

22.3.5 Mixing

Integral crystalline water proofing slurry / coating material should be mechanically mixed with clean water to a creamy consistency resembling to thick oil. Only that much material should be mixas can be used within 20 minutes and mixture should be stirred frequently. The mixture should not be allowed to set, it if happens, simply re-stir to restore workability but no more water should be added to it. The ratio of integral crystalline water proofing material with water should be as below

i) Vertical surface: - For applying with brush the ratio should be 5 parts of integral crystalline water proofing material to 2 parts of water. For applying with spray the ratio should be 5 parts of Integral Crystalline water proofing coating material to 2.75 – 3.25 parts of water.

ii) Horizontal surface: - The ratio should be 3 parts of Integral Crystalline water proofing coating material to 1 part of water. This should be applied by brush only.

22.3.6 Application Procedure

The slurry mix of the Integral Crystalline water proofing slurry material shall be applied in one or two coats as specified/ required according to work situation in the item. After preparation of surface as described in para 22.23.2, and making the surface saturated with water before application of Crystalline Slurry, then first coat of the slurry mix shall be applied by the brush or appropriate power spray equipment. The second coat as specified shall be applied while the first coat is still green. The other method of application known as Dry powder consistency can also be applied on horizontal surfaces only. The specified amount of integral crystalline water proofing material is distributed in powder form through a sieve or a semi mechanical barrow spreader and troweled intothe freshly placed concrete as this reaches the initial set. The integral crystalline water proofing material to be used shall be as following:-

i.) Vertical surface: - Two coats of integral crystalline water proofing material slurry coat shall be applied @ of 0.70 kg per sqm. per coat or as specified.

ii.) Horizontal surface: - One coat of integral crystalline water proofing material slurry coat shall be applied @ of 1.10 kg per sqm or as specified by the manufacturer to harden concrete. Alternatively integral crystalline water proofing material mix can be dry sprinkled @ of 1.00 kg per sqm and trowel applied to fresh concrete when it has reached initial set.

iii.) Construction joint: - Integral crystalline water proofing material mix shall be applied either as slurry coat or dry powder consistency immediately prior to placing the next lift/ bay of concrete @ 1.60 kg per sqm. or as specified.

iv.) Binding concrete: - Integral crystalline water proofing material mix shall be applied either as slurry coat or dry powder consistency immediately prior to placing the overlying concrete slab.

22.3.7 Curing

The treated surfaces should be kept damp for a period of five days and must be protected against direct sun, wind and frost, by covering with polyethylene sheeting, damp burlap or similar material.

22.3.8 Precaution / Special Consideration

Do not apply Integral Crystalline Slurry at temperatures at or below freezing or to frozen or freezing surfaces. Integral Crystalline slurry cannot be used as an additive to concrete or plasters. (Integral Crystalline Admixture should be considered for these applications).

22.3.9 Storage / Self Life

When properly stored in a dry place in unopened and undamaged original packaging its self life is 12 months.

22.4 Water Proofing Treatment with Integral Crystalline Water Proofing Dry-Shake

22.4.1 General

Crystalline water proofing dry-shake of hydrophilic in nature is a unique Integral Crystallinechemical treatment for the waterproofing and protection of concrete. Crystalline water proofing dry-shake has been formulated for dry-shake applications on horizontal concrete surfaces where greater impact and abrasion resistance is required.

22.4.2 Material

Integral Crystalline water proofing dry-shake (dry powder) compound consists of Portland cement, various active proprietary chemicals, and a synthetic aggregate hardener that has been crushed and graded to particle sizes suitable for concrete floors. Crystalline water proofing dry-shake becomes an integral part of the concrete surface, thereby eliminating problems normally associated with coatings (e.g. scaling, dusting, flaking and delaminating). The active chemicals react with the moisture in the fresh concrete causing a catalytic reaction that generates a non-soluble crystalline formation within the pores and capillary tracts of the concrete. Technical Specification/ Parameters: The Integral Crystalline dry shake material of hydrophilic in nature shall confirm to the following requirements:

1. Dosage as specified or at the manufacturer's recommended reduction of co-efficient of Permeability of concrete more than 90% at 56 days curing, when tested as per DIN 1048 Part 5 (after applying 4 cycle each 5 bar hydrostatic pressure) and permeability co-efficient calculated as per Valenta / Darcy's formula.

2. It shall confirm to EN1504-3 (For structural repairs-R3, Compressive Strength > 25 Mpa) supplied from an approved manufacturing unit having CE approval confirming to EN 1504-3R3.

3. The product has no corrosion effect on reinforcement steel according to test norm DINV18998. The maximum chloride content lies less than 0.1% and maximum alkali content less than 9.3%.

4. The material must be capable of withstanding high hydrostatic pressure. It is important that the civil contractor chooses the best material available in the market and the product meeting all the required criteria and demonstrating highest resistance to the high hydrostatic will be given preference.

5. The Integral Crystalline dry shake must be capable of self-healing of cracks up to a width of 0.5mm.

6. Product must possess third party assurance, confirming that the product, when used in the concrete, will have no detrimental side effects in terms of Alkali Silica Reaction (ASR), corrosion of Steel Reinforcement etc.

7. The product performance shall not be affected by wear abrasion of the treated concrete surface and crystalline treated concrete shall not require protection layer.

8. Potable Water Compatibility: Nontoxic & suitable for use in potable water facilities- NSF listed as per ANSI 61 listing.

Note: The manufacturer shall produce relevant test certificates of Integral Crystalline water proofing dry shake material from reputed laboratories as per relevant codes as state above. Total quantity of the Integral Crystalline water proofing dry shake material required shall be arranged only after obtaining the prior approval of the Engineer-in-Charge in writing. Materials shall be keptunder double lock and key and proper account of water proofing compound used in the work shall

be maintained. It shall be ensured that the consumption of the compound is as per specified requirements. Contractor shall associate himself with anyone of the specialist firms mentioned in approved list of specialized agencies for the work relating to the Water Proofing Treatment.

22.4.3 Application Rate (Dose)

Crystalline water proofing dry-shake material to be used under normal conditions should cover the application surface @ minimum 0.60 kg per sqm. Depending upon of the degree of abrasion resistance required. If the surface is to be used under heavy traffic conditions or where greater abrasion resistance is required, the manufacturers recommendation should be taken into account while applying crystalline water proofing dry-shake on the concrete surface.

22.4.4 Application Procedure

i.) After fresh concrete is placed. Consolidated and leveled, wait until concrete can be walked on leaving an indentation of 6-8mm.

ii.) Concrete should be free of bleed water and be able to support the weight of a power trowel. Float open the surface.

iiii. Immediately after floating open the surface and within one hour of finishing the concrete, apply one-half of the Integral Crystalline water proofing dry-shake material by hand or mechanical spreader, in one direction only. The Integral Crystalline water proofing dry-shake material must be spread evenly.

iii) As soon as the Integral Crystalline water proofing dry-shake material has absorbed moisture from the base slab, it should be power floated to the surface.

iii v. Immediately after power floating, apply remaining Integral Crystalline water proofing dry-shake material at right angles to the first application.

iv) Allow remaining Integral Crystalline water proofing dry-shake material to absorb moisture from the base slab and then power float the material into the surface. When concrete has hardened sufficiently, power trowel surface to the required finish.

22.4.5 Curing

Curing is important and shall begin as soon as final set has occurred but before surface starts to dry. Conventional moist curing procedures such as water spray, wet burlap or plastic covers may be used. Curing should continue for at least 48 hours. In hot dry sunny or windy conditions, it is advisable to use an evaporation retardant on the fresh concrete surface to prevent premature drying of the slab conditions due consultations should be made from the technical representatives of the manufacturer for specific instructions. In lieu or moist curing, concrete sealers and curing compounds may be used. In all cases, recommended guidelines for proper curing should be followed.

22.4.6 Recommended uses

Foundations /Rafts Slabs Water tank base slab Sewage and Water Treatment Plants below Grade Structures Warehouses Floors Traffic Bearing Surfaces Parking Structures

22.4.7 Precaution / Special Consideration

For the best results when applying Integral Crystalline dry shake materials, the air content of the concrete shall not exceed 3% (a high air content can make it difficult to achieve a proper application).

In hot, dry, or windy conditions, it is advisable to use an evaporation retardant on the fresh concrete surface to prevent premature drying of the slab. Chronic moving cracks or joints will require a suitable flexible sealant.

22.4.8 Storage / Self Life

Integral Crystalline dry shake must be stored dry at a minimum temperature of 7º C and its self life is one year when stored under proper conditions.

22.4.9 Measurement

The quantity of Integral Crystalline dry shake sprinkled area shall be calculated in sqm correct to two places of decimal. The Length & breadth/height of the Integral Crystalline dry shake sprinkled area shall be correct to a centimetre.

22.4.10 Rate

The rate shall include the cost of all the labour, material and equipments involved in all the operations described above.

22.5 Crystalline Water Proofing Mortar

22.5.1 Material

Crystalline water proofing mortar consists of Portland cement, specially treated quartz sand and a compound of active chemicals. The active chemicals react with moisture and the by-products of cement hydration to cause a catalytic reaction, which generates an insoluble integral crystalline complex. These crystalline complexes grow in the presence of water and block the capillaries of the concrete and minor shrinkage cracks, thus water proofing the concrete. Chemicals activation begins when the power is mixed with water and may take several days to completely block the capillaries, depending on ambient temperature and environmental conditions. It can be applied to the positive or negative water pressure sides of a structure.

22.5.2 Technical Specification/ Parameters:

1. It shall confirm to EN1504-3 (For structural repairs-R3, Compressive Strength > 25 Mpa) supplied from an approved manufacturing unit having CE approval confirming to EN 1504-3R3.

2. The product has no corrosion effect on reinforcement steel according to test norm DIN V 18998. The maximum chloride content lies less than 0.1% and maximum alkali content less than 9.3%.

3. Product must possess third party assurance, confirming that the product, when used in the concrete, will have no detrimental side effects in terms of Alkali Silica Reaction (ASR), corrosion of Steel Reinforcement etc.

4. The product performance shall not be affected by wear abrasion of the treated concrete surface and crystalline treated concrete shall not require protection layer.

5. Potable Water Compatibility: Nontoxic & suitable for use in potable water facilities- NSF listed as per ANSI 61 listing. (i) Installation of seal strips, reglets and coves at joints to assure water tightness

(ii) Patching and filling / sealing of routed out cracks.

(iii) Patching of tie holes and faulty construction joints.
(iv) Repairing of spalled and honeycombed area.

Note: The manufacturer shall produce relevant test certificates of Crystalline Mortar from reputed laboratories as per relevant codes as state above. Total quantity of the water proofing Crystalline Mortar material required shall be arranged only after obtaining the prior approval of the Engineer-in-Charge in writing. Materials shall be kept under double lock and key and proper account of water proofing compound used in the work shall be maintained. It shall be ensured that the consumption of the compound is as per specified requirements. Contractor shall associate himself with anyone of the specialist firms mentioned in approved list of specialized agencies for the work relating to the Water Proofing Treatment.

22.5.3 Recommended Uses

This material can be applied in conjunction with crystalline water proofing coating for:

22.5.4 Preparation of Surface

All surfaces to be patched, repaired or sealed with crystalline water proofing mortar must be clean and sound. Crack should be routed out to a U-shaped configuration, approximately 25 mm wide and a minimum of 25 mm deep. Tie holes should be roughened prior to filling. Spalled and honeycombed area must be thoroughly cleaned and chiseled back to sound concrete prior to repair. Remove all dirt, cement laitance, form release agents, curing compounds, paints, coating, etc. by means of wet or dry sandblasting, high pressure water jet or other approved mechanical means. Surfaces must be well moistened to a dull dampness at the time of application. The concrete should be damp with no wet sheen on the surface.

22.5.5 Mixing

i.) For routed cracks, coves and non-moving joints: Add water to crystalline water proofing mortar powder until a medium stiff, trowelable consistency reached. The texture of the mix should be pliable enough to be trowelled into the cracks with some pressure, but not so pliable that it would run out or sag out of the crack. Approximate mixing ratio (by volume) is 4.5 parts crystalline water proofing mortar powder to 1 part water. Alternatively, 450gm of crystalline water proofing mortar powder to 100 ml of water is to be mixed.
ii.) Tie holes and pointing applications: Add only a small amount of water. Mixed consistency should be that of "dry earth," holding a shape when squeezed in your hand but easily crumbled when pressed between fingers. Mix only as much material as can be used within 20 minutes.

22.5.6 Application Procedure

i.) For sealing cracks and faulty construction joints, routed out/making U-shape groove size 25x25mm and then priming the surface with integral crystalline slurry @0.05 kg per running meter and while the surface is tacky filled the cavity upto surface crystalline mortar @1.50 kg per running meter. Once crystalline mortar is touch dry then finally applying two coats of integral crystalline slurry @0.05 kg per running meter per coat.

ii.) For repairing spalled & honeycombed areas, prepared the surface and chiesel back upto sound concrete and then primed the area with integral crystalline slurry @0.70 kg per sqm. and while the surface is tacky repair and level the honeycomb area with crystalline mortar @ 22.70 kg per sqm. for an average thickness of 10mm. Once crystalline mortar is touch dry then finally two coats of integral crystalline slurry @ 0.70kg per sqm. per coat.

iii.) For patching of tie rod holes, prepared tie rod hole surface and primed the area with integral crystalline slurry @ 0.07 kg per sqm and while the surface is tacky repair and filled the tie rod holes with crystalline mortar @ 0.040 kg per hole. The crystalline mortar shall be tightly rodded into tie holes or packed tightly. For 25x25x25 mm hole, use 0.040 kg per hole to fill the tie hole.

22.5.7 Curing

Provide protection against extreme weather conditions, such as heavy rain or freezing conditions, during the setting period. Curing is not normally required except during hot, low humidity weather. In these conditions, a light mist of water approximately 25 hours after the repair is completed will help to ensure a controlled cure. In extreme dry heat, water misting may be carried out at required intervals more frequently.

22.5.8 Precaution / Special Consideration

Crystalline mortar shall not applied at temperatures below 40ºF (4ºC), to a frozen substrate or if temperatures will drop below freezing during the curing period (approximately 24 hours). This product is not recommended for use in expansion or construction joints. Crystalline mortar can be applied in (13 mm) layers not exceeding 2.5 inch (approximately 6.5 cm) to prevent shrinkage cracks in the mortar.

22.5.9 Storage / Self Life

Crystalline mortar shall be stored in a dry enclosed area off the ground at a minimum temperature of 7ºC. Self life when stored in proper conditions in unopened, undamaged packaging is 12 months.

22.5.10 Measurement

Faulty construction joint will be measured by measuring the length in running meter correct to a centimetre. Repair of honeycombed area will be measured in square meter correct to two places of decimal by measuring the length and width of treated area correct to a centimetre. Repair of tie rod holes will be measured in numbers.

22.5.11 Rate

The rate includes the cost of all the labour and material involved in all the operations described above.

22.6 Swellable Type Water Stop Tape

22.6.1 General

Swellable type water stop tape of size 19 mm x 25 mm is a unique sealing compound designed to expand rapidly when exposed to moisture, making it a self-healing joint material for construction joint applications /treatment.

22.6.2 Material

This is a swellable type sealing compound which expand rapidly after coming in to contact or exposed to moisture. This acts as a self healing material and is used for applications in construction joints.

22.6.3 Physical Properties

i.) Specific gravity (ASTM D71): 1.35 ± 0.05 (ASTM D-71)

ii.) Volatile matter: 1% maximum (ASTM D-6)

iii.) Penetration, 150g cone at 25°C, 5sec: 40 ± 5mm (ASTM D-217)

iv.) Rate of Rapid Expansion:

(a) Fresh Water Exposure: 24 Hours-140%, 48 Hours- 175%, 72 Hours-190% & 120 Hours-210%
(b) Salt Water Exposure: 24 Hours-7%, 48 Hours-12%, 72 Hours-14%, 120 Hours-18%
The water stop material should meet the requirements to EU REACH Regulation (EC) No 1907/2006.

22.6.4 Physical Properties of Swellable Waterstop Primer

i.) % Solid: Min 20%

ii.) Flash Point: 93 deg C

iii.) Dry Time: 25 deg C: 10 min

iv.) Dry Time: 4 deg C: 60 min

22.6.5 Recommended Uses

Typical applications for swellable type waterstop tape include building foundations, slabs, retaining walls, storage tanks, and similar non-moving cold construction joints

22.6.6 Application

i.) The entire surface length where the water stop is to be applied is cleaned thoroughly by using blower and brush. Apply one coat of required primer throughout the length of the joint @ 3.78 litre per 240 running metre. Allow the primer to dry for 10 to 15 minutes at the temperature of 25°C. This should be allowed to dry for some longer time in the areas where the temperature are low.

ii.) By using the heel of the hand and moderate pressure, press the self expanding joint material firmly into the position on the structure on the entire area which has been primed and dried. Make sure that the product has bonded with the primed area.

iii.) Where required, splice ends to from a continuous, uninterrupted seal. For best results, cut each end at opposite 45deg. angles and tightly butt ends together. DO NOR OVERLAP ENDS. Gently knead the spliced ends creating an uninterrupted seal.

iv.) Peel the protective covering from the exposed side of the installed expandable joints sealing compound.

v.) Pour the matting structural member in position.

Notes: Always use swellable type waterstop primer to avoid displacement of the swellable type waterstop tape during concrete pouring. It may be necessary to utilize masonry nails or other mechanical means to hold the sealant in place on vertical surfaces. Place swellable type waterstop tape so that it is not closer than 5 cm away from the outer surface of poured structure. If a Keyway is utilized, place the swellable type waterstop tape into the bottom of the formed Keyway area.

22.6.7 Precaution / Special Consideration

Always use swellable type waterstop primer to ensure tight adhesion and to aid in preventing swellable type waterstop tape from moving during the concrete pour. For vertical surfaces, nails may be used to hold the product in place in conjunction with swellable type waterstop primer. Swellable type waterstop tape shall be used at a minimum depth of 50 mm inside the concrete. When used on pipes and other structural penetrations, swellable type waterstop tape shall be cut to measured length and placed around the penetration with ends butted. In all cases, swellable type

waterstop tape shall be in direct contact with the substrate along the entire length of the installation. Swellable type waterstop tape is not an expansion joint sealant and only suitable for non-moving concrete joints. Swellable type waterstop tape should not be installed in standing water or on frozen or icy surfaces

22.6.8 Storage / Self Life

When stored in a dry enclosed area off the ground at a minimum temperature of 45°F (7°C) in unopened, undamaged cartons, its self life is unlimited.

22.6.9 Measurement

The measurement shall be taken by measuring the length of swellable type waterstop tape in running metre correct to a centimeter. Length shall be measured in metres correct to two places of decimal. Measurement shall be made in metres.

22.6.10 Rate

The rate shall include the cost of all labour and materials involved in all the operations described above.

22.6.11 Measurement

The quantity of Integral Crystalline Slurry coat will be measured in sqm. The Length & breadth/height of the coated area by Integral Crystalline Sully shall be measured correct to a centimeter. The area shall be calculated in sqm correct to two place of decimal.

22.6.12 Rate

The rate shall include the cost of all the labour, material and equipments involved in all the operations described above.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.1		Water proofing on horizontal surface, kota stone	Providing and laying integral cement based treatment for water proofing on horizontal surface at all depth below ground level for under ground structures as directed by Engineer-in-Charge consisting of: (i) Ist layer of 22mm to 25mm thick approved and specified rough stone slab over a 25mm thick base of cement mortar 1:3 (1 cement: 3 sand) mixed with water proofing compound conforming to IS: 2645 in the recommended proportion over the levelling course (levelling course to be paid separately). Joints sealed and grouted with cement slurry mixed with water proofing compound. (ii) 2nd layer of 25mm thick cement mortar 1:3 (1 cement: 3 sand) mixed with water proofing compound in recommended proportions. (iii) Finishing top with stone aggregate of 10mm to 12mm nominal size spreading @ 8 cudm/sqm thoroughly embedded in the 2nd layer. (with 5 years service guarantee) Using rough Kota stone.	sqm	916.00
22.2		Water proofing on vertical surface, kota stone	Providing and laying integral cement based treatment for water proofing on the vertical surface by fixing specified stone slab 22 mm to 25mm thick with cement slurry mixed with water proofing compound conforming to IS:2645 in recommended proportions with a gap of 20mm (minimum) between stone slabs and the receiving surfaces and filling the gaps with neat cement slurry mixed with water proofing compound and finishing the exterior of stone slab with cement mortar 1:3 (1 cement : 3 sand) 20 mm thick with neat cement punning mixed with water proofing compound in recommended proportion complete at all levels and as directed by Engineer-in-charge (with 5 years service guarantee) Using rough Kota stone.	sqm	1013.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.3		Water proofing treatment four course blown bitumen treatment	Providing and laying water proofing treatment to vertical and horizontal surfaces of depressed portions of W.C., kitchen and the like consisting of : **a)** Ist course of applying cement slurry @ 4.4 Kg/sqm mixed with water proofing compound conforming to IS 2645 in recommended proportions including roundingoff junction of vertical and horizontal surface. **b)** IInd course of 20mm cement plaster 1:3 (1 cement: 3 sand) mixed with water proofing compound in recommended proportion including rounding off junction of vertical and horizontal surface. **c)** IIIrd course of applying blown or residual bitumen applied hot at 1.7 Kg. per sqm of area. **d)** IVth course of 400 micron thick PVC sheet. (Overlaps at joints of PVC sheet should be 100 mm wide and pasted to each other with bitumen @ 1.7 Kg/sqm.) (with 5 years service guarantee)	sqm	440.00
22.4		PVC water stops for construction/ expansion joints	Providing and Placing in position suitable PVC water stops conforming to IS:12200 for construction/ expansion joints between two RCC members and fixed to the reinforcement with binding wire before pouring concrete etc. complete (with 5 years service guarantee)		
	22.4.1	225mm	Serrated with central bulb (225mm wide, 8-11mm thick).	metre	551.00
	22.4.2	180 mm	Dumb bell with central bulb (180mm wide, 8mm thick).	metre	499.00
	22.4.3	320 mm	Kickers (320mm wide, 5mm thick).	metre	522.00
22.5		Water proofing treatment two course cement mortar	Providing and laying water proofing treatment in sunken portion of WCs,bathroom etc., by applying cement slurry mixed with water proofing cement compound consisting of applying : **a)** First layer of slurry of cement @ 0.488 kg/sqm mixed with water proofing cement compound @ 0.253 kg/sqm. This layer will be allowed to air cure for 4 hours. **b)** Second layer of slurry of cement @ 0.242 kg/sqm mixed with water proofing cement	sqm	207.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			compound @ 0.126 kg/sqm. This layer will be allowed to air cure for 4 hours followed with water curing for 48 hours. The rate includes preparation of surface, treatment andsealing of all joints, corners, junctions of pipes and masonry with polymer mixedslurry.		
22.6		Water proofing treatment Fibre glass cloth and cement slurry	Providing and laying water proofing treatment on roofs of slabs by applying cement slurry mixed with water proofing cement compound consisting of applying : **a)** After surface preparation, first layer of slurry of cement @ 0.488 kg/sqm mixed with water proofing cement compound @ 0.253 kg/sqm. **b)** Laying second layer of Fibre glass cloth when the first layer is still green. Overlaps of joints of fibre cloth should not be less than 10 cm. **c)** Third layer of 1.5 mm thickness consisting of slurry of cement @ 1.289 kg/sqm mixed with water proofing cement compound @ 0.670 kg/sqm and sand @ 1.289 kg/sqm. This will be allowed to air cure for 4 hours followed by water curing for 48 hours. The entire treatment will be taken upto 30cm on parapet wall and tucked into groove in parapet all around.	sqm	338.00
22.7		Water proofing treatment brick coba	Providing and laying integral cement based water proofing treatment including preparation of surface as required for treatment of roofs, balconies, terraces etc consisting of following operations: **a)** Applying a slurry coat of neat cement using 2.75 kg/sqm. of cement admixed with water proofing compound conforming to IS. 2645 and approved by Engineer-in-charge over the RCC slab including adjoining walls upto 300mm height including cleaning the surface before treatment. **b)** Laying brick bats with mortar usingbroken bricks/brick bats 25 mm to 115mmsize with 50% of cement mortar 1:5 (1 cement : 5 sand) admixed with water proofing compound conforming to IS : 2645 and approved by Engineer-in-Charge over 20 mm thick layer of cement mortar of mix 1:5(1 cement :5 sand) admixed with water	sqm	885.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			proofing compound conforming to IS : 2645 and approved by Engineer-in-Charge to required slope and treating similarly the adjoining walls upto 300 mm height including rounding of junctions of walls and slabs. **c)** After two days of proper curing applying a second coat of cement slurry using 2.75kg/ sqm of cement admixed with water proofing compound conforming to IS : 2645 and approved by Engineer-in-Charge. **d)** Finishing the surface with 20 mm thick jointless cement mortar of mix 1:4 (1 cement: 4 sand) admixed with water proofing compound conforming to IS: 2645 and approved by Engineer-in-charge including laying glass fibre cloth of approved quality in top layer of plaster and finally finishing the surface with trowel with neat cement slurry and making pattern of 300x300 mm square 3mm deep. **e)** The whole terrace so finished shall be flooded with water for a minimum period of two weeks for curing and for final test. All above operations to be done in order and as directed and specified by the Engineer-in-Charge: (with 5 years service guarantee) With average thickness of 120mm and minimum thickness at khurra as 65 mm.		
22.8		Water proofing Four courses hessian base bitumen felt	Providing and laying four courses water proofing treatment with bitumen felt over roofs consisting of first and third courses of blown bitumen 85/25 or 90/15 conforming to IS : 702 applied hot @ 1.45 Kg per square metre of area for each course, second course of roofing felt type 3 grade-I (hessian based self finished bitumen felt) and fourth and final course of stone grit 6mm and down size or pea-sized gravel spread at 6 cubic decimeter per square metre including preparation of surface but excluding grading complete with : (with five years service gurantee). Bitumen felt (hessian base) type 3 grade I conforming to IS: 1322.	sqm	371.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.9		Water proofing hessian based bitumen felt	Providing and laying six courses water proofing treatment with bitumen felt over roofs consisting of first, third and fifth course of blown bitumen 85/25 or 90/15 conforming to IS : 702 applied hot @ 1.45, 1.20 and 1.45 Kg per square metre of area respectively, second and fourth courses of roofing felt type 3 grade I conforming to IS : 1322 (Hessian based self finished bitumen felt) conforming to IS : 1322 and sixth and final course ofstone grit 6 mm and down size or pea sized gravel spread at 6 cubic dm per sqm including preparation of surface butexcluding grading, complete. (With five yearsservice gurantee).	sqm	597.00
22.10		Water proofing Six courses fibre based grade-I bitumen felt	Providing and laying six courses water proofing treatment with bitumen felt over roofs consisting of first, third and fifthcourses of blown or / and residual bitumen applied hot at 1.45, 1.20 and 1.70 kg per square metre of area respectively, second and fourth courses of roofing felt type 2 grade I (fibre base self finished bitumen felt) six and final courses of stone grit 6 mm and down size or pea sized gravel spread at 6cu.dm per sqm including preparation of surface, excluding grading, complete. (With five yearsservice gurantee).	sqm	585.00
22.11		Water proofing Six courses fibre base grade-II bitumen felt	Providing and laying six courses water proofing treatment with bitumen felt over roofs consisting of first, third and fifthcourses of blow or/ and residual bitumen applied hot at 1.45, 1.20 and 1.70 kg per square metre of area respectively, second and fourth courses of roofing felt type 2 grade II (glass fibre base self finished bitumen felt) and sixth and final course of stone grit 6 mm and down size or pea sized gravel spread at 6 cubic dm per sqm including preparation of surface but excluding grading, complete. (With five years service gurantee)	sqm	586.00
22.12		Deduct for omitting water proofing treatment final course 6mm grit	Deduct for omitting in water proofing treatment final course of spreading stone grit 6mm down size or pea sized gravel at 6 cudm per sqm.	sqm	10.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.13		Water proofing Bituminous solution primer on roof and wall	Supplying and applying bituminous solution primer on roof and or wall surface at 0.24 litre per sqm.	sqm	23.00
22.14		Roof grading	Grading roof for receiving water proofing treatment with		
	22.14.1	C.C. 1:2:4	Cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 20 mm nominal size)	cum	6569.00
	22.14.2	CM 1:3	Cement mortar 1:3 (1 cement : 3 sand)	cum	9308.00
	22.14.3	CM 1:4	Cement mortar 1:4 (1cement : 4 sand)	cum	8437.00
22.15		Water proofing treatment six course (1.5mm thick APP membrane) bitumen based	Providing and laying insitu six course water proofing treatment with APP (Atactic poly-propylene) modified Polymeric membrane over roof consisting of first coat of bitumen primer @ 0.40Kg per sqm, 2nd, 4th and 6th courses of bonding material @ 1.20 Kg/sqm, which shall consist of blown type bitumen of grade 85/25 conforming to IS : 702, 3rd and 5th layers of roofing membrane APP modified Polymeric membrane 1.5mm thick of 2.25 Kg/sqm weight consisting of five layers prefabricated with centre core as 20micron HMHDPE film sandwiched on both sides with polymeric mix and the polymeric mix is protected on both side with 20 micron HMHDPE film. (with 5 years service guarantee)	sqm	504.00
22.16		Water proofing treatment four course (2.0mm thick APP membrane) bitumen based	Providing and laying in situ four course water proofing treatment with APP (Atactic Polypropylene) modified Polymeric membrane over roof consisting of first coat of bitumen primer @ 0.40Kg per sqm, 2nd and 4th courses of bonding material @ 1.20 Kg/sqm, which shall consist of blown type bitumen of grade 85/25 conforming to IS : 702, 3rd layer of roofing membrane APP modified Polymeric membrane 2.0mm thick of 3.00 Kg/sqm weight consisting of five layers prefabricated with centre core as 100micron HMHDPE film sandwiched on both sides with polymeric mix and the polymeric mix is protected on both side with 20micron HMHDPE film. (with 5 years service guarantee).	sqm	368.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.17		Water proofing treatment six course (2.0mm thick APP membrane) bitumen based	Providing and laying in situ six course water proofing treatment with APP (Atactic Polypropylene) modified Polymeric membrane over roof consisting of first coat of bitumen primer @ 0.40Kg per sqm, 2nd, 4th and 6th courses of bonding material @ 1.20 Kg/sqm, which shall consist of blown type bitumen of grade 85/25 conforming to IS : 702, 3rd and 5th layers of roofing membrane APP modified Polymeric membrane 2.0mm thick of 3.00 Kg/sqm weight consisting of five layers prefabricated with centre core as 100 micron HMHDPE film sandwiched on both sides with polymeric mix and the polymeric mix is protected on both side with 20micron HMHDPE film. (with five years service gurantee).	sqm	606.00
22.18		Water proofing treatment five layer (2.0mm thick APP membrane) reinforced with glass fibre mesh bitumen based	Providing and fixing APP (Atactic Polypropylene Polymer) modified prefabricated five layer 2mm thick water proofing membrane for corrugated roof sheets, black finished reinforced with glass fibre matt consisting of a coat of bitumen primer for bitumen membrane @ 0.40 ltr/sq. mtr. by the same membrance manufacture of density at 25°C, 0.87 - 0.89 kg/ ltr and viscocity 70 - 160 cps. Over the primer coat the layer of membrane shall be laid using Butane torch and sealing all joints etc., and preparing the surface complete. The vital physical and chemical parameters of the membrane shall be as under: Joint strength in longitudinal and transverse direction at 23°C as 350/300 N/ 5cm. Tear strength in longitudinal and transverse direction as 60/80N. Softening point of membrane notless than 150°C. Cold flexibility shall be upto -2°C when tested in accordance with ASTM, D - 5147. The laying of membrane shall be got done through the authorised applicator of the manufacture of membrane. (With five years service gurantee).	sqm	389.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.19		Water proofing treatment five layers (3mm thick APP membrane, glass fibre mesh) bitumen based	Providing and laying APP (Atactic Polypropylene Polymer) modified prefabricated five layer, 3mm thick water proofing membrane, black finished reinforced with glass fibre matt consisting of a coat of bitumen primer for bitumen membrane @ 0.40 ltr/sqm. by the same membrane manufactured of density at 25°C, 0.87 - 0.89 kg/ltr and viscocity 70 - 160 cps. Over the primer coat the layer of membrane shall be laid using butane torch and sealing all joints etc., and preparing the surface complete. The vital physical and chemical parameters of the membrane shall be as under:Joint strength in longitudinal and transverse direction at 23°C as 350/300 N/5cm. Tear strength in longitudinal and transverse direction as 60/80N. Softening point of membrane not less than 150°C. Cold flexibility shall be upto -2°C when tested in accordance with ASTM, D - 5147. The laying of membrane shall be got done through the authorised applicator of the manufacturer of membrane : (with 5 years service guarantee)	sqm	440.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.20		Water proofing treatment five layers (3mm thick APP reinforced with non-woven polyester matt) bitumen based	Providing and laying APP (Atactic Polypropylene Polymer) modified prefabricated five layer 3mm thick water proofing membrane, black finished reinforced with non-woven polyester matt consisting of a coat of bitumen primer for bitumen membrane @ 0.40 ltr/sqm. by the same membrane manufacture of density at 25°C, 0.87-0.89 kg/ltr and viscosity 70-160 cps. Over the primer coat the layer of membrane shall be laid using Butane Torch and sealing all joints etc., and preparing the surface complete. The vital physical and chemical parameters of the membrane shall be as under: Joint strength in longitudinal andtransverse direction at 23°C as 650/450N/5cm. Tear strength in longitudinal and transverse direction as 300/250N. Softening point of membrane not less than150°C. Cold flexibility shall be upto -2°C when tested in accordance with ASTM, D - 5147. The laying of membrane shall be got done through the authorised applicator of the manufacturer of membrane: (with 5 years service guarantee).	sqm	510.00
22.21		Extra for protective geotextile	Extra for covering top of membrane with Geotextile, 120gsm non woven, 100% polyester of thickness 1 to 1.25mm bonded to the membrane with intermittent touch by heating the membrane by Butane Torch as per manufactures recommendation. [for item No. 22.18 to 22.20]	sqm	59.00
22.22		Water proofing with fibre glass mesh and two component acrylic resin	Water proofing treatment over Roof, Wall, Chajjas,, Balcony with fibre glass mesh and Sealer coat at leakage/ seepage area consisting of **(i)** Surface preparation roughening of surface, opening cracks in 'V' groove in size of 5mmX10mm (WXD), filling of cracks with putty with laying fiber glass mesh, Cleaning of surface by scrubbing with steel wire/Nylon brush. Removing all dust particles and washing with adequate water to clean completely. **(ii)** Providing and applying 1st coat of fibre glass mesh (minimum thicknes 70-80 micron) and two component flexible waterproof and protective modified mortar, dry powder 80%	sqm	465.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			chemical 20% acrylic resin having 30% solid contents) with mix proportion of part 1 and 2 with proper mixing with laying of fiber glass mesh (of weaving size of 10X10 yarn/inch duly coated with alkaline resistant polymer). Allow the coating to air cure for minimum 2 Hrs. After two hours of 1st coat applying 2nd coat (minimum thickness 100 micron) of fibre glass mesh of with mix proporation of part 1 and 2 with proper mixing. Allow the coating to air cure for minimum 2 hrs. Consumption of two component should beacrylic resin @17.90 kg for 10 sqm area or as per manufacture instructions. (After two coats of diamond shield, surface is to be cured properly minimum for 4 hrs. before application of third coat). **(iii)** Providing and applying 1st coat of sealer coat 50-60 micron (single component Hight build elastomeric, fixable pure acrylic waterproofing membrane having solidcontent of 65%) and allow it to air cure for 4Hrs. minimum. After two hours minimum of1st coat applying 2nd and finals coat 120-140 micron of sealer coat and allow it to air cure for 4 Hrs mimimum. Consumption of Sealer coat should be @ 5.40 kg per 10 sqm area. **(iv)** The treated area should be cure with water for 48 hrs. by flooding the surface. **(v)** The final appearance of the coating willbe milky white. All above operations to be done in order.(five years guarantee)		
22.23		Waterproofing treatment with rubberized elastomeric membrane	Providing water proofing treatment over Roof, Wall, Chhajjas, Balcony with Rubberized elastomeric cold applied liquid waterproofing membrane blend of special resin and rubber with more than 1300% elongation (confirming ASTM-D2370-98), at leakage/seepage area consisting of the following operation. **(i)** Removal of damaged waterproofing treatment if any like bituminous membrane / plaster/ brick bat koba/screeding etc. **(ii)** Surface preparation: Roughening of surface by scrubbing with steel wire/ Nylon brush/ Removing all dust particles and washing with adequate water to clean completely, opening of cracks in 'V' groove	sqm	544.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			in size of 5mm x 10m (WxD), filling of cracks with elastomeric creak filler/PU sealent repair of potholes and holor makingon roof surface by polymer mortar with adding SBR bonding agent. **(iii)** Providing and applying 1st coating of rubberized elastomeric liquid membrane as primer (ratio 1:2 1 part liquid membrane and 2 part water) by brush/ Roller/ Spray having minimum thickness of 70-80 micron. Allow the coating to set in natural air for minimum 2 hrs, then put reinforcement by laying 45gsm fibre glass mesh and applying 2nd coat of rubberized elastomeric liquid membrane (Ratio: 1 part membrane and 20% water) with minimum thickness 100-120 micron. Allow the coating to set in natural airfor minimum 12 hrs. **(iv)** Providing and applying 3rd coat of rubberized elastomeric liquid membrane (Ratio: 1 part membrane and 10% water). Total DFT (Dry film thickness) thickness after 3rd coat must be 280-300 micron and consumption in three coats @ 6.5kg for 10sqm. **(v)** Providing and applying 4th and 5th coat of High quality waterproof membrane in white colour, single component special synthetic resin based flexible thixotropicwhite color coating with excellent UV resistance (SRI - 144) with more than 300% elongation (confirming ASTM D 412-98) in ratio: 1 part membrane and 10% water thickness of 200 micron DFT in two coats, with average consumption @ 5kg per 10 sqm. After application of fourth coat let it air dry for minimum 3 hrs before application of fifth coat. **(vi)** The treated area must be tested for water leakage/seepage by flooding with water for 48 hrs **(vi)** Application of membrane coating shall be got done by authorized applicator of manufacturer. **(vii)** 5 years of service guarantee to be provided. All above operations to be done in order as per direction on Engineer-in-Charge.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.24		Fibre reinforced liquid waterproofing	Supplying and applying fibre reinforced, UV resistant elastomeric liquid applied waterproofing & Heat reflective coating coating with PU Hybrid polymers & acrylic fibers. The coating shall resist 10 bars of water pressure as per DIN 1048 standards, with elongation of 350% as per ASTM D412 & crack bridging of upto 2.5mm conforming to ASNZ standard. The operation shall be carried out after scrapping and properlycleaning the surface to remove loose particles with wire brushes, complete in all respect as recommended by the manufacturer's specification and as per the direction of Engineer-in-Charge.	sqm	365.00
22.25		Acrylic polymer modified cementitious waterproofing	Supplying and applying two component elastomeric acrylic polymer modified cementitious waterproofing coating, having 80% elongation as per ASTM D412, resisting 5 bars hyrostatic pressure as per DIN 1048, reduction to rapid chloride shall be 70 as per DIN 1048, crack bridging upto 1.2mm as per ASTM 4548.5. The coating shall have food grade certificate from laboratories likeCFTRI. The mixing ratio of the coating shall be 2 parts of powder & 1 part of liquid without any water addition in the mixing, etc., complete as per manufacturers specifications and as per direction ofEngineer-in-Charge.	sqm	195.00
22.26		Polyurethane liquid water proofing membrane	Providing and applying UV Stable polyurethane liquid water proofing membrane having Sun Reflectivity Index (SRI) of 106on top of concrete roof / Retainning wall /Old layer of bituminous surfaces / Polyurethane foam and metal surfaces in three or more coats @20kg/ 10 sqm consisting of a coat of two component epoxy primer in ratio of 1:3(one part of resin and 3 parts of hardner) at a consumption of 2kg/10sqm. the vital physical and chemical parameters shall be as under : Min Elongation	sqm	750.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			at break of 600% as per ASTM D412 and tensile strength of 3.0 N/mm^2 as per ASTM D412. The operation shall be carried out after scrapping and properly cleaning the surface to remove loose particles with wire brushes, complete in all respect as recommended by the manufacturer's specification and direction of Engineer-in-Charge.		
22.27		Waterproofing of RCC base slab with 1.2 mm thick HDPE membrane	Supplying & installing waterproofing treatment by using 1.2 mm thick HDPE membrane before casting of the base RCCslab. The pre-applied membrane gets bonded to the underneath of the poured concrete used as base slab. The membrane shall consist of a thick HDPE film, a highly aggressive pressure sensitive adhesive and weather resisting coating which bonds integrally to poured concrete of base slab. The membrane shall conforms to basement waterproofing protection to grades 2, 3 & 4 as defined in BS 8102:2009. The waterproofing membrane shall have following minimum properties: **a)** Resist hydrostatic pressure of up to 70 m head of water as per ASTM D5385; **b)** Puncture Resistance of 950 N (as per ASTM E154) **c)** Elongation of >400 % (as per ASTM D412) **d)** Peel Strength of concrete: >1500 N/m (as per ASTM D903) **e)** Tensile strength of 28.2 MPA (as per ASTM D412).	sqm	1738.00
22.28		Waterproofing of RCC base slab and vertical wall with 2 mm thick PVC membrane	Providing and applying 2.0 mm thick twin colour flexible PVC waterproofingmembrane. Flexibility under lowtemperatures as per EN 495-5 should be < - 20degC. On all horizontal surfaces Geotextile300 Gsm to be laid over PCC, than laying PVC membrane over Geotextile and to be sealed with hot air welding method at all joints and over PVC membrane provide 300 Gsm Geotextile Then protection screed of	sqm	1670.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			suitable thickness. On all vertical surfaces Geotextile 300 Gsm to be laid along with Vertical wall, than laying PVC membrane over Geotextile and to be sealed with hot air welding method at all joints and over PVC membrane providing protection layer of Drain Board HDPE Dimple sheet before backfilling of soil.		
22.29		Waterproofing cum heat reflective roof coating	Providing and applying waterproofing & Heat reflecting roof coating with 100% acrylic elastomeric polymers. Contains 30% Volume solid, tensile strength 8 kg/cm², crack bridging, VOC 30 ± 5 gm/ltr upto 2 mm. then applying two coats with forced coverage of 20 sqft/ltr/two coats. The operation shall be carried out after scrapping & propely clening the surface to remove loose particles with wire brush/Jet washer. Apply 1 coat @1 ltr/10 sqm & 3 undiluted coats @ 8 ltr/10 sqm with 4 hrs interval of each coat.The work shall be carried out as recommended by the manufacturer's specification & the direction of the Engineer-in-Charge.	sqm	342.00
22.30		Crack repairing upto 5 MM crack	Providing & applying crack fill paste for crack filling up to 5 mm cracks.Widen the crack by putty knife in V/U shape & apply1st priming coat of crack fill paste dilutedwith water in 1:1 proportion over the crack bybrush. Apply 2nd coat with filling of crack fill paste when the surface is tacky but not dried completely with putty knife & spatula.After 4 hours of application 2nd coat apply 3rd coat of Crack fill paste. The work shall be carried out as recommended by the manufacturer's specification & the direction of the Engineer-in-Charge	metre	55.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.31		Waterproofing roof by waterbase polyurethane elastomeric liquid applied membrane	Providing & applying single pack component, solvent-free, cold applied, Polyurethane modified, elastomeric liquid applied waterproofing membrane- PU coat with excellent UV resistant capacity, Non-toxic with high SRI Index, eco friendly, low VOC & complies with the requirement as laid outin ASTMD 6083-97a.It can be applied on cementious substrates,brick & stone, Bituminous coatings, metals, wooden substrates, paint/coatings with application of two coats without dilution along with a self Priming Coat-50% dilution with water. Surface should be cleaned properly with required tools & crack repairing by cement mortar mixed with latex for waterproofing application. Apply in 3 coats (1st priming coat @ 2 kg/10 Sqm and 2nd & 3rd coat @ 4 kg/10 Sqm of each coat) for complete waterproofing with Interval of 4 hours between every coat. The material shall meet the requirement as specified in ASTM D2370-98 of elongation at break of film>350% with geotextile fleece =50% approx.The work shall be carried out as recommended by the manufacturer's specification & the direction of the Engineer-in-Charge. The product performance shall carry guaranteed for 5 year against any leakage.		
	22.31.1	With Geotextile fleece	For Horizontal surface (with Geotextile fleece) two coats without dilution & 1 Self priming coat- 50% dilution with water @ 10 kg/m^2	sqm	459.00
	22.31.2	Without Geotextile fleece	For Horizontal surface (without Geotextile fleece) two coats without dilution & 1 Self priming coat- 50% dilution with water @ 10 kg/m^2	sqm	390.00
22.32		Waterproofing RCC base slab and wall with Integral crystalline admixture	Providing and mixing integral crystalline admixture for water proofing treatment to RCC structures like basement raft, retaining walls, reservior, sewage & water treatment plant, tunnels / subway and bridge deck etc. at the time of transporting of concrete into the drum of the ready-mix truck, using integral crystalline admixture @ 0.80% (minimum) to	kg	239.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			the weight of cement content per cubic meter of concrete) or higher as recommended by the manufacturer's specification in reinforced cement concrete at site of work. The material shall meet the requirements as specified in ACI-212-3R- 2010 i.e. by reducing permeability of concrete by more than 90%, Compared with control concrete as per DIN 1048 and resistant to 16 bar hydrostatic pressure. The crystalline admixture shall be capable of self-healing of cracks up to a width of 0.50mm. The work shall be carried out all complete as per specification and the direction of the Engineer-in-Charge. The product performance shall carry guarantee for 10 years against any leakage.		
22.33		Waterproofing with Integral crystalline slurry of hydrophilic nature	Providing and applying integral crystalline slurry of hydrophilic in nature for waterproofing treatment to the RCCstructures like retaining walls of the basement, water tanks, roof slabs, podiums, reservior, sewage & water treatment plant,tunnels / subway and bridge deck etc., prepared by mixing in the ratio of 5 : 2 (5 parts integral crystalline powder: 2 parts water by weight) for vertical surfaces and 3 :1 (3 parts integral crystalline slurry : 1 part water by weight) for horizontal surfaces and applying the same from positive (internal) side with the help of synthetic fiber brush. The material shall meet the requirements as specified in ACI- 212-3R-2010 i.e by reducing permeability of concrete by more than 90% compared with control concrete as per DIN 1048 and resistant to 16 bar hydrostatic pressure on positive side. The crystalline slurry shall be capable of self-healing of cracks up to a width of 0.50mm. The work shall be carried out all complete as per specification and the direction of the engineer-in-charge. The product performance shall carry guarantee for 10 years against any leakage.		
	22.33.1	Vertical surface	For vertical surface two coats @ 0.70 kg per sqm	sqm	412.00
	22.33.2	Horizontal surface	For horizontal surface one coat @1.10 kg per sqm.	sqm	319.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
22.34		Waterproofing horizontal surface by Integral crystalline (dry shake) of hydrophilic nature	Providing and applying integral crystalline (dry shake) of hydrophilic in nature for waterproofing treatment to the RCCstructures like basement raft, foundation slab, sewage & water treatment plant slab, warehouses floor, parking structures and water tank base slab etc. sprinkled @ 0.60 kg per sqm or higher as recommended by the manufacturers specification over the leanconcrete of above cited structures. The material shall meet the requirements as specified in ACI-212-3R-2010 i.e. by reducing permeability of concrete by morethan 85% compared control concrete as per DIN 1048 and resistant to 16 bar hydrostatic pressure on negative side. The crystalline dryshake shall be capable of self-healing of cracks up to a width of 0.50mm. The work shall be carried out all complete as per specification and the direction of theEngineer-in-charge. The product Performanceshall carry guarantee for 10 years against any leakage.	sqm	249.00
22.35		Crystalline mortar 4:5 for the treatment of faulty construction joints/holes	Providing and applying crystalline mortar by mixing in the ratio of 4.5 : 1 (4.5 parts crystalline mortar : 1 part water) for the treatment of faulty construction joints, cracks, tie rod holes and spalled & honeycombed surface of RCC underground structures like basement, water tanks, bridge deck etc. to ensure water tightness. The crystallie mortar shall conform to the EN 1504-3 having compressive strength Class R4 > 45 MPa and adhesive bond strength Class R3 > 1.5 MPa. The work shall be carried out all complete as per specification and the direction of the Engineer-In-Charge.The product performance shall carry guarantee for 10 years against any leakage.		
	22.35.1	Construction joints	For sealing cracks and faulty construction joints, routed out/making U-shape groove size 25x25mm and then primed the area with integral crystalline slurry @0.05kg/ running metre and while the surface is tacky filled the groove upto surface with crystalline mortar @1.50kg/ running metre. Once crystalline mortar is touch dry then finally applied two	metre	449.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			coats of integral crystalline slurry @0.05kg/running metre per coat.		
	22.35.2	Tie rod hole	For patching of tie rod holes, prepared tie rod hole surface and then primed the area with integral crystalline slurry @ 0.070 kg/sqm and while the surface is tacky repair and then filled the tie rod holes with crystalline mortar @ 0.040 kg per hole. The crystalline mortar should be tightly rodded into tie rod holes or packed tightly (For 25x25x25 mm tie rod hole, use 0.040 kg to fill the hole)	each hole	16.00
22.36		Waterproofing of RCC base and wall using swellable type water stop tape	Providing and applying of swellable type water stop tape, 19mm x 25mm thick in linear meter (expansive nature) for construction joints treatment of RCC structure such as raft slab, retaining walls,water storage tank and at the junctions of raft slab with the retaining walls etc. Aftercleaning the surface, one coat of required primer for swellable water stop tape shall be applied throughout the length of the joint @3.78 litre per 240 running meter. Over the primed surface swellable type water stop tapeshall be placed. The work shall be carried out all complete as per specification and the direction of the Engineer-In-Charge. Theproduct performance shall carry guarantee for 10 years against any leakage.	metre	426.00
22.37		Waterproofing treatment of bathroom or tiled balcony area using waterproofing sheets	Providing and applying waterproofing treatment in bathroom or tiled balcony area without breaking or removing tiles, by using waterproofing sheets as per manufacturer's data sheet instructions. Permitting use of bathroom/balcony after 24 hours of application of treatement, Complete with materials and labour and gurantee of leakage for a period of 5 year.	sqm	653.00

CHAPTER – 23.0

MODULAR OPERATION THEATRE

PREAMBLE

MODULAR OPERATION THEATRE
(Chapter No. – 23)

23.1 Wall and ceiling panels

The main requirements for any operation theatre is wall and ceiling system that they should be impact resistant, show resistance to solvents & chemicals, prevent colonizing of micro-organisms or neutralize micro-organisms which come it contact with the wall surface. The surface should be stable prevent cracking and movement, scrub able-amenable to cleaning & completely sealed finish, biological attack resistant & have hygiene finish & lastly have hydrothermal performance. The exposed surface of panels should be easily washable, scratch free, dent free, flame resistant, scratch resistant as per BS 3900 Part E2 standard.

All wall and ceiling mounted equipment should be flush and sealed into theatre wall/ceiling by means of durable sterile jointing system. The wall and ceiling panel design and construction should allow for the installation and support of all equipments with provision of openings on back side required for repair and maintenance without affection rigidity and strength. In case required one should be able to easily remove the panel for service requirement.

The total distance between the inside and outside surfaces of the theatre should vary to suit the architect's layout.

The cavity between the inner and outer surface of walls and ceiling should be left with minimum obstruction for the possible addition of services, pipes, conduits etc, to be run within the cavity at later date.

23.2 Anti-bacterial paint:-

Anti-bacterial and anti- fungus sterile coating shall be applied on finished walls and ceiling with at 2-3 coats to achieve a thickness of minimum 200 micron and it shouldresistance to water, most chemicals commonly used in hospital departments. It should neutralize micro-organisms which come into contact and should prevent colonizing or sparing of micro-organisms.

The sterile coating should show resistance to a wide range of mixed species, including stubborn pathogens such as MRSA, Acinetobacter sp, Aerobatic aero genes, Bacillus subtitles, Escherichia coli, Listeria monocytogenes, Pseudomonas aeruginosa, Pseudomonas putida, Salmonella typimurium, Serratia marescens, Staphylococcus Aureus.

The sterile coating should remain unaffected by radiation, its anti-microbial system should not leach out and its anti-microbial properties should last for a minimum of 10 years.

The sterile coating should be easily maintained and should withstand repeated clearing with alkaline detergents, antiseptics and fumigation agents without any degradation to the surface finished or performances. Coating should conform to standards HTM 08-01/HTM 56

23.3 Surgeon's scrub sink (2 bay):-

There should be Surgeon's scrub sink of stainless steel grade SS 304, seamless welded constructions polished to a satin finished with hands free operation by infra red sensors with programmable adjustment for thermostatic mixing and food of water with foam flow attachment and beta dine dispensers, Knee/ foot operated switches should also be provideadditionally. It should also have wall mounted possibility.

23-.4 G.I. Puff panels for wall and ceiling:-

The OT wall shall be constructed using GI puff panels having 0.8 mm tick GI sheet on both sides. These panels should have been produced in a double band laminator, in which two continuously moving belts carrying 120 GSM galvanized steel of 0.80mm thickness are firmly bonded together by sprayed in insulating foam, which hardens during the process. The core consists of rigid polyurethane foam, which has been injected under high pressure, with a minimum density of 40kg/m3. The thickness of puff shall be minimum 50 mm. The puff panel shall be fixed to the floor with bottom track of 1.5 mm thick steel. U channel section having min size of 50 mmX50 mm. The bottom shall be fixed in the floor of the OT with 10 mm tick and min 60 mm long SS fasteners which shall be fully threaded and power drilled. The panels of walls and ceiling shall be tied together by suitable SS coving. The joints shall be filled with metal filler and shall be sand casted on site till smooth finish is obtained. Wall & panel joints shall not be visible after the final wall coating is applied. The gap between the masonry wall surfaces and the panel of the theatre shall be sufficient for the flues mounting of equipment's. This gap shall be provided to accommodate the equipment at a the date and to enable services, pipes, conduits etc, to be run within the gape. All wall mounted equipment shall be flush mounted and seated in operation theatre. All the sharp edges and corners shall be in radius to avoid bacterial contamination. Access boxes shall be fitted to the rear of all wall mounted equipment to enable maintenance to be carried out from outside the operation theatre. There shall be provision for accommodating 4 return riser air ducts with SS 304 grills to meet HVAC requirements as per HVAC layout. Product should have warranty of at least 10 years.

23.5 Toughened glass window

Modular OT should have a double isolated fixed glazing, with not less than 5 mm thick toughened glass window which have 12 mm air gap in between. Window frame should be power coated aluminium fixed flush with wall panels should have horizontal venetian blinds ofpower coated aluminium strips of vista level.

23.6 Storage unit

There should be a storage unit of stainless steel grade SS304 with 10mm thick toughened glass shelves supported with D type bracket (SS 304 grade)

23.7 Writing Board:

The writing unit/white Board of size 900/600mm flush mounted 1.60mm thick white lamination board, bonded to a 40mm high density fiber board sheet for additional rigidity. The unit should open to create a wall mounted writing surface within the operating room.

23.8 Hatch Box/ Pass Box:

The Hatch box manufactured with stainless steel grade. SS 304 provided in operation theatre to remove waste material from OT to dirty linen area just adjacent to operation theatre Hatch box equipped with electronically operated two nos doors. It is designed in such a way that only one door should be opened at a time.

The UV light should be so installed that it is kept on while both the doors are closed, this UV light has to be automatically turned off in case of opening of either of the doors. There should be indicators on both sides of the OT so that door open/close status can be monitored from both ends.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
23.1		Stainless steel grade SS304 O.T. WallPanels	Supply, installation, testing, commissioning of modular OT wall & Corner panels made of stainless steel (SS) of grade SS304 .The inner surface of wall should be constructed with 1.5 mm thick stainless steel grade SS304 backed by minimum 12 mm thick gypsum board fixed by grouting and suitable galvanized iron frame on the masonry walls of OT with stainless steel fasteners of 10 mm dia and 600 mm length at suitable intervals. The steel sheet panel shall be indigenous of reputed make and shall be provided with manufactures test certificate. The fastener shall be fully threaded and shall be power drilled The panels shall be fixed to the wall by a 1.5 mm thick U shape SS channel of minimum size 50 mm x 50 mm. The individual wall panels shall have folded edges which shall be fixed to S.S. U channel by welding and then shall be welded together. The inner surface of O.T. walls (facing inside the Modular should Be easily washable, scratch free, dent free, flame resistant, stretch resistant as per BS 3900 Part E2 standards. All joints shall be filled with metal casted and shall be sand flush on site till ready to receive the plastic finish. The individual wall panels should be perfectly sealed together without any gap and free of projections. Wall panel joints should be invisible when the OT is finished. The gap between masonry wall and the panel of the theatre shall be sufficient for the flush mounting of equipments and to enable services, pipes conduits etc., to run within the gap as per approved layout. All wall-mounted equipment shall be flush mounted and sealed into operation theatre. All the sharp edges and corners shall be in curve to avoid bacterial contamination. The wall panel design shall be such that it allows installation and provide support to all equipment without affecting rigidity and strength. All wall mounted equipment should be flush mounted and sealed into theatre wall by means of a sterile jointing system. There shall be provision for accommodating 4 return riser air ducts with per HVAC layout. Product should have warranty of at least 10 years. The work should be executedas per the directions of Engineer–in-Charge	sqm	9750.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
23.2		Stainless steel grade SS304 O.T. Ceiling Panels	Supply ,installation , testing ,commissioning of modular OT ceiling panels made of stainless steel (SS) of grade SS304 .The inner surface of panels should be constructed with 1.5 mm thick stainless steel grade SS304 backed by minimum 12 mm thick gypsum board fixed by grouting on the roof of OT . The sheet panel shall be indigenous of reputed make and shall be provided with manufactures test certificate. The outer surface of panels shall be fixed to the roof by essential fastener supports with stainless steel fastener of 10 mm dia and min. 60 mm length at suitable interval. The fastener shall be fully threaded and power drilled. Ceiling support element should be galvanized or powder coated steel suspension bracket with tension springs. Theminimum size of the tie/threaded rod shall be 10 mm dia and section as per site requirement. The individual fasteners except those at the edges should be removable individually. The inner surface of panels (facing inside the MOT) should be easily washable, scratch free, dent free, flame resistant, stretch resistant as per BS 3900 Part E2 Standards. All joints shall be filled with metalcasted and shall be sand flush on site till ready to receive the plastic finish. The individual SS panels should be perfectly sealed together withoutany gap and free of projections. SS ceiling panel joints should be invisible when the OT is finished. The gap between roof & panels the inside and outside surfaces of the theatre shall be variable to suit the architects' layout, but shall be sufficient for the flush mounting of equipment's. The gap between roof & panel shall be left to accommodate the equipment at a later date and to enable services, pipes, conduits etc., to run within the gap. All ceiling mounted equipment shall be flush mounted and sealed into operation theatre. All the sharp edges and corners shall be in curve to avoid bacterial contamination. The ceiling	sqm	10150.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			panel design shall be such that it allows installation and provide support to all equipment and the provision of openings required for the installations, without affecting rigidity and strength. All ceiling mounted equipment should be flush mounted and sealed by means of a sterile jointing system. The ceiling should be thermodynamically stable, not getting heated with electrical wiring or air flow and should not sag with weight of pendants. There shall be provision for accommodating room lighting, air supply inlet, ceiling service units, return air outlets etc and should be integrated with SS panel ceiling system. Product should have warranty of at least 10 years. The work should be executed as per the directions of Engineer–in-Charge.		
23.3		Anti bacterial paint on G.I.Puff panelor S.S. panel of O.T.	Providing and establishing anti-bacterial & anti fungicidal sterile coating of liquid plastic paint on GI Puff/SS panel of modular OT. The paint should be seamless, easy to clean, should have vapor permeability which allows substrate moisture to escape and protects against growth of bacteria, mound and yeasts, moisture resistance, freeze / thaw resistance. Anti-bacterial paint's anti-microbial quality should remain active throughout the life of the product. Vapors permeable structure should allow damp substratesto dry out without causing blistering. It should tolerate cleaning with hospital chemical agents. Filling of all joints and cavities should be with metallic epoxy filler and sand casted flush to provide a joint-less finish. This should include all fittings, accessories, hardware, support structures, etc. required for installation. Anti-bacterial andanti-fungus sterile coating shall be applied on the finished walls & ceiling with at least 2-3 coats to achieve a thickness of minimum 200 micron. Before application of sterile wall coating an appropriate primer shall be applied. The primer shall be supplied along with the paint and should be of the same make as that of paint. and shall be applied prior to coating of anti-bacterial & anti- fungus sterile paint on walls & ceiling 2-3 coats	sqm	1200.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			shall be applied to achieve a thickness of 200 microns. Sterile coating and primer shall be applied on the surface by air-less spray gun. Sterile coating shall withstand against application of commonly used cleaners in the operation rooms and shall withstand against regular steam cleaning. Product should have warranty of at least10 years. The work should be executed as per thedirections of Engineer–in-Charge.		
23.4		Surgeon's scrub sink (2 bay) in O.T.	Supplying, fixing installation, testing and commissioning of scrub sink with the following specifications:- Compact surgical scrub sink should be designed for use in OT complex for pre procedural scrub up. (Double sink combination size 1600 mm X 450mm X 1650 mm back height & 1200 mm and front heightas suitable) • Each fixture should be fabricated from heavy gaugetype 304 stainless steel (minimum thickness 1.5mm) and should be seamless welded construction, polished to a satin finish • The scrubsink should be provided with a front access panel which should be easily removed for access to the water controlled valve, waste connections,stoppers and strainers.• Hands free operation should include infra-red sensors with programmable adjustment.• Thermostatic mixing, valve control should be located behind the access panel and maintain constant water temperature.• Timing should be adjustable to meet individual application requirements.• Provided with infrared sensors, thermostatic control taps with feel safe temperature controls with foam flow attachments and provide with Betadine dispensers .• All units should have reduced anti- splash fronts. • Knee/foot operated switch should be provided additionally. The work should be executed as per the directions of Engineer–in-Charge.	each	86000.00
23.5		G.I. puff panels for O.T wall	Supply, installation, testing and commissioning of modular OT Walland corner panels with insulated prefabricated puff Panels. The OT wall shall be constructed using GI puff panels having 0.8 mm thick GI sheets on both sides. These panels should have been produced in a double band laminator, in which two continuously moving belts carrying 120 GSM galvanized steel sheets of	sqm	7000.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			0.80 mm thickness are firmly bonded together by sprayed -in insulating foam, which hardens during the process. The core consists of rigid polyurethane foam, which has been injected under high pressure, with a minimum density of 40kg/m3. The thickness of Puff shall be minimum 40 mm. The sheet panel shall be of indigenous reputed make and be provided with manufactures test certificate. The puff panel shall be fixed to the floor with bottom track of 1.5 mm thick steel U channel section having min size of 50mm X 50mm. The bottom shall be fixed in the floor of the OT with 10mm thicknessand min. 60 mmlong SS fasteners which shall be fully threaded and power drilled. The Panels of walls and ceilingshall be tied together by suitable SS coving. The individual wall panels of wall and ceiling shall be tied together by suitable SS coving. The individual wall panels shall be welded together bySS. All joints shall be filled with metal filler and shall be sand casted on site till smooth finish is obtained. Wall & panel joints shall not be visible after the final wall coating is applied. The gap between the masonry wall surfaces and the panel of the theatre shall be sufficient for the flush mounting of equipment's. This gap shall be provided to accommodate the equipment at a latter date and to enable services, pipes conduits etc, to be run within the gap. All wall-mounted equipment shall be flush mounted and sealed in operation theatre. All the sharp edges and corners shall be in radius to avoid bacterialcontamination. The wall panel design shall be such that it enables the installation and supports all equipment's. Provision of openings required for the installations shall not affect rigidity and strength of the panel. Access boxes shall be fitted to the rear of all wall-mounted equipment to enable maintenance to be carried out from outside the operation theatre. Product should have warranty of at least 10 years. Thework should be executed as per the directions of Engineer–in-Charge.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
23.6		G.I. puff panels for OT ceiling	Supply, installation, testing and commissioning of modular OT ceiling panels with insulated prefabricated puff Panels. The OT ceiling shall be constructed using GI puff panels having 0.8 mm thick GI sheets on both sides. These panels should have been produced in a double band laminator, in which two continuously moving belts carrying 120 GSM galvanized steel sheets of 0.80 mm thickness are firmly bonded together by sprayed -in insulating foam, which hardens during the process. The core consists of rigid polyurethane foam, which has been injected under high pressure, with a minimum density of 40kg/m3. The thickness of Puff shall be minimum 40 mm. The sheet panel shall be of indigenous reputed make and be provided with manufactures test certificate. The puff panel shall be fixed to the floor with bottom track of 1.5 mm thick steel U channel section having min size of 50mm X 50mm. The bottom shall be fixed in the floor of the OT with 10mm thicknessand min. 60 mm long SS fasteners which shall be fully threaded and power drilled. The Panels of walls and ceiling shall be tied together by suitable SS coving. The individual panels of wall and ceiling shall be tied together by suitable SS coving. The individual wall panels shall be welded together bySS. All joints shall be filled with metal filler and shall be sand casted on site till smooth finish is obtained. Wall & panel joints shall not be visible after the final wall coating is applied. The gap between the masonry wall surfaces and the panel of the theatre shall be sufficient for the flush mounting of equipment's. This gap shall be provided to accommodate the equipment at a latter date and to enable services, pipes conduits etc, to be run within the gap. All wall-mounted equipment shall be flush mounted and sealed in operation theatre. All the sharp edges and corners shall be in radius to avoid bacterial contamination. The wall panel design shall be such that it enables the installation and supports all equipment's. Provision of openings required for the installations shall not affect rigidity and strength of the panel. Access boxes shall be fitted to the rear of all wall-mounted equipment to enable maintenance to be carried out from outside the operation theatre. Product should have warranty of at least 10 years. The work should be executed as per thedirections of Engineer–in-Charge.	sqm	7500.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
23.7		Doubled toughened glass O.T. window	Providing and establishing the view windows of size 1.00 m x1.00 m. It should be provided with each operation theatre and should have double isolated fixed glazing with not less than 5 mm thick toughened glass on both sides with 12 mm air gap. Window frame should be of powdered coated aluminum of approved shape flush mounted with wall paneling should have horizontal venetian blinds of powder coatedaluminum strips of vista level or equivalent of approved shade including necessary accessories. Window (Size: - 1000 mm x 1000 mm) with both side glass with blinds sandwiched in between.The blinds should be openable from inside the O.T. The work should be executed as per the directions of Engineer–in-Charge.	sqm	25000.00
23.8		Storage unit in O.T.	Supply and installation of storage unit of SS 304 of reputed make and having size 1665 x 865 x 350 mm and having two or more glass shelves of toughened glass 10 mm thick, supported with D type bracket (SS 304 grade) with 10 mm thickness and rubber gasket of 2mm and mountedflush on wall. The work should be executed asper the directions of Engineer–in-Charge.	each	30000.00
23.9		Writing board in O.T.	Supply & Installation of writing unit/white board of size 900/600 mm having 1.60 mm thick white laminate board top, bonded to a 40 mm high density fiber board sheet for additional rigidity. The unit should open to create a wall mounted writing surface within the operating room. The work should be executed as per the direction of Engineer-in-Charge.	each	3700.00
23.10		Hatch box / pass box in O.T.	A hatch of size 600mm x 600mm manufactured by stainless steel grade SS 304, should beprovided in the operation theatre as specified in the scope of the work to remove waste materials from the operation theatre to dirty linen area (or dirty corridor) just adjacent to operation theatre. Each hatch box should be electronically operated.The hatch box should be designed in such a way that only one door should be opened at a time.The UV light should be so installed that it is kept on while both the doors are closed, this UV light has to be automatically turned off in case of opening of either of the doors. There should be indicators on both sides of the OT so that the dooropen/close status can be monitored from both ends. Hatch Box should be installed as per	each	26000.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			manufacturer's instructions manual. The work should be executed as per the direction of Engineer-in-Charge.		
23.11		Homogeneous Conductive PVC Vinyl Flooring in O.T.,ICU,OPD.	Providing and fixing 2 mm thick homogeneous conductive PVC (Poly Vinyl Flooring) EN (European Standard), total weight is 3250 g/m2, approx. PUR (Polyurethane Reinforced), abrasion group P: ≤2.00 mm, residual indentation approx. 0.03 mm, anti-bacterial and fungal resistance, castor wheel test qualified, dimensional stability ≤0.40%,static electrical charge < 2kv, all vinyl sheets must be linked crosswise with a continuous copper strip and finally connected to earth for grounding electrostatic electrical charges. impact soundreduction approx. +4db, good resistance to scratch, slip, chemical and fire.100% recyclable (eco-freindly) Low VOC emissions after 28days of installation (lower than ≤10µg/m3), Phthalate free, directional in design.The floor finish should be termite free at the room perimeter passing over a concealed cove forming and continuing up on the wall upto 100mm. Recommended to use inarea like ICU , Post operative, Recovery rooms, OPD, Patient rooms, Corridors ,passages etc.	sqm	2830.00
23.12		Homogeneous anti-static PVC vinyl flooringon O.T.,ICU & Server area	Providing and fixing permanently static dissipative pressed homogeneous conductive PVC (Poly Vinyl Flooring) EN (European Standard), anti static, wear layer thickness is 2mm (EN429), vinyl flooring having carbon backing with total weight 3000 g/m2 approx (EN 430), XR PUR (Polyurethane Reinforced) , anti- bacterial and fungal resistance, abrasion group P:≤4.0 mm, residual indentation approx. 0.02 mm, castor wheel test qualified, dimensional stability ≤0.40%, static electrical charge < 2kv, all vinyl sheets must be linked crosswise with a continuous copper strip and finally connected to earth for grounding electrostatic electrical charges. good resistance to scratch, slip, chemical and fire, electric resistance is as per EN 1081 and ESD approved, restorable properties,available in rolls. 100% recyclable (eco - freindly). Low VOC emission after 28days of installation (lower than ≤100µg/m3), Phthalate free non-directional in Design. Recommended to use in area like Operation theater, ICU, Server room, etc.	sqm	3455.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
23.13		Heterogeneous antifungal PVCvinyl 0.92mm thick wall coveringin OT,ICU,serve r room, etc.	Providing and fixing 0.92mm thick hetrogeneous compact PVC (Poly Vinyl Wall Covering) for Interior wall area, with wear layer thickness 0.12mm of PVC, EN(European Standard) total weight is 1500 g/m2, approx PUR (Polyurethane Reinforced), good resistance to chemical and fire, doesn't favour in growth of fungi and bacterial, 100% recyclable (eco-freindly), low VOC emissions after 28 days of installation (lower than ≤10µg/m3), phthalate free, directional in design, The wall finish should termite free at the room perimeter. Recommended to use in area like OT,ICU, Corridors, Patient Room etc.	sqm	1598.00
23.14		Homogeneous 1.3mm thick Antifungal Pvc Vinyl Wall Covering in OT, ICU, server room, etc.	Providing and fixing 1.3mm thick homogeneous PVC (Poly Vinyl Wall Covering) for interior area with wear layer thickness 1.3 mm of PVC, EN (European Standard) total weight is 2100 g/m2,approx. PUR (Polyurethane Reinforced), goodresistant to chemical and fire, doesn't favour in growth of fungi and bacterial, 100% recyclable (eco-freindly), low VOC emissions after 28days of installation (lower than ≤10µg/m3), phthalate free, directional in design.The wall finish should termite free at the room. Recommended to use in area like OT.ICU,Corridors, Passages , Patient Room Etc	sqm	1763.00
23.15		Homogeneous 2.00mm thick antibacterial PVC Vinyl Flooring in OT, ICU, server room, etc.	Supply and installation of 2mm thick homogeneous vinyl flooring, Highest abrasion wear rating Group T (EN660), binder content exceeds 55% of the flooring composition, no waste technology, consisting of PVC resin, plasticizers and stabilizers, type I. flooring (EN ISO 10581). UV cured, cross linked polyurethane reinforcement. PUR surface. Anti-bacterial, antistatic, non-directional design. 100% recyclable, no wax products, phthalate free, solvent free, roll dimensions 2x20mtr., (EN ISO 23997), Product specification (EN649), European classification EN685 class: 34/43, residual indentation EN433 max 0.1mm, Dimensional stability (EN ISO 23997) max 0.4%, Abrasion group (MSZ EN 660-1) T, Fire resistance EN 13501-1 BF1-s1, Slip resistance DIN 51130 R9, EN 13893 DS, electrical resistance EN 1815 <2 kv, Chemical resistance EN ISO 26987, Castor chair resistance EN ISO 4918, Hygenie Bacterial resistant, Seaming method - hot welding, finishing complete . Recommended to use in area like Operation theater, ICU, Server room, etc.	sqm	2100.00

CHAPTER – 24.0

STEEL STRUCTURE
PRE-FABRICATED BUILDINGS

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1.	IS 801-1975	Indian Standard code of practice for use of cold formed light gauge steel structural member's in general building construction.
2.	IS 800-2007	Indian Standard General Construction in steel Code of Practice.
3.	IS 875-1987-Part I	Indian Standard Code of Practice Design Loads-Dead Load
4.	IS 875-1987-Part II	Indian Standard Code of Practice Design Loads-Imposed Load
5.	IS 875-1987-Part III	Indian Standard Code of Practice Design Loads-Wind Load
6.	IS 875-1987-Part III	Indian Standard Code of Practice Design Loads- Special Load and combination.
7.	IS 1893-2002	Criteria for Earthquake Resistant Design of Structures.
8.	AISI	American Iron Steel Institute for CFS Member.
9.	IS 14862-2000	Cement fiber board
10.	IS 8183	Glass wool insulation
11.	IS 8386-1993	Non asbestos fiber Cement board

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
24.1		LGSF super structure (wall panel, trusses purling etc)	Designing, providing, installing and fixing factory finished custom designed cold form Light Gauge Steel Framed super structure comprising of steel wall panel, trusses, purlins etc manufactured out of minimum 0.75 mm thick steel sheet as per design requirements. The steel sheet shall be galvanized (AZ-150gms Aluminium Zinc Alloy coated steel having minimum yield strength 300-550 Mpa) conforming to AISI specifications and IBC 2009 for cold formed steel framing and construction and also as per IS: 875-1987, ISO 800-1984 and IS:801- 1975. The wind load shall be as per provisions of IS 875(part-III). LGSFS frame shall be designed as per IS: 801 using commercially available software such as Frame CAD Pro-11.7/ STAAD PRO-V8i/ArchitekV2.5.16/Revit architecture or equivalent. Proper usage of Connection Accessories like Heavy Duty Tension Ties, Light Duty Hold-ons, Twist Straps (to connect truss with wall frames), Strong Tie, Tie Rod, H-Brackets, Boxing Sections, L-Shaped Angles for better structural stability. The framing section shall be cold form C-type having minimum web depth 89 mm x 39mm flange x 11mm lip in required length as per structural design requirement duly punched with dimple/slot at required locations as per approved drawings. The slots will be along center line of webs and shall be spaced minimum 250mm away from both ends of the member. The frame can be supplied in panelized or knock down condition in specific dimensions and fastened with screws extending through the steel beyond by minimum of three exposed threads. All self-drilling tapping screws for joining the members shall have a Type II coating in accordance with ASTM B633(13) or equivalent corrosion protection of gauge 10 &12, TPL 16 & 8 of length 20mm. The frame shall be fixed to RCC slab or Tie beam over Neoprene	kg	235.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			rubber using self expanding carbon steel anchor bolt of dia as per approved drawings. design subject to minimum 12mm diameter and 121mm length conforming to AISI 304 and 316 at 500mm c/c with minimum embedment of 100mm in RCC (RCC to be paid separately) and located not more than 300mm from corners or termination of bottom tracks complete in all respects. The item also includes the submission of stability reports duly examined and issued by any NIT/IIT. The rate includes the concept design, detailed design, fabrication of sections, transportation, installation and all required fixing arrangement at site as described above.The main frame shall be fixed to RCC slab or Plinth Tie beam, over Neoprene rubber using self-expanding carbon steel anchor bolt of dia. as per design subject to minimum 12 mm diametre and minimum 121 mm length, confirming to AISI 304 and 316 at 500 mm c/c with minimum embedment of 100 mm in RCC (RCC to be paid separately) and located not more than 300 mm from corners or termination of bottom tracks completein all respects.Welding shall not be permitted to the member of structure having thickness less than 1.09 mm. Load-bearing steel framing members shall have a legible label, stamp, stencil or embossment, spaced at a minimum of 1200 mm on center along the length of the member, with the following minimum information: Manufacturer's Name; Minimum uncoated steel thickness in mm, Minimum coating designation and, Minimum yieldstrength in MPa.Rate is inclusive of designing charges contractor shall submit shop drawings and structural stability report from any structural engineer empanelled in o/o the chief architect, MPPWD or NIT/IIT.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
24.2		Pre-engineered steel work	Design, supply, fabrication, Transport and erection of structural Steel/Mild Steel built-up sections for Pre-Engineered steel building in column, Rafter, Truss, floor joist, Wind Bracings, Tie Beam, foundation bolts, Fasteners etc. including two coats of Epoxy paint and one coat of prime. These structural steel/built-up sections shall be manufactured using advanced CNC Plasma/Oxy Acetylene cutter and H Beam Welding line, Welding shall be in full depth on both sides of the plate adopting relevant I.S. Code, design of PEB Steel building to be proof checked by any NIT/IIT	kg	78.70
24.3		Trim for window and door	Providing and fixing Trim for window and door 1mm thick with zinc coating not less than 275gm/m^3 galvanizedsheet material Girth up to 400mm.	metre	592.00
24.4		Neoprene rubber	Providing and Fixing neoprene rubber in between wall frame and foundation.	sqm	292.00
24.5		External wall panels on LGSF work	Providing and fixing of external wall system on Light gauge steel frame work with . Outer face having 6mm thick heavy duty fiber cement board fixed on 9mm thick heavy duty fiber cement board confirming to IS 14862:2000, category IV type A (High pressure steam cured) as per standard sizes fixed with self-drilling / taping screws / fasteners @ 60cm c/c of approvedmake. A grove of 2 mm to 3mm shall be maintained and groves shall besealed with silicon based sealant. The board shall be fixed in a staggered pattern.Screws shall be of counter sunk rib head of 1.60mm to 4 mm thick of 8 to 10 gauge of length varying from 25to 45 mm and internal face 12.5mm thick gypsum plaster board fixed on 8mm thick fiber cement boardconfirming to IS 14862:2000 of category III type B (High pressure steam cured) as per standard sizes fixedwith self-drilling / taping screws / fasteners @ 60cm c/c of approved	sqm	2813.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			make, proper taping and jointing to be done using fiber mesh tape and epoxy and acrylic based jointing compound for seamless finish. Including breathable vapour barrier underneath the cement fiber board as per national building code 2009 complete as per direction of Engineer-in-Charge. (costof frame work to be paid for separately). **Note**: Insulating material is not required in the hollow wall space between two cement fiber boards.		
24.6		Internal wall panels on LGSF work	Providing and fixing internal wall panels on Light gauge steel frame work with 12.5mm thick gypsum plaster board conforming IS 2095:2011 fixed on 8mm thick fiber cement board conforming to IS 14862:2000 of category III type B (High pressure steam cured) as per standard sizes fixed with self-drilling / taping screws / fasteners @ 60cm c/c of approved make, Screws shall be of counter sunk rib head of 1.60mm to 4 mm thick of 8 to 10 gauge of length varying from 25 to 45 mm. Proper taping and jointing to be done using fiber mesh tape and epoxyand acrylic based jointing compound for seamless finish.(cost of frame work to be paid for separately) **Note**: Insulating material is not required in the hollow wall space between two cement fiber boards.	sqm	1988.00
24.7		Breathable vapor barrier on exterior face	Providing and fixing in all exterior face panels breathable vapor barrier underneath the cement fiber board asper National Building Code 2009 complete as per direction of Engineer- in-charge.	sqm	195.00
24.8		Heavy duty fiber cement board	Supplying and installation of moisture resistant/fire resistant 6 mm thick Heavy duty fiber cement board (High pressure steam cured) conforming to IS 14862:2000 of category III type B as	sqm	827.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			per standard sizes fixed with self-drilling / taping screws. Screws shall be of counter sunk rib head of 1.60mm to 4mm thick of 8 to 10 gauge of length varying from 25 to 45 mm.		
24.9		Thermal insulation	Providing and fixing thermal insulation with Resin Bonded glass wool conforming to IS: 8183, having density 48 kg/m^3. 50 mm thick, wrapped in 200 G Virgin Polythene Bags fixed to wall with screw, rawl plug and washers and held and in position by crisscrossing GI wire etc. complete as per directions of Engineer-in-Charge.		
	24.9.1	Glass wool thermal insulation	Providing and fixing insulation with resin bonded Glass wool	sqm	338.00
	27.9.2	Rock wool thermal insulation	Providing and fixing insulation with resin bonded Rock wool	sqm	238.00
24.10		Lap siding	Supply and installation of moisture/fire resistant lap siding 8 mm thick and width as per standard sizes made of cement fiber board with embossed decorative finish to give pleasing exterior finish fixed horizontally withstaggered joints including minimum overlap of 32 mm on external cement fiber board conforming to IS 14862:2000 fixed with self-drilling/tapping screws/fasteners of approved make.	sqm	702.00
24.11		Iron profilesheets	Providing and fixing Precoated galvanised iron profile sheets (size, shape and pitch of corrugation as approved by Engineer-in-Charge) 0.50 mm + 0.05 %, total coated thicknesswith zinc coating 120 gsm as per IS: 277 in 240 mpa steel grade, 5-7 micronsepoxy primer on both side of the sheet and polyester top coat 15-18 microns. Sheet should have protective guard film of 25	sqm	652.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			microns minimum to avoid scratches while transportation and should be supplied in single length upto 12 metre or as desired by Engineer-in- charge. The sheet shall be fixed using self-drilling /self-tapping screws of size (5.5x 55mm) with EPDM seal, complete upto any pitch in horizontal/ vertical or curved surfaces excluding the cost of purlins, rafters and trusses and including cutting to size and shape wherever required.		
24.12		Roofing accessories	Providing and fixing precoatedgalvanised steel sheet roofing accessories 0.50 mm + 0.05 % total coated thickness, Zinc coating 120gsm as per IS: 277 in 240 mpa steel grade, 5-7 microns epoxy primer on both side of the sheet and polyester top coat 15-18 microns using self-drilling/ self-tapping screws complete :		
	24.12.1	Ridge	Ridges plain (500 - 600mm).	metre	722.00
	24.12.2	Flashing/Aprons	Flashings/ Aprons.(Upto 600 mm)	metre	706.00
	24.12.3	North light curves	North light curves.	metre	777.00
	24.12.4	Barge board	Barge board (Upto 300 mm).	metre	669.00
	24.12.5	Crimp curve	Crimp curve	sqm	738.00
	24.12.6	Gutter	Gutter. (600 mm over all girth).	metre	848.00
24.13		Under deck insulation	Providing and fixing thermal insulation of ceiling (under deck insulation) with Resin Bonded Fiber glass wool conforming to IS: 8183 density 24kg/m^3, 50mm thick, wrapped in aluminium foil laminated placed over metal mesh between the gape of purlin and sheet with wire mesh of 12.5mm x24 gauge wire mesh, using self- drilling/ self-taping screws of size (5.5x55mm), EPDM seal.		
	24.13.1	Under deck insulation (Fiber glass wool)	Providing and fixing under deck fiber glass wool insulation.	sqm	368.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	24.13.2	Under deck insulation (Rock wool)	Providing and fixing under deck rock wool insulation.	sqm	268.00
24.14		PUF panels for roofs 30mm thick external face corrugated	Providing and fixing PUF insulated continuous sandwich panels for roofs of total thickness not less than 30 mm made out from continue line method panel shall have pre coated GI sheet on both side of Polyurethane foam with external face being corrugated in shape for GI and PU foam both material. The crest height of the panel shall be of 35 mm minimum with 250mm c/c pitch. The Precoated sheet shall be ofminimum 240mpa steel grade confirming to IS 14245:1995 and shall have zinc coating of 120 gsm as perIS:277, 5-7 microns epoxy primer onboth side of the sheet and polyester top coat 15-18 micron. The PPGI Sheet shall have protective guard film of minimum 25 microns to avoid scratches whiletransportation. The roof panels are laid over a frame work of trusses, columns and purlins fixed using 90 mm self-drilling bolts with rubber washer including all types of flashings. PU Foam must be self-extinguishing. Fire retardant type having minimum density of 40 Kg/ Cum.(±2 Kgs) including 0.25 mm craft paper edging, self-tapping screws of required length and nos. etc. complete as per structural design and direction of Engineer-in-charge.		
	24.14.1	PPGI sheet 0.5mm thick	PPGI sheet 0.5mm thick skin on both sides.	sqm	2555.00
	24.14.2	PPGI sheet 0.35mm thick	PPGI sheet 0.35mm thick skin on both sides.	sqm	2276.00
24.15		PUF panels for wall	Providing and fixing PUF insulated continuous sandwich panels for wall of total thickness not less than 50 mm or 40mm (as specified) and width 1.0m, made out from continue line method on automatic plant. Panel shall have pre		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			coated GI sheet on both side of Polyurethane foam confirming to IS12436:1988. The precoated sheet shall be of minimum 240 mpa steel grade confirming to IS 14246:1995 and shall have zinc coating of minimum 120 gsm as per IS:277, 5-7 microns epoxy primer on both side of the sheet andpolyester top coat 15-18 micron. ThePPGI Sheet shall have plastic protective guard film of minimum 25 microns to avoid scratches during transportation. The panels shall be vertically joined together by tongue and groove joints. The PU foam shall be self- extinguishing, fire retardant type havingminimum density of 40 kg/Cum. (±2 Kgs) including 0.25 mm thick craft paper edging on both edges. The panels shall be fixed to the steel frame structure with minimum 5mm thick craft paper edging on both edges. The panels shall be fixed to the steel frame structure with minimum 5mm thick self- tapping GI screws of required length and nos. with minimum spacing of 300 mm c/c. UUF panel 40 mm thick.		
	24.15.1	50mm thick wall	50 mm thick		
	24.15.1.1	PPGI sheet 0.5mm thick	PPGI sheet 0.5mm thick skin on both sides.	sqm	2566.00
	24.15.1.2	PPGI sheet 0.35mm thick	PPGI sheet 0.35mm thick skin on both sides.	sqm	2276.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	24.15.2	Wall 40 mm	PUF panels for 40 mm thick wall		
	24.15.2.1	PPGI sheet 0.5mm thick	PPGI sheet 0.5mm thick skin on both sides.	sqm	2496.00
	24.15.2.2	PPGI sheet 0.35mm thick	PPGI sheet 0.35mm thick skin on both sides.	sqm	2218.00
24.16		Solid wall panels 75mm thick AAC core sandwiched between cement board	Supply and installation of moisture/fire resistant solid wall 75mm thick non asbestos fiber reinforced aerated cement sandwich panel made out of light weight fiber reinforced aerated cement/ concrete core composed of Portland cement, fly ash, binders etc. and 4mm thick non asbestos fiber cement board conforming to IS 14862:2000and ISO : 8336:1993 (E) on either side of the core, having a tongue and groove on longitudinal side of the panels and fixed to steel structure frame by pre- coated water head self-drilling/tapping fasteners of minimum 100 mm length and 4 mm dia of approved make as per design. Rate include for supply and fixing of 15mm high density non asbestos fiber cement board conformingto IS 14862:2000 and ISO 8336:1993 (E) on exterior side of the steel structure frame with similar fastening arrangement. The joints of the cement boards must be sealed with polymer fortified thin set mortar and self- adhesive alkali resistant dry wall fiber mesh tape of minimum 100 mm width complete in all respects.	sqm	2234.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
24.17		U track for closing of end joints for cut PUF panels	Providing and fixing of Bottom and top U track for closing of end joints for cut PUF panels. The U track shall be made out of 1 mm thick pre coated GI sheet. The Precoated sheet shall be ofminimum 240 mpa steel grade confirming to IS 14246:1995 and shall have zinc coating of minimum 120 gsm as per IS:277: 1992 , 5-7 microns epoxyprimer on both side of the sheet andpolyester top coat 15-18 micron. ThePPGI Sheet shall have plastic protectiveguard film of minimum 25 microns to avoid scratches during transportation. The U track shall be fixed with suitable POP rivet with wall roof panels and plinth with Expandable fasteners withminimum spacing of 450 mm c/c. The Utrack shall be fixed to RCC Plinth or Tie Beams by suitable anchor bolts.		
	24.17.1	For 50mm thick panel	For panel of size of 52mmx30mm with 5mm lipping on both edges.	RM	338.00
	24.17.2	For 40mm thick panel	For panel of size of 42mmx30mm with 5mm lipping on both edges.	RM	323.00
24.18		Drywall partition (66mm thick)	Supplying and installation of magnesia block 66 mm thick drywall partition including Framework includes Floor/ Head GI channels having thickness 0.5mm, length 3600mm, equal flanges of 32mm and web of 50mm fixed tofloor and ceiling with suitable fasteners at 300mm centers staggered. Noise and fire rated silicon bead sealant to be caulked along the perimetre of thepartition frame before fixing channels. Then Stud channel having thickness 0.45mm, length 3600mm, unequal flanges of 34/36mm and web of 48mm should be placed into the floor/head channel positioned vertically at 600mm centers. Extra reinforcement to be provided at openings (doors, windows, etc.,)The first layer of magnesia oxide block 8mm thick should be fixed on oneside of the Stud framework in a staggered pattern by using suitable metal fasteners at 300mm centers. Second layer of magnesia oxide block	sqm	1578.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			8mm thick should be fixed other side of the frame work similar fashion as the first layer. 3mm gap should be maintained between boards.3 mm gap shall be filled with jointing compound. Magnesia oxide block shall have following physical characteristics. • Core - Magnesia • Fire Rating – upto 150mins • Acoustics – STC upto 44-50 • Climate (OC RH) – 50, 99 • Moisture Absorption - 6.8% after 2hrs and 11.3% after 24hrs soaking • Wet Expansion - < 0.02% from ambient to saturation • Dry Contraction - ≤ 0.02% from evaporation • Moisture Movement - 0.02% • Light reflectance – 80 % • Green (RC %) – 30 • Strength - Antisag • Impact - 4kg hammer 175mm		
24.19		Extra for thermal insulation 50mm thick synth PF	Add extra for providing and fixing Thermal insulation by providing 50mm thick synth PF (Phenolic Foam) having density 1000 GSM within the cavity held in position by using chicken wire mesh/ cross bracing/ pasting to the wall.	sqm	768.00
24.20		Solid wall 75mm panel thick, polystyrene sandwich between cement board	Supply and installation of moisture/fire resistant solid wall 75mm thick non asbestos fiber reinforced with expanded polysteryrene sandwich panel made out of light weight fiber cement facing board composed of Portland cement, fly ash, binders expanded polystyrenebeads etc. and 4mm thick non asbestos fiber cement board (category 3 type B) conforming to IS 14862:2000 and ISO : 8336:1993 (E) on either side of the core, having a tongue and groove on longitudinal side of the panels and fixed to steel structure frame by pre-coated water head self-drilling/tappingfasteners of minimum 100 mm length and 4 mm dia of approved make as per design. Rate include for supply andfixing of 15mm high density non asbestos fiber cement board conforming	sqm	1862.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			to IS 14862:2000 and ISO 8336:1993 (E) on exterior side of the steel structure frame with similar fastening arrangement. The joints of the cement boards must be sealed with polymer fortified thin set mortar and self-adhesivealkali resistant dry wall fiber mesh tape of minimum 100 mm width complete in all respects.		
24.21		Cement bonded particle board to wall /partition wall	Supply and installation of termite proof, moisture/fire resistant cement bonded particle board confirming to IS: 14276: 1995 in wall panel on structure to form wall/partition wall on external face of the frame fixed the self-drilling/tapping screws/fasteners of approved make. The hollow wall space between two cement particle board providing with 50mm thick resin bonded glass wool insulation conforming to IS 8183 of density not less than 48kg/cum (glass wool insulation shall be paid separately). All exterior faces of the wall/frame shall essentially be provided with breathable vapor barrier underneath of the cement bonded particle board. The inslattional of the vapor barrier should conforming to the provision of the international building code 2009 and as per the design, The joints of the cement board must be sealed with polymer modified thin set mortar conforming to ANSI A 118.4 standard and self-adhesivealkali resistant dry wall fiber mesh tap of minimum 100 mm width. All exterior wall corner of cement bonded particle board wall panel must be reinforced with galvanized metal angel of 100X100X1.09 mm thickness and fixed on both adjoining walls with self-drilling/tapping fasteners of approved make as per design including providing silicone gel caulking complete in all respects.		
	24.21.1	12mm thick	12mm thick cement bonded particle board as per IS: 14276	sqm	487.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	24.21.2	10mm thick	10mm thick cement bonded particle board as per IS: 14276	sqm	402.00
24.22		EPS panels	Providing and fixing in position factory made EPS cement sandwich wall/roof/floor light weight solid core panels made of core material of EPS granule balls/beads (conforming to IS 4671:1984 and shall have density notless than 15kg per cum) adhesive, cement, sand, flyash and other bonding material in mortar state processed to form in a preset mould. The outer face on both sides of the panels will be non asbestos fiber cement board confirming to IS 14862:2000 or Calcium silicate board confirming to EN 14306:2009 of 5mm thick each. Panel shall be laid on 6mm thick cement mortar (1 cement: 2 fine sand) mixed with chemical adhesive of 0.5kg per 50kg of cement or shall be preferably fixed into 'C' channel made of 1.2mm thick MS plate screwed/fastenened to the slab/column/beam etc. The panel shall fixed vertically with tongue and groove joint and horizontally locked with steel bar between each other and floors and filled with cement mortar and adhesive. Panels should be used as floor &roofing with additional structural support, steel or RCC depending upon the design. All the operation shall be completed in all respect as per drawings, Manufacturers specifications and under the overall direction of Engineer-in-Charge (Cost of all the material is included except "C channel" which will be paid separately).		
	24.22.1	50 mm thick	Non load bearing panels 50mm thick of required size	sqm	466.00
	24.22.2	60mm thick	Non load bearing panels 60mm thick of required size	sqm	523.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	24.22.3	75mm thick	Non load bearing panels 75mm thick of required size	sqm	674.00
	24.22.4	90mm thick	Non load bearing panels 90mm thick of required size	sqm	707.00
	24.22.5	100mm thick	Non load bearing panels 100mm thick of required size	sqm	870.00
24.23		Non asbestos aerated cement sandwich panels	Providing and fixing in position factory made non asbestos fibre reinforced aerated cement sandwich wall/roof/floor light weight solid core panels made of light weight cement concrete core composed of OPC cement, pulverized flyash, quick lime, cotton pulp & Gypsum in mortar state mixed with aeration agent in a preset mould. The outer face on both sides of the panels will be non asbestos fibrecement board confirming to IS 14862:2000. These solid wall panels are installed using Galvanized iron steel tracks/C channel of 1mm thick of required sizes as recommended by manufacturer's and fixed to floor andRCC soffit in plumb to each other with steel screw/fasteners. The panel shall befixed vertically with tongue & groove joint with cement based polymer modified jointing compound. The exposed surface finished with fibre mesh/glass fibre tape with polymerbased jointing compound having superior flexibility. Panels should be used as floor & roofing with additional structural support, steel or RCC depending upon the design. All the operation shall be completed in all respect as per drawings, Manufacturers specifications and under the overall direction of Engineer-in-Charge (Cost of all the material is included except "tracks/C channel" which will be paid separately).		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	24.23.1	50mm thick	Non load bearing panels 50mm thick of required size (minimum 4mm thickfibre cement board)	sqm	970.00
	24.23.2	75mm thick	Non load bearing panels 75mm thick of required size (minimum 5mm thickfibre cement board))	sqm	1183.00

CHAPTER – 25.0

SANITARY AND WATER SUPPLY (PREMIUM RANGE INSTALLATION)

SANITARY INSTALLATION AND WATER SUPPLY (PREMIUM RANGE)

CHAPTER-25

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
25.1		Orrisa Pan	Providing and fixing white vitreous china Anti germ Fluoro-Polymer Coated Water closet squatting pan with 100 mm sand cast Iron P and S- trap, and 7.2 liter Low level, dual flush, PP (Poly Propylene) made open slim wall hung cistern capacity 3 liter/6 liter flushing, with all fittings and fixtures complete including cutting and making good the wall and floors wherever required.		
	25.1.1	Size 440X575X250mm	White Vitreous china Anti germ Fluoro-Polymer coated Orissa Pattern W. C. Pan of Size 440X575X250mm With integral foot rests. (Premium Range)	each	4726.00
	25.1.2	Size 410X515X255 mm	White Vitreous china Anti germ Fluoro-Polymer coated Orissa pattern W. C. pan of size 410X515X255 mm with integral foot rests. (Premium Range)	each	4392.00
25.2		European Pedestal	Providing and fixing white vitreous china pedestal type Anti germ Fluoro-Polymer Coated Water Closet with soft closing seat cover, and 7.2 litre Low level slim dual flushing, PP (Poly Propeline) made 80 mm thick concealed flushing cistern, flushing capacity 3 litre/6 litre, with all fittings and fixtures complete including cutting and making good the wall and floors wherever required.		
	25.2.1	Size 360X545X395	Anti-germ Fluoro-Polymer Coated (nano coating) European W. C. pan Size 360X545X395 with solid poly propelyne made soft closing seat cover (Premium Range)	each	9788.00
	25.2.2	W. C. pan size 360X475X395 mm	Anti-germ Fluoro-Polymer Coated Water Closet European type W. C. pan size 360X475X395 mm with solid poly propelyne made soft closing seat cover (Premium Range)	each	8742.00
25.3		Wall Hung WC	Providing and fixing white vitreous china European Wall hung Anti germ Fluoro-Polymer Coated water closet of approved shape with soft closing seat cover, and 7.2 litre Low level slim dual flushing, PP (Poly Propylene) made 80 mm thick concealed flushing cistern, flushing capacity 3 litre/6 litre, with all fittings and fixtures complete including cutting and making good the wall and floors wherever required.		
	25.3.1	Size 380X520X375 mm	Anti germ Fluoro-Polymer Coated (Nano coating) Wall hung water closet Size 380X520X375 mm with solid poly propelyne made soft closing seat cover (Premium Range)	each	26412.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	25.3.2	Size 390X570X365mm	Anti germ Fluoro-Polymer Coated (Nano coating) Wall hung water closet size 390X570X365 mm with solid poly propelyne made, soft closing seat cover (Premium Range)	each	23559.00
	25.3.3	Size 360 x 580 x 360mm	Anti germ Fluoro-Polymer Coated (Nano coating) Wall hung water closet size 360 x 500 x 330mm with solid poly propelyne made, soft closing seat cover (Premium Range)	each	21099.00
	25.3.4	Size 360 x 500 x 330mm	Anti germ Fluoro-Polymer Coated (Nano coating) Wall hung water closet size 360 x 500 x 330mm or greater size with solid poly propelyne made, soft closing seat cover (Premium Range)	each	18838.00
25.4		European Couple closet :	Providing and fixing white vitreous china Anti germ fluoro-polymer coated couple water closet flushing capacity 3 and 6 litre of approved shape with 15mm quarter turn angular stop cock and pvc connection pipe i/c cutting and making good the wall and floors wherever required with all fittings and fixtures etc. complete .		
	25.4.1	Size 420X710X810mm	Anti germ Fluoro-Polymer Coated couple water closet Size 420X710X810mm with solid poly propelyne made soft closing seat cover. (Premium Range)	each	28103.00
	25.4.2	Size 380X610X805 mm	Anti germ Fluoro-Polymer Coated couple water closet Size 380X610X805mm with solid poly propelyne made soft closing seat cover. (Premium Range)	each	19275.00
	25.4.3	Size 370 x 640 x 760mm	Anti germ Fluoro-Polymer Coated couple water closet Size 370 x 640 x 760mm with solid poly propelyne made soft closing seat cover. (Premium Range)	each	18952.00
	25.4.4	Size 375X650X800mm	Anti germ Fluoro-Polymer Coated couple water closet Size 375X650X800 mm with solid poly propelyne made soft closing seat cover. (Premium Range)	each	18023.00
25.5		Eropean Single Piece Pedestal WC	Providing and fixing white vitreous china Anti germ fluoro-polymer coated single piece W.C. dual flushing capacity 3 and 6 litre of approved shape with 15mm quarter turn angular stop cock and pvc connection pipe i/c cutting and making good the wall and floors wherever required all fittings and fixtures etc. complete.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	25.5.1	Size 370X695X795mm	Anti germ Fluoro-Polymer Coated Single Piece W.C. Pan Size 370X695X795mm with solid poly propelyne made soft closing seat cover.(Premium Range)	Each	34317.00
	25.5.2	Size 395X710X730mm	Anti germ Fluoro-Polymer Coated Single Piece W.C. Pan Size 395X710X730mm with solid poly propelyne made soft closing seat cover.(Premium Range)	each	25954.00
	25.5.3	Size 385X720X710mm	Anti germ Fluoro-Polymer Coated Single PieceW.C. Pan Size 385X720X710mm with solid poly propelyne made soft closing seat cover.(Premium Range)	each	22237.00
	25.5.4	Size 360 x 640 x 685mm	Anti germ Fluoro-Polymer Coated Single Piece W.C. Pan Size 360 x 640 x 685mm with solid poly propelyne made soft closing seat cover.(Premium Range)	each	19450.00
25.6		Extended Wall Hung WC:	Providing and fixing white vitreous china made Anti germ fluoro-polymer coated extended wall hung type couple closet along with dual flushing cistern flushing capacity 3 & 6 litre approved shape with 15mm quarter turn angular stop cock and pvc connection pipe i/c cutting and making good the wall and floors wherever required all fittings and fixtures etc. complete.		
	25.6.1	Size 380X650X730 mm	Anti germ Fluoro-Polymer Coated extended wall hung type couple closet Size 380X650X730mm with solid poly propelyne made soft closing seat cover (Premium Range)	each	13365.00
	25.6.2	Size 375X630X750 mm	Providing and fixing white vitreous china made Anti germ fluoro-polymer coated extended wall hung type couple closet Size 375X360X750mm along with dual flushing cistern flushing capacity 3 & 6 litre. (Premium Range)	each	10670.00
25.7		Counter Top Wash Basin :	Providing and fixing white vitreous china type Anti germ Fluoro-Polymer Coated Wash Basin with C.I. Brackets, 15 mm C. P. wall mounted pressmatic pillar cock auto closing system (decol technology), 15mm quarter turn angular stop cock, 32 mm C.P. brass made waste coupling length 130mm and 32 mm C.P. Brass Made Bottle Trap size 200x300mm fittings and fixtures complete including Painting of fittings and cutting and making good the wall and floors wherever required.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	25.7.1	Size 595X435X185 mm	White Vetrious China Anti germ Fluoro-Polymer Coated Wash Basin Size 595X435X185 mm with basin mounted pressmatic pillar cock auto closing system (decol technology). (Premium Range)	each	16581.00
25.8		Counter Top Basin :	Providing and fixing white vitreous china type Anti germ fluoro-polymer coated counter top wash basin with 15 mm C. P. wall mounted pressmatic Pillar cock auto closing system with square wall flange and spout complete (decol technology), 32 mm C.P. brass waste coupling length 130mm and 32 mm C.P. brass bottle trap size 200x300mm i/c cutting and making good the wall and floors wherever required all fittings and fixtures etc. complete.		
	25.8.1	Size 570X820X195 mm	Providing and fixing white vitreous china type Anti germ fluoro-polymer coated counter top wash basin Size 570x415x200mm with 15 mm C. P. wall mounted pressmatic Pillar cock auto closing system with square wall flance & spout complete (decol technology). (Premium Range)	each	14543.00
	25.8.2	Size 570X820X195 mm	Providing and fixing white vitreous china type Anti germ fluoro-polymer coated counter top wash basin Size 570X820X195mm with 15 mm C. P. Table Top pressmatic Pillar cock auto closing system (decol technology). (Premium Range)	each	13614.00
25.9		Counter Top Basin :	Providing and fixing white vitreous china type Anti germ fluoro-polymer coated counter top wash basin with 15 mm C. P. wall mounted single lever concealed basin mixture, (cartridge size 32 mm) 32 mm C. P. brass waste coupling length 130mm and 32 mm C. P. brass bottle trap size 200x300mm i/c cutting and making good the wall and floors wherever required, all fittings and fixtures etc. complete. (Premium Range)		
	25.9.1	Size 615x415x150 mm	White Vetrious China Anti germ Fluoro-Polymer Coated Wash Basin Size 615x415x150mm with wall mounted single lever concealed basin mixture (cartridge size 32mm.) (Premium Range)	each	15591.00
	25.9.2	Size 600x400x185 mm	Providing and fixing white vitreous china type Anti germ fluoro-polymer coated counter top wash basin Size 600x400x185mm with 15 mm C. P. wall mounted single lever concealed basin mixture. (Premium Range)	each	14940.00
25.10		Counter Top Wash Basin	Providing and fixing white vitreous china type Anti germ Fluoro-Polymer Coated Wash Basin with 15 mm Table mounted single lever Tall boy basin mixture, (cartridge size 32 mm) 15mm quarter turn angular stop cock, 32 mm C. P. brass		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			made waste coupling length 130mm and 32 mm C. P. Brass Made Bottle Trap size 200x300mm fittings and fixtures complete including Painting of fittings and cutting and making good the wall and floors wherever required.		
	25.10.1	Size 595x435x185 mm	White Vetrious China Anti germ Fluoro-Polymer Coated Wash Basin Size 595x435x185mm with Table mounted single lever Tall boy basin mixture, (cartridge size 32mm.) (Premium Range)	each	18585.00
	25.10.2	Size 530x420x140 mm	White vitreous china type Anti germ fluoro-polymer coated counter top wash basin Size 530x420x140mm with tall boy basin mixture, (cartridge size 32 mm). (Premium Range)	each	16785.00
25.11		Flat Back Half Stall Urinal :	Providing and fixing white vitreous china flat back Anti germ fluoro-polymer coated urinal (Inbuilt spreaders, bottle trap and ceramic waste coupling) with 15mm brass pressmatic auto closing concealed urinal flush valve with wall flange (decol technology, 400000 operation warranted) i/c cutting and making good the wall and floors wherever required, all fittings and fixtures etc. complete .		
	25.11.1	Size 340x325x650 mm	White Vitreous China Anti germ Fluoro-Polymer Coated Urinal Size 340x325x650 mm. (Premium Range)	each	14544.00
	25.11.2	Size 475x315x715 mm	Providing and fixing white vitreous china Flat Back Anti germ Fluoro-Polymer Coated urinal size 475X315X715 mm top inlet with fixing accessories (Inbuilt spreaders, Inbuilt Bottle trape and inbuilt ceramic Waste coupling) with 15mm Brass made Pressmatic auto closing open urinal flush valve with wall flange (Decol technology, 400000 operation warranted) complete including painting of fittings and brakets, cutting and making good the wall and floors wherever required :	each	13336.00
	25.11.3	Size 385x325x635 mm.	Providing and fixing white vitreous china Flat Back Anti germ Fluoro-Polymer Coated urinal back inlet with fixing accessories (Inbuilt spreaders, Inbuilt Bottle trape and inbuilt ceramic Waste coupling) Size 385X325X635 mm with Concealed sensotronic flushing valve for urinal complete set with installation box with control cock complete including painting of fittings and brakets, cutting and making good the wall and floors wherever required : URS-WHT-13253N & PRS-CHR-073	each	10641.00
25.12		Flat Back Half Stall Sensor Urinal :	Providing and fixing white vitreous china Flat Back Anti germ Fluoro-Polymer Coated urinal (Inbuilt spreaders, Inbuilt Bottle trap and inbuilt ceramic Waste coupling) with sensotronic concealed type flushing valve battery & electrical		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			oprated type (9V transformer) for urinal complete set with installation box with control cock complete including painting of fittings and brackets, cutting and making good the wall and floors wherever required .		
	25.12.1	Size340x325x650 mm	White Vetrious China Anti germ Fluoro-Polymer Coated Urinal Size 340x325x650 mm with sensotronic concealed type flushing valve battery & electrical oprated type (9V transformer) for urinal complete set with installation box with control cock. (Premium Range)	each	23927.00
	25.12.2	Size 385x325x635 mm.	White Vetrious China Anti germ Fluoro-Polymer Coated Urinal size 385x325x635 mm with concealed sensotronic flushing valve for urinal complete set with installation box, (Premium Range)	each	20025.00
25.13		Vitreous china Flat Back Urinal	Providing and fixing white vitreous china Flat Back Anti germ Fluoro-Polymer Coated Urinal (Inbuilt spreaders, Inbuilt Bottle trap and inbuilt ceramic Waste coupling) with 15mm Brass made pressmatic (Decol technology, 400000 operation warranted) concealed flushing Stop cock complete including painting of fittings and brackets, cutting and making good the wall and floors wherever required .		
	25.13.1	Size 480 X 300 X740 mm	White Vetrious China Anti germ Fluoro-Polymer Coated Urinal Size 480 X 300X740 mm with 15mm Brass made pressmatic auto closing concealed urinal flush valve with wall flange (Decol technology, 400000 operation warranted). (Premium Range)	each	18491.00
25.14		Flat Back Anti germ Fluoro-Polymer Coated Urinal	Providing and fixing white vitreous china Flat Back Anti germ Fluoro-Polymer Coated Urinal (Inbuilt spreaders, Inbuilt Bottle trap and inbuilt ceramic Waste coupling)with sensor battery & electrical oprated type (9V transformer) complete including painting of fittings and brackets, cutting and making good the wall and floors wherever required.		
	25.14.1	Size340x325x650 mm	White Vetrious China made Flat Back Anti germ Fluoro-Polymer Coated Urinal (Inbuilt spreaders, Inbuilt Bottle trap and inbuilt ceramic Waste coupling) Size 340x325x650 mm with sensor battery & electrical oprated type (9V transformer) (Premium Range)	each	23927.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	25.14.2	Size 475x315x715 mm	Providing and fixing white vitreous china flat back Anti germ fluoro-polymer coated Urinal size 475x315x715 mm (Inbuilt spreaders, bottle trap and ceramic waste coupling) with inbuilt sensor battery & electrical oprated type (9V transformer) for flushing. (Premium Range)	each	22998.00
25.15		Urinal Glass Partition (Straight):	Providing & Fixing 8mm thick froasted Urinal glass partition size 900x450mm brass made bracket with chrome plating. (Premium Range)	each	6644.00
25.16		Urinal glass partition (Curve):	Providing & Fixing 8mm thick froasted Urinal glass partition size height 900 mm top width 450mm and bottom width 300mm brass made bracket with chrome plating. (Premium Range)	each	7631.00
25.17		Bath tub (plain) 180x80x47cm	Providing & Fixing plain bath tub of size 180x80x47cm body made of PMMA (Poly Methyl metha acrylate) with 20mm wooden reinforcement. (Premium Range)	each	33601.00
25.18		Pillar cock	Providing and fixing in position best Indian (bonut size 24mm, ceramic disc size 19mm and min. body thickness 2mm, nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn) Pillar cock. (Premium Range)	each	2326.00
25.19		Bib cock	Providing and fixing in position best Indian (bonut size 24mm, ceramic disc size 19mm and min. body thickness 2mm, nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn)Bib cock. (Premium Range)	each	2071.00
25.20		Angular stop cock	Providing and fixing in position best Indian (bonut size 24mm, ceramic disc size 19mm and min. body thickness 2mm, nickel plating. 10 micron and chrome plating 0.3 micron, quarter turn) Angular stop cock. (Premium Range)	each	1573.00
25.21		Concealed stop cock	Providing and fixing in position best Indian (bonut size 24mm, ceramic disc size 19mm and min. body thickness 2mm, nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn) Concealed stop cock. (Premium Range)	each	2015.00
25.22		Wall Mixer	Providing and fixing in position best Indian (bonut size 24mm, ceramic disc size 19mm and min. nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn) Wall Mixer with 115mm bend. (Premium Range)	each	10291.00
25.23		Wall Mixer (3-in1)	Providing and fixing in position best Indian (bonut size 24mm, ceramic disc size 19mm, nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn) 3-in1 Wall Mixer .(Premium Range Sanitary Ware)	each	10756.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
25.24		Flush cock quarter turn	Providing and fixing in position best Indian C. P. brass made quarter turn (Ceramic disc Size 25mm) Flush Cock for Indian W. C. (Orissa Pan). (Premium Range)	each	2858.00
25.25		Wall Mixer 40 mm (auto)	Providing and fixing in position best Indian single Lever Auto Mixing Wall Mixer (Cartridge Size 40mm, nickel plating 0.10 micron and chrome plating 0.3 micron) for Shower area use only. (Premium Range)	each	9920.00
25.26		Wall Mixer 32 mm (auto)	Providing and fixing in position best Indian single Lever Auto mixing Basin Mixer (Cartridge Size 32mm, nickel plating 0.10 micron and chrome plating 0.3 micron) For Wash Basin area use only. (Premium Range)	each	6528.00
25.27		Diverter (auto)	Providing and fixing in position best Indian single Lever Auto Mixing Diverter (Cartridge Size 47mm, nickel plating 0.10 micron and chrome plating 0.3 micron) for Shower area use only. (Premium Range)	each	7039.00
25.28		Spout	Providing and fixing in position best Indian C. P. brass made Spout (nickel plating 0.10 micron and chrome plating 0.3 micron). (Premium Range)	each	2199.00
25.29		Shower arm length 300mm	Providing and fixing in position best Indian C. P. brass made Shower Arm Length 300mm (nickel plating 0.10 micron and chrome plating 0.3 micron). (Premium Range)	each	1038.00
25.30		Shower arm length 400 mm	Providing and fixing in position best Indian C. P. brass made Shower Arm Length 400x25x25mm (nickel plating 0.10 micron and chrome plating 0.3 micron).(Premium Range Sanitary Ware)	each	2385.00
25.31		Shower arm length 600 mm	Providing and fixing in position best Indian C. P. brass made Shower Arm Length 600mm (nickel plating 0.10 micron and chrome plating 0.3 micron) .(Premium Range Sanitary Ware)	each	1502.00
25.32		Shower rose 300x300mm square	Providing and fixing in position best Indian shower rose 300x300mm square body and face chrome plated, brass made body, Silicon based Rubbit cleaning nozzles.(Premium Range Sanitary Ware)	each	10887.00
25.33		Shower rose 150mm	Providing and fixing in position best Indian shower rose 150mm dia, body and face chrome plated, brass made body, Silicon based Rubbit cleaning nozzles.(Premium Range)	each	3825.00
25.34		Shower rose 200 mm	Providing and fixing in position best Indian shower rose 200mm dia body and face chrome plated, brass made body, Silicon based Rubbit cleaning nozzles .(Premium Range)	each	4987.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
25.35		Shower rose 190x295mm	Providing and fixing in position best Indian shower rose 190x295mm body and face chrome plated, brass made body, Silicon based Rubbit cleaning nozzles.(Premium Range)	each	9261.00
25.36		2-way bib cock	Providing and fixing in position best Indian (bonut size 24mm, ceramic disc size 19mm and min. body thickness 2mm, nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn) 2-way Bib Cock .(Premium Range)	each	2687.00
25.37		Non-return valve	Providing and fixing in position best Indian Health Faucet with 8mm dia 1.2 meter long Flexible Tube and wall bracket with N. R. V. (non-return wall).(Premium Range Sanitary Ware)	each	1818.00
25.38		Pillar cock (wash basin)	Providing and fixing in position best indian Auto Closing Pillar Cock with 65 mm Extension body (with DECOL cartredge) for wash basin use.	each	3235.00
25.39		Bib cock (auto)	Providing and fixing in position best indian Auto Closing Bib Cock with Built in Control Cockwith Wall Flange (with DECOL cartredge) for comon area use.	each	2524.00
25.40		Urinal valve (auto)	Providing and fixing in position best indian Auto Closing Urinal valve open type with Built in Control Cock & Wall Flange (with DECOL cartredge) for urinal use only.	each	2571.00
25.41		Urinal valve (concealed)	Providing and fixing in position best Indian Auto Closing Urinal valve concealed type (with DECOL cartridge) for urinal use only. (Premium Range)	each	2835.00
25.42		Pillar cock wall mounted	Providing and fixing in position best indian Auto Closing wall mounted Pillar Cock with wall flange Rectangular (with DECOL cartredge) for wash basin use only.	each	4205.00
25.43		Flush valve 40 mm (auto)	Providing and fixing in position best Indian Auto Closing 40mm Flush valve (for W. C. European and Wall Hung Type) concealed type (with DECOL technology/ cartridge). (Premium Range)	each	4112.00
25.44		Flush valve 40 mm (dual)	Providing and fixing in position best Indian Auto Closing 40mm Dual Flow (3.0/6.0 liter) Flush valve (for W. C. European and Wall Hung Type) concealed type (with DECOL technology/ cartridge). (Premium Range)	each	4251.00
25.45		Flush valve 32 mm (concealed)	Providing and fixing in position best Indian Auto Closing 32mm, Flush valve (for W. C. European and Wall Hung Type) concealed type (with DECOL technology/cartridge). (Premium Range)	each	3833.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
25.46		Faucet Wash Basin	Providing and fixing in position best indian Sensotronic Sensor battery & electrical oprated type (9V transformer) Faucet For Wash Basin in Rectangular shape use only.	each	9734.00
25.47		Faucet For Urinal	Providing and fixing in position best Indian Sensotronic Sensor battery & electrical oprated type (9V transformer) Faucet For Urinal use only. (Premium Range)	each	14287.00
25.48		Faucet wall mounted	Providing and fixing in position best Indian Sensotronic Sensor concealed type Faucet For Wall Mounted Water Closet use only. (Premium Range)	each	12800.00
25.49		Towel rail 300 mm	Providing and fixing in position best indian Stainless steel made (0.3 micron Chrom and 10 micron Nickel plated) Single Towel Rail 300 mm long .	each	2097.00
25.50		Towel rail 600 mm	Providing and fixing in position best Indian Stainless steel made (0.3 micron Chrome and 10 micron Nickel plated) Single Towel Rail 600 mm long. (Premium Range)	each	2631.00
25.51		Towel holder	Providing and fixing in position best Indian Stainless steel made (0.3 micron Chrome and 10 micron Nickel plated) Swivel Towel holder Twin Type. (Premium Range)	each	2027.00
25.52		Towel ring square	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) Single Towel Ring Square. (Premium Range)	each	2167.00
25.53		Soap dish holder	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) Soap Dish Holder. (Premium Range)	each	1191.00
25.54		Soap dispenser	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) Soap Dispenser with Metallic Bottle. (Premium Range)	each	3653.00
25.55		Tumbler holder	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) Tumbler Holder. (Premium Range)	each	1214.00
25.56		Toilet paper holder	Providing and fixing in position best Indian Stainless steel made (0.3 micron Chrome and 10 micron Nickel plated) Toilet paper holder with Flap. (Premium Range)	each	1888.00
25.57		Toilet Roll Holder	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) Toilet Roll Holder. (Premium Range)	each	1354.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
25.58		Glass Self 600mm	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) Glass Self-size 600mm with toughened glass (Premium Range)	each	1934.00
25.59		Double hook	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) Double coat Hook. (Premium Range)	each	1005.00
25.60		Towel Rake 600 mm	Providing and fixing in position best Indian Stainless steel made (0.3 micron Chrome and 10 micron Nickel plated) Towel Rake 600 mm. (Premium Range)	each	4066.00
25.61		W. C. brush holder	Providing and fixing in position best Indian C. P. Brass made (0.3 micron Chrome and 10 micron Nickel plated) W. C. Brush Holder. (Premium Range)	each	2532.00
25.62		Urinal glass	Providing & Fixing 8mm thick froasted Urinal glass partition size 900x450mm brass made bracket with. (Premium Range)	each	6644.00
25.63		Wall hung basin	Wall Hung Basin: Providing and fixing white vitreous china type Anti germ fluoro-polymer coated wall mounted wash basin with C.I. brackets, C. P. brass single lever basin mixer, 15mm quarter turn angular stop cock, 32 mm C.P. brass waste coupling length 130mm and 32 mm C.P. brass bottle trap size 200x300mm i/c cutting and making good the wall and floors wherever required, all fittings and fixtures etc. complete . (Premium Range)		
	25.63.1	620x480x185 mm	white Vetrious china Anti germ fluoro-polymer coated wall mounted wash basin size 620x480x185 mm. (Premium Range)	each	14729.00
	25.63.2	570x470x190 mm	white Vetrious china Anti germ fluoro-polymer coated wall mounted wash basin size 570x470x190 mm. (Premium Range)	each	15100.00
	25.63.3	595x400x365 mm	white Vetrious china Anti germ fluoro-polymer coated wall hung intergrated wash basin size 495x400x365 mm. (Premium Range)	each	18490.00
	25.63.4	500x440x380 mm	white Vetrious china Anti germ fluoro-polymer coated wall hung intergrated wash basin size 500x440x385 mm. (Premium Range)	each	19234.00
	25.63.5	635x485x305 mm	white Vetrious china Anti germ fluoro-polymer coated wall hung intergrated wash basin size 635x485x305 mm. (Premium Range)	each	25273.00
25.64		Wall hung Wash basin with pillar cock	Wall Hung Basin ; Providing and fixing white vitreous china type Anti germ fluoro-polymer coated wall mounted wash basin with C.I. brackets, 15 mm C. P. brass pillar cock, 15mm quarter turn angular stop cock, 32 mm C.P. brass		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			waste coupling length 130mm and 32 mm C.P. brass bottle trap size 200x300mm i/c cutting and making good the wall and floors wherever required, all fittings and fixtures etc. complete . (Premium Range)		
	25.64.1	620x480x185 mm.	white Vetrious china Anti germ fluoro-polymer coated wall mounted wash basin size 620x480x185 mm. (Premium Range)	each	10617.00
25.65		Sink mixer	Providing and fixing in position best Indian, quarter turn table mounted sink mixer (bonut size 24mm, ceramic disc size 19mm and min. nickel plating 0.10 micron and chrome plating 0.3 micron). (Premium Range)	each	4264.00
25.66		Sink cock	Providing and fixing in position best Indian, quarter turn wall mounted sink cock (bonut size 24mm, ceramic disc size 19mm and min. nickel plating 0.10 micron and chrome plating 0.3 micron). (Premium Range)	each	2125.00
25.67		Shower glass cubical	Shower Glass Cubical: Providing and fixing toughen glass shower enclosure conformed Is 2553 (part 1. 1990) or IS 14900. 2000 or IS 2835. 1987 or IS 14450. 2000 with C. P. brass 5 Year warranted hardware, 19mm dia SS 304 grade stabilizer, all plastic Parts of food grade plastic and all glass shall be toughened at 700 degree Celsius and Anti lime scale treated including painting of fittings, cutting and making good the wall and floors wherever required, all fittings and fixtures etc. complete. (Premium Range)		
	25.67.1	1200mm x1200mm x 1950mm	Providing and fixing corner entry 6 mm toughen glass made to measurement shower enclosure size 1200mm x1200mm x 1950mm with aluminum profile on sides, shower enclosure with door sliding on both sides. (Premium Range)	each	46339.00
	25.67.2	Hight:2000 mm Width: 1500mm to 2000mm.	Providing and fixing 8 mm toughen glass wall to wall sliding shower enclosure with one fixed glass and one silding door with a Towel Rod Fixed on the fixed glass size Hight:2000 mm Width: 1500mm to 2000mm. (Premium Range)	each	46668.00
	25.67.3	Hight: 2000mm Width 800mm to 1200mm	Providing and fixing 8 mm toughen glass made to measurement door Openable (inside/outside) shower enclosure two parts Wall to Wall enclosure with one fixed glass and openable door fixed with wall with square hardware & profile glass size Hight: 2000mm Width 800mm to 1200mm. (Premium Range)	each	45410.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	25.67.4	Hight: 1950mm Width 1000mm x 1000mm	Providing and fixing Corner entry half round shower enclosure with curved sliding door - 4 parts Finish: Black Matt Glass: 6mm (Premium Range)	each	34376.00
	25.67.5	Hight: 1950mm Width 1200mm x 1200mm	Providing and fixing Corner entry half round shower enclosure with curved sliding door - 4 parts Finish: Black Matt Glass: 6mm (Premium Range)	each	36699.00
	25.67.6	Hight: 1950mm Width 900mm	Providing and fixing Specifications Free walk-in shower enclosure with stabilizer - 1 part Glass: 8mm / 10mm Finish: Chrome Size - H: 1950 W: 900mm (Premium Range)	each	20786.00
	25.67.7	Hight: 1950mm Width 1000mm	Providing and fixing Specifications Free walk-in shower enclosure with stabilizer - 1 part Glass: 8mm / 10mm Finish: Chrome Size - H: 1950 W: 1000mm (Premium Range)	each	21367.00
25.68		Shower panel anti finger	providing and fixing shower penal Anti finger point with ABS Movable tops Shower, Cascade shower, 2 moveable ABS body massage jets, Hand shower with fiexible hose,Brass Sout, Thermostatic mixer with diverter, 40 cm stainless steel braided hose, Minimum water pressure required 2 bar.	each	47845.00
25.69		White tempered glass shower penal	providing and fixing white tempered glass shower penal with special Surface body materials tempered glass with glass special surface PVC back. ABS Movable tops Shower, 4 moveable ABS body massage jets, Hand shower with fiexible hose,Brass Sout, shelf, Thermostatic mixer with diverter, 40 cm stainless steel braided hose, Minimum water pressure required 2 bar.	each	46683.00
25.70		Thermostat mixer	Thermostat Mixer : Providing and fixing in position best Indian, concealed high flow bath and shower mixer with thermostatic control cartridge (standard temperature range 15 degree c to 60 degree c, cartridge made of high performance thermostatic Polymer material with Anti scaled function). (Premium Range)	each	15123.00
25.71		Shower mixer exposed)	Providing and fixing in position best Indian, exposed wall mounted bath and shower mixer with thermostatic control cartridge with button attachment on spout with connecting legs and wall flange (standard temperature range 15 degree c to 60 degree c, cartridge made of high performance thermostatic Polymer material with Anti scaled function) .(Premium Range)	each	10756.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
25.72		Shower mixer (wall mounted)	Providing and fixing in position best indian, eko exposed shower mixer (wall mounted) with thermostatic control cartridge, 115mm long bend pipe on lower side, connecting legs and wall flange (standard temperature range 15 degree c to 60 degree c, cartridge made of high performance thermostatic Polymer material with Anti scaled function) . (Premium Range)	each	7457.00
25.73		Shower mixer (tub mounted)	Providing and fixing in position best Indian, exposed shower mixer (tub mounted) with thermostatic control cartridge, with button attachment on spout with straight legs (standard temperature range 15 degree c to 60 degree c, cartridge made of high performance thermostatic Polymer material with Anti scaled function). (Premium Range)	each	11313.00
25.74		4 - way diverter	Providing and fixing in position best Indian, 4 - way diverter consisting of concealed body and wall flange for multiple shower working with one shut-off position and 3 outlets for body shower, head shower and spout (standard temperature range 15 degree c to 60 degree c, cartridge made of high performance thermostatic Polymer material with Anti scaled function). (Premium Range)	each	7917.00
25.75		Control valve	Providing and fixing in position best Indian, thermostatic concealed high flow control valve with wall flange (standard temperature range 15 degree c to 60 degree c, cartridge made of high performance thermostatic Polymer material with Anti scaled function) . (Premium Range)	each	11185.00
25.76		Medical Faucets washbasin	Medical Faucets : Providing and fixing in position best indian, single lever surgical purpose elbow action basin mounted mixer with extended Operating lever without popup waste system with 450mm long braided hoses (cartage size32mm, ceramic disc size 19mm and min. nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn). (Premium Range)	each	3929.00
25.77		Pillar cock elbow action	Providing and fixing in position best Indian, surgical purpose elbow action pillar cock with extended Operating lever.(bonut size 24mm, ceramic disc size 19mm and min. nickel plating 0.10 micron and chrome plating 0.3 micron, quarter turn) (Premium Range)	each	2210.00
25.78		Pillar cock wall mounted	Providing and fixing in position best Indian, surgical purpose elbow action wall mounted pillar cock with extended Operating lever and wall flange. (bonut size 24mm, ceramic disc size 19mm and min. nickel plating 0.10 micron and chrome	each	2303.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			plating 0.3 micron, quarter turn). (Premium Range)		
25.79		Grab bar	Providing and fixing in position best Indian, 692mm long, satin finish C. P. Brass grab bar made of 304 grade ss i/c all nuts and screw (0.3 micron chrome and 10 micron nickel plated). (Premium Range)	each	7413.00
25.80		Grab bar with left/right	Providing and fixing in position best Indian, satin finish, C. P. Brass grab bar with left/right 90 degree angle, made of 304 grade ss i/c all nuts and screws. (0.3 micron chrome and 10 micron nickel plated). (Premium Range)	each	9504.00
25.81		Grab bar vertical swing	Providing and fixing in position best Indian, C. P. Brass grab bar vertical swing, 304 grade ss made all nuts and screws. (0.3 micron chrome and 10 micron nickel plated). (Premium Range)	each	18332.00
25.82		Urinal partition frosted glass	Providing and fixing 8mm thick frosted safety glass Urinal partition size 900mm height and 450mm width, fixed with C P brass bracket of appropriate size i/c cutting and making good the wall and floors wherever required with all fittings and fixtures etc. complete.(Premium Range)	each	7631.00
25.83		Full height urinal partition frosted glass	Providing and fixing 8mm thick frosted safety glass Full height urinal partition. Finish: Chrome Glass: 8mm. Height : 1950mm with C P brass bracket of appropriate size i/c cutting and making good the wall and floors wherever required with all fittings and fixtures etc. complete.(Premium Range)	each	11813.00
25.84		Hand dryer 1850 w	Providing and fixing Nuovo dualflow touch-free infrared hand dryer with intelligent LED display, 1850W rated power, DC brushless motor & double HEPA filter Material: ABS Finish: Silver (Premium Range)	each	47504.00
25.85		Hand dryer 1150 w	Providing and fixing Flash touch-free infrared hand dryer with 1150W rated power & carbon brush motor Material: 304 Stainless Steel Finish: Stainless Steel Finish (Premium Range)	each	9502.00
25.86		Paper Towel Dispensers Towels	Providing and fixing Paper Towel Dispensers Towels With C /Z Folds - Wall Mounted, Capacity: 400-600 C / Z towels, Material: AISI 304 Stainless Steel, Finish: Satin (Premium Range)	each	7371.00
25.87		Automatic Soap Dispensers	Providing and fixing Automatic Soap Dispensers, Capacity : 0.8L, Material: Aluminium /ABS, Finish : Chrome/Black (Premium Range)	each	3933.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
25.88		Single Piece slim in-wall cistern with Full Frame	Providing and fixing Single Piece Slim In-wall Cistern with Floor Mounting Frame, Installation Kit and "S-Type" Drain Pipe Connection Set for Wall Hung WC (with Flush Control Plate) (Premium Range)	each	12761.00
25.89		Single Piece slim in-wall cistern with Half Frame	Providing and fixing Single Piece Slim Concealed Cistern with Wall Mounting Frame, Installation Kit and "S-Type" Drain Pipe Connection Set for Wall Hung WC (with Flush Control Plate) (Premium Range)	each	10322.00
25.90		Single Piece slim in-wall cistern	Providing and fixing FLUSHING SYSTEM Single Piece Slim Concealed Cistern Body With Installation Kit (Premium Range)	each	6954.00

Note :- All the items and brand/make of chapter 25 (Sanitary installation and water supply (Premium Range)) Shall be executed only after the prior approval of Chief Engineer

CHAPTER – 26.0

FIRE FIGHTING SYSTEM

LIST OF BUREAU OF INDIAN STANDARD CODES
(CHAPTER -26: FIRE FIGHTING SYSTEM)

S. No.	I.S. No.	Subject
1.	IS 1239 (Part-I) :1990	Mild steel tubes, tubular and other wrought steel fittings.
2.	IS 3589-2001	Steel pipe for water and sewerage.
3.	IS 5290-1993 (Reaffirmed 1998)	Landing valves specification (Hydrant valve) Third Revision)
4.	IS 13095-1991 (Reaffirmed 2003)	Butterfly valves for General purposes.
5.	IS 5312 (Part-I) 2004	Swing check type reflux. (Non return) valve for water works purposes specification. (Part-1 Single door pattern) (Part-II Multiple door pattern)
6.	IS 636-1988	Non-Percolating flexible fire fighting Delivery hose.
7.	IS 884-1985	First aid hose reed for fire fighting.
8.	IS 12585 (1988)	Specification for thermoplastic hoses for water - General Purpose.
9.	IS 904-1983	Specification for 2-way and 3-way suction collecting, heads for fire fighting purposes.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
26.1		'C' class heavy duty MS pipe in underground	Providing laying, testing and commissioning of 'C' class heavy duty MS pipe confirming to IS 1239/3589 underground i/c fittings like elbows, tees, flanges, tapers, jointing with nuts bolts, gaskets and welding etc. underground including excavation and providing cement concrete blocks as supports, at prescribed intervals and anticorrosive treatment with coal tar/ asphalts tape as IS 10221, with 4mm thick fiber reinforced tape and 12mm overlap and refilling the trench etc. of the following sizes complete asdirected by Engineering-in-Charge.		
	26.1.1	200 mm dia	200 mm dia	metre	3098.00
	26.1.2	150 mm dia	150 mm dia	metre	2216.00
	26.1.3	100 mm dia	100mm dia	metre	1466.00
	26.1.4	80 mm dia	80mm dia	metre	1073.00
26.2		'C' class heavy duty MS pipe in on wall/ceiling	Providing laying, testing and commissioning of 'C' class heavy duty MS pipe confirming to IS 1239/3589 i/c fittings like elbows, tees, flanges, reducer, jointing with nuts bolts, gaskets and welding etc. and fixing the pipe on wall/ceiling with suitable hangers, clamps, supports, as required and painting with two or more coats of synthetic. Enamel paint of required shade complete as directed by Engineering-in-Charge.		
	26.2.1	200 mm dia	200 mm dia	metre	3086.00
	26.2.2	150 mm dia	150 mm dia	metre	2203.00
	26.2.3	100 mmdia	100 mm dia	metre	1527.00
	26.2.4	80 mm dia	80 mm dia	metre	1060.00
	26.2.5	65 mm dia	65 mm dia	metre	854.00
	26.2.6	50 mm dia	50 mm dia	metre	676.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	26.2.7	40 mm dia	40mm dia	metre	509.00
	26.2.8	32 mm dia	32mm dia	metre	451.00
	26.2.9	25 mm dia	25mm dia	metre	356.00
26.3		Single headed hydrant valve	Providing and fixing single headed hydrant valve with instantaneous female Gun metal couplings ofrequired dia with cast iron wheel ISI marked conforming to IS 5290 (Type -A) with blank gunmetal cap and chain,adopter as required complete as directed by Engineer-in-Charge.		
	26.3.1	63mm dia	63mm dia	each	6700.00
	26.3.2	52mm dia	52mm dia	each	4690.00
	26.3.3	38mm dia	38mm dia	each	2990.00
26.4		Double headed hydrant valve	Providing and fixing double headed hydrant valve with instantaneousfemale Gun metal couplings of required dia with cast iron wheel ISI marked conforming to IS 5290 (Type -A) with blankgunmetal cap and chain, adopter as required complete as directed by Engineer-in-Charge.		
	26.4.1	63mm dia	63mm dia	each	9980.00
	26.4.2	52mm dia	52mm dia	each	6890.00
	26.4.3	38mm dia	38mm dia	each	4750.00
26.5		Flanged type sluice valve	Providing fixing, including testing and commissioning the following size of flanged type sluice valve of cast iron body and bronze/ gun metal seat confirming to class PN 1.6 of IS: 14846/IS: 780 complete with bolts and nuts, rubber insertions etc. as per specification complete as required.		
	26.5.1	200mm dia	200mm dia	each	5996.00
	26.5.2	150mm dia	150mm dia	each	4497.00
	26.5.3	100mm dia	100mm dia	each	2998.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	26.5.4	80mm dia	80mm dia	each	2399.00
	26.5.5	65mm dia	65mm dia	each	1950.00
	26.5.6	50mm dia	50mm dia	each	1500.00
26.6		Butterfly valve	Providing, fixing, testing and commissioning of butterfly valve PN 1.6, with Bronze/Gunmetal seat duly ISI marked complete with Nuts, Bolts, washers, gaskets, confirming to IS 13095, of following sizes as required.		
	26.6.1	200mm dia	200 mm dia	each	8530.00
	26.6.2	150mm dia	150 mm dia	each	6815.00
	26.6.3	100mm dia	100 mm dia	each	4500.00
	26.6.4	80mm dia	80 mm dia	each	3400.00
	26.6.5	65mm dia	65 mm dia	each	2515.00
	26.6.6	50mm dia	50 mm dia	each	1925.00
26.7		Orifice plate	Providing and fixing orifice plate made of 6mm thick stainless steel withorifice of required size in between flanged and landing valve of external and internal hydrant to reduce pressureto working pressure of 3.5 kg/cm complete as per specification required.	each	1195.00
26.8		Dual plate non-return valve	Providing, installation, testing and commissioning of dual plate non-return valve of following sizes confirming to IS 5312 complete with rubber gasket, GI bolts, nuts, washers etc. as required.		
	26.8.1	200mm dia	200 mm dia	each	9775.00
	26.8.2	150mm dia	150 mm dia	each	7350.00
	26.8.3	100mm dia	100 mm dia	each	4900.00
	26.8.4	80mm dia	80 mm dia	each	4000.00
	26.8.5	65mm dia	65 mm dia	each	3250.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	26.8.6	50mm dia	50 mm dia	each	2499.00
26.9		Stainless steel strainer	Providing, installation, testing and commissioning stainless steel strainer fabricated out of 1.6 mm thick stainless steel sheet with 3 mm dia holes with stainless steel flange.		
	26.9.1	80mm dia	80 mm dia	each	4570.00
	26.9.2	100mm dia	100 mm dia	each	6631.00
26.10		Gun metal valves	Providing, installation, testing and commissioning of gun metal valves of following sizes as required.		
	26.10.1	100mm dia	100 mm dia	each	3240.00
	26.10.2	80mmdia	80 mm dia	each	2592.00
	26.10.3	65mm dia	65 mm dia	each	2106.00
	26.10.4	50mm dia	50 mm dia	each	1620.00
	26.10.5	25mm dia	25 mm dia	each	810.00
26.11		RRL hose pipe	Supplying and fixing dia,15m long, RRL hose pipe with following diamale and female gun metal couplings duly binded with GI wire, rivets etc. confirming to IS 636 (type-A)as required.		
	26.11.1	70mm dia	70 mm dia	each	6100.00
	26.11.2	63mm dia	63 mm dia	each	5510.00
	26.11.3	50mm dia	50 mm dia	each	4423.00
	26.11.4	38mm dia	38 mm dia	each	3324.00
26.12		First -Aid hose reel	Providing and fixing first -Aid hosereel with MS construction spray painted in post office red, confirming to IS 884 with up to date amendments,complete with the following asrequired.) **(a)** 30-36m long 20mm (nominal	each	6065.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			internal) dia water hose thermoplastic (textile reinforced) type -2 as per IS 12585 **(b)** 20mm (nominal internal) dia gun metal globe valve and nozzle. **(c)** Drum and brakets for fixing the equipments on wall. **(d)** Connections from riser with 40 mm dia stop valve (gun metal) and M.S. pipe		
26.13		Hose cabinet 900x600x500mm	Providing and fixing hose cabinet of size 900x600x500mm made of 2 mm thick MS sheet with 4 mm thick float glass doors in front painted "FIRE" in red paint i/c necessary locking arrangement suitable to accommodate external hydrant with butterfly valve 2 nos. 15m Long hose pipe, 1 no branch pipe mounted on wall or raised brick platform& duly painted with post office red externally and white internally with synthetic enamel paint complete in all respects for external hydrant, as directed by Engineer-in-Charge.	each	8570.00
26.14		Hose cabinet size 700x600x300mm	Providing and fixing hose cabinet of size 700x600x300mm made of 16 gauge thick MS sheet with 4 mm thick float glass doors in front painted "FIRE" in red paint i/c necessary locking arrangement suitable to accommodate external hydrant with butterfly valve 2 nos. 15m Long hose pipe, 1 no branch pipe mounted onwall or raised brick platform& dulypainted with post office red externally and white internally with synthetic enamel paint complete in all respects for external hydrant, as directed by Engineer-in-Charge.	each	4000.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
26.15		Gun metal branch pipe with nozzle	Providing and fixing 63mm gun metal branch pipe with 20mm (nominal internal diameter)size gun metal nozzle conforming to IS 903, suitable for instantaneous connection, to interconnect hose pipe coupling as required.	each	2510.00
26.16		4 way fire brigade connection	Providing and fixing 4way fire brigade connection (FBC)of cast iron body with 4 nos. gun metal male instantaneous inlet couplings complete with cap and chain as required. For 150mm dia MS pipe connection, confirming to IS 904 as required.	each	14150.00
26.17		2 way fire brigade connection	Providing and fixing 2 way fire brigade connection (FBC) of CI body with2 nos. gun metal male instantaneous inlet couplings complete with cap and chain as required. For 150mm dia MS pipe connection, confirming to IS 904 as required.	each	6000.00
26.18		Air vessel	Providing and fixing air vessel made of 250mm dia, 8mm thick MS sheet, 1200mm in height with air release valve on top and flanged connection to riser, drain arrangement with 25 mm dia Gunmetal wheel valve, with required accessories, pressure guage and painting with synthetic enamel paint of approved shade as required .	each	10150.00
26.19		Quartzoid bulb type sprinklers	Providing, fixing testing and commissioning of 15mm sizequartzoid bulb type sprinklers, set to operate at 68/79 degree Centigrade. Pendent type/upright type/side wall type with required accessories complete as directed by Engineer-in- Charge.	each	477.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
26.20		Control valve of cast iron body	Providing, fixing, testing and commissioning of installation control valve of Cast iron body and brass/bronze working parts comprising of water motor alarm, bronze seat clapper, and clapper arm, hydraulically driven mechanical going bell to sound continuous alarm when the Wet riser/Sprinkler system activates, pressure gauges, emergency, releases, strainer, pressure switch, cock valve complete with drain valve and bypass, test control box, ballvalves, MS pipe of required sizeflanges, orifice plate, gasket etc. of size 150 mm dia and obtaining NOC from local bodies etc required.	each	42500.00
26.21		Rosette plate for sprinkler	Providing and installation of adjustable Rosette plate for sprinkler as per direction of Engineer-in-Charge.	each	110.00
26.22		Electrically operated flow indicating switches in sprinkler branch line	Providing and fixing electrically operated flow indicating switches in sprinkler branch line on each floor with necessary junction box installedin accessible place. (Wiring from switches to panel not included).	each	3865.00
26.23		Pressure switch	Providing and fixing Pressure switch in the MS pipe Line including connection etc. as required. Capacity 4.5 kg.	each	1485.00
26.24		Flow switches	Providing and fixing flow switches in 100/150 mm dia MS pipe. Capacity 5.0 kg.	each	3525.00
26.25		Pressure transducer	Providing and fixing pressure transducer (dial type pressure gauge with isolation cock) in MS pipe line (dial diametre 100mm pressure range 0-10 kg/cm) suitable for 12 volts/24DC. Including connection etc. as required.	each	1510.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
26.26		Angle iron door frame and MS sheet cum glass shutter	Providing and fixing angle iron (40 x 40 x 5mm) door frame and M.S. sheet (2mm thick) cum glass shutter of size 1.2 x 2.1 m (N.S.) with 25 x 25 x 3 mm angle frame all around & stiffened in between including hinges, handle, locking arrangement, painting with approved synthetic enamel paint including sign writing on glass at internal hydrant including providing and fixing M.S. Sheet 2 mm thick on remaining portion above door to close opening including painting etc. as required.	each	7045.00
26.27		Carbon-di-oxide type fire extinguishers	Providing and fixing of carbon-di-oxide type fire extinguishers consisting of welded M.S. cylindrical body, squeeze lever discharge valve fitted with pressure indicating gauge internal discharge tube 30 cm long high pressure discharge hose,discharge nozzle, suspension bracketconforming to IS : 15683 finished externally with red enamel paint and fixed to wall with brackets complete with internal charge.		
	26.27.1	2 kg	Capacity 2.0 kg	each	6960.00
	26.27.2	3 kg	Capacity 3.0 kg	each	8160.00
	26.27.3	4.5 kg	Capacity 4.5 kg	each	11351.00
26.28		ABC Power type fire extinguishers	Providing and fixing of ABC Power type fire extinguishers consisting of welded M.S. cylindrical body, squeeze lever discharge valve fitted with pressure indicating gauge internal discharge tube 30 cm long high pressure discharge hose, discharge nozzle, suspension bracket. Conforming to IS: 15683 finished externally with red enamel paint fixed to wall with brackets complete with internal charge.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	26.28.1	1.0 kg	Capacity 1.0 kg	each	1560.00
	26.28.2	2.0 kg	Capacity 2.0 kg	each	1800.00
	26.28.3	4.0 kg	Capacity 4.0 kg	each	4091.00
	26.28.4	6.0 kg	Capacity 6.0 kg	each	4629.00
	26.28.5	9.0 kg	Capacity 9.0 kg	each	5493.00
	26.28.6	Refilling of extinguishers	Refill only 4 kg	each	1276.00
	26.28.7	Refilling of extinguishers	Refill only 6 kg	each	1753.00
	26.28.8	Refilling of extinguishers	Refill only 9 kg	each	2142.00
26.29		Portable Automatic Fire Prevention Apparatus	Providing and fixing of Portable (FCO) Automatic Fire Prevention Apparatus very high quality bio degradable, ozone friendly UL (Underwriters Laboratories)listed, built in HDPE (High-density polyethylene) container to automatically extinguish fire when comes in contact with fire light in weight, compact in size extremelyeasy to use ABC powder type withautomatic and manual operated CE certified with latest EN 1568-3:2008 & EN 1568-4:2008. The product shall be cover 5 years warranty as per guideline.		
	26.29.1	1 kg weight	1 Kg. weight Automatic Fire Extinguisher.	nos	5958.00
	26.29.2	1.5 kg weight	1.5 Kg. weight Automatic Fire Extinguisher.	nos	7486.00
	26.29.3	2.00 kg weight	2.00 Kg. weight Automatic Fire Extinguisher.	nos	8348.00

CHAPTER – 27.0

MISCELLANEOUS

MISCELLANEOUS
(CHAPTER NO. 27)

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
27.1		Video shooting of buildings	Providing and carrying out video shooting of buildings, including hiring of vehicle equipments for video shooting, titling, lightening, mixing, lettering, editing, including cost of two colour CDs of approved makes and quality etc. complete.		
	27.1.1	Within 50 kmsfrom dist. H.Q.	Within 50 kms from dist. H.Q.	per day	2500.00
	27.1.2	Beyond 50 kmsfrom dist. H.Q.	Beyond 50 kms from dist. H.Q.	per day	2700.00
	27.1.3	Work upto 5 hoursper day only	Work upto 5 hours per day only	per day	1500.00
27.2		Video shooting of buildings by drone in 4k resolution	Providing and performing video shooting and still photographs by drone in 4k resolution (FHD 1920X1080,24/25/30/ 48/50/60 frames to UHD 4096X2160, 24/25 frames) with 1.0 km horizontal and 300m vertical movement in any side of any building, ground or campusincluding hiring of vehicle equipments for video shooting, titling, lightening, mixing, lettering, editing, including cost of digital data (4k video) excluding costof hard copies etc. complete. (Pay for hard copies as per item no. 27.3.2)		
	27.2.1	Within 50 kms	Within 50 kms from dist. H.Q.	per sqkm	8700.00
	27.2.2	Beyond 50 kms	Beyond 50 kms from dist. H.Q.	per sqkm	8950.00
	27.2.3	Add extra for additional movement	Add extra for every additional 1.0 km horizontal movement at same site.	per sqkm	5500.00
27.3		Still photographs	Shooting of still photographs outdoor and indoor of construction and other works of building, bridges, roads etc. at any distances. By camera Post Card size		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	27.3.1	4"x6"	4"x6" size	3 nos.	75.00
	27.3.2	5"x7"	5"x7" size	3 nos.	135.00
	27.3.3	8"x12"	8"x12" Size (A-4 Size)	3 nos.	210.00
	27.3.4	20"x30"	20"x30" size (Banner Size)	1 no.	400.00
	27.3.5	Extra Copies	Extra Copies : Post Card size		
	27.3.5.1	4"x6"	4"x6" size	1 no	12.00
	27.3.5.2	5"x7"	5"x7" size	1 no	15.00
	27.3.5.3	8"x12"	8"x12" Size (A-4 Size)	1 no	50.00
	27.3.5.4	20'x30'	20'x30' size (Banner Size)	1 no	200.00
27.4		Name plate letter embossing on S.S.plate	Providing and fixing Stainless steel plate 304 grade of 1.50mm thick, including writing with Embossing of letters of required size including fixing at required placed with studs, screws and rawl plugs etc. compete as per direction of Engineer-in-Charge.	sqm	13652.00
27.5		Name plate computerized writing on S.S. plate	Providing and fixing of name plate for offices name of Stainless steel plate 304 grade of 1.50mm thick, including writing with computerized vinayle sheet letters of required size including fixing at required placed with studs, screws and rawl plugs etc. compete as per direction of Engineer- in-Charge.	sqm	11926.00
27.6		Name plate computerized writing on ACPfixing on pipe	Providing and fixing Aluminum composite panels Silver colour with matte finish, 3mm thick, (0.25mm 2.5mm and 0.25mm) sheet for name plate including writing with computerized vinayle sheet letters of required size including fixing at required placed with 1.25m Chromium plated Steel Pipe 15mm dia, compete as per direction of Engineer-in-Charge.	sqm	7287.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
27.7		Name plate computerized writing on ACPfixing with studand screws	Providing and fixing Aluminum composite panels Silver colour with matte finish, 3mm thick, (0.25mm 2.5mmand 0.25mm) sheet for name plate including writing with computerized vinayle sheet letters of required size including fixing at required placed with studs, screws and rawl plugs etc. competeas per direction of Engineer-in-Charge.	sqm	6180.00
27.8		S.S letters	Providing and fixing stainless steel English (Capital) and Hindi letters in standard pattern bellow type made with 1.20 mm thick stainless steel (ASI Grade-304) Depth of letter shall not be less than 30mm and rate includes welding/soldering of joints polishing, buffing and fixing on the wall, plate and beam surface at any height with necessary nut and bolts/fastener whereveris required complete as per direction of Engineer-in-Charge.	per letter per cm height	75.00
27.9		External thermal insulation	Providing and fixing of external thermal insulation and composite system with First layer of self-extinguishing type Expanded Polystyrene (EPS) insulation boards of 120 mm thick (max 1mX0.5m section), confirming to IS 4671:1984, having thermal conductivity of 0.034 W/mK, (measured as per IS 3346- 1980), density of 20-24 kg/m³ measured as per IS 5688-1982, Fire retardant property self-extinguishing type as per EN 13501- 1, bonded with special polymer modified cementitious adhesive confirming toEOTA ETAG 004 (European Technical Approval) formulated to bond polystyrene insulation boards to typical mineral substrate (according to ETAG004) and Polypropylene mechanical fasteners with plastic pin confirming to EOTA ETAG 014 (European Technical	sqm	3086.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Approval) having dia 10mm & L=200mm on finished level wall and the junction between two adjacent EPS boards to be sealed with low expansion moisture cure Polyurethane Foam.Second layer consists of Fiberglass mesh covered with alkali-resistant coating, mass per unit area > 145 gm/m2 , mesh size: 3.9x4.0 mm ±10% embedded in special polymer modified cementitious Base Coat with hydrophobes and the corners will be protected with Corner- beads with alkali-resistant mesh wings at least 10 cm wide, mesh mass per unit area min 145 gm/m2 . The surface will belevelled, finished, made smooth complete in all respect as per manufactures specification and as per directions of Engineer-in-Charge.		
27.10		Precast BoundaryWall	Providing and fixing of prestressed vertically casted concrete panels (both surfaces should be smooth and having grooves on both faces) of M-40 grade of concrete of size 2100x300x50 mm having minimum 4 nos of high tensilesteel wire as per IS 6006:2014 of dia 3 mm placed horizontally in full length of panel with prestressed precast concrete column of M-40 grade of size3000x150x150 mm having minimum 7 nos of high tensile steel wire as per IS 6006:2014 of dia 3 mm placed vertically in full length to erect precast boundary wall panels to height 2100 mm from GL including excavation of pit of size 750x750x900 mm in all kind of soil except rock for erection of columns and grouting the pit with M-20 grade of concrete. Manufacturing, transportation	metre	3478.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			and installation at site all complete as per the direction of Engineer-in-Charge.		
27.11		Add extra for additional height	Extra for every additional height of 0.3 mup to height of wall 3 m	metre	289.00
27.12		Acrylic multilingual Braille directional signboards	Providing and fixing acrylic multilingual Braille (raised dots) directional signboards of size 6" x9 " as per accessibility standards and having 3mm thick acrylic base plate with Upper Case San Serif words white acrylic non glare cut out letters of height 15mm raised not less than 0.8mm above base plate and the word written in Hindi with Devanagari non glare acrylic letters of height 15mm raised not less than above base plate and having a non-glare acrylic cut arrow pointing in the required direction. Each signboard as per manufacturer's specifications With 5 year replacement warranty, installed on the wall at the approved location and upto a height of 1200mm, as per prevailing NBC guidelines with entire satisfaction of the Engineer-in-charge.	each	1736.00
27.13		Tactile layout plan	Providing and fixing tactile layout plan of size 900mmX600mm as approved design with 5 year replacement warranty, made of 3mm thick acrylic non glare base plate and non-glare cut-out symbols with grade One/Two Braille on which individual facility points, like inquirey, reception, restrooms, medical rooms/first aid, cafeteria or breakout area, lounge, meeting room, conference room etc present the repective floors with pictograph wherever needed all in tactile raised by minimum of 1 to 2 mm in various colours to make it legible for seniors, partial vision impaired etc. All the facility names should be in	each	26101.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			braille, along with English. The either movable or fixed for easy reading. The map to be installed at specified locations and installed at a height of upto 1200mm from ground level, With 5 year replacement warranty as per prevailing NBC guidelines with entire satisfaction of the Engineer-in-charge.		
27.14		Tactile audio pictorial building layout	Providing and fixing of acrylic, non-glare, tactile audio pictorial building layout as per the approved size, design with 5 year replacement warranty, having 3mm thick acrylic non glare base plate of the approved colour and non-glare acrylic cut out symbols raised 3mm above base plate and Grade 2/ Grade 1 Braille to be integral with the sign face and be raised 0.5mm above base plate. The map to be installed at the specified location as per accessibility standards, shall be oriented to the position of the viewer and to be installed at a height of 800 mm from the finished ground/ floor level and using the necessary hardware specified by the manufacturer and to be installed up to the height upto 1200 mm from the finished ground/ floor level and using the necessary hardware specified by the manufacturer and to be installed complete as per design/specifications and as per prevailing NBC guidelines with entire satisfaction of the Engineer-in-charge.		38176.00
27.15		Fixing tactile of Photoluminescent Emergency evacuation map	Providing and fixing tactile of Photoluminescent Emergency evacuation map in Braille for each floor indicating the current position of the viewer saying "YOU ARE HERE" in Braille indicating the emergency exits, locations of fire extinguishers etc. The size of the board shall be 600 x 900mm. With 5 year replacement warranty and shall be installed	each	25521.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			at a height of 1200mm from FFL and shall be oriented to the position of the viewer as per the direction of the site engineer complete as per design/specifications and guidelines as per the entire satisfaction of the the Engineer-in-charge		
27.16		Conspicuity tape	Providing and fixing of conspicuity tape 50mm width, on the treads in staircase including, cutting, pasting etc. As per harmonized guidelines complete in all respect to the entire satisfaction of Engineer-in-charge.	metre	118.00
27.17			Gate Façade: Designing, Providing, Fixing/ Installation & Dismantling (If Required) of the Gate Façade signage as per the recce done. Items required for the signage include the following–WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degree bendable and should carry that warranty of minimum 10 years. Graphics film - Graphic film applied on ACP should be high quality Cast vinyl of thickness not more than 60 Microns without adhesive when measured using ASTM standard of measurements. The Adhesive of the film should be a Grey coloured pigmented adhesive and it shall stick, without the use of an activator such as solvents or heat on any metallic and painted, primed, polished and Clean surfaces free from any grease or silicone without producing wrinkles, rolling up, tearing or detaching. The adhesive of the film should be a pressure activated adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the	Sqm	26058.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC and its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The Name of the manufacturer / brand should be clearly displayed on the liner. Overlaminate - The graphic film must be protected using Matt finished high-performance cast over- laminate with not more than 2mil thickness. Laminate must be designed for Flat and smooth application. It should also possess Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product qualifies Flame Spread Index 15, smoke development index 10 as per product flammability test conducted by ASTM-E84 standard method. This product does not exceed the maximum concentration values (MCVs) for phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category 1-7, 10-11 products and after July		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. Printing - UV Wide Format Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6 colour printing process - Cyan, Magenta, Yellow, Black, Light Cyan, Light Magenta Inks to maintain excellent print quality and colour production consistency. Recommended printer must possess GREENGUARD / GREENGUARD GOLD Certified Inks from OEM Supplier. Ink Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3 years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage.		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Coloured Translucent Film for channel letters – Energy efficient or equivalent self-adhesive cast translucent vinyl of approved colours / design to be processed in a dust free application room. The translucent film should have clear adhesive and to be designed for backlit LED application. The coloured vinyl should of cast manufacturing and meeting to pantone / RAL shade confirmed by customer in reflected light conditions. The vinyl thickness (without adhesive) should not more than 60 Microns without adhesive when measured using IS / ASTM standards. The Adhesive of the film should clear and pressure sensitive. The liner of the Film should Synthetic, lay-flat liner that is not affected by water or humidity. The coloured vinyl should comply with UL 94 for Flammability & UL 746 Plastic Material Properties. Flammability - should confirm to class A rating for fire retardant as per ASTM E84 standard - (Standard Test Method for Surface Burning Characteristics of Building Materials) and pass DIN EN 13501-1 (DIN 4102) – (Fire Test to Building Material). The film should be ROHs complaint and should not Contain EU REACH Substances of Very High Concern (SVHC) at >0.1% (EU Directive 2011/65/EU). The surface preparation and the application should be done using manufacturers recommended procedures and the warranty should be provided by manufacturer directly. Powder Coated MS Square Iron Pipe of 1inch of 16-18 gauge for inner framing and strength; Acrylic 5mm-8mm shape (laser/router machine) cut letters along with all ISI approved electrical fittings; Silica		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			for ACP fixing; MS Square Iron Angle 35/5 would be used for strength to the signage. LED Module of 1.2watts per module; with efficacy greater than or equal to 100lm/watt; colour temperature should be 6500K~10000k; Flux per module should be greater than or equal to 120lm; beam angle should be greater than or equal to 160 degree; CRI is in colour range from R1 to R15 and should have minimum CRI of 70 for outdoor luminaire; Current type should be of constant voltage; voltage should be 12VDC; LED Make type should be of ABS body for proper heat dissipation; LED Modules must be connected with AWG 20 Wire and its IP rating should be greater than or equal to IP 66 rating and should have minimum of 3 years warranty. Power Supply Wattage Should be of 30W/60W/100W/150W/200W/300W only; Power Supply Input voltage range Should be between 180～250VAC; Power Supply Input frequency range Should be between 50～60HZ; Power Supply Efficiency Should be greater than or equal to 87%; Power Supply AC inrush current Should be less than or equal to 46A@230VAC; Power Supply Leakage current Should be less than or equal to 0.25mA@230VAC; Power Supply Output Voltage Should be of 12VDC±5%; Power Supply Case Should be of Aluminium casing only; Power Supply Waterproofing SMPS should be Completely Potted; Power Supply Waterproof level Should be minimum IP67 grade; Power Supply BIS Proper BIS of the make must be available; Power Supply Protection Should have Short circuit, Overload & Over temperature protection; Power Supply Safety & EMC IEC61347		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			A1 + A14; EN 55015;EN 61547; EN 61000-3-2; EN61000-3-3; Power Supply Withstand Voltage I/P-F-G: 1.5KV AC I/P-O/P: 3.0KV AC; Power Supply Isolation I/P-O/P:>100M Ohms/500VDC/25°C/75%RH; Power Supply Load should be of Max 80%; Power Supply Warranty should be of at least 3 Years. Both, LED Modules and Power Supply, should be manufactured from same principal manufacturer. Separate 3 - Year Warranty Certificate to be furnished from OEM for LED Modules and Power Supply for each signage		
27.18			Building / Wall Façade: Designing, Providing, Fixing/ Installation & Dismantling (If Required) of Building / Wall Façade signage. Items required for the signage include the following– WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degree bendable and should carry that warranty of minimum 10 years. Graphics film - Graphic film applied on ACP should be high quality Cast vinyl of thickness not more than 60 Microns without adhesive when measured using ASTM standard of measurements. The Adhesive of the film should be a Grey coloured pigmented adhesive and it shall stick, without the use of an activator such as solvents or heat on any metallic and painted, primed, polished and Clean surfaces free from any grease or silicone without producing wrinkles, rolling up, tearing or detaching. The adhesive of the film should be a pressure activated	Sqm	26058.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC and its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The name of the manufacturer / brand should be clearly displayed on the liner Over laminate - The graphic film must be protected using Matt finished high-performance cast over- laminate with not more than 2mil thickness. Laminate must be designed for Flat and smooth application. It should also possess Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product qualifies Flame Spread Index 15, smoke development index 10 as per product flammability test conducted by ASTM-E84 standard method. This product does not exceed the maximum concentration values (MCVs) for phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			As amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category 1-7, 10-11 products and after July 22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. Printing - UV Wide Format Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability. Recommended printer must possess GREENGUARD / GREENGUARD GOLD Certified Inks from OEM Supplier. Ink Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6 colour printing process -Cyan, Magenta, Yellow, Black, Light Cyan, Light Magenta Inks to maintain excellent print quality and colour production consistency.The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3 years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes,		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage. Coloured Translucent Film for channel letters – Energy efficient or equivalent self-adhesive cast translucent vinyl of approved colours / design to be processed in a dust free application room. The translucent film should have clear adhesive and to be designed for backlit LED application. The coloured vinyl should of cast manufacturing and meeting to pantone / RAL shade confirmed by customer in reflected light conditions The vinyl thickness (without adhesive) should not more than 60 Microns without adhesive when measured using IS / ASTM standards. The Adhesive of the film should clear and pressure sensitive. The liner of the Film should Synthetic, lay-flat liner that is not affected by water or humidity. The coloured vinyl should comply with UL 94 for Flammability & UL 746 Plastic Material Properties. Flammability - should confirm to class A rating for fire retardant as per ASTM E84 standard - (Standard Test Method for Surface Burning Characteristics of Building Materials) and pass DIN EN 13501-1 (DIN 4102) – (Fire Test to Building Material). The film should be ROHs complaint and should not Contain EU REACH Substances of Very High Concern (SVHC) at >0.1% (EU Directive 2011/65/EU). The surface preparation and the application should be done using manufacturers recommended procedures and the warranty should be provided by manufacturer directly. Powder Coated MS Square Iron Pipe of 1inch of 18 gauge for inner framing and strength; Acrylic 5mm-		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			8mm shape (laser/router machine) cut letters along with all ISI approved electrical fittings; Silica for ACP fixing; MS Square Iron Angle 35/5 would be used for strength to the signage. LED Module of 1.2watts per module; with efficacy greater than or equal to 100lm/watt; colour temperature should be 6500K~10000k; Flux per module should be greater than or equal to 120lm; beam angle should be greater than or equal to 160 degree; CRI is in colour range from R1 to R15 and should have minimum CRI of 70 for outdoor luminaire; Current type should be of constant voltage; voltage should be 12VDC; LED Make type should be of ABS body for proper heat dissipation; LED Modules must be connected with AWG 20 Wire and its IP rating should be greater than or equal to IP 66 rating and should have minimum of 3 years warranty. Power Supply Wattage Should be of 30W/60W/100W/150W/200W/300W only; Power Supply Input voltage range Should be between 180～250VAC; Power Supply Input frequency range Should be between 50～60HZ; Power Supply Efficiency Should be greater than or equal to 87%; Power Supply AC inrush current Should be less than or equal to 46A@230VAC; Power Supply Leakage current Should be less than or equal to 0.25mA@230VAC; Power Supply Output Voltage Should be of 12VDC±5%; Power Supply Case Should be of Aluminium casing only; Power Supply Waterproofing SMPS should be Completely Potted; Power Supply Waterproof level Should be minimum IP67 grade; Power Supply BIS Proper BIS of the make must be available; Power Supply Protection Should have Short circuit, Overload & Over temperature		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			protection; Power Supply Safety & EMC IEC61347 A1 + A14; EN 55015;EN 61547; EN 61000-3-2; EN61000-3-3; Power Supply Withstand Voltage I/P-F-G: 1.5KV AC I/P-O/P: 3.0KV AC; Power Supply Isolation I/P-O/P:>100M Ohms/500VDC/25°C/75%RH; Power Supply Load should be of Max 80%; Power Supply Warranty should be of at least 3 Years. Both, LED Modules and Power Supply, should be manufactured from same principal manufacturer. Separate 3 - Year Warranty Certificate to be furnished from OEM for LED Modules and Power Supply for each signage.		
27.19			Terrace Signage / Building Hoarding / Hoarding signage: Designing, Providing, Fixing/ Installation & Dismantling (If Required) of Terrace Signage / Building Hoarding / Hoarding signage. Items required for the signage include the following– WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years' warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degrees bendable and should carry that warranty of minimum 10 years. Graphics film - Graphic film applied on ACP should be high quality Cast vinyl of thickness not more than 60 Microns without adhesive when measured using ASTM standard of measurements. The Adhesive of the film should be a Grey coloured pigmented adhesive and it shall stick, without the use of an activator such as solvents or heat on any metallic and painted, primed, polished and Clean surfaces free from any grease or silicone	Sqm	26058.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Without producing wrinkles, rolling up, tearing or detaching. The adhesive of the film should be a pressure activated adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC and its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The name of the manufacturer / brand should be clearly displayed on the liner. Overlaminate - The graphic film must be protected using Matt finished high-performance cast over- laminate with not more than 2mil thickness. Laminate must be designed for Flat and smooth application. It should also possess Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product qualifies Flame Spread Index 15, smoke development index 10 as per product flammability test conducted by ASTM-E84 standard method. This product does not exceed the maximum concentration values (MCVs) for		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category 1-7, 10-11 products and after July 22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. Printing - UV Wide Format Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability. Recommended printer must possess GREENGUARD / GREENGUARD GOLD Certified Inks from OEM Supplier. Ink Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6 colour printing process - Cyan, Magenta, Yellow, Black, Light Cyan, Light Magenta Inks to maintain excellent print quality and colour production consistency. The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3 years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage. Coloured Translucent Film for channel letters – Energy efficient or equivalent self-adhesive cast translucent vinyl of approved colours / design to be processed in a dust free application room. The translucent film should have clear adhesive and to be designed for backlit LED application. The coloured vinyl should of cast manufacturing and meeting to pantone / RAL shade confirmed by customer in reflected light conditions. The vinyl thickness (without adhesive) should not more than 60 Microns without adhesive when measured using IS / ASTM standards. The Adhesive of the film should clear and pressure sensitive. The liner of the Film should Synthetic, lay-flat liner that is not affected by water or humidity. The coloured vinyl should comply with UL 94 for Flammability & UL 746 Plastic Material Properties. Flammability - should confirm to class A rating for fire retardant as per ASTM E84 standard - (Standard Test Method for Surface Burning Characteristics of Building Materials) and pass DIN EN 13501-1 (DIN 4102) – (Fire Test to Building Material). The film should be ROHs complaint and should not Contain EU REACH Substances of Very High Concern (SVHC) at >0.1% (EU Directive 2011/65/EU). The surface preparation and the application should be done using manufacturers recommended procedures and the warranty should be		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			provided by manufacturer directly. Powder Coated MS Square Iron Pipe of 1inch of 18 gauge for inner framing and strength; Acrylic 5mm-8mm shape (laser/router machine) cut letters along with all ISI approved electrical fittings; Silica for ACP fixing; MS Square Iron Angle 35/5 would be used for strength to the signage. LED Module of 1.2watts per module; with efficacy greater than or equal to 100lm/watt; colour temperature should be 6500K~10000k; Flux per module should be greater than or equal to 120lm; beam angle should be greater than or equal to 160 degree; CRI is in colour range from R1 to R15 and should have minimum CRI of 70 for outdoor luminaire; Current type should be of constant voltage; voltage should be 12VDC; LED Make type should be of ABS body for proper heat dissipation; LED Modules must be connected with AWG 20 Wire and its IP rating should be greater than or equal to IP 66 rating and should have minimum of 3 years warranty. Power Supply Wattage Should be of 30W/60W/100W/150W/200W/300W only; Power Supply Input voltage range Should be between 180～250VAC; Power Supply Input frequency range Should be between 50～60HZ; Power Supply Efficiency Should be greater than or equal to 87%; Power Supply AC inrush current Should be less than or equal to 46A@230VAC; Power Supply Leakage current Should be less than or equal to 0.25mA@230VAC; Power Supply Output Voltage Should be of 12VDC±5%; Power Supply Case Should be of Aluminium casing only; Power Supply		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Waterproofing SMPS should be Completely Potted; Power Supply Waterproof level Should be minimum IP67 grade; Power Supply BIS Proper BIS of the make must be available; Power Supply Protection Should have Short circuit, Overload & Over temperature protection; Power Supply Safety & EMC IEC61347 A1 + A14; EN 55015;EN 61547; EN 61000-3-2; EN61000-3-3; Power Supply Withstand Voltage I/P-F-G: 1.5KV AC I/P-O/P: 3.0KV AC; Power Supply Isolation I/P-O/P:>100M Ohms/500VDC/25°C/75%RH; Power Supply Load should be of Max 80%; Power Supply Warranty should be of at least 3 Years. Both, LED Modules and Power Supply, should be manufactured from same principal manufacturer. Separate 3 - Year Warranty Certificate to be furnished from OEM for LED Modules and Power Supply for each signage.		
27.20			Pylon: Designing, Providing, Fixing/ Installation & Dismantling (If Required) of the signage as per the recce done. Items required for the signage include the following– (Viz. Taking Size of Each Unit = (8*4) 32 Sqft which is equal to 2.97 Sq Meter on one side and 5.94 Sq Meter for both sides) WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degree bendable and should carry that warranty of minimum 10 years. Graphics film - Graphic film applied on ACP should be high quality Cast vinyl of	Each	177797.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Thickness not more than 60 Microns without adhesive when measured using ASTM standard of measurements. The Adhesive of the film should be a Grey coloured pigmented adhesive and it shall stick, without the use of an activator such as solvents or heat on any metallic and painted, primed, polished and Clean surfaces free from any grease or silicone without producing wrinkles, rolling up, tearing or detaching. The adhesive of the film should be a pressure activated adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC and its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The name of the manufacturer / brand should be clearly displayed on the liner. Over laminate - The graphic film must be protected using Matt finished high-performance cast over- laminate with not more than 2mil thickness. Laminate must		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			be designed for Flat and smooth application. It should also possess Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product qualifies Flame Spread Index 15, smoke development index 10 as per product flammability test conducted by ASTM-E84 standard method. This product does not exceed the maximum concentration values (MCVs) for phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category -7, 10-11 products and after July 22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. Printing - UV Wide Format Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability. Recommended printer must possess GREENGUARD / GREENGUARD GOLD Certified Inks from OEM Supplier. Ink Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6 colour printing process - Cyan, Magenta, Yellow, Black, Light Cyan, Light Magenta		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Inks to maintain excellent print quality and colour production consistency. The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3 years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage. Coloured / Printable Translucent Film for channel letters – Energy efficient or equivalent self-adhesive cast translucent vinyl of approved colours / design to be processed in a dust free application room. The translucent film should have clear adhesive and to be designed for backlit LED application. The coloured vinyl should of cast manufacturing and meeting to pantone / RAL shade confirmed by customer in reflected light conditions. The vinyl thickness (without adhesive) should not more than 60 Microns without adhesive when measured using IS / ASTM standards. The Adhesive of the film should clear and pressure sensitive. The liner of the Film should Synthetic, lay-flat liner that is not affected by water or humidity. The coloured vinyl should comply with UL 94 for Flammability & UL 746 Plastic Material Properties. Flammability - should confirm to class A rating for fire retardant as per ASTM E84 standard - (Standard Test Method for Surface Burning Characteristics of Building Materials) and		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			pass DIN EN 13501-1 (DIN 4102) – (Fire Test to Building Material). The film should be ROHs complaint and should not Contain EU REACH Substances of Very High Concern (SVHC) at >0.1% (EU Directive 2011/65/EU). The surface preparation and the application should be done using manufacturers recommended procedures and the warranty should be provided by manufacturer directly. The film to be protected using Matte Over Laminate as per the specification prescribed above. Powder Coated MS Square Iron Pipe of 1inch of 18 gauge for inner framing and strength to withstand the wind pressure as per IS standards. MS Square Iron Pipe of 100mm x 50mm steel structure to be designed and fabricated as per requirements. The top portion of to have 5mm PC / PET G from reputed make such as Bayer / Spectar and should be solar grade. Acrylic 5mm-8mm shape (laser/router machine) cut letters along with all ISI approved electrical fittings; Silica for ACP fixing; MS Square Iron Angle 35/5 would be used for strength to the signage. LED Module of 1.2watts per module; with efficacy greater than or equal to 100lm/watt; colour temperature should be 6500K~10000k; Flux per module should be greater than or equal to 120lm; beam angle should be greater than or equal to 160 degree; CRI is in colour range from R1 to R15 and should have minimum CRI of 70 for outdoor luminaire; Current type should be of constant voltage; voltage should be 12VDC; LED Make type should be of ABS body for proper heat dissipation; LED Modules must be connected with AWG 20 Wire and its IP rating should be greater than or equal to IP 66 rating and should		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Have minimum of 3 years warranty. Power Supply Wattage Should be of 30W/60W/100W/150W/200W/300W only; Power Supply Input voltage range Should be between 180～250VAC; Power Supply Input frequency range Should be between 50～60HZ; Power Supply Efficiency Should be greater than or equal to 87%; Power Supply AC inrush current Should be less than or equal to 46A@230VAC; Power Supply Leakage current Should be less than or equal to 0.25mA@230VAC; Power Supply Output Voltage Should be of 12VDC±5%; Power Supply Case Should be of Aluminium casing only; Power Supply Waterproofing SMPS should be Completely Potted; Power Supply Waterproof level Should be minimum IP67 grade; Power Supply BIS Proper BIS of the make must be available; Power Supply Protection Should have Short circuit, Overload & Over temperature protection; Power Supply Safety & EMC IEC61347 A1 + A14; EN 55015;EN 61547; EN 61000-3-2; EN61000-3-3; Power Supply Withstand Voltage I/P-F-G: 1.5KV AC I/P-O/P: 3.0KV AC; Power Supply Isolation I/P-O/P:>100M Ohms/500VDC/25°C/75%RH; Power Supply Load should be of Max 80%; Power Supply Warranty should be of at least 3 Years. Both, LED Modules and Power Supply, should be manufactured from same principal manufacturer. Separate 3 - Year Warranty Certificate to be furnished from OEM for LED Modules and Power Supply for each signage.		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
27.21			**Name Change Board** / Room No. Plates: Designing, Providing,Fixing/ Installation & Dismantling (If Required) of Name Change Board / Room No. Plate's signage as per the recce done. Items required for the signage include the following WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degree bendable and should carry that warranty of minimum 10 years. Graphic Film - Graphic film applied on ACP should be high quality Polymeric Calendar vinyl of thickness not more than 82 Microns without adhesive. The Adhesive of the film should be a Grey coloured pigmented adhesive and it should be a pressure activated adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC and its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with	Sqm	11262.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The name of the manufacturer / brand should be clearly displayed on the liner. Over Laminate - The graphic film must be protected using Matt finished calendar over-laminate with not more than 3.5 mils thickness. Laminate must be designed for Flat and smooth application. It also possesses Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product does not exceed the maximum concentration values (MCVs) for phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category 1-7, 10-11 products and after July 22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. LATEX Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability. Recommended printer must possess Certified Inks from OEM Supplier nk Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6 colour printing process - Cyan, Magenta, Yellow, Black,		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Light Cyan, Light Magenta Inks to maintain excellent print quality and colour production consistency. The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3 years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage.		
27.22			Lollipop: Designing, Providing, Fixing/ Installation & Dismantling of the signage as per the recce done. Items required for the signage include the following (Lollipop) – (Viz. Taking Size of Each Unit = (3*3) 9 Soft which is equal to 0.84 Sq. Meter) Powder Coated MS Square Iron Pipe Angle 2inch x 2inch of 18 gauge double density for framing and strength; MS Angle Pipe of 8inch x 4inch of 14 gauge; Permanent fixture with Cement Mortar; Provision for flexible substrate with L angle provided on both side of lollipop for tensioning. Flexible Substrate - Flexible substrate should be of high quality white pigmented vinyl and of GSM greater than 450. The thickness should be around 0.45mm and whiteness more than 100. Flexible Substrate should be translucent, flexible and with gloss finish surface and its property should be mechanically tension able/ digitally printable with Solvent Print and a warranty minimum 1 year. The Flexible substrate should offer typically	Each	51475.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP The Total solution (Flexible Substrate + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 1 years from manufacturer, recorded and supplied in E- document form valid for flexible substrate printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage. LED Module of 1.2watts per module; with efficacy greater than or equal to 100lm/watt; colour temperature should be 6500K~10000k; Flux per module should be greater than or equal to 120lm; beam angle should be greater than or equal to 160 degree; CRI is in colour range from R1 to R15 and should have minimum CRI of 70 for outdoor luminaire; Current type should be of constant voltage; voltage should be 12VDC; LED Make type should be of ABS body for proper heat dissipation; LED Modules must be connected with AWG 20 Wire and its IP rating should be greater than or equal to IP 66 rating and should have minimum of 3 years warranty. Power Supply Wattage Should be of 30W/60W/100W/150W/200W/300W only; Power Supply Input voltage range Should be between 180～250VAC; Power Supply Input frequency range Should be between 50～60HZ; Power Supply Efficiency Should be greater than or equal to 87%; Power Supply AC inrush current		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Should be less than or equal to 46A@230VAC; Power Supply Leakage current Should be less than or equal to 0.25mA@230VAC; Power Supply Output Voltage Should be of 12VDC±5%; Power Supply Case Should be of Aluminium casing only; Power Supply Waterproofing SMPS should be Completely Potted; Power Supply Waterproof level Should be minimum IP67 grade; Power Supply BIS Proper BIS of the make must be available; Power Supply Protection Should have Short circuit, Overload & Over temperature protection; Power Supply Safety & EMC IEC61347 A1 + A14; EN 55015;EN 61547; EN 61000-3-2; EN61000-3-3; Power Supply Withstand Voltage I/P-F-G: 1.5KV AC I/P-O/P: 3.0KV AC; Power Supply Isolation I/P-O/P:>100M Ohms/500VDC/25°C/75%RH; Power Supply Load should be of Max 80%; Power Supply Warranty should be of at least 3 Years. Both, LED Modules and Power Supply, should be manufactured from same principal manufacturer. Separate 3 - Year Warranty Certificate to be furnished from OEM for LED Modules and Power Supply for each signage.		
27.23			Door / Wall Header / Wall Signage / Staircase Signage / Lift Board Header / Lift Signages / Pillar Signage / Push Pull Signage / Notice Board signage : Designing, Providing, Fixing/ Installation & Dismantling (If Required) of Door / Wall Header / Wall Signage / Staircase Signage / Lift Board Header / Lift Signages/ Pillar Signage / Push Pull	Sqm	11262.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Signage / Notice Board signage as per the recce done. Items required for the signage include the following – WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degree bendable and should carry that warranty of minimum 10 years. Graphic Film - Graphic film applied on ACP should be high quality Polymeric Calendar vinyl of thickness not more than 82 Microns without adhesive. The Adhesive of the film should be a Grey coloured pigmented adhesive and it should be a pressure activated adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC and its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The name of the manufacturer / brand should be clearly displayed on the liner. Over Laminate - The graphic film must be protected using Matt finished		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Over-laminate with not more than 3.5 mils thickness. Laminate must be designed for Flat and smooth application. It also possesses Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product does not exceed the maximum concentration values (MCVs) for phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category 1-7, 10-11 products and after July 22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. LATEX Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability. Recommended printer must possess Certified Inks from OEM Supplier. Ink Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6 colour printing process - Cyan, Magenta, Yellow, Black, Light Cyan, Light Magenta Inks to maintain excellent print quality and colour production consistency.The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage		
27.24			Hanging Name Board/ Hanging Signage: Designing, Providing, Fixing/ Installation & Dismantling (If Required) of the signage as per the recce done. Items required for the signage include the following – WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degree bendable and should carry that warranty of minimum 10 years. Graphic Film - Graphic film applied on ACP should be high quality Polymeric Calendar vinyl of thickness not more than 82 Microns without adhesive. The Adhesive of the film should be a Grey coloured pigmented adhesive and it should be a pressure activated adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC	Sqm	12822.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			And its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The name of the manufacturer / brand should be clearly displayed on the liner. Over Laminate - The graphic film must be protected using Matt finished calendar over-laminate with not more than 3.5 mils thickness. Laminate must be designed for Flat and smooth application. It also possesses Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product does not exceed the maximum concentration values (MCVs) for phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category 1-7, 10-11 products and after July 22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. LATEX Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability.		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Recommended printer must possess Certified Inks from OEM Supplier. Ink Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6 colour printing process - Cyan, Magenta, Yellow, Black, Light Cyan, Light Magenta Inks to maintain excellent print quality and colour production consistency. The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3 years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage.		
27.25			Garden Signages: Designing, Providing, Fixing/ Installation & Dismantling (If Required) of the signage as per the recce done. Items required for the signage include the following– WASH COAT Aluminium Composite Panel (ACP) with thickness 3mm, coil thickness 0.25 (Top & Bottom) of 10 years warranty; ACP Alloy Grade AA1100; Tensile strength Min 110 Newton; ACP Core Spec should be of 1st recycled LDPE and 360 degree bendable and should carry that warranty of minimum 10 years	Sqm	15251.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Graphics film - Graphic film applied on ACP should be high quality Cast vinyl of thickness not more than 60 Microns without adhesive when measured using ASTM standard of measurements. The Adhesive of the film should be a Grey coloured mpigmented adhesive and it shall stick, without the use of an activator such as solvents or heat on any metallic and painted, primed, polished and Clean surfaces free from any grease or silicone without producing wrinkles, rolling up, tearing or detaching. The adhesive of the film should be a pressure activated adhesive providing repositionability and has in built air release channels to provide a bubble free application. The liner of the Film should be a high- quality Paper liner which has a polyethylene coating on both sides to provide excellent lay flatness while converting. The Graphic film should not contain any Heavy metals in its construction and should be compliant with the limits of RoHS Directive 2002/95/EC and its Subsequent amendments. The Film should not contain at greater than 0.1% by weight a Substance of Very High Concern (SVHC) substance identified according to Article 59 of REACH. The film should have fire retardant properties and should have a flame spread index of 5 or less and Smoke spread index of 0 when tested as per ASTM E 84 standard of flammability testing. The film should also comply with ASTM E 162-94 flammability standard and should have a Flame spread index of less than 1.90. The name of the manufacturer / brand should be clearly displayed on the liner. Over laminate - The graphic film must be		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Protected using Matt finished high-performance cast over- laminate with not more than 2mil thickness. Laminate must be designed for Flat and smooth application. It should also possess Resistance to mild alkalis, mild acids and salt, additionally resists occasional fuel spills. This product qualifies Flame Spread Index 15, smoke development index 10 as per product flammability test conducted by ASTM-E84 standard method. This product does not exceed the maximum concentration values (MCVs) for phthalates set under EU Directive 2011/65/EU(RoHS recast/RoHS 2), as amended by EU 2015/863, which applies to finished EEE after July 22, 2019 for Category 1-7, 10-11 products and after July 22, 2021 for Category 8 and 9 products. This means that each of the homogenous materials within this product does not exceed the MCV of 0.1% (by weight) for each of the following phthalates: DEHP, BBP, DBP and DIBP. Printing - UV Wide Format Printing with Inks manufactured by Media manufacturer or Media manufacturer's Co-branded Inks on Wide Format Digital Printer designed for indoor & outdoor application with excellent print quality and weather durability. Recommended printer must possess GREENGUARD / GREENGUARD GOLD Certified Inks from OEM Supplier. Ink Layer should not peel, crack or fade away from base film used in the intended print application. Inks must be designed for indoor/ outdoor application with excellent print durability. Print Substrate can be printed with 600 dpi printing mode for excellent quality printing. Ink configuration should be 6		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			colour printing process - Cyan, Magenta, Yellow, Black, Light Cyan, Light Magenta Inks to maintain excellent print quality and colour production consistency. The Total solution (Vinyl + Over Laminate) during the Performance Life as mentioned in the specification is backed by warranty for 3 years from manufacturer, recorded and supplied in E- document form valid for film printed using a recommended printing platform applied with a suitable over-laminate film for protection and then installed. The warranty covers failures against fading, cracking, peel off, blistering, excessive dimensional changes, manufacturing defect except for physical vandalism, arson, natural calamities, or intentional damage. Colour Translucent Film for channel letters – Energy efficient or equivalent self-adhesive cast translucent vinyl of approved colours / design to be processed in a dust free application room. The translucent film should have clear adhesive and to be designed for backlit LED application. The coloured vinyl should of cast manufacturing and meeting to pantone / RAL shade confirmed by customer in reflected light conditions. The vinyl thickness (without adhesive) should not more than 60 Microns without adhesive when measured using IS / ASTM standards. The Adhesive of the film should clear and pressure sensitive. The liner of the Film should Synthetic, lay-flat liner that is not affected by water or humidity. The coloured vinyl should comply with UL 94 for Flammability & UL 746 Plastic Material Properties. Flammability - should confirm to class A rating for fire retardant as per ASTM E84 standard - (Standard Test		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Method for Surface Burning Characteristics of Building Materials) and pass DIN EN 13501-1 (DIN 4102) – (Fire Test to Building Material). The film should be ROHs complaint and should not Contain EU REACH Substances of Very High Concern (SVHC) at >0.1% (EU Directive 2011/65/EU). The surface preparation and the application should be done using manufacturers recommended procedures and the warranty should be provided by manufacturer directly.		
27.26			Internal Glass Film : Designing, Providing, Fixing/ Installation of highly durable decorative translucent vinyl film for interior glass finish. It should come with pressure sensitive, acrylic and permanent adhesive. The dusted crystal film should have thickness of approx. 3.2 mils (81 microns) and frosted crystal film should have thickness of approx. 4.7 mils (120 microns). The liner of the film should be Silicone-coated polyester with thickness of approx. 3.6 mils (90 microns). The film should have adhesion strength of 18N/25mm on glass / polycarbonate / acrylics after 24 hours of application. The tensile strength of dusted crystal film should be 23N/25mm at 23°C and for frosted film tensile strength should be 15N / 25mm at 23°C. When used in interior applications on glass, products should have Class A rating as per ASTM E84 (as defined by NFPA 101 “Life Safety Code”). Dusted and frosted crystal glass finishes should give the uniform appearance with a dusted or frosted sparkle effect and suitable for interior and exterior glass surfaces. The crystal glass finishes should be used only on glass, acrylic, polycarbonate surfaces.	Sqm	2100.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			This film can be plotter cut in as per customized design. The film should have warranty of 15 years for non-perimeter glass and 5 years for perimeter glass (no warranty will be applicable on printed product).		
27.27			Window Film: Providing, Fixing/ Installation of solar window film. The film should be a high performance, abrasion resistant film which when applied to the interior window surface should reduce the gain of solar heat energy through the window. The film shall be an optically clear non- metalized polyester film manufactured with multi-layered optical film technology and incorporating an acrylic pressure sensitive adhesive on one side. The film shall have a nominal thickness of 2.0 mils (0.002 inches). The film should be multi-layered film with VLT: 69%, SHGC: 0.51, Exterior Visible Reflection: 8%, Interior Visible Reflection: 8%, TSER: 49%, U value: 0.99 (Btu/(hr.) (ft^2) (°F), UV rejection 99.9% and Glare reduction 22%. This film should have 10 years of warranty from OEM for vertical application	Sqm	9626.00
27.28			Outdoor Totem - Providing and fixing Totem with pattern less micro prismatic sheeting comprising of customised modular structure, aluminium composite panel as base, special fungus resistance rebound able prismatic retro sheeting confirming to S2 clauses of ASTM D4956 for flexibility and impact resistance of reloadable sheeting and UV overlay film. (Viz. Taking Size of Each Unit = (8*4) 32 Sqft which is equal to 2.97 Sq Meter) The aluminium composite panel (ACP) shall have a thickness of 3mm and shall be	Each	188912.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Fixed on surface along with structure. On the front face of the ACP, fungus resistance rebound able flexible micro prismatic sheet shall be pasted. The approved theme, graphics shall be printed on fungus resistance rebound able flexible micro prismatic sheet with green latex ink as per the recommendation and approved methodology of OEM. Powder Coated MS Square Iron Pipe of 1inch of 18 gauge for inner framing and strength to withstand the wind pressure as per IS standards. MS Square Iron Pipe of 100mm x 50mm steel structure to be designed and fabricated as per requirements. The top portion of to have 5mm PC / PET G from reputed make such as Bayer Spectar and should be solar grade. Acrylic 5mm-8mm shape (laser/router machine) cut letters along with all ISI approved electrical fittings; Silica for ACP fixing; MS Square Iron Angle 35/5 would be used for strength to the signage. LED Module of 1.2watts per module; with efficacy greater than or equal to 100lm/watt; colour temperature should be 6500K~10000k; Flux per module should be greater than or equal to 120lm; beam angle should be greater than or equal to 160 degree; CRI is in colour range from R1 to R15 and should have minimum CRI of 70 for outdoor luminaire; Current type should be of constant voltage; voltage should be 12VDC; LED Make type should be of ABS body for proper heat dissipation; LED Modules must be connected with AWG 20 Wire and its IP rating should be greater than or equal to IP 66 rating and should have minimum of 3 years warranty. Power Supply Wattage Should		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			be of 30W/60W/100W/150W/200W/300W only; Power Supply Input voltage range Should be between 180～250VAC; Power Supply Input frequency range Should be between 50～60HZ; Power Supply Efficiency Should be greater than or equal to 87%; Power Supply AC inrush current Should be less than or equal to 46A@230VAC; Power Supply Leakage current Should be less than or equal to 0.25mA@230VAC; Power Supply Output Voltage Should be of 12VDC±5%; Power Supply Case Should be of Aluminium casing only; Power Supply Waterproofing SMPS should be Completely Potted; Power Supply Waterproof level Should be minimum IP67 grade; Power Supply BIS Proper BIS of the make must be available; Power Supply Protection Should have Short circuit, Overload & Over temperature protection; Power Supply Safety & EMC IEC61347 A1 + A14; EN 55015;EN 61547; EN 61000-3-2; EN61000-3-3; Power Supply Withstand Voltage I/P-F-G: 1.5KV AC I/P-O/P: 3.0KV AC; Power Supply Isolation I/P-O/P:>100M Ohms/500VDC/25°C/75%RH; Power Supply Load should be of Max 80%; Power Supply Warranty should be of at least 3 Years. Both, LED Modules and Power Supply, should be manufactured from same principal manufacturer. Separate 3 - Year Warranty Certificate to be furnished from OEM for LED Modules and Power Supply for each signage. Design, Providing & Fixing the totem sign made out of specially designed backlit retro reflective type XI sheeting which shall be fixed over polycarbonate UV sheet and over the 3mm thick Aluminium composite panel. The		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			totem shall be backlit with the help of LED Module of 0.72 W- 12V		
27.29			Variable Massage Sign- Board- Supply, Installation and Commissioning for Single sided real time multi colour LED with DIP 10mm Pitch +/- 10% or better; having an active Area of 5.76m x 3.84m. IP65 Front & IP65Rear, Max Power Consumption in LED screens shall be 600w per cabinet with average power consumption of 300w-350w per cabinet. Operating temptature of the cabinets needs to be –20℃ to+60℃. Supplies for LED Screen to be complete with 4 sqmm n copper cables, CAT 6 network cables , Windows based media Player with 8GB RAM , 64GB storage, IP rated enclosure for equipment . Installation Commissioning of required accessories and player application to include the unique license for each screen location which shall communicate with central server over encrypted protocols to ensure secure remote operations and remote diagnostic. Local Player access to be enabled only on usage of encrypted Hardware lock. The real time LED shall be played with cloud based real time content through the software and not by any pen drive or local wired connection. All Supplied material of LED screens to have comprehensive 1 year warranty from date of supply. Remote monitoring of individual screen for FRC cable and SMPS faults by popups on server application UPS with 1 Hour Back-up Earth work in rough excavation, banking excavated earth in layers not exceeding 20cm in depth, breaking clods watering, rolling each layer with ½ tons roller or	Each	4907338.0

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			wooden or steel rammers and rolling every 3rd and top-most layer with power roller of minimum 8 tons and dressing up in embankments for roads, flood banks, Marginal banks and guide banks or filling up ground depressions, lead up to 50 m and lift up to 1.5m. Providing and laying in position cement concrete of specified grade excluding the cost of cantering and shuttering All work up to plinth level (Cement concrete grade M-20 (Nominal Mix) with 20 mm maximum size of stone aggregate) Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses/frame work etc. including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete. (Hot finished welded type tubes) Reinforcement for R.C.C. work including straightening, cutting, bending, placing in position and binding all complete. Thermo-Mechanically Treated bars. Finishing with Epoxy paint (two or more coats) at all locations prepared and applied as per manufacturer's specifications including appropriate priming coat, preparation of surface, etc. complete. On Steel Work		
27.30			Flexible Medium Marker:- Providing and Fixing of Flexible Median Markers as per IRC 79:2019 which shall be made of combination of tough, high impact resistant moulded thermoplastic body along with florescent yellow colour retro-reflective type XI sheeting with minimum exposed reflective area mm 90 sq. cm and with rebound property. The reflective sheeting should confirm to Type XI Florescent	Each	194.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Yellow sheeting as per IRC 67 2012 and ASTM D4956- 11. The flexible median marker shall have overall height of minimum 180 mm, width of minimum 120 mm, body Thickness of minimum 6.5 mm and shank depth of minimum 30mm. The retro-reflective sheeting shall be on both sides of the Flexible Median Marker and shall be edge protected with no exposed edges which will prevent edge lifting, vandalism, sheeting damage, etc. The edge of sheeting should not come out easily by putting nails, sharp objects etc. The logo of the manufacturer shall be embossed on either side of the body. The product design has to with stand all natural impacts and should not get damaged if any pedestrian stamp on top of it intentionally or unintentionally. The product should bend up to 90 degree on both sides without breakage or damage. The flexible median marker shall be tested from government lab for rebound ability and tensile test as per IRC 79-2019.		
27.31			Printable Micro prismatic Sheeting - Providing and fixing pattern less micro prismatic sheeting comprising of customised modular structure, aluminium composite panel as base including specialized back support frame, special fungus resistance rebound able prismatic retro sheeting Confirming to S2 clauses of ASTM D4956 for flexibility and impact resistance of reloadable sheeting and UV overlay film. The aluminium composite panel (ACP) shall have a thickness of 3mm and shall be fixed on surface along with structure. On the front face of the ACP, fungus resistance rebound able flexible micro prismatic sheet shall be pasted. The approved theme,	Sqm	8042.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Graphics shall be printed on fungus resistance rebound able flexible micro prismatic sheet with green latex ink as per the recommendation and approved methodology of OEM. The rate shall include all materials, labour, equipment's, fabrication, transportation, erection, installation and incidentals all complete as per direction of the Engineer-in-Charge		
27.32			Aluminium Flexible Prismatic – Providing and fixing Aluminium-backed flexible prismatic sheeting consisting of yellow coloured flexible prismatic sheeting with non-metallic prismatic lens as retro reflective elements and conforming to ASTM D 4956-09 Type VI specification values for rebound able sheeting. AFP sheeting shall be of 1ft width and laminated at the back with a 50 micron thick Aluminium (Al) foil with pressure sensitive adhesive and liner. AFP sheeting shall have screen printed arrow/slant line pattern in black colour on top & shall not crack when tested for flexibility as per section S2.2.2 of ASTM D 4956-09. The AFP sheeting shall be applied with a neoprene contact adhesive with Polychloroprene as base & the edges of the product shall be sealed all around with a two-part epoxy based structural adhesive and shall be extremely resistant to peel-off & shall confirm to IRC Accreditation. The AFP shall be provided with govt test report from ARAI/ CRRI and report confirming to coefficient of retro reflection, Flexibility and impact resistance test as per ASTM which shall be submitted to engineer-in-charge before installation/ award of work	Sqm	7535.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
27.33			Preformed Patterned Pavement Marking Tapes for Dry & Wet Reflective as per IRC 35 - Providing and laying of pre-formed patterned pavement marking tapes, with pre-coated self-adhesive, having abrasion-resistant high refractive index microcrystalline ceramic beads bonded to a highly durable top coat, having a nominal thickness of 2.0 mm at pattern heights, meeting the requirements specified in ASTM D4505-12. The patterned surface shall have approximately 50% ± 15% of the surface area raised -and presenting a near vertical face, angled from 0 degree to 60 degree, to traffic From any direction. All microcrystalline ceramic beads shall have a minimum index of refraction of 1.85 and/or 2.4 and ceramic beads shall show resistance to corrosion on exposure to a 1% solution of sulfuric acid. The applied marking surface to be level, uniform and free from streaks, cracks and holes. Markings shall have initial average skid resistance value of at least 45 BPN when tested according to ASTM E303 and meet the initial retro reflectance average values.	Sqm	6269.00
27.34		Installation of Tent with allied facilities	Providing and installation with complete item of tent for a day with safe erection, All responsibility of safety belongs to supplier.		
	27.34.1		MS/ Aluminum Dom water proof	Sqm	323
	27.34.2		Plane Tent	Sqm	194
	27.34.3		Kanat	Sqm	43
	27.34.4		Stage with ladder up to 2 m height	Sqm	215
	27.34.5		Back Drop	Sqm	215
	27.34.6		Fiber Chair with Handle	Each	12.00
	27.34.7		VIP Chair Iron	Each	40.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	27.34.8		Iron Sofa three seater	Each	450.00
	27.34.9		Chemical Bathroom (For VIP)	Each	5000.00
	27.34.10		VIP Chair (Foam Cushion)	Each	100.00
	27.34.11		Carpet Green and Red	Sqm	43.04
	27.34.12		VIP Green Room	Sqm	860.80
	27.34.13		MS Pipe Bericatting in three rows including post all complete	Sqm	193.68
	27.34.14		Center Table with Glass Top(2x4) and wardrob	Each	350.00
	27.34.15		Samai	Each	300.00
	27.34.16		Podiam with complete Mick system	Each	4000.00
	27.34.17		Welcome Gate	Each	2000.00
	27.34.18		Tent ceiling in white Colour and carpet	Sqm	64.56
	27.34.19		Pipe curtain	Sqm	43.04
	27.34.20		MS/WTakath	Each	60.00
	27.34.21		MS Table	Each	40.00
	27.34.22		Table Cover	Each	20.00
	27.34.23		Tin shed	Sqm	161.40
	27.34.24		Empty Drum with 200 Ltr capacity (PVC)	Each	50.00
	27.34.25		back stage decoration	Sqm	1076.00
	27.34.26		Decoration in front of stage by Pots with plants and Nursery	Each	20.00
	27.34.27		Garland for welcome	Each	30.00
	27.34.28		Hand Boque for Welcome	Each	150.00
	27.34.29		Water bottle 20 Ltr	Each	30.00
	27.34.30		Flex banner including printing.	sqm	130.00
27.35			Providing and fixing Inaugural stone / Foundation stone of granite stone/ marble stone (all type of colour and pattern) including embossing, lettering and painting complete as per direction of Engineer –In- Charge.	sqm	4300.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
27.36		50 litres capacity Industrial Model Reverse Osmosis Water-cum-purifier	Supplying ,instalation ,testing and commissioning in position of 50 litres capacity Industrial Model Reverse Osmosis Water-cum-purifier to deliver Potable Water with world Health Organisation (WHO) Standards with following accessories (a) MS Skid – 1No (b) Media 11054 No, Vessel with 20 bb Multiport Valve (with Media Pubble and Fine Sand and Activated Carbon Filling) set (c) Feed Pump 0.50 HP – 1 No. (d) Vertical Pump 2-9, 2.00HP / Single Phase – 1 No (e) Membrane 40 x 40. and Membrane Pressure Vessel 40 x 40 – 1 No. & Pressure Gauge – 2 Nos. (f) Flow Meter – 1 No. 20" Filter with Housing – 1 No., Low Pressure Switch – 1 No., Level Switch – 1 No. & Necessary CPVC Pipes and Fittings. The system shall be provide with Two Panel. Required to complete the Job along with one year warranty and 02 free services which should include mandatory replacement of filters etc.	Each	31540.00
27.37			Supplying installation ,testing and commissioning in position of 250 liters capacity Industrial Model Reverse Osmosis Water-cum-purifier to deliver Potable Water with world Health Organisation (WHO) Standards with following accessories a) MS Skid – 1No (b) Media 11054 No, Vessel with 20 bb Multiport Valve (with Media Pubble and Fine Sand and Activated Carbon Filling) set (c) Feed Pump 0.50 HP – 1 No. (d) Vertical Pump 2-9, 2.00HP / Single Phase – 1 No (e) Membrane 40 x 40 & Membrane Pressure Vessel 40 x 40 – 1 No. & Pressure Gauge – 2 Nos. (f) Flow Meter – 1 No. 20" Filter with Housing – 1 No., Low Pressure Switch – 1 No., Level Switch – 1 No. & Necessary CPVC Pipes and Fittings. The system shall	Each	96000.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			be provide with Two Panel. Required to complete the Job along with one year warranty and 02 free services which should include mandatory replacement of filters etc.		
27.38			Control System with necessary Stainless Supplying and erecting in position of 500 liters capacity Industrial Model Reverse Osmosis Water-cum-purifier to deliver Potable Water with world Health Organisation (WHO) Standards with following accessories (a) MS Skid – 1No (b) Media 11054 No, Vessel with 20 bb Multiport Valve (with Media Pubble and Fine Sand and Activated Carbon Filling) set (c) Feed Pump 0.50 HP – 1 No. (d) Vertical Pump 2-9, 2.00HP / Single Phase – 1 No (e) Membrane 40 x 40 and Membrane Pressure Vessel 40 x 40 – 1 No. & Pressure Gauge – 2 Nos. (f) Flow Meter – 1 No. 20" Filter with Housing – 1 No., Low Pressure Switch – 1 No., Level Switch – 1 No. & Necessary CPVC Pipes and Fittings. The system shall be provide with Two Panel. Required to complete the Job along with one year warranty and 02 free services which should include mandatory replacement of filters etc.	Each	182000.00
27.39			Supplying and erecting in position of 1000 liters capacity Industrial Model Reverse Osmosis Water-cum-purifier to deliver Potable Water with world Health Organisation (WHO) Standards with following accessories (a) MS Skid – 1No (b) Media 11054 No, Vessel with 20 bb Multiport Valve (with Media Pubble and Fine Sand and Activated Carbon Filling) set (c) Feed Pump 0.50 HP – 1 No. (d) Vertical Pump 2-9, 2.00HP / Single Phase - 1 No. (e) Membrane 40 x 40 and Membrane Pressure Vessel 40 x 40 – 1 No. & Pressure	Each	270000.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Gauge –2 Nos. (f) Flow Meter – 1 No. 20" Filter with Housing – 1 No., Low Pressure Switch – 1 No., Level Switch – 1 No. & Necessary CPVC Pipes and Fittings. The system shall be provide with Two Panel. Required to complete the Job along with one year warranty and 02 free services which should include mandatory replacement of filters etc.		
27.40			Control System with necessary Stainless Supplying and erecting in position of 2000 liters capacity Industrial Model Reverse Osmosis Water-cum-purifier to deliver Potable Water with world Health Organisation (WHO) Standards with following accessories (a) MS Skid – 1No (b) Media 11054 No, Vessel with 20 bb Multiport Valve (with Media Pubble and Fine Sand and Activated Carbon Filling) set (c) Feed Pump 0.50 HP – 1 No. (d) Vertical Pump 2-9, 2.00HP / Single Phase– 1 No (e) Membrane 40 x 40 Membrane Pressure Vessel 40 x 40 – 1 No. & Pressure Gauge – 2 Nos. (f) Flow Meter – 1 No. 20" Filter with Housing – 1 No., Low Pressure Switch – 1 No., Level Switch – 1 No. & Necessary CPVC Pipes and Fittings. The system shall be provide with Two Panel. Required to complete the Job along with one year warranty and 02 free services which should include mandatory replacement of filters etc.	Each	409000.00

CHAPTER – 28.0

LANDSCAPING AND HORTICULTURE

LANDSCAPING AND HORTICULTURE
(Chapter-28)

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
28.1		Trenching in ordinary soil	Trenching in ordinary soil up to a depth of60cm including removal and stacking of serviceable materials and then disposing of surplus soil, by spreading and neatly leveling within a lead of 50m and making up the trenched area to proper levels by filling with earth or earth mixed with sludge or/and manurebefore and after flooding trench with water (excluding cost of imported earth, sludge ormanure).	cum	186.00
28.2		Supplying and stacking of good earth	Supplying and stacking of good earth specific useful for plantation at site including royalty if any and carriage (earth measured in stacks will be reduced by 20% for payment).	cum	511.00
28.3		Supplying and stacking sludge	Supplying and stacking sludge at site including royalty and carriage (sludge measured in stacks will be reduced by 8% for payment).	cum	643.00
28.4		Supplying and stacking dung manure (fym) farm yard manure	Supplying and stacking at site dump manure from approved source, including carriage (manure measured in stacks will be reduced by 8% for payment) :	cum	734.00
28.5		Dressing of the trenched ground	Rough dressing the trenched ground including breaking clods.	100 sqm	445.00
28.6		Uprooting weeds	Uprooting weeds from the trenched area after 10 to 15 days of its flooding with water including disposal of uprooted vegetation.	100 sqm	151.00
28.7		Fine dressing of the ground	Fine dressing of the ground	100 sqm	113.00
28.8		Spreading of sludge, dung manure and / or good earth	Spreading of sludge, dump manure and / orgood earth in required thickness as per direction of Engineer-in-charge (Cost of sludge, dump manure and / or good earth to be paid separately).	cum	16.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
28.9		Mixing earth and sludge or manure (fym) farm yard manure	Mixing earth and sludge or manure in the required proportion specified or directed by the Engineer-in-charge.	cum	11.00
28.10		Planting of Grass	Grassing with Selection No.1 grass including watering and maintenance of the lawn for 30 days or more till the grass forms a thick lawn, free from weeds and fit for mowing including supplying good earth, if needed (the grass and good earth shall be paid for separately):		
	28.10.1	In rows 5 cm	In rows 15 cm apart in both directions.	100 sqm	1163.00
	28.10.2	In rows 7.5 cm	In rows 7.5 cm apart in both directions.	100 sqm	2404.00
	28.10.3	In rows 5 cm	In rows 5 cm apart in both directions.	100 sqm	3601.00
	28.10.4	With grass	With grass turf	100 sqm	2856.00
	28.10.5	Grassing with selected grass turf	Providing and laying Selected type of Grassturf with earth 50mm to 60mm thickness on existing ground prepared with proper level and ramming with required tools wooden and than rolling the surface with light roller, making the surface smoothen and light watering the same, as per direction of Engineer-in-Charge.	sqm	99.00
	28.10.6	Grassing with Mexican/Zyosia grass turf	Providing and Laying Mexican/Zyosia carpet grass Turf with earth 50mm to 60mm thickness on existing ground prepared with proper level and ramming with tools wooden (Durmuth) and then rolling the surface with light roller make the surface smoothen and light watering with sprinkler and maintenance for 30 days or more till the grass establishes properly, as perdirection of Engineer-in-Charge.	sqm	310.00
	28.10.7	Seeding beds in existing soil	Preparation of preplanting of seeding beds in existing native soil. The seed or plantation bed top to be thoroughly cleaned, preferably by stone burier machine and to be made free of any stone or foreign material.	sqm	5.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	28.10.8	Seeding by machine	Seeding to be done by high speed precision mechanism machine, calibrated to required seeding density or as per site requirement. The machine should have front end rear rollers. The sown grass seed bed to be maintained for 30 days or germination whichever is earlier. The grass bed to be prepared of Cynadone Dactylon (Bermuda Grass), variety Gobi.	sqm	102.00
	28.10.9	Maintenance of Grassing	Maintenance of Grassing ground done by Seeding Method as mentioned in Item No. 28.10.6 from the stage of germination to firstcut i.e. approximately 45 days.	sqm	40.00
28.11		Renovating lawns	Renovating lawns including weeding, cheeling the grass, forking the ground, top dressing with sludge or manure, mixing the same with forked soil, watering and maintaining the lawn for 30 days or more till the grass forms a thick lawn free from weeds and fit for mowing and disposal of rubbish as directed, including supplying good earth if needed but excluding the cost of sludge or manure (the good earth shall be paid for separately).	100 sqm	1081.00
28.12		Uprooting rank vegetation and weeds	Uprooting rank vegetation and weeds by digging the area to a depth of 60cm removing all weeds and other growth with roots by forking repeatedly, breaking clods, rough dressing, flooding with water, uprooting fresh growths after 10 to 15 days and then fine dressing for planting new grass, including disposal of all rubbish with all leads and lifts.	100 sqm	1861.00
28.13		Preparation of beds for hedging and shrubbery	Preparation of beds for hedging and shrubbery by excavating further 30cm deep of already prepared trench, refilling the excavated earth after breaking clods and mixing with sludge or manure in the ratio of 8:1 (8 parts of stacked volume of earth after reduction by 20% : one part of stacked volume of sludge or manureafter reduction by 8%), flooding with water, filling with earth if necessary, watering and finally fine dressing, levelling etc. including stacking and disposal of materials declared unserviceable and surplus earth by spreading and levelling as directed, within a lead of 50mlift upto 1.5 m complete (cost of sludge, manure or extra earth to be paid for separately).	cum	67.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
28.14		Digging holes refilling mixed with manure or sludge	Digging holes in ordinary soil and refilling the same with the excavated earth mixed with manure or sludge in the ratio of 2:1 by volume (2 parts of stacked volume of earth after reduction by 20% : 1 part of stacked volume of manure after reduction by 8%) flooding with water, dressing including removal of rubbishand surplus earth, if any with all leads and lifts (cost of manure, sludge or extra good earth if needed to be paid for separately) :		
	28.14.1	120 cm dia	Holes 120 cm dia and 120 cm deep.	each	141.00
	28.14.2	60 cm dia	Holes 60 cm dia and 60 cm deep.	each	61.00
	28.14.3	45 cm dia	Holes 45 cm dia and 45 cm deep.	each	19.00
	28.14.4	30 cm dia	Holes 30 cm dia and 30 cm deep.	each	9.00
28.15		Planting different variety of plants	Providing and planting different variety of plants of approved quality and sizes as mentioned including making pits of requiredsize at site, refilled with B.C. Soil mixture manuring and pesticide etc. complete (to be paid separately) including watering and 90 days maintenance from the date of final bill as per direction of Engineer-in-Charge complete in all respect (B.C. Mixture paid separately).		
	28.15.1	Shrubs (75-90 cm height)	Bahunia Tomentosa, Beloperone Species, Caselpinnia Pulcherima, Bird of Paradise, Callandra Emarginata, Callandra Hybrida, Cassia Bi flora, Cassia Laevigata, Cestrum Nocturnum, Dombeya Mastersii, Euphorbia Caracasana, Euphorbia Pulcherima, Excorea bi color, Excorea tri color, Ficuas Blackii, Ficus Reginold, Ficus Panda, Gardenia, Gulphinia, Jasminoides, Hamelia Pattens, Heliconia, Hibiscus Rosasinensis, Hibiscus Verigated, Hibiscus Viceroy, Jatropha Multifida, Largestomia Indica, Malpighia Coccigera, Murraya Exotica, Murraya Koeniggi, Murraya Sambucum, Mussaenda Erythrophylla, Nerium Oleander, Nyctanthus Arbortristis, Plumbego Capensis, Putranjeeva Roxburghii, Tabernaemontana Coronaria, Tabernaemontana Divaricata, Tecoma Gaudi Chaudi, Tecoma Stans, Thevetia Nerifolia, Thuja Compacta and equivalent plants.	sqm	80.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	28.15.2	Creeper plants (height :30cm to 45 cm)	Allamanda Cathartica, Allamanda Grandiflora, Allamanda Violacea, Begnonia Venustha, Boughan Villas (Variety : Butiana, Lady Mary Bearing, Mahaara, Mohan, Scarlet Queen Variagata, Glabra Formosa, Peruviana Odissi, Partha, Shubhra, Thimma, Spectabillis, L.N. Birla, Refulgens), Clerodendrum Splendens, Clerodendrum Thompsonae, Ipomea Purpurea, Jasminum Grandiflorum, Jasminum Humile (Yellow), Passiflora Caerulea (Rakhi Bel), Petrea Volubilis, Quis Qualis Indica, Tecoma Grandiflora, Veronia Elaegnifolia (Curtain Creeper), Mandvellia, Garlic Creeper, Callimitis	each	50.00
	28.15.3	Ground cover plants height: 30cm-45cm	Alpinia Verigated, Alternanthera species, Aspyragus Myerri, Aspyragus Springenii, Aspidistra, Canna (regular, Dwarf), Chlorodendron Inermii, Chlorophytum (Green), Chlorophytum verigated, Coffea Chinensis, Dianella Verigated, Durranta (Goldiana, Green, Verigated), Euphorbia Milli hybrid, Ipomea (Golden leaves), Iresine herbstii, Juniper Prostata, Juniper Africana, Ophiopogon plant, Ophiopogon jaburan, Portulacaria Afra (jade Plant), Schefflera Green, Schefflera verigated, Setcreasea Purpurea, Syngonium (Butterfly) species, Syngonium miniature, Syngonium Dwarf, Syngonium variegated, Tradescantia, Wadelia Trilobata, Zebrina Tradescantia, Pendanas, Spider Lily Black, Spider Lily verigated, Aclypha (Red, Green,Mini), Lantana (Red, Yellow, Purple, White, verigated), Haemalia Mini and similar plants.	each	50.00
	28.15.4	Winter seasonal height: 20-30 cm	Any of one from Alyssum, Anemone, Antirrhinium Hybrid, Aster Hybrid, Begonia and its different varieties, Calendula, Carnation,Coleus, Daisy, Dianthus, Fressia, Gazania, Impatiens, Kalanchoe, Marigold Inca, Missam branthemum, Nemasia, Nasturitium, Ornamental Kale, Pansy Hybrid Sakata,Petunia Hybrids such as Bravo, Star andPicotee, Salvia, Stock, Stock, Verbena	each	42.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	28.15.5	Summer and rainy seasonal plants	Any of one from Celosia, Caladium Hybrid, Cockscomb, Cosmos, Gaillardia, Gomphorena, Kochchea, Portulacca, Sunflower Hybrid, Sunflower Single, Tapiocca Variegated (Manihot Esculenta), Vinca, Zinnia Hybrids.	each	42.00
	28.15.6	Rose plants height: 30cm & above	Budded Roses of H.T variety and miniature Roses.	each	170.00
	28.15.7	Rose plants creepers	Plant Creeper Rose Varieties	each	212.00
	28.15.8	Rose plant H.T. standard	Standard Roses H.T. Variety	each	382.00
28.16		Tree guard (bricks)	Half brick circular tree guard in bricks, internal diametre 1.25 metre and height 1.2 metre above ground and 0.20 m below ground, bottom two courses laid dry and top three courses in cement mortar 1:6 (1 cement : 6 sand) and the intermediate courses being in dry honey comb masonry as per design complete:	each	800.00
28.17		Edging (bricks)	Edging with bricks laid dry length wise, including required excavation, refilling, consolidating with hand packing and spreading surplus earth neatly within a lead of 50 m:		
	28.17.1	Burnt clay	Common burnt clay modular bricks	meter	31.00
	28.17.2	Fly ash	Fly ash bricks	meter	34.00
28.18		Filling mixture of earth and sludge or manure	Filling mixture of earth and sludge or manure in the desired proportion in trenches, flooding with water and levelling (cost of supplying earth and sludge or manure and mixing excluded).	cum	16.00
28.19		Excavation in dumped stones or malba	Excavation in dumped stones or malba including stacking of serviceable and unserviceable material separately and disposal of unserviceable material lead upto 50 m andlift upto 1.5 m disposed material to be neatly dressed.	cum	155.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
28.20		Flooding water and making kiaries	Flooding the ground with water including making kiaries and dismantling the same.	100 sqm	102.00
28.21		Tree guard 45 cm square (M. S.)	Providing and fixing M. S. tree guard 45 cm square in plan, height 1.20 metre above ground level and 0.40 metre below ground level. The vertical members shall consist of four nos. angle iron of size 25x25x3 mm, 1.8 m long,one at Each corner and 8 nos. flat iron of size 25x3 mm, 1.2 m long. The vertical mambers shall be welded to 4 nos. 25x6 mm M. S. flats placed horizontally around the vertical member of the cage. One name plate of 1 mm thick M.S. sheet of size 250x100 mm shall be welded to the tree guard near the middle height and lettered PWD/ any other approved name. The tree guard shall be fixed to the ground by making suitable holes and by embedding four corners leg in the ground, including refillingthe earth, compaction etc. complete. The tree guard shall be painted with two or more coatsof synthetic enamel paint of approved brand and manufacture over a coat of primer, complete in all respect.	each	1426.00
28.22		Tree guard 50 cm square in M. S.	Providing and fixing M. S. tree guard 50 cm square in plan, height 1.40 metre above ground level and 0.50 metre below ground level. The vertical members shall consist of four nos. of angle iron of size 25x25x5 mm 1.9 long, one at Each corner and 8 nos. flat iron of size 25x5 mm 1.4 long. The vertical members shall be welded to 4 nos. 25x6 mm M. S. flats placed horizontally around the vertical member of the cage. One name plate of 1 mm thick M.S. sheet of size 250x100 mm shall be welded to the tree guard near the middle height and lettered PWD/ any other approved name. The tree guard shall be fixed to the ground by making suitable holes and by embedding four corners leg in the ground, including refilling the earth, compaction etc. complete. The tree guard shall be painted with two coats of paintof approved brand and manufacture over a coat of primer, complete in all respect.	each	2245.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
28.23		Preparation of mounds	Preparation of mounds of various size and shape by available excavated /supplied earth in layers not exceeding 20 cm in depth, breaking clods, watering of Each layer, dressing etc. leadupto 50 metre and lift upto 1.5 m complete as per direction of Engineer-in-Charge.	cum	138.00
28.24		Providing Circular Cement Concrete pots	Providing Circular Cement Concrete pots of specified size, cast with cement concrete of nominal mix 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 6 mm nominal size), reinforced with 7 Nos. (3 nos. horizontal and 4 nos. vertical "U" shape) M.S. wires of 3.5 mm diaas per design, including required form work, finishing with cement punning on exposed surface, curing for specified period and stacking in required rows and height, allcomplete as per direction of Engineer-in- Charge.		
	28.24.1	35cm dia	Top inside dia 35 cm, outer bottom dia 25 cm, total height 35 cm with wall thickness of 25.4 mm.	each	137.00
	28.24.2	30 cm dia	Top inside dia 30 cm, outer bottom dia 20 cm, total height 30 cm with wall thickness of 25.4 mm.	each	100.00
28.25		Providing square cement concrete pots	Providing Square Cement Concrete pots of specified size, cast with cement concrete of nominal mix 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 6 mm nominal size), reinforced with 7 Nos. (3 nos. horizontal and 4 nos. vertical "V" shape) M.S. wires of 3.5 mm diaas per design, including required form work, finishing with cement punning on exposed surface, curing for specified period and stacking in required rows and height, allcomplete as per direction of Engineer-in- Charge. Top inner width 35 cm, outer bottom width 25 cm, total height 35 cm and wall thickness 25.4 mm.	each	165.00
28.26		Rotomoulded Circular Planter	Providing of Rotomoulded circular planters of different sizes.		
	28.26.1	30 cm dia	Upper dia 30cm and height 24cm	each	405.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	28. 26.2	40 cm dia	Upper dia 40cm and height 31cm	each	675.00
	28. 26.3	50 cm dia	Upper dia 50cm and height 37cm	each	1147.00
	28. 26.4	60 cm dia	Upper dia 60cm and height 47cm	each	1665.00
	28. 26.5	70 cm dia	Upper dia 70cm and height 55cm	each	2700.00
	28. 26.6	80 cm dia	Upper dia 80cm and height 62cm	each	3510.00
	28. 26.7	90 cm dia	Upper dia 90cm and height 69cm	each	4552.00
	28. 26.8	100 cm dia	Upper dia 100cm and height 74cm	each	5805.00
	28. 26.9	120 cm dia	Upper dia 120cm and height 95cm	each	8910.00
28.27		Rotomoulded Square Planters	Providing of rotomoulded planters of square shape of different sizes.		
	28. 27.1	40cm x 40 cm x 37cm	40cm x 40 cm x 37cm	each	1260.00
	28. 27.2	50cm x 50cm x 45cm	50cm x 50cm x 45cm	each	2700.00
	28. 27.3	60cm x 60cm x 53cm	60cm x 60cm x 53cm	each	3600.00
	28. 27.4	70cm x 70cm x 65cm	70cm x 70cm x 65cm	each	5490.00
28.28		Asbestos cement pots	Providing circular Asbestos cement pots of specified size, stacking in required rows a height all complete as per direction of Engineer-in-Charge		
	28. 28.1	25cm dia	Upper dia: 25cm, Lower dia: 20cm, Height: 25cm	each	108.00
	28. 28.2	30cm dia	Upper dia: 30cm, Lower dia: 23cm, Height: 30cm	each	144.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	28. 28.3	35cm dia	Upper dia: 35cm, Lower dia: 26cm, Height: 35cm	each	270.00
	28. 28.4	40cm dia	Upper dia: 40cm, Lower dia: 30cm, Height:40cm	each	360.00
28.29		Floral foam bricks	Supplying of floral foam brick premium quality /high density (Oasis or equivalent) proper packing box (one box containing 20 bricks of size 22.5cmx10.5cmx7.5cm) at site of work including loading, unloading, carriage with all taxes paid etc.and as per direction of Engineer-in-Charge.	each	1080.00
	28.29.1	River pebbles	Providing and fixing of White River (stone) pebbles of size 2"to 2.50" dia in natural colour at site of work including loading, unloading, carriage with all taxes paid etc and as per direction of Engineer-in-Charge.	kg	10.00
	28.29.2	Marbles pebbles	Providing and fixing of White Marbles circular/oval in shape (stone) pebbles of size 2"to 2.50" dia in natural colour at site of work including loading, unloading, carriage with all taxes paid etc. and as per direction of Engineer-in-Charge.	kg	22.00
28.30		Floral stage decoration	Stage decoration with arrangement of cutflower height 15cm to 75cm with 6 to 8 rows offlowers and fillers on the oasis (flower foam) in presentable manner in the tray as per design approved by authority including clearing of site (Cost of Tray, Oasis and Cut flowers shall be paid separately) as per direction of Engineer in charge.	per metre	139.00
28.31		Anti termite treatment of trees	Providing and applying Anti Termite treatment of infected areas i/c branches/leaves by spraying chemical emulsion 0.5% (Chloropyriphos 20%EC) concentration through power spray machine (@100ml Chloropyriphos per Tree i/c cost of chemical) and as per the direction of Engineer-in-Charge.	each	70.00
28.32		Anti termite treatment of grass	Anti Termite treatment of Lawn area through premise 30.5%I.P, one liter premise diluted in 499 litres water and applying solution @1.00 liter solution per sq.m lawn or bed area. (Two application) i/c cost of chemical and as per direction of Engineer-in-Charge.	sqm	10.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
28.33		Green agro shade net	Providing and fixing of Green Agro Shade net of make: Tuflex or equivalent, of 75% permeabilty, of green colour having contents (Weight grams/sqm: 510 (+/-8%)) in width of 75cms with bamboo of 90 cm length.The bamboo should be painted with green colour paint of approved brand and manufacture (two or more coats) and fixed 30 cm below ground leveland 60 cm above ground level at adistance of1.5 mtrs. The net and bamboo should be binded with2 mm G.I wire at three places properly as per the direction of Engineer-in-Charge.	sqm	243.00
28.34		Artificial grass for landscape	Providing of Artificial Grass of make: Tiger Turf or equivalent with following specifications:		
	28.34.1	Artificial grass of 35mm	Yarn Characteristics: All yarns should be heavy metal free and UV Stabilised. Yarn Type: Texturised monofilament, Material: 100% Polyethylene, Standard Dtex: 7200, Colours: Field Green/Lime Green/Brown, Line Colours: Any Backing Characteristics: Primary Backing-100% woven polypropylene, UV resistant. Backing Compound: A concentrated latex compound optimised for tuft lock. Product Specification: (Pre Coating +/-10%), Pile Weight : 1361 gm/sq.m, Tufted Pile Height 35mm, Total product weight 2413 g/sq m, Machine Gauge: 3/8th, Roll Width: 4.00m, Shipping weight: 9.6kg/LM Particulate Infill: Sand Type- Sub Angular Silica, Sand Size: Upon request, Sand Infill Height: 15mm, Toatal Infill Height: 15mm. Recommended Base Construction: Timber/ Crushed Rock/ Asphalt/ Concrete(porous or non porous)	sqm	288.00
	28.34.2	Artificial grass of 25mm	Yarn Characteristics: All yarns should be heavy metal free and UV Stabilised. Yarn Type: Texturised monofilament, Material: 100% Polyethylene, Standard Dtex: 7200, Colours: Field Green/Lime Green/Brown, Line Colours: Any Backing Characteristics: Primary Backing-100% woven polypropylene, UV resistant. Backing Compound: A concentrated latex compound optimised for tuft lock. Product Specification: (Pre Coating +/-10%), Pile Weight : 1021 gm/sq.m, Tufted Pile Height 25mm, Total product weight 2073 g/sq.m, Machine Gauge: 3/8th, Roll Width: 4.00m,	sqm	261.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Shipping weight: 8.3kg/LM Particulate Infill: Sand Type-Sub Angular Silica, Sand Size: Upon request, Sand Infill Height: 10mm, Total Infill Height: 10mm. Recommended Base Construction: Timber / Crushed Rock / Asphalt / Concrete (porous or non porous)		
28.35		Stone bench	Providing and fixing of 3 seater stone bench made of 75mm thick stone slab finished by different numbers of ambry stone and two coats of 124 no sealer coat to prevent fromweather including fixing on a pedestal (cost of pedestal to be paid separately)etc. complete as per direction of engineer in charge. Size: length 1500mm, width 600mm, legs height 500 mm, back support height 500mm.	each	9700.00
28.36		Stone planter	Providing and fixing of stone planter (bench) made of 75mm thick stone slab finished by different numbers of ambry stone and two coats of 124 no sealer coat to prevent fromweather including fixing on a pedestal (cost of pedestal to be paid separately)etc. complete as per direction of engineer in charge. Size: length 1200mm, width 500mm, legs height 500 mm.	each	6200.00
28.37		Stone table	Providing and fixing of stone table made of 75mm thick stone slab finished by different numbers of ambry stone and two coats of 124 no sealer coat to prevent from weather including fixing on a pedestal (cost of pedestal to be paid separately)etc. complete as per direction of engineer in charge. Size: length1000mm, width 600mm, legs height 750 mm.	each	7900.00
28.38		M-25 C.C. Pavement	Providing and laying C.C. pavement of mix M-25 with ready mixed concrete from batching plant including centering and shuttering complete. The ready mixed concrete shall be laid and finished with screed board vibrator, vacuum dewatering process and finally finished by floating, with wire brush etc. complete as per specifications and directions of Engineer- in-Charge. **Note:** Cement content considered in this item is @ 330 kg/cum.	cum	6936.00

CHAPTER – 29.0

DPR PREPARATION

PREAMBLE
DPR PREPRATION
(CHAPTER-29)

SOR items in this chapter has been included to enable preparation of DPR including detailed estimate of a building project. The architect shall participate in the bidding process in which special condition appended with this chapter shall be made applicable. Standard format for preparation of estimate to be appended with the DPR as per Appendix-7. Terms and conditions for scope of work for work of preparation of DPR and bifurcation of fee payable to the architect at various stage of DPR work shall be as below:-

Terms and Conditions

(For Comprehensive Architectural Services and preparing DPR of a Building Project)

SCOPE OF WORK

The architect shall provide services for preparation of master plan of the project area,architectural working drawings, reports, estimates, DPR to obtain administrative sanction and thereafter statuary submission drawings, BOQ, detail drawings structural drawings and at last as constructed drawings in respect of following:

1.1 Taking client's instruction and preparation project brief & Finalise functional requirement.

1.2 Site visit and preparation of master plan of project area.

1.3 Details Survey & Investigation.

1.4 Architectural work

1.5 Foundation and Structural engineering work

1.6 Site development

1.7 Sanitary, plumbing, drainage, water supply, sewerage and STP work

1.8 Electrical, electronics, communication system

1.9 Lift, elevator, escalators etc

1.10 Heating, ventilation and air conditioning (HVAC)

1.11 Acoustical, light & sound

1.12 Fire detection, fire protection and security systems

1.13 Green building concept

1.14 Utility shifting

1.15 Interior designing and graphic signage

1.16 Landscaping

1.17 Construction stage planning

1.18 Building management system

1.19 Parking

1.20 Medical Gaspipeline, Operation theatres

1.21 Acces Controal, CCTV

1.22 Solar Power, Solar Heating

1.23 LAN, EPBX

1.24 Perparation of B/S Coordinated model to detect clashes

1.25 Any other services as desired by the client

2. SERVICES FOR DPR

For each of the items included in the scope of work the architect shall render the services sequentially as below:

2.1 Concept Design

2.1.1 Plans and concept design shall be prepared by the architect for approved of the department owning building. Modification in the concept plan that was presented before the selections jury, may be modified at the atage to accommodate functional requriment of the project.

Howrver plans and concept design, duly approves, shall be provided the DPR consultant, who is preparing DPR in accordance with SOR.

2.2 Master Planning

Visit site and prepare master plan of the project area (campus) i/c double line plan of existing buildings/structures to be retained / demolished, exact details of all features inside and adjoining the campus such as poles, over head lines, under ground utilities, cables, wells, handpumps, trees, roads, nallah, river, drains, under ground water lines, sewer lines, adjoining land owners, over head tanks etc. The architect shall verify contours and directions (E-W-N-S) etc at the project site and if contours are found not truly representing the ground realities, the architect shall engage a surveyor and get site plan prepared afresh for which payment shall be made separately according to relevant SOR item. Google plan showing project boundaries shall also be provided in support of master plan.

2.3 Architectural Working Drawing (For obtaining AA)

Prepare architectural working drawings as per **List-1** on scale 1:100 according to applicable standards /norms such as NBC, IS, MCI, M.P. Bhumi Vikas Adhiniyam, IPHS, local self govt. guidelines &

rules etc and submit as **DPR Appendix-1** Prepare guidelines and plans for the phased (staged) construction of the project , if required and submit as **DPR Appendix-1**

2.4 Estimating (for obtaining AA)

Prepare specifications and detailed estimate of cost based on current specifications and SOR Format – **T1, T2, T3** and **A, B, C, D, E, F & G**, **P1, P2** prescribed and as amended form time to time .Detail estimate in prescribed format shall be submitted in the SOR as **DPR Vol-1** Details of standard formate such need, contents, formatting & directives to use/preparation formats are also given in the SOR.

2.5 Utilities (for obtaining AA)

Prepare plans and estimates for utility shifting and tree cutting ,if required .Obtain estimates from other agencies such as electricity companies /telephone companies, PHE department, municipal governments forest department etc and include cost in DPR as external works .However, if required, utility plans & estimate shall be prepared separately also for submission to other department for the there permission or forgiving them these work as deposit work and submitted as **DPR Vol-2.**

2.6 Rate Analysis (for obtaining AA)

Prepare rate analysis in standard format of C.P.W.D. For non SOR items, if included, in the estimate and submit as **DPR Vol-3**

2.7 BOQ (After AA is received)

Prepare Bill of Quantities (BOQ) upon grant of administrative approval of the project by the administrative department comprising of two parts, '**BOQ Part A-SOR items'** and '**BOQ Part B-Non SOR items'** for inviting tenders. It is to be submitted as **DPR Vol-4.**

2.8 Statutory Approval (After AA is received)

Prepare, if required upon getting AA, all **Preliminary Working Drawings** as per List-**2** on scale 1:100 in required number of sets for clients/statutory approvals and ensure compliances with codes, standards and legislations, as applicable and assist the client in obtaining clearances /NOC/ approvals from statutory bodies such as municipal govt. environmental clearance bodies. In case of statuary approval not warranted for the project, Department's approval shall be granted at this stage. Drawings to be submitted as **DPR Appendix-2** mentioning **"Preliminary working For Statutory Submission Only and Not for Construction".** Coordinate with various agencies such as Town and Country Planning, Municipal Corporation, fire department, electricity board, P.H.E.D., EPCO, C.P.A. Airport Authority and planning agency, environmental clearance authority etc. and assist department in obtaining

approval. Necessary fee, if required to be paid to these agencies, shall be paid by thedepartment.However consultant engaged for preparing submission and making/presenting before these agencies shall bee paid by the architect and not the department.

2.9 Model (After AA is received):- Prepare model of building on a suitable made from fiber acrylic sheet. The model shall be enclosed by 5mm thick glass and mounted on portable base.

2.10 Detail Drawings (Construction Stage)

Prepare Architectural Detail Drawings as per **List 3** on scale 1:50 or larger and submit as **DPR Appendix-3.** Coordinate and prepare modified architectural working drawings if required, as per structural design.

2.11 Structural Design (Construction Stage)

Prepare structural design, drawings, Tables for bar-bending schedule, concrete consumption and steel consumption as per List-4 and submit as **DPR Appendix-4**

2.12 Completion Drawing (Project Completion Stage):-

Prepare as constructed drawings up to completion of the project but before preparation of final bill of the contractor.

3. Professional Fees Payable

3.1 Fees payable

Fees payable to the Architect/Consultant at various stages of all the services for DPR defined in clause-2 of this **Annexure** are shown in the '**Activity Sheet No. 1'** for DPR and Drawings and '**Activity Sheet-2'** for Structural Design and Drawings.

3.2 Total professional fees shall be computed on the basis of percentage of cost of project in fixed fee Format or percentage cost of DPR preparation quo ted in Design pmc format. However building area shall be basis for payment in case of assignment of DPR as SOR cost of project means project cost for which AA is granted.

3.3 If a working drawing such as standard drawing prepared/approved by the Chief Architect is made available to the consultant/architect for customization/modification customization work under this assignment, payment for activities/professional services shall be made only for the modification part (built-up area shall be the basis) of the project and not for the whole project.

3.4 Bifurcation of Fee:-

Bifurcation of professional fee related to various stages as per Activity Sheet No -1 & 2 shall be according to the drawings submitted as per List-1, List-2, and List-3 & List-4.

3.5 Schedule of Payment:-

The payment to the Architect/Consultant during various stages mentioned in Activity Sheet No 1. & 2 is subject to its prior scrutiny and approval by the competent officer of the department and shall be running account payments and adjusted in the final payment.

4. Time Allowed:-

Architectural/consulting services (scope of work) shall be rendered by the Architect/ Consultant as per Activity Sheet No 1. and 2. In case of delay in obtaining administrative approval and statutory approval, the time period will accordingly be relaxed. Once a project is awarded/assigned to a architect/consultant, work of all stages given in activity sheet shall be done by the same architect/consultant for smooth continuity and in the interest of the project and time allowed shall be deemed to have been extended proportionately.

ACTIVITY SHEET-1
(For item of prepration of DPR and Drawings as per SOR)

S. No.	Activity	Time Period	Start of Time Period	Action by	% of total Fee Payble for Activity	Cummulative fee payable upto Activity	Penalty for Late Submission (in Terms of % of Fee of col 6) Per day P
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
1	Make available concept design (concept plan)	Within one week of work order.	Activity-1	Department	Nil	Nil	Nil
2	Visit site and submission of master plan of project area based on concept plan	1 weeks	After concept plan is made available	Architect / Consultant	Nil	Nil	Nil
3	Comments/ suggestion on master plan	1 week	After submission of master plan	Department	Nil	Nil	Nil
4	Submission of master plan	1 week	After receiving comments/ suggestions on master plan	Architect / Consultant	**5% Ref List-1**	**5%**	5%
	Working Drawing Stage (for optaining AA)						

S. No.	Activity	Time Period	Start of Time Period	Action by	% of total Fee Payble for Activity	Cummulative fee payable upto Activity	Penalty for Late Submission (in Terms of % of Fee of col 6) Per day P
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
5	Submission of architectural working drawings as per "**List-1".**	3 weeks	After master plan is approved	Architect / Consultant	Nil	**5%**	Nil
6	Checking/ suggestions on working drawings	1 week	After submission of architectural working drawings	Department	Nil	**5%**	Nil
7	Submission of all architectural working drawings with compliances, if any, as per "List-1" as part of **DPR Appendix-1** in 6- sets (hard copies) and soft copy)	1 week	After receiving suggestions on working drawings	Architect / Consultant	20% Refer "List 1"	25%	5%
	DPR Stage (for obtaining AA):-						
8	Submission of specification, estimate & rate analysis of Non SOR items	2 week	After approval of working drawings	Architect / Consultant	Nil	25%	Nil
9	Checking /suggestions on specification, estimates & rate analysis of Non SOR items	1 week	After submission of specification, estimates and rate analysis	Department	Nil	25%	Nil
10	Submission of DPR as below:- **DPR Vol-1:-** • Front page	1 week	After approval of specification estimate and rate analysis	Architect / Consultant	Nil	25%	Nil

S. No.	Activity	Time Period	Start of Time Period	Action by	% of total Fee Payble for Activity	Cummulative fee payable upto Activity	Penalty for Late Submission (in Terms of % of Fee of col 6) Per day P
1	2	3	4	5	6	7	8
	• Index sheet • SFC/EFC Formats • Master plan of Project Area • Approved Concept plans & 3D views of building • Estimates in A, B, C, and D…. Formats A,B,C,D… format)						
	DPR Vol-2:- • Utility shifting estimates (if required separately despite included in site development component of estimate in **DPR Vol-3;-** • Rate Analysis of Non SOR items						
11	Technical Sanction	1 week	After submission of DPR (Vol-1 + Vol-2 +Vol-3)	Department	For DPR Vol-1 = 15% DPR Vol-2 = 2% DPR Vol-3=3% Refer List 1	40 % 42% 45%	5% 5% 5%

S. No.	Activity	Time Period	Start of Time Period	Action by	% of total Fee Payble for Activity	Cummulative fee payable upto Activity	Penalty for Late Submission (in Terms of % of Fee of col 6) Per day P
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
	BOQ Stage (After AA is received)						
12	Submission of BOQ	1 week	After AA	Architect / Consultant	Nil	45%	Nil
13	Checking/suggestion on BOQ	1 week	After submission of	Department	Nil	45%	Nil
14	Submission of BOQ (Part A-SOR items + Part B-Non SOR items) as part of **DPR**	1 week		Architect / Consultant	5% Refer List-1	50%	5%

	Statutory Approval Stage (After AA is received)					
15	Submission of working drawings, PPT as per**"List-2" as** part of **DPR Appendix- 2** for clearance/NOC/ approval from statutory bodies.	2 weeks	After AA	Architect/ Consultant	2% Refer "List-2"	52%
16	**Model stage (after AA is received)**					
	Preparation of one modelof building from fibre acrylic sheet mounted on asuitable portable base covered with 5mm thick glass on top & all sides.	4 weeks	After AA	Architect/ Consultant	1% Refer "List-2"	53%
	Detail Drawings Stage (During Construction)					
17	Submission of detail drawings as per list-3.	4 weeks	After AA	Architect/ Consultant	Nil	53%
18	Checking/Suggestions on detail drawings	2 weeks	After submission of detail drawings	Department	Nil	53%

19	Submission of detail drawings after compliances as per **"List-3" as** part of **DPR Appendix-3** in 6 sets.	1 weeks	After checking/ suggestion by department	Architect/ Consultant	37% Refer List-3	90%
	As Constructed Drawings (Project Completion Stage)					
20	Submission of 'As Constructed Drawings of all buildings as part of **DPR Appendix- As Built Drawings.**	4 weeks	On completion of building & before final bill payable to the contractor.	Architect/ Consultant	10%	100%

ACTIVITY SHEET-2
(For item of prepration of DPR and Drawings as per SOR)

S. No.	Activity	Time Period	Start of Time Period	Action by	% of total Fee Payable for activity	Cummulative fee payable upto Activity	Penalty for Late Submission (in Terms of % of Fee of col 6) Per day P
1	2	3	4	5	6	7	8
	Structural Design Stage(During Construction):-						
1	Submission of Structural Design & Drawings as per "List-4: along with structural calculationand analysis in Staad Pro or equivalent software.	2 weeks	Work orderor after AA is received	Architect/ Consultant	Nil	Nil	Nil
2	Proof checking	2 weeks	After submission of structural design &drawings	Department/ Structural Engineer (proof checking) on behalf of department	Nil	Nil	Nil
3	Compliance in Structural Drawings as required in proof checking and submission of final Structural Drawings as part of **DPR Appendix-4** in six sets as per "List-4" .All drawings & designs shall be duly signed by a empanelled structural engineer	1week	After submission of structural design drawings	Architect/ Consultant	100% Refer**"List-4"**	100%	5%

NOTE FOR ACTIVITY SHEET No 1& 2

1 The Architect shall submit DPR volumes and Appendices to the concerned Divisional Project Engineer, PWD, PIU as below:-

(a) Volumes:-
DPR Vol-1; (i) Front page
(ii) Index sheet
(iii) All SFC/EFC Formats
(iv) Master plan of Project Area
(v) Approved Concept plans & 3D views
(vi) Estimates in A, B, C, D…. Formats
DPR Vol-2 Utility shifting estimates
DPR Vol-3 Rate Analysis of Non SOR items
DPR Vol-4 BOQ (Part-A SOR items + Part-B Non SOR items)

b) Appendices
- DPR Appendix 1 – (Architectural Working Drawings as per **"List-1"**)
- DPR Appendix 2 – (Statutory Drawings as per **"List-2")**
- DPR Appendix 3 –(Detail Drawings as per **"List-3"**)
- DPR Appendix 4 –(Structural Design and Drawings as per **"List-4")**

2 All Architecture drawings shall be signed by the authorised/lead person of the consultant/architect mandatorily before submission otherwise payment shall not be released. Merely signing by any other person/employee of the consultant/architect shall not be accepted for release of payment.

3 All detailed drawings should be supported by fixing detail, plan, elevation, section and detailed information like specification etc. as per requirement.

4 Structural drawing & design calculation, analysis (In STAAD Pro or equivalent software) shall be duly signed by an empanelled Structural Engineer only.

5 Submission of approved architectural drawing of plan, section, and elevation and approved layout plan is compulsory with the submission of structural design for proper coordination

6 Plumbing drawing is needed before casting of plinth and floor beam with opening details.

7 Structural consultant / structural proof checking consultant should submit Structural Safety Certificate for each individual project as per National Building Code 2016. Annexure "C" and "J".

List-1

Fee Bifurcation (Ref Activity sheet No-1 and item for prepration of DPR)

S. No	Particular of drawings/Detail	Category of Empanelled Consultant Competent to sign drawings/document	% of total fee payable for activity
1	2	3	4
	Master Planning Stage:-		
1	Site visit the architect for verification of all features and contours. Submission of master plan of project area showing all existing features, structures to be dismantled/retained over head/underground utilities roads, directions, contours, adjoining land owners submission of google plans with project. Boundaries in support of master plan.	Architect	5%
	Architecture Working Drawing Stage:-		
2	Architectural working drawings having all floor plans, all elevations and all sectional elevations	Architect	5%
3	3D views, at least 4 number from all sides.	Architect	2%
4	Layout plan for internal electrical services	Electrical, Electronics, Communications and Security system	1%
5	Layout plan for external electrical services including tapping point, substation panels etc.	Electrical, Electronics, Communications and Security system	1%
6	Layout plan for mechanical services (HVAC, lift etc) with details	HVAC and other Mechanical System	1%
7	Layout plan for communication electronic.	Electrical, Electronics, Communications and Security system	1%
8	Layout plan for ceiling, paneling, acoustic, light & sound services	Electrical, Electronics, Communications and Security system	1%
9	Layout plan for roof drainage with water proofing, roof slopes, spout and rainwater pipes.	Sanitary, Plumbing Drainage, Water Supply and Sewerage	1%

10	Layout plan for rainwater harvesting	Sanitary, Plumbing Drainage,Water Supply and Sewerage	1%
11	Layout plan for fire fighting	Fire Detection and FireProtection	1%
12	Layout plan for water supply lines, storagetanks, source.	Sanitary, Plumbing Drainage,Water Supply and Sewerage	1%
13	Layout plan for sanitary pipes/lines & sewerage, septic tanks, STP/ETP, inspectionchambers, soakpit.	Sanitary, Plumbing Drainage,Water Supply and Sewerage	1%
14	Solid waste management and incinerator layout,if any	Sanitary, Plumbing Drainage, Water Supply and Sewerage / Environmental Engineer	1%
15	layout plan for furniture	Interior Architect	1%
16	Layout plan for parking and landscaping.	Architect/ Arboriculture/Landscape Architect	1%
	Total % of fee payable at Architectural Working Drawing Stage (item 2 to16)		**25%**

	DPR stage:-		
17	**For submission of DPR as below** **DPR vol-1:-** • Front page • Index sheet • SFC/EFC Formats • Master plan of Project Area • Approved Concept plans & 3D views of building • Estimates in A,B,C,D…. Formats **DPR Vol-2:-** • Utility shifting estimates (if required separately despite included in site development component of estimate in A,B,C,D… format) **DPR Vol-3;-** • Rate analysis of Non SOR items	Lead Person Lead Person Lead Person	10% 2% 3%
	Total % of fee payable at DPR stage(item 17)		**15%**
	BOQ Stage (after AA is received) :-		
18	**Submission of BOQ** :- BOQ (Part-A SOR items + Part-B Non SOR items) as **DPR Vol-4**	Lead Person	**5%**
	Total % of fee payable at master planning, architectural working drawings, DPR and BOQ stage(list-1,Item 1 to 18)		**50%**

List-2			
Fee Bifurcation (Ref Activity sheet No-1 and item for prepration of DPR)			
S. No	**Particular of Drawings/Detail-**	**Category of Empanelled Consultant Competent to sign drawings/document**	**% of total fee payable for activity.**
1	**2**	**3**	**4**
	Statutory Approval Stage:-		
1	Submission of all necessary drawings & details for approval of Govt. & Semi Govt.Departments, Municipal Corporations/ Municipality/Gram Panchayat / Town & Country Planning/M.P. Electricity DistributionCompanies / Environment Departments, SEIAA, SEAC /Airport Authority / Fire Department etcas **DPR Appendix-2.**	Lead person	Nil
2	After permission granted from all concerned authorities for construction of building.	Concerned authority	2%
	Model Stage :-		
3	Submission of one model on suitable scale made form acrylic sheet & enclosed with 5mm thick glasses	Lead person	1%
	Total % of fee payable at statutory approval and model stage as part of DPR-Appendix-2 (List -2)		**3%**

List-3			
Fee Bifurcation (Ref Activity sheet No-1 and item for prepration of DPR)			
	Construction stage:-		
1	Good for Construction (GFC) detail drawings (Scale 1:50) with all dimensions (refer Activity Sheet-1) as part of **DPR Appendix-3** after getting administrative approval as below:- Visit site to verify layout and submit GFC working drawings along with layout plan, all floor plans, all elevations, all sections special features and any other necessary drawings & information with all dimensions.	Architect	5%
2	3D Views, if different than that submitted at working drawing stage.	Architect	2%
3	GFC design and detail of Flooring, Dado, Sill & Skirting, Layout plan for labs and details of Platform etc.	Architect	5%
4	GFC details of doors, windows, ventilators, grills, railing and glazing.	Architect	2%
5	GFC Details of staircases, ramps with railing and handrails	Architect	2%
6	GFC Layout plan for roof drainage with, water proofing, slopes, spout and rainwater pipe details and Expansion Joint	Sanitary, Plumbing Drainage, Water Supply and Sewerage	3%
7	GFC Layout plan for rainwater harvesting with design and details	Sanitary, Plumbing Drainage, Water Supply and Sewerage	2%
8	GFC Layout plan for water supply & storage tanks design details	Sanitary, Plumbing Drainage, Water Supply and Sewerage	2%
9	GFC Layout plan for sanitary fittings & plumbing, sewerage, , inspection chamber with design & details such as slopes, dia etc.	Sanitary, Plumbing Drainage, Water Supply and Sewerage	2%
10	GFC Layout plan for, septic tanks, STP/ETP with design & details	Sanitary, Plumbing Drainage, Water Supply and Sewerage	1%

11	GFC Solid waste management details and incinerator details, if any	Sanitary, Plumbing Drainage, Water Supply and Sewerage / Environmental Engineer	1%
12	GFC Design and Details for fire fighting with layout.	Fire Detection and Fire Protection	3%
13	GFC Design and Detail of communication services , Layout and Detail of Security System Layout plan for electronics services with design and details	Electrical, Electronics, Communication and Security System	2%
14	GFC Design and Detail of internal electrical working drawing including electrical point wiring, circuit wiring, mains & sub-mains, electrical & power conduit plan, L.V. (CCTV, LAN, Telephone, Fire Alarm), S.L.D. for panels & DBs, electrical load calculation	Electrical, Electronics, Communication and Security System	3%
15	GFC Design and Detail of external electrical working drawing including substation design, H.T./L.T. panel S.L.D. & design, Outer H.T./L.T. cable layout, street light design layout, electrical feeder details for water pump, STP, WTP,	Electrical, Electronics, Communication and Security System	2%
	Total % of fee payable at Construction Stage (List-3, Item 1 to 15)		

	List-4		
Fee Bifurcation at structural design and drawing stage (Ref Activity sheet No-2 and item for prepration of DPR)			
S.No	**Particular of drawings/Detail**	**Category of Empanelled Consultant Competent to sign drawings/document**	**% of Total fee payable for activity.**
1	**2**	**3**	**4**
	Structural Design Stage:-		
1.	Visiting site for verification of S.B.C and contours at site. Getting soil testing done again if warranted by an empanelled geotechnical engineer (paid separately for soil testing)	Structural Engineer	10%
2.	Submission and obtaining approval of dasign Analysis and calculation in STAAD Pro or Equivalent software included all correction suggested (including wind and earthquake loading with static and dynamic analysis as applicable) and calculation of foundation in excel sheets or at appropriate software (with wind and earthquake loading) and there output in hard and soft copy.	Structural Engineer	25%
3	Submission and obtaining approval of structural drawing of Column index plan with all floor framing plans, details of column footing, foundation (combined raft etc.), piles, pile caps, plinth beam, grade beam, curtain walls and retaining walls etc. Drawing in six set in hard and soft copy (AUTOCAD files) in C.D., which shall be vetted/coordinated with sanitary and plumbing drawing.	Structural Engineer	15%
4.	Submission and obtaining approval of structural drawing of Ramps (internal/external), stair, stair cases, porch, lintel, chajjas , shear wall and tower, mumty floor, lifts block lift well and lift machine room according to the type of lift to be used. septic tank, sump well, overhead tank, in six set ofhard and soft copy (AUTOCAD files) in C.D	Structural Engineer	10%

	or Finalized Special provision if sloppy terrain or any other site specific problem and all related detailed drawing or Any other structural design drawing provided in the DPR and as and when Required/ desired by the Engineer in charge.		
5.	Submission and approval of structural drawing of All Floor beams and slabs at different level, with calculation of slab in excel at appropriate software, in six set of hard and soft copy (AUTOCAD files) in C.D. And vetted/coordinated with electrical layout of slab. or Truss analysis and detail's of component and there connection with all calculation detail in excel, STAAD Pro equivalent software and all related detailed drawing in in six set of hard and soft copy (AUTOCAD files) in C.D or Finalized Pre-stressing or Post tensioning (if required in any part of structure) with calculation details in excel, ADAPT, RAM CONCEPT P.T or equivalent software and Submission and approval of structural drawing in six set of hard and soft copy (AUTOCAD files) in C.D	Structural Engineer	15%
6.	Submission and approval of Bar Bending Schedule as per finalized structure design(after proof checking), in hard and soft copy (in any software or Excel programs) dully signed with structure engineer, and Finalized quantity of Concrete and steel in the preserved form of steel & concrete consumption table.	Structural Engineer	25%
	Total % of fee payable at structural design stage (List-4, item 1 to 6)	**Total**	**100%**

NOTE:- For SOR items for prepration of DPR

1. Competent officer to approve drawings shall be:

   i. Architectural Drawings : Chief Architect

   ii. Structural Drawings:

   - DPE for building with area upto 500 sqm.
   - APD for building with built-up area more than 500 sqm & upto 3000 sqm.
   - Design cell in the office of the chief architect for buildings with built up area >3000 sqm and any other building (even less than 3000 sqm.) referred to design cell

2. Structural design, drawings and calculations analysis (In STAAD Pro or Equivalent software) shall be duly signed by empanelled Structural Engineer only.

3. Submission of approved architectural plan, section, and elevation and layout plan is compulsory with the submission of structural design for good coordination.

4. Plumbing drawing with opening details is needed before casting of plinth and floor beams.

5. Structural consultant/ Structural proof checking consultant should submit structural safety certificate for each individual project as per National Building Code-2016 Annexure “C” and “J”.

6. If an architect/consultant is assigned a work of customisation of a project work in which a already prepared/approved drawing of repetitive project is used, then payment for only services provided by the architect/ consultant shall be made. Built up area shall be the basis for such calculation of fee payable.

Special Condition for Work/Assignment for inviting bid for preparation of D.P.R. as per Items in SOR

1. Bidder for work/assignment for preparation of D.P.R. as per Items in SOR shall be an Architect only, who is empanelled with the Chief Architect M.P. P.W.D., Bhopal.

2. In case bid document 2.10 of PWD Work Manual is used for inviting bid for prepration of DPR, then "work" shall be read as "assignement" and "bidder" or "contractor" shall be read as "architect".

3. This work/assignment includes several activities/components that can be done only by other consultants who are empanelled with the Chief Architect M.P. P.W.D., Bhopal and therefore an Architect bidding for this work/assignment is expected to have a legallybinding joint venture (J.V.) agreement with other consultants.

4. In case of Architect winning this assignment does not have J.V. before bidding and submit bid in lone capacity, other activities/works such as land survey, geotechnical investigation, preparation of geotechnical reports, estimation, quantity surveying, HVAC, sanitary, plumbing, acoustic, fire detection/protection, structural design etc. shall be got done by the Architect only through the consultants empanelled with the Chief Architect M.P. P.W.D., Bhopal. Under appropriate category given in schedule 1 of the "Document for Empanelment" issued by government of Madhya Pradesh vide letter No. F 17-1/2010/G/19 Bhopal dated 07.06.2014 and subsequent addendums.

5. In both the cases, J.V. prior to bidding (as per Clause 2 above) or engaging other consultants after winning the bid (as per Clause 3 above) provisions for black listing and suspension as per Clause 5 of letter No. F 17-1/2010/G/19 Bhopal, dated 07.06.2014 and subsequent addendums issued by Government of M.P. P.W.D. shall be applicable jointly and severally both successful bidding Architect and the Consultants.

6. Payment for services rendered by the other consultants shall be done by the Architect awarded the work at rates mutually agreed between them. The department shall not be held responsible for any dispute between them. Payment for work done by the department shall be made only to the Architect who has been awarded the work.

7. Payment to the Architect shall not be done for the non S.O.R. items for which rate analysis as per C.P.W.D. pattern has not been submitted along with the estimate. Similarly payment to the Architect for those works which are required to be done by the Consultants empanelled with the Chief Architect, M.P. P.W.D., Bhopal shall not be made if such documents, reports, drawings, estimates, design and etc. are not signed and sealed/stamped by the authorized persons of such empanelled Consultant.

8. The Architect is required to appoint a suitable technical person with computer, printer, stationery and wireless internet internet connection in the office of the Divisional Project Engineer (Nodal Agency) during office days to coordinate between the Architect andvarious consultants on his behalf and the department. This technical person shall have to working knowledge of computer and should be capable of taking instruction from the Divisional Project Engineer in regards to works to be done or compliances/correction in report drawing etc.

9. Zonal work/assignment for preparation of D.P.R. shall include several works located within the districts which may be assigned to the PIU by other departments in the financial year for

which bids have been invited. But this will not entitle the Architect to claim allotment of assignment which have been put for architectural competition or identified for in house design in the office of the Chief Architect.

10. In case the Architect is allotted a project of which architectural drawing are already prepared/approved by the Chief Architect, only prorated payment of modification work shall be made.

11. The D.P.R. consultant empanelled with the chief Architect shall have to submit all details drawings duly signed by him/her and also by their associate expert/consultant empanelled with Chief Architect in their respective field/stream/areas of expertise such as structural design/S.Q.C./landscape/water supply and sanitary/ electrical/ fire fighting/ interior design. Payment to the Architect shall be made only for percentage of fee (component of total fee) mentioned in preamble, which shall further be prorated to the extent of modification in allotted work/assignment while customizing previously/already prepared/approved project by the Chief Architect.

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
29.1		Site plan and location plan with total station	Preparation of site plan and location plan with total station surveying showing all individual structures and features like trees, roads, culverts, nallas, electric poles, telephonepoles, sewer lines, chambers, water supply lines, hand pumps, well etc. complete along with showinglocation of roads proposed in master plan. i/c Contours of entire area at 0.5 m. intervals using auto level, preparing of site plan in soft copy (using autocad) in suitable scale as desired by engineer-in-charge duly indicating north direction. Rate includes providing 1 digital copy and six hard copies, one print on tracing film, khasra map duly showing the land and location plan (index plan) and google image of the land. Rates are inclusive of transportation charges. (Work shallbe got done through consultants duly empanelled with chief Architect MP P.W.D.)		
	29. 1.1	Upto 1 Hectare	Upto 1 Hectare	each	13000.00
	29.1.2	1 to 2 Hectare	1 Hectare to 2 Hectare	each	16000.00
	29.1.3	Extra for additional Hectare	Add extra for each additional one Hectare or part there off	each	2250.000
	29.1.4	Trial Pit	Excavation of trial pit of size 1.20 m X 1.20 m with minimum depth of 0.9 m and maximum depth of 3.5m as directed by the Engineer in Charge. Trial pit shall be taken at suitable location/ locations as directed by Engineer in charge and shall be indicated on the site plan showing details of soil strata.	each	2450.00
29.2		Detail Double line plan of existing buildings structures	Preparation of double line plan after measurement of internal aswell as external dimensions of existing buildings/structures and plotting the same in suitable scale as directed by Engineer-in-charge.	per sqm	18.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
29.3		Double line plan for planning of addition and alteration work	Preparing of double line plan for planning of addition and alteration work after measurement of internal as well as external dimensions and details of existing door window, ventilator openings and partitions, addition /alteration done previously in buildings/structures and plotting the same separately showing sectional elevations (minimum two numbers) and all elevations along with all relevant details like doors, windows stairs etc in suitable scale as directed by Engineer-in-charge. The consultant shall also submit detailed condition survey report.	per sqm	45.00
29.4		S.B.C. by Plate load test	Performing Plate Load Test (P.L.T.) for Determining soil bearing capacity (SBC) at site including submission of Geotechnical Report and recommending bearing capacity in accordance with IS-1888-1982 (Without Conveyance).	per pit	32580.00
29.5		S.B.C. by S.P.T.	Performing Standard Penetration Test for determining soil bearing capacity (SBC) in accordance with IS-2131-1981 and submission of Geotechnical Report on bearing capacity based on SPT in accordance with IS-6403-1981 (Without Conveyance)	per bore	16497.00
29.6		Transportation charges for plate load test or S.P.T.	Transportation charges for transportation of Geotechnical testing equipments for plate load test or standard penetration test upto a distance of 100km. (upto 100km going and 100km returning from district H.Q. both way) Rates are inclusive of Loading and unloading of testing equipments and other incidental charges during both journeys to be paid only once forone set (separate for PLT or SPT) of testing.		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	29.6.1	Up to 100km	Up to 100km (both way)	upto 100km both way	3150.00
	29.6.2	Extra km from 100 to 150km	Add extra for distance beyond 100km and upto 150km. (both way)	per km.	27.00
	29.6.3	Additional km beyond 150km	Add extra for every additional km beyond 150km (both way)	per km.	22.00
29.7		Preparing detailed estimate based on working drawing of the department	Preparation of detailed estimate on the basis of the working drawing supplied by the department and current SOR applicable, including computation of bill of quantities (BOQ) of material and supplying the same in six hard copies and one soft copy to the Engineer-in-Charge and if required revision of the same without any additional charges. (Work shall be got done through consultants duly empanelled with chief Architect MP P.W.D.) The estimate shall be prepared only in standard format prescribed as per appendix 14 of this SOR.	per estimate	0.15% of the total estimated amount
29.8		DPR Preparation including site visit	Preparation of DPR including site visit to verify contours & soil reports and preparation of master plan of campus including architectural working drawings and estimates in prescribed format as per appendix 14 of this SOR to obtain AA; and there after architectural drawings for statutory approvals and detail drawings at construction stage; and as constructed drawings on completion of work the DPR (excluding structural design) shall be prepared according to "Terms & Conditions", Activity Sheet No-1" , List-1" , List-2 and List-3" given in Annexure-14 as amended from time to time appended with SOR. In addition to civil drawings, the	per sqm	160.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			architectural work shall include preparation of MEP drawings of Heating Ventilation and Air Conditioning (HVAC), Mechanical system (elevators/escalators etc), acoustics, sanitary, plumbing drainage, water supply, rain water harvesting, sewage, STP, ETP electrical, electronic, communication system, interior design, graphics signage, fire detection, fire protection, security systems, arboriculture, parking and landscaping based on the concept plan supplied by the department. All drawings shall be prepared according to soil condition, topographical condition surface & subsoil water levels at site, relevant codes/manuals, bylaws, green building and GRIHA norms. The work shall also include preparation of model of the building made of acrylic fiber sheet, on a suitable scale, mounted on a portable baseand 3D views on flex on plain sheet as directed by the Engineer in charge. It will be the responsibilityof the architect to collect the sub soil reports (Bearing Capacity for soil) from the department, if available. However, if bearing capacity of soil is not available with the department, then it will be responsibility of the architect to engage a geotechnical engineer empanelled with the Chief Architect and get the investigation done and collect the re4port for which payment shall be made separately according to geotechnical testing items in SOR. The work shall also include revision of thedrawings, obtaining staturary approvals, obtaining utilities shifting estimates from owners of the utilities at site and other approvals such as GRIHA rating.		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			Environmental clearance etc, if required, within time stipulated bythe Engineer in Charge. All drawings shall be duly stamped & signed by the Architects and Consultants empanelled with the Chief Architect MP PWD Bhopal in relevant category. The drawings shall be submitted to the Engineer- in-charge in six hard copies and one soft copy. Note:-1.Lead person for this work/assignment shall be only architect empanelled with the chief architect MP PWD Bhopal. The Architect may retain a in house team of other engineers/consultants or may hire other consultants, as required, for which no extra payment shall be made by the department. 2. Bifurcation of professional fee payable in stages as per this item shall be in accordance with List-1, List-2 & List-3. Given in preambles of this chapter 3.All work such as preparation of DPR,working drawings, statuary drawings, details drawings and as constructed drawings shall be done by the same architect/consultant to ensure consistency in services, even if time required to complete all stages as per Activity sheet exceeds the contract period.		
29.9		Structure Design & Drawings	Preparation of the structure design & drawing by structure engineer empanelled with the chief architect MP PWD on the basis of architectural drawing supplied / approved by the department in accordance with relevant code / manual and according to the “terms & conditions”, prescribed in activity		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			Sheet No-1", List-4" given in Annexure-14 appended with SOR. The work shall include site visit, verification of soil testing report, topography, contours etc. made available to the structural engineer The structural engineer is not satisfied with adequacy of geotechnical reports, he may engage a geotechnical engineer and get the soil investigation done, for which charges are payable separately, according to relevant item of SOR, to satisfy himself that the design is optimal. If design work pertains to a work to be done over an existing structural, the structural engineer shall get structural soundness study done for which payment shall be made separately. The structural engineer shall do revision/ compliances of the structural design and drawings, if required, without any additional cost with-in stipulated time and obtain approval from to the competent authority. Design and drawing shall also include 'Tables' for the bar bending schedules (BBS) and concrete/steel consumption. The design and drawing shall be submitted to the Engineer-in-Charge in six hard copies and one soft copy for execution. The payment shall be done in accordance with relevant stage in activity sheet and list.		
	29.9.1	First time design work	First time design work		
	29.9.1.1	Up to 500 sqm	Built up area upto 500 sqm	per sqm	55.00
	29.9.1.2	500sqm to 3000sqm	Built up area more than 500 sqm and upto 3000 sqm	per sqm	45.00
	29.9.1.3	More than 3000sqm	Built up area more than 3000 sqm.	per sqm	35.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	29.9.1.4	Design Of Retaining wall	Preparation of structure design for RCC Retaining wall (Retaining structures)	% basis	0.75% on cost of the work
	29.9.1.5	Design Of water retaining structure	Preparation of structure design of water retaining structure like STP/WTP/ETP/Water Tank	% basis	1.00% on cost of the work
	29.9.1.6	Design of Campus Entry Gate	Preparation of Campus Entry Gate Inc. Boundary wall, Gard roof.	Lump Sum	15,000.00
	29.9.1.7	Design of Steel Structure	Preparation of the steel structure design & drawing by structure engineer empanelled with the chief architect MP PWD on the basis of required Layout approved by the department. (Conventional Steel structure Like truss etc & Pre-Engineered Building)	Per KG	2.50 Rs.
	29.9.2	Repetition Design work	Repetition of structural design of a structural design done by the structural himself or by any other structural engineer at different site, where S.B.C. or seismic zone is different.(only plinth area paymentto be paid)		
	29.9.2.1	Up to 500sqm	Plinth area upto 500 sqm	per sqm	22.00
	29.9.2.2	500sqm to 3000sqm	Plinth area more than 500 sqm and upto 3000 sqm	per sqm	18.00
	29.9.2.3	More than 3000sqm	Plinth area more than 3000 sqm	per sqm	14.00
29.10		Load carrying capacity of existing R.C.C. structure	Assessment of load carrying capacity and structural soundness of existing building mainly comprising of R.C.C. structure using existing information such a existing drawing, design and footing details, if available. The consultant shall carry out destructive/non destructivetesting of structure at site including site investigation which may require opening of footings by digging adjoining ground fill and restore backfill structure after testing isdone. The work may include consultation with the geotechnical consultant/expert. The report shall contain express recommendation on capacity/ ability of the structure to	per sqm	260.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			carry extra loads of next floor proposed on the building. The report shall also include recommendation on rehabilitation measures like as strengthening of structural elements such as column, footing, beam, slab, lintels etc. This work shall be performed by a structural engineer (proof checking) empanelled with the chief Architect MP PWD Bhopal or any government engineering college. The rate shall include all charges for labour, testing Consultation of geotechnical consultant and site visit to carry out site testing Per Sqm (built up area of wing/block of existing building where construction is proposed).		
29.11		Drafting structural drawing	Drafting structural drawing on the basis of the structural design supplied by the department in accordance with all relevant codes/manuals and as per IS: 1893-2002 and IS: 13920-1993 with updation. The work shall include making revision/compliance if any and obtaining approval of drafting work, without any additional cost within time specified by the Engineer-in-Charge or from the competent authority. The drawings for approval shall be submitted to the Engineer- in- Charge in two hard copies (minimum size A-2) and there after fair drawings on 100 micron film along with seven copies of amonia prints (minimum size A- 1) and one soft copy. Drawings shallbe stamped and signed with the statement that "Drafted by me as per structural design done or got done by the department." **Note:-** This work can be done only through a person having qualification of Diploma in civil engineering/BE (civil engineering)	sqm	26.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			and having certificate in AutoCAD issued by AutoDesk or by it's authorized institution.		
29.12		BIM (Campus)	Providing and prebuilding information modeling (BIM) conversion of 2D or 3D engineering drawing(Hard or Soft copy) of a campus showing building, roads, tunnel culverts, landscapes over head/under ground services such as electrical lines, cables, water lines, sewer lines, gas pipe lines, telecom lines, optical fiber lines, file pipe lines, hydrants etc. to virtual 3D building model using Building modeling software **'REVIT'** and extraction of MEP 2D format drawing from this 3D model incorporating landscape details of the campus for visualization with finishing and true texture and colors and submission of soft copies(6 sets of CD) of the model as well as hard copies (6 sets of) 2-D drawings extracted from 2-D model.	hectare	35000.00
29.13		BIM (Buildings)	Building information modeling (BIM) conversion of architectural, structural and MEP (mechanical, electrical and plumbing) services 2-D/AutoCAD/pdf/hardcopy drawings to virtual 3 D building model using Building Modeling REVIT software and extraction of MEP 2-D format drawings from this 3-D model, incorporating interior as well as exterior architectural details of building for visualization with finishing and true texture and colors and submission of soft copies(6 sets of CD) of the model as well as hard copies (6 sets of) 2-D drawings extracted from 3-D model. Payment shall be based on builtup area.		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	29.13.1	Upto 10,000 sqm	Upto 10,000 sqm	sqm	35.00
	29.13.2	10,000 sqm to 25,000 sqm	10,000 sqm to 25,000 sqm	sqm	32.00
	29.13.3	25,000 sqm to 50,000 sqm	25,000 sqm to 50,000 sqm	sqm	30.00
	29.13.4	More than 50, 000 sqm	More than 50, 000 sqm	sqm	27.00

CHAPTER – 30.0

SEWAGE TREATMENT

CHAPTER NO – 30.0
SEWAGE TREATMENT
PREAMBLE

With the increase in urbanization, fresh water resources are becoming stressed due to increase in domestic, industrial and commercial usage and are attaining the maniacal proportion where as safe water supply and hygienic sanitation facilities are the two basic essential amenities the community needs on top priority for healthy living.

While provision of safe drinking water takes precedence in the order of provision of basicamenities to community, the importance of hygienic sanitation facilities through low cost on – site sanitation, conventional sewerage and sewerage treatment can no longer be allowed to lag behind, as about 80% of water used by the community comes out of houses in the form of waste water which unless properly collected, conveyed, treated and safely disposed off may eventually pollute our precious water resources and cause environmental degradation.

The national urban sanitation policy 2008, envisions, "All Indian cities and towns become totally sanitized, healthy and livable and ensure and sustain good public health and environmental outcomes for all their citizens, with a special focus on hygienic and affordable sanitation facilities for the urban poor and women".

Thus there is an urgent need to have proper sewage treatment at the point of generation thru De-Centralized sewage treatment plants based on effective and proven treatment process as defined in CPHEEO manual on sewage treatment, 2013 with effluent confirming to the defined Parameters as MOEFC, notification dated 13.10.2017 but according to the national green tribunal New Delhi guideline related to discharge standards for sewage treatment plant has been revised (dated 30-04-2019) and making it mandatory to recycle, reclaim and reuse of effluent for various end usage applications viz: gardening, flushing, irrigation, landscaping to name a few, which in turn will lead to reduction in precious fresh water intake along with scientific sludge handling, treatment and disposal .

Above will help

- To Protect Public Health
- To protect the Environment and States Water Resources
- To Promote Recycle, Reclaim and Reuse of Treated Sewage Water for Non-Potable Drinking Purposes
- Increase in the Adequate Sewerage Coverage to all Population.
- Reuse of treated sewerage can provide incremental supply for non – potable applications and thus reduce need for augmenting supplies. In other words, water reuse promotes environmental sustainability by reducing burden on already stressed basin and aquifers and preventing their depletion.

The major aim of waste water treatment is to remove as much of the suspended solids as possible before the remaining water, called effluent is discharged the environment. Treated wastewater to ported public health prevent disease causes environment, prevent pollution of water bodies and can be reused in flushing or horticulture purposes.

All electromechanical equipment (used in STP/ETP) shall not produce operational sound as per the noise pollution (Regulation and Control) Rules 2000.
(The Principal Rules were published in the Gazette of India, vide S.O. 123(E), dated 14.2.2000 and subsequently amended vide S.O. 1046(E), dated 22.11.2000, S.O. 1088(E), dated 11.10.2002, S.O. 1569 (E), dated 19.09.2006 and S.O. 50 (E) dated 11.01.2010 under the Environment (Protection) Act, 1986.) Ambient Air Quality Standards in respect of Noise)

TABLE – I
Proposed Sewage Treatment need to be designed based on the following Parameters

S. No	Parameter	Incoming sewage	Desired sewage discharge standard
A	pH	6.5 -9	5.5 -9
B	BOD	250-450	10
C	TSS	150– 400	20
D	COD	300-500	50
E	Fecal Coli form[MPN/ 100 ml]	-	<230
F	Nitrogen Total	40-100	10
G	Phosphorus Total (For Discharge into ponds lakes)	5-10	1.0

SEPTIC TANK SIZE -
TABLE - II
RECOMMENDED SIZES OF SEPTIC TANKS FOR 5-20 USERS

No. of users	Length	Breadth	Liquid depth (Cleaning interval of)	
			1 year	2 year
	M	M	M	m
05	1.5	0.75	1.0	1.05
10	2.0	0.90	1.0	1.40
15	2.0	0.90	1.3	2.00
20	2.3	1.10	1.3	1.80

NOTES:

1. The capacities are recommended on the assumption that discharges from only WC will be treated in the septic tank.
2. A provision of 300 mm should be madefor free board.
3. The sizes of septic tanks are based on certain assumptions, while choosing the size of septic tank exact calculation shall be made.

TABLE - III
RECOMMENDED SIZES OF SEPTIC TANKS FOR RESIDENTIAL COLONIES

No. of users	Length	Breadth	Liquid depth (Cleaning interval of)	
			1 year	2 year
	M	M	M	m
050	05.0	2.0	1.0	1.24
100	07.5	2.65	1.0	1.24
150	10.0	3.0	1.0	1.24
200	12.0	3.3	1.0	1.24
300	15.0	4.0	1.0	1.24

NOTES:
1. A provision of 300 mm should be made for free board.
2. The sizes of the septic tank are based on certain assumptions while choosing the size of septic tank, exact calculation shall be made.
3. For population over 100, the tank may be divided into independent parallel chambers for ease of maintenance and cleaning.

TABLE - IV
RECOMMENDED SIZES OF SEPTIC TANKS FOR HOSTEL AND BOARDING SCHOOLS

No. of users	Length	Breadth	Liquid depth (D) for stated Intervals of sludge withdrawal	
			Once in a year	Once in 2 year
	M	M	M	m
50	5.0	1.6	1.3	1.4
100	5.7	2.1	1.4	1.7
150	7.7	2.4	1.4	1.7
200	8.9	2.7	1.4	1.7
300	10.7	3.3	1.4	1.7

NOTES:

1- A provision of 300 mm should be made for freeboard.
2- The sizes of the septic tank are based on certain assumptions while choosing the size of septic tank exact calculation shall be made.
3- For population over 100, the tank may be divided into independent parallel chambers for ease of maintenance and cleaning.

STP (Sewage Treatment Plant) – In the Indian Environmental and Climatic Conditions, De-Centralized Sewage Treatment Plants are designed and installed based on depending on End Usage Applications. De-Centralized STP offers Added Advantages like –

- Minimum Turn Around Time for Quick Deploy ability
- Ease of Installation and Commissioning
- Less Treatment Cost /KLD
- Reduces the Cost of Sewerage and Energy consumed in Pumping to designated City STP.
- Ease of Maintenance and Monitoring
- Effluent can be Reclaimed and Reused for the Location specific End Usage Application.
- Minimum Civil Construction

Typical Sewage Treatment Scheme

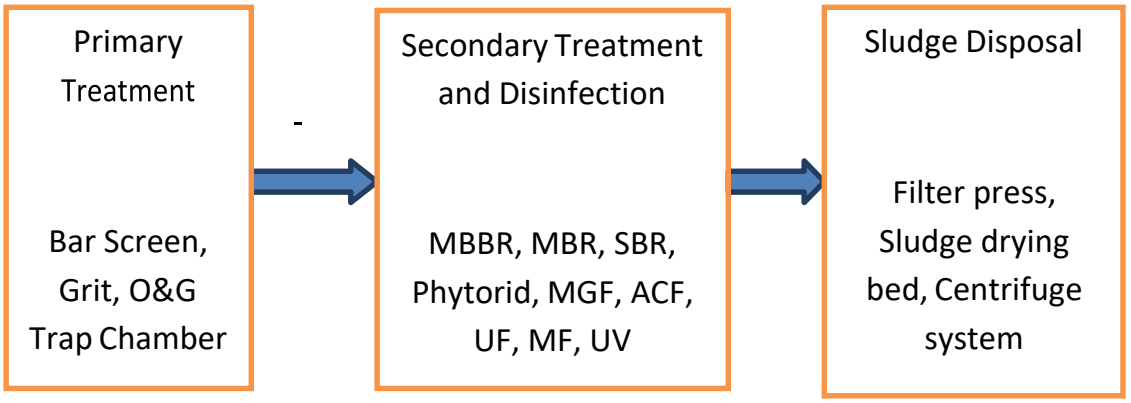


TABLE - V

Relationship between USER and KLD for STP -

Type of Building	Per Capita Load	Example		(LPD) Liter per day	KLD (Kilo liter per day)
For Residential Building	135L/PERSON	100 flats, consider 5 person in each flat	100 X 5 X135	67500	67.5

TABLE - VI

Typical Water Requirement/Supply across various END USERS -

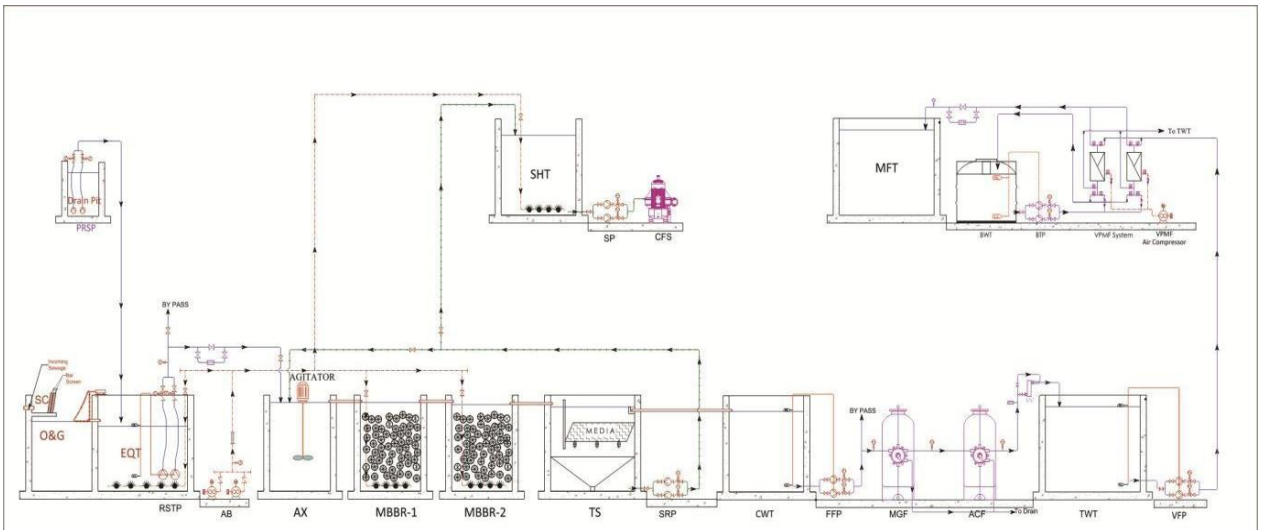
S. No.	Description	Tentative Water Requirement
a.	Office	45 lpcd (liter per capita per day)
b.	Factories –Where Bathrooms are provided	45 lpcd
	w/o Bathrooms	30 lpcd
c.	Boarding Schools	135 lpcd
d.	Day Schools/Colleges	45 lpcd
e.	Hostels	135 lpcd

f.	Hotels	180 l/bed
g.	Restaurants	70 l/seat

MBBR (Moving Bed Bio Reactor) –

Sewage from the collection tank is received to Equalization Tank (EQT) for dampening of flow variation and to equalize the concentration after passing through fine and coarse screen with suitable lifting arrangement. It shall be feasible to take out the screens from the chamber for periodical cleaning. Oil and grease trap air shall be supplied in EQT by means of diffused aeration to keep the sewage in suspension.

MBBR reactor for the biodegradation of dissolved organics by biomass attached on the surface of specific synthetic media (suitable capacity as per manufacturer recommendation). Air blowers Capable of delivering of free air at minimum 0.37 kg/cm2 driven through "V" belt or directly coupled through flexible coupling to a TEFC motor of suitable HP, 50 cycles a/c supply .The biodegraded effluent from MBBR shall be let in to a Secondary Clarifier (tube/lamella settling media to be installed in Secondary tube/lamella Settler tank along with suitable sludge removal arrangement either by pumps) for separation of scoured bio-film.



Flow Schematic for Sewage Treatment Plant

List Of Electromechanical Unit		
Mark	Description	Unit
RSTP	Raw Sewage Transfer Pump	2 No's
FFP	Filter Feed Pump	2 No's
PRSP	Plant Room Sump Pump	2 No's
TWTP	Treated Water Transfer Pump	2 No's
AB	Air Blower	2 No's
SRP	Sludge Recycling Pump	2 No's
MGF	Multigrade Filter	1 No
ACF	Activated Carbon Filter	1 No
SP	Screw Pump	2No's
CFS	Centrifuge System	1 No
UV	Ultra Violet System	1 No
VPF	VPMF Feed Pump	1 Set
VPMFBP	VPMF Backwash Pump	2No's
BWT	HDPE Backwash Tank	1 No

Legend	
Mark	Description
	Air Line
	Sludge Line
	Effluent Line
	Water Level
	Butterfly Valve
	Non Return Valve
	Pressure Gauge
	Pump
	Air Blower
	Ball Valve
	High Level
	Low Level
	Air Rota Meter

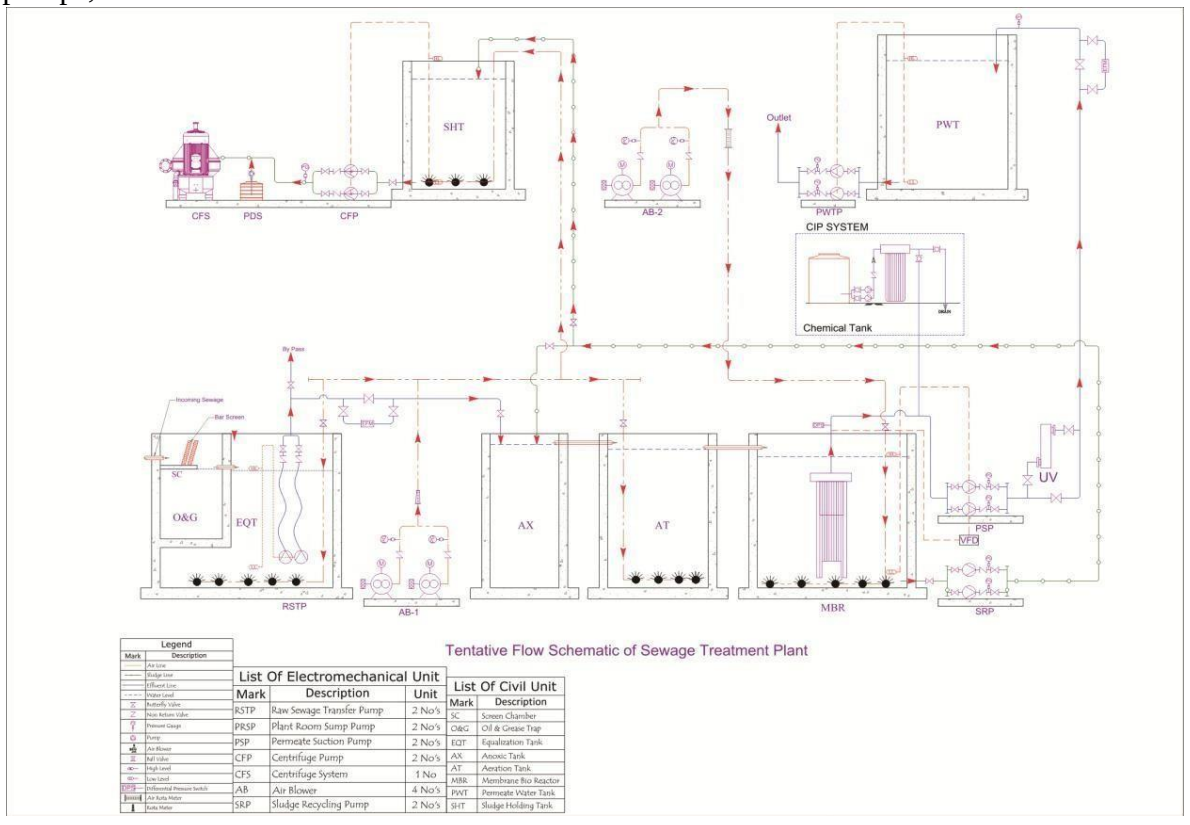
List Of Civil Unit	
Mark	Description
SC	Screen Chamber
O&G	Oil & Grease Trap
EQT	Equalization Tank
AX	Anoxic Tank
MBBR-1	Moving Bed Bio Reactor-1
MBBR-2	Moving Bed Bio Reactor-2
TS	Tube Settler
CWT	Clear Water Tank
TWT	Treated Water Tank
MFWT	Micro Filtration Water Tank
SHT	Sludge Holding Tank

MBR (Membrane Bio reactor) –
Membrane = A thin sheet of natural or synthetic material that is permeable to substances in solution and Bio Reactor = Reactors for Biological treatment.

Sewage from the collection tank is received to equalization tank for dampening of flow variation and to equalize the concentration after passing through fine and coarse screens with suitable lifting arrangement. It should be feasible to take out the screens from the chamber for periodical cleaning and oil and grease, trap along with mechanically pre-screened wastewater-automatic fine screen with 3mm pore size than sewage treated biologically and separated fromthe activated sludge by ultra filtration plate membranes.

The air-purged sewage from equalization tank shall be pumped (non clogging type pumps, having CI casing CI impeller & CI Shaft complete with all accessories, motor of required capacity). Delivery header with isolation valve, pressure gauge on delivery line with isolation cock level controller with wiring to control the level of sump automatically. Pump shall have following duty, min Head = 12 Mtr, anoxic tank. Air blowers capable of delivering free air at min 0.37 kg/cm2 driven through "V" belt or directly coupled through flexible coupling to a TEFC motor of suitable HP .The MBR system is a combination of biological wastewater treatment and high-efficient solids/liquid separation.

The surplus sludge removed from the filter chambers of the MBR system has a high concentration of approx. 1– 1.5 % MLSS (compared to 0.3 – 0.5 % MLSS achieved by conventional biological systems). This is a surplus sludge reduction of more than 50 %. Significantly reduced investment and operating costs for sludge thickening are therefore another positive aspect. . Instrument for the auto operation of system and safety of MBR modules, electro-magnetic flow meter of 40 mm at the sewage inlet and outlet of membrane suction pump of suitable size, pH meter at membrane suction pump outlet for measurement of pH, differential pressure switch for pressure alarm high and low, flow switch for flow alarm high and low, online turbidity meter for checking the health of membrane at the outlet of membrane suction pump, online rota meter for checking the air flow to the MBR module, VFD for permeate pumps, VFD & DO sensor for air blower.



Tentative Flow Schematic of Sewage Treatment Plant

Legend	
Mark	Description
	Air Line
	Sludge Line
	Effluent Line
	Water Level
	Butterfly Valve
	Non Return Valve
	Pressure Gauge
	Pump
	Air Blower
	Ball Valve
	High Level
	Low Level
DPS	Differential Pressure Switch
	Air Rota Meter
	Rota Meter

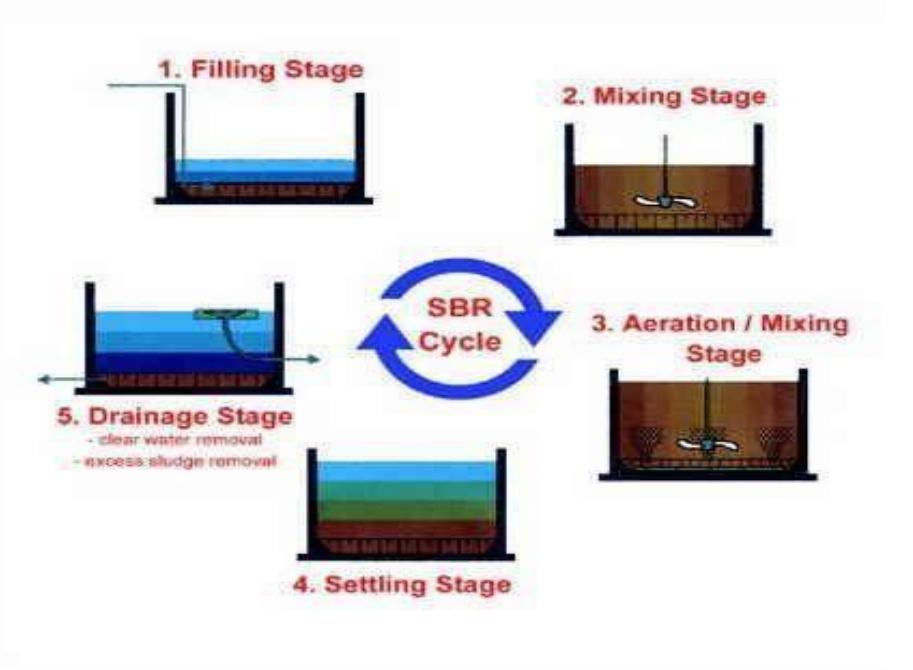
List Of Electromechanical Unit		
Mark	Description	Unit
RSTP	Raw Sewage Transfer Pump	2 No's
PRSP	Plant Room Sump Pump	2 No's
PSP	Permeate Suction Pump	2 No's
CFP	Centrifuge Pump	2 No's
CFS	Centrifuge System	1 No
AB	Air Blower	4 No's
SRP	Sludge Recycling Pump	2 No's

List Of Civil Unit	
Mark	Description
SC	Screen Chamber
O&G	Oil & Grease Trap
EQT	Equalization Tank
AX	Anoxic Tank
AT	Aeration Tank
MBR	Membrane Bio Reactor
PWT	Permeate Water Tank
SHT	Sludge Holding Tank

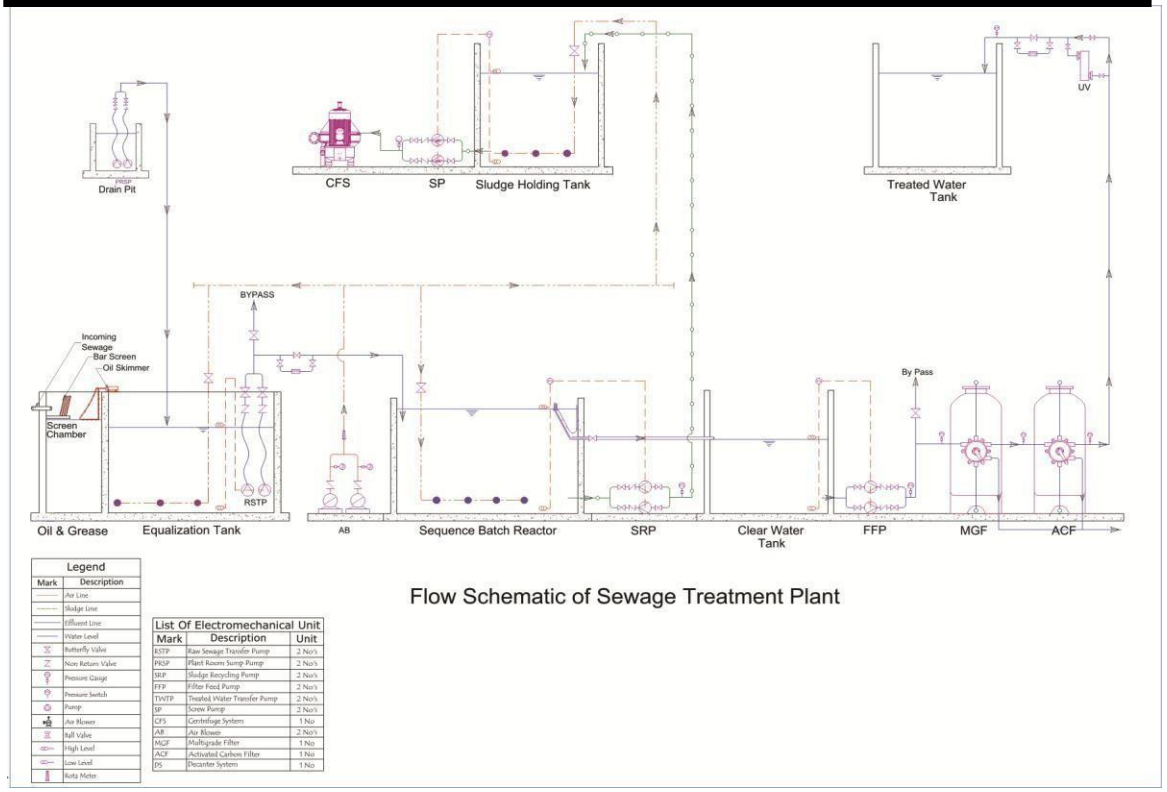
SBR (Sequential Batch Reactor) –

The process operates on the continuous inflow and batch outflow principle. The phases of aeration, settling, decantation occur sequentially and are controlled by PLC. The tank is divided into main aeration & settler tank.

The sewage flows continuously into main aeration tank, which acts as a biological selector enhancing the growth of bacteria Sewage from collection tank flows through openings at the bottom of the baffle wall and into the main aeration tank where BOD removal and nitrification occur. After aeration phase, sludge separation occurs in the settling phase. After settling, the clear water is removed. De-nitrification occurs during anoxic periods of settling and decants phases. The sludge is taken during decant phase.



Process Cycles - Anoxic Fill, Aerated Fill, React, Settle, Decant.



Flow Schematic of Sewage Treatment Plant

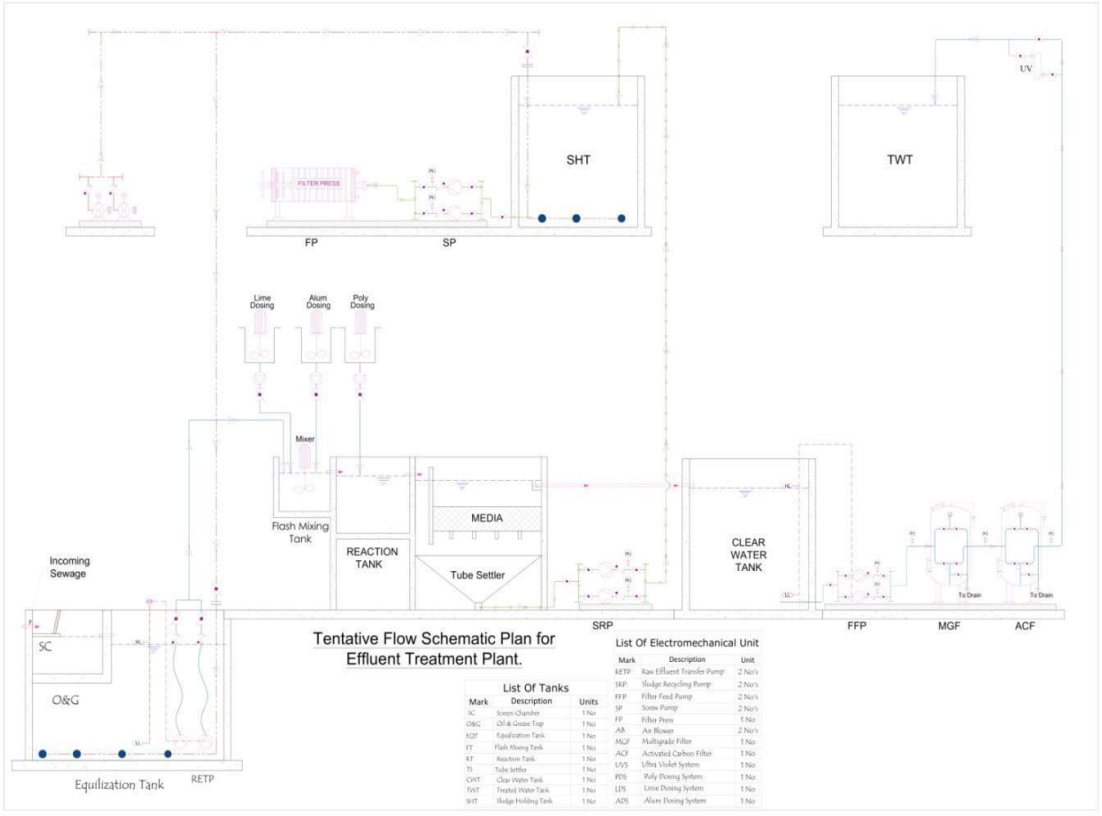
ETP (Effluent Treatment Plant) –

Effluent may contain large variety of potentially hazardous components including, microbiological pathogens, radioactive isotopes, disinfectants, sterilizant, drugs and their metabolites, chemical compounds, heavy metals and pharmaceuticals etc., at high concentrations.

The raw effluent will be first passed through a bar screen chamber where any extraneous/floating matter would be screened & trapped. Subsequently the raw effluent water shall be collected in oil & grease chamber wherein the floating oil shall be separated.

Acid dosing (depend upon capacity of plan as per design) since the effluent generated is mostly alkaline in nature an appropriate amount of acid is dosed for neutralizing the same in equalization tank. The effluent would then be collected in a receiving/equalization tank where the variations in flow and characteristics are dampened, which otherwise can lead to operational problems and moreover it allows a constant flow rate downstream. Flash Mixer is a chamber in which coagulants and other chemicals are blended with raw water prior to flocculation. The chemical and water mixture is vigorously agitated within the chamber resulting in almost instantaneous mixing. This tank should have 1-5 min of detention time with suitable velocity gradient.

Tube/Lamella Settler shall be provided for the effluent passed from flocculation tank along



with suspended solids to settle. The settled solids shall be recycled back to aeration tank as per requirement and excess biomass shall be transferred to sludge dewatering system.

Components of STP/ETP –

Bar screen: This is the first unit of the plant in which large or floating materials in the sewage gets arrested and blockage or choking of the downstream equipments can be avoided.

Set of (SS304 - suitable for handling peak flow of raw sewage into the equalization tank)coarse and fine bar screen (stainless steel perforated corrugated screen) consisting of 1st screen 10 mm and 2nd screen, 03 mm manually punched holes (In MBR technology 3rd screen 03 mm manually punched holes followed by automatic type fine screen, 03 mm perforation screen withautomatic self cleaning system) provide in bar screen chamber/tank to remove the large sized particles present in the sewage water associated with suitable SS angle/channel/flats of minimum 06 mm thickness mounted inside the bar screen chamber at an inclination. Bar rack fabricated from SS bars shall be provided to cover the width of the screen chamber with a racking arm for manual operation. Screen sizes are depends on the KLD.

Oil, grease trap and grit removal system - Oil & grease trap is provided after the bar screen, where free floating oil is arrested prior to entry in the plant. Accumulated oil will be removed periodically and disposed of properly. Electrically operated MOP/Tube/Belt, corrosion resistance static oil remover in oil/grease and grit chamber as per manufactures specification with suitable collection basin. Provide electronic type level indicator and controller for automatic operation of the system with high/low level alarm complete with auxiliary NO/NC contacts.

Equalization cum collection tank – To absorb variation in quantity and quality of sewage and to provide uniform flow at the downstream treatment process, a collection or equalization tank is provided.

Raw sewage transfer pump – To transfer sewage from equalisation tank to anoxic tank. Two numbers (one workingand one standby) submersible/non submersible centrifugal type (pumping head 8-10 mtr.) raw sewage pumps with cutter version to pump sewage from equalization tank to anoxic tank Pumps should have vertex/non clogging impeller of SS304, coupled to suitable motors with SS 304body with IP 68 protection. The pump shall have CI casing, CI impeller and SS shaft, TEFC induction motor suitable for 415+10% volt, 3 phase 50 cycle AC power supply, dry running protection NRV& vibration eliminators on delivery side of pumps. Capacity depends on KLD of plant.

Air blower – For supply of air into the diffusers. In STP two numbers (one working and one standby) twin type rotary acoustic type air blower complete with motors, pulleys, one piece rigid CI casing and side plates, silencer, base plate, anti-friction bearings, rotary oil sealing having horizontal configuration. Blower should be with acoustic enclose to accommodate two number of twin lobe air blower, noise level reduction to the tune of 25-30% should be achieved, rubber gaskets should be provided to all doors and external joints to resist weathering air blower having TEFC motor of suitable HP suitable for 415+- 10% volts, 3 phase, 50 cycle A/C supply with all accessories complete in all respects. Flow rate depend on KLD of plant.

Flash mixing tank with motor – Provide vertical mounted gear box, motor and MSEP agitator

(Capacity depend upon KLD of plant) mechanism in the flash mixing tank. The mixture is agitated quickly and thoroughly in a process, called flash mixing. The chemicals introduced into the water stream will attract any very fine particles, such as silt, that will not readily settle or filter out and make them clump together. These larger, heavier formations are called flock.

Dosing system with reaction tank – Reaction tank (It is made as per vender's design based on capacity of ETP) in which Alum/Poly/Lime dosing are take place.
Dosing system consisting of one HDPE tank of required capacity (100 litres) with a positive displacement diaphragm dosing pump having variable flow rate of 0-6 lph. The motor (vertical mounted gear motor) shall be suitable for operation at 240 V/single phase/50 Hz. supply. The pump shall be supplied complete with necessary polypropylene piping, valve, strainers, and low level switch and injection fittings. The pump shall be speed and stroke control.

Anoxic tank- Anoxic tank is provided for de-nitrification process where nitrogen removal takes place without supply of oxygen in anoxic condition. Provide MSEP agitator (capacity as per KLD of plant) with gear box and motor to be provided in the anoxic tank along with its suitable mounting frame.

MBBR Media – This is the main section of the plant where degradation of organic pollutants with the help of aerobic micro-organism takes place. To provide higher surface area (not less than 25100SQMT/CUM) for micro-organism floating media MBBR media is provided on which micro-organism growth take place and to support the biomass in the aeration tank. Prefabricated MBBR systems (in two reactors) should be made of min 06 mm thickness of MS plate supported and strengthened with required supporting structure to make it a self supported structure and capable to take required water pressure, inner side coated with 03 mm FRP/anti corrosivecoating.

Diffuser system –For supply of fine air into the tank—EQT, MBBR and SHT. Coarse and fine bubble diffusers UPVC header, disc type, consisting of set of EPDM membranes, non clog type air dispersion system, SS304 clips, UPVC support pipe (10Kg/Sq cm pressure rating and minimum wall thickness 3.0 mm), UPVC fitting (like - elbows, tees, collars etc.) complete. Air dispersion grid shall be assembled in modular form so that they can be replaced / repaired easily from top plate form.

Tube/Lamella settler tank with tube/lamella media – Tube/Lamella media (as per design) to be installed in tube/lamella settler tank along with retainer plates and suitable sludge removal arrangement either by pumps or any other suitable arrangement. Capacities, length, width, depth of the media (depends upon KLD of plant) shall be enough for proper functioning of STP. Gravity overflow from the bioreactor is collected in the tube/lamella settler tank. In this settling tank, generated sludge from the bioreactor undergoes a gravity settling. Clear supernatant from settling tank will flow by gravity to a chlorine contact tank.

Chlorine dosing system with Clear water tank -
Chlorine dosing system —installed in clear water tank in which chlorine dosing to take place. This system consisting of one HDPE tank of required capacity (as per capacity of plant) with a positive displacement diaphragm dosing pump having variable flow rate of 0-6lph. The motor shall be suitable for operation at 240Vsingle phase50Hz supply. The pump shall be supplied complete with necessary polypropylene piping, valves, strainers, and low level switch and injection fittings and pump shall be speed & stroke control.

Plant Room with Sump Pump – Two numbers pumps (one working and one standby) to pump back to equalization tank over flown sewage from any tank collected in sump. Cost shall be inclusive of PVC flexible pipe 80 mm dia., rope for lifting pump from sump to outside. Capacity of flow rate depends on KLD of plant having pumping head 12-15mtr. All motors, valves, level indicator/controller, blowers, filter press, control panel etc. shall be providing in plant room.

Plant interconnecting piping and valves – All piping works for the STP consisting of interconnecting piping, valves within the battery limit of the sewage treatment plant MOC of the Piping: GI/MS/UPVC piping with valves. All inclusive necessary to finish the piping of STP. UPVC for sewage, filter water, backwash water, treated water and submerged air line, fittings, specials and other appurtenances, auxiliaries and accessories required as per process design and scope of work. All the piping, valves, specials shall be designed for peak flow. In case of pumping mains thrust blocks shall be provided whenever required. In case of buried pipes warning tapes shall be provided of the appropriate colours.

Manual/PLC based electrical control panel – It contains (all the knobs, switches, automation, timers of all nature associated with STP) suitable for 415V, 3 phase, 4 wire, and 50 Hz distribution system. The panel shall be wall stand mounted type; construction of panel should be 2 mm CRCA sheet with duly powder coated with 07 tank pre-treatment process. Separate earth bus shall be provided throughout the length of the panel. The incoming and the outgoing feeder breakers, switch fuse units, change over switches, indicating lamps, meters etc. shall be accommodated in a modular multi tier arrangement.

Incoming/outgoing feeder – depend upon the capacity of plant.

Cable laying with cable tray – Perforated type hot dip galvanized sheet steel cable tray with cable lying (used in connections as per vender's design) of different widths and delivered in standard lengths along with coupler plates and requisite number of hard wares.

Earthing systems - GI earth conductor of suitable sizes as per specification along the cable trenches or cable trays or fixing to walls or structures or below the slab, burries individually in paved/unpaved areas including supply of earthing strip and minor civil work finishing after laying of earthing strips etc. The installation shall include drilling of holes, connection and crimping of adequate lugs, clamping, hardware material, etc. All hardware shall be as per the MOC of earthing strip

Pressure gauges - Chemical sealed, glycerine filled, min dia. size 4 (one no). For each rotating equipments and filters with all necessary hardware, valves, drain valves, snubber etc. and pH meter at membrane suction pump outlet for measurement of pH and correction. Piping (CPVC for tank overflow pipeline and MS epoxy for air piping & pumped effluent riser in non submerged and SS304 for submerged air piping) and valving as per design.

Level switches - Float type level switches with three level set points except for sludge recirculation pumps. Each level switch should be with 01 NO and 01 NC contacts with weather proof enclose. Also provide differential pressure switch transmitter and pressure alarm high and low, flow switch for flow alarm high and low.

Flow Meter – To check the flow of sewage. One number electromagnetic flow meter shall be providing for treated water flow measurement. Electro-magnetic flow meter cum totalize at the sewage inlet and outlet of membrane suction pump of suitable size location inlet andoutlet. Online rotameter for checking the air flow from air blower and for checking the air

flow to the MBR module. Air rotameter for biological air supply in MBBR, SBR tank and air scouring.

Sludge holding system— to dewater sludge.

Sludge recycling pump – Two numbers sludge recycling (one working and one standby) to pump the sludge from tube/lamella settler tank and from MBR tank (in MBR technology) and from SBR tank (in SBR technology) to sludge holding tank and transfer the sludge and recirculation of sludge to anoxic tank, necessary arrangement to be made in piping so that same pump can be use for both the purpose. The pump shall be horizontal centrifugal non- clog self priming pump with wetted parts in CI construction. . Capacity of flow rate depends on KLD of plant having pumping head 8-10mtr.The pump shall be coupled to the adequate
H.P. electrical motor, mounted on a common base frame

Sludge feed screw pumps – Two numbers sludge feed pumps (one working and one standby) screw type to pump the sludge from sludge holding tank to filter press or feed secondary sludge to mechanical dewatering device like - filter press/ centrifuge /sludge drying bed. The pump shall be of mono block screw type press feed pump coupled to the adequate H.P. electrical motor, mounted on a common base frame of suitable capacity with CI body, SS- Impeller suitable for handling biological sludge

Filter press system – Plate and frame type mechanical sludge de-watering system, it should be with hydraulic closing mechanism. Filter press of suitable capacity depending upon as per design for handling sludge generated from the STP.

Centrifuge system/Bag filter- Basket type centrifuge system designed dia. for the sludge handling as per required capacity.

Sludge drying bed –A sludge drying bed is a common method utilized to dewater sludge via filtration and evaporation. Perforated pipes situated at the bottom of the bed are used to drain seepage water or filtrate. A reduction of about 35% or less in moisture content is expected after drying**.** A sludge drying bed usually consists of a bottom layer of gravel of uniform size over which is laid a bed of clean sand. Open jointed the under drains are laid in the gravel layer to provide positive drainage as the liquid passes through the sand and gravel.

Filter feed pump – Transfer water from clear water tank into filter vessels. Two numbers of centrifugal filter feed pumps (head 25-30mtrs- and capacity depend upon the KLD of plant - one working and one standby), CI body, and CI impeller motor, pressure gauge with isolation cock, isolation valve, NRV on delivery line. Isolation valve, stainer at suction. The pump shall be suitable for 415±10% volts 3 phase AC supply.

Multi grade sand filter (MGF) – Mild steel epoxy (MSEP) pressure type multi grade sand pressure filter (FRP) for filtration of suspended impurities in treated sewage water.
The filter shall be fitted with inlet distribution and bottom collecting system. Filter shall include media such as - multi grade sand, shellac and complete set of valves, standard fittings like – cover, sampling cock, rinse drain (backwash) valve, air vent valve with piping, bolts, nuts and rubber gaskets, vacuum breaker, pressure gauge with needle valve and frontal pipe work done with GI 'B' class pipes. Flow rate depend upon KLD of plant and filtration velocity 18cumtr/hr/sqmtr. Filtering sand, anthracite, pebbles and gravel media.

Activated carbon filter (ACF) –Mild steel epoxy (MSEP) pressure type activated carbon filter (FRP) for filtration and removal of colour/odour in sewage water. The sizing/diameter of ACF shall be designed based on specific velocity. The vessel is painted externally and internally

with epoxy paint with primer with frontal piping & valves, back wash system & filter media like – activated carbon, multi grade sand and shellac etc. complete with standard fittings like cover, pressure gauge at inlet & outlet, sampling cock, rinse drain valve, air vent valve with piping, bolts, nuts & rubber gaskets. Flow rate depend upon KLD of plant and filtration velocity 14cumtr/hr/sqmtr. Filtering sand, anthracite, pebbles and gravel media.

Treated water transfer pump –
Two numbers of treated water transfer pumps (one working and one standby) to pump the Water from treated water tank to external drainage or outlet.

Treated water tank –
To collect treated water from activated carbon filter with inlet, outlet, and drain.

Micro and Ultra filtration system –Micro filtration and ultra filtration are both separation processes by which a contaminated liquid is passed through a semi permeable membrane that removes solids too large to fit through the membrane's pore size, yielding a purified liquid stream. In Micro filtration system piping and valves with all the required accessories & chemical cleaning system, complete in all respects. Module (fully back washable top to bottom filtration) including SS 304, Pneumatic Valve, PLC Electrical Panel, flow meter, compressor and all other accessories making the system complete.
The difference between microfiltration and ultra filtration is in the membrane pore size. Microfiltration pores are on the order 0.1 – 10 µm in size and are able to completely hold back bacteria and partially hold back viruses. Ultra filtration pores are on the order of 0.005 – 0.1 µm in size and can also completely remove viruses from the water.

Ultra violet system - UV dis-infection system for the complete disinfection of the finaleffluent of the following make or equal approved. The unit shall have over 99.9% bacterial reductions from inlet to outlet. The disinfection chamber shall be constructed of SS 316L on all welted parts. The UV lamp shall be of low pressure mercury vapor type with hard glass enclosure, the sockets shall be water tight & vibration resistant. Flow rate depends upon the KLD of plant.
The UV unit shall have with reactor, cabinet housing, cabinet cooling, treatment chamber, electrical panel, temperature safety control, lamp out alert, UV radiometer along with UV monitoring system and UV monitoring readout panel. The UV dosage shall be > 60,000 uW Sec / sq.cm.

MBR Module – MBR module/package ultra filtration membranes with SS–304 skid for housing the membranes & air grid for scouring the sludge, permeate S.S.-304 piping and air blower etc. complete as required. Flux- 30 LMH, surface area (as per design) of each membrane. Membranes shall be reinforced hollow fiber capable of regular backwashing with and without cleaning chemicals to minimize pore fouling, respectively also having membrane lifting mechanism and membrane permeate pumps (1w+1s).

The material used for the manufacturing of the membrane in fibers chemically resistant to high concentrations of chlorine, low pH and high pH wash solutions for up to 24 hours.

Membrane clean-in-place (CIP) system- To clean the MBR membrane by back wash. Provided solution dosing pump with metering (0-100 LPH) and fixing of citric solution dosing pump with metering (0-50 LPH) comprising of required capacity FRP tank inlet/outlet connection, piping & valves along with pump for membrane cleaning system complete in all respects. Its design depends upon the KLD of plant.

Decanter – Decanter (as per design) for required capacity STP (in KLD) MOC- SS-304 (PLC controlled automatic decant mechanism) shall be floating type & decanter will be passed over the design top water level during aeration and settling phase, thereby eliminating any possibility of solid carry over during this phase.

SBR process shall work on batch mode in single step. It shall perform biological organic removal, nitrification, de-nitrification and phosphorous removal. It shall be capable of simultaneous sludge Stabilization.

The oxygen required shall be supplied through diffused aeration system with auto control of oxygen level in tank. The system shall be designed in such a way that growth of filamentous bacteria is restricted. Complete operation of SBR process including decanting rate, sludge recirculation and wasting of excess sludge shall controlled by PLC. Treated Sewage from SBR process units shall be collected in chlorination tank for its disinfection.
The maximum decanter loading at average dry water flow condition, shall not exceed 1.5to1.8m3/m/min. The drive mechanism will be designed for continuous duty, variable speed mode of operation thereby producing a uniform effluent flow throughout the decant phase. The decanter drive will be configured such that the decanter weir reaches bottom water level at the end of the decant phase thereby maximizing settling time. Providing & fixing of motorized valve for decanters including required accessories (for decantation of water from SBR Tank) include timer controlled type system having required actuator and automation to decant treated water.

Constructed Wetland Based Technology -

Phytorid/constructed wetland technology is an up gradation of the existing technological concepts with respect to Indian climatic conditions.

This technology is based on principle of constructed wetland technology, these plant beds along the banks of the lake/river or as Sewage Treatment Plants would achieve waste water treatment by filtration, adsorption, nutrient uptake and decomposition.

Phytorid (developed by CSIR-National Environmental Engineering Research Institute(NEERI) that works on the principles of natural wetland) and Constructed wetland Technologyis a self-sustaining technology and has been researched extensively at premium institution like IITs and implemented at multiple placed across India and globally to treat the sewage effectively without any requirement of electricity except pumping from collection tank to constructed wetland bed which also can be avoided if the site terrain permit for the same.
Phytorid/ constructed wetland systems are based on natural treatment methods and have distinct advantages over conventional treatment plants. In this technology, treatment occurs vianatural methods such as filtration, sedimentation, nutrient uptake by plants and microbial action in a constructed system which is filled with gravels. Specifically identified different species of plants which are known to have good nutrient uptake rates are planted in the gravel bed.
Phytorid/constructed wetland unit is essentially a civil structure. It comprises of screen chambers, collection chamber, Plant bed, treated water storage tank. The system includes screen cum sedimentation chamber and modified Plant beds. Solids are separated in the screen cum sedimentation chamber and broken down an aerobically. Wastewater from screen cum sedimentation chamber further flows into modified Plant bed by gravity. Plant bed is provided with different gradation of gravels/stone aggregate and hydrophytes.

The Plant bed is divided into compartments with baffles provided in such a way that the flow of wastewater is in sinusoidal manner. This unique design provides both the anaerobic and aerobic zones in the same Phytorid/ constructed wetland bed. Aerobic zone is near roots of the plants, as plants transports oxygen from air to the roots and In-turn into water for biochemical oxidation. The flow of wastewater is 6 inch below the top gravel layer and therefore no sewage is exposed thereby escaping mosquito problems. The plants in thePhytorid/ constructed wetland bed uptake nitrates and phosphates from the wastewater. Dueto several passages through gravel beds with both aerobic and anaerobic zones faecal coliformis also reduced by more than 99%. Phytorid/ constructed wetland do not require any electricity and skilled operators for its operation.

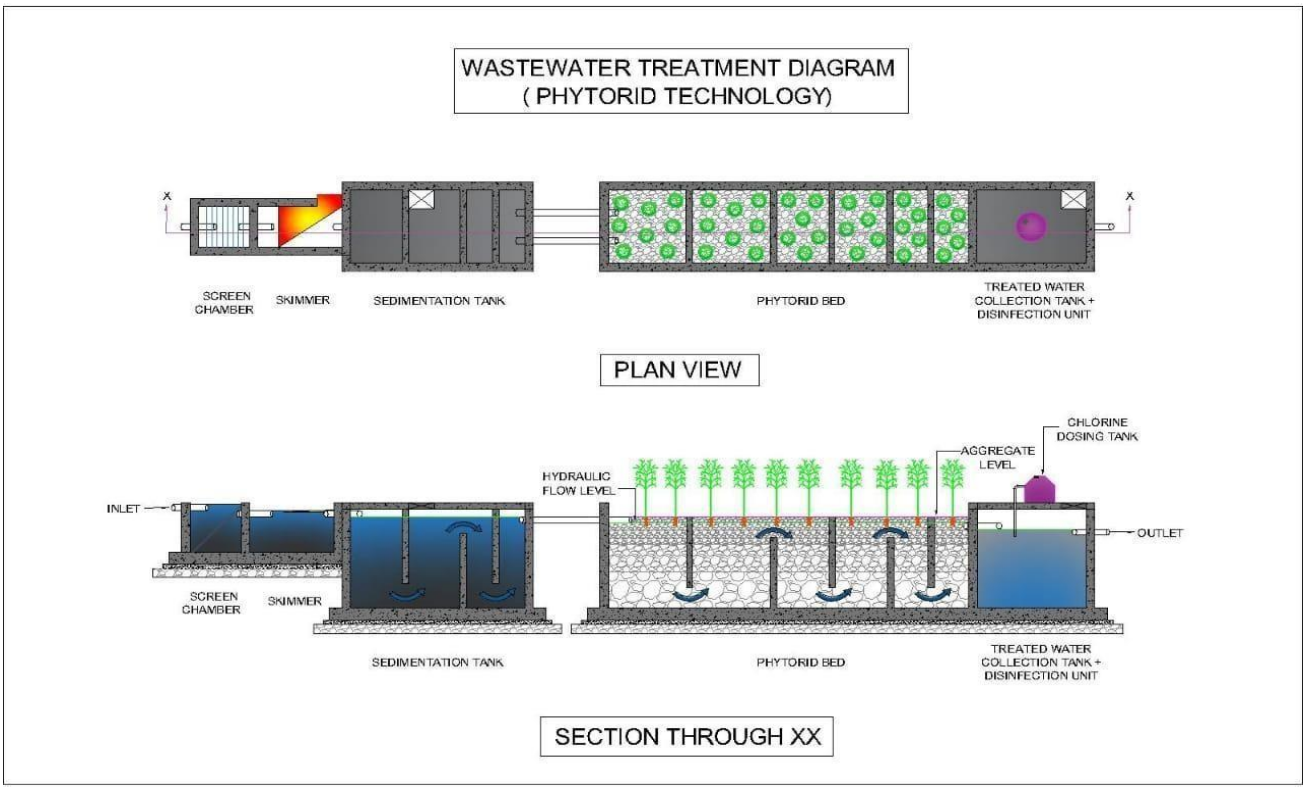
Collection-cum-Sedimentation tank –

The specified design of collection-cum-sedimentation system allows suspended solids tosettle down in the tank and achieve simultaneous removal of BOD (by more than fifty per cent). This would require no pH adjustment and clarification system to remove settled solids if domestic wastewater treatment is being targeted.

Phytorid/constructed wetland chamber –

Phytorid/ constructed wetland bed comprises of gravel bed sown with wetland plants. The system is based on the specific natural "Wetlands plants" like Typh, Canna, etc. The plants in the Phytorid bed uptake nitrates and phosphates from the wastewater. Aerobic zone is near roots of the plants, as plants transports oxygen from air to the roots and in-turn into water for biochemical oxidation. Filling media i.e. the gravel bed serves the purpose of filtration.

Capacity of Plant (KLD)	Total Area required of phytorid technology (Sqm)
10	17.27
50	66.36
100	127.71
150	188.57
200	250.33
250	316.69
300	378.04
350	439.30
400	500.66
450	562.01
500	623.37
550	684.73
600	745.99
650	807.34
700	868.70



Use of Bio-culture in Treatment:

Phytorid/ constructed wetland is a natural non-conventional treatment phenomenon to treat the wastewater using plants and combined working of their root system in a designed ecosystem, along with the natural attenuation processes. Sewage itself has inherent bacteria's which have properties to form consortia that helps reduce Biological Oxygen Demand (BOD) when provided with particular condition. However, the quantity of these bacteria is not sufficient enough as well as these bacteria's are more or less anaerobic in nature due to the prevailing septic condition in sewage.

Anaerobic process usually being a very slow process; a Phytorid/ wetland system can't be dependent only on this process and existing bacteria as it will take a very long time for these bacteria to act on the required quality of purification of sewage.

This technology being a natural system, operation is mostly passive and requires least operator's intervention. Operational aspects include harvesting of the roots once in 6 months and loosening of the gravel after a year.

Types of plant species: -

A range of plants has shown their ability to breakdown wastewater contaminants. Some of the species are listed below:

Colocasiaesculenta / Canna indica / Cyperusalternifolius / Yellow Cana / Red Cana / Pampas Grass / Cyperus Spp. / Spider Lily / Canna /Verigated / Typha spp. / Soft Rush / Bull Rush Spp. / Elephant Ear / Phragmitesaustralis / Phragmites Australis /Typha Latifolia / Arundo donax / Calamagrostis / Scirpus Lacustris / Juncus (Bulrush)

BENEFITS OF IMPLEMENTING THE SYSTEM: -

- Negligible consumption of electric power.
- The technology is very simple in design and operation therefore needs no skilled manpower for operation and maintenance.
- Negligible sludge production
- No odour
- No application of chemicals for treatment process
- Lower costs of treating water
- Extra gardens and landscaping features not required for beautification

DRDO BIO - DIGESTER TECHNOLOGY

DRDO Bio Digester technology is developed by DRDO wing of Ministry Of Defense, Government of India.

DRDO has perfected an eco-friendly biodegradation technology for human waste disposal for armed forces deployed at high altitude locations and glaciers. The sub-zero temperature does not allow the natural biodegradation of organic matter leading to accumulation of the human waste over the years, contaminating the ice which is the only source of drinking water and poses a great health risk. Further, melted ice contaminates the rivers and other water sources and ultimately spreading the contamination downstream. DRDO developed Bio Digester technology is environmental friendly, maintenance free and efficient without depending on conventional energy sources. The effluent is odorless and gets rid-off most of the pathogens.

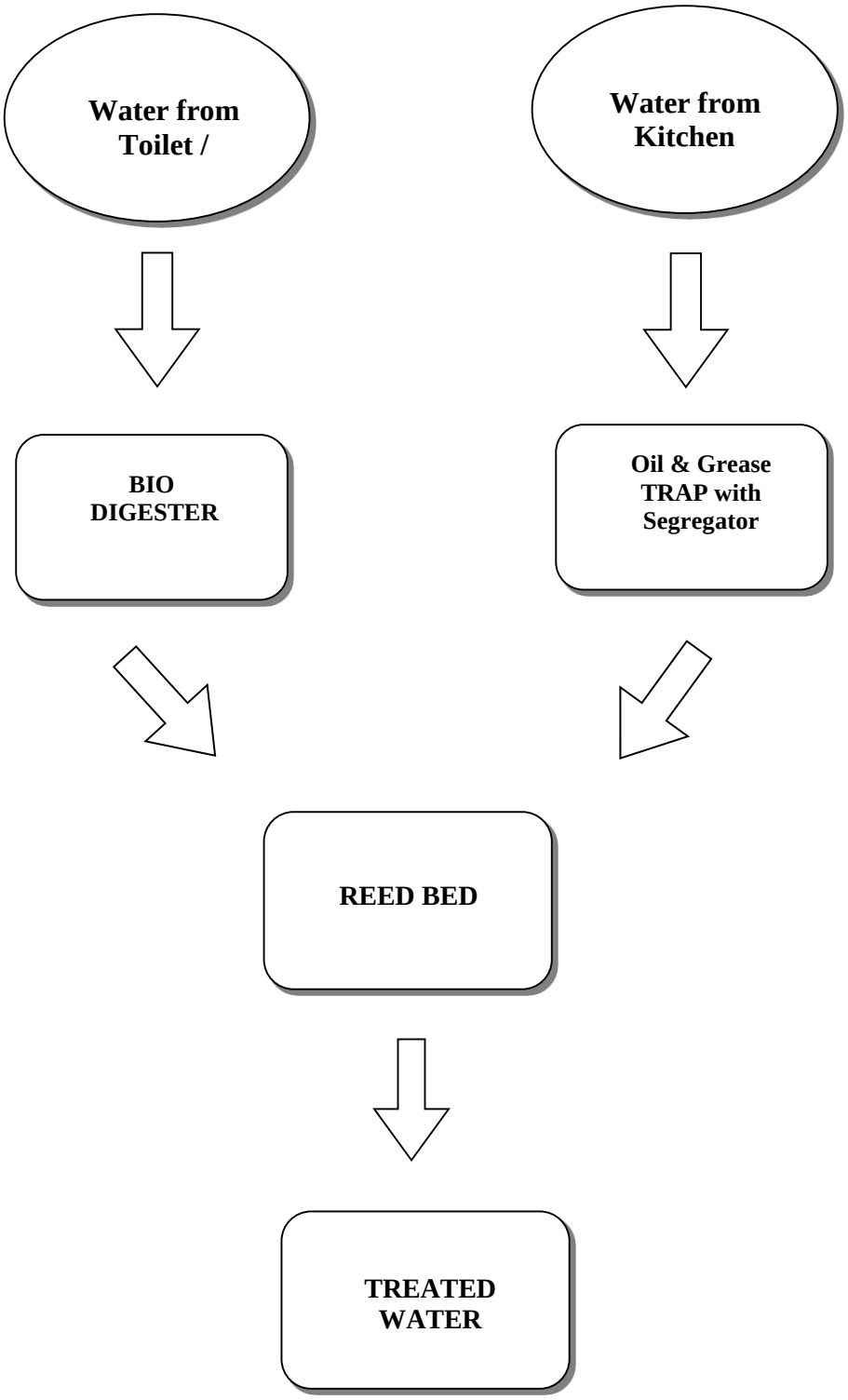
Hundreds of such Bio Digester have been installed at various locations in J&K, Sikkim and Arunachal Pradesh. The Bio Digester technology has two components: anaerobic microbial consortium and specially designed fermentation tank. The microbial consortium has been made by acclimatization, enrichment and bioaugmentation with the coldactive bacteria collected from Antarctica and low temperature areas. It is composed of four clusters of bacteria belonging to hydrolytic, acidogenic, acetogenic and methenogenic groups with high efficiency of biodegradation. Fermentation tank is made of FRP and has the provision of immobilizing the bacteria in large numbers.

Subsequently, the technology has been extended to Indian Railways for on-board treatment of human waste. Hundreds of stainless steel made Bio Digesters are in operation in different trains of IR. Under a Memorandum of Understanding (MoU) with Railways, it is planned to install these biodigesters in all trains in the years to come. Lakshadweep administration has ordered 12000 biodigesters for all houses of its ten islands.

Bio Digester – Special Features :-

- Technology developed by DRDO – Ministry Of Defense – Government Of India
- Anaerobic microbial inoculums
- Specially designed tank for efficient biodegradation
- Cold tolerant microbial consortium
- Suitable for mobile and stationary platforms
- Required one time inoculum (bacteria) charging
- Permits use of toilet cleansing agents
- Tailor made in respect to number of users, materials, situation and condition.
- Immobilization matrix for retaining higher microbial mass for survival in adverse conditions.

Bio Digester Flow Chart :



Bio Digester RCC Tank Details:

STP - Bio Digester : FRP Tank & RCC Tank Details					
Bio Digester FRP Tank Size					RCC Tank Details
Sr. No.	**No. of Users**	**Dimensions (LXWXD)**	**Capacity (in Liter)**	**Bacteria Quantity (in Liter)**	**Length x Breadth x (Depth+A)**
1	**10**	1.0m X 1.0m X 0.70m	700	200	1.5m X 1.5m X 0.70m
2	**15**	1.0m X 1.0m X 1.0 m	1000	400	1.5m X 1.5m X 1.0 m
3	**25**	1.5m X 1.0m X 1.0m	1500	400	2.0m X 1.5m X 1.0m
4	**40**	2.0m X 1.0m X 1.0m	2000	600	2.5m X 1.5m X 1.0m
5	**60**	2.4m X 1.25m X 1.0m	3000	800	2.9m X 1.75m X 1.0m
6	**80**	2.7m X 1.5m X 1.0m	4000	1200	3.2m X 2.0m X 1.0m
7	**100**	2.5m X 2.0m X 1.0m	5000	1400	3.0m X 2.5m X 1.0m
8	**150**	3.2m X 2.0m X 1.25m	8000	2400	3.7m X 2.5m X 1.25m
9	**200**	4.0m X 2.0m X 1.25m	10000	3000	4.5m X 2.5m X 1.25m
10	**300**	4.8m X 2.0m X 1.25m	12000	3600	5.3m X 2.5m X 1.25m
11	**400**	5.0m X 2.0m X 1.5m	15000	4400	5.5m X 2.5m X 1.5m
12	**500**	6.0m X2.0m X 1.67m	20000	6000	6.5m X2.5m X 1.67m

Note : A is Distance from final ground level , to bottom of sewage outlet pipe

RCC Tank for FRP Bio Digester :
(Sizes Clear from Inside – After PCC / RCC / Plaster) - Top View

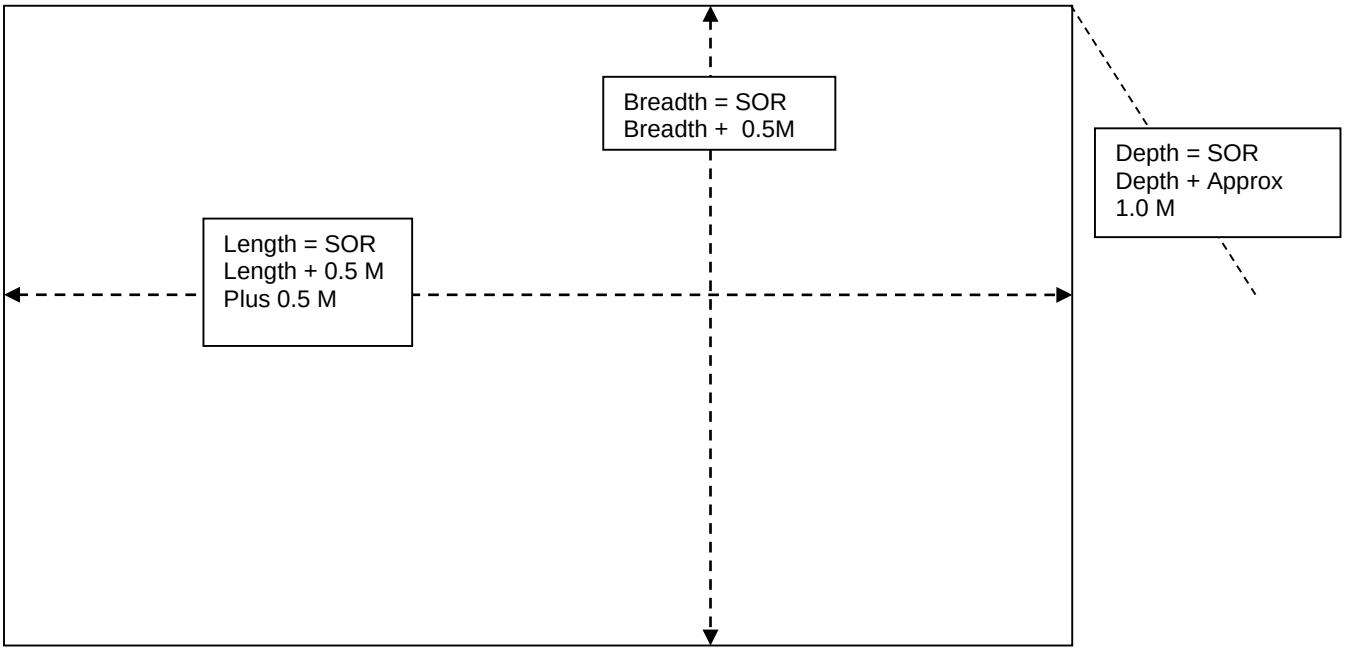
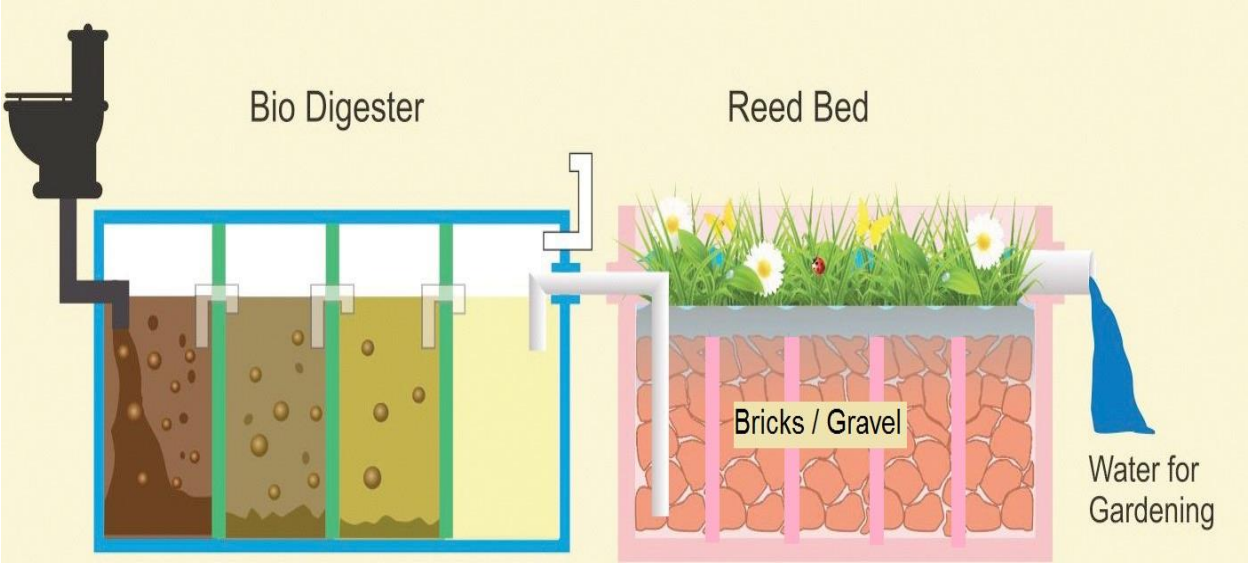
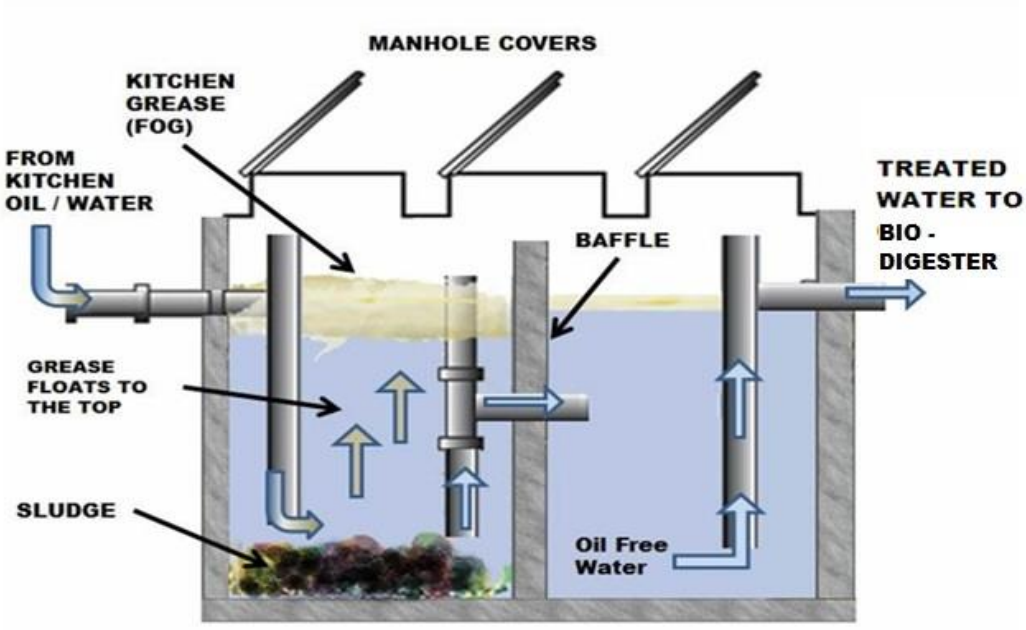


Diagram - Bio Digester Process :



Oil & Grease Trap for Kitchen Waste Water



Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
30.1		Sewage Treatment Plant(STP)/Effluent Treatment Plant (ETP)	Designing and preparing good for construction civil, mechanical and electrical shop drawings of conventional/package unit of a Sewage Treatment Plant(STP)/ Effluent Treatment Plant (ETP) as per manufacturer/vender's design and specification consisting of treatment units complete for meeting up to date/ target parameters of treated effluent conforming to Madhya Pradesh Pollution Control Board (MPPCB), MoEF GOI guidelines, National Green Tribunal guideline 30.4.2019 and technical specifications codes of practices as published by the Bureau of Indian Standard (BIS) and latest version of "Manual on Sewerage and Sewage Treatment" published by the Central Public Health And Environmental Engineering Organization (CPHEEO). The work shall include supplying, installing, testing and commissioning of electro-mechanical equipments, instruments, pipes, valves, pumps, and gauges etc. as per approved technical specifications, layout and drawing of STP/ETP. Most of the electro-mechanical unit will be placed below ground under road/greenarea/as per site condition, space available, equipment and pump room will be above the ground level and civil work RCC (M25) according to the approved shop drawings and direction of Engineer in charge. Civil work shall be paid separately. Rates are inclusive of cost of operation and maintenance (O&M) for one year after completion of works which include periodical /monthly checking quality of treated sewage, checking of safety arrangements and routine maintenance of plant such as oiling, greasing, cleaning, operating, providing Consumable replacement of components within guarantee period, disposal of dried sludge and also ensuring safety against theft. The STP/ETP shall comprise of the units of required size as per detail process and hydraulic design.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			NOTE :- 1- If STP and ETP plant are situated in the same campus then only one O&M Contract shall be applicable. 2- Treated water effluent testing shall be got done by the vender from PollutionControl Board (PCB) approved laboratory only covering parameterssuch as - BOD, COD, TSS, PH, O&G, TKN, TP and E COLI etc. 3- Main power supply with earthing upto main control panel of STP/ETP shall be provided by the owner/department. The electricalpanel which will be supplied by the vender will also include all the required NO/NC contacts and provision for BMS connectivity. 4- Detail specifications of components shall be as per manufacturer stipulation and preamble both. 5- Tanks shall be made of MSFRP (mild steel fiber reinforced polymer/plastic)/ HDPE (High Density Polyethylene)/FRP (Fiber reinforced plastic) anti corrosive in packaged unit.		
	30.1.1	STP MBBR Conventional Technology	Conventional MBBR STP Technology Primary Treatment – -Manual Bar Screen -Oil, Grease Trap and Grit Removal System Secondary Treatment – -Raw Sewage Transfer Pump (01set) - Air Blower (01set) and Diffuser System - Media -Tube Media/Lamella -Chlorine Dosing system with Chlorine Dosing Pump(01set) -Sump Pump (01set) -Plant/Interconnecting Piping And Valves -Manual/PLC Based Electrical Control Panel -Incoming/Outgoing Feeder -Cable Laying With Cable Tray -Earthing Systems		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			-Pressure Gauges -Level Switches -Online Rota meter -Air Rota meter -Flow Meter -Filter Feed Pump (01set) -Multi Grade Sand Filter -Activated Carbon Filter -Micro Filtration system Sludge Handling System – -Sludge Recycling Pump (01set) -Sludge Feed Screw Pumps (01set) -Filter Press System		
	30.1.1.1	50 KLD	50 KLD	each	2398521.00
	30.1.1.2	100 KLD	100 KLD	each	2910978.00
	30.1.1.3	150 KLD	150 KLD	each	3412593.00
	30.1.1.4	200 KLD	200 KLD	each	3795593.00
	30.1.1.5	250 KLD	250 KLD	each	4516459.00
	30.1.1.6	300 KLD	300 KLD	each	4978910.00
	30.1.1.7	400 KLD	400 KLD	each	5961113.00
	30.1.1.8	500 KLD	500 KLD	each	6152390.00
	30.1.2	STP MBBR Package Technology	Package MBBR STP Technology Primary treatment – - Manual Bar Screen -Oil, Grease Trap and Grit Removal System Secondary treatment – -Raw Sewage Transfer Pump (01set) - Air Blower(01set) and Diffuser System - Anoxic Tank with Agitator - MBBR Tank with Media - Tube/Lamella settler tank with Tube Media -Clear water Tank With Chlorine Dosing system and Chlorine Dosing Pump(01set) -Sump Pump (01set) -Plant/Interconnecting Piping And Valves -Manual/PLC Based Electrical Control Panel		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			-Incoming/Outgoing Feeder -Cable Laying With Cable Tray -Earthing Systems -Pressure Gauges -Level Switches -Online Rota meter -Air Rota meter -Flow Meter -Filter Feed Pump (01set) -Multi Grade Sand Filter -Activated Carbon Filter -Micro Filtration system Sludge Handling System- -Sludge Recycling Pump (01set) -Sludge Feed Screw Pump (01set) -Filter Press System		
	30.1.2.1	10KLD	10KLD	each	1968323.00
	30.1.2.2	30KLD	30KLD	each	2796448.00
	30.1.2.3	50 KLD	50 KLD	each	3025423.00
	30.1.2.4	100 KLD	100 KLD	each	4214291.00
	30.1.2.5	150 KLD	150 KLD	each	5263199.00
	30.1.2.6	200 KLD	200 KLD	each	6354750.00
	30.1.3	STP MBR ConventionalTechnology	Conventional MBR STP Technology Primary treatment – - Manual Bar Screen -Oil, Grease Trap and Grit Removal System Secondary treatment – -Raw Sewage Transfer Pump (01set) - Air Blower (02set) and Diffuser System - Agitator -MBR Module -Membrane Clean-in-place (CIP) system -Chlorine Dosing system with Chlorine		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			Dosing Pump(01set) -Permeate Suction and Transfer Pump (01set) -Sump Pump (01set) -Plant/Interconnecting Piping And Valves -Manual/PLC Based Electrical Control Panel -Incoming/Outgoing Feeder -Cable Laying With Cable Tray -Earthing Systems -Pressure Gauges -Level Switches -Online Rota meter -Air Rota meter -Flow Meter Sludge Handling System – -Sludge Recycling Pump (01set) -Sludge Feed Screw Pump (01set) -Filter Press System		
	30.1.3.1	50 KLD	50 KLD	each	2737457.00
	30.1.3.2	100 KLD	100 KLD	each	4627286.00
	30.1.3.3	150 KLD	150 KLD	each	5758429.00
	30.1.3.4	200 KLD	200 KLD	each	6736457.00
	30.1.3.5	250 KLD	250 KLD	each	7489171.00
	30.1.3.6	300 KLD	300 KLD	each	8527134.00
	30.1.3.7	400 KLD	400 KLD	each	9964529.00
	30.1.3.8	500 KLD	500 KLD	each	12407050.00
	30.1.4	STP MBR Package Technology	Package MBR STP Technology Primary treatment – - Manual Bar Screen -Oil, Grease Trap and Grit Removal System Secondary treatment – -Raw Sewage Transfer Pump (01set) -Air Blower (02set) and Diffuser System - Anoxic Tank with Agitator -Aeration tank		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			- MBR tank with MBR Module -Membrane Clean-in-place (CIP) system -Chlorine Dosing Tank with Chlorine Dosing system and Pump(01set) -Permeate Suction and Transfer Pump (01set) -Sump Pump (01set) -Plant/Interconnecting Piping And Valves -Manual/PLC Based Electrical Control Panel -Incoming/Outgoing Feeder -Cable Laying With Cable Tray -Earthing Systems -Pressure Gauges -Level Switches -Online Rota meter -Air Rota meter -Flow Meter Sludge Handling System- -Sludge Recycling Pump (01set) -Sludge Feed Screw Pump (01set) -Filter Press System		
	30.1.4.1	30 KLD	30 KLD	each	3297700.00
	30.1.4.2	50 KLD	50 KLD	each	3651200.00
	30.1.4.3	100 KLD	100 KLD	each	5847367.00
	30.1.4.4	150 KLD	150 KLD	each	7663700.00
	30.1.4.5	200 KLD	200 KLD	each	9152683.00
	30.1.5	STP SBR ConventionalTechnology	Conventional SBR STP Technology Primary treatment – -Manual Bar Screen -Oil, Grease Trap and Grit Removal System Secondary treatment – -Raw Sewage Transfer Pump (01set) -Air Blower (01set) -Decanting mechanism -Diffuser system -Chlorine Dosing system and Chlorine Dosing Pump(01set) -Sump Pump (01set) -Plant/Interconnecting Piping And Valves -PLC Based Electrical Control Panel -Incoming/Outgoing Feeder -Cable Laying With Cable Tray -Earthing Systems -Pressure Gauges		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			-Level Switches -Online Rota meter -Air Rota meter -Flow Meter -Dissolved oxygen sensor and SS (Suspended solids sensor) -Filter Feed Pump (01set) -Multi Grade Sand Filter -Activated Carbon Filter Sludge Handling System - -Sludge Recycling Pump (01set) -Sludge Feed Screw Pump (01set) -Filter Press System		
	30.1.5.1	50 KLD	50 KLD	each	2629971.00
	30.1.5.2	100 KLD	100 KLD	each	3070481.00
	30.1.5.3	150 KLD	150 KLD	each	3435509.00
	30.1.5.4	200 KLD	200 KLD	each	4157702.00
	30.1.5.5	250 KLD	250 KLD	each	5010374.00
	30.1.5.6	300 KLD	300 KLD	each	5352656.00
	30.1.5.7	400KLD	400KLD	each	6776663.00
	30.1.5.8	500 KLD	500 KLD	each	8610543.00
	30.1.6	STP SBR Package Technology	Package SBR STP Technology Primary treatment – - Manual Bar Screen -Oil, Grease Trap and Grit Removal System Secondary treatment – -Raw Sewage Transfer Pump (01set) -Air Blower (01set) -SBR tank with decanting mechanism (MS) -Diffuser System -Clear Water Tank With Chlorine Dosing system and Chlorine Dosing Pump(01set) -Sump Pump (01set) -Plant/Interconnecting Piping And Valves -PLC Based Electrical Control Panel -Incoming/Outgoing Feeder -Cable Laying With Cable Tray- Earthing System -Pressure Gauges		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			-Level Switches -Online Rota meter -Air Rota meter -Flow Meter -Filter Feed Pump (01set) -Multi Grade Sand Filter -Activated Carbon Filter Sludge Handling System – -"Sludge Recycling Pump (01set) -Sludge Feed Screw Pump (01set) -Filter Press System		
	30.1.6.1	30 KLD	30 KLD	each	2280214.00
	30.1.6.2	50 KLD	50 KLD	each	3261445.00
	30.1.6.3	100 KLD	100 KLD	each	4575171.00
	30.1.6.4	150 KLD	150 KLD	each	6277751.00
	30.1.6.5	200 KLD	200 KLD	each	7048325.00
	30.1.7	ETP-Package Technology	Primary treatment – - Manual bar screen - Oil, Grease Trap and Grit Removal System Secondary treatment – -Raw Sewage/Effluent Transfer Pump (01set) -Air blower (01set) -Flash Mixing tank with Motor -Alum/poly/lime Dosing System with Reaction Tank -Anoxic Tank with Agitator -Lamella Clarifier/Tube Settler Tank with Media -Diffuser System -Clear Water Tank with Chlorine Dosing System and pump(01set) -Sump Pump(01set) -Plant/Interconnecting Piping And Valves -Manual/PLC Based Electrical Control Panel -Incoming/Outgoing Feeder -Cable Laying With Cable Tray -Earthing Systems -Pressure Gauges -Level Switches		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			-Online Rota meter -Air Rota meter -Flow Meter -Filter Feed Pump (01set) -Multi Grade Sand Filter -Activated Carbon Filter Sludge Handling System - -Sludge Recycling Pump (01set) -Sludge Holding Tank -Sludge Feed Screw Pump (01set) -Filter Press System		
	30.1.7.1	5 KLD	5 KLD	each	1911829.00
	30.1.7.2	10 KLD	10 KLD	each	2013307.00
	30.1.7.3	20 KLD	20 KLD	each	2216235.00
	30.1.7.4	30KLD	30KLD	each	2431618.00
	30.1.7.5	40KLD	40KLD	each	2681893.00
	30.1.7.6	50 KLD	50 KLD	each	3035463.00
30.2		Ultra Violet System	Supply and fixing of ultra violet disinfection unit. The unit shall have over 99.9 %bacterial reduction from inlet to outlet. The disinfection chamber shall be constructed of SS 316L on all welted parts. The UV lamp shall be of low pressure mercury vapor type with hard glass enclosure, the sockets shallbe water tight & vibration resistant. The unit shall be complete with temperature safety control, lamp out alert circuit & UV radiometer with 4 – 20 mA output. The UV unit shall have with reactor, cabinet housing, cabinet cooling, treatment chamber, electrical panel, temperature safety control, lamp out alert, UV radiometer along with UV monitoring system and UV monitoring readout panel. The lamps should be selected based upon the flow requirement of respective unit. As recommended by manufacturer complete in all respects.		
	30.2.1	30 KLD	30 KLD	each	41625.00
	30.2.2	50 KLD	50 KLD	each	61075.00
	30. 2.3	100 KLD	100 KLD	each	65250.00
	30. 2.4	150 KLD	150 KLD	each	73250.00
	30. 2.5	200 KLD	200 KLD	each	118000.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	30. 2.6	250 KLD	250 KLD	each	123250.00
	30. 2.7	300 KLD	300 KLD	each	130250.00
	30.2.8	400 KLD	400 KLD	each	135800.00
	30.2.9	500 KLD	500 KLD	each	143700.00
30.3		O&M-STP&ETP (Non-Comprehensive)	Operation and maintenance of Sewage Treatment Plant and Effluent Treatment Plant after successful commissioning as per item number 30.1 and sub item number 30.1.1, 30.1.2, 30.1.3, 30.1.4, 30.1.5, 30.1.6 and 30.1.7 for a period of one year, including cost of providing swift operators, periodical/monthly checking of the treated water quality and safety arrangement of plant, oiling, greasing, routine maintenance, necessary etc. O&M shall include all cost involved in maintenance except for electricity, consumable, components (if to be changed for operation) and disposal of dried sludge. If STP and ETP plant in the same campus then only one O&M contract shall be applicable. Treated water testing report shall be got done by the vender from PCB (Pollution Control Board) and approved laboratory. Parameters covered BOD, COD, TSS, PH, O&G, TKN, TP and E COLI etc.	per year	3,80,000.00
30.4		STP-Phytorid Bed/Constructed wetland	STP With Phytorid Bed/Constructedwetland:-Designing and preparing good for construction (GFC) civil, mechanical and electrical drawings of phytorid/constructed wetland sewage treatment plant as per National Environmental Engineering Research Institute (NEERI) or specialized in constructed wetland technology, design, drawing and specification complete for meeting up to date design (target) parameters of treated effluent conforming to Madhya Pradesh Pollution Control Board (MPPCB), MoEF GOI guidelines, technical specifications codes of practices as published by the Bureau of Indian Standard (BIS) and latest version of "Manual on Sewerage and Sewage Treatment" published by the Central Public Health And Environmental Engineering Organization (CPHEEO).		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			The work shall include supplying, installing, testing and commissioning of bio culture, plants (including replacement dead plants up to one year) etc. but cost of filter media, electro-mechanical equipments, instruments, pipes, valves and other civil works shall be paid separately, details specifications of components shall be as per preamble. -Bar screen -Oil, grease trap and grit removal system -Bio culture -Plants		
	30.4.1	10 KLD	10 KLD	each	135450.00
	30.4.2	50 KLD	50 KLD	each	197250.00
	30.4.3	100 KLD	100 KLD	each	274500.00
	30.4.4	150 KLD	150 KLD	each	351650.00
	30.4.5	200 KLD	200 KLD	each	428900.00
	30.4.6	250KLD	250 KLD	each	508150.00
	30.4.7	300 KLD	300 KLD	each	585400.00
	30.4.8	350KLD	350 KLD	each	662550.00
	30.4.9	400KLD	400 KLD	each	739800.00
	30.4.10	450KLD	450 KLD	each	825050.00
	30.4.11	500KLD	500 KLD	each	902300.00
	30.4.12	550KLD	550 KLD	each	989550.00
	30.4.13	600KLD	600 KLD	each	1066700.00
	30.4.14	650KLD	650 KLD	each	1143950.00
	30.4.15	700KLD	700 KLD	each	1221200.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
30.5		Bio culture	Supplying and spreading of bio culture at 0.1liter/sqm in phytorid bed/constructed wetland for additional requirement/ maintenance purposes.	litre	6750.00
30.6		Plantation	Plantation (minimum 05 plants/sqm) in phytorid bed/ constructed wetland for replacement /maintenance purposes.	sqm	1000.00
30.7		Bio Digester	Providing and Installing bio digester unit based on DRDO technology made of fiber reinforced plastic having composition and the technical specifications such as:Parentage of fiber glass (glass content in fiber glass should be minimum 30%) :- 33.33%Water absorption (as per ASTM) : - 0.30%Flammability :- as per ASTMTensile strength :-115 MPA Specific gravity at 25 degree centigrade :-1.45 IZOD impact strength :- 65 KJ/M2 Heat defection/distort temperature :-65 deg C Corrosion resistance (non corrosive) fitted with Inlet pipe size rigid PVC :-100mmOutlet pipe size rigid PVC :-80mmGas outlet rigid PVC with valve :- 20mm Bacteria immobilization PVC matrix(on all partition) and capacity :-		
	30.7.1	UPTO 10 USER	UPTO 10 USER SIZE (1.0m X 1.0mX0.70m) Capacity of bio-digester :- 700 litersArea of top, bottom and side walls 4.80 sq.m Area of partition walls : 1.40 sq.m Total weight of FRP: 33.48 kg Bacteria inoculums quantity:- 200liters Minimum thickness of side walls/topand bottom:- 3mm Number of inside partitions :- 2 Minimum thickness of partition wall:- 3mm	each	46650.00
	30.7.2	UPTO 15 USER	UPTO 15 USER SIZE (1.0m X 1.0mX 1.0m) Capacity of bio-digester :- 1000 litersArea of top, bottom and side walls 6.00 sq.m. Area of partition walls : 2.00 sq.m Total weight of FRP: 54.00 kg Bacteria inoculums quantity:-400liters Minimum thickness of side walls/topand bottom :- 4mm Number of inside partitions :- 2 Minimum thickness of partition wall :- 3mm	each	78597.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	30.7.3	UPTO 25 USER	UPTO 25 USER SIZE (1.5m X 1.0mX 1.0m) Capacity of bio-digester :- 1500 litersArea of top, bottom and side walls 8.00 sq.m. Area of partition walls : 2.00 sq.mTotal weight of FRP: 82.80 kg Bacteria inoculums quantity :- 400 liters Minimum thickness of side walls/topand bottom :-5mm Number of inside artitions:- 2 Minimum thickness of partition wall:-3mm	each	107151.00
	30.7.4	UPTO 40 USER	UPTO 40 USER SIZE (2.0m X 1.0mX 1.0m) Capacity of bio-digester :- 2000 litersArea of top, bottom and side walls 10.00 sq.m. Area of partition walls : 3.00 sq.m Total weight of FRP: 106.20 kg Bacteria inoculums uantity :-600liters Minimum thickness of side walls/ top and bottom :- 5mm Number of inside partitions:- 3 Minimum thickness of partition wall :- 3mm	each	142575.00
	30.7.5	UPTO 60 USER	UPTO 60 USER SIZE (2.4m X 1.25mX 1.0m) Capacity of bio-digester :-3000 liters Area of top, bottom and side walls 13.30 sq.m. Area of partition walls : 3.75 sq.m Total weight of FRP: 139.95 kg Bacteria inoculums quantity :- 800litres Minimum thickness of side walls/top and bottom :- 5mm Number of inside partitions :-3 Minimum thickness of partition wall:- 3mm	each	187750.00
	30.7.6	UPTO 80 USER	UPTO 80 USER SIZE (2.7m X 1.5mX 1.0m) Capacity of bio-digester :- 4000 litersArea of top, bottom and side walls 16.50sq.m Area of partition walls : 4.50 sq.m Total weight of FRP: 202.50 kg Bacteria inoculums quantity :- 1200litres Minimum thickness of side walls/top and bottom :- 6mm Number of inside partitions :-4 Minimum thickness of partition wall:-3mm	each	271667.00
	30.7.7	UPTO 100 USER	UPTO 100 USER SIZE (2.5m X2.0m X 1.0m) Capacity of bio-digester :- 5000 liters Area of top, bottom and side walls:- 19.00 sq.m. Area of partition walls :- 8.00 sq.m Total weight of FRP:- 248.40 kg Bacteria inoculums quantity :- 1400liters Minimum thickness of side walls/topand bottom :- 6mm Number of inside partitions :- 4 Minimum thickness of partition wall:-3mm.	each	334491.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	30.7.8	UPTO 150 USER	UPTO 150 USER SIZE (3.2m X2.0m X 1.25m) Capacity of bio-digester :- 8000 liters Area of top, bottom and side walls:- 25.80 sq.m. Area of partition walls :- 10.00 sq.m Total weight of FRP:- 397.08 kg Bacteria inoculums quantity :- 2400 liters Minimum thickness of side walls/topand bottom :- 7mm Number of inside artitions:- 4 Minimum thickness of partition wall:-mm	each	536907.00
	30.7.9	UPTO 200 USER	UPTO 200 USER SIZE (4.0m X2.0m X 1.25 m) Capacity of bio-digester :- 10,000liters Area of top, bottom and side walls:-31.00 sq.m.Area of partition walls :- 10.00 sq.mTotal weight of FRP:- 462.60 kg Bacteria inoculums quantity :- 3000liters Minimum thickness of side walls/topand bottom :- 7mm Number of inside partitions:- 4 Minimum thickness of partition wall :-4mm.	each	632426.00
	30.7.10	UPTO 300 USER	UPTO 300 USER SIZE (4.8m X2.0m X 1.25 m) Capacity of bio-digester :- 12000 liters Area of top, bottom and side walls:- 36.20 sq.m. Area of partition walls : - 10.00 sq.m Total weight of FRP: - 593.28 kg Bacteria inoculums quantity :- 3600 liters Minimum thickness of side walls/top and bottom:- 8mm Number of inside partitions:-4 Minimum thickness of partition wall:-4mm.	each	792550.00
	30.7.11	UPTO 400 USER	UPTO 400 USER SIZE (5.0m X 2.0mX .5m) Capacity of bio-digester :- 15000liters Area of top, bottom and side walls:-41.00sq.m. Area of partition walls:- 12.00 sq. mTotal weight of FRP :- 676.80 kg Bacteria inoculums quantity :- 4400 liters Minimum thickness of side walls/topand bottom:- 8mm Number of inside partitions:- 4 Minimum thickness of partition wall:-4mm.	each	920176.00
	30.7.12	UPTO 500 USER	UPTO 500 USER SIZE (6.0m X2.0m X 1.67m) Capacity of bio-digester :- 20,000 liters Area of top, bottom and side walls:- 50.72 sq.m.Area of partition walls:- 13.36 sq.m.Total weight of FRP :- 917.86 kg Bacteria inoculums quantity:- 6000 liter Minimum thickness of side walls/top and bottom :- 9mmNumber of inside partitions :- 4 Minimum thickness of partition wall:- 4mm	each	1243438.00

CHAPTER – 31.0

NEW TECHNOLOGY AND MATERIAL

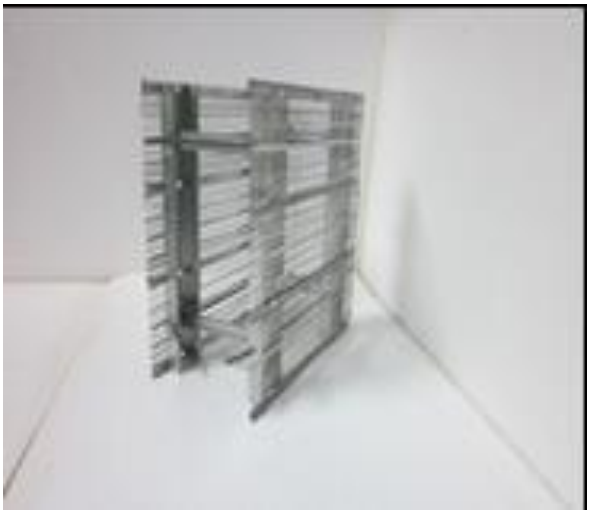
PREAMBLE
NEW TECHNOLOGIES AND MATERIAL
(CHAPTER: 31.0)

About – Structural Stay-in-Place Formwork Systems:

Structural Stay in place formwork system are used to build concrete load bearing monolithic wall structures like building , water tank retaining wall structure, drainage and all type of RCC structures. The formwork system comprises of two filtering grids (made from 0.42mm thick high galvanized sheet having 180gm/m2 /275gm/m2 zinc coating) reinforced by vertical stiffeners called 'C' channel made from 0.6mm Hot dip galvanized steel having 180 gm/m2/25gm/m2 z8nc coating , c channel acts as equivalent to 8mm bar . All member of this has to confirm Hot dip galvanized steel in coil shall be as per IS 277:2003/ASTM A653/ASTM A653 M-00.The grids are connected by rebar which act as horizontal stiffeners and tie up with connector which act as a shear link. The grids on both faces act as sacrificial formwork in which concrete is poured in-situ. The vertical steel channels and horizontal steel bars act as steel reinforcement for load bearing wall. The connectors help to fold the formwork for easy transportation. Rebar are horizontal stiffeners placed at every 200/100mm centre to centre, these are 5mm Ø MS or galvanized bars and used as distribution bars. Structural design and analysis of the formwork shall be based on relevant Indian and International standards. The panel construction assembly shall be used for free standing walls when designed. Panels shall be tied up with scaffolding materials to maintain alignment. If necessary Additional reinforcement and cement plaster shall be provided as required by the design. Foundation shall be specifically designed in accordance with provisions given in IS 1904:2005. The details of construction methodology are available with BMPTC web site.



Standard Single Panel



Standard Double Panel

The incorporation of the openings make it possible to better fix the joinery to the structure ,the joinery is fixed with the aid of reinforced lugs or braces, with lengths such that the anchoring in the concrete extend beyond the vertical stiffening sections of the contiguous panel rim .

EPS CORE PANEL SYSTEM

Expanded Polystyrene (EPS) core Panel system is a modern, efficient, safe and economic construction system for the construction of buildings. These panels can be used both as load bearing as well as non-load bearing elements.

EPS core panel is a 3D panel consisting of 3- dimensional welded wire space frame provided with the polystyrene insulation core. Panel is placed in position and shotcrete on both the sides.

The EPS panels consist of a 3-dimensional welded wire space frame utilizing a truss concept for stress transfer and stiffness as shown in Fig. 1.1.

EPS panel includes welded reinforcing meshes of high strength wire, diagonal wire and self – extinguishing expanded polystyrene uncoated concrete, manufactured in the factory and shotcrete is applied to the panel

assembled a the construction site, which gives the bearing capacity of the structure.
EPS panel after shotcrete has the following five components (as Fig. 1.1).

i. The outer layer of shotcrete.
ii. Welded reinforcing mesh of high wire.
iii. The core of expanded polystyrene sheet.
iv. Diagonal wire (stainless or galvanized wire).
v. The inner layer of shotcrete.

The welded mesh fabric connected piercing polystyrene with truss of steel wire welded to the welded fabric at an angle. It gives a rigidity spatial structure and simultaneously prevents polystyrene core shifting. Individually welded internal strut wires of diagonals extend through the panel core between each surface.

These galvanized strut wires are welded continuously in the required spacing so they form, with the welded wire fabric, into a triangulated truss system which greatly increases the panel strength.

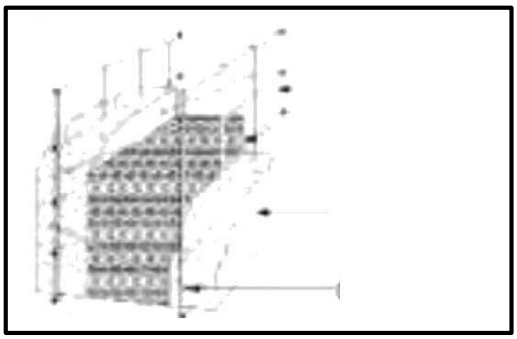


Fig. 1.1 : Typical cross section of wall panels.

EPS panel is a versatile structural element designed for floors, walls, partitions, roof and stairs. Fig. 1.2 & Fig. 1.3 shows the welded reinforcing mesh of the EPS panels at different cross-sections.

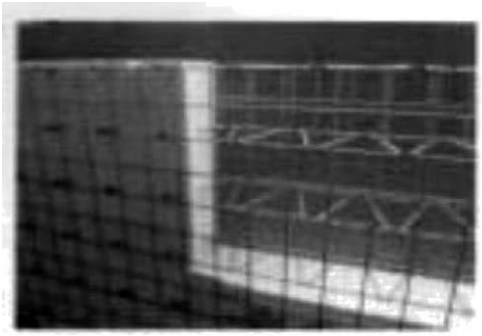


Fig. 1.2 : Reinforcing mesh expanded polystyrene core and diagonal wire.



Fig. 1.3 : Welded reinforcing mesh 3-D panel without expanded polystyrene core.

The typical EPS panel is generally manufactured with dimensions of 1200 m width, 3000 mm length and over all thickness range of 80-230 mm. The panels are finished at the site using minimum 30 mm thick shotcreting of cement & coarse sand in the ratio of 1:4 applied under pressure. The shotcreting coat encases the EPS Core with centrally placed steel welded wire mesh.

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
31.1		Phenol bonded Bamboowood flooring	Providing & fixing in position Phenol bonded Bamboowood flooring withplanks of sizes 14 mm thick, 1800 mm length (minimum) and 130 mm wide (minimum), in approved colour, texture and finish, having Performance Appraisal Certificate (PAC) issued by Building Materials & Technology Promotion Council (BMTPC). The flooring shall be fixed with tongue and groove interlocking system, with underlayment of 4 mm thick expanded polyethylene foam sheets having density 40 kg/cum, over prepared surface with necessary quarter round planks of size 1900 mm x 18mm and door reducer of size 1900 mm x 44 mm, wherever required. The bamboowood planks shall have minimum density of 1000 Kg/cum & minimum Hardness 1000 kgf. with Eco friendly UV coating, all complete as per direction of the Engineer in-charge.	sqm	4297.00
31.2		Phenol bonded Bamboo wood wall skirting	Providing & fixing in position Phenol bonded Bamboo wood in wall skirting with planks of sizes 14 mm thick, 1900 mm length (minimum) and 85 mm wide (minimum), in approved colour, texture and finish, having Performance Appraisal Certificate (PAC) issued by Building Materials & Technology Promotion Council (BMTPC). The skirting shall be fixed with SS screws & rawl plugs, over underlayment of 4 mm thick, expanded polyethylene foam sheets having 40 kg/cum density over prepared surface. The bamboowood planks shall have minimum density of 1000 kg/cum & minimum Hardness 1000 kgf. with Eco friendly UV coating, all complete as per direction of the Engineer in-charge.	Sqm	4215.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
31.3		Phenol bonded Bomboowood wall cladding.	Providing & fixing in position Phenol bonded Bamboowood wall cladding at all height with planks of sizes 10 mm thick, 1800 mm length (minimum) and 130 mm wide (minimum), in approved colour, texture and finish, having Performance Appraisal Certificate (PAC) issued by Building Materials & Technology Promotion Council (BMTPC), with necesasary profiled edges fixed with 40 mm SS screws 5 nos in each tile to frame work made of second class teak wood of size 20 x 15 mm in centre of each tile and bottom and top of work height, 40 x 15 mm placed at ends of each tile. The cladding shall be laid over backlayment of 1.00 mm thick expanded polyethylene foam of density 40 kg/cum in two layers, first layer on wall surface before fixing wooden frame and second layer on frame under cladding. The bamboowood planks shall have minimum density of 1000 Kg/cum & minimum Hardness 1000 kgf. with Eco friendly UV coating, all complete as per direction of the Engineer in-charge.	sqm	4546.00
31.4		Phenol bonded Bomboowood Shutters	Providing & fixing in position Phenol bonded Bamboowood panelled or panelled and glazed shutters for doors windows, clerestorey windows with pre-molded 30mm thick planks, in approved colours, texture & finishe. It shall have 10mm wide, 25 mm deep grove to fit in panels. The bamboo wood shall have minimum density of 1000 Kg/cum, minimum Hardness 1000 kgf. All styles and rails shall have profiled interlocking system locked in place by bamboo pins, all complete as per direction of Engineer in charge. (The panelling will be paid for separately).	sqm	4248.22

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
31.5		Phenol bonded Bamboo wood panelling	Providing & fixing in position Phenol bonded Bamboo wood panelling of 10 mm thick, in 25 to 40 mm thick panelled or panelled & glazed shutters for doors, windows, clerestorey windows, in approved colour, texture & finish. The bamboowood planks shall have minimum density of 1000 Kg/cum & minimum Hardness 1000 Kgf. The panels shall have profiled interlocking system locked in place with bamboo pins all complete as per direction of the Engineer incharge. (area of opening for panel inserts excluding portion inside grooves or rebates to be measured)	sqm	2546.22
31.6		Phenol bonded Bamboo wood door frame	Providing & fixing in position 65 mm thick factory made door frame of Phenol bonded Bamboo wood (superior class, interior use), in approved colour, texture and finish.The bamboo wood shall have minimum density of 1000 Kg/cum, minimum hardness 1000 Kgf. The door frame shall have tenon & mortise interlocking system, to be fixed to the wall with 100 mm size G.I screws all a complete as per direction of Engineer-in charge.	sqm	197.00
31.7		Bamboo mat corrugated sheet (BMCS)	Providing, erecting, laying and fixing in position in 3.5 to 4 mm thick bamboo mat corrugated sheet (BMCS) as per IS: 15476-2004 in roofing with self drilling screws along with EPDM washers complete or with galvanized iron J or L hooks 8 mm dia G.I. plain and bitumen washers etc, all complete as per direction of Engineer-in-Charge.	sqm	4065.00
31.8		Bamboo mat ridge cap (BMRC)	Providing and fixing in position ridges of 3.5 to 4 mm thick bamboo mat ridge cap (BMRC) as per IS: 15476-2004 in roofing with self drilling screws along with EPDM washers complete or with galvanized iron J or L hooks 8 mm dia G.I. plain and bitumen washers etc, all complete as per direction of Engineerin-Charge	meter	3331.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
31.9		Phenol bonded Bamboo Mat board false ceiling	Providing and fixing at all height false ceiling of 4mm thick phenol bonded Bamboo Mat board (595x595mm) conforming to IS:13958-1994 includingm providing and fixing of frame work made of GI angle 25x25x0.4 mm thick all around suitably fixed to wall with the help of dash fastener and hanger frame (600x600 c/c) made GI slotted Tee having powder coating on bottom side (30x25x0.3 mm thick for main member & 25x25x0.3 mm for cross member) connected to ceiling with 2.64 mm GI wire and anchor fastener at every junction and also including cost of making openings for light fittings, grills, diffusers, cut outs made with frame of perimeter channels suitably fixed all complete as per direction of Engineer-in-charge.	sqm	2529.00
31.10		Bamboo Mat board for partition	Providing and fixing at Bamboo Mat board conforming to IS:13958-1994 for partition to frame by bucking or studding with screws etc. complete (Frames, backing or studding to be paid separately)		
	31.10.1	3 mm thickness	3 mm thickness	sqm	1847.00
	31.10.2	4 mm thickness	4 mm thickness	sqm	2173.00
	31.10.3	6 mm thickness	6 mm thickness	sqm	2672.00
	31.10.4	9 mm thickness	9 mm thickness	sqm	3175.00
	31.10.5	12 mm thickness	12 mm thickness	sqm	3810.00
31.11		Phenol bonded Bamboo Mat board all height wall panelling	Providing and fixing at all height wall panelling with phenol bonded Bamboo Mat board conforming to IS:13958-1994 including providing and fixing to frame work made of 50mm x 50mm hardwood plugs including cutting brick work and fixing in cement mortar and making good the wall etc. and also providing and fixing wooden moulded corner beading of triangular shape to the junction of panelling etc. with iron screws all complete as per direction of Engineer -in-Charge		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	31.11.1	9 mm thickness	9 mm thickness	sqm	3427.00
	31.11.2	12 mm thickness	12 mm thickness	sqm	4063.00
31.12		Structural stay in place panels	Design, Supplying & Installation of factory made Structural stay in place panels of design specified thickness having width 1100 mm and height 3000 mm. Panel is composed of two filtering grids made of Net reinforced by vertical stiffeners called 'C' channel. The grids are connected by rebar act as horizontal stiffeners & connectors act as a shear link. Design specified concrete will be poured in between panels from the top of the wall and finally finished with 12 mm to 15 mm plaster. Net shall be made from 0.42 mm thick and 342 mm width 275 GSM galvanized plain steel with Net at 100 mm interval in horizontal directions ,mesh further strengthen by "C" channel made from 0.6 mm thick and 101 width 275 GSM galvanized plain steel with 8 mm hole at every 200 mm or 100 mm intervals along the length of the channel, combination of Net and "C" channel called single panels which remains on both face of the wall connected with the help of 5 mm dia 100 GSM galvanized wire at every 200 mm interval in vertical direction and connector made from 1.5 mm thick and 19 mm width 120 GSM galvanized plain steel placed at every 400 mm intervals in horizontal direction and 200 mm intervals in vertical direction in zig zag pattern. In wall elements structural stay in place double panels installed with "C" channel in vertical direction, Door and window lintel and bottom double panels with "C" in either vertical or horizontal direction. All corner covers with corner piece of "L" shape made with single panels. In slab, structural stay in place single panels used in replacement of shuttering plates supported by wood or steel runners with steel props. Both the outer faces of the panel shall be finished by applying the layer of 12 mm to 15 mm thick cement mortar 1:6 (1 cement: 6 coarse sand). Fixing operations of		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			wall panels shall be completed in all respect as per drawings and specifications and under the overall direction of the Engineer-in-charge. Fixing of structural stay in place Panels of any thickness(110, 140, 160 mm) in walls in all floors using suitable scaffolding material as per instructions of Engineer-in-Charge, as per cutting drawings and structural drawings, in perfect line and plumb, above RCC plinth beam or Raft foundation and provide necessary lateral/ slanting support to keep the wall panel in safe position, providing & tieing of Reinforcement in all junctions, door and window lintels as per structural drawings and make ready for RCC work Including cost of shuttering material, scaffolding wherever required. (Cost of reinforcement, Reinforced Cement Concrete as per Structural details shall be paid separately).		
	31.12.1	Double Panel	Stay in Place Double Panel of Width 1100 mm, Height 3000 mm,		
	31.12.1.1	110 mm thickness	110 mm thickness	Sqm	3649.00
	31.12.1.2	140 mm thickness	140 mm thickness	Sqm	3661.00
	31.12.1.3	160 mm thickness	160 mm thickness	Sqm	3672.00
	31.12.1.4	200 mm thickness	200 mm thickness	Sqm	3706.00
	31.12.1.5	250 mm thickness	250 mm thickness	Sqm	3736.00
	31.12.2	Single Panel	Stay in Place single Panel Width 1100 mm, Height 3000 mm,	Sqm	1865.00
	31.12.3	'L' shape	Stay in Place corner piece of 'L' shape of Height 3000 mm		
	31.12.3.1	Width: (110+110) mm	Width: (110+110) mm	Sqm	2373.00
	31.12.3.2	Width: (140+140) mm	Width: (140+140) mm	Sqm	2538.00
	31.12.3.3	Width :(160+160) mm	Width :(160+160) mm	Sqm	2337.00
	31.12.3.4	Width :(200+200) mm	Width :(200+200) mm	Sqm	2218.00
	31.12.3.5	Width :(250+250) mm	Width :(250+250) mm	Sqm	2185.00

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
31.13		Expanded Polystyrene (EPS) core Panel system	Providing Expanded Polystyrene (EPS) core Panel system complete building elements, it is a modular panel composed of two electro welded galvanized steel meshes, reciprocally joined by and in the middle of which is a suitabley shaped foam polystyrene sheet as per ASTM STD satisfying NBC Norms or having equivalent specification manufactured by ISO Certified company satisfying density, self extinguishing test as well as impact test, thermal insulation, termite proof, compressive strength satisfying IS 12436-1988 as per given ASTM Norms. E.P.S. panel are energy saving and comes under the Green Building concept and is also CFC free, (Expanded Polystyrene) EPS panel with salient features Heat resisttant, Water resistant, Non-Toxi, Sound resistant, Self Extinguishing material fo having panels Standard width of 1200mm standard length upto 5000mm. For Roofing panelare having provision for hidden beam profile shape and generally used 50mm×50mm wiremesh with 2.5 mm thicknes. With this panel as per your requirement you can create all architectural features and designs. This panel can be used for G+3 structure by minimizing colomn, beam,steel and labour. This product is also available in ibn 100mm×100mm wiremesh, this will be useful for partition wall, fencing wall etc. EPS slab is having thickness of 50mm,80mm,100mm, 120mm and 150mm bothside welded wiremesh have a gap of 12mm to 15mm either side, total thickness with wiremesh is approx. 50mm is 80mm and 80mm is 110mm etc. and after covering plaster 50mm will be 100mm and 80mm will be 130mm etc. These panels are tied with galvanized ancillaries and complete structure become a monolithic connected with each other and this structure is safe during earthquake and tested up to 7.2 Richter scale. All work done as per direction of Engineer in Charge.		

Item No.	Sub Item No.	Item name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	31.13.1	Panel Without Load	EPS panel size 100mm including GI wiremesh bothside 2mm (Wiremesh Net 100×100mm)	Sqm	2972.00
	31.13.2	Panel Without Load	EPS panel size 100mm including GI wiremesh bothside 2mm (Wiremesh Net 100×100mm)	Sqm	3317.00
	31.13.3	Panel With Load	EPS panel size 100mm including GI wiremesh bothside 2.5mm (Wiremesh Net 50×50mm)	Sqm	3346.00
	31.13.4	Panel With Load	EPS panel size 150mm including GI wiremesh bothside 2.5mm (Wiremesh Net 50×50mm)	Sqm	3743.00
	31.13.5	Roofing Panel For Slab	EPS panel size 80mm including GI wiremesh bothside 2.5mm (Wiremesh Net 50×50mm)	Sqm	3321.00
	31.13.6	Roofing Panel For Slab	EPS panel size 100mm including GI wiremesh bothside 2.5mm (Wiremesh Net 50×50mm)	Sqm	3471.00
	31.13.7	Roofing Panel For Slab	EPS panel size 150mm including GI wiremesh bothside 2.5mm (Wiremesh Net 50×50mm)	Sqm	3885.00
	31.13.8		EPS panel 'C' Accessories for opening	Sqm	626.00

ANNEXURE – 1 TO 7

Table-A
Weight of Materials of Construction

S.No.	Material	Unit			Weights	
		British	Metric	Kg./Metric Unit	Lbs British Unit	Kg/british Unit
1	2	3	4	5	6	7
1.	Cement	Cft	Cum.	1440	89.8422	40.752
2.	Steel					
	(a)Mild Steel	Cubic inch	Cu.cm.	0.0078	0.2830	0.12883
	(b) Cast Steel	-”-	-”-	0.0078	0.2830	0.1283
	(c) H.T. Steel	-”-	-”-	0.0078	0.2830	0.1283
3.	Stainless Steel	-”-	-”-	0.0078	0.2830	0.1283
4.	Lead	-”-	-”-	0.0113	0.4100	0.1850
5.	Plain cement concrete or R.C.C.	Cubic foot	Cum. m	2306 to 2400	144 to 150	66.5 to 68.0
6.	Stones					
	(a) Lime	Cubic foot	Cum m	2650	162	74.00
	(b) Sand	-”-	-”-	2800	174	79.2
7.	Stone masonry rubble	-”-	-”-	2100	131	59.43
8.	Bricks	-”-	-”-	1600	100	45.50
9.	Brick masonry	-”-	-”-	1920	100 to 120	45.50 to 54.33
10.	Timber	-”-	-”-	650 to 720	40.5 to 45.00	18.4 to 20.4
11.	Copper alloy (70:5:25)	Cubic Inch	Cubic Cm.	0.0101	0.3660	0.1660

Table-B Useful Conversion Factors				
S.No.	***To convert***	***Into***	***Multiply by***	
1	Meters	Yards	1.09631	0.9144
2	Meters	Feet	0.28083	0.3048
3	Meters	Inches	39.37014	0.0254
4	Kilometers	Miles	0.62137	1.609344
5	Milimeters	Inches	0.03937	25.4
6	Square Kilometers (100 mects)	Squaare Miles	0.386101	2.58999
7	Square Meters	Square yard	1.19599	0.83613
8	Square Meters	Square Feet	10.7639	0.092903
9	Square Meters	Square inches	1550.00	0.00064516
10	Square Centimeters	Square inches	0.155	9.4516
11	Hectare (10000 Sq. Meters)	Acres	2.47105	0.404686
12	Cubic Meters	Cubic Feet	35.3147	0.028317
13	Cubic Meters	Gallons (Imp.)	219.969	0.0045609
14	Cubic Centimeters	Cubic inches	0.061024	16.3871
15	Liters	Gallons	0.219976	4.54596
16	Kilograms	Pounds (Avoirdupois)	2.20162	0.4535924
17	(100 Kgs.) Quintal	Maund	2.67923	0.373242
18	(100 Kgs.) Quintal	Hundred weight	1.9684	0.50802
19	Tonnes (Metric ton)	Tons (British)	0.9842	1.01605
20	Tonnes (Metric ton)	Maunds	26.7923	0.0373242
21	Kilometers per Hour	Miles per hour	0.62137	1.60934
22	Kilometers per Hour	Feet per second	0.91134	1.09728
23	Gram per Cubic Cm.	Pounds per Cu. Foot	62.428	0.0160185
24	Gram per Cubic Cm.	Pounds per Cu. Inch	0.0361273	27.6799
25	Gram per Litre	Pounds per Cu. Foot	0.062426	16.0189
26	Kilograms per Cum.	Pounds per Cu. Foot	0.0624	16.018
27	Kilograms per Litre	Pounds per Cu. Foot	62.426	0.0160189
28	Kilograms per Metre	Pounds per Foot	0.67179	14.8816
29	Tonne per Sqm.	Ton per Sq. inch	0.09143	10.937
30	Kilograms pe Sq. Cm	Pounds per Sq. inch	14.2233	0.07037
31	Kilograms pe Sq. Metre	Pounds per Sq. Foot	0.20482	4.8825
32	Kilograms Metre	Foot pounds	7.231	0.1382
33	Inch pounds	Kilogram centimeters	1.152	0.867962
34	Ton foot	Ton metre	0.3098	3.2278
35	Pounds per Cu. Ft.	Kg. per cu. M.	16.0185	0.06243

Table-C

Weight, Perimeter and sectional Area of Metric Steel Bars.

Diameter (mm)	Weight per meter in kg	Perimeter Cm	Sectional Area of steel bars in Sq. Cms.
5	0.154	1.571	0.196
6	0.222	1.887	0.282
7	0.302	2.2	0.385
8	0.395	2.514	0.503
10	0.617	3.143	0.786
12	0.888	3.771	1.131
14	1.208	4.4	1.54
16	1.578	5.028	2.011
18	1.998	5.657	2.546
20	2.466	6.286	3.143
22	2.984	6.914	3.803
24	3.551	7.543	4.526
25	3.853	7.857	4.911
26	4.163	8.171	5.311
28	4.834	8.8	6.16
30	5.549	9.428	7.071
32	6.31	10.057	8.046
34	7.13	10.686	9.083
36	7.99	11.314	10.183
40	9.87	12.571	12.571
45	12.5	14.143	15.911
50	15.41	15.714	19.643

Table-D

Weight of Flats per meter

Width in mm.	Thickness in MM.														
	5	5.5	6	7	8	9	10	11	12	14	16	18	20	22	25
1	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***11***	***12***	***13***	***14***	***15***	***16***
12	0.471	0.518	0.565	0.659	0.753	0.848	0.942	1.036	1.130	1.319	1.507	1.695	1.884	2.072	2.355
16	0.628	0.691	0.753	0.879	1.005	1.130	1.256	1.381	1.507	1.759	2.009	2.261	2.512	2.763	3.140
20	0.785	0.863	0.942	1.099	1.256	1.413	1.570	1.727	1.884	2.198	2.512	2.826	3.140	3.454	3.925
25	0.981	1.079	1.177	1.374	1.570	1.766	1.962	2.158	2.355	2.747	3.140	3.532	3.924	4.317	4.906
32	1.256	1.381	1.507	1.758	2.009	2.360	2.512	2.763	3.014	3.516	4.019	4.521	5.023	5.526	6.279
40	1.570	1.727	1.884	2.198	2.512	2.826	3.140	3.453	3.768	4.395	5.023	5.651	6.279	6.907	7.849
50	1.962	2.158	2.355	2.747	3.140	3.532	3.924	4.317	4.709	5.494	6.279	7.064	7.849	8.634	9.811
63	2.472	2.720	2.967	3.461	3.956	4.450	4.945	5.439	5.934	6.923	7.912	8.901	9.890	10.879	12.362
80	3.139	3.454	3.768	4.395	5.023	5.651	6.279	6.907	7.535	8.791	10.470	11.303	12.585	13.814	15.398
100	3.924	4.317	4.709	5.494	6.279	7.064	7.849	8.634	9.419	10.989	12.558	14.128	15.698	17.268	19.622
125	4.905	5.396	5.887	6.868	7.849	8.830	9.811	10.792	11.773	13.736	15.698	17.660	19.622	21.585	24.520
160	6.275	6.907	7.536	8.791	10.047	11.303	12.558	13.814	15.07	17.582	20.093	22.605	25.117	27.628	31.396

Appendix2

Quantity of materials required for various building item of works.

S.No.	Description of Items (Mortar)	Materials	Unit	Quantity per unit
1	*2*	*3*	*4*	*5*
1	Cement Mortar (1.2)	Sand Cement	Cum. Cum.	1.00 Cum. 14.40 Bags
2	Cement Mortar (1.3)	Sand Cement	Cum. Cum.	1.00 Cum. 9.40 Bags
3	Cement Mortar (1.4)	Sand Cement	Cum. Cum.	1.00 Cum. 7.20 Bags
4	Cement Mortar (1.5)	Sand Cement	Cum. Cum.	1.00 Cum. 5.70 Bags
5	Cement Mortar (1.6)	Sand Cement	Cum. Cum.	1.00 Cum. 4.70 Bags
6	Cement Mortar (1.8)	Sand Cement	Cum. Cum.	1.00 Cum. 3.50 Bags
7	1 Cement : 1 Lime : 6 Sand	Sand Hydrated lime Cement	Cum. Cum. Cum.	1.0Cum. 0.166 cum. 4.74 Bags
8	1 Cement : 2 Lime : 9 Sand	Sand Hydrated lime Cement	Cum. Cum. Cum.	1.0Cum. 0.222 cum. 3.17 Bags
9	1 Cement : 3 Lime : 12 Sand	Sand Hydrated lime Cement	Cum. Cum. Cum.	1.00 Cum. 0.249 cum. 2.37 Bags
10	1 Cement : 1 Lime : 12 Surkhi : 18 Sand	Sand Hydrated lime Surkhi Cement	Cum. Cum. Cum. Cum.	1.00 Cum. 0.055 cum. 0.110 Cum. 1.57 Bags
11	1 Cement : 4 Lime : 6 Surkhi : 18 Sand	Sand Hydrated lime Surkhi Cement	Cum. Cum. Cum. Cum.	1.00 Cum. 0.220 cum. 0.330 Cum. 1.57 Bags
12	Lime Mortar 1:2	Sand Unslaked Lime	Cum. Cum.	1.00 cum. 0.50 cum.
13	Lime Mortar with R.O.H. Lime	Sand Hydrated lime	Cum. Cum.	1.00 cum. 0.33 cum.
14	Filling foundations: Rammed moorum and metal	Metal Moorum	Cum. Cum.	1.00 cum. 0.33 cum.
15	Rammed moorum and boulders	Boulders Moorum	Cum. Cum.	1.00 cum. 0.40 cum.
16	Lime Concrete	Metal Lime Mortar	Cum. Cum.	1.00 cum. 0.50 cum.
17	Brick Bat lime Concrete	Lime Mortar Brick Bats	Cum. Cum.	0.50 cum. 1.00 cum.

18	Cement Concrete 1:1:2	Metal Sand Cement	Cum. Cum. Cum.	0.775 cum. 0.387 cum. 11.15 Bags.
19	Cement Concrete 1:1 1\2:3	Metal Sand Cement	Cum. Cum. Cum.	0.856 cum. 0.428 cum. 8.20 Bags.
20	Cement Concrete 1:1:2	Metal Sand Cement	Cum. Cum. Cum.	0.810 cum. 0.540 cum. 7.75 Bags.
21	Cement Concrete 1:2:4	Metal Sand Cement	Cum. Cum. Cum.	0.880 cum. 0.450 cum. 6.35 Bags.
22	Cement Concrete 1:2½ :5	Metal Sand Cement	Cum. Cum. Cum.	0.905 cum. 0.452 cum. 5.20 Bags.
23	Cement Concrete 1:3:6	Metal Sand Cement	Cum. Cum. Cum.	0.906 cum. 0.453 cum. 4.35 Bags.
24	Cement Concrete 1:4:8	Metal Sand Cement	Cum. Cum. Cum.	0.960 cum. 0.480 cum. 3.45 Bags.
25	Cement Concrete 1:5:8	Metal Sand Cement	Cum. Cum. Cum.	0.85 cum. 0.530 cum. 3.20 Bags.
26	Cement Concrete 1:6:10	Metal Sand Cement	Cum. Cum. Cum.	0.900 cum. 0.540cum. 2.60 Bags.
27	Cement Concrete 1:8:15	Metal Sand Cement	Cum. Cum. Cum.	0.900 cum. 0.480 cum. 1.75 Bags.

PLUM CEMENT CONCRETE

28	Plum cement concrete 1:5:8 with 60% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.60 cum. 0.34 cum. 0.21 cum. 1.20 Bags.
29	Plum cement concrete 1:5:8 with 40% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.40 cum. 0.51 cum. 0.32cum. 1.85 Bags.
30	Plum cement concrete 1:5:8 with 25% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.25 cum. 0.64 cum. 0.40 cum. 2.40 Bags.
31	Plum cement concrete 1:4:8 with 60% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.60 cum. 0.34 cum. 0.17 cum. 1.20 Bags.

32	Plum cement concrete 1:4:8 with 40% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.40 cum. 0.51 cum. 0.26cum. 1.85 Bags.
33	Plum cement concrete 1:4:8 with 25% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.25 cum. 0.64 cum. 0.32cum. 2.40 Bags.
34	Plum cement concrete 1:3:6 with 60% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.60 cum. 0.36 cum. 0.18cum. 1.75 Bags.
35	Plum cement concrete 1:3:6 with 40% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.40 cum. 0.54 cum. 0.27 cum. 2.60 Bags.
36	Plum cement concrete 1:3:6 with 25% plum	Plum Metal Sand Cement	Cum. Cum. Cum. Cum.	0.25 cum. 0.67 cum. 0.34 cum. 3.55 Bags.

MASONRY

37	Random rubble masonry	Stones Mortar	Cum. Cum.	1.00 cum. 0.33 cum.
38	Coursed rubble masonry	Stones Mortar	Cum. Cum.	1.00 cum. 0.33 cum.
39	Fine ashlars Stone masonry	Stones Mortar	Cum. Cum.	1.00 cum. 0.12 cum.
40	Cut Stone masonry	Stones Mortar	Cum. Cum.	1.00 cum. 0.16 cum.
41	Cut Stone work for lintels beams etc.	Stones Mortar	Cum. Cum.	1.00 cum. 0.75 cum.
42	Brick Masonry	Bricks Mortar	Cum. Cum.	500 Nos. 0.75 cum.
43	Honey comb brick masonry	Bricks Mortar	Cum. Cum.	360 Nos. 0.05 cum.

PLASTERING AND POINTING

44	6mm. thick plastering	Mortar	100 sqm.	0.90 cum.
45	13mm. thick plastering	Mortar	100 sqm.	1.50 cum.
46	20mm. thick plastering	Mortar	100 sqm.	2.40 cum.
47	Neat Finishing	w.w. lime	100 sqm.	4.28 Bags
48	Neat cement plaster	Cement	100 sqm.	4.28 Bags
49	Rough cast cement plaster			
	(a)Base Coat	Mortar	100 sqm.	1.377 cum.
	(b)Rough cast mixture	Aggregate Sand Cement	100 sqm. 100 sqm. 100 sqm.	1.377 cum. 0.688 cum. 13.11 Bages
50	Pebble dashing cement rendering	Mortar Pebble	100 sqm. 100 sqm.	0.30 cum. 1.50 cum.
51	Sand faced plastering base coat 13 mm thick Finishing coat 8mm. thick Sand facing	Mortar Pebble	100 sqm. 100 sqm.	2.70 cum. 4.28 Bags.
52	Rubbing on brick work	Mortar	100 sqm.	0.60 cum.
53	Grooved or flush pointing			
	(a) Brick work	Mortar	100 sqm.	0.90 cum.
	(b) Stone work	Mortar	100 sqm.	0.45 cum.
54	Tuck pointing			
	(a) Brick work	Mortar	100 sqm.	0.90 cum.
	(b) Stone work	Mortar	100 sqm.	0.45 cum.
55	Raised Pointing to stone work	Mortar	100 sqm.	0.50 cum.
56	Flush Cum. raised pointing to stone work	Mortar	100 sqm.	0.40 cum.
57	Pointing to flag stone	Mortar	100 sqm.	0.30 cum.
58	Drip course			
	(a) 4 cm.x10 cm	Bricks Mortar	100 R.M. 100 R.M.	0.40 cum. 0.37 Cum.
	(b) 10cm. x10 cm.	Bricks Mortar	100 R.M. 100 R.M.	1.00 cum. 0.70 Cum.
59	25x20 mm. Projected Patta	Mortar	100 R.M.	0.075 cum.
60	25x25 mm. groove	Mortar	100 R.M.	0.12 cum.
61	Throating of chajja	Mortar	100 R.M.	0.06 cum.
62	20 mm. thick Architectural molding or cut back Plaster	Mortar	100 sqm.	3.00 cum.

FLOORING

63	Brick on edge paving with moorum or cement mortarbedding & Joint	Bricks Moorum Mortar	100 sqm. 100 sqm. 100 sqm.	10.00 cum. 5000 Nos. 4.50 cum.
	C.M. pointing for a above	Mortar	100 sqm.	0.90 cum.
64	Flag stone Flooring with moorum or mortar bedding	Flag Stone Moorum or mortar	100 sqm. 100 sqm.	120 sqm. 2.00 cum.
	C.M. pointing for a above	Mortar	100 sqm.	0.30 cum.
65	Machine Polished Red Stone slabs Flooring	Red Stone Slabs Mortar for Bedding Cement for Slurry	100 sqm. 100 sqm. 100 sqm.	120 sqm. 2.00 sqm. 8.50 cum.
66	15 C.M. thick masonry stone paving	Masonry Stone Moorum Mortar for Bedding and joints Mortar for Pointing	100 sqm. 100 sqm. 100 sqm.	15.00 cum. 7.00 cum. 0.45 cum.
67	20 C.M. thick masonry stone paving	Masonry Stone Moorum Mortar for Bedding and joints Mortar for Pointing	100 sqm. 100 sqm. 100 sqm.	20.00 cum. 8.60 cum. 0.45 cum.
68	25 to 30 C.M. thick grouted Pitching	Rubble Stones Mortar	Cum. Cum.	1.00 cum. 0.20 Cum.
69	I.T. Flooring 75 mm thick with C.C. 1:3:6	Metal Sand Cement	100 sqm. 100 sqm. 100 sqm.	6.80 cum. 3.40 cum. 36.86 Bags
70	I.T. Flooring 75 mm thick with C.C. 1:2:4	Metal Sand Cement	100 sqm. 100 sqm. 100 sqm.	6.60 cum. 3.40 cum. 51.84 Bags
71	I.T. Flooring 40 mm thick with C.C. 1:3:6	Metal Sand Cement	100 sqm. 100 sqm. 100 sqm.	3.60 cum. 1.80 cum. 21.59 Bags
72	I.T. Flooring 40 mm thick with C.C. 1:2:4	Metal Sand Cement	100 sqm. 100 sqm. 100 sqm.	3.50 cum. 1.80 cum. 29.55 Bags
73	I.T. Flooring 40 mm thick with C.C. 1:2:3	Metal Sand	100 sqm. 100 sqm.	3.24 cum. 2.16 cum.
		Cement	100 sqm.	35.12 Bags
74	I.T. Flooring 25 mm thick with C.C. 1:2:4	Metal Sand Cement	100 sqm. 100 sqm. 100 sqm.	2.20 cum. 1.12 cum. 20.09 Bags
75	I.T. Flooring 25 mm thick with C.C. 1:2:3	Metal Sand Cement	100 sqm. 100 sqm. 100 sqm.	2.02 cum. 1.35 cum. 25.88 Bags
76	6 mm. thick mosaic topping with 40 mm thick under layer C.C. 1:2:4	Mosaic Chips Marble Power 12.5mm Metal Sand Cement	10 sqm. 10 sqm. 10 sqm. 10 sqm. 10 sqm.	108 Kg. 0.368 Bags 0.35 Cum. 0.18 Cum. 3.65 Bags

77	10 mm. thick mosaic topping with 40 mm thick under layer C.C. 1:2:4	Mosaic Chips Marble Power 12.5mm Metal Sand Cement	10 sqm. 10 sqm. 10 sqm. 10 sqm. 10 sqm.	180 Kg. 0.61 Bags 0.35 Cum. 0.18 Cum. 4.38 Bags
78	6 mm. thick mosaic or terrazo for dado/ skirting on 13 mm thick Plaster	Mosaic Chips Marble Power 12.5mm Metal Cement Topping	10 sqm. 10 sqm. 10 sqm. 10 sqm. 10 sqm.	108 Kg. 0.368 Bags 0.368 Cum. 0.15 Cum. 1.11 Bags
79	10 mm. thick mosaic or terrazo for dado/ skirting on 13 mm thick Plaster	Mosaic Chips Marble Power Mortar Cement for Topping	10 sqm. 10 sqm. 10 sqm. 10 sqm.	180 Kg. 0.61 Bags 0.15 Cum. 0.84 Bags
80	20 mm. thick precast coloured cement titles/mosaic/terra marble titles on 20 mm. thick bedding and cement slurry	Precast Title Mortar Cement for slurry and joint	10 sqm. 10 sqm. 10 sqm.	170 Nos. 0.24 cum. 1.72 bags
81	Laying Glazed ceramic tiles 152x152x6mm. on C.M. bedding	Ceramic Tile Mortar for Bedding Cement for Slurry	1 sqm. 1 sqm. 1 sqm.	48 Nos. 0.015 cum. 0.080 Bags
82	Laying Glazed ceramic tiles 108x1190x4mm. on C.M. bedding	Ceramic Tile Mortar for Bedding Cement for Slurry	1 sqm. 1 sqm. 1 sqm.	110 Nos. 0.015 cum. 0.080 Bags
83	Kotah/Ginno/Red sand for dado/Skirting over 13 mm thick plaster	Stones Mortar Cement for Slurry and joint	10 sqm. 10 sqm. 10 sqm.	11 sqm. 0.15 cum. 1.72 Bags
84	Kotah/Ginnor/Cuddapah stone Flooring over 20 mm thick mortar bedding	Stones Mortar Cement for Slurry and joint	10sqm. 10 sqm. 10 sqm.	11 sqm. 0.24 Cum. 1.72 Bags
85	Marble stone for flooring on 20 mm. thick mortar bedding	Marble Stones Mortar Cement for Slurry and joint	10sqm. 10 sqm. 10 sqm.	11 sqm. 0.24 Cum. 1.30 Bags
86	Marble stone for dado/skirting on 13 mm. thick plaster.	Marble Stones Mortar Cement for Slurry and joint	10 sqm. 10 sqm. 10 sqm.	11 sqm. 0.15 Cum. 1.30 Bags
87	Polished red stone slab for shelves etc.	Stone slab Mortar	1 sqm 1 sqm.	1.1 sqm. 0.025 bags.

ROOFING

88	Single wheel Tiling	Tiles Mortar	10 sqm 10 sqm	1300 Nos. 0.030 cum.
89	Double wheel Tiling	Tiles Mortar	10 sqm 10 sqm	2600 Nos. 0.060 cum.
90	Mangalore Pattern Tiling	Tiles Mortar Hip or Ridge	10 sqm 10 sqm	172 Nos. 0.096 cum.
91	Mangalore Pattern hip or Ridge Tiling	Tiles Mortar	10 sqm 10 sqm	92 Nos. 0.121 cum.
92	Cut stone slab roofing	Cut stone slab Mortar for Joints Mortar for pointing	1 cum. 1 cum. 1 cum.	1.00 cum. 0.20 cum. 0.04 cum.
93	Fixing 100 mm. thick precast R.C.C. Jali	Cement Sand	10 sqm. 10 sqm.	2.90 Bags 0.30 cum.
94	Fixing 40 mm. thick R.C.C. Jali	Cement Sand	10 sqm. 10 sqm.	1.62 Bags 0.15 cum.

PAINTING WHITE WASHING AND DISTEMPERING

95	White washing 3 coats	W.W. lime	10 sqm.	0.03 Qtl.
96	White washing 2 coats	W.W. lime	10 sqm.	0.02 Qtl.
97	White washing 1 coats	W.W. lime	10 sqm.	0.01 Qtl.
98	Colour washing	Add. Ramraj equal to 10% W.W. lime		
99	Chalk washing 3 coats	Chalk whiting	10 sqm.	0.03 Qtl.
100	Chalk washing 2 coats	Chalk whiting	10 sqm.	0.02 Qtl.
101	Chalk washing 1 coats	Chalk whiting	10 sqm.	0.01 Qtl.
102	Cement washing 3 coats	Cement	10 sqm.	3.21 kg.
103	Cement washing 2 coats	Cement	10 sqm.	2.14 kg.
104	Cement washing 1 coats	Cement	10 sqm.	1.07 kg.
105	Distempering washing 2 coat	Dry. Distempering	10 sqm.	1.00 kg
106	Distempering washing 1 coat	Dry. Distempering	10 sqm.	0.50 kg
107	Distemper	Distemper Primer	10 sqm.	0.81 Lit
108	Oil Bound Distempering 2 coat	Distemper	10 sqm.	1.50 kg.
109	Oil Bound Distempering 1 coat	Distemper	10 sqm.	0.75 kg.
110	Water Proofing cement paint 2 coats	W.P. cement paint	10 sqm.	3.00 Kg.
111	Water Proofing cement paint 1 coats	W.P. cement paint	10 sqm.	1.80 Kg.
112	Plaster Emulsion paint 2 coat	Plastic Emulsion paint	10 sqm.	1.21 Litr.
113	Plaster Emulsion paint 2 coat	Plastic Emulsion paint	10 sqm.	0.73 Litr.
114	Priming coat over steel and other metal surface with red oxide paint.	Ready mixed Red lead Paint	10 sqm.	0.90 Lit.

115	Priming coat with Zinc Chromite over steel and other surface.	Zine Chromite Paint	10 sqm.	0.54 Lit.
116	Priming coat with wood pink primer	Primer	10 sqm.	0.90 Lit.
117	Priming coat with Aluminum brushing primer	Aluminum paint primer	10 sqm.	1.00 Lit.
118	Ready mixed primer coat on concrete/masonry/A.C./Plastered surface.	Ready mixed Primer	10 sqm.	0.84 Lit.
119	Priming coat with linseed oil on concrete/masonry etc.	Linseed Oil	10 sqm.	1.16 Lit.
120	Steel/concrete/masonry etc. surface 2 coats.	Enamel paint	10 sqm.	1.25 Lit.
121	Steel/concrete/masonry etc. surface 2 coats.	1 cost	10 sqm.	0.60 Lit.
122	Synthetic enamel painting on old steel/concrete/masonry etc. surface 2 coats.	enamel paint	10 sqm.	1.00 Lit.
123	Varnishing 2 cost on new wood work.	Varnish	10 sqm.	1.45 Lit.
124	Do. 1 cost on old wood work.	Varnish	10 sqm.	1.45 Lit.
125	Polishing with French polish on new wood work.	French polish	10 sqm.	1.60 Lit.
126	Do on old wood work.	French polish	10 sqm.	0.95 Lit.
127	Linseed oiling on new wood work	Linseed oil	10 sqm.	1.16 Lit.
128	Linseed oiling on old wood work	Linseed oil	10 sqm.	0.7 Lit.
129	Coal tarring 2 coat.	Coal tar Kerosen oil	10 sqm. 10 sqm.	2.80 Lit. 0.50 Lit.
130	Coal tarring 1 coat.	Coal tar Kerosen oil	10 sqm. 10 sqm.	1.20 Lit. 0.20 Lit.
131	Fixing R.C.C. N.P. 2 pipes and collars in cement mortar			
	a) Diameter up to 30 C.M.	Mortar	each collar	0.0054 cum.
	b) Do above 35 and up to 40 cm	Mortar	each collar	0.0072 cum.
	c) Do above 40 cm up to 40 cm	Mortar	each collar	0.0090 cum.
	d) Do above 50 cm up to 60 cm	Mortar	each collar	0.0108 cum.
	e) Do above 60 cm up to 70 cm	Mortar	each collar	0.0126 cum.
	f) Do above 70 cm up to 80 cm	Mortar	each collar	0.0144 cum.
	g) Do above 80 cm up to 95 cm	Mortar	each collar	0.0171 cum.

TABLE -A

STEEL TUBES FOR STRUCTURAL PURPOSES

Nominal Bore (mm)	Outside Diameter (mm)	Class	Wall Thickness (mm)	Weight (kg/m)
15	21.3	H	3.2	1.44
20	26.9	H	3.2	1.87
25	33.7	M	3.2	2.41
		H	4.0	2.93
32	42.4	L	2.60	2.54
		M	3.2	3.10
		H	4.00	3.79
40	48.3	L	2.90	3.23
		M	3.2	3.56
		H	4.0	4.37
50	60.3	L-1	2.9	4.08
		M	3.6	5.03
		H	4.5	6.19
65	76.1	L	3.2	5.71
		M	3.6	6.42
		H	4.50	7.93
80	88.9	L	3.2	6.72
		M	4.00	8.36
		H	4.8	9.9
90	101.6	L	3.6	8.7
		M	4.0	9.63
		H	4.8	11.50
100	114.3	L	3.6	9.75
		M	4.5	12.2
		H	5.4	14.5
110	127.0	L	4.5	13.6
		M	4.8	14.50
		H	5.4	16.2
125	139.7	L	4.50	15.00
		M	4.8	15.90
		H	5.4	17.90
135	152.4	L	4.50	16.40
		M	4.8	17.50
		H	5.40	19.60
150	165.1	L	4.50	17.80
		M	4.8	18.90
		H	5.40	21.30
150	168.3	L	4.50	18.20
		M	4.8	19.40
		H-1	5.40	21.70
		H-2	6.3	25.20
175	193.7	L	4.8	22.40
		M	5.40	25.10
		H	5.9	27.30
200	219.1	L	4.8	25.40
		M	5.60	29.50
		H	5.90	31.00

225	244.5	H	5.90	34.70
250	273	H	5.90	38.90
300	323.90	H	6.30	49.30
350	355.60	H	8.00	68.60

TABLE –B

SECTION PROPERTIES			
Properties of Tata Structural (Square Hollow Sections) YST 310 Grade			
SHS B x B mm	**Thickness mm**	**Sec Area A cm2**	**Unit W kg/m**
25 x 25	2.00	1.74	1.36
	2.60	2.16	1.69
	3.20	2.53	1.98
32 x 32	2.00	2.30	1.80
	2.60	2.88	2.26
	3.20	3.42	2.69
38 x 38	2.60	3.51	2.75
	3.20	4.19	3.29
	4.00	5.03	3.95
40 x 40	2.60	3.72	2.92
	3.20	4.45	3.49
	4.00	5.35	4.20
50 x 50	2.60	4.76	3.74
	2.90	5.25	4.12
	3.60	6.35	4.98
	4.50	7.67	6.02
60 x 60	2.60	5.80	4.55
	3.20	7.01	5.50
	4.00	8.55	6.71
	4.80	10.01	7.85
72 x 72	3.20	8.54	6.71
	4.00	10.47	8.22
	4.80	12.31	9.66
80 x 80	3.20	9.57	7.51
	4.00	11.75	9.22
	4.80	13.85	10.87
91.5 x 91.5	3.60	12.32	9.67
	4.50	15.14	11.88
	5.40	17.85	14.01
100 x 100	4.00	14.95	11.73
	5.00	18.36	14.41

SECTION PROPERTIES			
Properties of Tata Structural (Square Hollow Sections) YST 310 Grade			
SHS B x B mm	**Thickness mm**	**Sec Area A cm2**	**Unit W kg/m**
	6.00	21.63	16.98
113.5 x 113.5	4.80	20.28	15.92
	5.40	22.60	17.74
132 x 132	4.80	23.83	18.71
	5.40	26.60	20.88
150 x 150	4.00	22.95	18.01
	5.00	28.36	22.26
	6.00	33.63	26.40
	8.00	43.79	34.38
180 x 180	4.00	27.75	21.78
	5.00	34.36	26.97
	6.00	40.83	32.05
	8.00	53.39	41.91
220 x 220	5.00	42.36	33.25
	6.00	50.43	39.59
	8.00	66.19	51.96
	10.00	81.43	63.92
250 x 250	6.00	57.63	45.24
	8.00	75.79	59.50
	10.00	93.43	73.34

TABLE –C

SECTION PROPERTIES			
Properties of Tata Structural (Rectangular Hollow Sections) YST 310 Grade			
RHS D x B mm	**Thickness mm**	**Sec Area A cm2**	**Unit W kg/m**
50 x 25	2.00	2.74	2.15
	2.60	3.46	2.71
	3.20	4.13	3.24
	4.00	4.95	3.88
60 x 40	2.60	4.76	3.73
	2.90	5.25	4.12
	3.60	6.35	4.98
	4.50	7.67	6.02
66 x 33	2.60	4.70	3.69
	2.90	5.19	4.07
	3.60	6.28	4.93
	4.50	7.58	5.95
80 x 40	2.60	5.80	4.55
	3.20	7.01	5.50
	4.00	8.55	6.71
	4.80	10.01	7.85
96 x 48	3.20	8.54	6.71
	4.00	10.47	8.22
	4.80	12.31	9.66
122 x 61	3.60	12.32	9.67
	4.50	15.14	11.88
	5.40	17.85	14.01
145 x 82	4.80	20.28	15.92
	5.40	22.60	17.74
172 x 92	4.80	23.83	18.71
	5.40	26.59	20.88
200 x 100	4.00	22.95	18.01
	5.00	28.36	22.26
	6.00	33.63	26.40
	8.00	43.79	34.38

TABLE –C			
240 x 120	4.00	27.75	21.78
	5.00	34.36	26.97
	6.00	40.83	32.05
	8.00	53.39	41.91
300 x 150	6.00	51.63	40.53
	8.00	67.79	53.22
	10.00	83.43	65.49
300 x 200	6.00	57.63	45.24
	8.00	75.79	59.50
	10.00	93.43	73.34

TABLE –D

SECTION PROPERTIES			
Properties of Tata Structural		**(Circular Hollow Sections) YST 310 Grade**	
Nominal Bore	**Outside Diameter**	**Thickness**	**Weight**
mm	**mm**	**mm**	**mm**
15	21.3	2.00	0.95
		2.60	1.21
		3.20	1.44
20	26.9	2.30	1.38
		2.60	1.56
		3.20	1.87
25	33.7	2.60	1.98
		3.20	2.41
		4.00	2.93
32	42.4	2.60	2.54
		3.20	3.01
		4.00	3.79
40	48.3	2.90	3.23
		3.20	3.56
		4.00	4.37
50	60.3	2.90	4.08
		3.60	5.03
		4.50	6.19
65	76.1	3.20	5.71
		3.60	6.42
		4.50	7.93
80	88.9	3.20	6.72
		4.00	8.36
		4.80	9.90
100	114.3	3.60	9.75
		4.50	12.20
		5.40	14.50
125	139.7	4.50	15.00
		4.80	15.90
		5.40	17.90
150	165.1	4.50	17.80
		4.80	18.90
		5.40	21.30

200	219.1	4.80	25.38
		6.00	31.51
		8.00	41.67
		10.00	51.59
250	273	6.00	39.51
		8.00	52.30
		10.00	64.89
300	323.9	6.30	49.36
		8.00	62.35
		10.00	77.44

Table-E

Sections and weights of I.S. Rolled Steel Beams Dimension

Designation		Weight per Metre W Kg.	Sectional Area a Cm. 2	Depth of Section h mm.	Width of Flange b mm.	Thickness Flange tf. Mm.	Thickness of Web. Tw mm.
1		2	3	4	5	6	7
I.S.J.B.	150	7.1	9.01	150	50	4.6	3.0
I.S.J.B.	175	8.1	10.28	175	50	4.8	3.2
I.S.J.B.	200	9.9	12.64	200	60	5.0	3.4
I.S.J.B.	225	12.8	16.28	225	80	5.0	3.7
I.S.L.B.	75	6.1	7.71	75	50	5.0	3.7
I.S.L.B.	100	8.0	10.21	100	50	6.4	4.0
I.S.L.B.	125	11.9	15.12	125	75	6.5	4.4
I.S.L.B.	150	14.2	18.08	150	80	6.8	4.8
I.S.L.B.	175	16.7	21.30	175	90	6.9	5.1
I.S.L.B.	200	19.8	25.27	200	100	7.3	5.4
I.S.L.B.	225	23.5	29.92	225	100	8.6	5.8
I.S.L.B.	250	27.9	35.53	250	125	8.2	6.1
I.S.L.B.	275	33.0	42.02	275	140	8.8	6.4
I.S.L.B.	300	37.7	48.08	300	150	9.4	6.7
I.S.L.B.	325	43.1	54.90	325	165	9.8	7.0
I.S.L.B.	350	49.5	63.01	350	165	11.4	7.4
I.S.L.B.	400	56.9	72.43	400	165	12.5	8.0
I.S.L.B.	450	65.3	83.14	450	170	13.4	8.6
I.S.L.B.	500	75.0	95.50	500	180	14.1	9.2
I.S.L.B.	550	86.3	109.97	550	190	15.0	9.9
I.S.L.B.	600	99.5	126.69	600	210	15.5	10.5
I.S.M.B.	100	11.5	14.60	100	75	7.2	4.0
I.S.M.B.	125	13.0	16.60	125	75	7.6	4.4
I.S.M.B.	150	14.9	19.00	150	80	7.6	4.8
I.S.M.B.	175	19.3	24.62	175	90	8.6	5.5
I.S.M.B.	200	25.4	32.33	200	100	10.8	5.7
I.S.M.B.	225	31.2	39.72	225	110	11.87	6.5
I.S.M.B.	250	37.3	47.55	250	125	12.5	6.9
I.S.M.B.	300	44.2	56.26	300	140	12.4	7.5
I.S.M.B.	350	52.4	66.71	350	140	14.2	8.1
I.S.M.B.	400	61.6	78.46	400	140	16.0	8.9
I.S.M.B.	450	72.4	92.27	450	150	17.4	9.4
I.S.M.B.	500	86.9	110.74	500	180	17.2	10.2
I.S.M.B.	550	103.7	132.11	550	190	19.3	11.2
I.S.M.B.	600	122.6	156.21	600	210	208	2.0
I.S.W.B.	150	17.0	21.67	150	100	7.0	5.4
I.S.W.B.	175	22.1	28.11	175	125	7.4	5.8
I.S.W.B.	200	28.8	36.71	200	140	9.0	6.1
I.S.W.B.	225	33.9	43.24	225	150	9.9	6.4
I.S.W.B.	250	40.9	52.05	250	200	9.0	6.7

Designation		Weight per Metre W Kg.	Sectional Area a Cm. 2	Depth of Section h mm.	Width of Flange b mm.	Thickness Flange tf. Mm.	Thickness of Web. Tw mm.
1		2	3	4	5	6	7
I.S.W.B.	300	48.1	61.33	300	200	10.0	7.4
I.S.W.B.	350	56.9	72.50	350	200	11.4	8.0
I.S.W.B.	400	66.7	85.01	400	200	13.0	8.6
I.S.W.B.	450	79.4	101.15	450	200	15.4	9.2
I.S.W.B.	500	95.2	121.22	500	250	14.7	9.9
I.S.W.B.	550	112.5	143.34	550	250	17.6	10.5
I.S.W.B.	600	133.7	170.38	600	250	21.3	11.2
I.S.W.B.	600	145.1	184.86	600	250	23.6	11.8
I.S.H.B.	150	27.1	34.48	150	150	9.0	5.4
I.S.H.B.	150	30.6	38.98	150	150	9.0	8.4
I.S.H.B.	150	34.6	44.08	150	150	9.0	11.8
I.S.H.B.	200	37.3	47.54	200	200	9.0	6.1
I.S.H.B.	200	40.0	50.94	200	200	9.0	7.8
I.S.H.B.	225	43.1	54.94	225	225	9.1	6.5
I.S.H.B.	225	46.8	59.66	225	225	9.1	8.6
I.S.H.B.	250	51.0	64.96	250	250	9.7	6.9
I.S.H.B.	250	54.7	69.71	250	250	9.7	8.8
I.S.H.B.	300	58.8	74.85	300	250	10.6	7.6
I.S.H.B.	300	63.0	80.25	300	250	10.6	9.4
I.S.H.B.	350	67.4	85.91	350	250	10.6	8.3
I.S.H.B.	350	72.4	92.21	350	250	11.6	10.1
I.S.H.B.	400	77.4	98.66	400	250	12.7	9.1
I.S.H.B.	400	82.2	104.66	400	250	12.7	10.6
I.S.H.B.	450	87.2	111.14	450	250	13.7	9.8
I.S.H.B.	450	92.5	117.89	450	250	13.7	11.3

Table-F

Sizes and weights of I.S. Rolled Steel channels

Designation		Weight per Metre W Kg.	Sectional Area a Cm. 2	Depth of Section h mm.	Width of Flange b mm.	Thickness Flange tf. Mm.	Thickness of Web. Tw mm.
1		2	3	4	5	6	7
I.S.J.C.	100	5.8	7.41	100	45	5.1	3.0
I.S.J.C.	125	7.9	10.07	125	50	6.6	3.0
I.S.J.C.	150	9.9	12.65	150	55	6.9	3.6
I.S.J.C.	175	11.2	14.24	175	60	6.9	3.6
I.S.J.C.	200	13.9	17.77	200	70	7.1	4.1
I.S.L.C.	75	5.7	7.26	75	40	6.0	3.7
I.S.L.C.	100	7.9	10.02	100	50	6.4	4.0
I.S.CL.C.	125	10.7	13.67	125	65	6.6	4.5
I.S.L.C.	150	14.4	18.36	150	75	7.8	4.8
I.S.L.C.	175	17.6	22.40	175	75	9.5	5.1
I.S.L.C.	200	20.6	26.22	200	75	10.8	5.5
I.S.L.C.	225	24.0	30.53	225	90	10.2	5.8
I.S.L.C.	250	28.0	35.65	250	100	10.7	6.1
I.S.L.C.	300	33.1	42.11	300	100	11.6	6.7
I.S.L.C.	350	38.8	49.47	350	100	12.5	7.4
I.S.L.C.	400	45.7	58.25	400	100	14.0	8.0
I.S.M.C.	75	6.8	8.67	75	40	7.3	4.1
I.S.M.C.	100	9.2	11.70	100	50	7.5	4.7
I.S.M.C.	125	12.7	16.19	125	65	8.1	5.0
I.S.M.C.	150	16.4	20.88	150	75	9.0	5.4
I.S.M.C.	175	19.1	24.38	175	75	10.2	5.7
I.S.M.C.	200	22.1	28.21	200	75	11.4	6.1
I.S.M.C.	225	25.9	33.01	225	80	12.4	6.4
I.S.M.C.	250	30.4	38.67	250	80	14.1	7.1
I.S.M.C.	300	35.8	45.64	300	90	13.6	7.6
I.S.M.C.	350	52.1	43.66	350	100	13.5	8.1
I.S.M.C.	400	49.4	62.93	400	100	15.3	8.6

Appendix G
Weights and Sizes of I.S. Rolled Steel Equal Angles

Designation	Size AXB mm.	Thickness T mm	Sectional area A Cm.2	Weight Per Metre W Kg.
1	2	3	4	5
I.S.A. 2020	20x20	3.0 4.0	1.12 1.45	0.0 1.1
I.S.A. 2525	25x25	3.0 4.0 5.0	1.41 1.84 2.25	1.1 1.4 1.8
I.S.A. 3030	30x30	3.0 4.0 5.0	1.73 2.26 2.77	1.4 1.8 2.2
I.S.A. 3535	35x35	3.0 4.0 5.0 6.0	2.03 2.66 3.27 3.86	1.6 2.1 2.6 3.0
I.S.A. 4040	40x40	3.0 4.0 5.0 6.0	2.34 3.07 3.78 4.47	1.8 2.4 3.0 3.5
I.S.A. 4545	45x45	3.0 4.0 5.0 6.0	2.64 3.47 4.28 5.07	2.1 2.7 3.4 4.0
I.S.A. 5050	50x50	3.0 4.0 5.0 6.0	2.95 3.88 4.79 5.68	2.3 3.0 3.8 4.5
I.S.A. 5555	55x55	5.0 6.0 8.0 10.0	5.27 6.26 8.18 10.02	4.1 4.9 6.4 7.9
I.S.A. 6060	60x60	5.0 6.0 8.0 10.0	5.75 6.84 8.96 11.00	4.5 5.4 7.0 8.6
I.S.A. 6565	65x65	5.0 6.0 8.0 10.0	6.25 7.44 9.76 12.00	4.9 5.8 7.7 9.4
I.S.A. 7070	70x70	5.0 6.0 8.0 10.0	6.77 8.06 10.58 13.02	5.3 6.3 8.3 10.2
I.S.A. 7575	75x75	5.0 6.0 8.0 10.0	7.27 8.66 11.38 14.02	5.7 6.8 8.9 11.0

Designation	Size AXB mm.	Thickness T mm	Sectional area A Cm.2	Weight Per Metre W Kg.
1	2	3	4	5
I.S.A. 8080	80x80	6.0	9.29	7.3
		8.0	12.21	9.6
		10.0	15.05	11.8
		12.0	17.81	14.0
I.S.A. 9090	90x90	6.0	10.47	8.2
		8.0	13.79	10.8
		10.0	17.03	13.4
		12.0	20.19	15.8
I.S.A. 100100	100x100	6.0	11.67	9.2
		8.0	15.39	12.1
		10.0	19.03	14.9
		12.0	22.59	17.7
I.S.A. 110110	110x110	8.0	17.02	13.4
		10.0	21.06	16.5
		12.0	25.02	19.6
		15.0	30.81	24.2
I.S.A. 130130	130x130	8.0	20.22	15.9
		10.0	25.06	19.7
		12.0	29.82	23.4
		15.0	36.81	28.9
I.S.A. 150150	150x150	10.0	29.03	22.8
		12.0	34.59	27.2
		15.0	42.78	33.6
		18.0	50.79	39.9
I.S.A. 200200	200x200	12.0	46.61	36.4
		15.0	57.80	45.0
		18.0	68.81	54.6
		25.0	93.80	73.6

Table- H

Sizes and Weights of I.S. Rolled Steel Unequal Angles

Designation	Size AXB mm.	Thickness T mm	Sectional area A Cm.2	Weight Per Metre W Kg.
1	2	3	4	5
I.S.A. 3020	30x20	3.0	1.41	1.1
		4.0	1.84	1.4
		5.0	2.25	1.8
I.S.A. 4025	40x25	3.0	1.88	1.5
		4.0	2.46	1.9
		5.0	3.02	2.4
		6.0	3.56	2.8
I.S.A. 4530	45x30	3.0	2.18	1.7
		4.0	2.86	2.2
		5.0	3.52	2.8
		6.0	4.16	3.3
I.S.A. 5030	50x30	3.0	2.34	1.8
		4.0	3.07	2.4
		5.0	3.78	3.0
		6.0	4.47	3.5
I.S.A. 6040	60x40	5.0	4.76	3.7
		6.0	5.65	4.4
		8.0	7.37	5.8
I.S.A. 6545	65x45	5.0	5.26	4.1
		6.0	6.25	4.9
		8.0	8.17	6.4
I.S.A. 7045	70x45	5.0	5.52	4.3
		6.0	6.56	5.2
		8.0	8.58	6.7
		10.0	10.52	8.3
I.S.A. 7550	75x50	5.0	6.02	4.7
		6.0	7.16	5.6
		8.0	9.38	7.4
		10.0	11.52	9.0
I.S.A. 8050	80x50	5.0	6.27	4.9
		6.0	7.46	5.9
		8.0	9.78	7.7
		10.0	12.02	9.4
I.S.A. 9060	90x60	6.0	8.65	6.8
		8.0	11.37	8.9
		10.0	14.01	11.0
		12.0	16.57	13.0
I.S.A. 10065	100x65	6.0	9.55	7.5
		8.0	12.57	9.9
		10.0	15.51	12.2
I.S.A. 10075	100x75	6.0	10.14	8.0
		8.0	13.36	10.5
		10.0	16.50	13.0
		12.0	19.56	15.4
I.S.A. 12575	125x75	6.0	11.66	9.2
		8.0	15.38	12.1
		10.0	19.02	14.9

Designation	Size AXB mm.	Thickness T mm	Sectional area A Cm.2	Weight Per Metre W Kg.
1	2	3	4	5
I.S.A. 12595	125x95	6.0 8.0 10.0 12.0	12.86 16.98 21.02 24.98	10.1 13.3 16.5 19.6
I.S.A. 15075	150x75	8.0 10.0 12.0	17.42 21.56 25.62	13.7 16.9 20.10
I.S.A. 150115	150x115	8.0 10.0 12.0 15.0	20.58 25.52 30.38 37.52	16.2 20.0 23.8 29.5
I.S.A. 200100	200x100	10.0 12.0 15.0	29.03 34.59 42.78	22.8 27.2 33.6
I.S.A. 200150	200x150	10.0 12.0 15.0 18.0	34.00 40.56 50.25 59.76	26.7 31.8 39.4 46.9

Table-I

Weights and Sizes of Rolled Steel Tee Bars

Designation		Weight per Metre W Kg.	Sectional Area a Cm. 2	Depth of Section h mm.	Width of Flange b mm.	Thickness Flange tf. Mm.	Thickness of Web. Tw mm.
1		2	3	4	5	6	7
I.S.N.T.	20	0.9	1.13	20	20	3.0	3.0
I.S.N.T.	30	1.4	1.75	30	30	3.0	3.0
I.S.N.T.	40	3.5	4.48	40	40	6.0	6.0
I.S.N.T.	50	4.5	5.70	50	50	6.0	6.0
I.S.N.T.	60	5.4	6.90	60	60	6.0	6.0
I.S.N.T.	80	9.6	12.25	80	80	8.0	8.0
I.S.N.T.	100	15.0	19.10	100	100	10.0	10.0
I.S.N.T.	150	22.8	29.08	150	150	10.0	10.0
I.S.H.T.	75	15.3	19.49	75	150	9.0	8.4
I.S.H.T.	100	20.0	25.47	100	200	9.0	7.8
I.S.H.T.	125	27.4	34.85	125	250	9.7	8.8
I.S.H.T.	150	29.4	37.42	150	250	10.6	7.6
S.S.T.	1001	8.1	10.37	100	50	10.00	5.8
I.S.S.T.	150	15.7	19.96	150	75	11.6	8.0
I.S.S.T.	200	28.4	36.22	200	165	12.5	8.6
I.S.S.T.	250	37.5	47.75	250	180	14.1	9.2
I.S.L.T.	50	4.0	5.11	50	50	6.4	4.0
I.S.J.T.	75	7.01	9.04	75	80	6.8	4.7
I.S.L.T.	100	12.7	16.16	100	100	10.8	5.8
I.S.L.T.	75	3.5	4.50	75	50	4.6	3.0
I.S.J.T.	87.5	4.0	5.14	87.5	50	4.8	3.2
I.S.J.T.	100	5.0	6.32	100	60	5.0	3.4
I.S.J.T.	112.5	6.4	8.14	112.5	80	5.0	3.7

Appendix 4

SHAPES, WEIGHTS AND DESIGNATION OF MS ROLLED STEEL SECTION

Designation	Wt kg/m	Situation of Use of Section
T2	1.036	Vertical and horizontal glazing bars for doors and shashes : windows, ventilators glazing bars for door side lights sub dividing bars for fixed length, sash bars for doors, windows and ventilators wheel steel aluminium or wooden beading is used for fixing glasses.
T3	1.14	Vertical glazing bar for FZ 7 frame
T 6	0.839	Vertical and horizontal glazing bar for standard windows and ventilators.
F 2	1.46	Inner frames for open-in windows
F 3	2.28	Outer frames for open in windows
F 5	1.55	(a) Inner and middle frames in centre-hung ventilators (b) F5 is some times used as inner frames for open-out windows. Also used as inner frame for bottom hung ventilators (c) F8 is also used as outer frame for bottom hung ventilators.
F 4 B	2.28	Central mullion (meeting bar for shutters) for windows and ventilators using F7D as inner frames, outer frames for open- in windows in rainy areas, subdividing bars for openable windows and top- hung ventilators.
F 7 D	1.419	Inner and outer frames for windows and top hung
		ventilators, for inner frames for centre-hung ventilators and outer frames for door sidelights.
FX 6	2.52	Inner frame for doors
F Z 7	1.90	Used as outer frame for industrial shashes. Also used for outer frame for wooden doors
FX 8	2.31	Outer frames for doors
FZ5	2.52	Inner frames for doors
K 11B	1.80	(a) Vertical coupling mullion for standard windows (b) Can be used as horizontal coupling bar when openable windows are to be coupled above fixed ones or between two fixed windows. (c) Can also be used as horizontal coupling mullion where windows are not exposed toweather.
K 12 B	2.30	Horizontal coupling mullion, also known as weather bar, Especially used when the coupled unit is exposed to rain.

Appendix 5

Table-A

PARTICULARS OF MEDIUM GRADE G.I. PIPES

Nominal bore	Dimension of pip s			Weight of pipe	
	Outside diameter		Thickness	Plain end	Screwed end socket
	Max.	Min.			
mm	mm	mm	mm	Kg/m	Kg/m
6	10.6	9.8	2.0	0.404	0.407
8	14.0	13.2	2.3	0.641	0.645
10	17.5	16.7	2.3	0.839	0.845
15	21.8	21.0	2.6	1.21	1.22
20	27.3	26.5	2.6	1.56	1.57
25	34.2	33.3	3.2	2.41	2.43
32	42.9	42.0	3.2	3.10	3.13
40	48.8	47.9	3.2	3.56	3.60
50	60.8	59.7	3.6	5.03	5.10
65	76.6	75.3	3.6	6.42	6.54
80	89.5	88.0	4.0	8.36	8.53
100	115.0	113.1	4.5	12.2	12.50
125	140.8	138.5	4.8	15.90	16.40
150	166.5	163.9	4.8	18.90	19.50

Tolerance in Thickness and Weight

A) Thickness

1. Butt welded medium tubes + not limited
   - 10 per cent
2. Seamless tubes + not limited
   - 12.5 per cent

B) Weight

1. Single tube (light series) + 10 percent
   - 8 per cent
2. Single tube (medium andheavy series) ± 10 per cent
3. For quantities per load of 10 tonnes, min (light series) ± 5 per cent
   - 8 per cent
4. For quantities per load of 10 tonnes, min (medium and heavy series) ± 7.5 per cent

Table-B

Stoneware Pipes

Internal Diameter (mm)	Mean Thickness of the Barrel and Socket (mm)
100	12
150	15
200	16
230	19
250	20
300	25
350	30
400	35
450	37

Appendix 6

STANDARD FORMAT FOR STRUCTURAL AUDIT REPORT

1.	**Name of Building**	
2.	**CTS No. / Ward**	
3.	**No. of Storey**	
4.	**Year of Construction**	
5.	**User Department**	
6.	**Mode of construction of existing bldg**	
	i.Foundation	
	ii.Floors	
	iii.Walls	
	iv.Beams	
	v.Columns	
	vi.Roof	
7.	**History of Repairs done year-wise-** a) Slab recasting b) Column Jacketing	
	i.Structural Repairs	
	ii.Tenantable Repairs	
	iii.Roof / Waterproofing	
	iv.Plumbing	
	v.Additions/Alterations if any	
8.	**Date of Inspection by Consultant -**	
9.	**Condition of -**	
	i.Internal plaster	
	ii.External plaster	
	iii.Plumbings	
	iv.Drains lines / chambers	
10.	**Observations -**	
	a)Doors & windows don't close	
	b)Columns & steel exposed	
	c)Settlement uneven flooring gaps between and skirting & floor	

	d)Foundation settlement	
	e)Deflections / sagging	
	f)Major cracks in column / beams	
	g)Seepages / Leakages	
	h)Staircase area / column condition	
	i)Lift walls	
	j)U.G.Tank	
	k)OHT / Column condition	
	l)Parapet at terraces	
	m)Chhajas	
	n) Common areas	
	o) Toilet blocks	
	p) Terrace / Water proofing	

11.	**Test carried on structural observations thereof – (Note: Please see detailed report for interpretation & explanation).**		
	NDT	**Findings**	**Range as per is Code**
	a) Ultrasonic pulse velocity test		3 to 3.5 km/sec concrete quality grading – Medium
	b) Rebound Hammer Test		15 N/mm^2 (original design)
	c) Half Cell Potential Test		No IS code available for this test
	d) Carbonation Depth Test		No I. S. Code available for this test
	e) Core Test (3 nos)		11.25 (N/mm^2)
			11.25 (N/mm^2)
			11.25 (N/mm^2)
	Average value		12.75 (N/mm^2)

	f)Chemical Analysis - Chloride		Shall be less than 0.6 kg/cu.m.
	- Sulphates		Shall be less than 4%

	- pH		No I. S. Code available.
	g)Cement aggregate ratio		No I. S. Code available.
12.	**Distress Mapping Plan & photographs with caption below about description of structural member and its location**		
13.	**Brief Description of repairs to be done**		
	a)Water proofing		
	b)External plaster		
	c)Structural repairs		
	i.Column jacketing		
	ii.Slab recasting		
	iii.RCC cover to be replaced		
	iv.Beam recasting		
	d)Partial Evacuation during repairs needed		
	e)propping		
14.	**Conclusions of Consultants -**	**Observations**	**Key reason**
	i. Whether structural is livable / or whether it is to be evacuated / pulled down		
	ii. Whether structural requires tenantable repairs / Major structural repairs & its time frame		
	iii.Whether structure can be allowed to occupy during course of repairs		
	iv.Nature / Methodology of repairs		
	v. Whether structure requires immediate propping. If so. Its propping plan / methodology given		
	vi.Whether other immediate safety measures required – What is specific recommendation?		
	vii.Enhancement in life of structure after repairs / frequency of repairs required in extended life period		
	viii.Projected repairs cost / Sq. ft of built up area		
	ix.Projected reconstruction cost / Sq. ft		
	x. Specific remarks. Whether building needs to be vacated / demolished / repairable		

	xi.Whether structural in extremely critical condition		
15.	**Critical Observation**		
16.	**Classification of Bldgs**	**Category**	**Auditors final conclusion**
		C1	To be evacuated / demolition immediately
		C2-A	To be evacuated and/or partial demolition requiring major structural repairs
		C2-B	No eviction only structural repairs
		C-3	No eviction needs minor repairs only.

Signature

( )
Structural Auditor
Licensed Structural Engg. Consultant

(Lic. No.)

STANDARD FORMATS A,B,C,D ……..
FOR
PREPARATION OF ESTIMATE FOR DPROF
A BUILDING PROJECT

Appendix-7

STANDARD FORMATS A,B,C,D ..... FOR PREPARATION OF ESTIMATE FOR DPR OF A BUILDING PROJECT

Preparation of accurate detail estimate is key for successful completion of abuilding project. It is, therefore, necessary that the cost estimate attached with a Detail Project Report (DPR) of a building work should be prepared accurately in standard formatswhich fulfil intended functional and structural requirement of the building. DPR for large number of building projects for various user departments are prepared by the Public Works Department for their administrative and financial sanctions. It is, therefore, very important that the project cost in a DPR should discretely display gross as well as individual cost ofeach major and minor work constituents of the building project to enable technical & financial project sanctioning authorities to take informed decisions.

For clarity and simplicity in understanding of a building project, standardformats for estimating has been devised in which "Project Cost" has been divided into"Project Cost Center" (PCC) and in turn each "Project Cost Center" have been further subdivided into "Cost Centers" (CC). Each "Project Cost Center" represents a major project activity namely individual building, campus services or campus development. Similarly "Cost Centers" of each "Project Cost Center" represents distinct components of "Project Cost Center" (PCC) such as RCC structure below plinth, RCC structure above plinth, internal electrification, sewage treatment, lifts, HVAC, internal roads, boundary wall etc. With this motive, templates such as Technical Report (Format 'A'), Project Abstract (Format 'B'), Building Work Abstract (Format 'C'), External Work Abstract (Format 'D'), Cost Abstract (Format 'E') and Quantity Sheet (Format 'F') have been standardized.

To enable project sanctioning authority to take informed decisions, standard formats for granting technical sanctions by competent authority of civil & electrical worksand presenting all information of technical sanctions have been standardized as formats T2, T3, & T1 respectively.

For good understanding of a building by the authorities at the time of granting sanction and execution and taking over a constructed building by user department, it is recommended thatat finishing information of the building should be provided in a standard format and should be appended with the estimate and hand over documents as well. With this motive a standard format of schedule of finishes has been devised as Format 'SOF' for civil works, toilets and electrical works.

After preparation of estimate for a building project, a very well prepared Bill of Quantity (BOQ) is a very important document for successful execution of work throughcontract. With this motive, a format for BOQ Schedule "Q" has been standardized.

During construction stage monitoring of work progress (quantity & cost of work completed) with respect to stipulated quantities & time for completion is very essential for successful completion the project and with this motive a format Chart 'P' has been standardized.

In case of a DPR for a revised sanction, a revised cost estimate of the building project is required to be prepared in which component wise (cost centre wise) cost excess statement with reasons is always essential to enable project sanctioning authority to take informed decision on revised cost. With this motive & for clarity a standard template Format 'RA' has been devised which gives details of access cost with reasons for each cost centres. Further details of all above standard formats are as below:-

I. Format "A" Technical Report:-

All important information and guidelines about estimate is included in Format "A" which contains following details-

1. Administrative & General Information:-

Describes name, location, project cost and ownership of the project and project land .It also describes about project area ,adjoining land details with the help of Google index plan , drone photos, khasra,map ,survey plans,etc

2. Executive Summary:-

Describes history of existing campus and necessity of the project.

3. Functional Spaces and Facilities Provided in Project :-

Describes present and future project requirement to meet intended use of building. It also describes space (capacity) to be provided for intended building use, plinth area, built up area, number of stories of each building, type of buildings, number of buildings, future project needs and provision made for the same, etc.

4. Salient Features of Project :-

Describes salient features such as total Built up Area, Plinth Area, Number of Floors, Future expansions (provision horizontal and vertical) and concept Design/DrawingNumber etc.

5. Design Basis Report (DBR):-

Describes in technical terms, all physical elements of the project which are basis for inclusion of items in the estimates. Description of physical element in technical terms should be in group such that each group represents a Cost Centre belonging to aspecific Project Cost Centre Each Cost Centre has been assigned a unique number for future reference. Prepare DBR for each building separately and also for campus services and campus development separately for clarity to quantity surveyor while preparing estimate cost centre wise for monitoring quantity by project and engineers while executing project.

6. Finishes :-

Standard format for finishing of civil work 'SOF – Civil', finishing of toilets 'SOF- Toilets' and finishing of Electrical works 'SOF- Electrical' have been standardized for good understanding of finishes of building project.

7. Program Chart ' P1' & 'P2' :-

Standard format Chart 'P1'(Project Program Chart) displaying projected time line for completion each "Project Cost Centre" is prepared only one for whole project, showing stipulated time line for all major construction activity "Project Cost Centers" such as a building or campus services or campus development works. However in a major construction project Chart "P2" is also prepared to display time lines for all components ('Cost Centers') of "Project Cost Centers".

8. Revised Project Cost :-

Standard Format 'RA' for representing revised cost of project, if required, is preparedto provide Project Cost Centre wise and Cost Centre wise cost comparison.

9. Rates:-

Describes about SOR applicable and market rates (Non SOR terms) if any adopted.

10. Standards applicable:-

Describes about standards (NBC, IPHS, MCI, BIS, Bhumi Vikas Adhiniyam, AICTE, PCB, NEC, AAI etc.) that are followed in planning and designing theproject.

11. Specification:-

Describes specifications (CPWD, BIS, MPPWD, MORTH etc.) followed in designing all elements of the project.

12. Green Building Rating, if targeted:-

Describes GRIHA or IGB rating, if targeted.

II. Format "B" (Project Abstract):-

Is prepared only one for whole project and displays project cost inclusive of supervision charges and GST saperately at the end of summation cost of "Building numbers -1"," Building number- 2" , "Building number -N" and so on and also " Campus services " and "Campus Development" which are designated as "Project Cost Centers".
Format "B" also displays project cost center values which comprise of apportionedcost (% of Net Project Cost ie. project cost without supervision charges and GST), unit and quantity. Supervision charges and GST are added at last to arrive at Project Cost. Apprortioned cost is very useful data to know cost impact of each "Project Cost Centers" on total project cost.

III. Format "C" (Building Work Abstract)-:-

Is prepared individually for each "Project Cost Centers" in building work category. These Building Work Abstracts in building work category are Format C1 for of Buildings no 1, C2 for Building no 2, C3 for Building no 3 and CN for Building no
N. Each "Project Cost Centers" are further subdivided into "Cost Centers" undervarious building sub head (I) Civil Works, (II)- Internal Services , (III) Internal Special Services and (IV) Equipment & Furniture.
Each "Cost Center" represents a constituent of the "Project Cost Center", Examples of a "Cost Center" of "Project Cost Center" namely " building No.-1" are works suchas civil works below plinth , civil works above plinth , internal electrification,Internal water supply ,HVAC, lift, firefighting, etc. Amount of "Project Cost Center" of each building is arrived at by summing estimated amount of all "Cost Centers" asper DBR from S.No. CC 1 to CC 30 The Supervision Charges and GST are not addedat this stage.
Format "C" also displays apportioned cost which is % of "Project Cost Center" cost Format 'C' other cost center values such as units, quantity as per unit cost. Apprortioned cost is a very useful data to know impact of cost center on project cost center.

IV. Format "D" (External Work Abstract):-

Is prepared only one for whole project. Total amount of External Work Abstract is arrived at by summing estimated amount of "Project Cost Center" namely campus services and campus development which are further sub divided into "Cost Centers" from s.no. CC 31 to CC 55 as per DBR.
Examples of "Cost Center" of "Project Cost Center" namely 'Campus services' arerain water harvesting, external electrification, external water supply, sewerage, sewage treatment plants etc. Similarly examples of "Cost Centers" of "Project Cost Center" namely ''Campus Development" are works such as roads, culverts, boundary walls, gates, plantation, open drains, nalla training, landscaping, open parking, utility shifting, tree cutting, soil investigation, play fields, demolition, statuary approval fees etc.
Format "D" also displays apportioned cost (% of Project Cost Center cost) and other values such as units, quantity as per unit cost.

V. Format "E" (Cost Abstract):-

Is prepared individually in traditional way for each "Cost Center" (S.No. CC 1 to S.No. CC 30 of DBR) for each building of the project and these costs are filled up in Format 'C' to arrive at Format C1, Format C2 and Format CN etc. for Building No-1,

Building No-2 and Building No-N respectively.

Format 'E' is also prepared individually only once for each "Cost Center" of Campus services and external development work (S.No CC 31 to CC 55 of DBR) and these costs are filled up in Format 'D' for whole project. A "Cost Center", if not relevant toa building work or external work of the project, shall be bypassed (not prepared) but unique serial number and name of the "Cost Center" prescribed in DBR shall not be altered to maintain uniformity in the estimate. New "Cost Center" for a building and external work can always be added, as misc-1, misc-2, misc. -3 etc., on requirement basis against item no. 24 and 55 to capture any imagination in planning and design ofa project. Description of SOR items in full are optional in format E for ease of reading the cost abstract. Therefore, full description of SOR items in Format E can be hidden and can display only BOQ serial number, SOR Item Number, SOR Sub Item Number and SOR Item Name.

VI. Format "F" (Quantity Sheet):-

Is prepared traditional in support of each sheet of Format "E". Full description of SOR items are always written in full text in quantity sheet Format-'F'

VII. Standard Formats "T1", "T2", "T3" (Technical Sanction):-

Format T2 (Technical Sanction Civil Works) and Format T3 (Technical SanctionElectrical & Mechanical Works) are prescribed for granting technical sanctions by the competent authorities. Amount of technical sanctions do not include supervision charges and GST. Format T1 (Technical Sanction Details) is a prescribed format to display summation of technically sanctioned cost (T2+T3) and supervision charges and GST added at the end. Standard format T2 & T3 also display "Cost Center"number belonging to civil and E/M component. This is as guide only however technical sanction granting competent authority may further bifurcate cost centers.

VIIl. Format "SOF"- (Schedule of Finishes):-

For clarity in understanding of final built product during project sanction, execution and building handing over, standard formats 'SOF-Civil' for civil work , 'SOF –Toilets' for toilet finishes and 'SOF- Electrical' for Electrical work are preprared after writing DPR and framing the estimate.

IX. Schedule 'Q' (BOQ):-

BOQ is a final outcome of estimating and costing and is a most important documentfor execution of works at site through legally binding contract and is also a basic document for making payment to the contractors. Therefore BOQ has to be prepared with utmost accuracy and clarity. Standard Format Schedule 'Q' has been prescribed for preparing BOQ. Schedule 'Q' should be prepared individually for each "Project Cost Centers" known as Project Cost Centre BOQ and then added horizontally to form a Project 'BOQ'. In standard format "Q" each BOQ items is assigned a unique BOQ Serial Number. Each BOQ item is further identified by Unique SOR Item number, Unique SOR Sub Item number, Unique SOR name, and Unique SOR description. Each BOQ item has Quantity, Measuring Units, Rates and Amount. Order of placement of SOR items in Schedule 'Q'BOQ has been prescribed as Civil SOR items, Civil Non SOR items, Electrical SOR items, and Electrical Non SORitems in descending order. Standard Format Schedule 'Q' (BOQ) is same for both Project 'BOQ' & "Project Cost Center" BOQ. Project 'BOQ' is obtained by horizontal summation of "Project Cost Center" BOQ.

X. Chart 'P' (Program Chart):-

Project Program Chart 'P' is prepared one for whole project in which target of each "Project Cost Center" are set against milestones set in the project. This Chart 'P1'

(Project Program Chart) is prepared by the architect/engineer who prepares D.P.R. In cases where the contractor is required to submit construction program, Chart 'P1'(Project Program Chart) shall be modified by the contractor and submitted for approval of the Engineer-in-charge. In case of a complex project Chart 'P2' (P.C.C. Program Chart) may be prepared for each "Project Cost Center" individually in which targets shall be set for each "Cost Center" also. Progress for work during construction shall be monitored by observing achievement data filled by the field staff.

XI. Format 'RA' (Revised Cost):-

After preparation of Revised Cost estimate, if prepared for revised 'AA' of the project standard template RA-'B' is prepared to display variation at P.C.C level. Standard template RA-'C' is prepared to display variation in each building.Revised template RA-'D' is prepared to display variation in external works.

Flow Chart For Estimating

PROJECT COST (FORMAT 'B') Including GST

↓

NET PROJECT COST (FORMAT 'B') Excluding GST

↓

BUILDING WORKS + **EXTERNAL**

↓

Format'C1'
Name / Description of "Project Cost Center" (PCC) Building No.1 + **Format'C2'** (PCC) Building No.2 + **Format'CN'** (PCC) Building No.N

↓

(I) Civil Works

Cost Center No (CC No.)	Name of Cost Center (As per DBR In Format "A")	Cost
1	Concrete in R.C.C. Frame Structure up to Plinth Level	0
2	Steel in R.C.C. Frame Structure work up to Plinth Level	0
3	Civil Work Other Than R.C.C Frame Structure upto plinth Level	0
4	Concrete in R.C.C. Frame Structure Above Plinth Level	0
5	Steel in R.C.C. Frame Structure Above Plinth Level	0
6	Civil Work Other Than R.C.C Frame Structure Above Plinth Level	0

+

(II) Internal Services

Cost Center No (CC No.)	Name of Cost Center (As per DBR In Format "A")	Cost
7	Internal Water Supply & Plumbing and Sanitary	0
8	Internal Electrification	0

+

(III) Internal Special Services

Cost Center No (CC No.)	Name of Cost Center (As per DBR In Format "A")	Cost
9	HVAC	0
10	Lifts/Elevators	0
11	Solar Water Heating	0
12	Solar Power	0
13	Fire Fighting	0
14	Fire Detection	0
15	Audio Visual System	0
16	Access Control	0
17	C.C.T.V.	0
18	Telephony (EPABX, Nurse call System)	0
19	Acoustic & Raise Ceiling	0
20	Interior Decoration & Art Work	0
21	Information Communication (IT)	0
22	Internal signage	0
23	Building Management System (BMS)	0
24	Miscellaneous Special Services such as Lighting Conductor, Façade light, Water Purifier, Incinerator, if Any	0

+

(IV) Equipment & Furniture

Cost Center No (CC No.)	Name of Cost Center (As per DBR In Format "A")	Cost
25	Furnitures	0
26	Lab Equipment	0
27	Medical Equipment & Accessories AMGPS,OT,PTS,etc	0
i	MGPS	0
ii	OT (High End/Low End)	0
iii	PTS (Pneumatic Tube System)	0
28	Kitchen Equipments	0
29	Laundary Equipments	0
30	Computer Hardware/Software	0

Same as Building No.-1

Same as Building No.-1

(E1) Campus Services

Cost Center No (CC No.)	Name of (As per DBR
31	Rain Water Ha
32	External Wate
33	Sewerage Net
34	Effluent Treat Tank / STP/ETP
35	External Feede
36	Substation (Ou
37	L.T. Distributio
38	Street Lighting
39	D.G. Sets

+

(E2) Campus Development

Cost Center No (CC No.)	Name of (As per DBR
40	Dismantling an
41	Trees Cutting
42	Utilities shifting
43	Boundary Wall
44	Main Ent /Boundary wal
45	Roads (Interna
46	Road (External
47	Culverts
48	Drainage (Su Ground)
49	Landscaping Plantations)
50	Art & Sculptur
51	Parking
52	Play Fields
53	Play Equipmen
54	Statutory appr
55	Miscellaneous Development

TECHNICAL REPORT					
Item No.		**Particular**			**Description**
1		**ADMINISTRATIVE & GENERAL INFORMATION**			
	1.1	**PROJECT NAME :-**			
	1.2	**PROJECT LOCATION :-** (Place and District)			
	1.3	**DEPARTMENT OWNING BUILDING:-**			
	1.4	**PROJECT COST :-**			
	1.5	**PROJECT BUILT UP AREA :-** (In.Sq.m.)			
	1.6	**AUTHORITY COMPETENT TO GRANT ADMINISTRATIVE APPROVAL OF PROJECT :-**			
	1.7	**PROJECT LAND :-**			
		1.7.1	Project Plot/Land Area (in hectares) :-		**0**
		1.7.2	Topography :- (Plain /Undulated) :-		
		1.7.3	Adjoining Landmark (Nalla, Road, Building, Pipe Lines , Electrical Line, private land , Basti , orchord , garden etc) :-		
			1.7.3.1	North Side :-	
			1.7.3.2	South Side :-	
			1.7.3.3	East Side :-	
			1.7.3.4	West Side :-	
		1.7.4	Adjoining Land use and ownership :-		
			1.7.4.1	North Side :-	
			1.7.4.2	South Side :-	
			1.7.4.3	East Side :-	
			1.7.4.4	West Side :-	
		1.7.5	Old/ Existing buildings and other structures in the Project Land:-		
		1.7.6	Distance From Nearest Town/Village:-		
		1.7.7	/Index Plan /Photo / Map / Plan/ Drawing/Bore logs/T.P. :-		
			1.7.7.1	Google index Map (Yes/No):-	
			1.7.7.2	Drone Photo/Video(Yes / No) :-	
			1.7.7.3	Khasra Map (Yes/No):-	
			1.7.7.4	Engineer Survey plan (Yes/No.):-	

			1.7.7.5	Master Plan of Compus (Yes/No.):-	
			1.7.7.6	Concept / Working Drawing (Yes /No.):-	
			1.7.7.7	Borelogs (Yes /No.):-	
2		**EXECUTIVE SUMMARY :-**			
	2.1	**HISTORY OF EXISTING FACILITY AT PROJECT SITE :-** (Add extra sheet , if required to respond in details)			
	2.2	**NECESSITY OF PROJECT :-** (Reason/Justification/Need for proposed construction) (Add extra sheet , if required ,to respond in details)			
3		**(FUNCTIONAL SPACES AND FACILITIES TO BE PROVIDED IN PROJECT)**			
		PARTICULERS			**Functional Spaces & Facilities Provided in Project** (Attach extra column or sheet, if required to respond in detail)
	3.1	**Building Spaces :-** (e.g.Classrooms/ Office/ Reception/ Waiting/lab/ Meeting hall / Computer room/Library/Dormitory/Room/Dinning / Common Room/ Kitchen/ Store /Pantry/ Laundry/ Wards/ OT/OPD/ Labour room/ICCU / Court rooms/ Malkhana/Bed rooms/Balcony/ Suites/ Basement use, If anyetc.) (Add extra sheet , if requried ,to respond in details)			
	3.2	**Internal Special Services :-** (e.g.,HVAC/Lifts/Elevators/Solar Water Heating /Solar Power/Fire Fighting/Fire Detection/ Audio Visual System/Access Control & Low Voltage System/Acoustic & False Ceiling/Interior Decoration & Art Work/Information Communication (IT)/Internal signage/Building Management System (BMS) ,etc)			
	3.3	**Equipment & Furniture:-** (eg. Furnitures/Medical Equipment & Accessories such as (MGPS,OT,Nurse call , PTS Medical/Kitchen / Laundary , etc)			

	3.4	**Campus Services :-** (eg. Rain Water Harvesting/External Water Supply/Sewerage Net Work/Effluent Treatment (Septic Tank / STP/ETP /Bio Digester/External Feeder Line Extension /Substation:- (Outdoor/Indoor)/L.T. Distributions /Street Lighting/D.G. Set, etc):- (Add extra sheet , if required ,to respond in details)	
	3.5	**Campus Development :-** (e.g.,Dismantling and Site Clearing/Trees Cutting /Utilities shifting /Boundary Wall /Main Entrance Gate /Boundary wall Gate / Wicket Gate/security room/Roads (internal Campus)/Road (External Approach)/Culverts /Drainage (Surface / Under Ground)/Landscaping (Horticulture, Plantations)/Art & Sculpture/Parking /Play Fields/Statutory approval fees.,etc):- (Add extra sheet , if required ,to respond in details)	
4		**SALIENT FEATURES OF PROJECT :-** (Add extra column or sheet , if requried for each building)	
	4.1	Built Up Area (Sq.m.) :-	
	4.2	Plinth Area (Sq.m.)	
	4.3	Type of Building Construction (Permanent / Temporary /LGSF / Pre.FabBuilding)	
	4.4	Number of Stories (Floors) in each building :-	
	4.5	Future Expansion Provision if any :-	
		4.5.1 Vertical Number of (Floor)	
		4.5.2 Horizontal (Area)	
	4.6	Concept Design/ Drawing Number :-	

1		**DESIGN BASIS REPORT (DBR)** **(** Describe all physical elements of the project which are basis for inclusion of items in the estimates.) PerpareDBR for each building saperately and only one for external works of the campus.			
		Particulars			**Technical provisions made for each building** (If required attach extra column or sheets, to respond in detail for Building No-1, Building No-2, Building No-3 and BuildingNo-n)
		(I) Civil Works			
	1	**Concrete in R.C.C. Frame Structure up to Plinth Level**			
		1.1	**Unit Quantity of R.C.C. 0.50 Cum/Sq.m. of B.U.A :-** (Cum/Sq.m. of Plinth Area)		
		1.2	**Foundation Details :-**		
			1.2.1	Average Depth of Foundation :- (Mts.)	
			1.2.2	Founding Strata (Hard soil / Moorum / Soft rock / Hard rock /Largeboulders/ sand) & SBC	
			1.2.3	Type :- (Open Column footing / Pile/ Raft/Strip etc.)	
			1.2.4	Grade of Concrete in Footings :- (e.g. M 15 /M 20/ M 25/M30 etc.)	
	2	**Steel in R.C.C. Frame Structure work up to Plinth Level**			
		(Unit Quantity of steel) 75 Kg/ Sq.m. of B.U.A. :- (Kg./Sq.m. of Plinth Area)			
	3	**Civil Work Other Than R.C.C Frame Structure upto plinth Level**			
		3.1	**Basement :-**		
			3.1.1	Area :- (Sqm.)	
			3.1.2	Ceiling Height :- (m)	
			3.1.3	Flooring :- (e.g. CC / Kota/ Tiles etc.)	
			3.1.4	Water proofing, if any :- (e.g. Kota,Membrane,Crystallization of concrete etc.)	
			3.1.5	Water Proofing of Horizontal Surfaces :- (+Ve or - Ve Water Proofing Area)	

			3.1.6	Water Proofing of Vertical Surfaces:- (+Ve or - Ve Water Proofing Area)	
			3.1.7	Construction joint water Proofing:-	
		3.2	**Plinth/Under Floor Filling :-**		
			3.2.1	Average Plinth Height :- (Mts.)	
			3.2.2	Non B.C. Soil filling from available excavated earth. (thickness in mm)	
			3.2.3	Imported hard muram/hard copra filling having MDD not Less than 2.0 t/m ^{3} (thickness in mm)	
			3.2.4	Imported non B.C. Soil having MDD not Less than 1.85 t/m ^{3} (thickness in mm)	
			3.2.5	Cru-sher stone dust (thickness in mm)	
			3.2.6	P.C.C. (M:10 Or M:15) thickness inmm	
			3.2.7	Plinth Protection (Width & thickness in mm)	
		3.3	**Anti Termite Treatment**		
			3.3.1	**Treatment Process :-** (Spreading /Grouting/Piping,etc) Sq.m./Lit./Nos,etc	
			3.3.2	Chemical used (Fipronil/ Lmidaclopride /Cloropyrifos/ Cypermethrin/Bifenthrin,etc):-	
	4	**Concrete in R.C.C. Frame Structure Above Plinth Level**			
		4.1	**Unit Quantity of R.C.C in /Cum./Sq.m. of Built Up Area**		
		4.2	**Super Structure :-**		
			4.2.1	Type :- (e.g. Load Bearing ,RCC Frame etc.)	
			4.2.2	Grade of Concrete :- (e.g. M: 20 , M: 25, M:30 , M:40 etc.)	

	5	**Steel in R.C.C.Frame Structure Above Plinth Level**			
		(Kg./Sq.m. of Built Up Area) Assumed or actual Quantity **)**			
	6	**Civil Work Other Than R.C.C Frame Structure Above Plinth Level**			
		6.1	**Stilt Parking, if any :-**		
			6.1.1	Area :- (Sqm.)	
			6.1.2	Parking Capacity : (No. of car and bike parking)	
			6.1.3	Ceiling Height :- (Mts.)	
			6.1.4	Flooring :- (e.g. CC/ CC tiles/ Endura tiles/ Kota/Stone Flooring/ Paver Block etc.)	
		6.2	**Walls :-**		
			6.2.1	Material :- (e.g. Bricks/Fly Ash Bricks/ AAC Blocks/ ACC panels etc.)	
			6.2.2	Jointing Mortar :- (e.g. C.M. 1:4/1:6 or Adhesive etc.)	
		6.3	**Roof :-** (e.g. Soft roof (AC/ GI/ Acrylic/ PUF/ Galvalumium) Clay tile/ RCC/Prestressed/ Precast		
		6.4	**Inner Plaster :-**		
			6.4.1	Type :- (e.g. Cement sand Mortar/Synthetic polymermodified Mortar /Textured/Calcium silicate etc.)	
			6.4.2	Thickness :- (e.g. 6 mm , 8 mm , 12 mm/ 15 mm ,18 mm etc.)	
			6.4.3	Mix-Ratio :- (e.g. 1:4 /1:6 etc.)	
		6.5	**Outer Plaster :-**		
			6.5.1	Type :- (e.g. Cement sand Mortar/Sand faced/Synthetic polymer modified Mortar Washed stone grit/ Synthetic/Texturedetc.)	

			6.5.2	Thickness :- (e.g. 2-6 mm, 8-10 mm , 12 mm/ 15 mm ,18 mm , 20 mm etc.)	
			6.5.3	Mix-Ratio :- (e.g. 1:4 / 1:6/ as per item etc.)	
			6.5.4	Fibre mesh on external face of outer wall only (5 mm x 5 mm x45 GSM) ifany	
			6.5.5	Application Method (By Machine or Hand)	
		6.6	**Wall Finishing (Inner):-** (e.g. Putty/Distemper/Paint/Texture etc.)		
		6.7	**Finishing (Outer):-** (e.g. Putty/Distemper/ Paint/ Texture/Stone dry Cladding/Stone vet Cladding/ Plain Glazing/ Structural Glazing / Spider Glazing / ACP/Cement Board /sand faced / Solid Surfaceetc.)		
		6.8	**Windows and Ventilators :-**		
			6.8.1	Frames :- (e.g. (Powder Coated/Anodised) Aluminium / UPVC/ MS - Tubular/ Pressed steel frame/ Wooden/ Angle ironetc.)	
			6.8.2	Shutters :- (e.g. Wooden/ (Powder Coated/Anodised) Aluminium/ UPVC/ MS tubular/ Glazed/ Panelled/ steel etc.)	
			6.8.3	Shutter Panels :- (e.g. Glass/ MS sheet/ Cement board/APP board/ Wooden Ply/ACPetc.)	
			6.8.4	Grills :- (e.g. Pipe tubes/ Square Bars/ Flats/ Aluminium etc.)	
			6.8.5	Fly proof mesh :- (Steel /HDPE Net Fabrics)	
			6.8.6	Windows Sill & Jam Finishes (eg. Plaster , stone , etc)	
		6.9	**Doors :-**		
			6.9.1	Frames :- (e.g. MS Angle iron/Wood/Plastic wood /Granite/Concrete frame /M.S. Tublar/UPVC/ Pressed steel/ Aluminium etc.)	

			6.9.2	Shutters :- (e.g. Flush /PVC/ Aluminium/wooden/ MS sheet/ Grill/ Rolling/ Collapsible/ Rolling Grill etc.)	
			6.9.3	Shutter Thickness :- (mm)	
			6.9.4	Door Jam Finishes (eg. Plaster , Stone , etc)	
		6.10	**Flooring Types :-** (e.g. IT/Kota/Ceramic/Vitrified tiles/Granite/ Marble/Wooden/PVC/ Vinyl/Anti-Static Flooring/Mosaic/ Terrazzo/ Sand stone/Cement Tilesetc.)		
			6.10.1	In Circulation Area :- (e.g. Corridor/Reception/ Waiting/ Lift Lobby /Dado etc.)	
			6.10.2	In Rooms and Usable Spaces :- (e.g. Principal /Warden/ classrooms/Dormitory / Laboratory/Multipurpose Hall/Kitchenetc.)	
			6.10.3	Stair Cases :-	
			6.10.4	Toilets :- (e.g. Floor etc.)	
		6.11	**Dado / Skirting :-** (e.g. Kota/Rectified Ceramic Tiles/Vitrified tiles/Ultrathin Vitrified Tiles /Granite/ mosaic Marble/ Sand stone/Cement Tiles etc.)		
			6.11.1	In Circulation Area :- (e.g. Corridor/Reception/ Waiting/ Lift Lobby /Dado etc.)	
			6.11.2	In Rooms and Usable Spaces :- (e.g. Ward/Dormitory / Laboratory/Dining Hall/ Kitchen/O.T. etc.)	
			6.11.3	Staircase :-	
			6.11.4	Toilets :- (e.g. Walls etc.)	
		6.12	**Roof Treatment(Water Proofing) :-**		
			6.12.1	**Treatment Type :-** Grading Concrete/ APP/ Chemical treatment etc.	
			6.12.2	**Water ingress protection :-** (e.g. Gola,Khurra,Drip Course)	
		6.13	**Roof Treatment(Heat insulation) :-**		

			6.13.1	Top of the roof:- (e.g. PUF, Heat Treatment Chemical , Heat reflecting , Tiles Polyurethane foam , Extruded polystyrene board , etc)	
			6.13.2	Under Deck :- (Fiberglass, Mineral wool ,Celluose, Polystyrene Over False Ceiling ,etc)	
		6.14	**Expansion Joints Treatment (if any)**		
			6.14.1	Roof :- (Yes/No)	
			6.14.2	Outer wall side :-(Yes/No)	
			6.14.3	Inner wall Side :-(Yes/No)	
			6.14.4	Intermediate Slab:- (Yes/No)	
			6.14.5	Joint Treatment Type :- (Bitumen mix filler, silicon sealant, Fibre board ,Stainless steel grade 304 plate , Polysulphide filler, Aluminium Strip,Aluminium Extruded sections,etc	
		6.15	**Railing (Stair Case / Porch / Balcony / Terrace :-**		
			6.15.1	Quantity :- (Rm/Sq.m.)	
			6.15.2	Material :- (M.S./Steel/Wooden/Glass/Acrylic	
		6.16	**False Ceiling** **(**e.g.Gypsum ceiling ,Plaster of Paris ceiling , Fiber Mesh ceiling,Wooden ceiling , Glass ceiling, ,Metal ceiling , Metalic sheet ceiling , Mineral ceiling , Synthetic leather or cloth ceiling , Antibacterial ceiling ,PVC ceiling , Aluminum ceiling , Thermocol ceiling, etc)		
		6.17	**False Ceiling Area**		
			6.17.1	Quantity in Sq.m. :-	
			6.17.2	Quantity in % of Built Up Area :-	
		(II) Internal Services			
	7	**Internal Water Supply & Plumbing and Sanitary**			
		7.1	**Sanitary Fittings :-** (e.g. .Premium/Normal/CP/PTMT etc.)		
		7.2	**Water Supply Pipes(Internal):-** (e.g. GI/CPVC/PE-AL-PE etc.)		

		7.3	**Drainage (Internal Soil & Waste water) :-** (e.g. UPVC/ CI/AC/PVC/DWC etc.)	
		7.4	**Drainage (Rain water) :-** (e.g. UPVC/AC/CI/PVC etc.)	
	8	**Internal Electrification**		
		8.1	**POINT Wiring in pvc conduit surface or concealed with pvc insulated copper conductor wire:-** **(e.g. Modular/Normal Flush Type)**	
		8.2	**Distribution Boxes IP43 Powder Coated:-** (e.g. Single Door/ Double door)	
		8.3	**MS Switch Boxes :-** (e.g. Modular/ Non modular)	
		8.4	**Fans:-** (e.g. 5 Star rating Energy saver50 W 1200mm/ Ornamental etc.)	
		8.5	**Tube lights:-** (e.g. T5/ Led type/ CFL type etc.)	
		8.6	**Lumanaries :-** (e.g. CFL/ LED etc.)	
		8.7	**Down lighter(Size & Type) :-** (e.g. T5/Led type/CFL type etc.)	
		8.8	**Electrical Load (KVA) :-**	
		8.9	**UPS & Inverter** (Capacity / Member / KVA)	
		(III) Internal Special Services		
	9	**HVAC**		
		9.1	**Types :-** (e.g. Window/Split/wall mounted/ ceiling mounted cassette,/duct able /VRV/Water Chiller, etc.)	
		9.2	**Name of spaces**:- (e.g. Officers/Judge/OT/ICU/ etc.)	
		9.3	**Area of Spaces Covered (sq.m)**	
		9.4	**Percentage of area covered w.r.t total (Builtup Area (%)**	
		9.5	**HVAC Load** :- (tonnes)	
		9.6	**Electrical Load (KVA)** :-	
	10	**Lifts/Elevators**		
		10.1	**Types :-** (e.g. passenger, Stretcher, Goods etc.)	
		10.2	**Capacity**:- (e.g. No.of Passenger/Stretcher/load)	
		10.3	**Number of Lifts** :-	

		10.4	**Nos. Of Floors** :-	
		10.5	**Speed (metre/Sec)**	
		10.6	**Machine room :-** (machine room less or with machine room)	
		10.7	**Vision Panels (provided or not)**	
	11	**Solar Water Heating**		
		11.1	**Location :-** (Yes/No)	
		11.2	**Capacity:-** (In Litres)	
	12	**Solar Power**		
		12.1	**Location:-**	
		12.2	**Capacity (KVA):-**	
	13	**Fire Fighting**		
		13.1	**Fire Fighting system :-** (Hydrant/wet riser/wet dowlomer,Hozereel , wall mounted/ trolley mounted/fire extinguisher/Gas based fire fighting system, sprinklers etc.)	
	14	**Fire Detection**		
		14.1	**Fire Detection System :-** (Fire detection System(alarming)/Smoke Detectors)	
		14.2	**Spaces Covered by Sprinklers ,if any**	
	15	**Audio Visual System**		
		15.1	Light (Material Type & Make):-	
		15.2	Sound (Material Type & Make):-	
		15.3	Wall Acoustics (Material & Type & Make):-	
	16	**Access Control**		
		16.1	Technical Details (Biometric or Electronic):-	
		16.2	Location Served :-	
	17	**C.C.T.V.**		
		17.1	CC TV system and locations capacity	
		17.2	LAN	
		17.3	LAN System Type	
		17.4	Location Served :-	
	18	**Telephony** **(EPABX, Nurse call System)**		
		18.1	EPABX	
		18.2	System Type	

		18.3	Location Served :-	
		18.4	Nurse Call System :-	
	19	**Acoustic & False Ceiling**		
		19.1	Acoustic Material Type :-	
		19.2	Area (in Sq.m.)	
	20	**Interior Decoration & Art Work**		
		20.1	Material Type:-	
		20.2	Location:-	
	21	**Information Communication (IT)**		
		21.1	Raceway in flooring ,if provided (Yes/No):-	
		21.2	Other Details :-	
	22	**Internal signage**		
	23	**Building Management System (BMS)**		
		23.1	Location :	
		23.2	Technical Details :	
	24	**Miscellaneous Special Services such as Lighting Conductor, Façade light, Water Purifier, Incinerator, if Any**		
		(IV) Equipment & Furniture		
	25	**Furnitures**		
		25.1	Material Type & Make :- (e.g. Steel , wooden, M.S. , Plastic,etc)	
		25.2	Use Type :- (Office, Class Room, Hospital , R.H.,C.H.,Laboratory, Kitchen, Dining, Library, Out door, etc.)	
	26	**Lab Equipment**		
		26.1	Material Type : (e.g. Steel , wooden, M.S. , Plastic,etc)	
		26.2	Type Laboratories/Lab :- (Physics , Chemistry ,Analytical and Quality Laboratories,BiosafetyLaboratories,Clinical and Medical Laboratories,IncubatorLaboratories,Production Laboratories,Research& Development (R&D) Laboratories.etc)	
	27	**Medical Equipment & Accessories ,MGPS,OT,PTS,etc**		
		i	**Medical Gas Pipe Line:-** (Material Type and system)	
		ii	**Operation Theater:-** (e.g. Modular or Non modular)	

		iii	Pneumatic Tube system:-		
		iv	Medical Equipment & Accessories :-		
	28	**Kitchen Equipments**			
	29	**LaundaryEquipments**			
	30	**Computer Hardware/Software**			
		(e.g. Computer , T.V. Video wall etc)			
		Particular			**Technical Provisions for whole campus**
		(E1) Campus Services			
	31	**Rain Water Harvesting**			
		31.1	**Type of Recharge structures** (e.g. Recharge through Abandoned Dug Well /Hand pump/Recharge Pit/Trench/Gravity Head Recharge Tube Well /Recharge Shaft)		
		31.2	**Dia of Pipe** (e.g. 50mm@6.6sq.m/ 75mm @ 20.4 sq.m/ 100mm@42.70sq.m/125mm @80.5 sq.m):-		
		31.3	**Nos. of down Pipe and Location :-**		
		31.4	**Filter Kit for down pipe :-** (for each down pipe or combined for two or more down pipe)		
	32	**External Water Supply**			
		32.1	**Under Ground Storage :-**		
			32.1.1	Size of tank :-	
			32.1.2	Capacity of tank :-	
			32.1.3	Location :- In Campus & Distance	
			32.1.4	Type of Water Proofing :- (e.g. Kota/ Ceramic Tile/ Vitrified Tile or Crystillazation of Concrete etc)	
			32.1.5	Water Proofing :- Phase (+ve or - ve)	
		32.2	**Over Head Storage :-**		
			32.2.1	Size :-	
			32.2.2	Capacity:-	
			32.2.3	Location :-	
			32.2.4	Type of Tank:- (e.g. Sintex/ RCCetc)	

			32.2.5	Type of Water Proofing :- (e.g. Kota/ Ceramic Tile/ Vitrified Tile or Crystillazation of Concrete etc)	
			32.2.6	Water Proofing :- Phase (+ve or - ve)	
		32.3	**Type of water Source :-** (e.g. Tube well /Well/Municipal Pipeline etc.)		
		32.4	**Water Supply Pipes made of :-** (e.g. GI/ Steel/ RCC etc.)		
		32.5	**Distance of water Source (Tapping Point) :-** (mts)		
		32.6	**Submersible Pump Details :-** (e.g. Deep Well , Open Well, Centrifugal etc.)		
	33	**Sewerage Net Work**			
		Drainage Pipe (e.g. SW /RCC / UPVC / CI etc.)			
	34	**Effluent Treatment (Septic Tank / STP/ETP /Bio Digester)**			
		34.1	**Type :-** (Septic Tank/STP/ETP /Bio Digester etc.)		
		34.2	**Technology :-** (MBBR/MBR/SBR /ETU /REED BED / ETP/KIT etc.)		
		34.3	Design User Number and Capacity :- (No.) & Technology Used		
		34.4	Distance of plant from building :- (mts)		
		34.5	Recommended /Provided Size :- KLD		
	35	**External Feeder Line Extension**			
		35.1	**Source Lines:-** (11kv,33kv)		
		35.2	**Distance from source(Tapping Point):-** (km)		
		35.3	**Connecting Feeder Lines:-** (11kv,33kv)		
		35.4	**Poles For Connecting Feeder Line :-** (e.g RS Joint/ H Beam/Rail, Concrateect.)		
	36	**Substation (Outdoor/Indoor)**			
		36.1	**Electrical Load of Campus (KVA):-**		

		36.2	**Substation Type:-** (e.g. conventional type/ Compact type/11kv /33 kv etc.)	
		36.3	**Transformer:-** (Capacity in KVA and NO.)	
		36.4	**Panels:-** (e.g. LT/MV/ Indoor/ Outdoor)	
	37	**L.T. Distributions**		
		37.1	**H.T. / Cables Type & Length :-** (km)	
		37.2	**L.T.Panels Type :-**	
		37.3	**Location for panels :-**	
	38	**Street Lighting**		
		38.1	**Street Light Poles:-** (e,g, Tubular/Octogonal/Decorative)	
		38.2	**Steet Lights Lumanaries :-** (e.g. MH/LED etc.)	
	39	**D.G. Sets**		
		39.1	**Diesel Generator (DG) Set:-** (e.g. AMF/Without AMF)	
		39.2	**DG Capacity and Numbers :-** (Kva)	
		39.3	**DG Emergency Load :-**	
		(E2) Campus Development		**Technical Provisions For Whole Campus**
	40	**Dismantling and Site Clearing**		
	41	**Trees Cutting**		
	42	**Utilities shifting**		
	43	**Boundary Wall**		
		43.1	**Length :-** (mts)	
		43.2	**Height :-** (mts)	
		43.3	**Retaining Wall, If any :-**	
	44	**Main Entrance Gate /Boundary wall Gate / Wicket Gate/Security Room**		
		44.1	**Area :-**	
		44.2	**Height :-** (mts)	
		44.3	**Guard Room :-** (Yes/NO)	
	45	**Roads (Internal Campus)**		

		45.1	**Internal Road :-** (e.g. WBM/CC/BT etc.)		
			45.1.1	Length :- (mts)	
			45.1.2	Width of Carriageway :- (mts)	
			45.1.3	Hard Shoulder Width and Type :- (Murrum/Paver block etc.)	
	46	**Road (External Approach)**			
		46.1	**Approach Road External :-** (e.g. WBM/CC/BT etc.)		
			46.1.1	Length :- (mts)	
			46.1.2	Width of Carriageway :- (mts)	
			46.1.3	Hard Shoulder Width and Type :- (Murrum/Paver block etc.)	
	47	**Culverts**			
		47.1	Length :- (mts)		
		47.2	Width of culvert :- (mts)		
		47.3	Type :(R.C.C/Pipe)		
	48	**Drainage (Surface / Under Ground)**			
		48.1	**Surface Drain (Storm Water) :-** (e.g.Open/Hume pipe/ RCC etc.)		
			48.1.1	Length :- (mts)	
			48.1.2	Size of opening :- (BXD/Dia.)	
		48.2	**Nalla Training :-**		
			48.2.1	Length :- (mts) & size	
			48.2.2	Size :- (BXD/Dia.)	
	49	**Landscaping (Horticulture, Plantations)**			
		49.1	**Hardscaping :-** (e.g. paver block/ Stone Flooring/ Cement Concrete etc.)		
		49.2	**Softscaping :-** (e.g. Soil/Murrum/Grass etc.)		
		49.3	**Types and No.s of Plantation :-** (e.g. Neem/ Gulmmohar/jatropha/karanj etc.)		
		49.4	**Overground :-** (Electricity ,Cables)		
		49.5	**Underground :-** (Sewerage water supply, Fibre Optics		
	50	**Art & Sculpture**			
		50.1	**Material:-**		
		50.2	**Locations / Area :-**		

	51	**Parking**			
		51.1	**Open**		
			51.1.1	Four Wheeler (Nos) :-	
			51.1.2	Two Wheeler (Nos) :-	
		51.2	**Underground/Covered :-**		
			51.2.1	Four Wheeler (Nos) :-	
			51.2.2	Two Wheeler (Nos) :-	
	52	**Play Fields**			
		(names of sports and type of play fields)			
	53	**Play Equipment**			
		(Type of Equipment)			
	54	**Statutory approval fees.**			
		(For GRIHA Rating Building Permission, Environmental Clearance etc.)			
	55	**Miscellaneous Campus Development**			
6		**SCHEDULE OF FINISHES :-** (Attached Yes/No):-			
		6.1	Civil Work (Format SOF- Civil Attached Yes/No):-):-		
		6.2	Toilet Work (Format SOF- Toilet Attached Yes/No):-):-		
		6.3	Electrical Work (Format SOF-Electrical Attached Yes/No):-):-		
7		**PROGRAM CHART:-**			
		7.1	CHART "P1" (Project Programme Chart) Attached Yes/No :-		
		7.2	CHART "P2" (P.C.C. Programme Chart) Attached Yes/No (Optional) :-		
8		**REVISED COST CHART , IF APPLICABLE**			
		8.1	Revised Cost Chart (Format RA-B,RA-C,RA-D Attached Yes/No):-		
9		**RATES ADOPTED:-** (MPPWD SOR, Non SOR etc.)			
10		**STANDARDS APPLICABLE :-** (e.g .NBC, IPHS, MCI, MP Bhumi Vikas Adhiniyam,Kendriya Vidhyalaya certification, AICTE,GAD,AAI,ISHRAE,PCB,NEC, GRIHA, AERB etc.)			

11		**SPECIFICATIONS :-** (CPWD,BIS,MPPWD SOR etc.)	
12		**GREEN BUILDING RATING ,IF TARGETED** (Yes / No/ Rating Grade)	

Consultants	Signature :-	
	Date :-	
	Name :-	
	Consultant:-	
	Company :-	
	Phone No. :-	
	Email Id:-	
	Consultant Empanelment No :-	
Department	Signature :-	
	Date :-	
	Name :-	
	Designation :-	
	Department :-	

FORMAT -B

(One for WholeProject)

Name of Project :-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.
Project Builtup Area (in sq.m.)	∑ Total BU
Total Campus Area (in Hectare) :-	0

PROJECT ABSTRACT

(SUM OF ALL MAJOR COST CENTER)

All Amount in Rs.

Project Sub Head	Project Cost Center (PCC)		Cost			Project Cost Center Values				Referance Format
	Project Cost Center No. (PCC No.)	(Name of Building Work or External Work)	SOR	NON SOR	Total	Unit (Such as Number /Sq.M./ Hectares/ M)	Quantity (Such as Counted Number, Built Up Area or Campus Area or Length)	Per Unit Cost	Apportioned cost (% of Net Project cost)	
1	2	3	4	5	6	7	8	9	10	11
Building Work	B1	Building No. 1 :-					0			Format C - 1
	B2	Building No. 2 :-								Format C - 2
	BN	Building No. N :-								Format C -N
		Sub Total (B1 +B2 +.....BN) (All BuildingWorks)	**Σ above**	**Σ above**	**Σ above**		**Σ Total BU**	**Average (per sqm.)**	**Σ above %**	
External Works	E1	Campus Services					0			Format D - 1
	E2	Campus Development					0			Format D - 1
		Sub Total (E1+E2) (All External Works)	**Σ above**	**Σ above**	**Σ above**			**Average (per Hectare)**	**Σ above %**	
		NET PROJECT COST (B1 +B2 + ... BN + E1+E2)	**Σ above**	**Σ above**	**Σ above**				100%	
		Add Supervision Charges @.....								
		GRAND TOTAL								
		Add GST Charges @.....								
		PROJECT COST								

Consultants	Signature :-	
	Date :-	
	Name :-	
	Designation :-	
	Consultant :-	
	Phone No. :-	
	Email Id:-	
	Consultant Empanelment No :-	
Department	Signature :-	
	Date :-	
	Name :-	
	Designation :-	
	Department :-	

FORMAT-C

(Such as C1, C2…..Cn For Each "Building Works" Or "Project Cost Center")

Name of Project:-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.	
Project Cost Center (As per Column 2, 3 & 8 of Format - B)	PCC No. (As per Column 2 of Format B)	B1
	Name of Project Cost Center (PCC) (As per Column 8 of Format B)	Building No. 1 :- …….
	Built Up Area of Building (in Sqm.) :-	0

Building Work Abstract

(One For each "Building Works" or "Project Cost Center")

Building Sub Heads	Name of Cost Center (As per DBR In Format "A")		Cost (INR)			Cost Center Values				ReferanceFormat
	Cost Center No (CC No.)	Building Work	SOR	NON SOR	Total	Unit (Sq.m./Cum/ M/Nos/KLD/ Hectares/Lump Sum)	Quantity	Per Unit Cost	Apportioned Cost (% of Total Cost of PCC)	
1	2	3	4	5	6	7	8	9	10	11
(I) Civil Works	1	Concrete in R.C.C. Frame Structure up to Plinth Level								Format -E
	2	Steel in R.C.C. Frame Structure work up to Plinth Level								Format -E
	3	Civil Work Other Than R.C.C Frame Structure upto plinth Level								Format -E
	4	Concrete in R.C.C. Frame Structure Above Plinth Level								Format -E
	5	Steel in R.C.C.Frame Structure Above Plinth Level								Format -E
	6	Civil Work Other Than R.C.C Frame Structure Above Plinth Level								Format -E
	Sub Total	**(I) Civil Works**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Average (per sqm.)**	**Σ above %**	
(II) Internal Services	7	Internal Water Supply & Plumbing and Sanitary								Format -E
	8	Internal Electrification								Format -E
	Sub Total	**(II) Internal Services**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Average (per sqm.)**	**Σ above %**	
(III) Internal Special Services	9	HVAC								Format -E
	10	Lifts/Elevators								Format -E
	11	Solar Water Heating								Format -E
	12	Solar Power								Format -E
	13	Fire Fighting								Format -E
	14	Fire Detection								Format -E
	15	Audio Visual System								Format -E
	16	Access Control								Format -E
	17	C.C.T.V.								Format -E
	18	Telephony (EPABX, Nurse call System)								Format -E
	19	Acoustic & False Ceiling								Format -E
	20	Interior Decoration & Art Work								Format -E
	21	Information Communication (IT)								Format -E
	22	Internal signage								Format -E
	23	Building Management System (BMS)								Format -E
	24	Miscellaneous Special Services such as Lighting Conductor, Façade light, Water Purifier, Incinerator, if Any								Format -E

FORMAT-

(Such as C1, C2.....Cn For Each "Building Works" Or "Project Cost Center")

Name of Project:-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.	
Project Cost Center (As per Column 2, 3 & 8 of Format - B)	PCC No. (As per Column 2 of Format B)	B1
	Name of Project Cost Center (PCC) (As per Column 8 of Format B)	Building No. 1 :- .....
	Built Up Area of Building (in Sqm.) :-	0

Building Work Abstract

(One For each "Building Works" or "Project Cost Center")

Building Sub Heads	Name of Cost Center (As per DBR In Format "A")		Cost (INR)			Cost Center Values				Referance Format
	Cost Center No (CC No.)	Building Work	SOR	NON SOR	Total	Unit (Sq.m./Cum/ M/Nos/KLD/ Hectares/Lump Sum)	Quantity	Per Unit Cost	Apportioned Cost (% of Total Cost of PCC)	
1	2	3	4	5	6	7	8	9	10	11
	Sub Total	**(III) Internal Special Services**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Average (per sqm.)**	**Σ above %**	
(IV) Equipment & Furniture	25	Furnitures								Format -E
	26	Lab Equipment								Format -E
	27	Medical Equipment & Accessories ,MGPS,OT,PTS,etc								Format -E
	i	MGPS								
	ii	OT (High End/Low End)								
	iii	PTS (Pneumatic Tube System)								
	iv	Medical Equipment & Accessories								
	28	Kitchen Equipments								Format -E
	29	LaundaryEquipments								Format -E
	30	Computer Hardware/Software								Format -E
Sub Total IV :- (Equipment & Furniture)			**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Average (per sqm.)**	**Σ above %**	
Cost Center Cost (Sub Total I+II+III+IV) :-			**Σ above**	**Σ above**	**Σ above**					
Consultants	Signature :-									
	Date :-									
	Name :-									
	Designation :-									
	Consultant :-									
	Phone No. :-									
	Email Id:-									
	Consultant Empanelment No :-									
Department	Signature :-									
	Date :-									
	Name :-									
	Designation :-									
	Department :-									

FORMAT-D
(OneforAllExternalWorks)

Name of Project:-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.
Campus Area (in Hect.) As per column 8 Format B :-	0

External Work Abstract

(One for All External Works)

External Sub Head	Name of Cost Center (As per DBR In Format "A")		Cost (INR)			Cost Center Values				ReferanceFormate
	Cost Center No (CC No.)	External Works	SOR	NON-SOR	Total	Unit (Sq.m./Cum/ M/Nos/KLD/ Hectares/Lump Sum)	Quantity	Per Unit Cost	Apportioned Cost (% of Total Cost of PCC)	
1	2	3	4	5	6	7	8	9	10	11
(E1) Campus Services	31	Rain Water Harvesting								Format -E
	32	External Water Supply								Format -E
	33	Sewerage Net Work								Format -E
	34	Effluent Treatment (Septic Tank / STP/ETP /Bio Digester)								Format -E
	35	External Feeder Line Extension								Format -E
	36	Substation (Outdoor/Indoor)								Format -E
	37	L.T. Distributions								Format -E
	38	Street Lighting								Format -E
	39	D.G. Sets								Format -E
	Sub Total	**(E1) Campus Services**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Average (per sqm.)**	**Σ above %**	
(E2) Campus Development	40	Dismantling and Site Clearing								Format -E
	41	Trees Cutting								Format -E
	42	Utilities shifting								Format -E
	43	Boundary Wall								Format -E
	44	Main Entrance Gate /Boundary wall Gate / Wicket Gate/Security Room								Format -E
	45	Roads (Internal Campus)								Format -E
	46	Road (External Approach)								Format -E
	47	Culverts								Format -E
	48	Drainage (Surface / Under Ground)								Format -E
	49	Landscaping (Horticulture, Plantations)								Format -E
	50	Art & Sculpture								Format -E
	51	Parking								Format -E
	52	Play Fields								Format -E
	53	Play Equipment								Format -E
	54	Statutory approval fees.								Format -E
	55	Miscellaneous Campus Development								Format -E
	Sub Total	**(E2) Campus Development**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Average (per sqm.)**	**Σ above %**	
All External work Total E1+ E2 :- (External Services + Campus Devlopment)			**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Σ above**	**Average (per sqm.)**	**Σ above %**	

Consultants	Signature :-									
	Date :-									
	Name :-									
	Designation :-									
	Consultant :-									
	Phone No. :-									
	Email Id:-									
	Consultant Empanelment No :-									
Department	Signature :-									
	Date :-									
	Name :-									
	Designation :-									
	Department :-									

RMAT -E
(Abstract of cost)

Name of Project:-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.
S.No. and Name of Project cost center as per column 2&3 of Format -B	
S.No.and Name of Cost Center As per Column 2 & 3 of Format C or Format D :-	
Note (1) :-	Prepare Format E for each Cost Center separately and fill up value in Format C & Format D
Note (2) :-	Description of item in Format E (Abstract of Cost) is optional . The same can be hidden.

Cost Abstract
(Separate for each "Cost Center")

All Amount in Rs.

S.No.	SOR Item NO.	SOR sub item No, if any	Item Name	Description of Item	Unit	Quantity	Rate	Amount
1	2	3	4	5	6	7	8	9
SOR ITEMS								
1								
2								
n								
							SUB TOTAL SOR ITEMS	
NON-SOR ITEMS								
1								
2								
n								
							SUB TOTAL NON SOR ITEMS	
							TOTAL SOR + NON SOR ITEMS	

Consultants	Signature :-		
	Date :-		
	Name :-		
	Designation :-		
	Consultant :-		
	Phone No. :-		
	Email Id:-		
	Consultant Empanelment No :-		
Department	Signature :-		
	Date :-		
	Name :-		
	Designation :-		
	Department :-		

FORMAT -F
(Quantity Sheet)

Name of Project:-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.
S.No. and Name of Project Cost Center as per column 2 and 3 of format -B	
S.No. and Name of Cost Center as per Column 2 & 3 of 'Format C' or 'Format D ':-	

Quantity Sheet

(Separate in Support of each Format 'F ')

S.No.	SOR ITEM NO.	SOR sub item No, if any	Items Name	Description of Item	Unit	Nos	Length L	Breadth B	Depth D	Quantity
1	2	3	4	5	6	7	8	9	10	11
SOR ITEMS										
1										
2										
n										
NON-SOR ITEMS										
1										
2										
n										

Consultants	Signature :-	
	Date :-	
	Name :-	
	Designation :-	
	Consultant :-	
	Phone No. :-	
	Email Id:-	
	Consultant Empanelment No :-	
Department	Signature :-	
	Date :-	
	Name :-	
	Designation :-	
	Department :-	

Σ

FORMAT -T1
(One for Whole Project)

Name of Project:-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.
Project Builtup Area (in sq.m.)	
Total Campus Area (in Hectare)	

(TECHNICAL SANCTION DETAILS)

All Amount in Rs.

S.No.	Description	Amount SOR	Amount NON SOR	Amount TOTAL	T/S No. & date	Reference Format
1	2	3	4	5	6	7
1	Civil Work of Project :-					Format T2
2	Electrical Work of Project :-					Format T3
	Project Cost (Net) :-	**Σ above**	**Σ above**	**Σ above**		
	AddSupervisioncharges ……..% (DoesnotrequiredT.S.)					
	Sub Total :-	**Σ above**	**Σ above**	**Σ above**		
	AddG.S.T.....% (DoesnotrequiredT.S.)					
	Project Cost Proposed for AA (in Rs.)	**Σ above**	**Σ above**	**Σ above**		
	Project Cost Proposed for AA (in Rs. Lakh)					

						FORMAT -T2 **(One for Whole Project**
Name of Project:-	**CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.**					
Total Builtup Area (Sq.m) :-		**Total Campus Area (Hect) :-**				
TECHNICAL SANCTION OF CIVIL WORKS (TO BE ACCORDED BY THE COMPETENT AUTHORITY FOR TECHNICAL SANCTION OF CIVIL WORKS)						
S.No.	**Sub item no.**	**Items**	**Amount SOR**	**Approved Amount NON SOR**	**Amount TOTAL**	**Format**
1	2	3	4	5	6	8
	CIVIL WORKS OF BUILDINGS (Repeat information of sub item no. I, II , III & IV for each building or group of buildings)					
1	**Name of buildings No.1:-…….**					
	I	**Civil Works** :-				
		1(R.C.C. Frame Structure up to plinth level)+ **2** (Steel for R.C.C. Frame Structure work up to Plinth Level)+ **3**(Civil Work Other Than R.C.C Frame Structure upto plinth Level)+ **4**(Concrete in R.C.C. Frame Structure Above Plinth Level)+ **5**(Steel for R.C.C. Frame Structure Above Plinth Level)+ **6**(Civil Work Other Than R.C.C Frame Structure Above Plinth Level)				Format C - 1
	II	**Internal Services** :-				
		7 (Internal Water Supply & Plumbing and Sanitary)				Format C - 1
	III	**Special Services :-**				
		13(Fire Fighting)+ **19**(Acoustic & False Ceiling)+ **20**(Interior Decoration & Art Work)				Format C - 1
	IV	**Equipment & Furniture**				
		25 (Furnitures)				Format C - 1
		Sub Total Amount :- I+II+III+IV	**∑above**	**∑above**	**∑ above**	
2	**Name of buildings No.2:-…….**					
	I	Civil Works :-				Format C - 2
	II	Services :-				Format C - 2
	III	Special Services :-				Format C - 2
	IV	Equipment & Furniture				Format C - 2
		Sub Total Amount :- I+II+III+IV	**∑above**	**∑above**	**∑ above**	
3	**Name of buildings No.N:-…….**					

	I	Civil Works :-				Format C - N
						Format C - N
	II	Services :-				Format C - N
	III	Special Services :-				Format C - N
	IV	Equipment & Furniture				Format C - N
		Sub Total Amount :- I+II+III+IV	**∑ above**	**∑ above**	**∑ above**	
		Sub Total Amount Building No. 1+2+……N :-				
		CIVIL WORKS OF EXTERNALWORKS (ONLY ONE FOR WHOLE PROJECT)				
	V	**External Services :-**				
		31(Rain Water Harvesting)+ **32**(External Water Supply)+ **33**(Sewerage Net Work)+ **34**(Effluent Treatment (Septic Tank / STP/ETP /Bio Digester)				Format D- 1
	VI	**Campus Development :-**				
		40(Dismantling and Site Clearing))+ **41**(Trees Cutting)+ **42**(Utilities shifting)+ **43**(Boundary Wall)+ **44**(Main Entrance Gate /Boundary wall Gate / Wicket Gate/Security Room)+ **45**(Internal Campus Road)+ **46**(External Approach Road)+ **47**(Culverts)+ **48**(Drainage (Surface / Under Ground)+ **49**(Landscaping (Horticulture, Plantations)+ **50**(Art & Sculpture)+ **51**(Parking)+ **52**(Play Fields)+ **53**(Play Equipment)+ **54** (Statutory approval fees. Civil)+ **55**(Miscellaneous Campus Development)				Format D - 1
		Sub Total Amount:-				
		Sub Total	**∑above**	**∑above**	**∑above**	
		TOTAL CIVIL WORK T2				
		Amount of Technical Sanction of civil works				
		SOR (Rs)……				
		NON SOR (Rs)……				
		(Signature & Seal)				
		Competent Authority For Technical Sanction of Civil Works				

FORMAT -T3
(One for WholeProject

Name of Project:-	**CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.**		
Project Builtup Area (In sq.m)		**Project Campus Area (In Hect)**	

TECHNICAL SANCTIONS OF ELECTRICAL WORKS
(TO BE ACCORDED BY THE COMPETENT AUTHORITY TECHNICAL SANCTION OF E/M WORKS)

S.No.	Sub item no	Items	Amount SOR	Approved Amount NONSOR	Amount TOTAL	Format	Page No.
1	2	3	4	5	6	8	9
	ELECTRICAL WORKS OF BUILDINGS (Repeat information of sub item no. I, II , III & IV for each buildings)						
1	**Name of building No.1:-……**						
	II	**Internal Services :-**					
		8 (Internal Electrification)				Format C - 1	
	III	**Special Services :-**					
		9(HVAC) + **10**(Lifts/Elevators)+ **11**(Solar Water Heating)+ **12**(Solar Power)+ **14**(Fire Detection + **15**(Audio Visual System)+ **17**(C.C.T.V.)+ **18**(Telephony (EPABX, Nurse call System)+ **21**(Information Communication (IT)+ **22**(Internal signage)+ **23**(Building Management System (BMS) + **24** (Miscellaneous Special Services such as Lighting Conductor, Water Purifier, Incinerator, if Any)				Format C - 1	
	IV	**Equipment & Furniture**					
		26(Lab Equipment)+ **27**(Medical Equipment & Accessories ,MGPS,OT,PTS,etci+ii+iii+iv)+ **28**(Kitchen Equipments)+ **29** (Laundary Equipments)+ **30** (Computer Hardware/Software)				Format C - 1	
		Sub Total Amount :	**∑ above**	**∑ above**	**∑ above**		
2	**Name of buildingNo.1..... 2**						
	II	Services :-				Format C - 2	
	III	Special Service :-				Format C - 2	
	IV	Equipment &Furniture				Format C - 2	
		Sub Total Amount :	**∑ above**	**∑ above**	**∑ above**		
3	**Name ofbuildingNo.1..... N**						
	II	Services :-				Format C - N	
	III	Special Service :-				Format C - N	
	IV	Equipment & Furniture				Format C - N	
		Sub Total Amount :	**∑ above**	**∑ above**	**∑ above**		
		Sub Total Amount Building or group of building No. 1+2+……N :-					

	ELECTRICAL WORKS OF EXTERNAL WORKS (ONLY ONE FOR WHOLE PROJECT)						
	V	**External Service of Campus :-**					
		35(External Feeder Line Extension)+ **36**(Substation (Outdoor/Indoor)+ **37** (L.T. Distributions)+ **38**(Street Lighting)+ **39**(D.G. Sets) (Add all E/M Works)				Format D- 1	
	VI	**Campus Development :-**					
		54 (Statutory approval fees. Electrical)				Format D- 1	
		Sub Total	**∑ above**	**∑ above**	**∑ above**		
		TOTAL ELECTRICAL WORK T3					
		Amount of Technical Sanction of E/M works					
		SOR (Rs)……					
		NON SOR (Rs)……					
		(Signature & Seal)					
		Competent Authority For Technical Sanction of E/M Works					

PROJECT SCHEDULE-Q
(One for Each PCC)

Name of Project:-	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.	
Project Cost Center (As per Column 2, 3 & 8 of Format -B)	PCC No. As per Column 2 of Format B	BN
	Name of "Project Cost Center" (PCC) (As per Column 8 of Format B	Building No. N :-……
	Built Up Area of Building (in Sqm.) :-	0

BILL OF QUANTITIES (BOQ)

All Amount in Rs.

Unique BOQ No.	Item No.	Sub Item No.	Items Name	Description of Item	Quantity	Unit	Rate	Amount
1	2		3	4	5	6	7	8
CIVIL WORKS SOR ITEMS (Q1)								
1								
2								
3								
n1								
							Q1=SUBTOTALCIVILWORKS(SORITEMS)	
CIVIL WORKS NON-SOR ITEMS (Q2)								
n1+1								
n1+2								
n1+3								
n1+n=n2								
							Q2=SUBTOTALCIVILWORKS(NONSORITEMS)	
							Q1+Q2=TOTALCIVILWORKS(SOR+NONSOR)	
ELECTRICAL & MECHANICAL WORKS SOR ITEMS (Q3)								
n2+1								
n2+2								
n2+3								
n2+n=n3								
							Q3=SUBTOTALELECTRICAL&MECHANICALWORKS(SORITEMS)	
ELECTRICAL & MECHANICAL WORKS NON-SOR ITEMS (Q4)								
n3+1								
n3+2								
n3+3								
n3+n=n4								
							Q4=SUBTOTALSELECTRICAL&MECHANICALWORKS(NONSORITEM)	
							Q3+Q4=TOTALELECTRICAL&MECHANICALWORKS(SOR+NONSOR)	
							Q = (Q1+Q2)+(Q3+Q4) GRAND TOTAL (=PAC):-	

Consultants	Signature :-	
	Date :-	
	Name :-	
	Consultant:-	
	Company :-	
	Phone No. :-	
	Email Id:-	
	Consultant Empanelment No :-	
Department	Signature :-	
	Date :-	
	Name :-	
	Designation :-	
	Department :-	

CHART "P1" (Project Program Chart)

Name of Project/Department Name/built-up Area Sq.m. / Cost in Lakhs			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																	
Stipulated Period in Month	……	Start Date (Zero date)	……/……/…..	Stipulated Target Date		……/……/ …..		Revised Target Date		……/……/ …..		Revised period in Month		……/……/… ..		Date of Achievement		……/……/… ..		
Project Sub Head	S.No.	Project Cost Center	Target \\Achievement \\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks
			Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.	
Building Works	1	Building No.-1																		
			\\\\\																	
			\\\\\\\\\\\\\\\\ \\\\\																	
	2	Building No.-2																		
			//////////////// ////																	
			\\\\\\\\\\\\\\\\ \\\\\																	
	3	Building No.-N																		
			//////////////// ////																	
			\\\\\\\\\\\\\\\\ \\\\\																		
External work	4	External Services																		
			//////////////// ////																	
			\\\\\\\\\\\\\\\\ \\\\\																		
	5	Campus Development																		
			//////////////// ////																	
			\\\\\\\\\\\\\\\\ \\\\\																		

CHART "P2" (Project Cost Center Programme Chart)

Name of Project / Department Name / Built up Area Sq.m. / Cost inLakhs			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																		
Stipulated Period inMonth		Start Date (Zero date)	...../...../.....	Stipulated Traget Date	...../...../.....			Revised TargetDate	...../...../.....			Revised period inMonth	...../...../.....			Date of Achievement	...../...../.....				
Building Sub Heads	S. No.	Description	Target Achievement \\\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)					Remarks
			Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.		
(I) Civil Works	1		Concrete in R.C.C. Frame Structure up to Plinth Level																		
		i	Excavation in Foundation	//////////////																	
				\\\\\\\\\\\\\\																	
		ii	Footing	//////////////																	
				\\\\\\\\\\\\\\																	
		iii	R.C.C. Work	//////////////																	
				\\\\\\\\\\\\\\																	
	2		Steel in R.C.C. Frame Structure work up to Plinth Level	//////////////																	
				\\\\\\\\\\\\\\																	
	3		Civil Work Other Than R.C.C Frame Structure upto plinth Level	//////////////																	
				\\\\\\\\\\\\\\																	
	4		Concrete in R.C.C. Frame StructureAbove Plinth Level																		
		i	R.C.C. Work	//////////////																	
				\\\\\\\\\\\\\\																	
		ii	Brick Work	//////////////																	
				\\\\\\\\\\\\\\																	
		iii	Door	//////////////																	
Name of Project / Department Name / Built up Area Sq.m. / Cost inLakhs			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																		

Stipulated Period inMonth			Start Date (Zero date)	...../...../.....	Stipulated Traget Date	...../...../.....	Revised TargetDate	...../...../.....	Revised period inMonth	...../...../.....	Date of Achievement	...../...../.....	
Building Sub Heads	S. No.		Description	Target Achievement \\\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks
				Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.	
				\\\\\\\\\\\\\\\																	
		iv	Windows	///////////////																	
				\\\\\\\\\\\\\\\																	
		v	Plastering Work	///////////////																	
				\\\\\\\\\\\\\\\																	
		vi	Flooring Work	///////////////																	
				\\\\\\\\\\\\\\\																	
		vii	False Ceiling Work	///////////////																	
				\\\\\\\\\\\\\\\																	
		viii	Painting Work	///////////////																	
				\\\\\\\\\\\\\\\																	
	5		Steel in R.C.C.Frame Structure Above Plinth Level	///////////////																	
				\\\\\\\\\\\\\\\																	
	6		Civil Work Other Than R.C.C Frame Structure Above Plinth Level	///////////////																	
				\\\\\\\\\\\\\\\																	
(II) Internal Services	7		Internal Water Supply & Plumbing and Sanitary																		
		i	Water Supply Line	///////////////																	
				\\\\\\\\\\\\\\\																	
		ii	Plumbing Work	///////////////																	

Name of Project / Department Name / Built up Area Sq.m. / Cost inLakh:			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																			
Stipulated Period inMonth			Start Date (Zero date)	...../...../.....	Stipulated Traget Date		...../...../.....		Revised TargetDate		...../...../.....		Revised period inMonth		...../...../.....		Date of Achievement		...../...../.....			
Building Sub Heads	S. No.		Description	Target Achievement \\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks	
				Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.		
				\\\\\\\\\\\\\\\\																		
		iii	Sanitary Fittings	////////////////																		
				\\\\\\\\\\\\\\\\																		
	8		Internal Electrification																			
		i	Electrical Line (Open & Concealed)	////////////////																		
				\\\\\\\\\\\\\\\\																		
		ii	Electrical Fixture	////////////////																		
				\\\\\\\\\\\\\\\\																		
(III) Special Services	9		HVAC	////////////////																		
				\\\\\\\\\\\\\\\\																		
	10.00		Lifts/Elevators	////////////////																		
				\\\\\\\\\\\\\\\\																		
	11		Solar Water Heating	////////////////																		
				\\\\\\\\\\\\\\\\																		
	12		Solar Power	////////////////																		
				\\\\\\\\\\\\\\\\																		
	13		Fire Fighting	////////////////																		
				\\\\\\\\\\\\\\\\																		
	14		Fire Detection	////////////////																		
				\\\\\\\\\\\\\\\\																		
	15		Audio Visual System	////////////////																		
				\\\\\\\\\\\\\\\\																		

Name of Project / Department Name / Built up Area Sq.m. / Cost inLakh:			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																			
Stipulated Period inMonth		Start Date (Zero date)	../...../.....	Stipulated Traget Date		../...../.....		Revised Targe Date		...../...../.....		Revised period In Month		...../...../.....		Date of Achievement		../...../.....				
Building Sub Heads	S. No.	Description	Target Achievement \\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks		
			Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.			
	16	Access Control	////////////////																			
			\\\\\\\\\\\\\\\\																			
	17	C.C.T.V.	////////////////																			
			\\\\\\\\\\\\\\\\																			
	18	Telephony (EPABX, Nurse call System)	////////////////																			
			\\\\\\\\\\\\\\\\																			
	19	Acoustic & False Ceiling	////////////////																			
			\\\\\\\\\\\\\\\\																			
	20	Interior Decoration & Art Work	////////////////																			
			\\\\\\\\\\\\\\\\																			
	21	Information Communication (IT)	////////////////																			
			\\\\\\\\\\\\\\\\																			
	22	Internal signage	////////////////																			
			\\\\\\\\\\\\\\\\																			
	23	Building Management System (BMS)	////////////////																			
			\\\\\\\\\\\\\\\\																			

Name of Project / Department Name / Built up Area Sq.m. / Cost inLakh:			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																		
Stipulated Period inMonth			Start Date (Zero date)	...../...../.....	Stipulated Traget Date		...../...../.....		Revised TargetDate		...../...../.....		Revise perio inMonth		...../...../.....		Date o Achievemen		...../...../.....		
Building Sub Heads	S. No.		Description	Target Achievement \\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks
				Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.	
	24		Miscellaneous Special Services such as Lighting Conductor, Façade light, Water Purifier, Incinerator, ifAny	///////////////																	
				\\\\\\\\\\\\\\\																	
(IV) Equipment & Furniture	25		Furnitures	///////////////																	
				\\\\\\\\\\\\\\\																	
	26		Lab Equipment	///////////////																	
				\\\\\\\\\\\\\\\																	
	27		Medical Equipment & Accessories ,MGPS,OT,PTS,etc	///////////////																	
				\\\\\\\\\\\\\\\																	
	28		Kitchen Equipments	///////////////																	
				\\\\\\\\\\\\\\\																	
	30		Computer Hardware/Software	///////////////																	
				\\\\\\\\\\\\\\\																	
(V) External Services	31		Rain Water Harvesting	///////////////																	
				\\\\\\\\\\\\\\\																	
	32		External Water Supply	///////////////																	
				\\\\\\\\\\\\\\\																	
	33		Sewerage Net Work	///////////////																	

Name of Project / Department Name / Built up Area Sq.m. / Cost inLakhs			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																		
Stipulated Period inMonth			Start Date (Zero date)	...../...../.....		Stipulated Traget Date		...../...../.....		Revised TargetDate		...../...../.....		Revised period inMonth		...../...../.....		Date of Achievemen		...../...../.....		
Building Sub Heads	S. No.		Description	Target Achievement \\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks
				Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.	
				\\\\\\\\\\\\\\\\																	
	34		Effluent Treatment (Septic Tank / STP/ETP /Bio Digester)	////////////////																	
				\\\\\\\\\\\\\\\\																	
	35		External Feeder Line Extension	////////////////																	
				\\\\\\\\\\\\\\\\																	
	36		Substation (Outdoor/Indoor)	////////////////																	
				\\\\\\\\\\\\\\\\																	
	37		L.T. Distributions	////////////////																	
				\\\\\\\\\\\\\\\\																	
	38		Street Lighting	////////////////																	
				\\\\\\\\\\\\\\\\																	
	39		D.G. Sets	////////////////																	
				\\\\\\\\\\\\\\\\																	
(V) Campus Developme nt	40		Dismantling and Site Clearing	////////////////																	
				\\\\\\\\\\\\\\\\																	
	41		Trees Cutting	////////////////																	
				\\\\\\\\\\\\\\\\																	
	42		Utilities shifting	////////////////																	
				\\\\\\\\\\\\\\\\																	

Name of Project / Department Name / Built up Area Sq.m. / Cost inLakh:			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																		
Stipulated Period inMonth		Start Date (Zero date)	...../...../.....	Stipulated Traget Date		...../...../.....		Revised TargetDate		...../...../.....		Revise period inMonth		...../...../.....		Date of Achievemen		...../...../.....			
Building Sub Heads	S. No.	Description	Target Achievement \\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks	
			Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.		
	43	Boundary Wall	//////////////																		
			\\\\\\\\\\\\\\\																		
	44	Main Entrance Gate /Boundary wall Gate / Wicket Gate/Security Room	//////////////																		
			\\\\\\\\\\\\\\\																		
	45	Roads (Internal Campus)	//////////////																		
			\\\\\\\\\\\\\\\																		
	46	Road (External Approach)	//////////////																		
			\\\\\\\\\\\\\\\																		
	47	Culverts	//////////////																		
			\\\\\\\\\\\\\\\																		
	48	Drainage (Surface / Under Ground)	//////////////																		
			\\\\\\\\\\\\\\\																		
	49	Landscaping (Horticulture, Plantations)	//////////////																		
			\\\\\\\\\\\\\\\																		
	50	Art & Sculpture	//////////////																		
			\\\\\\\\\\\\\\\																		
	51	Parking	//////////////																		
			\\\\\\\\\\\\\\\																		

Name of Project / Department Name / Built up Area Sq.m. / Cost inLakh:			CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.																		
Stipulated Period inMonth			Start Date (Zero date)	...../...../.....	Stipulated Traget Date		...../...../.....		Revised TargetDate		...../...../.....		Revise period inMonth		...../...../.....		Date of Achievemen		...../...../.....		
Building Sub Heads	S. No.		Description	Target Achievement \\\\\\\	Milestone :- (M-I)				Milestone :- (M-II)				Milestone :- (M-III)				Milestone :- (M-IV)				Remarks
				Period Elapsed from Start date Zero date	1/16.	2/16.	3/16.	4/16.	5/16.	6/16.	7/16.	8/16.	9/16.	10/16.	11/16.	12/16.	13/.16.	14/16.	15/16.	16/.16.	
	53		Play Equipment	///////////////																	
				\\\\\\\\\\\\\\\																	
	54		Statutory approval fees.	///////////////																	
				\\\\\\\\\\\\\\\																	
	55		Miscellaneous Campus Development	///////////////																	
				\\\\\\\\\\\\\\\																	

(FORMAT -'RA'-B)

REVISED COST CHART							
Name of Project / Department Name / Built up Area Sq.m. / Cost in Lakhs							
Sanction Project Cost (in Lacs)							
Project cost (As Per Tender (± %) Cost)							
Project Sub Head	**Project Cost Center No. (PCC No.)**	**(Name of Building Work or External Work)**	**As per sanction Estimate**	**As per Revised /Actual**	**Cost**		**Remark**
			Amount	**Amount**	**Excess**	**Saving**	
Building Work	B1	Building No. 1 :-					
	B2	Building No. 2 :-					
	BN	Building No. N :-					
Sub Total (B1 +B2+ .....BN) (All BuildingWorks)			**Σ above**	**Σ above**	**Σ above**	**Σ above**	
External Works	E1	Campus Services					
	E2	Campus Development					
Sub Total (E1+E2) (All External Works)							
NET PROJECT COST (B1 +B2+.....BN + E1+E2)							
Add Supervision Charges@.....							
GRAND TOTAL							
Add GST Charges@.....							
PROJECT COST							

(FORMAT-'RA'-C)

(I) CivilWork+(II)InternalServices+(III)InternalSpecialServices+(IV)Equipment&Furniture

REVISED COST CHART

NameofProject/DepartmentName/BuiltupArea Sq.m./CostinLakhs	
Sanction Project Cost (in Lacs)	
Project cost (As Per Tender (± %) Cost)	

Building Sub Heads	S. No.		Name of Cost center	As per sanction Estimate	As per Revised /Actual	Cost		Remark
				Amount	Amount	Excess	Saving	
(I) Civil Works	**1**		**Concrete in R.C.C. Frame Structure up to Plinth Level**					
		i	Excavation in Foundation					
		ii	Footing					
		iii	R.C.C. Work					
	2		Steel in R.C.C. Frame Structure work up to Plinth Level					
	3		Civil Work Other Than R.C.C Frame Structure upto plinth Level					
	4		Concrete in R.C.C. Frame Structure Above Plinth Level					
		i	R.C.C. Work					
		ii	Brick Work					
		iii	Door					
		iv	Windows					
		v	Plastering Work					
		vi	Flooring Work					
		vii	False Ceiling Work					
		viii	Painting Work					
	5		Steel in R.C.C.Frame Structure AbovePlinth Level					
	6		Civil Work Other Than R.C.C Frame Structure Above Plinth Level					
(II) Internal Services	7		Internal Water Supply & Plumbing and Sanitary					
		i	Water Supply Line					
		ii	Plumbing Work					
		iii	Sanitary Fittings					
	8		Internal Electrification					
		i	Electrical Line (Open & Concealed)					
		ii	Electrical Fixture					

REVISED COST CHART

NameofProject/DepartmentName/BuiltupArea Sq.m./CostinLakhs	
Sanction Project Cost (in Lacs)	
Project cost (As Per Tender (± %) Cost)	

Building Sub Heads	S. No.		Name of Cost center	As per sanction Estimate	As per Revised /Actual	Cost		Remark
				Amount	Amount	Excess	Saving	
(III) Internal Special Services	9		HVAC					
	10		Lifts/Elevators					
	11		Solar Water Heating					
	12		Solar Power					
	13		Fire Fighting					
	14		Fire Detection					
	15		Audio Visual System					
	16		Access Control					
	17		C.C.T.V.					
	18		Telephony (EPABX, Nurse call System)					
	19		Acoustic & False Ceiling					
	20		Interior Decoration & Art Work					
	21		Information Communication (IT)					
	22		Internal signage					
	23		Building Management System (BMS)					
	24		Miscellaneous Special Services such as Lighting Conductor, Façade light, Water Purifier, Incinerator, ifAny					
(IV) Equipment &Furniture	25		Furnitures					
	26		Lab Equipment					
	27		Medical Equipment & Accessories ,MGPS,OT,PTS,etc					
	i		MGPS					
	ii		OT (High End/Low End)					
	iii		PTS (Pneumatic Tube System)					
	iv		Medical Equipment & Accessories					
	28		Kitchen Equipments					
	30		Computer Hardware/Software					
TOTAL								

(FORMAT-'RA'-D)

(E1)CampusServices+(E2)CampusDevelopment

REVISED COST CHART

NameofProject/DepartmentName/BuiltupArea Sq.m./CostinLakhs	
Sanction Project Cost (in Lacs)	
Project cost (As Per Tender (± %) Cost)	

Building Sub Heads	S. No.		Name of Cost center	As per sanction Estimate	As per Revised /Actual	Cost		Remark
				Amount	Amount	Excess	Saving	
(E1) Campus Services	31		Rain Water Harvesting					
	32		External Water Supply					
	33		Sewerage Net Work					
	34		Effluent Treatment (Septic Tank / STP/ETP/Bio Digester)					
	35		External Feeder Line Extension					
	36		Substation (Outdoor/Indoor)					
	37		L.T. Distributions					
	38		Street Lighting					
	39		D.G. Sets					
(E2) Campus Developme nt	40		Dismantling and Site Clearing					
	41		Trees Cutting					
	42		Utilities shifting					
	43		Boundary Wall					
	44		Main Entrance Gate /Boundary wall Gate / Wicket Gate/Security Room					
	45		Roads (Internal Campus)					
	46		Road (External Approach)					
	47		Culverts					
	48		Drainage (Surface / Under Ground)					
	49		Landscaping (Horticulture, Plantations)					
	50		Art & Sculpture					
	51		Parking					
	53		Play Equipment					
	54		Statutory approval fees.					
	55		Miscellaneous Campus Development					
			TOTAL					

SCHEDULE OF FINISHES
(FORMAT - SOF CIVIL WORK)

Name of Project / Department Name / Built up Area Sq.m. / Cost in Lakhs	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.

S.No.	Space Details				Wall	Flooring	Skirting	Wall/Dado	Ceiling	Platfrom/Counter/ Almirah	Door				Windows					Ventilations			Type of Furniture	Remarks
	BUILDING NAME	WING/BLOCK	FLOOR	Space Name							Jamb &Sofit	Frames	Shutter	Fittings	Jamb &Sofit& Sill	Frame	Shutter	Fittings	Grill Work	Jamb &Sofit& Sill	Frames / Shutters/Loures	Grill Work		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

SOF -Toilet

SCHEDULE OF FINISHES
(FORMAT - SOF FOR TOILET WORK)

NameofProject/DepartmentName/BuiltupAreaSq.m./Cost inLakhs	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.

S.No.	Space Details				Male Toilet			Sanitary Fitings	Female Toilet		Sanitary Fitings	Handicapted Toilet			Sanitary Fitings	Remarks
	BUILDING NAME	WING/BLOCK	FLOOR	Space Name	W.C. (Indain /Europan)	Urinal/Urinal Partitions	Wash Basin /Counter/Sink	Premium/ Normal/CP /PTMT	W.C. (Indain /Europan)	Wash Basin /Counter/Sink	Premium/ Normal/CP /PTMT	W.C. (Indain /Europan)	Urinal/Urinal Partitions	Wash Basin /Counter/Sink	Premium/ Normal/CP /PTMT	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

SCHEDULE OF FINISHES
(FORMAT - SOF FOR ELECTRICAL WORK)

Name of Project / Department Name / Built up Area Sq.m. / Cost in Lakhs	CONSTRUCTION OF BUILDING PROJECT OF "PROJECT X" AT " PLACE Y" IN "DISTRICT Z" OF M.P.

S.No.	Space Details				Fans	Tube light	Bulbs	Down Lighter (Size/Type)	Remarks
	BUILDING NAME	WING/BLOCK	FLOOR	Space Name	5-Star rating energy saver 50 w/1200 mm / ornamental	T5/Led type/ CFL type	CFL/LED	T5/Led Type/ CFL Type	
1	2	3	4	5	6	7	8	9	10

M.P.P.W.D. SCHEDULE OF TESTING CHARGES (2013)

S.No.	Material of Test	Specified Test Performed	Reference of Test Method	Quantity of Sample	Testing Charges Per Test (Rs.)
1	2	3	3	4	5
1	Cement	Standard Consistency	IS:4031(Part-4) -1988	Min. 1 Sealed Bag of 50 Kgs.	385.00
		Initial And Final Setting Time	IS:4031(Part-5) - 1988		385.00
		Compressive Strength of Cement (For 2 Ages)	IS:4031(Part-6) - 1988		855.00
		Soundness Test By Le Chatelier Method	IS:4031(Part-3) -1988		620.00
		Fineness Test By Blains Air Permiability Method	IS:4031 (Part-2) - 1999		650.00
		Adultration by Bromoform Method	--		- 665.00
2	Aggregate	Sieve Analysis / Fineness Modulus (Coarse Aggregate)	IS:2386 (Part-1) - 1963	Min. 25 Kgs.	690.00
		Sieve Analysis / Fineness Modulus (Fine Aggregate)	IS:2386 (Part-1) - 1963	Min. 5 Kgs.	520.00
		Flakiness and Elongation Index	IS:2386 (Part-1) - 1963	Min. 50 Kgs.	420.00
		Water Absorption (Coarse Aggregate)	IS:2386 (Part-3) - 1963	Min. 5 Kgs.	265.00
		Water Absorption (Fine Aggregate)	IS:2386 (Part-3) - 1963	Min. 2 Kgs.	265.00
		Specific Gravity (Coarse Aggregate)	IS:2386 (Part-3) -1963	Min. 5 Kgs.	265.00
		Specific Gravity (Fine Aggregate)	IS:2386 (Part-3) -1963	Min. 2 Kgs.	265.00
		Aggregate Impact Value	IS:2386 (Part-4) - 1963	Min. 10 Kgs.	455.00
		Crushing Value	IS:2386 (Part-4) - 1963	Min. 20 Kgs.	455.00
		Bulk Density (Coarse Aggregate)	IS: 2386 (Part-3)-1963	Min. 30 Kgs.	200.00
		Bulk Density (Fine Aggregate)	IS: 2386 (Part-3)-1963	Min. 7 Kgs.	265.00
		Soundness (Coarse Aggregate) (5 Cycle)	IS: 2386 (Part-5) - 1963	Min. 10 Kgs.	2125.00
		Soundness (Fine Aggregate) (5 Cycle)	IS: 2386 (Part-5) - 1963	Min. 10 Kgs.	1875.00
		Delitrious Material (Coarse Aggregate)	IS: 2386 (Part-2) - 1963	Min. 10 Kgs.	705.00

Research Officer*
Central Research Laboratory
M.P. P.W.D., BHOPAL
Ph. No. 0755-2776494

T/C
31/7/19
संयुक्त परियोजना संचालक
कार्यालय परियोजना संचालक
लोक निर्माण विभाग (डी.आई.यू.)
निर्माण भवन, अरेरा हिल्स, भोपाल

6/7/19
मुख्य वास्तुविद
सह मुख्य अभियंता
(अनुसंधान विकास एवं रूपांकन)
लोक निर्माण विभाग, म.प्र., भोपाल

S.No.	Material of Test	Specified Test Performed	Reference of Test Method	Quantity of Sample	Testing Charges Per Test (Rs.)
1	2	3	3	4	5
		Delitirious Material (Fine Aggregate)	IS: 2386 (Part-2) - 1963	Min. 5 Kgs.	590.00
		Material Finer Than 75 μ	IS:2386 (Part-1) - 1963	Min 25 Kgs.	280.00
		10% Fine Value	IS: 2386 (Part-4) - 1963	Min 20 Kgs.	615.00
		Bulking of Fine Aggregate	IS:2386 (Part-3) -1963	Min. 10 Kgs.	460.00
		Los Angeles Abrasion	IS: 2386 (Part-4) - 1963	Min 20 Kgs.	560.00
		Silt Content by Volume	CPWD Speci. Vol. I	Min 2 Kgs.	390.00
		Stripping Value of Agreegate	IS: 6241 - 1971	Min. Agg. 5 Kgs + Bit. 1 Kg.	325.00
3	Bricks	Compressive Strength	IS:3495(Part-1) - 1992	Min. 5 Nos.	550.00
		Water Absorption	IS:3495(Part-2) - 1992	Min. 5 Nos.	290.00
		Efflorescence	IS:3495(Part-3) - 1992	Min. 5 Nos.	235.00
		Dimensions	IS:1077 - 1992	Min. 10 Nos.	180.00
4	Soil	Sieve Analysis	IS: 2720 (Part-4) - 1985	Min. 7 Kgs.	655.00
		Optimum Moisture Content and Maximum Dry Density	IS: 2720 (Part-7) - 1980 / IS: 2720 (Part-8) - 1983	Min. 10 Kgs.	1270.00
		Liquid Limit	IS 2720 (Part-5) - 1985	Min. 5 Kgs.	550.00
		Plastic Limit	IS 2720 (Part-5) - 1985	Min. 5 Kgs.	290.00
		Shrinkage Limit	IS 2720 (Part-6) - 1972	Min. 5 Kgs.	540.00
		C.B.R. (Soaked)	IS 2720 (Part-16) - 1987	Min. 35 Kgs.	2560.00
		C.B.R. (Un Soaked)	IS 2720 (Part-16) - 1987	Min. 35 Kgs.	1645.00
		Classification of Soil by Seive Analysis and Determination of Coefficient of Uniformity and Coefficient of Curvature	IS 1498 - 1970	Min 10 Kgs	1040.00
		Free Swelling Index	IS 2720 (Part-40) - 1977	Min. 1Kgs.	310.00

Research Officer*
Central Research Laboratory
M.P. P.W.D., BHOPAL
Ph. No. 0755-2776494

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निर्माण भवन, अरेरा हिल्स, भोपाल

6.7.13
मुख्य वास्तुविद
सह मुख्य अभियंता
(अनुसंधान विकास एवं रूपांकन)
लोक निर्माण विभाग, म.प्र., भोपाल

S.No.	Material of Test	Specified Test Performed	Reference of Test Method	Quantity of Sample	Testing Charges Per Test (Rs.)
1	2	3	3	4	5
		Specific Gravity	IS 2720 (Part-3) - 1980	Min. 1Kgs.	310.00
		Natural Moisture Content	IS 2720 (Part-2) -1973	Min. 1 Kgs.	310.00
5(A)	Bitumen	Penetration	IS: 1203 - 1978	Min. 1 Kgs.	480.00
		Absolute Viscosity	IS:1206 (Part-2)	Min. 1 Kgs.	540.00
		Ductility	IS:1208 - 1978	Min. 1 Kgs.	800.00
		Specific Gravity	IS:1202 -1978	Min. 1 Kgs.	295.00
		Softening Point	IS:1205 - 1978	Min. 1 Kgs.	480.00
		Flash Point & Fire Point	IS:1209 - 1978	Min. 1 Kgs.	480.00
		Rate of Spray of Binder	SP: 11-1976	--	365.00
5(B)	Bitumen Mix	Bitumen Binder Content	SP: 11-1988, ASTM D 2172.	Min. 5 Kgs.	1115.00
		Marshal Stability	ASTM D 1559	Min 20 Kgs.	1540.00
		Bituminus Mix Design (Job Mix Formula for Bituminus Mix) by Marshal Stability Method	IRC 29- 1988	Min Bit. 20 Kgs + Agg. 50 Kgs.	10835.00
6	Steel	Determination of Yeild Strength, Tensile Strength and Elongation of Steel Bar of All Diameter with the Help of UTM	IS 1608: 2005	3 Nos. of Min. 1000 mm Long	2170.00
		Weight Per Meter	IS:1786 - 2008	3 Nos. of Min 1000 mm Long	235.00
		Bend & Rebend Test	IS:1599 - 1985, IS:1786 - 2008	3 Nos. of Min 1500 mm Long	875.00
7	Concrete	Mix Design Upto M25 Grade	IS 10262-1982, IS: 456, SP 23	Min. Cement 50 Kgs + Sand 80 Kgs + Agg (10 mm) 80 Kgs. +Agg (20 mm) 120 Kgs.	6890.00

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Central Research Laboratory
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कार्यालय परियोजना संचालक
लोक निर्माण विभाग (पी आई यू)
निर्माण भवन, अरेरा हिल्स, भोपाल

6.7.13
मुख्य वास्तुविद
एवं मुख्य अभियंता
(अनुसंधान विकास एवं मूल्यांकन)
लोक निर्माण विभाग, म.प्र., भोपाल

No.	Material of Test	Specified Test Performed	Reference of Test Method	Quantity of Sample	Testing Charges Per Test (Rs.)
1	2	3	3	4	5
		Mix Design Above M25 Grade	IS 10262 - 1982	Min. Cement 50 Kgs. + Sand 80 Kgs + Agg (10 mm) - 80 Kgs. + Agg (20 mm). 120 Kgs.	7860.00
		Compressive Strength of Cube (15x15x15 Cms)	IS 516 - 1959	3 Nos.	380.00
		Compresive Strength of Cylinderical Concrete Specimen	IS 516 - 1960	3 Nos.	380.00
		Determination of Ratio of Cement, Sand and Aggregate in Hardened Concrete.	IS: 1199 - 1959	Min. Concerete: 10 Kgs. + Cement Used: 1Kg. + Sand Used 1 kg.	3880.00
8	Cement Mortar	Determination of Ratio of Cement and Sand in Hardened Mortar	IS: 1199 - 1959	Min. Mortar: 5 Kgs. + Cement Used: 1Kg. + Sand Used 1 kg.	3500.00
9	Concre Paver Block	Water Absorption	IS:15658:2006	3 Nos.	290.00
		Abrasion Resistance Value	IS:15658:2006	8 Nos.	1035.00
		Compressive Strength	IS:15658:2006	8 Nos.	495.00
10	Concrete / Terrazo / Chequered Flooring Tiles	Water Absorption	IS 1237-1980 / IS 13801	6 Nos.	290.00
		Resistance to Wear	IS:1237-1980 / IS 13801	6 Nos.	800.00
		Wet Transverse Strength	IS:1237-1980 / IS 13801	6 Nos.	480.00
11	Glazed Ceramic Vetrified Tiles	Water Absorption	IS:13630 (Part-2)-2006	5 Nos.	290.00
		Modulus Of Rupture	IS 13630 (Part-6)-2006	5 Nos.	480.00
		Scratch Hardness According To MOHS	IS:13630 (Part-13)-2006	3 Nos.	760.00
		Dimension/ Surface Quality	IS:1360 (Part-1)-2006	10 Nos.	265.00
		Abrasion	IS 13630 (Part 11): 2006	15 Nos.	800.00
		Chemical Resistance For Stains	IS:13630 (Part-2)-2006	5 Nos.	690.00

Research Officer*
Central Research Laboratory
M.P. P.W.D., BHOPAL
Ph. No. 0755-2776494

T/C
31/12/18
मुख्य परियोजना संचालक
कार्यालय परियोजना संचालक
लोक निर्माण विभाग (पी.आई.यू.)
निर्माण भवन, अरेरा हिल्स, भोपाल

6.7.13
मुख्य वास्तुविद
सह मुख्य अभियंता
(अनुसंधान विकास एवं रूपांकन)
लोक निर्माण विभाग, म.प्र., भोपाल

S.No.	Material of Test	Specified Test Performed	Reference of Test Method	Quantity of Sample	Testing Charges Per Test (Rs.)
1	2	3	3	4	5
12	Marble/Granite / Kota Stone	Scratch Hardness According To MOHS	IS:13630 (Part-13)-2006	3 Nos.	760.00
13	Timber	Moisture Content	IS:1708 (Part-1) - 1986	1 No. of 5 x 5 x 2.5 Cms.	415.00
		Flush Door End Emersion Test	IS: 4020 (Part-13) -1986	1 Door Shutter	520.00
		Flush Door Knife Test	IS: 4020 (Part-14) -1986	1 Door Shutter	520.00
		Flush Door Adhesion Test	IS: 4020 (Part-15) - 1986	1 Door Shutter	520.00
14	Hollow Concrete Block	Compresive Strength	IS : 2185 (Part I) - 1979	8 Nos.	495.00
		Water Absorption	IS : 2185 (Part I) - 1979	3 Nos.	290.00
		Block Density	IS : 2185 (Part I) - 1979	3 Nos.	345.00
15	Field Test	Plate Load Test (SBC) (Without Lab Conveyance)	IS: 1888 - 1982	--	15225.00
		Standard Penetration Test (Without Lab Conveyance)	IS: 2131-1981	--	8330.00
		Field density by Sand Replacement Method (Without Lab Conveyance)	IS: 2720 (Part-28) - 1974	--	790.00
		Field density by Core Cutter Method (Without Lab Conveyance)	IS-2720 (Part 29) - 1975	--	420.00
		Slump Test (3 Trials)	IS. 1199 - 1959	--	380.00
		3 Nos. Core of 15 Cms Dia and Max 30 Cms in Length Obtained from Cutting of Bituminus Pavement (Without Lab Conveyance)	--	--	3000.00
		3 Nos. Core of 15 Cms Dia and Max 30 Cms in Length Obtained from Cutting of Concrete Pavement (Without Lab Conveyance)	--	--	3000.00
16	Collection of Sample	Upto 8 Kms (Without Lab Conveyance)	--	--	800.00
		Beyond 8 Kms (Without Lab Conveyance)	--	--	1600.00

Research Officer *
Central Research Laboratory
M.P. P.W.D., BHOPAL
Ph. No. 0755-2776494

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31/1/13
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निर्माण भवन, अरेरा हिल्स, भोपाल

6.2.13
मुख्य वास्तुविद्
सह मुख्य अभियंता
(अनुसंधान विकास एवं रूपांकन)
लोक निर्माण विभाग, म.प्र., भोपाल


JABALPUR


UJJAIN


DATIA


REWA


CHHINDWARA



DATIA